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Extract from "Introduction to Quantum Information Theory"

by Armin Tavakoli and Gabriele Cobucci

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Introduction to Quantum Information Theory

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CONTENTS

Dedication	4
Acknowledgments	5
Preface	5
Foreword	6
Some useful preliminary relations	7
I Basics of quantum theory	8
1. Elementary concepts	9
1.1. States, transformations and measurements	9
1.2. The qubit	11
1.3. Density operators	13
1.4. Purity	14
1.5. Bloch sphere	15
2. Composition and distance measures	18
2.1. Purifications	19
2.2. Distance measures	19
3. Quantum measurements	24
3.1. Projective measurements	24
3.2. Positive operator-valued measures	26
3.3. Quantum state discrimination	28
3.4. Naimark dilation	30
II Core topics	32
4. Entanglement	33
4.1. Entanglement in pure states	33
4.2. Local operations and classical communication	35
4.3. Entanglement in mixed states	38
4.4. Quantifiers	40
4.5. Entanglement witness	41
4.6. Three-particle entanglement	43

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5. Bell nonlocality	45
5.1. The Einstein-Podolsky-Rosen paradox	45
5.2. Local hidden variable theories	45
5.3. Violating Bell inequalities	49
5.4. Entanglement versus nonlocality	51
5.5. No-signaling theories	53
6. Quantum information primitives	56
6.1. Dense coding	56
6.2. Quantum teleportation	58
6.3. Entanglement swapping	59
6.4. Random access code	62
6.5. Quantum cloning	63
7. Quantum channels and operations	65
7.1. Completely positive trace-preserving maps	65
7.2. Choi representation	66
7.3. Kraus representation	68
7.4. Schrödinger and Heisenberg picture	69
7.5. Stinespring representation	70
7.6. Quantum instruments	71
7.7. Information-disturbance relations	73
8. Entropic concepts	76
8.1. Shannon entropy	76
8.2. Quantum entropy	78
8.3. Data compression	81
8.4. Channel capacity	84
III Specialised topics	89
9. Device-independent certification	90
9.1. The idea of device-independence	90
9.2. Certification of measurements	90
9.3. Certification of states	92
10. Quantum cryptography	95
10.1. Noiseless BB84 protocol	96
10.2. Noisy BB84 protocol	97
10.3. Eavesdropping attacks	98
10.4. Entanglement-based protocols	100
11. Markovian master equations	102
11.1. The Lindblad master equation	102
11.2. Lindblad equation from Kraus representation	103
11.3. Spontaneous emission	106
11.4. Steady state properties	107
12. Quantum estimation theory	108
12.1. Estimators	108
12.2. Cramér-Rao bound and Fisher information	110
12.3. Quantum metrology	112
12.4. The Heisenberg limit	114
IV Exercises	115

A. Basics of quantum theory - Exercises	116
A.1. Elementary concepts	116
A.2. Purification and distance measures	117
A.3. Quantum measurements	119
B. Core topics - Exercises	139
B.1. Entanglement	139
B.2. Bell nonlocality	141
B.3. Quantum information primitives	144
B.4. Quantum channels and operations	147
B.5. Entropic concepts	150
C. Specialised topics - Exercises	197
C.1. Device-independence	197
C.2. Quantum cryptography	198
C.3. Markovian master equations	199
C.4. Quantum estimation theory	201
Index	220

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PREFACE

Purpose. The purpose of this textbook is to provide a broad, self-contained and above all student-friendly introduction to the field of quantum information. It is aimed at serving as an accessible entry-point to quantum information for readers with an academic background in physics or a neighbouring field, but no prior experience in this topic. Therefore, it is written with the intention of neither being too lengthy nor too technical, and to focus on the essential concepts and results as opposed to providing a comprehensive coverage. We hope that this textbook will contribute toward facilitating education in quantum information.

Quantum information and quantum computation emerged side-by-side in the 1990s. Now, in the mid 2020s, they have matured in increasingly independent directions and are often seen as modules within the more general context of quantum science and technology. In our textbook, we have chosen to not discuss quantum computation or the broader matters within quantum technology but to focus on the theoretical aspects of quantum information. For the larger scope, there are several well-written books that they interested reader may consult.

Structure. This textbook is divided into four parts.

1. *Basics of quantum theory*
Introduces quantum theory and provides elementary tools and concepts.
2. *Core topics*
Covers the essential topics in quantum information theory.
3. *Specialised topics*
An introduction to more specialised contemporary topics at the frontier of the field.
4. *Exercises*
Provides a number of exercises and detailed solutions for every chapter.

Throughout the book, important concepts are highlighted in green boxes (G) and important results are highlighted in red boxes (R). Blue boxes (B) provide the interested reader with a glance beyond the main focus of this book. Each chapter ends with a brief bullet point summary of its take-home message. Sometimes we also suggest references for further reading. The exercises are labelled by one of three colours. Exercises marked ● are direct calculations based on discussed results (but they can be lengthy). They are intended to create familiarity with the material and routine for calculation. Those marked ● require more than superficial understanding (you need to think). They are intended to encourage independent thinking and sometimes to educate the reader beyond the discussion in the main text. Finally, exercises marked ● are meant to be challenging (you need to Think). Sometimes, omitted details of arguments and theorems in the main text are given as step-by-step exercises for the reader.

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Foreword

WHEN I was an undergraduate student in the 1970's, I had never heard of information theory. At that time, it was already a well-established concept in mathematics, but most physicists considered it useless for physics. I had also never heard of entanglement and the bizarre correlations it can generate. Indeed, the very notion of *quantum information*, namely the use of these fundamental and striking features of quantum mechanics to deal with information, did not even exist.

My first encounter with concepts such as entanglement and Bell inequalities came when, as a PhD student, I was reading papers during holidays. The existing literature was limited and rather sparse, but I was totally fascinated. At the time there was a great deal of scepticism about these topics and the study of the foundational aspects of quantum mechanics was just an obscure branch of physics, more of a curiosity than anything useful. Professors used to say “Entanglement is as elusive as a dream, as soon as you think about it, it’s gone; you should not work on that”. The general tendency was to avoid these fundamental questions: “Bohr had solved it all” was the slogan.

This continued until 1991, when a young polish PhD student at Oxford University realised that the bizarre quantum correlations generated by entanglement could actually be useful for cryptography. Suddenly, the “elusive” entanglement and Bell inequalities became of great interest: they were not just a curiosity, but tools with enormous potential applications. It was the beginning of what is known as the *second quantum revolution*. A few years later, quantum teleportation was invented and the power of quantum computers understood, paving the way for the first steps toward what we now call *quantum technology*.

What do we mean by quantum information, and what makes it so special? Perhaps we should first ask ourselves what *information* is, and that is not an easy question to answer. After some thought, I suspect that many clever people would suggest, as a good starting point, that information is something one can copy. Well, quantum information is something that one cannot copy! And a full understanding of what it really is goes far beyond this observation...

Just like *classical information*, quantum information can be encoded, transmitted and processed. Yet, it can also exhibit phenomena that defy classical intuition: superposition of different states, mysterious links across distances due to entanglement and teleportation are just a few examples. But the study of quantum information has far more to reveal than its powerful practical applications! It sheds light on the question at the heart of all physics: how does Nature do it?

If you find the above paragraph intriguing, possibly even exciting, then you hold in your hands the right book. Read it, study it, and read it again. It introduces fascinating science and is an excellent book with which to begin discovering quantum information theory.

Quantum information is a young science, born from the synergy between classical information theory and the foundations of quantum physics - two of the main scientific revolutions of the 20th century. Today, one talks a lot about the practical power of quantum technologies and entanglement, forgetting that the most interesting concepts lie in-between. Therefore, allow me to advise all readers to keep a good balance between the technology on one side and the mathematical formalism on the other side. This is advice that the author, Armin Tavakoli, my former student, has heard more than once.

October 2025 — Nicolas Gisin

SOME USEFUL PRELIMINARY RELATIONS

- **Orthonormality**

$$\langle e_i | e_j \rangle = \delta_{i,j}, \quad \text{for orthonormal vectors } \{|e_i\rangle\}_i.$$

- **Completeness**

$$\sum_i |e_i\rangle\langle e_i| = \mathbb{1}, \quad \text{for an orthonormal basis } \{|e_i\rangle\}_i.$$

- **Trace**

$$\text{tr}(A) = \sum_i \langle i | A | i \rangle, \quad \text{is the trace of a matrix } A.$$

- **Transpose**

$$A^T = \sum_{i,j} a_{ji} |i\rangle\langle j|, \quad \text{is the transpose of a matrix } A = \sum_{i,j} a_{ij} |i\rangle\langle j|.$$

- **Projectivity**

$$\Pi^2 = \Pi, \quad \text{for a projection matrix } \Pi.$$

- **Hermiticity**

$$A = (A^*)^T = A^\dagger, \quad \text{for a Hermitian matrix } A.$$

- **Unitarity**

$$UU^\dagger = U^\dagger U = \mathbb{1}, \quad \text{for a finite-dimensional unitary matrix } U.$$

- **Spectral decomposition**

$$A = \sum_k \lambda_k |\lambda_k\rangle\langle \lambda_k|, \quad \text{for an Hermitian matrix } A$$

where $\{|\lambda_k\rangle\}_k$ and $\{\lambda_k\}_k$ are the eigenvectors and eigenvalues of A , respectively.

- **Polar decomposition**

$$A = UP, \quad \text{for any square-matrix } A, \text{ some unitary } U \text{ and } P = \sqrt{A^\dagger A}.$$

- **Convex combination**

$$y = q_1 x_1 + \cdots + q_n x_n, \quad \text{for } \{x_1, \dots, x_n\} \text{ in a vector space and } \{q_i \geq 0 : q_1 + \cdots + q_n = 1\}.$$

- **Convex function**

$$f \text{ is a convex function if } f(px + (1-p)y) \leq pf(x) + (1-p)f(y), \quad \forall p \in [0, 1]$$

- **Cauchy-Schwarz inequality**

$$|\langle v | w \rangle|^2 \leq \langle v | v \rangle \langle w | w \rangle, \quad \text{for any two vectors } |v\rangle \text{ and } |w\rangle.$$

- **Cauchy-Schwarz inequality for matrices**

$$|\text{tr}(A^\dagger B)|^2 \leq \text{tr}(A^\dagger A) \text{tr}(B^\dagger B), \quad \text{for any two matrices } A \text{ and } B.$$

Part I**Basics of quantum theory**

1. ELEMENTARY CONCEPTS

To pave the way for our journey into the field of quantum information, we must first lay the necessary groundwork. That is the purpose of this Chapter. Here, we introduce the basic concepts from quantum theory that we will rely on throughout this book.

1.1. States, transformations and measurements

Quantum theory takes place in *Hilbert spaces*. These spaces are usually denoted by $\mathcal{H}$. A Hilbert space is a complex-valued complete¹ vector space equipped with an inner product. This means that the vectors can have complex-number entries and the inner product ensures that we can talk about angles and distances inside the space. To our Hilbert space we can associate a dimension, $d = \dim(\mathcal{H})$, which is simply the number of complex-valued coordinates needed to specify a point. To make it concrete, we represent a vector in a d -dimensional Hilbert space as a column matrix with respect to a selected basis,

$$|v\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_d \end{pmatrix}, \quad (1.1)$$

where each α_i is a complex number. Hence, the Hilbert space is that which describes tuples of d complex numbers, i.e. $\mathcal{H} = \mathbb{C}^d$. The vector $|v\rangle$ in Hilbert space is called a *ket*. In the standard bra-ket notation of quantum theory, we associate to each ket another vector, $\langle v|$, called a *bra*. It is represented by the Hermitian conjugate of the ket, i.e.

$$\langle v| = |v\rangle^\dagger = (\alpha_1^* \ \alpha_2^* \ \cdots \ \alpha_d^*). \quad (1.2)$$

The Hermitian conjugate means a transposition followed by an element-wise conjugation, as shown above.

The reason that we are interested in these Hilbert space vectors is because of the first postulate of quantum theory. It tells us how vectors are used to describe the state of a quantum system.

G1: Postulate - Quantum states

The state of a closed quantum system is represented by a ray in a Hilbert space. A ray is defined by a normalised vector up to a global phase factor.

The postulate says that states are *normalised*. This means that the inner product of a state with itself is unity, namely

$$1 = \langle v|v\rangle = (\alpha_1^* \ \alpha_2^* \ \cdots \ \alpha_d^*) \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_d \end{pmatrix} = |\alpha_1|^2 + |\alpha_2|^2 + \dots + |\alpha_d|^2. \quad (1.3)$$

The reason why states must be normalised in quantum theory is that this property is needed later when we introduce measurements. For this brief moment, let us accept it. A consequence is that we can take any vector and simply normalise it to obtain a quantum state: if $|v\rangle$ is some vector then $|\tilde{v}\rangle \equiv \frac{|v\rangle}{\sqrt{\langle v|v\rangle}}$ is a state because it is now normalised: $\langle \tilde{v}|\tilde{v}\rangle = 1$.

The state postulate also says that quantum states are defined by *rays* rather than vectors. Rays are vectors too, but an equivalence class of them in which global phases are considered irrelevant. In other words, if $|\psi\rangle$ and $|\phi\rangle$ both are normalised vectors, but related through a relation of the type $|\psi\rangle = e^{i\varphi}|\phi\rangle$, for some phase φ , then we consider $|\psi\rangle$ and $|\phi\rangle$ to be the same ray and therefore the same state. The reason why we wish to ignore global phase factors when describing quantum states is that they are physically irrelevant, in the sense that they will not lead to any measurable consequences. Soon, we will be ready to motivate this claim.

Now that we know what a state is, we can ask another elementary question. How are states transformed into other states? The answer is given by the second postulate of quantum theory.

¹ Completeness means that the limit of every Cauchy sequence in $\mathcal{H}$ also belongs to $\mathcal{H}$. One can think of this as there being “no holes” in the space. The reason we need this property is to be able to deploy the tools of calculus.

G2: Postulate - Quantum evolution

The evolution of a quantum state of a closed system is described by a unitary transformation.

The evolution postulate tells us that if $|\psi\rangle$ and $|\phi\rangle$ belong to the same Hilbert space and we want to evolve the former into the latter, there exists a unitary operator, U , that does the job², i.e. $|\phi\rangle = U|\psi\rangle$. Any complex-valued operator U is called *unitary* if³

$$UU^\dagger = U^\dagger U = \mathbb{1}. \quad (1.4)$$

The key feature of a unitary operator is that it preserves the inner product. To see this, simply compare the inner product between two states $|\psi\rangle$ and $|\phi\rangle$ to that of their evolution $U|\psi\rangle$ and $U|\phi\rangle$,

$$(\langle\psi|U^\dagger)(U|\phi\rangle) = \langle\psi| \underbrace{(U^\dagger U)}_{=\mathbb{1}} |\phi\rangle = \langle\psi|\phi\rangle. \quad (1.5)$$

Why is this important for quantum theory? Because, if the evolution would not preserve inner product then U would take a state $|\psi\rangle$ to an unnormalised vector $U|\psi\rangle$, which would therefore no longer be a state. Hence, unitaries are the general way to transform states into other states.

To keep building our theory, we need to know how observers can read out physical properties from a state. In other words, we must introduce measurements to quantum theory. It is achieved through another postulate.

G3: Postulate - Quantum measurement

A measurement on a closed quantum system is described by a complete set of orthogonal projectors, $\{E_a\}_a$. The probability of observing outcome a when measuring state $|\psi\rangle$ is given by *Born's rule*,

$$p(a) = \langle\psi|E_a|\psi\rangle. \quad (1.6)$$

When observing outcome a the state is updated to

$$|\psi_a\rangle = \frac{E_a|\psi\rangle}{\sqrt{\langle\psi|E_a|\psi\rangle}}. \quad (1.7)$$

Thus, for each possible outcome of a measurement, we associate a corresponding measurement operator E_a . The postulate tells us that the measurement operator must be *projective*. It means that $E_a^2 = E_a$. The only operators that satisfy this are those whose eigenvalues are only zeros and ones. Consequently, projective operators are positive semidefinite, $E_a \geq 0$. This is important because it ensures that the probabilities in Eq (1.6) are non-negative. Furthermore, the postulate stipulates that the collection of measurement operators must be *complete*. Completeness means that

$$\sum_a E_a = \mathbb{1}. \quad (1.8)$$

The reason this property is needed is to guarantee that the sum of the probabilities that we assign to all the possible outcomes of the measurement equals one, i.e. that if we perform the measurement one of the outcomes always happens. This is essential for the physical interpretation to make sense. We can see it directly from Born's rule since

$$\sum_a p(a) \stackrel{(1.6)}{=} \sum_a \langle\psi|E_a|\psi\rangle = \langle\psi| \underbrace{\sum_a E_a}_{=\mathbb{1}} |\psi\rangle = \langle\psi|\psi\rangle \stackrel{(1.3)}{=} 1. \quad (1.9)$$

This also reveals why the quantum state postulate insisted on quantum states being normalised: we need it here in the last step to ensure that the probabilities sum to one. Another relevant consequence of the quantum measurement postulate is that

² In introductory quantum mechanics textbooks, we learn that time evolution is obtained from the Schrödinger equation and generated by the system's Hamiltonian according to $U = \exp(-iHt)$, where we have set $\hbar = 1$. In our context, we work directly with U instead of H .

³ In countable-dimensional Hilbert spaces, the property $UU^\dagger = \mathbb{1}$ is equivalent to $U^\dagger U = \mathbb{1}$ but in uncountably infinite-dimensional Hilbert spaces, this is no longer always the case. We will not deal with such cases here.

measurements cannot have more outcomes than the Hilbert space dimension. This follows from the fact that one cannot have more than d orthogonal directions in a d -dimensional Hilbert space. Furthermore, the measurement postulate also explains why quantum state postulate speaks about rays instead of normalised vectors. We previously claimed that global phases are physically irrelevant and this is now seen from Born's rule: if we were to append to a state a global phase factor, i.e. $|\psi\rangle \rightarrow e^{i\varphi}|\psi\rangle$, then it does not affect the probability of the possible outcomes because $p(a) = \langle\psi|e^{-i\varphi}E_a e^{i\varphi}|\psi\rangle = \langle\psi|E_a|\psi\rangle$.

In addition, the measurement postulate tells us that upon observing the outcome a , the state of our system jumps from $|\psi\rangle$ just before the measurement to $|\psi_a\rangle$ just after the measurement. In Eq (1.7) we see that the new state is the projection of $|\psi\rangle$ onto the relevant projector E_a , which we have re-normalised to ensure that it is still a state. However, once the measurement has been performed and the post-measurement state $|\psi_a\rangle$ is obtained, the measurement can be repeated without changing the state any further. In other words, the update of $|\psi_a\rangle$ is again $|\psi_a\rangle$. This becomes a consequence of the measurement postulate because from Born's rule the probability of again observing outcome a is

$$\langle\psi_a|E_a|\psi_a\rangle \stackrel{(1.7)}{=} \frac{1}{\langle\psi|E_a|\psi\rangle} \underbrace{\langle\psi|E_a E_a E_a|\psi\rangle}_{=E_a} = 1. \quad (1.10)$$

Hence, the state update in Eq (1.7) becomes just

$$\frac{E_a|\psi_a\rangle}{\sqrt{\langle\psi_a|E_a|\psi_a\rangle}} = E_a|\psi_a\rangle \stackrel{(1.7)}{=} \underbrace{E_a E_a}_{=E_a} \frac{|\psi\rangle}{\sqrt{\langle\psi|E_a|\psi\rangle}} \stackrel{(1.7)}{=} |\psi_a\rangle. \quad (1.11)$$

What we have seen here has a simple intuition: the first measurement has already projected $|\psi\rangle$ onto the space of E_a , so making yet another measurement guarantees the outcome a since the state now is already in the space of E_a .

Let us conclude with two remarks. Firstly, in contrast to most physics theories, we are building quantum theory through mathematical postulates without guidance from observation. The skeptical reader might for instance wonder ask why Nature would use Born's rule to determine the outcome of measurements. In Box B3 in Chapter 3 we briefly discuss this matter. Secondly, it is important to appreciate that the concepts defined here, namely rays, unitaries and complete sets of projectors, are fundamental only for closed (i.e. isolated) quantum systems. When we depart from these ideal conditions and venture into open systems, the description of quantum states, transformations and measurements will become richer and more powerful.

1.2. The qubit

It is important to appreciate that the postulates of quantum theory provide an abstract formalism for representing quantum states, transformations and measurements in real-life physical systems. The simplest quantum systems are those described by a ray a in two-dimensional Hilbert space⁴. These states are famously known as *qubits* (quantum bits). The qubit can be used in many ways: they can represent superpositions between the ground state and the excited state of a hydrogenic electron, the two possible paths taken by a quantum particles in the double-slit experiment, the horizontal and vertical polarisation of a photon, whether or not Cooper pairs have tunneled through a Josephson junction in a superconducting circuit, etc. Following tradition, we denote the quantum state vector by the Greek letter ψ and write the qubit as

$$|\psi\rangle = \underbrace{\begin{pmatrix} \alpha \\ \beta \end{pmatrix}}_{=|0\rangle} = \alpha \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{=|0\rangle} + \beta \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{=|1\rangle} = \alpha|0\rangle + \beta|1\rangle. \quad (1.12)$$

It is standard to use the notation $|i\rangle$ for the vector that is all zeros except at position i where the entry is one⁵, as illustrated in the above equation. The basis $\{|i\rangle\}_i$ is called the *computational basis*. To make sure $|\psi\rangle$ is a state, we need also to normalise our vector. Since α and β are complex numbers, let us write them in polar representation as $\alpha = r_0 e^{i\varphi_0}$ and $\beta = r_1 e^{i\varphi_1}$. Since the quantum state postulate in Box G1 tells us that global phases do not matter, let us multiply $|\psi\rangle$ with the global phase factor $e^{-i\varphi_0}$. Hence, our state becomes $|\psi\rangle = r_0|0\rangle + r_1 e^{i\varphi}|1\rangle$, where we put $\varphi = \varphi_1 - \varphi_0$. Next, we must normalise the state. This means that $r_0^2 + r_1^2 = 1$. We recognise this as the defining equation of a circle, so we can define $r_0 = \cos \frac{\theta}{2}$ and $r_1 = \sin \frac{\theta}{2}$ for some parameter θ . Thus, we have parameterised qubit states in the following convenient way,

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad (1.13)$$

⁴ Note that one-dimensional Hilbert space only contains a single ray, which is merely the number 1. No interesting physics happens here.

⁵ One of the quirks of quantum information theorists is that they, unlike most people, prefer to start counting from zero instead of from one.

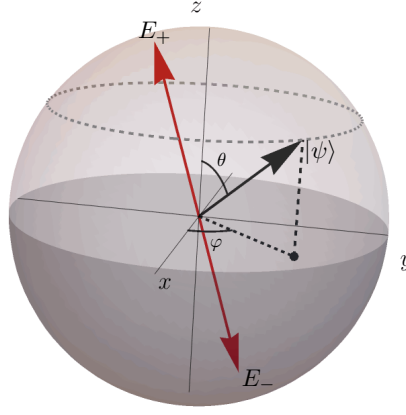


FIG. 1.1. Qubit states can be represented as vectors on and inside the Bloch sphere. States are therefore characterised by the pair of angles (θ, φ) as well as the length of the vector. A basis measurement is represented by two anti-podal unit-vectors (red).

where $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi]$. We observe that this is the mathematical definition of a unit-radius sphere, where θ is the polar angle and φ is the azimuthal angle; see the black arrow in Fig 1.1. This lets us geometrically think of qubits as points on a sphere specified by the pair of angles (θ, φ) . Since this sphere is important, it has its own name: the *Bloch sphere*.

We now look into how states, transformations and measurements can be interpreted on the Bloch sphere. To this end, we must first remind ourselves of the three Pauli operators,

$$\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |0\rangle\langle 1| + |1\rangle\langle 0|, \quad \sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = |1\rangle\langle 0| - i|0\rangle\langle 1|, \quad \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1|. \quad (1.14)$$

These operators are Hermitian and their eigenvalues are $+1$ and -1 . Since non-degenerate eigenvalues of Hermitian matrices are associated with orthogonal eigenvectors (see Exercise A.1.2), it means that each Pauli operator is the difference of two orthogonal projectors. For σ_Z since this is already given above by $\sigma_Z = |0\rangle\langle 0| - |1\rangle\langle 1|$. For σ_X and σ_Y , we have $\sigma_X = |+\rangle\langle +| - |-\rangle\langle -|$, where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ and $\sigma_Y = |+\rangle\langle +'| - |-\rangle\langle -'|$, where $|\pm'\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$. Hence, we can view the three sets $\{|+\rangle\langle +|, |-\rangle\langle -|\}$, $\{|+\rangle\langle +'|, |-\rangle\langle -'|\}$ and $\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$ as qubit measurements associated with σ_X , σ_Y and σ_Z respectively. They all satisfy the completeness relation (1.8). We can label the first and second outcome in each measurement as “+” and “−” respectively. Then, if we compute the expectation value of any one of the Pauli operators, σ_i , we obtain

$$\langle \sigma_i \rangle_\psi = \langle \psi | \sigma_i | \psi \rangle = p(+)-p(-), \quad (1.15)$$

where we used Born’s rule to interpret the expectation value as the probability difference between the two possible outcomes.

Let us now apply the Pauli operators to represent states on the Bloch sphere. For that, we compute all three expectation values for the state in Eq (1.13). A little simplification gives us

$$\langle \sigma_X \rangle_\psi = \langle \psi | \sigma_X | \psi \rangle = \sin \theta \cos \varphi, \quad \langle \sigma_Y \rangle_\psi = \langle \psi | \sigma_Y | \psi \rangle = \sin \theta \sin \varphi, \quad \langle \sigma_Z \rangle_\psi = \langle \psi | \sigma_Z | \psi \rangle = \cos \theta. \quad (1.16)$$

Then, we combine these three numbers into a vector,

$$\hat{n} \equiv (\langle \sigma_X \rangle, \langle \sigma_Y \rangle, \langle \sigma_Z \rangle) = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta). \quad (1.17)$$

This vector has unit-length because

$$|\hat{n}|^2 = \langle \sigma_X \rangle_\psi^2 + \langle \sigma_Y \rangle_\psi^2 + \langle \sigma_Z \rangle_\psi^2 = \sin^2 \theta \cos^2 \varphi + \sin^2 \theta \sin^2 \varphi + \cos^2 \theta = \sin^2 \theta (\cos^2 \varphi + \sin^2 \varphi) + \cos^2 \theta = 1, \quad (1.18)$$

where we used that $\sin^2 x + \cos^2 x = 1$. Hence, $\hat{n}$ is a coordinate tuple that points to the surface of the Bloch sphere. In other words, instead of working with the qubit state vector in Eq (1.12), we can work with the *Bloch vector* $\hat{n}$ to describe the state.

Because the two descriptions are equivalent, we can also think of qubit unitaries as operations that transform one Bloch vector into another Bloch vector. For example, the unitary $U = \exp(i\frac{\theta}{2}\sigma_Z)$ constitutes a rotation of angle θ on the Bloch sphere about the Z -axis (north-south pole). This is illustrated by the dashed circle in Fig 1.1. In general, as is shown in Exercise A.1.3, any qubit unitary can be written on the form

$$U = e^{i\varphi} (\cos \theta \mathbb{1} + i \sin \theta \hat{n} \cdot \vec{\sigma}), \quad (1.19)$$

where $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$. In this formula, φ is a global phase, while $(\cos \theta \mathbb{1} + i \sin \theta \hat{n} \cdot \vec{\sigma})$ is the expression for a clockwise rotation of angle 2θ about the $\hat{n}$ axis.

Also measurements can be accommodated in the Bloch sphere representation. Since qubits are two-dimensional, we focus on measurements that have two possible outcomes. Like in Eq (1.15) we call one outcome “+” and the other “−”. Following the measurement postulate, we represent them with projectors of the form $E_+ = |\phi_+\rangle\langle\phi_+|$ and $E_- = |\phi_-\rangle\langle\phi_-|$. According to the quantum measurement postulate in Box G3 the projectors must be orthogonal, namely $\langle\phi_+|\phi_-\rangle = 0$. Since the basis vectors $|\phi_+\rangle$ and $|\phi_-\rangle$ each can be thought of as a quantum state, we can associate to each a Bloch vector, $\hat{n}_+$ and $\hat{n}_-$ respectively. In Exercise A.1.4, we show that the condition $\langle\phi_+|\phi_-\rangle = 0$ is equivalent to the Bloch vector condition $\hat{n}_+ \cdot \hat{n}_- = -1$. The latter means that $\hat{n}_- = -\hat{n}_+$. Hence, when represented on the Bloch sphere, we have the somewhat counter-intuitive picture that orthogonal vectors correspond to anti-podal Bloch vectors, as is illustrated by the red pair of arrows in Fig 1.1.

1.3. Density operators

So far, we have only discussed closed quantum systems, but what if the system is not isolated? Answering this question will give us a more powerful toolbox. Here, we focus on quantum states, and delay the discussion of measurements and transformations to Chapter 3 and Chapter 7 respectively.

In an open system, an external influence can probabilistically select how to prepare our system. Consider that with probability p_i we generate a state $|\psi_i\rangle$. Given our ignorance of the label i , we must describe the state of our system that we obtain on average. To represent that, we consider the operator

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|. \quad (1.20)$$

The operator ρ is called a *density operator* (or commonly also density matrix). The probabilities $\{p_i\}_i$ need only to satisfy the basic requirements $p_i \geq 0$ and $\sum_i p_i = 1$, so we can say that ρ is a convex combination of states. Note that if we would remove the probabilistic choice of i , so that only one value of i , say i^* , would be possible, then $\rho = |\psi_{i^*}\rangle\langle\psi_{i^*}|$ becomes the projector onto the state $|\psi_{i^*}\rangle$, which is what we have seen for closed systems. Density operators have three defining properties.

- **Hermiticity.** If ρ is a density operator, then $\rho = \rho^\dagger$. To show this, we apply the Hermitian conjugate to Eq. (1.20) and obtain

$$\rho^\dagger = \left(\sum_i p_i |\psi_i\rangle\langle\psi_i| \right)^\dagger = \sum_i p_i (|\psi_i\rangle\langle\psi_i|)^\dagger = \sum_i p_i |\psi_i\rangle\langle\psi_i| = \rho. \quad (1.21)$$

- **Unit trace.** If ρ is a density operator then its trace is unity. This also follows from (1.20) since

$$\text{tr}(\rho) = \text{tr} \left(\sum_i p_i |\psi_i\rangle\langle\psi_i| \right) = \sum_i p_i \text{tr}(|\psi_i\rangle\langle\psi_i|) = \sum_i p_i = 1. \quad (1.22)$$

- **Positive semidefinite.** If ρ is a density operator then it is positive semidefinite. For positive semidefiniteness⁶, it must hold that $\langle\varphi|\rho|\varphi\rangle \geq 0$ for every vector $|\varphi\rangle$. The property holds because

$$\langle\varphi|\rho|\varphi\rangle = \langle\varphi| \left(\sum_i p_i |\psi_i\rangle\langle\psi_i| \right) |\varphi\rangle = \sum_i p_i \langle\varphi|\psi_i\rangle\langle\psi_i|\varphi\rangle = \sum_i p_i \underbrace{|\langle\varphi|\psi_i\rangle|^2}_{\geq 0} \geq 0. \quad (1.23)$$

In fact, every operator, ρ , that is Hermitian, has unit-trace and is positive semidefinite is a valid density operator. This follows from the fact that ρ has a spectral decomposition, namely $\rho = \sum_{i=1}^d \lambda_i |\lambda_i\rangle\langle\lambda_i|$, where $\{\lambda_i\}_i$ are the eigenvalues and $\{|\lambda_i\rangle\}_i$ are the eigenvectors. This has the form in Eq (1.20) since we can interpret $\{\lambda_i\}_i$ as a probability distribution.

Density operators can for all purposes replace rays as the description of the quantum state. Therefore, from now on when we speak of quantum states, we typically use density operators. Notice also that the global phases that we deemed irrelevant for rays do not appear for density operators since $e^{i\varphi}|\psi\rangle\langle\psi|e^{-i\varphi} = |\psi\rangle\langle\psi|$.

G4: Density operator representation

The state of a quantum system (open or closed) is fully described by its density operator.

⁶ Commonly positive semidefinite is a property defined only for Hermitian matrices. It is not the same as a generic operator having non-negative eigenvalues.

Having upgraded rays to density operators, we must similarly upgrade Born's rule to take as input a density operator instead of a ray. For this we can rely on the following intuition. Consider that we perform the measurement $\{E_a\}_a$ on ρ . Since we thought of ρ as an ensemble obtained from creating state $|\psi_i\rangle$ where the label i is drawn from a distribution p_i , the outcome of the measurement must be that obtained from applying Born's rule to $|\psi_i\rangle$ and averaging over the probabilities of the labels. The probability of observing outcome a becomes

$$p(a) = \sum_i p_i \langle \psi_i | E_a | \psi_i \rangle = \sum_i p_i \text{tr}(\langle \psi_i | E_a | \psi_i \rangle) = \sum_i p_i \text{tr}(|\psi_i\rangle \langle \psi_i | E_a) = \text{tr} \left(\underbrace{\sum_i p_i |\psi_i\rangle \langle \psi_i |}_{=\rho} E_a \right) = \text{tr}(\rho E_a), \quad (1.24)$$

where we used that the trace of a scalar is the same scalar, together with the cyclicity and linearity of the trace (see Exercise A.1.1). In this way, we have eliminated the rays from Born's rule in favour of the density operator. We can now write down Born's rule and a state update rule for density operators.

G5: Born's rule & state update

The probability of obtaining outcome a when performing the measurement $\{E_a\}_a$ on density operator ρ is

$$p(a) = \text{tr}(\rho E_a). \quad (1.25)$$

The updated state when observing outcome a is

$$\rho_a = \frac{E_a \rho E_a}{\text{tr}(\rho E_a)}. \quad (1.26)$$

We now see why the density operator must have its defining properties. The unit trace property ensures that probabilities sum to unity when performing a measurement, because $\sum_a p(a) = \text{tr}(\rho \sum_a E_a) = \text{tr}(\rho) = 1$. The positive semidefinite property guarantees that the probabilities are non-negative, because otherwise there would exist a projector E_a such that $\text{tr}(\rho E_a) < 0$.

1.4. Purity

The advantage in representing quantum states as density operators instead of as rays is that it permits us to include classical ignorance in the description of quantum states. States (i.e. density operators) that feature classical ignorance are called *mixed states* whereas states that do not (as in the closed picture) are called *pure states*. A common way to quantify purity is via the so-called purity function, $T(\rho)$. It is defined as

$$T(\rho) = \text{tr}(\rho^2) = \begin{cases} 1 & \text{if } \rho \text{ is pure} \\ < 1 & \text{if } \rho \text{ is mixed.} \end{cases} \quad (1.27)$$

To compute the purity, we use the spectral decomposition of the density operator, $\rho = \sum_{i=1}^d \lambda_i |\lambda_i\rangle \langle \lambda_i|$. By definition, the eigenvalues are non-negative and sum to unity. Using the orthogonality of the eigenvectors, we obtain

$$T(\rho) = \text{tr} \left(\left(\sum_{i=1}^d \lambda_i |\lambda_i\rangle \langle \lambda_i| \right)^2 \right) = \sum_{i,j} \lambda_i \lambda_j \text{tr}(|\lambda_i\rangle \langle \lambda_i| \lambda_j \langle \lambda_j|) = \sum_{i,j} \lambda_i \lambda_j \underbrace{|\langle \lambda_i | \lambda_j \rangle|^2}_{=\delta_{i,j}} = \sum_i \lambda_i^2. \quad (1.28)$$

The only way in which the right-hand-side can equal unity is if there is only one non-zero value of λ_i , which would mean that ρ is a projector onto a ray. Thus, as soon as classical ignorance is present in the state, some degree of mixedness is necessary. It then seems natural to ask how mixed a state possibly can be. In other words, what is the smallest value of the purity function? Answering this means solving the following optimisation problem

$$\min_{\{\lambda_i\}} \sum_{i=1}^d \lambda_i^2 \quad \text{s.t.} \quad \sum_i \lambda_i = 1, \quad \text{and} \quad \lambda_i \geq 0. \quad (1.29)$$

This problem can be solved by the method of Lagrange multipliers. Define $f = \sum_i \lambda_i^2 - \mu(\sum_j \lambda_j - 1)$. Then

$$\frac{df}{d\mu} = 0 \Rightarrow \sum_j \lambda_j = 1, \quad (1.30)$$

$$\frac{df}{d\lambda_i} = 0 \Leftrightarrow 2\lambda_i - \mu = 0 \Rightarrow \lambda_i = \frac{\mu}{2}. \quad (1.31)$$

We see that all λ_i are equal. Substituting into the first equation, we have $\frac{1}{2} \sum_i \mu = 1$ which implies $\mu = \frac{2}{d}$ and hence $\lambda_i = \frac{1}{d}$. For the purity function, this means that

$$T(\rho) \geq \frac{1}{d}, \quad (1.32)$$

and this is saturated if and only if the density operator only has eigenvalues $\frac{1}{d}$. This defines a unique state, because

$$\rho = \sum_{i=1}^d \frac{1}{d} |\lambda_i\rangle\langle\lambda_i| = \frac{1}{d} \underbrace{\sum_{i=1}^d |\lambda_i\rangle\langle\lambda_i|}_{=\mathbb{1}} = \frac{\mathbb{1}}{d}. \quad (1.33)$$

Therefore, the state is appropriately named the *maximally mixed state* and it has a fully classical interpretation: it represents flipping an unbiased d -sided coin and forgetting the result.

G6: Maximally mixed state

The maximally mixed state of dimension d is the density operator

$$\rho = \frac{\mathbb{1}}{d}. \quad (1.34)$$

The maximally mixed state illustrates a key property that concerns all mixed states: they can be realised in many different ways. Specifically, there exists infinitely many different selections of $\{p_i, |\psi_i\rangle\}_i$ that all lead to the same mixed state ρ . Also, there is no relationship between the dimension, d , and the number of states (indexed by i) used to realise the mixed state. This non-uniqueness feature is a key distinction between mixed states and pure states.

In Figure 1.2 we illustrate qualitatively the space of density operators. This space is convex, because if we take any two density matrices ρ and ρ' and mix them together by selecting the former with probability q and the latter with probability $1 - q$, we obtain the $q\rho + (1 - q)\rho'$ which is still Hermitian, positive semidefinite and has unit-trace, i.e. it defines a new density operator. Therefore, the interior of the space is only populated by mixed states while all pure states are located on the boundary. However, as seen in Fig 1.2, also some mixed states can in general be found on the boundary⁷. As we will see next, the case of qubits presents a simpler picture where all mixed states are located in the interior.

1.5. Bloch sphere

Equipped with the notion of mixed states, we can continue the discussion of qubit systems from Chapter 1.2 by asking how mixed states are represented on the Bloch sphere. To this end, consider a generic 2×2 density operator. Because density operators are Hermitian, we can write them as a linear combination of a basis of Hermitian operators. Such a basis is provided by the identity matrix together with the three Pauli operators (1.14). Hence,

$$\rho = c_0 \mathbb{1} + c_1 \sigma_X + c_2 \sigma_Y + c_3 \sigma_Z, \quad (1.35)$$

for some coefficients (c_0, c_1, c_2, c_3) . To make sure that this is a valid density operator, it must have unit trace and be positive semidefinite. For the unit-trace property, note that Paulis are traceless, which means that $c_0 = \frac{1}{2}$. To impose the positive semidefinite property, let us write the state as

$$\rho = \frac{1}{2} (\mathbb{1} + \vec{n} \cdot \vec{\sigma}), \quad (1.36)$$

⁷ Mixed states with a zero-determinant sit on the boundary because they have at least one zero-eigenvalue and can therefore by an infinitesimal perturbation in the direction of that associated eigenvector be pushed out of the space of density operators. For the special case of qubits, no such mixed states exist.

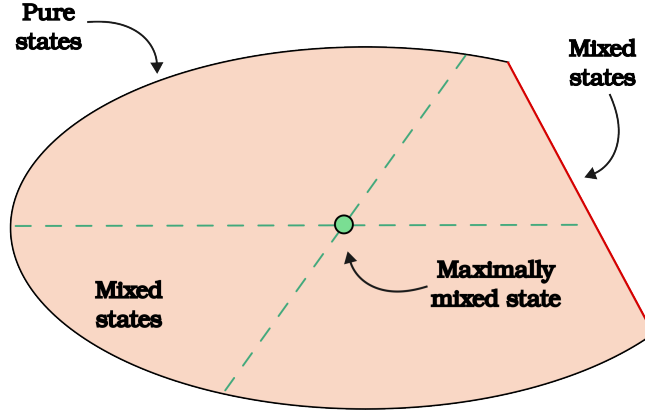


FIG. 1.2. The set of density operators is convex, because a convex combination of density operators yields another density operators. Furthermore, all pure states sit on the boundary, because they have unique realisations (1.20). Since every mixed state is a convex combination of pure states, it means that the space of density matrices is the convex hull of the pure states. Thus, the interior is only mixed states. One can think of the maximally mixed state as sitting in the “middle” of the set. However, there are still some mixed states on the boundary (even though these can be decomposed into convex combinations of pure states).

where $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$ and $\vec{n} = 2(c_1, c_2, c_3)$. We now need to select $\vec{n}$ so that for every vector $|\varphi\rangle$, it holds that $\langle\varphi|\rho|\varphi\rangle \geq 0$. This means that

$$\langle\varphi|\rho|\varphi\rangle \stackrel{(1.36)}{=} \frac{1}{2} + \frac{1}{2}\langle\varphi|\vec{n} \cdot \vec{\sigma}|\varphi\rangle \geq 0. \quad (1.37)$$

In other words, we must show that $\langle\varphi|\vec{n} \cdot \vec{\sigma}|\varphi\rangle \geq -1$. Notice now that $\vec{n} \cdot \vec{\sigma}$ is traceless and has two eigenvalues. Therefore, its eigenvalues must have the same magnitude but have opposite signs, implying $\vec{n} \cdot \vec{\sigma} = \lambda|\lambda_+\rangle\langle\lambda_+| - \lambda|\lambda_-\rangle\langle\lambda_-|$ where $\lambda \geq 0$ is the positive eigenvalue and $|\lambda_\pm\rangle$ are the positive and negative eigenvectors. We minimise $\langle\varphi|\vec{n} \cdot \vec{\sigma}|\varphi\rangle$ by selecting $|\varphi\rangle = |\lambda_- \rangle$, giving $\langle\varphi|\vec{n} \cdot \vec{\sigma}|\varphi\rangle = -\lambda$. This renders our positive semidefinite condition as $\lambda \leq 1$. To translate this into a constraint on the Bloch vector, $\vec{n}$, we somewhat excessively write $\lambda^2 = \langle\lambda_-|(\vec{n} \cdot \vec{\sigma})^2|\lambda_- \rangle$. If we simplify the operator $(\vec{n} \cdot \vec{\sigma})^2$ we obtain

$$(\vec{n} \cdot \vec{\sigma})^2 = \sum_{i,j=X,Y,Z} n_i n_j \sigma_i \sigma_j = \sum_i n_i^2 \underbrace{\sigma_i^2}_{=1} + \sum_{i \neq j} n_i n_j \sigma_i \sigma_j = |\vec{n}|^2 \mathbb{1} + \sum_{i < j} n_i n_j \underbrace{\{\sigma_i, \sigma_j\}}_{=0 \text{ if } i \neq j} = |\vec{n}|^2 \mathbb{1}. \quad (1.38)$$

Hence, we conclude that $\lambda^2 = \langle\lambda_-|(|\vec{n}|^2 \mathbb{1})|\lambda_- \rangle = |\vec{n}|^2$ and since $\lambda \leq 1$, the final result becomes $|\vec{n}| \leq 1$. This has given us the complete representation of qubit states as Bloch vectors.

G7: Bloch vector

Every qubit density operator is written

$$\rho = \frac{1}{2} (\mathbb{1} + \vec{n} \cdot \vec{\sigma}), \quad (1.39)$$

where $\vec{n}$ is called the Bloch vector. The space of Bloch vectors is fully characterised by

$$|\vec{n}| \leq 1. \quad (1.40)$$

Previously, we said that qubit rays (now pure qubit density operators) correspond to the shell of the Bloch sphere because they were characterised by Bloch vectors with a unit length, $|\vec{n}| = 1$. What we have seen now is that the mixed states occupy the interior of the sphere: their Bloch vectors have length $|\vec{n}| < 1$. To verify this, we compute the purity function and find

$$T(\rho) = \frac{1}{4} \text{tr} \left((\mathbb{1} + \vec{n} \cdot \vec{\sigma})^2 \right) = \frac{1}{4} \text{tr} \left(\mathbb{1} + 2\vec{n} \cdot \vec{\sigma} + \underbrace{(\vec{n} \cdot \vec{\sigma})^2}_{(1.38)} \right) = \frac{1}{2} + \frac{1}{4} \text{tr} \left(|\vec{n}|^2 \mathbb{1} \right) = \frac{1 + |\vec{n}|^2}{2}. \quad (1.41)$$

The maximally mixed state has the shortest Bloch vector, namely $\vec{n} = (0, 0, 0)$, and it therefore sits at the origin of the Bloch sphere. The longer is the Bloch vector, the more pure is the state.

Now that we know that qubit density matrices can be geometrically represented as a ball, one might ask whether density matrices of a higher dimension than two somehow also correspond to an elegant geometric picture. In short, the answer is negative, and the reason is discussed in Box B1. Now, and on many other occasions, we will see that the properties of qubits are a convenient accident in quantum theory.

B1. A glance beyond

State space beyond qubits

We have seen that the space of qubit density operators is a unit-radius ball. How can we represent the geometry of density operators when the dimension is greater than two? In Exercise A.1.6, it is shown that the real-space dimension of the state space is $D = d^2 - 1$; note that for qubits ($d = 2$) it returns $D = 3$ which matches the three-dimensional Bloch sphere. A naive guess could therefore be that the state space is a D -dimensional ball. Unfortunately, the situation is much more complicated.

We can always write a density operator over d -dimensional Hilbert space as $\rho = \frac{1}{d} (\mathbb{1} + \vec{n} \cdot \vec{\sigma})$, where $\vec{n}$ is the D -dimensional counterpart to the Bloch vector. Here, $\vec{\sigma}$ is an orthonormal set of D Hermitian and traceless $d \times d$ operators satisfying $\text{tr}(\sigma_i \sigma_j) = d \delta_{i,j}$. By construction, ρ is Hermitian and $\text{tr}(\rho) = 1$. Therefore, to make a valid density operator, we need only to select $\vec{n}$ so that ρ is positive semidefinite. For that, consider the purity of the density operator. It becomes

$$\text{tr}(\rho^2) = \frac{1}{d^2} \text{tr}((\mathbb{1} + \vec{n} \cdot \vec{\sigma})^2) = \frac{1}{d^2} \left(\underbrace{\text{tr}(\mathbb{1})}_{=d} + 2 \underbrace{\text{tr}(\vec{n} \cdot \vec{\sigma})}_{=0} + \text{tr}((\vec{n} \cdot \vec{\sigma})^2) \right) = \frac{1}{d} + \frac{1}{d^2} \text{tr}((\vec{n} \cdot \vec{\sigma})^2) = \frac{1 + |\vec{n}|^2}{d}, \quad (1.42)$$

where in the last step we used that $\text{tr}((\vec{n} \cdot \vec{\sigma})^2) = \sum_i n_i^2 \text{tr}(\sigma_i^2) + \sum_{i \neq j} n_i n_j \text{tr}(\sigma_i \sigma_j) = d |\vec{n}|^2$. To make a valid density operator the purity cannot exceed one. This implies that $|\vec{n}| \leq \sqrt{d-1}$, which indeed describes a D -dimensional ball. However, we have not shown that this is the state space, but only that whatever the state space actually is it must be contained inside this ball. The case of $d = 2$ is special because then the only non-trivial equation is the purity condition $\text{tr}(\rho^2) \leq 1$. For $d > 2$ we must consider the characteristic polynomial $\det(z\mathbb{1} - \rho) = \sum_{k=0}^d (-1)^k a_k z^{d-k}$, where $\{a_k\}_k$ are the coefficients. By Descartes' rule of signs, the positivity of ρ means that $a_k \geq 0$ for every k . In general, solving all these equations is challenging and a visual representation of their solutions is not convenient.

Chapter summary.

- Quantum systems are represented on Hilbert spaces. For closed systems, states correspond to rays, $|\psi\rangle$. Measurements correspond to a set of projection operators, $\{E_a\}_a$, that satisfies the completeness relation $\sum_a E_a = \mathbb{1}$. Transformations correspond to operators, U , that satisfy the unitarity property $UU^\dagger = U^\dagger U = \mathbb{1}$.
- Born's rule is $p(a) = \langle \psi | E_a | \psi \rangle$. It gives the probability, $p(a)$, of observing outcome a when the measurement $\{E_a\}_a$ is performed on the state $|\psi\rangle$.
- In open systems, quantum states are represented by density operators. A density operator is Hermitian, positive semi-definite and has unit trace.
- Qubit states can be represented as vectors on a unit-radius ball. The shell of the ball corresponds to states that do not feature classical probability. These are called pure states. The interior of the ball corresponds to states with some degree of classical ignorance and these are called mixed states. The origin of the ball represents the maximally mixed state $\frac{1}{2}$.

2. COMPOSITION AND DISTANCE MEASURES

Our tour of the postulates of quantum theory is not yet complete, because we have not had the theory tell us how to describe the joint state of two or more separate systems. That knowledge will pave the way for our discussion of entanglement in Chapter 4. To this end, the composition of quantum systems requires its own postulate.

G8: Postulate - Composition

Let ρ_{AB} be a density operator over both the Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$ representing two independent quantum systems with respective states ρ_A and ρ_B . The joint state is given by,

$$\rho_{AB} = \rho_A \otimes \rho_B. \quad (2.1)$$

The symbol $\otimes$ denotes a *tensor product*, which more specifically is a Kronecker product. The Kronecker product between an $n_1 \times n_2$ matrix $A = [a_{ij}]_{i,j}$ and an $m_1 \times m_2$ matrix $B = [b_{kl}]_{k,l}$ is defined as

$$A \otimes B = \begin{pmatrix} a_{11} & \dots & a_{1n_2} \\ \vdots & \ddots & \vdots \\ a_{n_1 1} & \dots & a_{n_1 n_2} \end{pmatrix} \otimes B = \begin{pmatrix} a_{11}B & \dots & a_{1n_2}B \\ \vdots & \ddots & \vdots \\ a_{n_1 1}B & \dots & a_{n_1 n_2}B \end{pmatrix}, \quad (2.2)$$

where each entry on the right-hand-side, namely $a_{11}B, \dots, a_{n_1 n_2}B$, now is a matrix. Thus, the Kronecker product produces a matrix of size $n_1 m_1 \times n_2 m_2$. Below, we state a few selected properties of the Kronecker product which are going to be useful for our discussions.

- **Bilinearity.** $A \otimes (B + C) = A \otimes B + A \otimes C$ and $(A + B) \otimes C = A \otimes C + B \otimes C$
- **Scalar multiplication.** $(cA) \otimes B = A \otimes (cB) = c(A \otimes B)$ for any $c \in \mathbb{C}$.
- **Associativity.** $(A \otimes B) \otimes C = A \otimes (B \otimes C)$.
- **Non-commutativity.** $A \otimes B \neq B \otimes A$.
- **Product.** $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$.
- **Adjoint.** $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$.
- **Trace.** $\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B)$.

All of these properties can be verified directly from the definition in Eq (2.2). Moreover, it is more convenient to re-write the matrices A and B in the bra-ket notation, namely $A = \sum_{i,j} a_{ij} |i\rangle\langle j|$ and $B = \sum_{k,l} b_{kl} |k\rangle\langle l|$. The tensor product then becomes

$$A \otimes B = \left(\sum_{i,j} a_{ij} |i\rangle\langle j| \right) \otimes \left(\sum_{k,l} b_{kl} |k\rangle\langle l| \right) \stackrel{(2.2)}{=} \sum_{i,j,k,l} a_{ij} b_{kl} |i, k\rangle\langle j, l|, \quad (2.3)$$

where we use the short-hand notation $|i\rangle \otimes |k\rangle = |i\rangle|k\rangle = |i, k\rangle$ for the computational basis in the composed Hilbert space.

While the composition postulate tells us how to join quantum systems, it is also relevant to reduce a joint system to its individual parts. This is achieved using the *partial trace* operation. Consider that we have density operators ρ_A and ρ_B on the respective Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$. Their joint state is $\rho_{AB} = \rho_A \otimes \rho_B$. Now, if we want to discard system B and consider only the state of system A , we perform the partial trace operation over system B ⁸.

$$\text{tr}_B(\rho_{AB}) = \rho_A \otimes \underbrace{\text{tr}(\rho_B)}_{=1} = \rho_A \otimes 1 \stackrel{(2.2)}{=} \rho_A, \quad (2.4)$$

where we have used that the tensor product of an operator and a number is just the operator multiplied with the number. The difference between the partial trace and the regular trace is that the former operates only over the selected Hilbert space (B) whereas the latter operates over the entire Hilbert space. Thus, the output of the former is an operator (on system A) whereas the output of the latter is a scalar.

⁸ The colloquial expression is to “trace out” system B .

2.1. Purifications

The quantum state postulate in Box G1 told us that states of closed systems are rays, but then we have used density operators to describe states in open systems. How do we reconcile the two descriptions? The answer is that we can always enlarge our system to include the quantum state of its environment and then we can expect the postulates for closed systems once again apply. In other words, the classical randomness that appears in the density operator formalism must be possible to see as emerging from a pure state that lives in a larger Hilbert space. This view, that all forms of classical noise are due to not having access to all necessary parts of a pure quantum state, is sometimes referred to as the Church of Larger Hilbert Space. Let us now see how this point of view can explain mixedness in quantum states.

Consider a mixed density operator ρ on system A . We ask whether there exist a pure state $|\psi\rangle_{AB}$ that lives both on system A and an ancillary system B , such that the reduced state on system A is precisely ρ ? Such a state always exist, and we say that $|\psi\rangle$ is a *purification* of ρ . To construct a purification, let us first write ρ in its spectral decomposition,

$$\rho = \sum_{i=1}^d \lambda_i |\lambda_i\rangle\langle\lambda_i|. \quad (2.5)$$

Now, consider the pure state

$$|\psi\rangle_{AB} = \sum_{i=1}^d \sqrt{\lambda_i} |\lambda_i\rangle_A \otimes |\varphi_i\rangle_B, \quad (2.6)$$

where $\{|\varphi_i\rangle\}_i$ is some orthonormal basis of $\mathcal{H}_B$. To check that the reduced state is the correct one, we trace out system B ;

$$\text{tr}_B(|\psi\rangle\langle\psi|) \stackrel{(2.6)}{=} \text{tr}_B\left(\sum_{i,j} \sqrt{\lambda_i} \sqrt{\lambda_j} |\lambda_i, \varphi_i\rangle\langle\lambda_j, \varphi_j|\right) \stackrel{(2.4)}{=} \sum_{i,j} \sqrt{\lambda_i} \sqrt{\lambda_j} |\lambda_i\rangle\langle\lambda_j| \underbrace{\text{tr}(|\varphi_i\rangle\langle\varphi_j|)}_{=\delta_{ij}} = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i| = \rho, \quad (2.7)$$

where δ_{ij} is the Kronecker delta, defined as $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise. This shows us that pure states on composite Hilbert spaces can have mixed reduced states on the individual Hilbert spaces.

In the above, we can equally well think of the partial trace operation as us measuring system B in the basis $\{|\varphi_i\rangle\}_i$ but ignoring the outcome. The probability of observing outcome i is λ_i and this would leave the system A in the state $|\lambda_i\rangle$, thus on average corresponding to the mixture in Eq (2.5). Moreover, notice that the purification is not unique: we have the freedom to choose the basis $\{|\varphi_i\rangle\}_i$. This freedom corresponds precisely to performing an arbitrary unitary transformation on system B , i.e. all states $\mathbb{1} \otimes U|\psi\rangle_{AB}$ are valid purifications of ρ for any unitary U .

In the above, we have only used the spectral decomposition so that we can limit ourselves to just d different states $|\lambda_i\rangle$ in the realisation (2.5) of ρ . This is useful because it only required us to use a d -dimensional purification system B . However, if we were to decompose ρ as some other mixture of m pure (not necessarily orthogonal) states, the purification procedure works analogously, with the only difference that system B now becomes m -dimensional. In view of that, let $\rho = \sum_{i=1}^m p_i |\phi_i\rangle\langle\phi_i|$ and $\rho = \sum_{i=1}^m p'_i |\phi'_i\rangle\langle\phi'_i|$ be two different convex combinations of pure states that both yield the same density operator ρ . Following the recipe in (2.6), let us write their respective purifications as

$$|\psi\rangle_{AB} = \sum_{i=1}^m \sqrt{p_i} |\phi_i\rangle_A \otimes |\varphi_i\rangle_B, \quad \text{and} \quad |\psi'\rangle_{AB} = \sum_{i=1}^m \sqrt{p'_i} |\phi'_i\rangle_A \otimes |\varphi'_i\rangle_B, \quad (2.8)$$

where $\{|\varphi_i\rangle\}_i$ and $\{|\varphi'_i\rangle\}_i$ are orthonormal bases of m -dimensional Hilbert space. However, since the purifications are related by a unitary, $|\psi'\rangle_{AB} = \mathbb{1} \otimes U|\psi\rangle_{AB}$, we can obtain either one of the realisations of ρ from suitably measuring system B of $|\psi\rangle_{AB}$. Clearly, to obtain the former realisation from $|\psi\rangle_{AB}$ we need only to measure in the basis $\{|\varphi_i\rangle\}_i$. Thanks to unitarity, we can also obtain the latter realisation from $|\psi\rangle_{AB}$ by measuring in the rotated basis $\{U^\dagger|\varphi_i\rangle\}_i$. This means that every mixture that realises ρ can be obtained by a suitable measurement on the purification system⁹.

2.2. Distance measures

Hilbert spaces are equipped with inner products. This means that we can meaningfully speak about “how close” two different quantum states are to each other. To this end, we need to define measures of distance, $d(\rho, \sigma)$, between density operators, ρ and σ , in a given Hilbert space. If we overlook the physics and think only about the mathematical aspects of distance measures, we can identify three reasonable properties

⁹ This was proven already by Schrödinger in 1936 but was later re-discovered in 1993, around the time the field of quantum information was born.

1. $d(\rho, \sigma) \geq 0 \quad \forall \rho, \sigma$, with equality if and only if $\rho = \sigma$,
2. $d(\rho, \sigma) = d(\sigma, \rho), \quad \forall \rho, \sigma$,
3. $d(\rho, \sigma) \leq d(\rho, \tau) + d(\tau, \sigma) \quad \forall \rho, \tau, \sigma$.

The first property tells us that distances are positive, and zero only when the states are identical. The second property tells us that the distance from ρ to σ is the same as from σ to ρ . The third property is the triangle inequality; it says that the distance only can increase if we add a “layover”, τ , on the journey from ρ to σ .

Now, let us bring in the physics by adding to our list properties that are motivated by quantum theory. These properties are¹⁰

4. $d(\rho, \sigma) = d(U\rho U^\dagger, U\sigma U^\dagger), \quad \forall \rho, \sigma$,
5. $d(\rho \otimes \tau, \sigma \otimes \tau) = d(\rho, \sigma), \quad \forall \rho, \sigma, \tau$.

The former property says that distances do not change under unitary transformations. This is because unitaries preserve the inner product; they only constitute a change of basis, and hence we do not expect them to change the distance between states. If this was not true, applying a unitary could bring states closer together or further away from each other. The latter property says that distances do not change if we compose both the states with an independent state τ . While there is a whole zoo of distance measures used in various aspects of quantum information science, we will first focus on a particularly common one called the trace-distance and then discuss another popular measure called the fidelity (which as we shall see is actually not a proper distance measure).

2.2.1. Trace-distance

The first distance measure that we discuss is called the *trace-distance*, which we denote by setting $d(\rho, \sigma) = D(\rho, \sigma)$.

G9: Trace-distance

The trace-distance, D , between two quantum states ρ and σ defined over the same Hilbert space is

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1, \quad (2.9)$$

where $\|\cdot\|_1$ is called the trace-norm (or alternatively the one-norm). It is defined as

$$\|A\|_1 = \text{tr} \sqrt{A^\dagger A}, \quad (2.10)$$

which can be understood as the sum of all the singular values of A .

Property 1 is satisfied because the trace-norm is non-negative. Property 2 follows from the symmetry of the trace-norm, namely

$$\|\rho - \sigma\|_1 \stackrel{(2.10)}{=} \text{tr} \sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)} = \text{tr} \sqrt{(-1)(\rho - \sigma)^\dagger (-1)(\rho - \sigma)} = \text{tr} \sqrt{(\sigma - \rho)^\dagger (\sigma - \rho)} = \|\sigma - \rho\|_1. \quad (2.11)$$

Similarly, properties 4 and 5 can also be derived directly from the definition of the trace-norm in Eq (2.10); see Exercise A.2.2.

However, toward showing property 3 we need to gain better understanding of the trace-distance, namely that it can be expressed as¹¹

$$D(\rho, \sigma) = \max_{0 \leq \Pi \leq \mathbb{1}} \text{tr} (\Pi(\rho - \sigma)). \quad (2.12)$$

To see that this is indeed the trace-distance defined in Eq (2.9), we can see how to optimally select Π . Let us name $A = \rho - \sigma$ and spectrally decompose it as $A = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$, where $|\lambda_i| \leq 1$ due to ρ and σ being density operators. We can then split the sum into the terms that are positive and negative.

$$A = \sum_{\lambda_i > 0} \lambda_i |\lambda_i\rangle\langle\lambda_i| + \sum_{\lambda_i < 0} \lambda_i |\lambda_i\rangle\langle\lambda_i|. \quad (2.13)$$

¹⁰ Actually, we want $d(\rho, \sigma)$ to be monotone under more operations than just unitaries, but here we overlook that aspect in favour of a pedagogical presentation.

¹¹ $0 \leq \Pi \leq \mathbb{1}$ means that all eigenvalues of Π lie between zero and one. More generally, a matrix inequality $A \leq B$ means that $B - A$ is positive semidefinite.

Evidently, if we want to maximise $\text{tr}(\Pi A)$, we should select Π as $\Pi_{\text{opt}} = \sum_{\lambda_i > 0} |\lambda_i\rangle\langle\lambda_i|$. This is the projector onto the positive eigenspace of A , and projectors always satisfy the required condition $0 \leq \Pi \leq \mathbb{1}$. Consequently, $\text{tr}(\Pi_{\text{opt}} A) = \sum_{\lambda_i > 0} \lambda_i$. However, since in our case $\text{tr}(A) = \text{tr}(\rho) - \text{tr}(\sigma) = 1 - 1 = 0$, it means $\sum_{\lambda_i > 0} \lambda_i = -\sum_{\lambda_i < 0} \lambda_i$. Therefore, the sum of singular values becomes

$$D(\rho, \sigma) \stackrel{(2.9)}{=} \frac{1}{2} \|\rho - \sigma\|_1 = \frac{1}{2} \|A\|_1 = \frac{1}{2} \left(\sum_{\lambda_i > 0} \lambda_i - \sum_{\lambda_i < 0} \lambda_i \right) = \sum_{\lambda_i > 0} \lambda_i = \text{Eq (2.12)}. \quad (2.14)$$

Thus, we see that the two formulations (2.9) and (2.12) of the trace-distance are equivalent. In Exercise A.2.2, we use the reformulation of the trace-distance from Eq (2.12) to prove property 3.

Let us consider the special case in which both states are pure, namely when $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\phi\rangle\langle\phi|$. We evaluate Eq (2.12). Since $\psi - \phi$ ¹² is traceless and rank-two for $\psi \neq \phi$, it must have two non-zero eigenvalues that are equal in magnitude but with opposite signs. Thus, the optimal Π in (2.12) is a rank-one projector onto the positive eigenvector of $\psi - \phi$. We thus have $D = \lambda_{\max}(\psi - \phi)$ ¹³. Since trace-distance is invariant under a unitary (property 4), let us apply a unitary that maps $|\psi\rangle \rightarrow |0\rangle$ and $|\phi\rangle \rightarrow \cos\theta|0\rangle + \sin\theta|1\rangle$ for some $\theta \in [0, \pi]$. Hence we have

$$\psi - \phi = \begin{pmatrix} 1 - \cos^2\theta & -\cos\theta\sin\theta \\ -\cos\theta\sin\theta & -\sin^2\theta \end{pmatrix}. \quad (2.15)$$

The eigenvalues are calculated to be $\pm \sin\theta$. Since $\sin\theta = \sqrt{1 - |\langle\psi|\phi\rangle|^2}$, we have obtained

$$D(\psi, \phi) = \sqrt{1 - |\langle\psi|\phi\rangle|^2}. \quad (2.16)$$

We now see that the trace-distance between two pure states is fully determined by the probability of taking e.g. $|\psi\rangle$ and successfully projecting it onto the state $|\phi\rangle$.

Finally, let us remark that the trace-distance is more than just a function satisfying the properties 1-5. In Chapter 3.3 we will show that it also has an interpretation in terms of a natural quantum information task. This will give the trace-distance an appealing physical relevance.

2.2.2. Fidelity

Another common quantifier of closeness between quantum states is called the *fidelity*. We discuss it here due to its importance in quantum information theory, even though it is not going to satisfy all the discussed properties of a distance measure.

G10: Fidelity

The fidelity, F , between two quantum states ρ and σ defined over the same Hilbert space is

$$F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1^2. \quad (2.17)$$

Actually, the distance corresponds not to F but to $1 - F$, because identical states evidently have $F(\rho, \rho) = \|\sqrt{\rho}\sqrt{\rho}\|_1 = \|\rho\|_1 = 1$ whereas for orthogonal states we have $F(\rho, \sigma) = 0$. Property 1 follows because

$$F(\rho, \sigma) \stackrel{(2.17)}{=} \|\sqrt{\rho}\sqrt{\sigma}\|_1^2 \leq \underbrace{\left(\sqrt{\text{tr}(\sqrt{\rho}^\dagger \sqrt{\rho})} \right)^2}_{=1} \underbrace{\left(\sqrt{\text{tr}(\sqrt{\sigma}^\dagger \sqrt{\sigma})} \right)^2}_{=1} = 1, \quad (2.18)$$

where we have used Hölder's inequality¹⁴. The properties 2, 4 and 5 also hold: these are all proven in Exercise A.2.3 by a working directly from the definition in Eq (2.17). This leaves us with proving property 3, namely the triangle inequality. However, this property is impossible to prove because it actually does not hold. In Exercise A.2.4 we show this through an explicit counterexample to property 3, thereby concluding that fidelity is not a proper distance measure. This fact has not prevented it from being widely used in quantum information.

¹² Sometimes when we write a ray $|\psi\rangle$ we directly write ψ for its associated density operator $\psi = |\psi\rangle\langle\psi|$.

¹³ We denote the largest eigenvalue of a matrix A by $\lambda_{\max}(A)$.

¹⁴ A variant of Hölder's inequality states that $\|AB\|_1^2 \leq \text{tr}(A^\dagger A) \text{tr}(B^\dagger B)$.

An important special case for the fidelity is when one of the states is pure, e.g. $\sigma = |\phi\rangle\langle\phi|$. Then the fidelity simplifies as

$$F(\rho, \sigma) \stackrel{(2.17)}{=} \|\sqrt{\rho}\sqrt{\sigma}\|_1^2 \stackrel{(2.10)}{=} \left(\text{tr} \sqrt{(\sqrt{\rho}\sqrt{\sigma})^\dagger (\sqrt{\rho}\sqrt{\sigma})} \right)^2 = \left(\text{tr} \sqrt{|\phi\rangle\langle\phi| \rho |\phi\rangle\langle\phi|} \right)^2 = \langle\phi|\rho|\phi\rangle \text{tr}(|\phi\rangle\langle\phi|)^2 = \langle\phi|\rho|\phi\rangle, \quad (2.19)$$

where we have used that $\sqrt{|\phi\rangle\langle\phi|} = |\phi\rangle\langle\phi|$. This relation is appealing because it admits a nice interpretation in terms of Born's rule: the fidelity between a pure and a mixed state is precisely the probability of successfully projecting ρ onto $|\phi\rangle\langle\phi|$. In particular, if both states are pure, namely $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\phi\rangle\langle\phi|$, the fidelity further simplifies to

$$F(\psi, \phi) = |\langle\psi|\phi\rangle|^2. \quad (2.20)$$

Like the case of trace-distance in Eq (2.16), the fidelity between two pure states is fully determined by their modulus overlap. Hence, there is a one-to-one correspondence between trace-distance and fidelity, which means that we can think of them as equivalent distance measures for pure states.

Returning to the general case of mixed states, the fidelity has an important feature which allows it to be connected to the notion of purification. This connection is given by *Uhlmann's theorem*.

R1: Uhlmann's theorem

The fidelity between two states ρ and σ defined over the same Hilbert space is given by

$$F(\rho, \sigma) = \max_{\psi_\rho, \phi_\sigma} |\langle\psi_\rho|\phi_\sigma\rangle|^2, \quad (2.21)$$

where $|\psi_\rho\rangle$ and $|\phi_\sigma\rangle$ are purifications of ρ and σ respectively.

Let $\rho = \sum_k \lambda_k |\lambda_k\rangle\langle\lambda_k|$ and $\sigma = \sum_j \mu_j |\mu_j\rangle\langle\mu_j|$ be the spectral decompositions of ρ and σ . Define $|\psi_\rho\rangle$ and $|\phi_\sigma\rangle$ to be their respective purifications. Following (2.6), these purifications are

$$|\psi_\rho\rangle = \sum_k \sqrt{\lambda_k} |\lambda_k\rangle \otimes |\tilde{\lambda}_k\rangle, \quad (2.22)$$

$$|\phi_\sigma\rangle = \sum_j \sqrt{\mu_j} |\mu_j\rangle \otimes |\tilde{\mu}_j\rangle, \quad (2.23)$$

where $\{|\tilde{\lambda}_k\rangle\}_k$ and $\{|\tilde{\mu}_j\rangle\}_j$ are orthonormal bases of the ancillary Hilbert space. We then have that

$$\langle\psi_\rho|\phi_\sigma\rangle = \sum_{j,k} \sqrt{\lambda_k \mu_j} \langle\lambda_k|\mu_j\rangle \langle\tilde{\lambda}_k|\tilde{\mu}_j\rangle. \quad (2.24)$$

Let us now define the following operator

$$U = \sum_{i,l} |\mu_i\rangle\langle\tilde{\lambda}_l|\tilde{\mu}_i\rangle\langle\lambda_l|. \quad (2.25)$$

As suggested by our choice of letter, U is actually a unitary operator. To verify this, we check the condition (1.4)

$$\begin{aligned} UU^\dagger &\stackrel{(2.25)}{=} \left(\sum_{i,l} |\mu_i\rangle\langle\tilde{\lambda}_l|\tilde{\mu}_i\rangle\langle\lambda_l| \right) \left(\sum_{i',l'} |\mu_{i'}\rangle\langle\tilde{\lambda}_{l'}|\tilde{\mu}_{i'}\rangle\langle\lambda_{l'}| \right)^\dagger = \left(\sum_{i,l} |\mu_i\rangle\langle\tilde{\lambda}_l|\tilde{\mu}_i\rangle\langle\lambda_l| \right) \left(\sum_{i',l'} |\lambda_{l'}\rangle\langle\tilde{\mu}_{i'}|\tilde{\lambda}_{l'}\rangle\langle\mu_{i'}| \right) \\ &= \sum_{i,l,i',l'} \langle\tilde{\lambda}_l|\tilde{\mu}_i\rangle\langle\tilde{\mu}_{i'}|\tilde{\lambda}_{l'}\rangle \underbrace{\langle\lambda_l|\lambda_{l'}\rangle}_{=\delta_{l,l'}} |\mu_i\rangle\langle\mu_{i'}| = \sum_{i,l,i'} \langle\tilde{\lambda}_l|\tilde{\mu}_i\rangle\langle\tilde{\mu}_{i'}|\tilde{\lambda}_l\rangle |\mu_i\rangle\langle\mu_{i'}| \\ &= \sum_{i,i'} \underbrace{\langle\tilde{\mu}_{i'}|\left(\sum_l |\tilde{\lambda}_l\rangle\langle\tilde{\lambda}_l| \right)|\tilde{\mu}_i\rangle}_{=\mathbb{1}} |\mu_i\rangle\langle\mu_{i'}| = \sum_{i,i'} \underbrace{\langle\tilde{\mu}_{i'}|\tilde{\mu}_i\rangle}_{=\delta_{i,i'}} |\mu_i\rangle\langle\mu_{i'}| = \sum_i |\mu_i\rangle\langle\mu_i| = \mathbb{1}. \end{aligned} \quad (2.26)$$

Now, we can see why this unitary is useful. Use the spectral decompositions and (2.25) to compute the expression $\text{tr}(\sqrt{\rho}\sqrt{\sigma}U)$.

$$\text{tr}(\sqrt{\rho}\sqrt{\sigma}U) = \sum_{j,k} \sqrt{\lambda_k} \sqrt{\mu_j} \text{tr}(|\lambda_k\rangle\langle\lambda_k|\mu_j\rangle\langle\mu_j|U) = \sum_{j,k,i,l} \sqrt{\lambda_k \mu_j} \text{tr}(|\lambda_k\rangle\langle\lambda_k|\mu_j\rangle\langle\mu_j|\underbrace{|\tilde{\lambda}_l\rangle\langle\tilde{\mu}_i\rangle\langle\lambda_l|}_{=\delta_{i,j}}) \quad (2.27)$$

$$= \sum_{j,k,l} \sqrt{\lambda_k \mu_j} \text{tr}(\langle\lambda_k|\mu_j\rangle\langle\tilde{\lambda}_l|\tilde{\mu}_j\rangle \underbrace{\langle\lambda_l|\lambda_k\rangle}_{=\delta_{k,l}}) = \sum_{j,k} \sqrt{\lambda_k \mu_j} \langle\lambda_k|\mu_j\rangle\langle\tilde{\lambda}_k|\tilde{\mu}_j\rangle = \text{Eq (2.24)}. \quad (2.28)$$

Hence we conclude that

$$\langle \psi_\rho | \phi_\sigma \rangle = \text{tr}(\sqrt{\rho} \sqrt{\sigma} U). \quad (2.29)$$

As shown in Exercise A.2.5, for any matrix A it holds that $|\text{tr}(AU)|^2 \leq \|A\|_1^2$. Therefore, we get

$$|\langle \psi_\rho | \phi_\sigma \rangle|^2 = |\text{tr}(\sqrt{\rho} \sqrt{\sigma} U)|^2 \leq \|\sqrt{\rho} \sqrt{\sigma}\|_1^2 = F(\rho, \sigma), \quad (2.30)$$

which proves the Uhlmann's theorem. $\square$

2.2.3. Fidelity vs trace-distance

Before we saw that trace-distance and fidelity have a one-to-one correspondence for pure states, namely via Eqs (2.16) and (2.20). From Uhlmann's theorem, we can see how this equivalence is broken down for mixed states. Consider two mixed state ρ and σ and let $|\psi_\rho\rangle$ and $|\phi_\sigma\rangle$ be their optimal purifications in Uhlmann's theorem. Then, we have

$$D(\rho, \sigma) \leq D(\psi_\rho, \phi_\sigma) = \sqrt{1 - F(\psi_\rho, \phi_\sigma)} \stackrel{(2.21)}{=} \sqrt{1 - F(\rho, \sigma)}. \quad (2.31)$$

Here, we have first used that the trace-distance can only decrease under a partial trace¹⁵, then the equivalence between D and F for pure states and finally Uhlmann's theorem. Thus, the trace-distance can be upper bounded by the fidelity. In Exercise A.2.6 we use yet a different formulation of the fidelity to show also an inequality in the other direction. Putting these two together, we have the general relations between trace-distance and fidelity for mixed states,

$$1 - \sqrt{F(\rho, \sigma)} \leq D(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)}, \quad (2.32)$$

which are called the *Fuchs-van de Graaf inequalities*. In general, the inequivalence between trace-distance and fidelity means that they can disagree on whether a given mix state ρ is closed to the state σ_1 or the state σ_2 . An explicit example is found in Exercise A.2.7. This should not be a surprise: there is no absolute way of measuring distances in Hilbert space.

Chapter summary.

- Two independent quantum systems, ρ and σ , are composed via their tensor product, $\rho \otimes \sigma$.
- Every mixed quantum state is the reduced state of a suitably chosen pure state on a bipartite Hilbert space. Therefore, classical randomness can be viewed as a lack of access to all parts of a pure quantum state.
- There are many different ways of measuring the distance between quantum states. Two important but inequivalent examples are the trace-distance and the fidelity, although the latter is not a proper distance measure.

¹⁵ We have not proven this yet. It will be shown in Chapter 8.2.

3. QUANTUM MEASUREMENTS

Measurements are essential components of quantum theory. They tell us how quantum states are transformed into the classical outcomes produced by our measuring devices. This chapter is an introduction to the abstract treatment of quantum measurements and their role in quantum information.

3.1. Projective measurements

The quantum measurement postulate G3 introduced in Chapter 1.1 told us that a measurement on Hilbert space $\mathcal{H}$ is represented by a complete set of projectors, $\{E_a\}$, where a denotes the outcome. The conditions that we call projectivity and completeness were defined as

$$E_a^2 = E_a \quad (\text{projectivity}), \quad (3.1)$$

$$\sum_a E_a = \mathbb{1} \quad (\text{completeness}). \quad (3.2)$$

Taken together, projectivity and completeness imply that the projectors must be pairwise orthogonal. To see this implication, let us assume the opposite, namely that there exists non-orthogonal E_a and E_b ($a \neq b$) that represent mutually exclusive outcomes a and b . Non-orthogonality means that there exists a normalised eigenvector of E_b , call it $|\varphi\rangle$, such that $\langle\varphi|E_a|\varphi\rangle > 0$. By Born's rule, the probability of obtaining either a or b when measuring $|\varphi\rangle$ is $p(b) + p(a) = 1 + \langle\varphi|E_a|\varphi\rangle$, but because this number exceeds one it contradicts the basic laws of probability. Hence, we must conclude that the projectors are orthogonal, i.e.

$$E_a E_b = \delta_{a,b} E_a. \quad (3.3)$$

Alternatively, it is common for quantum physicists to associate measurements with *observables*. This is equivalent to the notion of complete sets of projectors. An observable, E , is a Hermitian operator: its eigenvalues label the possible outcomes of the measurement and the corresponding eigenspaces are associated with the projectors. Thus, we can think of the eigenvalues as our outcome index, a , and E_a as the projector onto the a 'th eigenspace. The eigenspaces of Hermitian operators associated to distinct eigenvalues are well-known to be orthogonal (see Exercise A.1.2), thereby the projectors satisfy the condition (3.3). For non-degenerate observables, all eigenvectors are orthogonal, meaning that the measurement is associated with an orthonormal basis¹⁶ of Hilbert space. In our formalism, this means selecting $E_a = |e_a\rangle\langle e_a|$ for $a = 1, \dots, d$, where $\langle e_a|e_b\rangle = \delta_{a,b}$. We refer to these as *basis measurements*, or sometimes as *rank-one projective measurements* or *von Neumann measurements*. Basis measurements are only a particular class of quantum measurements, albeit an important one. In Exercise A.3.2 we will encounter a simple example of their limitations.

We typically represent measurements not with observables but with the set of measurement operators $\{E_a\}_a$. However, it is sometimes convenient to make an exception when our measurement only has two possible outcomes. We have seen this exemplified in Chapter 1.2 where we used the observables σ_X , σ_Y and σ_Z . In general, we associate to any binary-outcome measurement $\{E_0, E_1\}$ an observable $E = E_0 - E_1$. We can easily recover the measurement operators from the observable because

$$E_0 = \frac{\mathbb{1} + E}{2}, \quad E_1 = \frac{\mathbb{1} - E}{2}, \quad (3.4)$$

which follows from the completeness relation $E_0 + E_1 = \mathbb{1}$. Furthermore, these observables can be characterised by the fact that their squares always equal identity. This is again a consequence of projectivity and completeness,

$$E^2 = (E_0 - E_1)^2 = \underbrace{E_0^2}_{=E_0} + \underbrace{E_1^2}_{=E_1} - \underbrace{E_0 E_1}_{=0} - \underbrace{E_1 E_0}_{=0} \stackrel{(3.1)}{=} E_0 + E_1 \stackrel{(3.2)}{=} \mathbb{1}. \quad (3.5)$$

It is important to appreciate that our abstract treatment of quantum measurements is applicable to potentially very different real-life physical systems. Two examples are illustrated in Fig 3.3. In part a), we illustrate a Stern-Gerlach magnet. It creates an inhomogeneous magnetic field. If we fire an electron through this field it will be deflected upwards or downwards depending on its spin, which we can then read out on the screen behind it. Thus, this setup constitutes a measurement of the spin along the vertical axis and can therefore be thought of as a σ_Z measurement in the basis of the spin. If we were to rotate the magnet,

¹⁶ Since we are essentially always only interested in bases that are orthonormal, we will from now on simply write “basis” when meaning “orthonormal basis”.

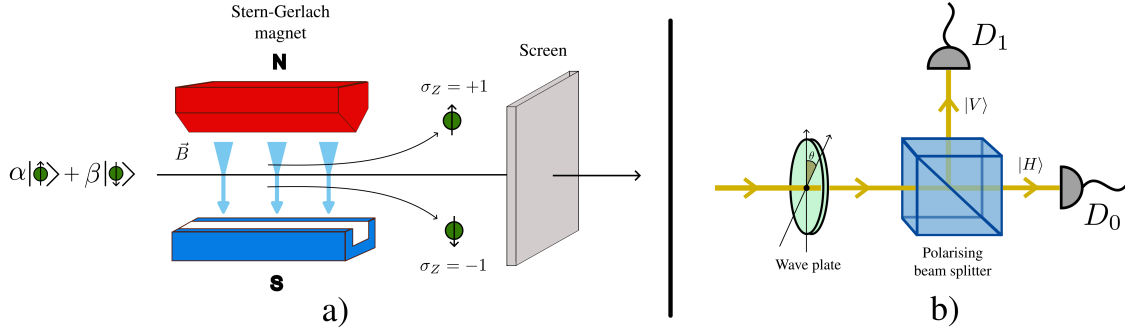


FIG. 3.3. Examples of quantum measurements. a) Stern-Gerlach magnet for measuring the spin of an electron. b) Polarising beam splitter and wave plate for measuring the polarisation of a photon in any basis. Both can be described by qubit observables which can be represented on the Bloch sphere.

it would correspond to measuring in a different basis. In part b) we have an example from polarisation optics. A single photon whose state is in a superposition of horizontal, $|H\rangle$, and vertical, $|V\rangle$, polarisation passes through a wave plate whose optical axis is at angle θ and then hits a polarising beam splitter. In the horizontal and vertical paths of the polarising beam splitter we have placed detectors D_0 and D_1 . If we set $\theta = 0$ the setup represents a σ_Z measurement in the horizontal-vertical polarisation basis. If we instead begin to rotate the wave plate then we can change the measurement basis. A wave plate is made of quartz which is a birefringent material. Because of its different indices of refraction, light that oscillates in parallel with its optical axis picks up a phase shift compared to light that oscillates perpendicular to the axis. This allows us to implement a unitary that rotates θ degrees about a given axis in the Bloch sphere. The thickness of the wave plate determines the accessible Bloch sphere observables. Naturally, there are countless other examples of how the abstract formalism applies to concrete quantum devices. It is important to have the real-world connections in mind, as one otherwise risks losing sight of the physics. A more exotic example of how a quantum measurement can be implemented is given in Box B2.

Finally, let us briefly discuss the relationship between distinct quantum measurements. From introductory quantum mechanics it is well-known that the properties measured by two different quantum measurements cannot always be known simultaneously. This is manifested through the Heisenberg uncertainty relation. In its original form, it tells us that for any state the variance of a position measurement and the variance of a momentum measurement cannot both be small. From this, one concludes that position and momentum are fundamentally incompatible quantities, i.e. they cannot be simultaneously known. Often, this is seen as a special case of the more general *Robertson uncertainty relation* which applies to generic observables.

R2: Robertson uncertainty relation

Let A and B be observables and let $|\psi\rangle$ be a state. Then,

$$\sigma(A) \sigma(B) \geq \frac{1}{2} |\langle \psi | [A, B] | \psi \rangle|, \quad (3.6)$$

where $\sigma^2(X) = \text{tr}(X^2\psi) - \text{tr}(X\psi)^2$ is the variance, with $X \in \{A, B\}$.

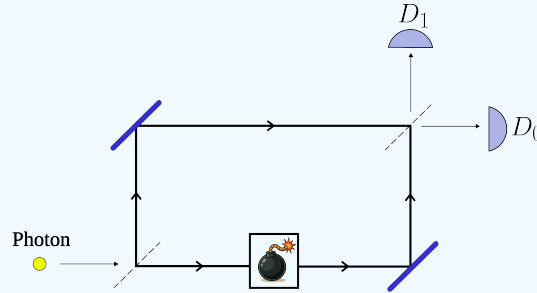
The proof of the Robertson uncertainty relation closely follows that of the Heisenberg uncertainty relation and we omit it for our discussion here¹⁷. The relation depends on the state $|\psi\rangle$ but we see that unless $[A, B] = 0$, there will always exist a state $|\psi\rangle$ so that the right-hand-side is non-zero, namely $AB|\psi\rangle \neq BA|\psi\rangle$. Therefore, in order to enable simultaneous knowledge of the outcomes of both A and B , the observables have to *commute*, namely $[A, B] = 0$. Phrased in terms of complete sets of projection operators, $\{A_a\}_a$ and $\{B_b\}_b$, every pair of measurement operators must satisfy $[A_a, B_b] = 0$ in order for the two physical quantities to be compatible. Thus, to unlock the fundamental features of quantum theory, we need measurements that fail to commute.

¹⁷ The proof can be found in most introductory quantum mechanics textbooks.

B2. A glance beyond

Quantum bomb tester

Consider that we have a large stockpile of bombs with a rather unusual trigger mechanism: each bomb has a single-photon sensor and if it detects a photon the bomb explodes. The stockpile has gotten old so many of the bombs do not work. We plan to use one of our bombs, but for that we need to first make sure that it works without exploding it in the process. In other words, we need an interaction-free measurement of whether the bomb works. Classically, we cannot solve this problem but quantum measurements can. For this, we insert the bomb in one of the paths of a Mach-Zehnder interferometer, as illustrated below. A single photon enters, passes through a beam splitter, is reflected by a mirror (one per path) and the paths are recombined on a second beam splitter. We put a detector at the end of each path.



Suppose the bomb does not work. In that case, it does not interact with the photon and our experiment is a standard (bomb-free) Mach-Zehnder interferometer. In quantum optics, one learns that constructive path-interference causes the photon to always end up in D_1 . Thus, $p(D_1|\text{dud}) = 100\%$. Now, suppose instead that the bomb works. Then the sensor is effectively performing a σ_Z measurement in the photon number basis, i.e. $|0\rangle$ means no photon and $|1\rangle$ means a photon. However, since the first beam splitter puts the state of the photon in a uniform superposition of the two paths, the probability of being detected by the sensor is only 50%. If the photon is detected, the bomb explodes. However, if it is not detected, the bomb does not explode, implying that the photon took the upper path and is no longer in a superposition. Then, once the photon arrives on the final beam splitter, it could either go into D_0 or D_1 . In total we will see $p(D_0|\text{bomb}) = 25\%$. Thus, whenever D_0 fires, we are guaranteed to have a working bomb that has not yet exploded. To achieve this, we needed only to be willing to sacrifice some of the other working bombs in our stockpile. Actually, more complicated setups are possible where the probability that the bomb explodes tends to zero.

3.2. Positive operator-valued measures

In analogy with how density operators generalised rays as the representation of quantum states, open quantum theory needs us to provide a more general notion of measurement than that given to us by the quantum measurement postulate [G3](#). Let us showcase the need for a more general measurement concept through an example. Consider that we receive a qubit state ρ and that our measurement apparatus simply discards ρ and flips a coin to determine whether to output $a = 0$ or $a = 1$. Thus, we expect $p(a) = 1/2$ independently of ρ . According to Born's rule, this is only possible if our measurement operators are $E_0 = E_1 = \frac{1}{2}$ because $\text{tr}(\frac{1}{2}\rho) = 1/2$, but these operators are not projective. The reason is that by introducing the coin, we have effectively opened the quantum system and we therefore cannot expect the quantum measurement postulate to hold. In order to accommodate these types of measurements, we need a more general concepts. The most general notion of a quantum measurement is called *positive operator-valued measure*¹⁸ (POVM).

G11: Positive operator-valued measures

A quantum measurement on a Hilbert space $\mathcal{H}$ is represented by a set of operators $\{E_a\}_a$ where a indexes the outcome and the operators satisfy $E_a \geq 0$ and $\sum_a E_a = \mathbb{1}$. Outcome probabilities are given by Born's rule, namely $p(a) = \text{tr}(\rho E_a)$ for any state ρ .

¹⁸ This is, admittedly, an awful name. We have got to live with it.

We see that the only difference between POVMs and projective measurements is that the operators now no longer need to be projective but only positive semidefinite. From a consistency point of view, POVMs are well-motivated because $E_a \geq 0$ is necessary and sufficient to ensure that $p(a) = \text{tr}(\rho E_a)$ is non-negative for every ρ . Therefore, from now on, we think of quantum measurements as POVMs and we keep in mind projective measurements as an important special case. A key conceptual difference between projective measurements and general POVMs is that the formers are repeatable (as we saw in Chapter 1.1) but the latter are not, i.e. if we perform the same POVM twice we are not guaranteed to get the same outcome and we may require further updates to the post-measurement state. State updating rules for POVMs are less straightforward than for projective measurements and we withhold that discussion until Chapter 7.6.

3.2.1. Qubits

Let us focus on describing POVMs over a qubit Hilbert space. Since POVM elements are positive semidefinite, they are also Hermitian. A basis of Hermitian 2×2 matrices is given by the identity together with the three Pauli operators. Therefore, we can write the measurement operators as

$$E_a = \lambda_a (\mathbb{1} + \vec{n}_a \cdot \vec{\sigma}). \quad (3.7)$$

Notice the similarity to the form of a qubit density operator in Eq (1.36). Recycling the argument we used for the positive semidefiniteness of qubit density operators in Chapter 1.5, we ensure $E_a \geq 0$ by imposing that $|\vec{n}_a| \leq 1$ together with $\lambda_a \geq 0$. In addition, we need to satisfy the completeness condition. This means that

$$\mathbb{1} \stackrel{(3.2)}{=} \sum_a E_a \stackrel{(3.7)}{=} \mathbb{1} \sum_a \lambda_a + \vec{\sigma} \cdot \sum_a \lambda_a \vec{n}_a \quad \Rightarrow \quad \begin{cases} \sum_a \lambda_a = 1 \\ \sum_a \lambda_a \vec{n}_a = 0 \end{cases}. \quad (3.8)$$

From this, we can recover the case of basis measurements as a special case. Recall that these were represented by anti-podal unit-length Bloch vectors, i.e. $\vec{n}_0 = -\vec{n}_1$ with $|\vec{n}_0| = 1$. Hence, the second condition obtained in (3.8) becomes $\lambda_0 \vec{n}_0 + \lambda_1 (-\vec{n}_0) = 0$, which means $\lambda_0 = \lambda_1$. The first condition obtained in (3.8) then says $\lambda_0 + \lambda_1 = 2\lambda_0 = 1$, and thus $\lambda_0 = \lambda_1 = \frac{1}{2}$. This makes E_a the type of projection operator that we have previously encountered.

However, qubit POVMs can go beyond this description. We exemplify this through two geometrically simple constructions where the number of outcomes exceeds the dimension. Firstly, we can imagine a three-outcome qubit measurement in which the three Bloch vectors all live in the XZ -disk of the Bloch sphere, on which they form an equilateral triangle,

$$\vec{n}_0 = (1, 0, 0), \quad \vec{n}_1 = \left(-\frac{1}{2}, 0, \frac{\sqrt{3}}{2}\right), \quad \vec{n}_2 = \left(-\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}\right). \quad (3.9)$$

Clearly, $\vec{n}_0 + \vec{n}_1 + \vec{n}_2 = 0$ so in order to satisfy the conditions in Eq (3.8) we must select $\lambda_0 = \lambda_1 = \lambda_2 = \frac{1}{3}$. In Exercise A.3.4 we will build POVMs from Bloch vectors that form triangles that are isosceles instead of equilateral. Let us now go further and make a four-outcome qubit POVM. This POVM has four Bloch vectors that point to the vertices of a regular tetrahedron inscribed in the Bloch sphere, namely

$$\vec{n}_0 = \frac{1}{\sqrt{3}}(1, 1, 1), \quad \vec{n}_1 = \frac{1}{\sqrt{3}}(1, -1, -1), \quad \vec{n}_2 = \frac{1}{\sqrt{3}}(-1, 1, -1), \quad \vec{n}_3 = \frac{1}{\sqrt{3}}(-1, -1, 1). \quad (3.10)$$

Again, we see that $\vec{n}_0 + \vec{n}_1 + \vec{n}_2 + \vec{n}_3 = 0$ so we can satisfy the two conditions in Eq (3.8) by selecting $\lambda_0 = \lambda_1 = \lambda_2 = \lambda_3 = \frac{1}{4}$. This POVM is important enough to have its own name: *symmetric informationally complete*. It is symmetric because $\vec{n}_a \cdot \vec{n}_b = -\frac{1}{3}$ for all $a \neq b$, and it is informationally complete because the four Bloch vectors span the Bloch sphere. The latter property is important because it means that any Bloch vector can be written as a linear combination of (3.10). Therefore, if we were to measure some unknown qubit state ρ , we could use the outcome statistics $p(a) = \text{tr}(\rho E_a)$ to precisely reconstruct the density operator ρ , which is impossible with any single projective measurement. For symmetric informationally complete POVMs, we will show how this is done in Exercise A.3.6.

The two examples of POVMs that we have now discussed highlight a conceptual difference between on the one hand pure and mixed states, and on the other hand projective and non-projective measurements. Mixed states can always be decomposed into a convex combination of pure states, but non-projective measurements cannot always be reduced to convex combinations of projective measurements. Therefore, non-projective measurements can be useful on their own for quantum information processing and in our upcoming discussion of unambiguous state discrimination we will encounter an example of that.

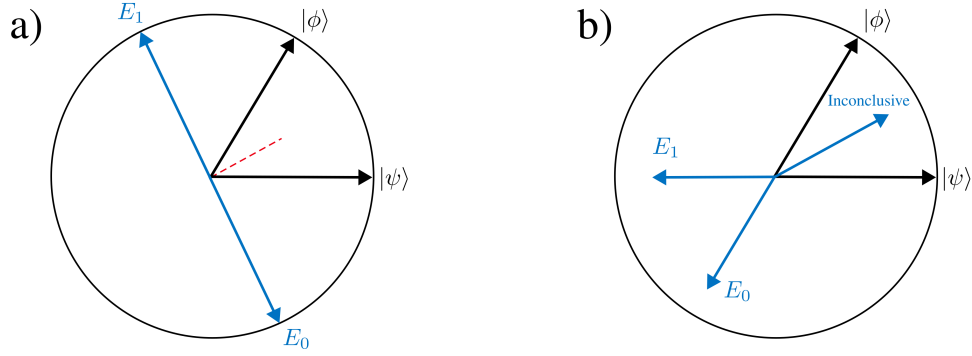


FIG. 3.4. State discrimination. a) Minimum error state discrimination of two states $|\psi\rangle$ and $|\phi\rangle$ represented in the XZ-disk of the Bloch sphere. The optimal measurement is projective and represented by the blue diagonal. b) Unambiguous state discrimination in the XZ-plane of the Bloch sphere. Blue arrows correspond to the effective Bloch vectors of the optimal three-outcome POVM. Their length is re-scaled to reflect the normalisation factor in Eq (3.17).

3.3. Quantum state discrimination

Quantum state discrimination is the simplest task in which the importance of appropriately selecting quantum measurements is essential. Consider that we randomly sample $x \in \{0, 1\}$. If $x = 0$ we prepare the qubit state $|\psi\rangle$ whereas if $x = 1$ we prepare the qubit state $|\phi\rangle$. Our job is to tell the value of x from the prepared quantum state. Only when the states are orthogonal is it possible to perfectly distinguish the two states, by measuring in the basis $\{|\psi\rangle, |\phi\rangle\}$. When the states are not orthogonal we seek to determine the best possible measurement for guessing the state correctly as often as possible, i.e. to guess $a = 0$ if the state was $|\psi\rangle$ and $a = 1$ if the state was $|\phi\rangle$. To quantify our success, we consider the average discrimination probability,

$$p_{\text{win}} = \frac{1}{2} p(a = 0|\psi) + \frac{1}{2} p(a = 1|\phi). \quad (3.11)$$

3.3.1. Minimum error discrimination

To make the best possible state discrimination, we must select the measurement that maximises p_{win} . Fortunately, this can be solved straightforwardly. We denote our measurement as $\{E_0, E_1\}$ and use Born's rule to write Eq (3.11) as

$$p_{\text{win}} = \frac{1}{2} \text{tr}(E_0\psi + E_1\phi). \quad (3.12)$$

The completeness of measurements allows us to write $E_1 = \mathbb{1} - E_0$, which gives us

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{2} \text{tr}(E_0(\psi - \phi)). \quad (3.13)$$

In order to be a valid POVM element, we must constrain E_0 to satisfy $0 \leq E_0 \leq \mathbb{1}$. Now, we should recognise the second term in the formula as the expression appearing inside the maximisation used to define the trace-distance in Eq (2.12). Therefore, by the same reasoning we used in Chapter 2.2 for the trace-distance, the optimal E_0 is the projector onto the eigenvector of $\psi - \phi$ corresponding to its positive eigenvalue. This defines our optimal measurement and it is illustrated in Fig 3.4a. The optimal success probability becomes

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{2} D(\psi, \phi). \quad (3.14)$$

Thus, the larger the distance between the states, the better we can discriminate them. In particular, when the trace-distance vanishes ($D = 0$) we get $p_{\text{win}} = 1/2$ which we could achieve by just selecting a at random. For pure states, we can use the expression (2.16) to write

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{2} \sqrt{1 - |\langle\psi|\phi\rangle|^2}. \quad (3.15)$$

This is not just a solution to the minimum error state discrimination problem but also an operational interpretation for the trace-distance: we can now think of it as the quantity that tells us how well a pair of quantum states can be discriminated. In Exercise A.3.7 we will extend this discussion to discriminating mixed states and we will show that the trace-distance remains the relevant quantity of interest.

3.3.2. Unambiguous discrimination

We found that the optimal measurement for minimum error state discrimination is projective. Are POVMs that are not projective at all interesting for state discrimination? A natural variation of the discrimination task, called *unambiguous state discrimination*, will give us an affirmative answer. The task is again to maximise p_{win} , but this time we are not allowed to make faulty guesses. In other words, we require that $p(a = 0|\phi) = p(a = 1|\psi) = 0$. Hence, if we are not sure which state we have been given, we must declare it inconclusive ($a = \emptyset$). This means that the measurement has three possible outcomes, either we output 0, 1 or $\emptyset$, corresponding to the POVM $\{E_0, E_1, E_\emptyset\}$. Let us first examine how projective measurements can solve the task. We will always assume that $\langle\psi|\phi\rangle \neq 0$ since otherwise the task is trivial.

Projective strategy. Projective measurements limit us to measure a basis of qubit Hilbert space. However, there exists no single basis $\{|u\rangle, |u^\perp\rangle\}$ such that $|\langle u|\psi\rangle|^2 = 0$ (in which case observer excludes the possibility of being given $|\psi\rangle$) and $|\langle u^\perp|\phi\rangle|^2 = 0$ (in which case the observer excludes the possibility of being given $|\phi\rangle$). What we may do instead is to randomly perform either a projective measurement in the basis $\{|\psi\rangle, |\psi^\perp\rangle\}$ or a projective measurement in the basis $\{|\phi\rangle, |\phi^\perp\rangle\}$. In the former case, we can exclude the state $|\psi\rangle$ if we observe the outcome associated with $\psi^\perp$, so we output $a = 1$. In the latter case, we can exclude the state $|\phi\rangle$ if we observe the outcome associated with $\phi^\perp$, so we output $a = 0$. Whenever we obtain an outcome that does not exclude a state, we must output $a = \emptyset$. This guarantees that the wrong state is never declared. The winning probability is

$$\begin{aligned} p_{\text{win}} &\stackrel{(3.11)}{=} \frac{1}{4} (|\langle\psi|\phi^\perp\rangle|^2 + |\langle\phi|\psi^\perp\rangle|^2) = \frac{1}{4} (\langle\psi|\phi^\perp\rangle\langle\phi^\perp|\psi\rangle + \langle\phi|\psi^\perp\rangle\langle\psi^\perp|\phi\rangle) \\ &= \frac{1}{4} (\langle\psi|(\mathbb{1} - |\phi\rangle\langle\phi|)|\psi\rangle + \langle\phi|(\mathbb{1} - |\psi\rangle\langle\psi|)|\phi\rangle) = \frac{1 - |\langle\psi|\phi\rangle|^2}{2}, \end{aligned} \quad (3.16)$$

where we have used the completeness relation. As the states become more distinguishable our success probability becomes larger. However, because we had to choose the basis randomly, we cannot exceed the limit $p_{\text{win}} = \frac{1}{2}$ for any non-orthogonal states, even if they are arbitrarily close to orthogonal.

Non-projective strategy. We now show how to beat the projective strategy by using non-projective POVMs. Let us select the discrimination measurement $\{E_0, E_1, E_\emptyset\}$ as

$$E_0 = \frac{\mathbb{1} - |\phi\rangle\langle\phi|}{1 + |\langle\psi|\phi\rangle|}, \quad E_1 = \frac{\mathbb{1} - |\psi\rangle\langle\psi|}{1 + |\langle\psi|\phi\rangle|}, \quad E_\emptyset = \mathbb{1} - E_\phi - E_\psi. \quad (3.17)$$

The reason we make this choice is that it satisfies our condition of unambiguity, namely $\langle\psi|E_1|\psi\rangle = \langle\phi|E_0|\phi\rangle = 0$. However, we must check that $\{E_0, E_1, E_\emptyset\}$ actually is a valid POVM. We see that E_0 and E_1 are rank-1 but not projective and that the completeness condition is satisfied by construction. Also, the operators E_0 and E_1 are clearly positive semidefinite. Still, we also need to check that $E_\emptyset$ is positive semidefinite and that is shown to be the case in Exercise A.3.8. Therefore, we conclude that (3.17) is a valid POVM. Evaluating the success probability of the state discrimination, we get

$$p_{\text{win}} = \frac{1}{2} \langle\psi|E_0|\psi\rangle + \frac{1}{2} \langle\phi|E_1|\phi\rangle = \frac{1 - |\langle\psi|\phi\rangle|^2}{1 + |\langle\psi|\phi\rangle|} = 1 - |\langle\phi|\psi\rangle|. \quad (3.18)$$

Comparing this result with that obtained by projective measurements in Eq (3.16), we see that the winning probability using the non-projective measurement always is larger, for $|\langle\psi|\phi\rangle| \notin \{0, 1\}$. Moreover, we also have no problem exceeding the limit $p_{\text{win}} = \frac{1}{2}$ when $|\psi\rangle$ and $|\phi\rangle$ are sufficiently distinguishable.

Consider Fig 3.4b to better understand the unambiguous discrimination measurement in Eq (3.17). We want the Bloch vector associated to outcome $a = 1$ to be anti-podal to the Bloch vector of the state $|\psi\rangle$. Analogously, we want the Bloch vector associated to outcome $a = 0$ to be anti-podal to the Bloch vector of the state $|\phi\rangle$. Without loss of generality, we can set the Bloch vectors of the states $|\psi\rangle$ and $|\phi\rangle$ to be $(1, 0, 0)$ and $(\cos \theta, 0, \sin \theta)$, where θ determines their overlap. Now, our measurement Bloch vectors become $\vec{n}_1 = -(1, 0, 0)$ and $\vec{n}_0 = -(\cos \theta, 0, \sin \theta)$. From (3.8) we have

$$\lambda_0 + \lambda_1 + \lambda_\emptyset = 1, \quad \text{and} \quad -\lambda_1(1, 0, 0) - \lambda_0(\cos \theta, 0, \sin \theta) + \lambda_\emptyset \vec{n}_\emptyset = 0 \quad (3.19)$$

By symmetry, we should take $\lambda_0 = \lambda_1 \equiv \lambda$. Then the first condition gives $\lambda_\emptyset = 1 - 2\lambda$. Inserted into the second condition, we obtain $(1 - 2\lambda)\vec{n}_\emptyset - \lambda(1 + \cos \theta, 0, \sin \theta) = 0$. From this we can determine the vector $\vec{n}_\emptyset$ to be

$$\vec{n}_\emptyset = \frac{\lambda}{1 - 2\lambda} ((1 + \cos \theta), 0, \sin \theta), \quad (3.20)$$

but note that we must still choose λ so that $\vec{n}_\varnothing$ is a valid Bloch vector, i.e. so that $|\vec{n}_\varnothing| \leq 1$. To maximise the discrimination probability, we want to make λ as large as possible. This means solving $|\vec{n}_\varnothing| = 1$, namely

$$1 = \frac{\lambda^2}{(1-2\lambda)^2} ((1 + \cos \theta)^2 + \sin^2 \theta) = \frac{4\lambda^2}{(1-2\lambda)^2} \cos^2 \left(\frac{\theta}{2} \right) \xrightarrow{\text{Quadratic equation}} \lambda = \frac{1}{4 \cos^2 \left(\frac{\theta}{4} \right)}. \quad (3.21)$$

This gives us a complete Bloch sphere characterisation of the optimal measurement for unambiguous state discrimination.

3.4. Naimark dilation

In Chapter 2 we showed that mixed states can be purified on a larger Hilbert space. The analogous relation exists between projective and non-projective POVMs. This is necessary in order to reconcile the notion of POVMs for open systems with the quantum measurement postulate G3 for closed systems. Specifically, we are going to show that POVMs can be viewed as projective measurements that are performed on a suitably enlarged Hilbert space. This process is called *Naimark dilation*.

R3: Naimark's theorem

Every n -outcome POVM on a Hilbert space $\mathcal{H}_A$ can be realised as a projective measurement on the extended Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$ where $\dim(\mathcal{H}_B) = n$.

To prove Naimark's theorem, let $\{E_a\}_{a=0}^{n-1}$ be the POVM on $\mathcal{H}_A$ that we wish to apply to a state $|\psi\rangle_A$. Introduce an ancillary state initialised in the state $|0\rangle_B$. To the joint state, $|\psi, 0\rangle_{AB} = |\psi\rangle_A \otimes |0\rangle_B$, we apply a unitary evolution U defined such that

$$U|\psi, 0\rangle_{AB} = \sum_a \sqrt{E_a} |\psi\rangle_A \otimes |a\rangle_B. \quad (3.22)$$

To check that this is unitary, we show that the inner product between two arbitrary pure states $|\psi\rangle$ and $|\phi\rangle$ is preserved,

$$\langle \phi, 0 | U^\dagger U | \psi, 0 \rangle \stackrel{(3.22)}{=} \sum_{a,a'} \langle \phi, a' | \sqrt{E_{a'}} \sqrt{E_a} \otimes \mathbb{1} | \psi, a \rangle = \sum_{a,a'} \langle \phi | \sqrt{E_{a'}} \sqrt{E_a} | \psi \rangle \delta_{a,a'} = \langle \phi | \sum_a E_a | \psi \rangle \stackrel{(3.2)}{=} \langle \phi | \psi \rangle. \quad (3.23)$$

Now, consider that we apply the projective measurement $\{\mathbb{1} \otimes |a\rangle\langle a|\}_a$ to the joint Hilbert space. This effectively means that we measure system B in the computational basis while leaving the system A untouched. The resulting probability distribution is given by Born's rule

$$p(a) = \sum_{j,k} \langle \psi, j | (\sqrt{E_j} \otimes \mathbb{1}) (\mathbb{1} \otimes |a\rangle\langle a|) (\sqrt{E_k} \otimes \mathbb{1}) | \psi, k \rangle = \sum_{j,k} \langle \psi | \sqrt{E_j} \sqrt{E_k} | \psi \rangle \underbrace{\langle a | k \rangle \langle j | a \rangle}_{=\delta_{a,k} \delta_{j,a}} = \langle \psi | E_a | \psi \rangle, \quad (3.24)$$

which is indeed Born's rule on the Hilbert space $\mathcal{H}_A$. Note that this procedure allows us to write the POVM in the form

$$E_a = \text{tr}_B (U^\dagger (\mathbb{1}_A \otimes |a\rangle\langle a|_B) U), \quad (3.25)$$

which is a projective measurement from which a subsystem has been discarded. $\square$

Naimark's theorem tells us that POVMs are an effective mathematical description of how measurements on closed systems should be represented if we are only interested in a part of the system we operate on. The physics of this statement can be taken literally: if an experimenter wants to measure a qubit POVM, a possible implementation is via a Naimark dilation. Specifically, the experimenter introduces an ancilla qubit in the state $|0\rangle$, performs a joint unitary operation on the system and the ancilla and finally makes a projective measurement on the ancilla. Naimark's theorem guarantees that the result of this measurement corresponds to that predicted by Born's rule when a POVM is applied to the original system.

It is helpful to explicitly illustrate the unitary that acts on the two systems in the Naimark dilation. Consider that we are given a d -dimensional n -outcome POVM and want to construct U according to Eq (3.22). Since $|\psi\rangle_A$ is a linear combination of computational basis states $\{|i\rangle\}_{i=0}^{d-1}$, we can equally well characterise U through the relation $U|i, 0\rangle_{AB} = \sum_a \sqrt{E_a} |i, a\rangle_{AB} \forall i \forall a$. Multiplying from the left with the bra $\langle j, k|$, we recover some of the matrix elements of U as $U_{(j,k),(i,0)} = \langle j, k | U | i, 0 \rangle = \sum_a \langle j, k | \sqrt{E_a} | i, a \rangle$, where the pairs (j, k) and $(i, 0)$ are the row and column indices respectively. Hence, out of the dn columns of U , d of them are fixed (those indexed by $(i, 0)$) whereas those columns corresponding to pairs (i, l) with $l \neq 0$ are unspecified. We are free to complete our unitary matrix with these missing columns and still be sure that the Naimark dilation ends up implementing our POVM. Hence, the dilation is far from unique: the same POVM can be implemented in infinitely many different ways.

B3. A glance beyond

Why Born's rule?

We have so far seen no argument for why Nature should respect Born's rule. A partial understanding was provided in 1957 through Gleason's theorem. Gleason assumed that quantum measurements are complete sets of projectors, $\{E_a\}$, on a Hilbert space. His theorem identifies the most general rule for mapping such operators to real numbers such that

$$1) \quad p(a) \geq 0, \quad 2) \quad \sum_a p(a) = 1, \quad 3) \quad \text{that the map depends only on the specific outcome operator } E_a.$$

The first two are basic probability requirements. The third means that E_a is independent of the other outcome operators E_b for $b \neq a$. The theorem states that Born's rule is the only rule consistent with conditions (1-3), i.e. that every mapping is associated with a unit-trace positive semidefinite operator, ρ , such that $p(a) = \text{tr}(\rho E_a)$. However, the theorem comes with a limitation: it holds only for Hilbert spaces of dimension $d \geq 3$. In more recent times, Gleason's theorem has been generalised to when $\{E_a\}$ is a POVM. This implies additional constraints since the probability assignment rule now must hold over a larger space than the projective measurements. This lets one obtain Born's rule also for qubits ($d = 2$). However, the justification of assumption (3) as an axiom is now questionable, as one may argue that POVMs are not fundamental. In that view, POVMs are merely effective descriptions of projective measurements on larger systems. Gleason's theorem is part of the larger question of deriving the axioms of quantum theory from physical principles.

Chapter summary.

- A complete set of projectors is called a projective measurement. Pairs of projective measurements that fail to commute are associated with uncertainty relations.
- In open systems, measurements are described by positive operator-valued measures. These are complete sets of positive semidefinite operators, and projective measurements constitute a special case of them.
- Quantum state discrimination is a task in which we want to learn the label of a randomly drawn state. Minimum error discrimination gives an operational meaning to the trace-distance. Unambiguous state discrimination reveals the advantage of non-projective measurements over projective measurements.
- Every non-projective measurement can be implemented as a Naimark dilation. In this process, we append an ancilla and perform a projective measurement on the joint system.

Further reading. For an in-depth discussion of the abstract formalism of quantum measurements, see the textbook [Quantum measurement](#) by Busch, Lahti, Pellonpää and Ylinen.

Part II

Core topics

4. ENTANGLEMENT

Quantum states with multiple subsystems can exhibit a property called entanglement. In a loose sense, a pair of systems being entangled means that the physics of the whole system goes beyond the physics of the individual parts: to understand the state we must access relational information between its parts. Schrödinger famously referred to entanglement as “the characteristic trait of quantum mechanics, the one that enforces its entire departure from classical lines of thought”. From a quantum information science perspective, entanglement is a resource for a multitude of quantum technologies. Understanding entanglement is therefore ubiquitous to both conceptual and applied aspects of quantum theory.

4.1. Entanglement in pure states

Consider that we have a pure state comprised of two distinct subsystems, $|\psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$. These are called *bipartite* states. A bipartite state that can be factorised over the two Hilbert spaces is called a *product state*. Thus, product states are written as

$$|\psi\rangle_{AB} = |\phi\rangle_A \otimes |\varphi\rangle_B, \quad (4.1)$$

for some arbitrary states $|\phi\rangle_A$ and $|\varphi\rangle_B$. Some examples of product states are

$$\begin{aligned} |\psi\rangle_{AB} &= |00\rangle_{AB}, & |\psi\rangle_{AB} &= \frac{1}{\sqrt{2}} (|00\rangle_{AB} + |01\rangle_{AB}) = |0\rangle_A \otimes \frac{|0\rangle_B + |1\rangle_B}{\sqrt{2}}, \\ |\psi\rangle_{AB} &= \frac{1}{2\sqrt{2}} |00\rangle_{AB} + \frac{\sqrt{3}}{2\sqrt{2}} |01\rangle_{AB} - \frac{1}{2\sqrt{2}} |10\rangle_{AB} - \frac{\sqrt{3}}{2\sqrt{2}} |11\rangle_{AB} = \frac{|0\rangle_A - |1\rangle_A}{\sqrt{2}} \otimes \left(\frac{1}{2} |0\rangle_B + \frac{\sqrt{3}}{2} |1\rangle_B \right). \end{aligned} \quad (4.2)$$

In the last example, we see that it is not always entirely obvious whether even a two-qubit state admits a product form. In contrast, if a pure state is not a product state, then it is called *entangled*. Entanglement is therefore a negative definition; it is simply the absence of a factorisability.

G12: Product states and entangled states

A pure bipartite state $|\psi\rangle_{AB}$ is called a product state if it can be factored as in Eq (4.1). Any pure bipartite state that is not a product state is called entangled.

A key feature of product states is that their reductions are always pure. That is, if we define the bipartite density operator $\rho_{AB} = |\psi\rangle\langle\psi|$ and select $|\psi\rangle_{AB}$ as a product state, then both the reductions $\rho_A \equiv \text{tr}_B(\rho_{AB})$ and $\rho_B \equiv \text{tr}_A(\rho_{AB})$ are pure states. This follows directly from Eq (4.1) because

$$\rho_A = \text{tr}_B(|\psi\rangle\langle\psi|) = \text{tr}_B(|\phi\rangle\langle\phi| \otimes |\varphi\rangle\langle\varphi|) = |\phi\rangle\langle\phi| \otimes \text{tr}(|\varphi\rangle\langle\varphi|) = |\phi\rangle\langle\phi|, \quad (4.3)$$

and analogously for ρ_B . However, entangled states are different: they have mixed reductions. As an example, consider the so-called *maximally entangled state* of two qubits,

$$|\phi^+\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle). \quad (4.4)$$

If we compute the reduced state we obtain

$$\text{tr}_B(\phi_{AB}^+) = \frac{1}{2} \text{tr}_B(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|) = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{\mathbb{1}}{2}, \quad (4.5)$$

which is the maximally mixed state. Another way of viewing this is that the maximally entangled state is a purification of the maximally mixed state. The reason why the reductions are mixed is that a share of the information is stored in the joint Hilbert space of A and B . Consequently, if we discard a subsystem, we must irreversibly lose some information. To exemplify this quantitative loss of information, consider the family of bipartite states

$$|\psi_\theta\rangle = \cos \theta |00\rangle + \sin \theta |11\rangle, \quad (4.6)$$

for $\theta \in [0, \frac{\pi}{4}]$. Clearly, at $\theta = \frac{\pi}{4}$ the state is maximally entangled and at $\theta = 0$ it is product. We expect the entanglement to gradually get weaker as we decrease θ from $\frac{\pi}{4}$ down to 0. The reason we claim that the entanglement becomes weaker is seen by considering the reduced state. A direct calculation gives

$$\rho_A = \text{tr}_B(|\psi_\theta\rangle\langle\psi_\theta|) = \cos^2 \theta |0\rangle\langle 0| + \sin^2 \theta |1\rangle\langle 1|. \quad (4.7)$$

The purity of this state, defined in Eq (1.27), is $T(\rho_A) = \cos^4 \theta + \sin^4 \theta$, which is a monotone for $\theta \in [0, \frac{\pi}{4}]$. Thus, as θ gets smaller, the purity gets larger. We can think of this as more information being made available in the reduced state, or equivalently as less information being inherently bipartite.

When analysing pure bipartite states, it is useful to be able to represent them in a systematic and compact way. This is possible by a tool called *Schmidt decomposition*.

R4: Schmidt decomposition

Let $|\psi\rangle_{AB}$ be a pure state defined on a bipartite Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$ with local dimensions d_A and d_B respectively. There exists orthonormal sets of vectors $\{|\alpha_i\rangle\}_i$ and $\{|\beta_i\rangle\}_i$ of $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively, such that

$$|\psi\rangle_{AB} = \sum_{i=1}^{d_{\min}} \sqrt{\lambda_i} |\alpha_i\rangle_A \otimes |\beta_i\rangle_B, \quad (4.8)$$

for some coefficients $\lambda_i \geq 0$ called *Schmidt coefficients*. We have defined $d_{\min} = \min(d_A, d_B)$.

To prove the Schmidt decomposition we focus on the case when $d_A = d_B = d$ and leave the case of $d_A \neq d_B$ as a straightforward extension. The central tool is the so-called singular-value decomposition of a complex-valued matrix, C , of size $d \times d$. This decomposition allows us to write any C as

$$C = V D W^\dagger, \quad (4.9)$$

where V and W are unitary matrices and D is a diagonal matrix whose entries, σ_i , are the square-roots of the eigenvalues of $C^\dagger C$. To prove the singular-value decomposition, we first observe that for any C the matrix $C^\dagger C$ is positive semidefinite. Let its eigenvalues and eigenvectors be w_i and $|w_i\rangle$ respectively. Due to Hermiticity, the set $\{|w_i\rangle\}_i$ forms a basis. We can therefore define W as the unitary matrix whose columns are $|w_i\rangle$. We also define D as the diagonal matrix whose entries are $\sqrt{w_i}$. Next, define vectors $|v_i\rangle = \frac{1}{\sqrt{w_i}} C |w_i\rangle$. These form a basis because

$$\langle v_i | v_j \rangle = \frac{1}{\sqrt{w_i w_j}} \langle w_i | C^\dagger C | w_j \rangle = \frac{w_j}{\sqrt{w_i w_j}} \langle w_i | w_j \rangle = \delta_{i,j}. \quad (4.10)$$

Thus, we can define V as the unitary matrix whose columns are $|v_i\rangle$. By construction, we now have $V = C W D^{-1}$ which is equivalent to Eq (4.9).

Equipped with the singular-value decomposition, we can now prove the Schmidt decomposition. The state $|\psi\rangle_{AB}$ expanded in the bipartite computational basis becomes

$$|\psi\rangle_{AB} = \sum_{j,k} C_{jk} |j\rangle_A |k\rangle_B, \quad (4.11)$$

where C_{jk} are complex expansion coefficients. We interpret C_{jk} as the elements of the $d \times d$ matrix C . Using the decomposition in Eq (4.9) we can write $C_{jk} = \sum_i V_{ji} D_{ii} W_{ik}^\dagger$ from which we obtain

$$|\psi\rangle_{AB} = \sum_{j,k} \sum_i V_{ji} D_{ii} W_{ik}^* |j\rangle_A |k\rangle_B. \quad (4.12)$$

We can now define $|\alpha_i\rangle_A = \sum_j V_{ji} |j\rangle$ and $|\beta_i\rangle_B = \sum_k W_{ki}^* |k\rangle$. Importantly, $\{|\alpha_i\rangle_A\}_i$ and $\{|\beta_i\rangle_B\}_i$ are both orthonormal sets because V and W are unitary and therefore preserve the orthonormality of the computational basis. Inserting these definition into Eq (4.12) we obtain

$$|\psi\rangle_{AB} = \sum_i D_{ii} |\alpha_i\rangle_A |\beta_i\rangle_B. \quad (4.13)$$

Recall now that by singular-value decomposition the diagonal entries D_{ii} are non-negative. Hence, we need only to change the label $D_{ii} = \sqrt{\lambda_i}$ to obtain the final result in Eq (4.8). $\square$

The proof shows us that Schmidt decomposition is essentially a re-phrasing of the singular-value decomposition. Although the proof is constructive, it is useful to explicitly spell out the recipe for how to find the Schmidt decomposition of a given state $|\psi\rangle_{AB}$. The procedure is as follows.

1. Define $\rho = |\psi\rangle\langle\psi|$ and compute $\rho_A = \text{tr}_B(\rho)$.
2. Compute the eigenvalues of ρ_A . These eigenvalues are the Schmidt coefficients, $\{\lambda_i\}_i$.
3. Compute the eigenvectors of ρ_A . These eigenvectors are the orthonormal vectors $\{|\alpha_i\rangle\}_i$ on $\mathcal{H}_A$.
4. Compute the eigenvectors of $\rho_B = \text{tr}_A(\rho)$. These eigenvectors are the orthonormal vectors $\{|\beta_i\rangle\}_i$ on $\mathcal{H}_B$.

This procedure might give the impression that the eigenvalues of ρ_A are privileged over the eigenvalues of ρ_B but this is not the case. For any pure bipartite state, one can prove that the eigenvalues of ρ_A and ρ_B are identical. That is shown in Exercise B.1.2.

A consequence of Schmidt decomposition is that all pure bipartite states, up to some local unitaries $U \otimes V$, take the form

$$|\psi\rangle_{AB} \simeq \sum_i \sqrt{\lambda_i} |\alpha_i\rangle |\beta_i\rangle. \quad (4.14)$$

For this, we need only to select U as the unitary that takes us from $\{|\alpha_j\rangle\}_j$ to $\{|j\rangle\}_j$ and V as the unitary that takes us from $\{|\beta_j\rangle\}_j$ to $\{|j\rangle\}_j$. Hence, if local unitaries are considered unimportant, we can restrict the state space to the form (4.14). Note that for two qubits this simplifies into the one-parameter family of states considered in Eq (4.6). This lets us once again to see the connection between local mixedness and entanglement. To this end, we can compute from Eq (4.14) the purity of ρ_A and obtain $T(\rho_A) = \sum_i \lambda_i^2$. Since normalisation requires $\sum_i \lambda_i = 1$, the reduction is mixed if and only if the number of non-zero Schmidt coefficients is greater than one. The number of non-zero Schmidt coefficients is called the *Schmidt rank* and from the Schmidt decomposition (4.8) we see that states with Schmidt rank one are equivalent to product states.

4.2. Local operations and classical communication

Consider that we have an entangled state of two particles. Quantum theory does not say that the entanglement is in any fundamental way affected by how far apart the particles are. Therefore, we can imagine that they are placed in two distant labs; one run by Alice and one run by Bob. Now, Alice and Bob want to perform transformations on the entangled state. Naturally, the separation of the labs prevents them from making arbitrary transformations. The standard paradigm for the available transformations is called *local operations and classical communication* (LOCC). Local operations means that Alice and Bob can perform any quantum operation that they want, but only in their respective laboratories. Classical communication means that Alice and Bob also can talk to each other and agree on what operations they are going to perform at any given time.

Having selected LOCC as our allowed operations, we can ask ourselves whether one state is more entangled than another. The meaning of this is the following: if a state $|\psi\rangle_{AB}$ can be converted to another state $|\phi\rangle_{AB}$ by using LOCC, then we consider the former to be at least as entangled as the latter. In other words,

$$|\psi\rangle_{AB} \xrightarrow{\text{LOCC}} |\phi\rangle_{AB} \quad \Rightarrow \quad |\psi\rangle_{AB} \text{ is at least as entangled as } |\phi\rangle_{AB}. \quad (4.15)$$

Consider that we are given a bipartite Hilbert space. Does there exist an entangled state from which all other states can be obtained by means of LOCC? Such a state would then be at least as entangled as every other state in the Hilbert space, so we could justifiably call it *maximally entangled*. For a pair of qubits, it turns out that such a state exists, and we have indeed already encountered it before but without justifying its name.

R5: The maximally entangled two-qubit state

The two-qubit state $|\phi^+\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$ is maximally entangled. It can be transformed into any other two-qubit state by using LOCC.

To prove this, we want to show that

$$|\phi^+\rangle \xrightarrow{\text{LOCC}} |\psi\rangle = \alpha|00\rangle + \beta|11\rangle, \quad (4.16)$$

for any two-qubit state $|\psi\rangle$. The reason we can restrict $|\psi\rangle$ to the above form, where α and β are real, is due to two facts. Firstly, if we can generate any pure two-qubit state by LOCC then we can also generate any mixed two-qubit state by LOCC. This follows from the possibility to decompose the mixed state as convex combination of pure states. Alice and Bob can use classical communication to select which pure state in the decomposition that they want to generate, and since they can generate any such state then they can also generate the mixed state. Secondly, local unitaries are just local operations and any pure two-qubit state

R6: Nielsen's majorisation theorem

Let $\vec{r}_\psi$ and $\vec{t}_\phi$ be the descendingly ordered vector of Schmidt coefficients for the state $|\psi\rangle_{AB}$ and $|\phi\rangle_{AB}$ respectively. It is possible to transform $|\psi\rangle_{AB}$ into $|\phi\rangle_{AB}$ by LOCC if and only if $\vec{r}_\psi$ is majorised by $\vec{t}_\phi$.

is with appropriate local unitaries equivalent to its Schmidt decomposition. Therefore, we need only to consider the Schmidt decomposition of $|\psi\rangle$, which is what appears in Eq (4.16) and it justifies the restriction to real-valued α and β .

Now we show how to implement the transformation (4.16) via LOCC. To this end, we first let Alice generate the auxiliary state $|0\rangle_{A'}$ in her lab. The total state becomes

$$|0\rangle_{A'} \otimes |\phi^+\rangle_{AB} = \left(\frac{|00\rangle|0\rangle + |01\rangle|1\rangle}{\sqrt{2}} \right)_{A'AB}, \quad (4.17)$$

where we have written $|00\rangle|0\rangle$ and $|01\rangle|1\rangle$ to emphasise which part of the state is in Alice's lab and which part is in Bob's lab. Next, let Alice apply a unitary to her two qubits $A'A$ such that

$$|00\rangle \rightarrow \alpha|00\rangle + \beta|11\rangle, \quad |01\rangle \rightarrow \beta|01\rangle + \alpha|10\rangle. \quad (4.18)$$

This is a legitimate unitary transformation because the inner product is preserved between the two input states and the two output states. Naturally, the transformation must also transform $|10\rangle$ and $|11\rangle$ but that part is irrelevant to our proof and it can therefore be chosen freely (as long as it preserves the inner product). After the unitary, the state becomes

$$\frac{1}{\sqrt{2}} [(\alpha|00\rangle + \beta|11\rangle)_{A'A} |0\rangle_B + (\beta|01\rangle + \alpha|10\rangle)_{A'A} |1\rangle_B] = \frac{1}{\sqrt{2}} \left[|0\rangle_{A'} \underbrace{(\alpha|00\rangle + \beta|11\rangle)}_{=|\psi\rangle}_{AB} + |1\rangle_{A'} \underbrace{(\beta|10\rangle + \alpha|01\rangle)}_{=\mathbf{1} \otimes \sigma_X |\psi\rangle}_{AB} \right]. \quad (4.19)$$

Alice now measures σ_Z on her qubit A' and classically communicates the outcome to Bob. Clearly, the post-measurement state is either $\alpha|00\rangle + \beta|11\rangle$ or $\beta|10\rangle + \alpha|01\rangle$ depending on whether the outcome is 0 or 1 respectively. If the outcome is zero, Bob does nothing and the state of system AB is precisely $|\psi\rangle$. If the outcome is one, Bob applies σ_X to his qubit which leaves system AB again in the state $|\psi\rangle$. This completes the LOCC transformation. $\square$

Actually, the notion of a maximally entangled state can be generalised from the case of two qubits to a pair of d -dimensional systems. As one might now guess, and as is shown in Exercise B.1.1, the d -dimensional maximally entangled state is

$$|\phi^+\rangle_{AB} = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |ii\rangle. \quad (4.20)$$

It is instructive to note that the state remains maximally entangled also if we apply arbitrary local unitaries, i.e. if U and V are unitaries then all the maximally entangled states are written $U \otimes V |\phi^+\rangle$. Let us highlight two important properties of the d -dimensional maximally entangled state. These are

$$(O \otimes \mathbf{1})|\phi^+\rangle = (\mathbf{1} \otimes O^T)|\phi^+\rangle, \quad (4.21)$$

$$(U \otimes U^*)|\phi^+\rangle = |\phi^+\rangle, \quad (4.22)$$

for all linear operators O and all unitaries U . The first property can be thought of as “action at a distance” in the sense that the operator O can be moved from system A to system B at the price of a transposition. The second property tells us that applying any unitary and its conjugate to system A and B respectively leaves the maximally entangled state invariant. These properties are proved in Exercise B.1.3.

What if we want to LOCC-transform between two states when neither of them is maximally entangled? Let us start from some arbitrary state $|\psi\rangle_{AB}$ and our goal is to use LOCC to end up in the state $|\phi\rangle_{AB}$. In order to decide whether the LOCC transformation is possible, we need a tool called *majorisation*. Consider two vectors, $\vec{r} = (r_1, \dots, r_N)$ and $\vec{t} = (t_1, \dots, t_N)$ where $r_1 \geq r_2 \geq \dots \geq r_N$ and $t_1 \geq t_2 \geq \dots \geq t_N$. It is said that $\vec{t}$ majorises $\vec{r}$ if

$$\sum_{i=1}^k r_i \leq \sum_{i=1}^k t_i, \quad \forall k \in \{1, 2, \dots, N\}. \quad (4.23)$$

Nielsen's majorisation theorem uses this concept to answer our question.

We will not prove Nielsen's theorem here because it is rather lengthy¹⁹. Instead, we focus on understanding its implications. As a start, we can select $|\psi\rangle_{AB}$ as the maximally entangled state. Since the Schmidt coefficients are the eigenvalues of the reduction and the reduction is $\psi_A = \frac{1}{d}$, we simply have $\vec{r}_\psi = (\frac{1}{d}, \dots, \frac{1}{d})$. Inspecting the definition (4.23), we see that this is trivially majorised by any other same-length vector of eigenvalues. Hence, we have again found the result we showed earlier: that every pure state can be obtained from the maximally entangled state by using LOCC.

However, what if we want to convert a two-qubit maximally entangled state into an entangled state of two three-dimensional systems? We now show that this is not always possible, even in the special case in which the latter state is arbitrarily close to being a two-qubit state. Let us select

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad \text{and} \quad |\phi\rangle = \frac{1}{\sqrt{1+\epsilon^2}} \left[\frac{4}{5}|00\rangle + \frac{3}{5}|11\rangle + \epsilon|22\rangle \right], \quad (4.24)$$

where $\epsilon > 0$ is small. If we view also $|\psi\rangle$ as embedded in the larger Hilbert space, their respective reductions become

$$\psi_A = \begin{pmatrix} \frac{1}{2} & & \\ & \frac{1}{2} & \\ & & 0 \end{pmatrix}, \quad \text{and} \quad \phi_A = \frac{1}{25(1+\epsilon^2)} \begin{pmatrix} 16 & & \\ & 9 & \\ & & 25\epsilon^2 \end{pmatrix}. \quad (4.25)$$

Note that the Schmidt coefficients are already appearing on the diagonals and that they are already sorted in descending order for small ϵ . Next, we check the majorisation conditions from Eq (4.23),

$$k=1: \quad \frac{1}{2} \stackrel{?}{\leq} \frac{16}{25(1+\epsilon^2)}, \quad \Rightarrow \text{True!} \quad (4.26)$$

$$k=2: \quad 1 \stackrel{?}{\leq} \frac{25}{25(1+\epsilon^2)}, \quad \Rightarrow \text{False!} \quad (4.27)$$

$$k=3: \quad 1 \stackrel{?}{\leq} 1, \quad \Rightarrow \text{True!} \quad (4.28)$$

For small ϵ , it is clear that the first inequality is true, but it is also clear that the second inequality is false. We must therefore conclude that neither state majorises the other, i.e. we cannot LOCC-transform $|\psi\rangle$ to $|\phi\rangle$ nor can we LOCC-transform $|\phi\rangle$ to $|\psi\rangle$. We must therefore conclude that neither state is more entangled than the other, i.e. they are simply incomparable.

Majorisation is both transitive and reflexive, but the fact that it is possible to have two objects so that neither majorises the other means that it only defines a *pre-order*. For entanglement, the consequence is that there is no way to fundamentally compare the entanglement of two generic pure states under LOCC. In Box B4, we will discuss a way to try to overcome this limitation, by assuming that $|\psi\rangle$ and $|\phi\rangle$ are available in infinitely many copies.

¹⁹ The interested reader can for instance find a complete proof in the [lecture notes of Watrous](#).

B4. A glance beyond

Asymptotic entanglement conversions

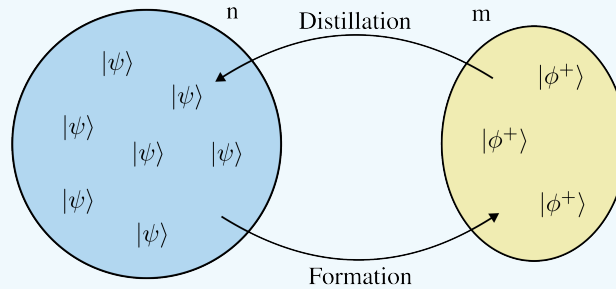
Nielsen's theorem showed us that it is not always possible to use LOCC to convert one pure state into another or vice versa. Also, if we start with the maximally entangled state, $|\phi^+\rangle$, which can be LOCC-converted into any other pure state of the same dimension, $|\psi\rangle$, we cannot reverse the conversion because some entanglement is lost in the process. In trying to circumvent these facts, let us ask a somewhat different question: how many “units” of $|\psi\rangle$ can we get for each unit of $|\phi^+\rangle$? Imagine that we have m copies of $|\phi^+\rangle$. We are going to use LOCC on all these copies jointly in order to produce a larger number of copies, $n(m)$, of $|\psi\rangle$. We then say that the ratio $\frac{m}{n(m)}$ is the cost of forming $|\psi\rangle$ and we want to know its smallest possible value in the limit of many copies. The resulting quantity is called the *entanglement cost*,

$$E_C = \lim_{m \rightarrow \infty} \min_{\text{LOCC}} \frac{m}{n(m)}, \quad \text{such that } |\phi^+\rangle^{\otimes m} \xrightarrow{\text{LOCC}} |\psi\rangle^{\otimes n}. \quad (4.29)$$

It is natural to also consider the reverse process, namely when we want to LOCC-convert n copies of $|\psi\rangle$ into a smaller number of copies, $m(n)$, of $|\phi^+\rangle$. This process is called *entanglement distillation* and its asymptotic rate is

$$E_D = \lim_{n \rightarrow \infty} \max_{\text{LOCC}} \frac{m(n)}{n}, \quad \text{such that } |\psi\rangle^{\otimes n} \xrightarrow{\text{LOCC}} |\phi^+\rangle^{\otimes m}. \quad (4.30)$$

The formation and distillation processes are illustrated here below. For pure entangled states, the two quantities are known to coincide, namely $E_C = E_D$. The value is obtained by evaluating over the reduced state a quantity called the von Neumann entropy, which we will encounter in Chapter 8.2. This means that we can convert our asymptotically many copies of $|\phi^+\rangle$ into a larger share of copies of $|\psi\rangle$, and then convert them back to the same share of $|\phi^+\rangle$. However, when extending these concepts to mixed entangled states the reversibility between asymptotic formation and distillation is no longer true.



4.3. Entanglement in mixed states

We have seen that a pure bipartite state is entangled if and only if its reduction is mixed. However, how can we know if a mixed state is entangled? Mixed states require different entanglement criteria and they are in general much more challenging to classify than pure states. To exemplify this, consider the two-qubit state $\rho_{AB} = \frac{1}{2} \otimes \frac{1}{2}$. Its reduction is $\rho_A = \frac{1}{2}$, which is a mixed state. Nevertheless, it is obvious that ρ_{AB} is not entangled. The reason for this failure is that since pure states feature no classical randomness, any local mixedness must be due to quantum features. In contrast, since mixed states already contain classical randomness, it becomes unclear whether it is this randomness or entanglement that is responsible for their mixed reductions.

For pure states, we defined entanglement as the state not admitting a factorisation, as in Eq (4.1). To also deal with mixed states, we must extend the definition so that it applies to density operators instead of rays. This extended definition can be motivated by LOCC. Alice and Bob can use local operations to prepare states in their respective labs. Moreover, they can use classical communication to coordinate their state preparations and also to classically mix these preparations together. This means that with probability q_i they prepare the pure states ϕ_i and φ_i respectively. The total state generated becomes

$$\rho_{AB} = \sum_i q_i \phi_i \otimes \varphi_i. \quad (4.31)$$

This state is mixed and it is a convex combination over product states $\phi_i \otimes \varphi_i$. In particular, if ρ_{AB} is pure then $\{q_i\}_i$ is deterministic, because this reduces the right-hand-side to the product state definition (4.1). Any state that admits the form (4.31)

is called *separable*. A state that is not separable is called *entangled*. This is the general definition of entanglement that we will use from now on.

G13: Separability and entanglement

A bipartite state is separable if and only if it is a convex combination of product states, as in Eq (4.31). States that are not separable are entangled.

Say that we are given some state ρ_{AB} . How can we know whether it is separable or entangled? Actually, no general solution is known²⁰. Only in the two simplest cases, namely when the system is a pair of qubits, or when it is a qubit and a qutrit, is a necessary and sufficient condition known. For all other Hilbert spaces we have only partial criteria. The most broadly known entanglement criterion is called the *positive partial-transpose (PPT) criterion*. In analogy with the partial-trace operation, the partial-transpose operation applies the usual transposition map in the computational basis (T) to only one share of the system.

R7: Positive partial-transpose criterion

If a bipartite state ρ_{AB} is separable then its partial transposition is positive semidefinite, i.e.

$$\rho_{AB} \text{ is separable} \Rightarrow \rho_{AB}^{T_A} \geq 0. \quad (4.32)$$

Moreover, when $d_A d_B \leq 6$, ρ_{AB} being PPT is also a sufficient condition for separability.

We will not prove the sufficiency part (which is harder and less educational) and instead focus on the more important direction: all separable states are PPT. In other words, if a state is not PPT then it is entangled. To this end, let us apply the partial-transpose to the separable state (4.31),

$$\left(\sum_i q_i \phi_i \otimes \varphi_i \right)^{T_A} = \sum_i q_i \phi_i^T \otimes \varphi_i. \quad (4.33)$$

Transposition does not change the eigenvalues of a Hermitian operator. Therefore, ϕ_i^T is positive semidefinite and has unit trace, i.e. it is a valid state. Hence the right-hand-side is a separable state, which by definition is positive semidefinite. $\square$

The key insight behind the PPT criterion is that some entangled states are not PPT. To showcase this, let us apply the PPT criterion to the maximally entangled state $|\phi^+\rangle_{AB}$. A partial-transpose in the computational basis gives

$$(\phi_{AB}^+)^{T_A} = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)^{T_A} = \frac{1}{2}(|00\rangle\langle 00| + |10\rangle\langle 01| + |01\rangle\langle 10| + |11\rangle\langle 11|). \quad (4.34)$$

Consider now the singlet state $|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$. We verify that this is an eigenstate of $(\phi^+)^{T_A}$,

$$\begin{aligned} (\phi_{AB}^+)^{T_A} |\psi^-\rangle &= \frac{1}{2\sqrt{2}}(|00\rangle\langle 00| + |10\rangle\langle 01| + |01\rangle\langle 10| + |11\rangle\langle 11|)(|01\rangle - |10\rangle) \\ &= \frac{1}{2\sqrt{2}}(|10\rangle - |01\rangle) = -\frac{1}{2\sqrt{2}}(|01\rangle - |10\rangle) = -\frac{1}{2}|\psi^-\rangle. \end{aligned} \quad (4.35)$$

The eigenvalue is negative. Hence $(\phi^+)^{T_A}$ is not PPT and we therefore conclude that $|\phi^+\rangle_{AB}$ is entangled.

Admittedly, we already knew that $|\phi^+\rangle$ is an entangled state, because it is pure and its marginals are mixed. In view of that, let us consider a more interesting example. Imagine that we flip a biased coin: with probability p it gives heads and with probability $1-p$ it gives tails. In case of heads we generate ϕ^+ and in case of tails we generate the maximally mixed two-qubit state $\frac{1}{2} \otimes \frac{1}{2} = \frac{1}{4}$, where the dimension of $\mathbb{1}$ is left implicit. Our density operator becomes

$$\rho_{AB} = p \phi_{AB}^+ + (1-p) \frac{\mathbb{1}_{AB}}{4}, \quad (4.36)$$

²⁰ Probably, we will never have a complete solution. The reason is that there are proofs that deciding whether a generic state is separable or entangled is computationally hard.

which we call the *isotropic state*. Clearly, at $p = 1$ the isotropic state is maximally entangled, and at $p = 0$ it is separable. This suggests that there is a critical value of p below which the state is “too noisy” to be entangled. Since the PPT criterion is necessary and sufficient for two-qubit states, we can use it to determine this critical value. Applying the partial transpose as before, we get

$$\rho_{AB}^{T_A} = p (\phi_{AB}^+)^{T_A} + \frac{1-p}{4} \mathbb{1}_{AB}^{T_A}. \quad (4.37)$$

The expression for $(\phi_{AB}^+)^{T_A}$ is already given in Eq (4.34) and we trivially have $\mathbb{1}_{AB}^{T_A} = \mathbb{1}_{AB}$. We have already shown that $|\psi^-\rangle$ is an eigenstate of $(\phi_{AB}^+)^{T_A}$. Furthermore, every state is an eigenstate of $\mathbb{1}$. Hence, $|\psi^-\rangle$ is an eigenstate of $\rho_{AB}^{T_A}$. We can compute the critical value of p for satisfying the PPT criterion by solving the equation

$$\langle \psi^- | \rho_{AB}^{T_A} | \psi^- \rangle = 0. \quad (4.38)$$

Using Eq (4.35) this becomes $-\frac{p}{2} + \frac{1-p}{4} = 0$ which has the solution $p = \frac{1}{3}$. Hence, for $p > \frac{1}{3}$ the isotropic state is entangled and for $p \leq \frac{1}{3}$ it is separable. In Exercise B.1.6 we will solve the same problem but for the d -dimensional maximally entangled state.

4.4. Quantifiers

Nielsen’s theorem taught us that there is no global order over the space of entangled states under LOCC operations. However, there still exists pre-orders. These are inequivalent ways of quantifying entanglement and we call them *entanglement measures*. An entanglement measure, $E(\rho_{AB})$, should map a bipartite density operator into a non-negative number that quantifies its entanglement. Let us list a few properties that we would like in an entanglement measure²¹,

1. $E(\rho) = 0$ for every separable ρ . This property is called faithfulness.
2. The state that maximises $E(\rho)$ is maximally entangled, namely $\rho = \phi^+$.
3. If σ can be obtained from ρ by LOCC then $E(\sigma) \leq E(\rho)$.

The first property says that separable states have zero entanglement. The second property says that the measure is maximised by the maximally entangled state. The third property says that we cannot increase the entanglement by LOCC. However, the converse of the third property is not true: it does not hold that $E(\rho) \geq E(\sigma)$ implies that ρ can be obtained from σ by LOCC. This is a consequence of the pre-order property: ρ can be more entangled than σ only according to the specific measure E .

We now survey the two most famous entanglement measures. They are known as the *negativity* and the *concurrence*. The negativity is directly based on the PPT criterion because it concerns the magnitude of negativity in the partially transposed state.

G14: Entanglement negativity

The negativity of a bipartite density operator, ρ_{AB} , is defined as

$$\mathcal{N}(\rho_{AB}) = \frac{1}{2} \sum_i (|\lambda_i| - \lambda_i), \quad (4.39)$$

where $\{\lambda_i\}_i$ are the eigenvalues of $\rho_{AB}^{T_A}$.

The connection to the PPT criterion consists in the negativity being precisely the magnitude of the sum of all negative eigenvalues in the partial transposition of a state. For separable states it holds that $\rho_{AB}^{T_A} \geq 0$, which implies $\lambda_i \geq 0$ and hence $\mathcal{N}(\rho) = 0$. Thus, the entanglement measure is faithful. Furthermore, for the maximally entangled state of two qubits we have $\mathcal{N}(\rho_{AB}) = \frac{1}{2}$, which is the maximal value of the negativity. Thus, also the second property is satisfied. The third property also holds, but we do not prove it here as it would require us to introduce several more tools.

An equivalent way to state the negativity is to use that $\text{tr}(\rho_{AB}^{T_A}) = 1$ to express it in terms of the trace-norm defined in Eq (2.10),

$$\mathcal{N}(\rho_{AB}) = \frac{\sum_i |\lambda_i| - 1}{2} = \frac{\|\rho^{T_A}\|_1 - 1}{2}. \quad (4.40)$$

²¹ There are several other properties that are convenient to have in entanglement measures but we do not discuss them here.

In this formulation, it follows from the factorisation of the trace-norm that $\mathcal{N}(\rho \otimes \sigma) = \frac{\|\rho^{TA}\|_1 \|\sigma^{TA}\|_1}{2}$. For this reason, it is sometimes common to speak about the *logarithmic negativity*, defined as $\log_2(\|\rho^{TA}\|_1)$, because this measure now becomes additive over tensor product, i.e. $\log_2(\|\rho^{TA} \otimes \sigma^{TA}\|_1) = \log_2(\|\rho^{TA}\|_1) + \log_2(\|\sigma^{TA}\|_1)$.

A second entanglement measure for two qubits is called concurrence.

G15: Concurrence

Let ρ_{AB} be a two-qubit state and denote by $\{\lambda_i\}_i$ the decreasingly ordered eigenvalues of $R = \sqrt{\sqrt{\rho} \tilde{\rho} \sqrt{\rho}}$ where $\tilde{\rho} = (\sigma_Y \otimes \sigma_Y) \rho^* (\sigma_Y \otimes \sigma_Y)$. The concurrence of ρ_{AB} is

$$\mathcal{C}(\rho_{AB}) = \max\{\lambda_1 - \lambda_2 - \lambda_3 - \lambda_4, 0\}. \quad (4.41)$$

We have first defined the operator $\tilde{\rho} = (\sigma_Y \otimes \sigma_Y) \rho^* (\sigma_Y \otimes \sigma_Y)$. Since ρ^* still is a density operator and $\sigma_Y \otimes \sigma_Y$ is a local unitary, it follows that $\tilde{\rho}$ is a state. The product of two positive semidefinite operators is no longer Hermitian, but it still has non-negative eigenvalues. Note that for the maximally entangled state, $\phi^+ = \tilde{\phi}^+$ and $R = \phi^+$ and therefore $\mathcal{C} = 1$.

Every two-qubit entangled state can be detected both by negativity and concurrence. However, the two entanglement measures do not agree on how to order the entangled states. In other words, they can disagree on whether one state is more entangled than another. As an example of this, consider the family of states $|\psi_\theta\rangle = \cos \theta |00\rangle + \sin \theta |11\rangle$ for $\theta \in [0, \frac{\pi}{4}]$. One can compute that $\mathcal{N}(\psi_\theta) = \frac{1}{2} \mathcal{C}(\psi_\theta) = \frac{1}{2} \sin(2\theta)$. Then, consider for instance the states

$$\rho(\alpha) = \begin{pmatrix} 0.2 & 0 & 0 & 0 \\ 0 & 0.4 & \alpha & 0 \\ 0 & \alpha & 0.4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (4.42)$$

for $0 \leq \alpha \leq 0.4$. Computing the negativity and the concurrence we find $\mathcal{N}(\rho) = (\sqrt{1 + 100\alpha^2} - 1)/10$ and $\mathcal{C}(\rho) = 2\alpha$. Let us for instance select $\alpha = 0.04$ and $\theta = 0.01$. Then, concurrence claims that the second state is more entangled than the first because $2\alpha = 0.08 > \sin(2\theta) \approx 0.02$. However, negativity claims that the first state is more entangled than the second because $\frac{1}{2} \sin(2\theta) \approx 0.010 > (\sqrt{1 + 100\alpha^2} - 1)/10 \approx 0.008$. Thus, we see that the question of which state is more entangled may depend on how we choose to quantify entanglement.

4.5. Entanglement witness

Beyond qubit-qubit and qubit-qutrit states, PPT is necessary but not sufficient for separability. This means that some entangled states need other entanglement criteria to be detected. The PPT criterion is also not particularly practical: to use it we must know the whole density operator, but learning it for an initially unknown quantum state often requires a large number of measurements which we usually want to avoid. It would be more convenient to detect entanglement by only making a few measurements on the system, without ever learning the whole density operator. This can be achieved by using an *entanglement witness*.

The idea behind an entanglement witness is to measure the state with respect to a suitably chosen quantity and based on its value inferring if the state is entangled. The quantity of interest is represented by an observable which is a Hermitian operator H . To use H as an entanglement witness we must first determine its largest expectation value compatible with separable states. Then, we must show that there exists an entangled state that violates this limitation. Formally,

$$\text{tr}(\rho H) \leq \alpha, \quad \forall \text{ separable } \rho \quad \text{and} \quad \exists \text{ entangled } \sigma \quad \text{s.t.} \quad \text{tr}(\sigma H) > \alpha. \quad (4.43)$$

A central property of entanglement witnessing is that every entangled state can be detected through a suitable witness. This is the consequence of the Hahn-Banach theorem: if S is a convex set and P is a point outside it, then there exists a hyperplane that separates P from S ; see Fig 4.5. In our case, the set of separable states is convex and the points outside it in the bipartite quantum state space are precisely the entangled states. Knowing this, we are left with two challenges: (i) how should we choose H to detect the entanglement of a given state, and (ii) how do we determine the bound α ? We cannot expect there to exist a simple and general answer, because that would mean that we could easily decide between separable and entangled states which we know to be difficult. However, this limitation does not prevent the entanglement witness method from being useful.

Let us consider what is perhaps the simplest entanglement witness. For a pair of qubits, let us measure the observable

$$H = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} = \sigma_X \otimes \sigma_X + \sigma_Z \otimes \sigma_Z. \quad (4.44)$$

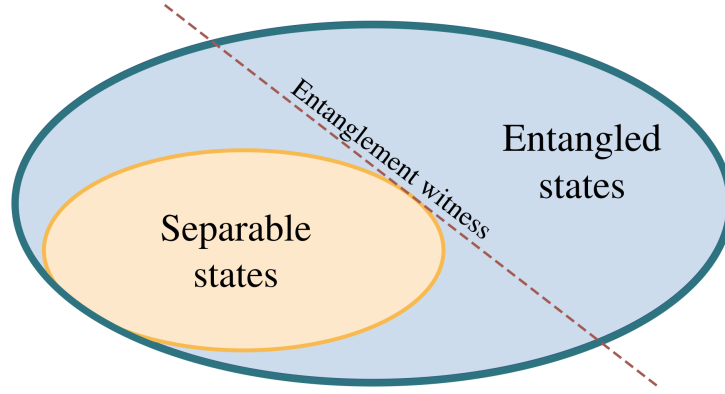


FIG. 4.5. The set of of separable states is a convex subset of the set of all states. Any state that is not separable is entangled. An entanglement witness is a hyperplane that separates the separable set from some of the entangled states.

First, we must determine its maximal expectation value over all separable states, namely

$$\text{tr}(\rho H) \stackrel{(4.31)}{=} \sum_i q_i \text{tr}(\phi_i \otimes \varphi_i (\sigma_X \otimes \sigma_X + \sigma_Z \otimes \sigma_Z)) = \sum_i q_i (\langle \phi_i | \sigma_X | \phi_i \rangle \langle \varphi_i | \sigma_X | \varphi_i \rangle + \langle \phi_i | \sigma_Z | \phi_i \rangle \langle \varphi_i | \sigma_Z | \varphi_i \rangle). \quad (4.45)$$

We use the Cauchy-Schwarz inequality to bound this expression. We use an inner product notation to make it clear how to apply the inequality,

$$\begin{aligned} \text{tr}(\rho H) &= \sum_i q_i (\langle \phi_i | \sigma_X | \phi_i \rangle \langle \phi_i | \sigma_Z | \phi_i \rangle) \cdot \left(\frac{\langle \varphi_i | \sigma_X | \varphi_i \rangle}{\langle \varphi_i | \sigma_Z | \varphi_i \rangle} \right) \\ &\stackrel{\text{CS-ineq}}{\leq} \sum_i q_i \sqrt{\langle \phi_i | \sigma_X | \phi_i \rangle^2 + \langle \phi_i | \sigma_Z | \phi_i \rangle^2} \sqrt{\langle \varphi_i | \sigma_X | \varphi_i \rangle^2 + \langle \varphi_i | \sigma_Z | \varphi_i \rangle^2}. \end{aligned} \quad (4.46)$$

Next, consider the quantity under the first square-root. In Bloch vector representation we have $\phi_i = \frac{1}{2} (\mathbb{1} + \vec{n}_i \cdot \vec{\sigma})$. We then have $\langle \phi_i | \sigma_X | \phi_i \rangle = n_{X,i}$ where $n_{X,i}$ is the X -component of $\vec{n}_i$. Similarly we have $\langle \phi_i | \sigma_Z | \phi_i \rangle = n_{Z,i}$ where $n_{Z,i}$ is the Z -component of $\vec{n}_i$. Putting it together, we get

$$\langle \phi_i | \sigma_X | \phi_i \rangle^2 + \langle \phi_i | \sigma_Z | \phi_i \rangle^2 = n_{X,i}^2 + n_{Z,i}^2 \leq 1, \quad (4.47)$$

where the last step follows because Bloch vectors must satisfy $|\vec{n}| \leq 1$. We can treat the second square-root in (4.46) analogously. Recalling that $\sum_i q_i = 1$, this brings us to the entanglement witness

$$\langle \sigma_X \otimes \sigma_X + \sigma_Z \otimes \sigma_Z \rangle^{\text{SEP}} \leq 1. \quad (4.48)$$

This inequality is violated by many entangled states. In particular the maximally entangled state $|\phi^+\rangle$ achieves $\langle \sigma_X \otimes \sigma_X + \sigma_Z \otimes \sigma_Z \rangle_{\phi^+} = 2$ because it is an eigenstate of both $\sigma_X \otimes \sigma_X$ and $\sigma_Z \otimes \sigma_Z$ with eigenvalue +1. Hence, we have met the conditions for an entanglement witness defined in Eq (4.43).

Let us apply the entanglement witness to the isotropic state defined in Eq (4.36). A direct calculation gives

$$\text{tr} \left((\sigma_X \otimes \sigma_X + \sigma_Z \otimes \sigma_Z) \left(p \phi^+ + (1-p) \frac{\mathbb{1}}{4} \right) \right) = p \underbrace{\text{tr}((\sigma_X \otimes \sigma_X + \sigma_Z \otimes \sigma_Z) \phi^+)}_{=2} + \frac{1-p}{4} \text{tr}(\sigma_X \otimes \sigma_X + \sigma_Z \otimes \sigma_Z) = 2p, \quad (4.49)$$

where we used that the Paulis are traceless. In order to violate the separable bound of the witness we require $2p > 1$, which means $p > \frac{1}{2}$. Compare this result, $p > \frac{1}{2}$, with the result $p > \frac{1}{3}$, that we got earlier from the PPT criterion. The PPT criterion is clearly better at detecting the entanglement of the isotropic state, but our witness was able to detect entanglement for a large portion of the relevant values of p by using only knowledge of the quantity H instead of the whole density operator ρ . In this spirit, one can detect entanglement using more modest resources, but these criteria are naturally less powerful than those that rely on knowledge of the entire density operator.

B5. A glance beyond

Entanglement witnessing via fidelity

Imagine an experimenter who wants to prepare the bipartite entangled state $|\psi\rangle$. Since he knows that he does not perfectly control all the processes in his laboratory, he expects to realistically end up preparing a different state, ρ , which hopefully approximates $|\psi\rangle$ reasonably well. To assess the quality of the preparation, he measures the fidelity with the target state, namely $F(\psi, \rho) = \langle\psi|\rho|\psi\rangle$. However, even if the fidelity is nearly unity, how can we be sure that the state ρ is at all entangled? The answer is that the fidelity itself can serve as an entanglement witness. To get the witness criterion, we need to determine

$$F_{\text{SEP}} \equiv \max_{\rho \in \text{SEP}} \langle\psi|\rho|\psi\rangle = \max_i \sum_i q_i \langle\psi|\phi_i \otimes \varphi_i|\psi\rangle. \quad (4.50)$$

This is a convex combination over i . Therefore, since the expression is also linear, we are guaranteed to reach the optimal value for a deterministic distribution $\{q_i\}_i$. Hence, it simplifies to $F_{\text{SEP}} = \max_{\phi, \varphi} F(\psi, \phi \otimes \varphi) = \max_{\phi, \varphi} \langle\psi|\phi \otimes \varphi|\psi\rangle$ where ϕ and φ are pure states. Let us now use that the fidelity between two pure bipartite states only can increase if we discard one subsystem, namely $F(\rho_{AB}, \sigma_{AB}) \leq F(\rho_A, \sigma_A)$. This can be seen as an immediate consequence of Uhlmann's theorem [R1](#) because ρ_{AB} and σ_{AB} are just one of all the possible purifications of ρ_A and σ_A respectively. Using this and writing $\psi_A = \text{tr}_B(\psi)$, we obtain

$$F_{\text{SEP}} \leq \max_{\phi} F(\psi_A, \phi) = \max_{\phi} \text{tr}(\psi_A \phi) = \lambda_{\max}(\psi_A). \quad (4.51)$$

Thus, we see that the largest eigenvalue of the reduced target state determines the separable limit of our entanglement witness. A larger fidelity than this value implies that the prepared state ρ must be entangled.

4.6. Three-particle entanglement

Entanglement becomes a richer concept when the state has more than two subsystems. An important difference with the bipartite case is that there no longer exists a maximally entangled state, i.e. there is no state from which we can obtain all other states in the same Hilbert space via LOCC. Consider for instance the following two entangled states of three qubits, called the Greenberger-Horne-Zeilinger state and the W -state respectively,

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle_{ABC} + |111\rangle_{ABC}), \quad |W\rangle = \frac{1}{\sqrt{3}}(|001\rangle_{ABC} + |010\rangle_{ABC} + |100\rangle_{ABC}). \quad (4.52)$$

Not only can we not LOCC-transform between these two states, but we cannot even do it if we accept that the transformation only succeeds with some non-zero probability. In contrast, in the bipartite case we can convert any pure entangled state into any other pure entangled state at some (perhaps very small) success probability; see Exercise [B.1.10](#). What this means is that the GHZ and W -state represent two distinct forms of entanglement²².

The difference between these two states is manifest in their entanglement structure. If we were to measure σ_Z on the first qubit in the GHZ-state and obtain outcome “0”, we would be left with the product state $|000\rangle$. However, were we to do the same for the W -state, the state would become $|0\rangle \otimes (|01\rangle + |10\rangle)/\sqrt{2}$, which is still entangled over the second and third qubit. We can go further and trace out the third qubit from both the states in Eq (4.52). The reduced GHZ state becomes

$$\rho_{AB}^{\text{GHZ}} = \text{tr}_C(|GHZ\rangle\langle GHZ|) = \frac{1}{2} \begin{pmatrix} 1 & & \\ & 0 & \\ & & 0 \end{pmatrix} = \frac{1}{2} |00\rangle\langle 00| + \frac{1}{2} |11\rangle\langle 11|. \quad (4.53)$$

This state is clearly separable. It can be interpreted as flipping an unbiased coin and then preparing the outcome in both subsystems, i.e. it is a bipartite coin-flip. We can think of this as entanglement of the GHZ-type truly needing all the three qubits. In contrast, a direct calculation (see Exercise [B.1.9](#)) shows that tracing out one of the qubits from the W -state leads to a very

²² While three qubits can be entangled in two different ways, it turns out that four-qubit entanglement can be sorted into nine different classes (some of these classes are infinitely large), and five-qubit entanglement needs infinitely many classes. Entanglement between many particles is a messy business.

different reduced state, specifically

$$\rho_{AB}^W = \text{tr}_C(|W\rangle\langle W|) = \frac{2}{3}|\psi^+\rangle\langle\psi^+| + \frac{1}{3}|00\rangle\langle 00|, \quad (4.54)$$

where $|\psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. To determine if ρ_{AB}^W is entangled, we need only to check if the partial-transpose is positive. This is addressed in Exercise B.1.9 and the result is that $\rho_{AB}^{T_B} \not\geq 0$. In conclusion, the entanglement of the W -state survives the loss of one subsystem, at the cost of introducing an amount of noise.

Given this feature of the W -state, it seems reasonable to ask whether we can construct a pure three-qubit state, ρ_{ABC} , such that all the two-qubit reduced states are maximally entangled. In other words, can we find a pure state ρ_{ABC} such that $\text{tr}_k(\rho_{ABC}) = \phi^+$ for any $k \in \{A, B, C\}$? The answer is negative and there are many reasons for it. A particularly simple reason comes from purification. Say that we pick $k = C$. Then we can think of ρ_{ABC} as a purification of the two-body marginal ϕ_{AB}^+ . However, since ϕ_{AB}^+ is itself pure, its purification is trivial, namely $\rho_{ABC} = \phi_{AB}^+ \otimes \tau_C$, where τ_C is some pure state. This follows from the fact that if we momentarily think of systems A and B as a single system, then in order for $\text{tr}_C(\rho_{ABC})$ to be pure, ρ_{ABC} must be product between AB versus C . This leads us to a contradiction, because if we now select $k = B$ we find that $\text{tr}_B(\rho_{ABC}) = \frac{1}{2} \otimes \tau$, which is clearly not maximally entangled. The lesson of this example is that the amount of entanglement between some qubits limits how entangled they can be with other qubits. This feature is known as *entanglement monogamy*: if one qubit “commits” to being strongly entangled with another qubit, it cannot be strongly entangled also with a third qubit.

Chapter summary.

- If a bipartite pure state can be factored as $|\psi\rangle_{AB} = |\phi\rangle_A \otimes |\varphi\rangle_B$ it is called a product state. Mixtures of product states, $\rho_{AB} = \sum_i q_i \phi_i \otimes \varphi_i$, are called separable states. States that are not separable are called entangled.
- Every pure bipartite state admits a Schmidt decomposition. While generic bipartite rays can be written as $|\psi\rangle = \sum_{j,k} c_{jk} |j, k\rangle$, Schmidt decomposition allows us to express them as $|\psi\rangle = \sum_i \sqrt{\lambda_i} |\alpha_i, \beta_i\rangle$, for appropriate orthonormal vectors $\{|\alpha_i\rangle\}_i$ and $\{|\beta_i\rangle\}_i$.
- A pure bipartite state is entangled if and only if its reduced state is mixed. A mixed two-qubit state is separable if and only if it has a positive partial-transpose.
- Nielsen’s majorisation theorem gives a necessary and sufficient condition for whether one pure state can be converted into another pure state via local operations and classical communication.
- Entanglement measures quantify the amount of entanglement in a given state. Since there is no global ordering of entangled states, different measures can disagree on whether one state is more entangled than another.
- Entanglement witnesses detect entanglement by measuring a suitably chosen physical quantity. Every entangled state can be detected through some entanglement witness.

Further reading. For a more extensive introduction, see for example [An introduction to entanglement measures](#) by Plenio and Virmani, Part II of [Modern Quantum Theory](#) by Bertlmann & Friis and the [lecture notes](#) of Watrous.

5. BELL NONLOCALITY

Quantum theory is famous for its counter-intuitive features (randomness, uncertainty relations, superposition, entanglement etc.). It is therefore a natural fundamental question whether we could hope to develop a new theory that stays much closer to traditional ideas of how Nature works but still reproduces the main body of physical predictions made by quantum theory²³. Strikingly, the answer is negative, and this plays a key role for understanding quantum theory. This result and its consequences for quantum information is the topic of this Chapter.

5.1. The Einstein-Podolsky-Rosen paradox

While discussions on the fundamental merits of quantum theory date back to its birth, a particularly insightful critique of the theory was leveraged in 1935 by Einstein, Podolsky and Rosen. Today, this critique is known as the EPR paradox. The essence of the paradox is as follows. Consider a pair of photons. We may think of the polarisation state as a qubit: horizontal (H), vertical (V), or a superposition thereof. We prepare them in the maximally entangled state $|\phi^+\rangle = \frac{1}{\sqrt{2}} [|HH\rangle + |VV\rangle]$. Quantum theory tells us that the state of each photon is maximally mixed and therefore the outcome of local single-photon polarisation measurements are completely random; the other photon could not know it in advance. However, if we measure the first photon and were to find polarisation H , then, according to quantum theory, we must update the state of the other photon so that it too has polarisation H . The theory does not provide a physical mechanism for this influence but simply postulates that such a seemingly immediate propagation of information must take place between the photons²⁴. This apparent violation of the idea that objects should only be influenced directly by their immediate surroundings lead to the criticism that quantum theory may be incomplete. What is meant by this is that the theory would be failing to describe all relevant aspects of Nature, i.e. some important elements of reality are left out of quantum theory.

In modern language, elements of reality are called *hidden variables*, i.e. a physically relevant parameter that our theory has overlooked. But why would a hidden variable be able to do away with some of the unorthodox features of quantum theory? Let us illustrate the answer through an example. Consider that two people observe an experiment in which a die is rolled. One observer is an average citizen and has no idea what the die will give. His theory is that the result is random. The second observer is a physicist. She takes note of the force of the throw, the angle of impact, the surface friction etc. Her theory for the outcome of the experiment is Newtonian mechanics and using it she can predict the outcome deterministically. Thus, what is random to the first observer is not random to the second observer. Why is this happening? Because the first observer's theory is incomplete: it is missing many variables that had they been known would have eliminated the randomness and replaced it with determinism. EPR suggested that a similar lack of knowledge is behind the seemingly random outcomes of the quantum polarisation measurement which triggers the seemingly faster-than-light state update: it only looks random because our theory does not know any better.

How could a hidden variable replace the quantum description of our two-photon polarisation experiment? Entertain the idea that quantum theory is wrong and what is actually going on is that some hidden variable is randomly preparing the photons either in the product state $|HH\rangle$ or in the product state $|VV\rangle$ with 50/50 probability before sending them apart. Thus, just as in quantum theory, the outcome of a polarisation measurement will be random and the photon polarisations remain always identical. However, no entanglement is needed and no immediate state update takes places upon measuring the polarisation. What is updated is only our knowledge of the hidden variable which determined the state of both polarisations in advance. Why should we adopt quantum theory when such a simple explanation is possible?

5.2. Local hidden variable theories

Let us formalise experiments of the type discussed in the EPR paradox. In Fig 5.6 we have a source that distributes one particle to Alice and another to Bob. Alice and Bob each privately choose to measure a property labelled by x and y respectively. Their respective measurement outcomes are denoted a and b . Thus, given the measurement choices (x, y) the outcomes (a, b) are observed. This is associated with a conditional probability distribution $p(a, b|x, y)$. Our goal is to model this conditional probability distribution using hidden variables that respect a classical view of physics. Let us denote the hidden variable by λ . We imagine that whatever it may be, it travels with the particles as they undergo separation. When Alice and Bob perform their measurements, their apparatuses can read λ and then select the output based on their respective inputs. Crucially, following EPRs argument, classicality is introduced via the principle of locality, namely that the outcome of Alice (Bob) is decided in her (his) lab alone, without any faster-than-light external influence. Therefore, for a given λ the probability of Alice and Bob's

²³ One can argue that quantum theory and its subsequent extensions into quantum field theories have a predictive power which is outstanding among scientific theories. For example, quantum electrodynamics' prediction of the electron's magnetic moment has been experimentally verified with an accuracy of 0.13 parts in a trillion. It lends remarkable credit to quantum theory.

²⁴ This was famously referred to by Einstein as a "spooky action at a distance".

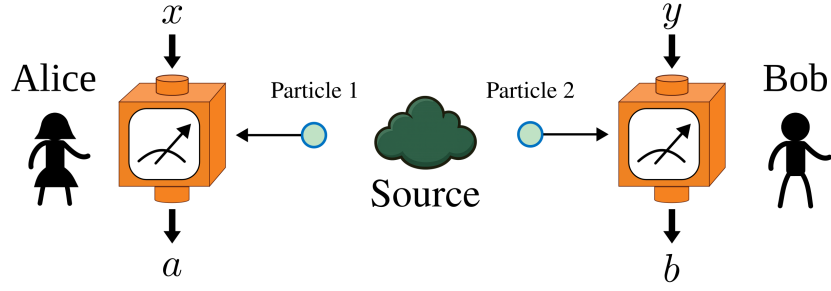


FIG. 5.6. A source generates a pair of particles and distributes them to two observers called Alice and Bob. Alice and Bob privately and independently select x and y and perform corresponding measurements on their particles. The outcomes of these measurements are denoted a and b respectively.

outcome is independent, $p(a, b|x, y, \lambda) = p(a|x, \lambda)p(b|y, \lambda)$, but the nature of these single-observer probability distributions remains unknown to us. Furthermore, the hidden variable can have some arbitrary probability density $q(\lambda)$ ²⁵. This leads us to a concise formulation of the predictions $p(a, b|x, y)$ made by hidden variable theories that respect the principle of locality.

G16: Local hidden variable model

A probability distribution $p(a, b|x, y)$ admits a local hidden variable model if it can be written on the form

$$p(a, b|x, y) = \int d\lambda q(\lambda) p_A(a|x, \lambda) p_B(b|y, \lambda), \quad (5.1)$$

where λ is the hidden variable and $q(\lambda)$ is some probability density function over its possible values.

While the response functions of Alice and Bob, $p_A(a|x, \lambda)$ and $p_B(b|y, \lambda)$, are in principle stochastic, one can show that any randomness appearing in them can be absorbed into the local hidden variable distribution $q(\lambda)$. Therefore, we may without loss of generality consider that Alice and Bob deterministically assign the outcomes a and b to the input pairs (x, λ) and (y, λ) respectively. To make that assignment, they can each use some function from the input to the output. Of course, for given finite sets of inputs and outputs, there are (finitely) many possible functions to use and all of them are candidates. Therefore, we can view λ as an index running over all the possible function pairs. For this, we write the functions as $f_\lambda : x \rightarrow a$ and $g_\lambda : y \rightarrow b$. The local hidden variable correlations in Eq (5.1) now take the form

$$p(a, b|x, y) = \sum_\lambda q(\lambda) D_\lambda^A(a|x) D_\lambda^B(b|y), \quad (5.2)$$

where we have written $D_\lambda^A(a|x) = \delta_{a, f_\lambda(x)} \in \{0, 1\}$ and $D_\lambda^B(b|y) = \delta_{b, g_\lambda(y)} \in \{0, 1\}$ to emphasise that the responses are deterministic. This notation simply means that Alice outputs $a = f_\lambda(x)$ and Bob outputs $b = g_\lambda(y)$ with unit probability. The integral in Eq (5.1) has been replaced with a sum since λ now takes only finitely many values.

There exists a geometric interpretation of the set of correlations that admit a local hidden variable model. For each given λ , the correlations between Alice and Bob are independent and given by $p_\lambda(a, b|x, y) \equiv D_\lambda^A(a|x) D_\lambda^B(b|y)$ which are deterministic since $p_\lambda(a, b|x, y) \in \{0, 1\}$. Since we have one such distribution for every λ , we can think of the set $\{p_\lambda(a, b|x, y)\}_\lambda$ as points in the space of conditional probability distributions, i.e. the space in which $p(a, b|x, y)$ lives. Moreover, since $q(\lambda)$ is a probability distribution, Alice and Bob can use it in Eq (5.2) to generate any convex combination over the set $\{p_\lambda(a, b|x, y)\}_\lambda$. Geometrically, the set of distributions $p(a, b|x, y)$ reachable in this way forms a polytope (a higher-dimensional convex polyhedron) whose vertices are $\{p_\lambda(a, b|x, y)\}_\lambda$; see Fig 5.7. Since this polytope is important to us, we give it a name: the *local polytope*. Any conditional probability distribution, $p(a, b|x, y)$, that belongs to the local polytope admits a local hidden variable model. All other $p(a, b|x, y)$ do not admit a local hidden variable model and are therefore called *nonlocal*²⁶. However, a different but ultimately equivalent characterisation of the local polytope is possible. We could describe it as a collection of linear inequalities, with each such inequality delimiting a part of the polytope's boundary; see the dashed line in Fig 5.7. For any given polytope, the inequalities that delimit its boundary are called *facets*, but for the local polytope specifically, the inequalities

²⁵ We assume that λ does not depend on x or y , i.e. that Alice and Bob can select their measurements independently of the hidden variable. This absence of metaphysical determinism, namely a freedom of choice, is arguably necessary in order to meaningfully test generic physical models. It can be partially relaxed, but not completely.

²⁶ We use the word nonlocality out of convenience to mean the failure of local hidden variable models. This clarification is necessary because nonlocality can have other meanings in different branches of physics, to which our use of the word is not related.

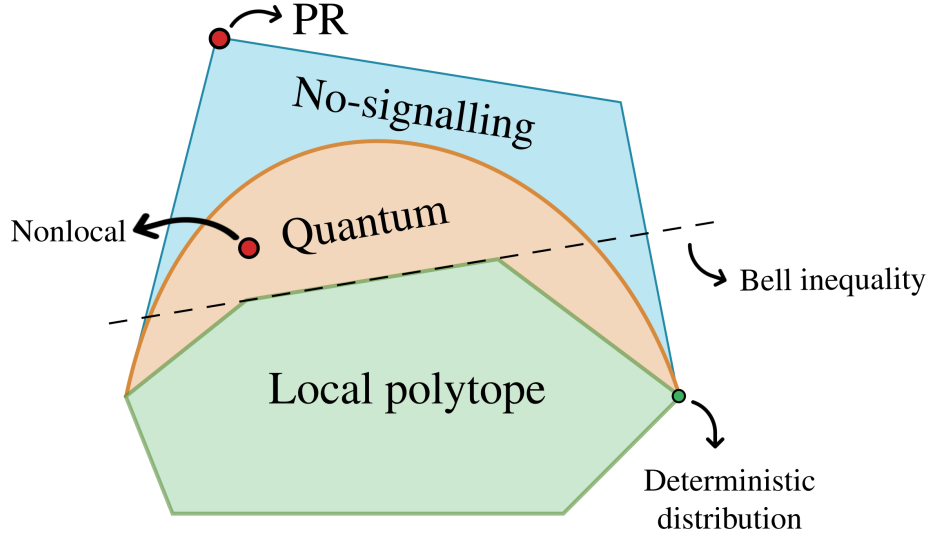


FIG. 5.7. The set of conditional probability distributions that admit a local hidden variable model forms a polytope. The vertices of this polytope are deterministic distributions, i.e. Alice and Bob each use a function to map their inputs to their outputs. The facets of the local polytope are called Bell inequalities. A conditional probability distribution that does not belong to the local polytope is called nonlocal. Bell's theorem tells us that quantum theory predicts distributions $p(a, b|x, y)$ that are nonlocal. The set of quantum conditional probability distributions is convex, as shown in Exercise B.2.1. The Popescu-Rohrlich probability distribution (discussed in Chapter 5.5) shows that there exists $p(a, b|x, y)$ that are no-signaling but incompatible with quantum theory.

are more famously known as *Bell inequalities*²⁷. Thus, a given $p(a, b|x, y)$ has a local hidden variable model if and only if it satisfies all the Bell inequalities associated with the local polytope. In other words, to falsify the possibility of a local hidden variable model, we must violate one of the Bell inequalities.

We now set out to identify a Bell inequality. To arrive at it, we consider a game played between Alice and Bob. Alice and Bob randomly and independently generate $x \in \{0, 1\}$ and $y \in \{0, 1\}$ respectively. Then, they are each asked to produce outputs $a \in \{0, 1\}$ and $b \in \{0, 1\}$ respectively. The game is won/lost if their outcomes satisfy the relation

$$\text{Winning condition:} \quad a \oplus b = xy, \quad (5.3)$$

$$\text{Losing condition:} \quad a \oplus b \neq xy, \quad (5.4)$$

When the winning condition is satisfied, Alice and Bob earn a point, and if they instead satisfy the losing condition they lose a point. In other words, the goal of the game is to output $a = b$ if $(x, y) \neq (1, 1)$ and to output $a \neq b$ if $(x, y) = (1, 1)$. This game is played many times, and in the limit of many rounds, the relative frequency of (a, b) given (x, y) becomes a probability distribution $p(a, b|x, y)$. We can now quantify the performance of Alice and Bob by looking at their point tally, namely

$$\mathcal{S} = \sum_{x, y} [p(a \oplus b = xy|x, y) - p(a \oplus b \neq xy|x, y)]. \quad (5.5)$$

The rules of the game require that Alice and Bob do not communicate with each other (otherwise they just exchange x and y and always win). This is enforced by placing Alice and Bob far away from each other and requiring them to perform their respective input/output assignments in space-like separated events. In other words, the time from Alice's generation of x to her production of a must be short enough so that no light-speed-travelling information about x can reach Bob in time to influence his outcome, and vice versa. In essence, Alice and Bob's local events are outside each other's lightcones.

Our goal is to determine the best performance possible in the game under the assumption that it is governed by a local hidden variable theory. To that end, we must determine the largest possible value of $\mathcal{S}$. A little trying and testing will suggest that the best Alice and Bob can do is $\mathcal{S} = 2$. One strategy that saturates this value is very simple: Alice and Bob simply ignore x and y and answer $a = 0$ and $b = 0$ all the time. In the three cases $(x, y) \in \{(0, 0), (0, 1), (1, 0)\}$ in which $xy = 0$ they win the game, while in the single case $(x, y) = (1, 1)$ in which $xy = 1$ they lose the game. Thus, their score in Eq (5.5) adds up to $\mathcal{S} = 1 + 1 + 1 - 1 = 2$. To make this rigorous, we now prove that in local hidden variable theories, one can never perform any better. This result is known as the *Clauser-Horne-Shimony-Holt (CHSH) inequality*.

²⁷ Some Bell inequalities are trivial from the physics point of view, for example that the relevant probabilities are non-negative. In Fig 5.7 this is for instance represented by the bottom facet of the local polytope.

R8: Clauser-Horne-Shimony-Holt inequality

Every local hidden variable theory respects the inequality

$$\mathcal{S} \equiv \langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle \stackrel{\text{LHV}}{\leq} 2, \quad (5.6)$$

where $\langle A_x B_y \rangle = \sum_{a,b=0,1} (-1)^{a+b} p(a, b|x, y)$. This inequality is a facet of the local polytope.

We now prove the CHSH inequality. To this end, notice first that the formulation of the parameter $\mathcal{S}$ in Eq (5.6) is equivalent to that in Eq (5.5). We can write it in terms of the probabilities $p(a, b|x, y)$ as $\mathcal{S} = \sum_{x,y} \sum_{a,b} (-1)^{a+b+xy} p(a, b|x, y)$. Now, we substitute $p(a, b|x, y)$ with the local hidden variable model in Eq (5.2) to obtain

$$\begin{aligned} \mathcal{S} &= \sum_{x,y} \sum_{a,b} (-1)^{a+b+xy} \sum_{\lambda} q(\lambda) D_{\lambda}^A(a|x) D_{\lambda}^B(b|y) = \sum_{\lambda} q(\lambda) \sum_{x,y} (-1)^{xy} \underbrace{\left(\sum_a (-1)^a D_{\lambda}^A(a|x) \right)}_{=A_x^{\lambda}} \underbrace{\left(\sum_b (-1)^b D_{\lambda}^B(b|y) \right)}_{=B_y^{\lambda}} \\ &= \sum_{\lambda} q(\lambda) (A_0^{\lambda}(B_0^{\lambda} + B_1^{\lambda}) + A_1^{\lambda}(B_0^{\lambda} - B_1^{\lambda})) \leq \max_{\lambda} (A_0^{\lambda}(B_0^{\lambda} + B_1^{\lambda}) + A_1^{\lambda}(B_0^{\lambda} - B_1^{\lambda})). \end{aligned} \quad (5.7)$$

In the last step, we see that one may restrict to a deterministic response function, namely the specific λ associated with the most successful deterministic strategy. By construction we have $A_x^{\lambda} \in \{+1, -1\}$ and $B_y^{\lambda} \in \{+1, -1\}$. This leads us to our result,

$$\mathcal{S} \leq A_0^{\lambda}(B_0^{\lambda} + B_1^{\lambda}) + A_1^{\lambda}(B_0^{\lambda} - B_1^{\lambda}) \leq |B_0^{\lambda} + B_1^{\lambda}| + |B_0^{\lambda} - B_1^{\lambda}| \leq 2. \quad (5.8)$$

This proves that local hidden variable theories cannot exceed $\mathcal{S} = 2$ but it does not prove that the CHSH inequality is a facet of the local polytope. The latter is shown in Exercise B.2.2. $\square$

The CHSH inequality is the simplest non-trivial Bell inequality. It is the simplest because Alice and Bob only have two possible inputs and outputs. One might consider that having only one input each would be simpler, but in such a scenario there is no interesting physics to learn: an argument of the type raised for the EPR paradox can be used to model any relevant conditional probability distribution with local hidden variables. However, if one ventures beyond the CHSH scenario by considering more than two inputs and/or outputs, the problem of identifying Bell inequalities becomes much more complicated (see Box B6).

B6. A glance beyond

The zoo of Bell inequalities

The CHSH inequality is the simplest non-trivial Bell inequality. It is also the only relevant Bell inequality when Alice and Bob only have two possible inputs and outputs. However, finding all relevant Bell inequalities becomes much more complicated if we start increasing the number of possible inputs and/or outputs. This is illustrated in the Table below. It gives the number of distinct Bell inequalities when Alice (Bob) has n_A (n_B) input choices and m_A (m_B) output choices respectively. The scenario is labelled (n_A, n_B, m_A, m_B) . We see that having three inputs adds just one new Bell inequality. As shown in Exercise B.2.6, this new Bell inequality turns out to reveal features of quantum nonlocality that are not within reach of the CHSH inequality. Next, going to four inputs for Alice and Bob leads to 175 Bell inequalities. For five inputs, there are many thousands more. We quickly reach beyond what is known in the literature.

#inputs & outputs	# Bell inequalities
(2, 2, 2, 2)	CHSH
(3, 3, 2, 2)	2
(2, 3, 3, 2)	4
(4, 4, 2, 2)	175
(4, 5, 2, 2)	≥ 18277
(2, 2, d , d)	unknown

5.3. Violating Bell inequalities

Let us now examine the CHSH inequality from the point of view of quantum theory. In quantum theory, the particles sent to Alice and Bob are described by a quantum state $|\psi\rangle_{AB}$. Importantly, this state can be entangled. Alice and Bob can each use x and y to choose a quantum measurement to apply to their share of $|\psi\rangle_{AB}$ and return the outcome as their respective answers a and b . For example, the measurement associated with $x = 0$ could be σ_Z and the measurement associated with $x = 1$ could be σ_X . The output $a \in \{0, 1\}$ would in the former case be associated with the positive and negative eigenvalues of σ_Z and the corresponding eigenvalues of σ_X in the latter case. In general, we denote Alice and Bob's measurements by $\{A_{a|x}\}_{a,x}$ and $\{B_{b|y}\}_{b,y}$ respectively. This should be read as follows: for every given x the operators in the set $\{A_{a|x}\}_a$ constitute a POVM, and analogously for Bob. Hence, $\{A_{a|x}\}_{a,x}$ ($\{B_{b|y}\}_{b,y}$) is not a POVM but a collection of POVMs, with the specific POVM labelled by x (y). The joint probabilities of Alice and Bob follow Born's rule, namely

$$p(a, b|x, y) = \langle \psi | A_{a|x} \otimes B_{b|y} | \psi \rangle. \quad (5.9)$$

The key question we must ask is whether there exists a conditional probability distribution predicted by quantum theory which achieves a value of $\mathcal{S}$ exceeding the limitation identified for local hidden variable theories. In other words, can quantum theory violate the CHSH inequality? The answer is positive. It means that there exists correlations in quantum theory that cannot be described by any conceivable local hidden variable model. This statement is known as *Bell's theorem*.

R9: Bell's theorem

Local hidden variable theories cannot reproduce all predictions of quantum theory.

Bell's theorem tells us that some predictions of quantum theory cannot be decomposed on the form in Eq (5.1). The simplest example of such nonlocality is appearing in the CHSH scenario. Consider that Alice and Bob share the maximally entangled two-qubit state $|\phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Let Alice measure the observable²⁸ $A_0 = \sigma_Z$ when $x = 0$ and $A_1 = \sigma_X$ when $x = 1$. Similarly, let Bob measure the observable $B_0 = \frac{\sigma_Z + \sigma_X}{\sqrt{2}}$ when $y = 0$ and the observable $B_1 = \frac{\sigma_Z - \sigma_X}{\sqrt{2}}$ when $y = 1$. We need to compute the four expectation values corresponding to all pairs (x, y) . These are

$$\begin{aligned} \langle \phi^+ | A_0 \otimes B_y | \phi^+ \rangle &= \frac{1}{\sqrt{2}} \underbrace{\langle \phi^+ | \sigma_Z \otimes \sigma_Z | \phi^+ \rangle}_{=1} + (-1)^y \frac{1}{\sqrt{2}} \underbrace{\langle \phi^+ | \sigma_Z \otimes \sigma_X | \phi^+ \rangle}_{=0} = \frac{1}{\sqrt{2}}, \\ \langle \phi^+ | A_1 \otimes B_y | \phi^+ \rangle &= \frac{1}{\sqrt{2}} \underbrace{\langle \phi^+ | \sigma_X \otimes \sigma_Z | \phi^+ \rangle}_{=0} + (-1)^y \frac{1}{\sqrt{2}} \underbrace{\langle \phi^+ | \sigma_X \otimes \sigma_X | \phi^+ \rangle}_{=1} = \frac{(-1)^y}{\sqrt{2}}. \end{aligned} \quad (5.10)$$

Inserting this in the CHSH expression (5.6) we obtain

$$\mathcal{S} = \sum_{x,y} (-1)^{xy} \underbrace{\langle \phi^+ | A_x \otimes B_y | \phi^+ \rangle}_{=\frac{(-1)^{xy}}{\sqrt{2}}} = \left(4 \times \frac{1}{\sqrt{2}} \right) = 2\sqrt{2}. \quad (5.11)$$

This is evidently larger than the limit $\mathcal{S} = 2$ respected by local hidden variable models. $\square$

We can illustrate the above constructed nonlocal distribution in Fig 5.7 as a point located outside the local polytope but inside the set of distributions achievable with quantum theory. Knowing that quantum theory can violate Bell inequalities, it seems natural to ask how large these violations can be. Our above example, based on a simple quantum state and measurements, has already reached the violation $\mathcal{S} = 2\sqrt{2} \not\leq 2$. Actually, it turns out that no larger CHSH violation is possible in quantum theory, even if we were to involve much more complex quantum states and measurements. This limitation is called *Tsirelson's bound*.

R10: Tsirelson's bound

The largest violation of the CHSH inequality possible in quantum theory is $\mathcal{S} = 2\sqrt{2}$.

²⁸ Recall that the observable defines the POVM elements through Eq (3.4).

To prove Tsirelson's bound, let $|\psi\rangle_{AB}$ be some arbitrary shared state and let us define vectors

$$|\alpha_x\rangle = (A_x \otimes \mathbb{1})|\psi\rangle_{AB}, \quad |\beta_y\rangle = (\mathbb{1} \otimes B_y)|\psi\rangle_{AB}, \quad (5.12)$$

where $A_x = A_{0|x} - A_{1|x}$ and $B_y = B_{0|y} - B_{1|y}$ are the observables of Alice and Bob. From Born's rule, the CHSH quantity $\mathcal{S}$ defined in Eq (5.6) now becomes

$$\mathcal{S} = \langle \alpha_0 | (|\beta_0\rangle + |\beta_1\rangle) + \langle \alpha_1 | (|\beta_0\rangle - |\beta_1\rangle). \quad (5.13)$$

Next, we apply the Cauchy-Schwarz inequality separately to each of the above terms. This gives us

$$\mathcal{S} \leq (\|\alpha_0\| \|\beta_0 + \beta_1\| + \|\alpha_1\| \|\beta_0 - \beta_1\|). \quad (5.14)$$

Before evaluating the norms, we can without loss of generality assume that the measurements are projective. This is because we have made no assumption on the local Hilbert space dimensions of Alice and Bob. Because of this, any non-projective POVM that Alice and Bob might choose to perform can be implemented as a projective POVM in a larger Hilbert space via the Naimark dilation encountered in Chapter 3.4. We use this together with the completeness relation to evaluate the norm

$$\begin{aligned} \|\alpha_x\|^2 = \langle \alpha_x | \alpha_x \rangle &\stackrel{(5.12)}{=} \langle \psi | A_x^2 \otimes \mathbb{1} | \psi \rangle = \langle \psi | \left(\underbrace{A_{0|x}^2}_{=A_{0|x}} + \underbrace{A_{1|x}^2}_{=A_{1|x}} - \underbrace{\{A_{0|x}, A_{1|x}\}}_{=0} \right) \otimes \mathbb{1} | \psi \rangle = \langle \psi | \mathbb{1} \otimes \mathbb{1} | \psi \rangle = 1. \end{aligned} \quad (5.15)$$

By an analogous argument, it also holds that $\|\beta_y\|^2 = 1$. Furthermore,

$$\|\beta_0 \pm \beta_1\|^2 = \langle \beta_0 | \beta_0 \rangle + \langle \beta_1 | \beta_1 \rangle \pm \langle \beta_0 | \beta_1 \rangle \pm \langle \beta_1 | \beta_0 \rangle = 2 \pm 2 \operatorname{Re}(\langle \beta_0 | \beta_1 \rangle). \quad (5.16)$$

We can therefore simplify our bound in Eq (5.14) to

$$\mathcal{S} \leq \sqrt{2} \left[\sqrt{1 + \operatorname{Re}(\langle \beta_0 | \beta_1 \rangle)} + \sqrt{1 - \operatorname{Re}(\langle \beta_0 | \beta_1 \rangle)} \right] \leq 2\sqrt{2}. \quad (5.17)$$

To motivate the final step, we need only to note that the maximum of the function $f(x) = \sqrt{1+x} + \sqrt{1-x}$ for $x \in [-1, 1]$ occurs at $x = 0$ ²⁹. $\square$

Our explicit violation of the CHSH inequality in Eq (5.11) used the maximally entangled state. However, the sizable gap in the value of $\mathcal{S}$ between local hidden variable theories and quantum theory indicates that many other quantum states should also be able to violate the CHSH inequality. Consider that we are given some arbitrary two-qubit state ρ_{AB} . By appropriately selecting the measurements of Alice and Bob, what is the largest possible value of $\mathcal{S}$? To answer this, let us use that $\{\sigma_i \otimes \sigma_j\}_{i,j}$ for $i, j = 0, X, Y, Z$ where $\sigma_0 = \mathbb{1}$, is a basis of Hermitian operators on the Hilbert space of two qubits. We can therefore express the state as a linear combination of these 16 basis elements, which we for convenience write as

$$\rho_{AB} = \frac{1}{4} \left(\mathbb{1} \otimes \mathbb{1} + \vec{r} \cdot \vec{\sigma} \otimes \mathbb{1} + \mathbb{1} \otimes \vec{s} \cdot \vec{\sigma} + \sum_{i,j=X,Y,Z} T_{ij} \sigma_i \otimes \sigma_j \right), \quad (5.18)$$

where $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$. Note that since all terms but the first are traceless, we have fixed the coefficient in front of the first term to ensure the normalisation $\operatorname{tr}(\rho_{AB}) = 1$. Furthermore, we require that individual states of both qubits are normalised positive semidefinite operators. This corresponds to choosing $\vec{r}$ and $\vec{s}$ as Bloch vectors for Alice and Bob respectively. Finally, the correlations between Alice and Bob are accounted for by the correlation tensor $T_{ij} = \operatorname{tr}(\sigma_i \otimes \sigma_j \rho_{AB})$. We can use this representation to determine the largest possible value of $\mathcal{S}$ for any given two-qubit state ρ_{AB} . This value is

$$\mathcal{S}(\rho_{AB}) = 2\sqrt{\lambda_1 + \lambda_2}, \quad (5.19)$$

where λ_1 and λ_2 are the two largest eigenvalues of $T^T T$. This is known as the *Horodecki criterion* and it is derived in Exercise B.2.5. In particular, the criterion implies that a given two-qubit state violates the CHSH inequality if and only if $\lambda_1 + \lambda_2 > 1$. Also, to saturate Tsirelson's bound we need $\lambda_1 = \lambda_2 = 1$. As expected, this is achieved by selecting $\rho_{AB} = \phi^+$. This follows from the fact that $T^T T = \mathbb{1}$ because $|T_{ij}| = |\operatorname{tr}(\sigma_i \otimes \sigma_j \phi^+)| = \frac{1}{2} |\operatorname{tr}(\sigma_i \sigma_j^T)| = \delta_{ij}$, where the second step uses the property of the maximally entangled state from Eq (4.21).

Although we have focused on the CHSH inequality, we know from Box B6 that there are many other Bell inequalities and in Exercises B.2.6 we see that some of these reveal the quantum nonlocality of entangled states that cannot violate the CHSH inequality. In Exercise B.2.7 and B.2.8 we will show yet another manifestation of quantum nonlocality: that it can provide solutions to logical problems where local hidden variables cannot.

²⁹ Consider $f(x) = \sqrt{f(x)^2} = \sqrt{(1+x) + (1-x) + 2\sqrt{(1+x)(1-x)}} = \sqrt{2 + 2\sqrt{1-x^2}}$. From here, it is clear that the optimum is at $x = 0$.

5.4. Entanglement versus nonlocality

It is not possible to violate a Bell inequality without Alice and Bob sharing an entangled state. This can be seen directly from applying Born's rule to a separable state $\rho_{AB} = \sum_{\lambda} q_{\lambda} \phi_{\lambda} \otimes \varphi_{\lambda}$. Denoting Alice and Bob's POVMs by $\{A_{a|x}\}_{a,x}$ and $\{B_{b|y}\}_{b,y}$ respectively, the probabilities become

$$\begin{aligned} p(a, b|x, y) &= \text{tr}((A_{a|x} \otimes B_{b|y}) \rho_{AB}) = \sum_{\lambda} q_{\lambda} \text{tr}((A_{a|x} \otimes B_{b|y}) (\phi_{\lambda} \otimes \varphi_{\lambda})) \\ &= \sum_{\lambda} q_{\lambda} \underbrace{\text{tr}(A_{a|x} \phi_{\lambda})}_{=p(a|x, \lambda)} \underbrace{\text{tr}(B_{b|y} \varphi_{\lambda})}_{=p(b|y, \lambda)} = \sum_{\lambda} q_{\lambda} p(a|x, \lambda) p(b|y, \lambda) = \text{Eq (5.1)}. \end{aligned} \quad (5.20)$$

Thus, we have on the right-hand-side recovered a local hidden variable model. This should come as no surprise: entanglement was the fuel behind the EPR paradox. One might reasonably expect that nonlocality is the essential manifestation of entanglement, i.e. that every entangled state reveals its nonclassical nature by violating some Bell inequality. Remarkably, this is actually not the case. Entanglement and nonlocality are distinct features of quantum theory.

In discussing the relationship between entanglement and nonlocality, let us focus on a family of states that we have already encountered, namely the two-qubit isotropic state

$$\rho_p = p |\phi^+\rangle\langle\phi^+| + \frac{1-p}{4} \mathbb{1}, \quad (5.21)$$

parameterised by $p \in [0, 1]$. Back in Chapter 4.3 we used the PPT criterion to show that this state is entangled if and only if $p > \frac{1}{3}$. This prompts us to ask for which p can we violate the CHSH inequality. We have already seen that a maximally entangled state can give a violation $S = 2\sqrt{2}$. By applying the Horodecki criterion to ρ_p we find that

$$S = 2\sqrt{2}p \quad \Rightarrow \quad \text{The CHSH inequality is violated when } p > \frac{1}{\sqrt{2}}. \quad (5.22)$$

Hence, in the interval $\frac{1}{3} < p \leq \frac{1}{\sqrt{2}}$ the state is entangled but fails to violate the CHSH inequality. What should we make of this observation? It is not a proof that entanglement and nonlocality are inequivalent because the CHSH inequality is only the simplest Bell inequality among countless more (see Box B6). It could be that one of the other Bell inequalities is violated also for smaller values of p than $\frac{1}{\sqrt{2}}$. While such better Bell inequalities indeed exist, none of them can be violated for every $p > \frac{1}{3}$. In other words, there exists a local hidden variable model that can reproduce every quantum prediction when performing arbitrary local projective measurements³⁰ on some entangled isotropic states. This is known as *Werner's local hidden variable model*.

To derive Werner's model, consider arbitrary projective measurements for Alice and Bob described by Bloch vectors $\vec{n}_A$ and $\vec{n}_B$ respectively. A direct evaluation of Born's rule shows that the quantum correlations obtained from measuring ρ_p are

$$p(a, b|\vec{n}_A, \vec{n}_B) = \frac{1}{4} (1 + (-1)^{a+b} p \vec{n}'_A \cdot \vec{n}_B), \quad (5.23)$$

where $\vec{n}' = (n_X, -n_Y, n_Z)$. The task of the local hidden variable model is to generate these probabilities. To show how it is done, we proceed in steps.

1. Since we are aiming to simulate qubit correlations, let the source distribute a local hidden variable that emulates a Bloch vector. We choose it as the vector $\vec{\lambda} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ sampled over the surface of a unit-sphere with uniform probability density $q(\vec{\lambda}) = \frac{1}{4\pi} \sin \theta d\theta d\phi$. This is merely a vector and not a representation of a quantum state.
2. Alice takes the Bloch vector of her measurement, $\vec{n}_A$, and computes the inner product of it with the local hidden variable, namely $\vec{n}_A \cdot \vec{\lambda}$. Since $|\vec{n}_A \cdot \vec{\lambda}| \leq 1$, she can use this to define her local response function as

$$p_A(a|\vec{n}_A, \vec{\lambda}) = \frac{1}{2} (1 + (-1)^a \vec{n}_A \cdot \vec{\lambda}). \quad (5.24)$$

3. Bob takes the Bloch vector of his measurement, $\vec{n}_B$, and computes $\vec{n}_B \cdot \vec{\lambda}$. If $\vec{n}_B \cdot \vec{\lambda} \geq 0$ he outputs $b = 0$ and otherwise he outputs $b = 1$. Hence, he determines his outcome based on whether or not $\vec{\lambda}$ belongs to the hemisphere defined by $\vec{n}_B$. We write this as

$$p_B(b|\vec{n}_B, \vec{\lambda}) = \frac{1}{2} (1 + (-1)^b \text{sgn}(\vec{n}_B \cdot \vec{\lambda})). \quad (5.25)$$

³⁰ Since consider a specific state, the local Hilbert space dimensions are fixed. Therefore, projective measurements are a strict subset of all POVMs. Although we focus on projective measurements, the result that some entangled isotropic states admit a local hidden variable model holds also for POVMs.

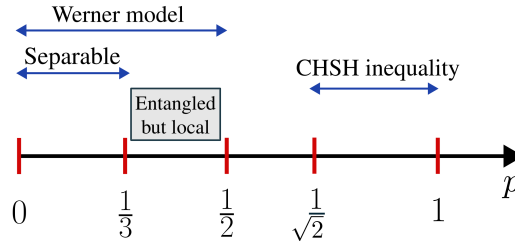


FIG. 5.8. The isotropic state is entangled for $p > 1/3$, admits a local hidden variable model for $p \leq \frac{1}{2}$ and violates the CHSH inequality for $p > \frac{1}{\sqrt{2}}$.

To compute the probabilities associated with this local hidden variable model, let us for simplicity focus on $b = 0$, i.e. the sign of $\vec{n}_B \cdot \vec{\lambda}$ is positive; Bob's response is only supported when $\vec{\lambda}$ belongs to the hemisphere defined by $\vec{n}_B$. Spherical symmetry permits us to choose $\vec{n}_B = (0, 0, 1)$ to simplify the calculation. This makes $b = 0$ associated with the northern Bloch-hemisphere, which is defined by polar angle $\theta \in [0, \frac{\pi}{2}]$. Consequently,

$$\begin{aligned} p(a, 0 | \vec{n}_A, \vec{n}_B) &= \int d\lambda q(\lambda) p_A(a | \vec{n}_A, \lambda) p_B(0 | \vec{n}_B, \lambda) = \frac{1}{4\pi} \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} d\theta \sin \theta \frac{1}{2} (1 + (-1)^a \vec{n}_A \cdot \vec{\lambda}) \\ &= \frac{1}{4} + \frac{(-1)^a}{8\pi} \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} d\theta \sin \theta \vec{n}_A \cdot (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta). \end{aligned} \quad (5.26)$$

The integral on the right-hand-side splits into three terms. However, the first and second terms are zero because the ϕ -integrals vanish. Computing the third term gives us the result

$$p(a, 0 | x, y) = \frac{1}{4} + n_{A,Z} \underbrace{\frac{(-1)^a}{8\pi} \int_0^{2\pi} d\phi}_{=2\pi} \underbrace{\int_0^{\frac{\pi}{2}} d\theta \sin \theta \cos \theta}_{=1/2} = \frac{1}{4} \left(1 + \frac{1}{2} \vec{n}_A \cdot (0, 0, 1) \right). \quad (5.27)$$

For our selected $\vec{n}_B = (0, 0, 1)$, this is precisely the quantum prediction in Eq (5.23) if we choose the noise parameter as $p = \frac{1}{2}$. Since the analogous argument can be made for any $\vec{n}_B$ and also $b = 1$, we must conclude that the local hidden variables can model all quantum theory's predictions on the isotropic state for $p \leq \frac{1}{2}$. Importantly, this includes a range in which the isotropic state is entangled³¹. The relationship between the entanglement of the isotropic state and its nonlocality is illustrated in Fig 5.8.

B7. A glance beyond

Revisiting the Einstein-Podolsky-Rosen paradox

The critique of EPR concerned how quantum theory lets Alice immediately update the remote state of Bob. If ρ_{AB} is the shared state and Alice measures the POVMs $\{A_{a|x}\}_{a,x}$, we can effectively describe Bob's local state with $\sigma_{a|x} = \text{tr}_A (A_{a|x} \otimes \mathbb{1} \rho_{AB})$. Note that this is just his updated state multiplied with the probability of the specific update, $p(a|x) = \text{tr}(\sigma_{a|x})$. Now, instead of going the route through Bell nonlocality, we could directly try to describe the set $\{\sigma_{a|x}\}_{a,x}$ using local variables. To that end, the local variable λ has some unknown probability density $q(\lambda)$. It would have to determine Alice's outcome together with her input, x , according to some response $p(a|x, \lambda)$. Furthermore, the local variable must supply Bob with a quantum state τ_λ . This *local hidden state* model becomes

$$\sigma_{a|x} = \int d\lambda q(\lambda) p(a|x, \lambda) \tau_\lambda, \quad \forall a, x. \quad (5.28)$$

If such a model exists then we can explain Alice's update of Bob's state without breaking with the principle of locality. However, if no such model exists we have a manifestation of nonlocality that is called *quantum steering*. Quantum steering is a strictly weaker notion than quantum nonlocality, i.e. some entangled states that cannot violate any Bell inequality can still achieve quantum steering.

³¹ We did not address the range $\frac{1}{2} < p \leq \frac{1}{\sqrt{2}}$, where the state is entangled but Werner's model does not apply. At the time of writing (2026), local hidden variable models are known up to $p \approx 0.6875$ and Bell inequality violations have been found above $p \approx 0.6964$. Thus, only a small window of uncertainty remains. However, beyond the two-qubit isotropic state considered here, little is known about which entangled states that can violate Bell inequalities.

5.5. No-signaling theories

If we momentarily forget the physics, we may note that the mathematically largest possible value of the parameter in the CHSH test is $\mathcal{S} = 4$. Nevertheless, Tsirelson's bound tells us that quantum theory cannot exceed $\mathcal{S} = 2\sqrt{2}$. Why should the amount of allowed nonlocality in Nature be limited in this seemingly arbitrary way? One may hypothesise that the answer is causality: perhaps Tsirelson's bound can be justified directly from the cornerstone principle that Alice and Bob's space-like separated events are independent? If true, this would be remarkable, because we would obtain the quantum prediction directly from causality without any further assumptions. While this is not how things are going to play out, we will nevertheless see that a world in which nonlocality can be very strong comes with some radical consequences.

Causality stipulates that if Alice and Bob's input/output processes are space-like separated they must be independent, as Alice cannot send information to Bob, and vice versa. In Bell scenarios, this limitation takes the form of the *no-signaling principle*.

G17: No-signaling principle

If Alice and Bob's measurement events are space-like separated, the input of Alice cannot influence the outcome of Bob, and vice versa.

We can formalise the no-signaling principle in the following equations,

$$\sum_b p(a, b|x, y) = p(a|x, y) \stackrel{\text{G17}}{=} p(a|x), \quad \forall a, x, y \quad (5.29)$$

$$\sum_a p(a, b|x, y) = p(b|x, y) \stackrel{\text{G17}}{=} p(b|y), \quad \forall b, x, y. \quad (5.30)$$

i.e. the marginal probability of Alice (Bob) does not depend on the input of Bob (Alice). Importantly, this is not the same as the principle of locality. In the EPR argument, quantum theory breaks with locality because remote state updates are possible, but since the outcomes of Alice are random she cannot control how Bob's state is updated, i.e. there is no transmission of information but only an action at a distance. Indeed, as shown in Exercise B.2.10, both local hidden variable models and quantum theory respect the no-signaling principle.

Let us now consider the parameter $\mathcal{S}$ in any theory limited only by the no-signaling principle. To that end, imagine that there exist a pair of devices for Alice and Bob that can generate the following probability distribution

$$p(a, b|x, y) = \frac{1}{4} [1 + (-1)^{a+b+xy}]. \quad (5.31)$$

A quick check shows that this distribution obeys all the conditions we require, namely $p(a, b|x, y) \geq 0$ (non-negativity of probability), $\sum_{a,b} p(a, b|x, y) = 1$ (normalisation), and $\sum_b p(a, b|x, y) = \sum_a p(a, b|x, y) = 1/2$ (no-signaling). This distribution is known as a *Popescu-Rohrlich no-signaling box* and it gives the mathematically maximal violation of the CHSH inequality,

$$\mathcal{S} \stackrel{(5.6)}{=} \sum_{a,b,x,y} (-1)^{a+b+xy} \left(\frac{1}{4} [1 + (-1)^{a+b+xy}] \right) = 4. \quad (5.32)$$

Hence, we must conclude that no-signaling theories can make predictions that go beyond what is possible in quantum theory. The relations between the correlations obtained from local hidden variable theories, quantum theory and no-signaling theories is a strict hierarchy, as illustrated in Fig 5.7.

5.5.1. The strangeness of a no-signaling world

Since no-signaling does not forbid theories with larger-than-quantum nonlocality, we might wonder whether Nature would be better described by a completely different no-signaling theory³². In other words, what distinguishes quantum theory? While a complete answer is not known, good partial answers are available and we will discuss one of them. Our purpose is to show that even though a no-signaling theory respects causality, it could still lead to predictions that we might deem too radical.

Imagine that Alice and Bob want to communicate. Alice holds two randomly selected bits, $x_0, x_1 \in \{0, 1\}$. Bob wants to learn either x_0 or x_1 . He marks the bit he is interested in by selecting $y \in \{0, 1\}$. However, Alice does not know y . One option is that

³² There are many arguments for why quantum theory ought not to be the ultimate fundamental theory of physics. We do not pursue that discussion here.

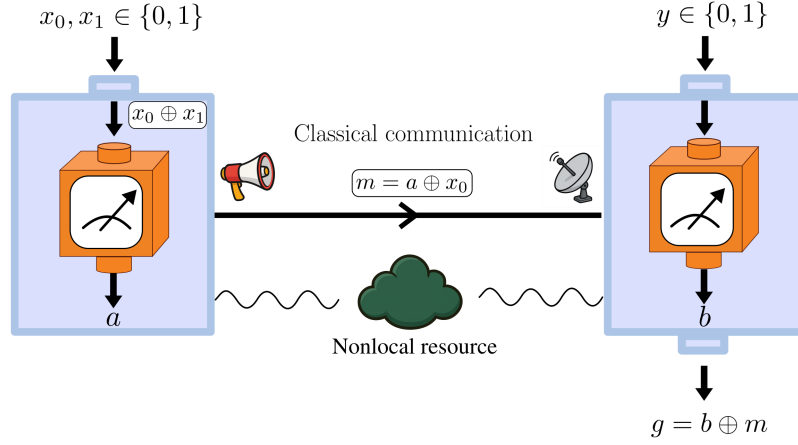


FIG. 5.9. Alice has two bits, (x_0, x_1) , which she encodes into a binary message m . From this message, Bob must retrieve either one of the bits. To facilitate this, Alice and Bob share a nonlocal resource which can advise their communication strategy.

Alice sends both her bits to Bob and he selects bit number y , but in our scenario we are going to forbid this by restricting Alice to send only one bit to Bob. Now, it looks impossible for Bob to always learn the bit of his interest. Classically, this intuition is entirely correct, and it is in fact also true in quantum theory, but in a no-signaling world it is not. Consider the scenario illustrated in Fig 5.9. Let Alice define the bit $x = x_0 \oplus x_1$. She and Bob use x and y as settings in a test of the CHSH inequality. This gives them outputs $a \in \{0, 1\}$ and $b \in \{0, 1\}$ respectively. Then, Alice encodes the binary message $m = a \oplus x_0$ and sends it to Bob. Once received by Bob, he computes $g = m \oplus b = a \oplus b \oplus x_0$ and outputs it as his guess for x_y . The average success probability is

$$p_{\text{suc}} = \frac{1}{8} \sum_{x_0, x_1, y} p(x_y = a \oplus b \oplus x_0 | x, y) = \frac{1}{4} \sum_{x, y} \underbrace{p(a \oplus b = xy | x, y)}_{=2+\mathcal{S}/2 \text{ via Eq (5.5)}}, \quad (5.33)$$

where in the last step we have identified the $\mathcal{S}$ -parameter after a minor re-writing of Eq (5.5) by using that $p(a \oplus b \neq xy) = 1 - p(a \oplus b = xy)$. Thus, we find that the average success probability in our communication task is in one-to-one correspondence with $\mathcal{S}$. Since the CHSH inequality limits local hidden variable theories to $\mathcal{S} = 2$, it follows that the communication cannot exceed the success probability $p_{\text{suc}} = \frac{3}{4}$. However, if Alice and Bob share entanglement they can violate the CHSH inequality and reach $\mathcal{S} = 2\sqrt{2}$, which implies the enhanced success probability $p_{\text{suc}} \approx 0.853$. While clearly better than in the classical case, quantum theory still does not allow Bob to deterministically learn the bit of his interest. To make that possible, we imagine that Alice and Bob live in a no-signaling world in which Popescu-Rohrlich boxes are available. This nonlocal device achieves the maximal value $\mathcal{S} = 4$, which therefore gives us $p_{\text{suc}} = 1$, implying that the communication is always successful. The conclusion is somewhat peculiar: in a no-signaling world Bob still cannot learn both bits at the same time, but he can freely choose to learn either one of them.

One may argue that the quantum enhancement of this task is already counter-intuitive, and that the use of no-signaling correlations simply makes the effect stronger. Therefore, let us build further on the above task to reveal a dramatic distinction between quantum theory and no-signaling theories. Consider that Alice and Bob are each given a random string of n bits, namely $x = x_1 \dots x_n \in \{0, 1\}^n$ and $y = y_1 \dots y_n \in \{0, 1\}^n$. Their task is to compute a distributed boolean function $f(x, y) \in \{0, 1\}$. For example, f could be the inner product function, defined as $f(x, y) = \text{IP}(x, y) = \bigoplus_{i=1}^n x_i y_i$. It seems intuitive that in order to compute the inner product, Alice must send her entire data set x to Bob, i.e. a message of n bits, so that he can evaluate $\text{IP}(x, y)$. As it turns out, the same is also true if Alice and Bob first share entanglement and generate quantum nonlocality. But what if Alice and Bob live in a no-signaling world? We now show that this would allow them to solve the task by using only a single bit of communication, independently of n . This is certainly radical, because it means that the cost of communication is entirely collapsed, regardless of how large the data set actually is.

R11: No-signaling correlations trivialise communication complexity

In a world in which nonlocality is only limited by the no-signaling principle, communication complexity is trivial.

In order to prove this, we use that every boolean function can be decomposed on the following form,

$$f(x, y) = \bigoplus_{i=1}^{2^n} g_i(x) h_i(y), \quad (5.34)$$

where g_i are polynomials and h_i are monomials³³. With this, we can represent the most general boolean function as an inner product function between a 2^n bit-string $\{g_i(x)\}_i$ and another 2^n bit-string $\{h_i(y)\}_i$. Notice that Alice and Bob locally can compute their respective bit-strings since they can locally determine g_i and h_i from f . Now, we let Alice and Bob share 2^n Popescu-Rohrlich boxes where they use the bits $g_i(x)$ and $h_i(y)$ as their respective inputs for the i 'th box. Their outputs are denoted (a_i, b_i) and they are guaranteed to satisfy $a_i \oplus b_i = g_i(x) h_i(y)$. Hence, we have

$$f(x, y) = \bigoplus_{i=1}^{2^n} g_i(x) h_i(y) = \bigoplus_{i=1}^{2^n} (a_i \oplus b_i) = \left(\bigoplus_{i=1}^{2^n} a_i \right) \oplus \left(\bigoplus_{i=1}^{2^n} b_i \right). \quad (5.35)$$

Alice can now locally compute the bit $a = \bigoplus_{i=1}^{2^n} a_i$ and send it to Bob. He then outputs $g = a \oplus \left(\bigoplus_{i=1}^{2^n} b_i \right)$ which is guaranteed to be equal to the value of $f(x, y)$. Thus, every distributed boolean function, of any input size, can be computed with only one bit of communication in a world restricted only by the no-signaling principle. $\square$

This result shows us that theories that respect no-signaling still can have extreme predictions, that at least appear to defy basic conditions of the world we live in. There are also other examples in this spirit, further revealing the strangeness of a no-signaling world. Therefore, in a quest to narrow down the space of possible theories of physics, one way forward is to supplement the no-signaling principle with postulates of this type, such as “no collapse of communication complexity”³⁴, in hope to better understand what makes quantum theory our currently leading answer to how Nature does it all.

Chapter summary.

- A local hidden variable theory uses a shared common cause to locally determine how the inputs of Alice and Bob respectively transform into their outputs.
- Bell's theorem states that not all predictions of quantum theory can be accounted for in local hidden variable theories. The simplest demonstration of this is the violation of the CHSH inequality where local hidden variables achieve at most a value of 2 but quantum theory can reach $2\sqrt{2}$.
- Entanglement and nonlocality are distinct phenomena. Entanglement is necessary for nonlocality but some entangled states cannot violate any Bell inequality.
- The no-signaling principle says that information cannot be transmitted between space-like separated events. It is a minimal requirement in theories for Bell scenarios. No-signaling theories predict stronger forms of nonlocality than does quantum theory.

Further reading. A more comprehensive coverage of Bell nonlocality is found in the book [Bell Nonlocality](#) by Scarani.

³³ The reason the summation runs from 1 to 2^n is that there are 2^n possible monomials, $y_1, y_1 y_2, y_1 y_2 y_4, \dots, y_1 \dots y_n$.

³⁴ If one adopts this as a postulate for no-signaling theories, one can prove that the inequality $S \leq 4 \frac{\sqrt{2}}{\sqrt{3}}$ is always satisfied. It eliminates a swath of no-signaling theories.

6. QUANTUM INFORMATION PRIMITIVES

We are now ready to start constructing useful quantum information protocols. This chapter discusses a selection of five protocols that are conceptually important and often used as building-blocks for solving problems in quantum information.

6.1. Dense coding

In classical systems, all pure states are fully distinguishable. This means that the pure state space of a classical d -dimensional system can be viewed as a set of symbols, $\{1, \dots, d\}$. In a quantum information notation, this can be written as the computational basis states $\{|i\rangle\}_{i=1}^d$. Clearly, such a system can transmit at most d -valued message. The question that we are interested in is whether it is possible to transmit more information by substituting the d -valued classical message for a d -dimensional quantum system? The answer is actually negative. One way to show this is through quantum state discrimination. Consider that Alice selects a classical symbol $x \in \{1, \dots, n\}$ with probability $p(x) = 1/n$. She aims to encode x into a classical message m_x that is limited to a smaller alphabet, $m_x \in \{1, \dots, d\}$, for some $d < n$. She communicates m_x to a separate party, Bob, who reads it and then aims to select an output b that is equal to Alice's symbol, namely $b = x$. His average success probability is therefore

$$p_{\text{win}} = \frac{1}{n} \sum_{x=1}^n p(b = x|x). \quad (6.1)$$

Since a classical message only allows Bob to distinguish d of Alice's symbols, we immediately get the classical limit $p_{\text{win}}^C \leq \frac{d}{n}$. In the quantum case, we substitute m_x for a d -dimensional quantum state ρ_x and Bob uses a POVM $\{M_b\}_b$ to decide his guess. Evidently, by simply choosing a classical encoding, $\rho_x = |m_x\rangle\langle m_x|$, we could achieve the classical value $p_{\text{win}}^C = \frac{d}{n}$. However, what is more interesting is that there exists no possible selection of quantum messages and POVMs that improves on this strategy. To prove this, consider the following argument

$$p_{\text{win}} \stackrel{(6.1)}{=} \frac{1}{n} \sum_{x=1}^n \text{tr}(\rho_x M_x) \leq \frac{1}{n} \sum_{x=1}^n \lambda_{\max}(M_x) \leq \frac{1}{n} \sum_{x=1}^n \text{tr}(M_x) = \frac{1}{n} \text{tr}\left(\sum_{x=1}^n M_x\right) \stackrel{(1.8)}{=} \frac{1}{n} \text{tr}(\mathbb{1}) = \frac{d}{n}. \quad (6.2)$$

Here, we have used (i) Born's rule, (ii) that the optimal choice of ρ_x is the eigenstate of M_x corresponding to the largest eigenvalue³⁵, (iii) that the trace of a positive operator is no smaller than its largest eigenvalue, and (iv) the completeness relation. This is an example where upgrading classical to quantum system is of no advantage.

In view of this, our quest for finding a quantum advantage must turn to different quantum resources. Suppose that Alice's message remains classical, i.e. $x \in \{1, \dots, n\}$, and is encoded into $m \in \{1, \dots, d\}$, where we for simplicity have left the message's dependence on x implicit. However, this time Alice and Bob can share an entangled state, ρ_{AB} , to assist their communication. Hence, for each x Alice can select a POVM $\{A_{m|x}\}_m$ and measure her share of ρ_{AB} to determine her message m . Once Bob receives m , he can select a POVM $\{B_{b|m}\}_b$ where the outcome b is his guess for the value of x . By Born's rule, the average success probability in Eq (6.1) becomes

$$p_{\text{win}} = \frac{1}{n} \sum_{x=1}^n \sum_{m=1}^d \text{tr}(\rho_{AB} A_{m|x} \otimes B_{x|m}). \quad (6.3)$$

We want to determine how large p_{win} can become for the best choice of state and measurements. To this end, we use the following algebraic relation. Consider that σ_{AB} is a state and $0 \leq M, N \leq \mathbb{1}$ are positive semidefinite operators. It then holds that

$$\text{tr}(\sigma_{AB} M \otimes N) \leq \lambda_{\max}(M) \text{tr}(\sigma_{AB} N), \quad (6.4)$$

where $\sigma_B = \text{tr}_A(\sigma_{AB})$. To show this, use a spectral decomposition $M = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$. Then

$$\text{tr}(\sigma_{AB} M \otimes N) = \sum_i \lambda_i \text{tr}(\sigma_{AB} |\lambda_i\rangle\langle\lambda_i| \otimes N) \leq \lambda_{\max}(M) \text{tr}(\underbrace{\sigma_{AB} \sum_i |\lambda_i\rangle\langle\lambda_i|}_{=\mathbb{1}} \otimes N) = \lambda_{\max}(M) \text{tr}(\sigma_B N), \quad (6.5)$$

³⁵ To see this, consider a Hermitian operator H in spectral decomposition, $H = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$. If we want to select a state $|\psi\rangle$ so that $\langle\psi|H|\psi\rangle$ is maximal, then we have $\langle\psi|H|\psi\rangle = \sum_i \lambda_i |\langle\psi|\lambda_i\rangle|^2$. Note that $|\langle\psi|\lambda_i\rangle|^2 = p(i)$ is a probability distribution. To maximise the expression, we want to put 'all our probability' in front of the largest eigenvalue, $\lambda_{\max} = \max_i \lambda_i$. To do that, we must select $|\psi\rangle = |\lambda_{\max}\rangle$.

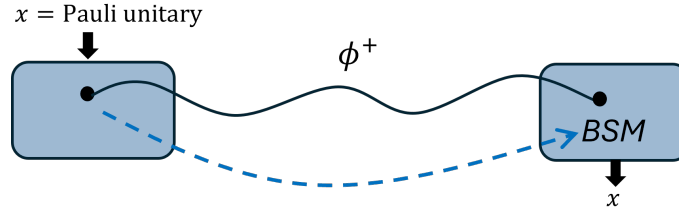


FIG. 6.10. In the dense coding protocol Alice and Bob share a maximally entangled state ϕ^+ . Alice encodes two bits of information by implementing one of the four unitaries $\mathbb{1}, \sigma_X, \sigma_Z$ or $i\sigma_Y$. Her qubit is then sent to Bob who jointly measures both qubits using the Bell state measurement. Bob's outcome recovers both Alice's bits.

where we used that $\lambda_i \leq \lambda_{\max}$ by definition. Applying this term-wise to the winning probability in Eq (6.3) we find

$$p_{\text{win}} \stackrel{(6.4)}{\leq} \frac{1}{n} \sum_{x=1}^n \sum_{m=1}^d \underbrace{\lambda_{\max}(A_{m|x})}_{\leq 1} \text{tr}(\rho_B B_{x|m}) \leq \frac{1}{n} \sum_{x=1}^n \sum_{m=1}^d \text{tr}(\rho_B B_{x|m}) \stackrel{(1.8)}{=} \frac{1}{n} \sum_{m=1}^d \text{tr}(\rho_B) = \frac{d}{n}, \quad (6.6)$$

where $\lambda_{\max}(A_{m|x}) \leq 1$ follows from the completeness condition for POVMs. In conclusion, we have once again failed to find a quantum advantage: the result is still identical to the classical case. Thus, neither quantum messages nor entanglement has managed to unlock a quantum advantage for us³⁶.

There is one natural way to proceed: we can combine quantum messages and shared entanglement. As we now show, this leads to a quantum advantage and the protocol that does the job is called *dense coding*. For simplicity, we focus on the case of qubits ($d = 2$) and leave the higher-dimensional case as an Exercise B.3.1.

R12: Dense coding

Qubit messages assisted by shared entanglement can be used to communicate two bits of information.

The essence of the dense coding protocol is that Alice encodes her part of a shared entangled state into a qubit message that is sent to Bob. Bob then holds both the quantum message and his share of the entangled state, and he extracts the information through a joint measurement on both. The protocol is illustrated in Fig 6.10 for the case when Alice has two bits of information ($n = 4$).

We select the shared entangled state as maximally entangled, $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. When Alice receives her share of the state, which is a single qubit, she performs a unitary transformation on it depending on x . Specifically, she acts as follows

$$\begin{aligned} x = 1 &\implies \mathbb{1} \\ x = 2 &\implies \sigma_X \\ x = 3 &\implies \sigma_Z \\ x = 4 &\implies i\sigma_Y. \end{aligned} \quad (6.7)$$

After performing the unitary, she communicates her qubit to Bob. The two-qubit state now held by Bob is the following

$$\begin{aligned} x = 1 &\implies (\mathbb{1} \otimes \mathbb{1}) |\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\phi^+\rangle \\ x = 2 &\implies (\sigma_X \otimes \mathbb{1}) |\phi^+\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle) = |\psi^+\rangle \\ x = 3 &\implies (\sigma_Z \otimes \mathbb{1}) |\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |\phi^-\rangle \\ x = 4 &\implies (i\sigma_Y \otimes \mathbb{1}) |\phi^+\rangle = \frac{1}{\sqrt{2}}(-|10\rangle + |01\rangle) = |\psi^-\rangle. \end{aligned} \quad (6.8)$$

The states on the right-hand-side are all maximally entangled (their reductions are maximally mixed) and together they are called the *Bell states*. The four Bell states have the key property that they are all pairwise orthogonal. This means that they form a basis of two-qubit Hilbert space. This makes them important, because we can think of $\{|\phi^+\rangle, |\phi^-\rangle, |\psi^+\rangle, |\psi^-\rangle\}$ as a rank-1 projective measurement in a two-qubit maximally entangled basis.

³⁶ There is a very basic lesson to learn here: not everything that is quantised becomes better.

G18: Bell state measurement

The Bell state measurement is the projection of two qubits onto the following basis of maximally entangled states

$$|\phi^\pm\rangle = \frac{1}{\sqrt{2}} (|00\rangle \pm |11\rangle), \quad |\psi^\pm\rangle = \frac{1}{\sqrt{2}} (|01\rangle \pm |10\rangle). \quad (6.9)$$

In the dense coding protocol, we let Bob perform the Bell state measurement. It allows him to deterministically distinguish between the four Bell states and hence he learns the value of x , thereby achieving $p_{\text{win}} = 1$. $\square$

Dense coding has shown us that entanglement can be used to increase the amount of information that can be transmitted in a single qubit. Importantly, the role of entanglement is not just in the state but also in the measurement: in Exercise B.3.2 we show that if we replace the Bell state measurement with a separable two-qubit measurement we would not beat the classical case.

6.2. Quantum teleportation

Dense coding is about sending classical information using quantum resources. What if we want to instead send quantum information (i.e. a quantum state)? Consider that Alice can choose any possible qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ which she then wants to send to Bob. This means that Bob must output a quantum state, ρ , which ideally should be precisely that held by Alice, namely $\rho = |\psi\rangle\langle\psi|$. We can quantify how well Bob's output state matches Alice's state by considering the average fidelity between them,

$$F = \int d\psi \langle\psi|\rho|\psi\rangle, \quad (6.10)$$

where $d\psi$ represents a uniform density over all pure qubit states. In the integrand we have used that the fidelity between a pure and a mixed state takes the form in Eq (2.19). Using our knowledge of the Bloch sphere from Eq (1.13), we can parameterise $\alpha = \cos \frac{\theta}{2}$ and $\beta = \sin \frac{\theta}{2} e^{i\varphi}$ and write the fidelity as

$$F = \frac{1}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \langle\psi|\rho|\psi\rangle. \quad (6.11)$$

Consider first that Alice does not communicate to Bob at all. Bob can then only make random guesses for her state. Since there is no fixed reference frame, and the Bloch sphere is spherically symmetric, we can simply let Bob always guess $\rho = |0\rangle\langle 0|$. This simplifies the fidelity to $\langle\psi|\rho|\psi\rangle = \frac{1+\cos\theta}{2}$ and its average becomes

$$F = \frac{1}{8\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta (1 + \cos \theta) = \frac{1}{2}. \quad (6.12)$$

Therefore, our first task is to use classical communication in such a way that we can increase this average fidelity.

For this purpose, let Alice measure her state $|\psi\rangle$ with the observable σ_Z . This will give Alice information about the hemisphere to which the Bloch vector of $|\psi\rangle$ likely belongs. She sends this information to Bob who chooses ρ as the state with the corresponding north-or-south pole Bloch vector, i.e. either $|0\rangle$ or $|1\rangle$. Thus, with probability $|\langle\psi|0\rangle|^2$ Alice gets the northern hemisphere and Bob outputs $\rho = |0\rangle\langle 0|$ and with probability $|\langle\psi|1\rangle|^2$ she gets the southern hemisphere and Bob outputs $\rho = |1\rangle\langle 1|$. The fidelity for a given state $|\psi\rangle$ becomes

$$\langle\psi|\rho|\psi\rangle = |\langle\psi|0\rangle|^4 + |\langle\psi|1\rangle|^4 = \left(\frac{1+\cos\theta}{2}\right)^2 + \left(\frac{1-\cos\theta}{2}\right)^2, \quad (6.13)$$

where we used the trigonometric half-angle formulas. This leads to the average fidelity in Eq (6.11) becoming

$$F = \frac{1}{16\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \left((1+\cos\theta)^2 + (1-\cos\theta)^2 \right) = \frac{2}{3}, \quad (6.14)$$

where the θ -integral can be solved through the integration by parts method. In conclusion, by sending one bit from Alice to Bob, we have improved on the result in Eq (6.12). Still, we are far from faithfully transmitting the quantum information, because that would require $F = 1$.

To achieve a perfect transmission of $|\psi\rangle$ from Alice to Bob via classical communication, we must introduce shared entanglement between the parties. The protocol that does the job is called *quantum teleportation*³⁷.

³⁷ We emphasise that what is teleported is the information (the quantum state) and not the physical object that carries that information.

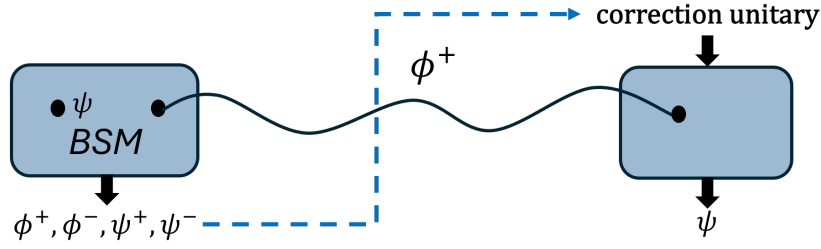


FIG. 6.11. In the quantum teleportation protocol Alice and Bob share a maximally entangled state, ϕ^+ . Alice performs a Bell state measurement jointly on her share and the qubit she wants to teleport, ψ . The classical outcome is communicated to Bob who performs a corresponding unitary on his share of ϕ^+ . Bob's state after performing his unitary is Alice's initial state ψ . In this process, the initial information ψ in Alice's system is erased and moved to Bob's system.

R13: Teleportation protocol

Any qubit state can be communicated by sharing a maximally entangled state and sending two classical bits.

The teleportation protocol is illustrated in Fig 6.11 where Alice and Bob share the two-qubit maximally entangled state $|\phi^+\rangle_{AB}$. The qubit Alice wants to teleport is called A' and has the state $|\psi\rangle_{A'}$. The combined system of three qubits is

$$|\psi\rangle_{A'} \otimes |\phi^+\rangle_{AB} = \frac{\alpha}{\sqrt{2}} (|00\rangle_{A'A} \otimes |0\rangle_B + |01\rangle_{A'A} \otimes |1\rangle_B) + \frac{\beta}{\sqrt{2}} (|10\rangle_{A'A} \otimes |0\rangle_B + |11\rangle_{A'A} \otimes |1\rangle_B), \quad (6.15)$$

where we have written $|00\rangle_{A'A} \otimes |0\rangle_B$ etc to emphasise that the qubits A' and A belong to Alice and the qubit B belongs to Bob. Let us now re-write Alice's part of the state in the basis of the four Bell states (6.9). Inspecting the Bell states shows that the basis transformation is given by

$$|00\rangle = \frac{1}{\sqrt{2}} (|\phi^+\rangle + |\phi^-\rangle), \quad |01\rangle = \frac{1}{\sqrt{2}} (|\psi^+\rangle + |\psi^-\rangle), \quad |10\rangle = \frac{1}{\sqrt{2}} (|\psi^+\rangle - |\psi^-\rangle), \quad |11\rangle = \frac{1}{\sqrt{2}} (|\phi^+\rangle - |\phi^-\rangle). \quad (6.16)$$

Expressed in the Bell basis, our state in Eq (6.15) becomes

$$\frac{1}{2} \left(|\phi^+\rangle_{A'A} \otimes \underbrace{(\alpha|0\rangle + \beta|1\rangle)}_{=|\psi\rangle}_B + |\phi^-\rangle_{A'A} \otimes \underbrace{(\alpha|0\rangle - \beta|1\rangle)}_{=\sigma_Z|\psi\rangle}_B + |\psi^+\rangle_{A'A} \otimes \underbrace{(\beta|0\rangle + \alpha|1\rangle)}_{=\sigma_X|\psi\rangle}_B + |\psi^-\rangle_{A'A} \otimes \underbrace{(\beta|0\rangle - \alpha|1\rangle)}_{=\sigma_Z\sigma_X|\psi\rangle}_B \right), \quad (6.17)$$

where we have observed that qubit B in each of the four terms can be expressed as an appropriate Pauli unitary acting on the state $|\psi\rangle$. With those substitutions inserted, we arrive at

$$|\psi\rangle_{A'} \otimes |\phi^+\rangle_{AB} = \frac{1}{2} (|\phi^+\rangle_{A'A} \otimes |\psi\rangle_B + |\phi^-\rangle_{A'A} \otimes \sigma_Z|\psi\rangle_B + |\psi^+\rangle_{A'A} \otimes \sigma_X|\psi\rangle_B + |\psi^-\rangle_{A'A} \otimes \sigma_Z\sigma_X|\psi\rangle_B). \quad (6.18)$$

Now we can see how quantum teleportation works. Alice has to measure her initial qubit (A') jointly with her share of the maximally entangled state (A) by projecting the two qubits onto the Bell basis. This gives one of four possible outcomes, which we for simplicity can classically label as ϕ^+ , ϕ^- , ψ^+ or ψ^- , and it leaves the post-measurement state in the product state between Alice and Bob corresponding to the appropriate term in Eq (6.18). Alice relays the classical outcome to Bob who can then undo the relevant unitary on qubit B . This means that Bob applies $\mathbb{1}, \sigma_Z, \sigma_X, \sigma_Z\sigma_X$ for the outcomes $\phi^+, \phi^-, \psi^+, \psi^-$ respectively, and then can be sure that his qubit (B) now is in the state $|\psi\rangle$. $\square$

Notice that the actions of Alice and Bob do not depend on $|\psi\rangle$. Hence, the qubit state that is teleported can be completely unknown. However, once the protocol is completed, the state of Alice's original qubit, A' , is no longer $|\psi\rangle$ but instead it is maximally mixed. The information has been erased in Alice's lab and moved to Bob's lab. In analogy with dense coding, the teleportation protocol succeeds thanks to the use of entanglement in the state and in the measurement. In Exercise B.3.3 we will examine the teleportation fidelity when the maximally entangled state is substituted with a partially entangled state and compute how the average fidelity degrades with the entanglement.

6.3. Entanglement swapping

There is a particularly remarkable type of quantum teleportation that is called *entanglement swapping*. Entanglement swapping is a process in which we start with two initially independent pairs of particles, then we make an interaction involving one particle

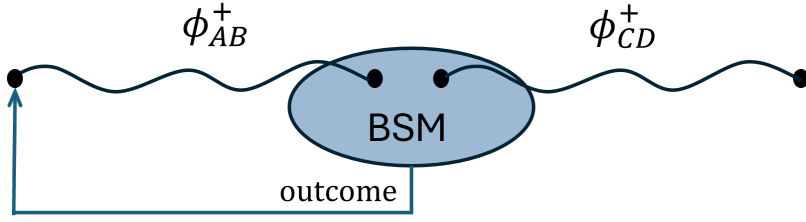


FIG. 6.12. In entanglement swapping a Bell state measurement is used to teleport one share of an entangled state into a particle with which the other share has never interacted.

from each pair and end up with the two remaining particles becoming entangled. What is remarkable here is that the two final particles become entangled without ever having interacted with each other.

In entanglement swapping, we begin with two qubits A and B that are in the state $|\phi^+\rangle$ and another pair of qubits, C and D , that are also in the state $|\phi^+\rangle$. Our goal is to entangle the qubits A and D without ever making an interaction between them. To achieve that, we will teleport the state of qubit C into qubit A ; see Fig 6.12. The initial state of the four-qubit system is

$$|\psi\rangle_{ABCD} = |\phi^+\rangle_{AB} \otimes |\phi^+\rangle_{CD} = \frac{1}{2}(|0000\rangle_{ABCD} + |0011\rangle_{ABCD} + |1100\rangle_{ABCD} + |1111\rangle_{ABCD}) \quad (6.19)$$

Just as in teleportation, we use the change of basis in Eq (6.16); from the computational basis to the Bell states. We apply this basis change to system B and C and it leaves us with

$$|\psi\rangle_{ABCD} = \frac{1}{2}(|\phi^+\rangle_{BC} \otimes |\phi^+\rangle_{AD} + |\phi^-\rangle_{BC} \otimes |\phi^-\rangle_{AD} + |\psi^+\rangle_{BC} \otimes |\psi^+\rangle_{AD} + |\psi^-\rangle_{BC} \otimes |\psi^-\rangle_{AD}). \quad (6.20)$$

Thus, by measuring in the Bell basis on qubits B and C , the post-measurement state between A and D is one of the four Bell states. The specific Bell state depends on the outcome of the measurement. If we know the outcome of the measurement, we can determine which of the four maximally entangled states we have between A and D . Alternatively, if the outcome is communicated to A and D , they can perform local unitaries as in Eq (6.8) so that they always end up sharing the state $|\phi^+\rangle$. This is analogous to the two bits of communication used in the teleportation protocol. Actually, as seen in Eq (6.8), it is sufficient if the outcome is communicated to just one of the two parties. In contrast, if we do not use the outcome at all, the state of the system AD becomes the mixture corresponding to tracing out system BC from Eq (6.20), which

$$\text{tr}_{BC}(|\psi\rangle\langle\psi|) = \frac{1}{4}(|\phi^+\rangle\langle\phi^+| + |\phi^-\rangle\langle\phi^-| + |\psi^+\rangle\langle\psi^+| + |\psi^-\rangle\langle\psi^-|)_{AD} = \frac{1}{4}\mathbb{1}_{AD}. \quad (6.21)$$

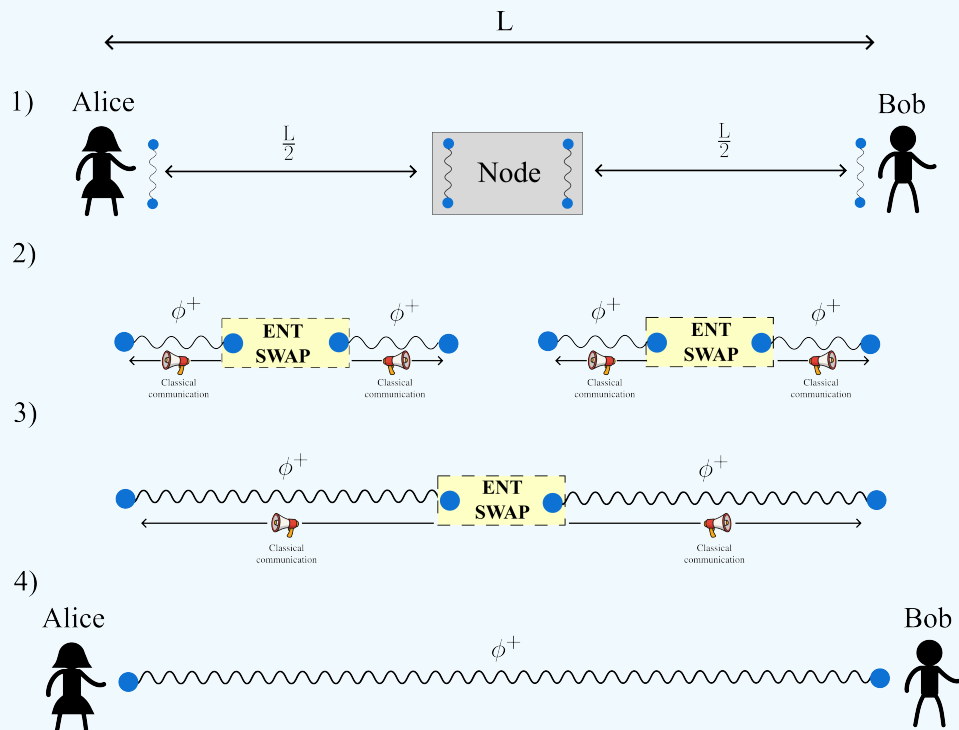
This means that without keeping the classical outcome we cannot perform entanglement swapping. In Box B8 we combine quantum teleportation and entanglement swapping to describe the basic idea behind a so-called quantum repeater. This is a key quantum technology for reliably distributing entanglement over large distances.

Note that in the above we have specifically shown how a maximally entangled state can be swapped into systems AD by means of a remote Bell state measurement. However, entanglement swapping is not limited to this specific state: in Exercise B.3.5 we show that any state can be swapped into systems AD by appropriately adapting the joint measurement.

B8. A glance beyond

Quantum repeaters

The ability to communicate quantum states over large distances is a central goal for applied quantum information science. The teleportation protocol has taught us that we do not need to physically transport the carrier of the quantum information, but we can teleport that information at the cost of classical communication and a shared maximally entangled state. However, this faces major practical obstacles. Photons are natural candidates for long-distance quantum communication, but how do we distribute a photonic maximally entangled state over a large distance? The transmittance of an optical fiber decreases exponentially with its length. For example, a common optical fiber can lose circa 90% of all the photons for every 50 kilometers. To overcome this issue, one can rely on a combination of teleportation and entanglement swapping, as illustrated here below.



Alice and Bob are separated by a distance L but the protocol's idea is to reduce the travel distance of the photons by placing multiple entanglement swapping stations along the way. The wavy lines represent maximally entangled states and the yellow boxes are Bell state measurement stations. 1) A node is placed half-way between Alice and Bob. Maximally entangled pairs are locally generated for Alice, Bob and the node. 2) An entanglement swapping station is placed half-way between Alice and the node as well as half-way between Bob and the node. One share from each pair is sent to the respective station where a Bell state measurement is performed. 3) The node now shares a maximally entangled state with Alice and Bob independently. It performs yet another Bell state measurement to swap the entanglement to Alice and Bob. 4) Alice and Bob now have a maximally entangled state which they can use to teleport quantum information to each other. Each traveling photon has only travelled $\frac{L}{4}$ units of distance. Moreover, the protocol could be straightforwardly extended with more nodes and more entanglement swapping stations to further reduce the travel distance per photon.

Naturally, it is not possible to perfectly implement the entanglement swapping procedure nor to traverse the shortened distance units without loss. This means that in practice Alice and Bob end up with some noisy version of the maximally entangled state. There are several ways to overcome practical issues of this sort. One option is for Alice and Bob to use an entanglement distillation protocol (see Box B4) to enhance the quality of their final entangled state.

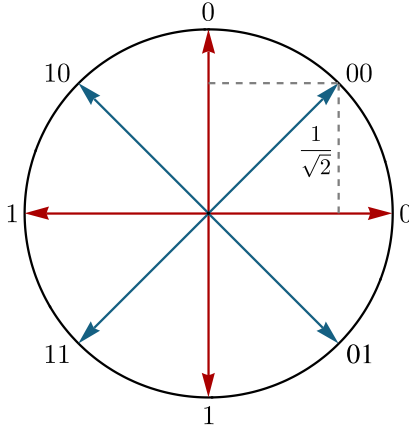


FIG. 6.13. In a quantum random access code Alice prepares four qubit states so that their Bloch vectors (blue) form a rotated square in the XZ -plane of the Bloch sphere. Bob performs a pair of measurements corresponding to σ_X and σ_Z , represented by the red Bloch vectors. The projection of any one of Alice's states onto either one of the outcomes x_0 for $y = 0$ and x_1 for $y = 1$ is $1/\sqrt{2}$, as exemplified for the state ρ_{00} .

6.4. Random access code

In section 6.1 we learned that qubits (without entanglement) have no advantage over bits when communicating a classical message from Alice to Bob. However, this does not mean that there are no other interesting tasks where qubits outperform bits. This is demonstrated by the *random access code*, which we already encountered in Chapter 5.5.

Imagine that Alice selects a pair of bits $x_0, x_1 \in \{0, 1\}$. Bob is interested to learn one of them, but Alice does not know which one. Bob privately selects his bit of interest by choosing $y \in \{0, 1\}$. Alice wants to help Bob to access the random bit of his interest, but in doing this she is only allowed to communicate a single bit to Bob. This prevents her from simply sending both bits and letting Bob select x_y . Let us consider the average success probability that Bob learns the right information, namely

$$p_{\text{suc}} = \frac{1}{8} \sum_{x_0, x_1, y} p(b = x_y | x_0, x_1, y). \quad (6.22)$$

In a classical approach, Alice uses an encoding function, E , to construct her message bit $m = E(x_0, x_1)$. Depending on y , Bob chooses some decoding function, D_y , to construct his output as $b = D_y(m)$. A particularly simple example is that Alice chooses $E(x_0, x_1) = x_0$ and Bob chooses $D_y(m) = m$. In other words, Alice always sends x_0 and Bob just outputs the message. Whenever $y = 0$ Bob successfully finds $b = x_0$, but when $y = 1$ he only has a probability of 50% to find $b = x_1$. Hence, the average success probability becomes $p_{\text{suc}}^C = \frac{3}{4}$. With some trial and error, we should be able to convince ourselves that no better classical strategy is possible³⁸.

Let us compare this to the situation in which the bit message is substituted with a qubit message. This means that Alice encodes (x_0, x_1) into a qubit state $\rho_{x_0 x_1}$ that is sent to Bob. Bob selects y and decodes the qubit state by performing a quantum measurement $\{M_{b|y}\}_b$ where $b \in \{0, 1\}$ is the outcome. Our strategy is to select Alice's four different states as pure, and in such a way that they form a square in the XZ -disk of the Bloch sphere. Specifically, we associate the Bloch vectors

$$\vec{n}_{00} = \frac{1}{\sqrt{2}}(1, 0, 1), \quad \vec{n}_{01} = \frac{1}{\sqrt{2}}(1, 0, -1), \quad \vec{n}_{10} = \frac{1}{\sqrt{2}}(-1, 0, 1), \quad \vec{n}_{11} = \frac{1}{\sqrt{2}}(-1, 0, -1). \quad (6.23)$$

Recall that the state is given by $\rho_{x_0 x_1} = \frac{1}{2}(\mathbb{1} + \vec{n}_{x_0 x_1} \cdot \vec{\sigma})$. For decoding, let Bob measure σ_X when $y = 0$ and σ_Z when $y = 1$. Both the states and the measurements are illustrated in Fig 6.13 where we see that all the relevant Bloch vector overlaps are identical and equal to $1/\sqrt{2}$. This means that all the winning probabilities are the same and are given by $p(b = x_y | x_0, x_1, y) = \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}}\right)$. Hence, we conclude that the use of qubits in the random access code leads to the average success probability

$$p_{\text{suc}} = \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 0.854, \quad (6.24)$$

which exceeds the classical limit. We therefore conclude that even though qubits cannot carry no more than one bit of information, they can nevertheless be used to enhance the success probability of probabilistically sending one of two different pieces of information. This means that qubits alone can give rise to nonclassical phenomena in communication tasks.

³⁸ We omit the proof here since it is not essential for our purpose.

6.5. Quantum cloning

Classical information is routinely cloned. In contrast, cloning of quantum information is not possible, and there are many good reasons for it. Let us give a couple of examples of how the possibility to clone information would impact quantum theory.

- Consider that we are given an unknown qubit state $|\psi\rangle$. If we could clone it, then we could keep cloning it until we have a very large number of copies. Then, we could perform the Pauli measurements on all the copies (except one) and thereby learn the Bloch vector of $|\psi\rangle$. Learning the Bloch vector means exactly knowing what is $|\psi\rangle$, and we still have our original copy of the state left. In conclusion, cloning would let us reconstruct quantum states without consuming them.
- Imagine that Alice and Bob share the Popescu-Rohrlich no-signaling box from Chapter 5.5. Recall that it satisfies the relation $a \oplus b = xy$. Imagine further that Bob could clone his part, so he could generate another input and output y' and b' satisfying $a \oplus b' = xy'$. Now, Bob can compute $b \oplus b'$ which is guaranteed to satisfy $b \oplus b' = (a \oplus b) \oplus (a \oplus b') = xy \oplus xy' = x(y \oplus y')$. Since Bob knows b, b', y, y' , he can gain information about x . This is violating the no-signaling principle!³⁹

The fact that quantum cloning is not possible is therefore essential to the physics of the theory.

R14: No-cloning theorem

There exists no quantum transformation that takes as input any state $|\psi\rangle$ and outputs $|\psi\rangle \otimes |\psi\rangle$.

Let system A have the state, $|\psi\rangle_A$. We want to copy it into system B , which we initialise in $|0\rangle_B$. The cloning transformation is some unitary, U , such that for all states $|\psi\rangle$ its action is $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$. To show its impossibility, assume that the cloning transformation exists and select two states $|\psi_1\rangle$ and $|\psi_2\rangle$. Applying the cloning machine gives

$$\langle\psi_1|\psi_2\rangle^2 = \langle\psi_1, \psi_1|\psi_2, \psi_2\rangle = \langle\psi_1, 0|\underbrace{U^\dagger U}_{=1}|\psi_2, 0\rangle = \langle\psi_1|\psi_2\rangle, \quad (6.25)$$

but the only numbers satisfying $x^2 = x$ are $x = 0$ and $x = 1$. Hence, when the states are neither orthogonal nor identical, we arrive at a contradiction. In conclusion, quantum cloning is impossible. $\square$

To further appreciate the importance of the no-cloning theorem, let us highlight its connection to the concept of entanglement monogamy which we briefly discussed in Chapter 4.6. Monogamy prevented a qubit A from simultaneously being maximally entangled with qubit B and another qubit C , i.e. A could not fully “commit” its entanglement to different systems simultaneously. Let us momentarily imagine that this is not true, i.e. that A can be maximally entangled with both B and C separately. Then, let Alice prepare a qubit A' in state $|\psi\rangle$. Since she is maximally entangled with both B and C , she can use the teleportation protocol to teleport $|\psi\rangle$ to both Bob and Charlie. In the teleportation protocol, we saw that this means erasing the information of $|\psi\rangle$ for Alice, but creating $|\psi\rangle$ in both the labs of Bob and Charlie individually. Thus, we now have two copies of $|\psi\rangle$ which is a successful instance of quantum cloning. We can therefore conclude that monogamy is a necessary feature of entanglement.

Knowing that quantum cloning is forbidden, it is natural to ask how well a cloning transformation can be approximated. To quantify this, we need a notion of distance and for that we may use the fidelity between quantum states. We answer the question for a pure qubit, which we describe with the unit Bloch vector $\hat{n}$. Our task is to start with any $\hat{n}$ and create two copies of the non-unit Bloch vector $\eta\hat{n}$, for the largest possible $\eta \in [0, 1]$, which correspond to a fidelity $F = \frac{1+\eta}{2}$. The protocol that achieves this is called the *universal quantum cloning machine*.

R15: Universal quantum cloning machine

There exists a quantum transformation whose input is any qubit state $|\psi\rangle$ and whose output is a two-qubit state χ_{AB} such that its reductions are identical, $\rho = \text{tr}_A(\chi) = \text{tr}_B(\chi)$ and their fidelity with the input state is

$$F = \langle\psi|\rho|\psi\rangle = \frac{5}{6}. \quad (6.26)$$

³⁹ This example is not from quantum theory but in a very similar way, cloning in quantum nonlocality can imply signaling.

Let system A hold the input qubit, $|\psi\rangle$, let system B hold the register into which we want to produce the clone and let system C be an ancilla. We initialise both systems B and C in the state $|0\rangle$. The quantum transformation is a unitary, U , acting jointly on systems A , B and C . Select the unitary such that

$$\begin{aligned} U|0\rangle_A|0\rangle_B|0\rangle_C &= \sqrt{\frac{2}{3}}|000\rangle + \sqrt{\frac{1}{3}}|\psi^+\rangle|1\rangle, \\ U|1\rangle_A|0\rangle_B|0\rangle_C &= \sqrt{\frac{2}{3}}|111\rangle + \sqrt{\frac{1}{3}}|\psi^+\rangle|0\rangle, \end{aligned} \quad (6.27)$$

where $|\psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ is one of the Bell states. It is clear that the two outputs are orthogonal since their supports are disjoint. Applying this to an arbitrary input state, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we obtain

$$|\chi\rangle \equiv U|\psi\rangle|0\rangle|0\rangle \stackrel{(6.27)}{=} \alpha \left(\sqrt{\frac{2}{3}}|000\rangle + \sqrt{\frac{1}{3}}|\psi^+\rangle|1\rangle \right) + \beta \left(\sqrt{\frac{2}{3}}|111\rangle + \sqrt{\frac{1}{3}}|\psi^+\rangle|0\rangle \right). \quad (6.28)$$

Now, we discard the ancilla system by tracing it out,

$$\begin{aligned} \chi_{AB} = \text{tr}_C(\chi) &= \alpha^2 \frac{2}{3}|00\rangle\langle 00| + \beta^2 \frac{2}{3}|11\rangle\langle 11| + \frac{\alpha^2 + \beta^2}{3}\psi^+ + \alpha\beta\frac{\sqrt{2}}{3}(|00\rangle\langle\psi^+| + |\psi^+\rangle\langle 00|) + \alpha\beta\frac{\sqrt{2}}{3}(|11\rangle\langle\psi^+| + |\psi^+\rangle\langle 11|) \\ &= \frac{2}{3}\psi \otimes \psi + \frac{1}{6}(|\psi, \psi_\perp\rangle + |\psi_\perp, \psi\rangle)(\langle\psi, \psi_\perp| + \langle\psi_\perp, \psi|), \end{aligned} \quad (6.29)$$

where $|\psi_\perp\rangle = \beta^*|0\rangle - \alpha^*|1\rangle$ is orthogonal to $|\psi\rangle$. This state is evidently symmetric in its two subsystems, so the states of system A and B are identical. To access each copy, we trace out also system B (or A) to get

$$\rho = \text{tr}_{BC}(\chi) = \frac{5}{6}\psi + \frac{1}{6}\psi_\perp. \quad (6.30)$$

This gives the cloning fidelity in Eq (6.26). □

Observe that the output of the universal quantum cloning machine is not a product state of the imperfect copies ($\rho \otimes \rho$), but an entangled state whose reductions are the imperfect copy ρ .

It is relevant to ask whether it is possible to clone an unknown qubit at a fidelity larger than that achieved above, namely $F = \frac{5}{6}$. The answer is negative. An insightful way to show that is to consider that Alice and Bob share a maximally entangled state and that Alice measures either the Z -basis or the X -basis. This induces an update in Bob's local state. Then, we consider that Bob clones his local state, $\hat{n}$, into two imperfect copies $\frac{1+\eta}{2}\hat{n} + \frac{\eta}{2}\vec{\sigma}$, stored in the marginals of the state $\rho_{BB'}(\hat{n})$. One may then ask what is the largest η such that Bob's state on BB' is independent of Alice's measurement choice, as required by the no-signaling principle. In Exercise B.3.10 we will show that the no-signaling principle is violated if and only if $\eta \leq \frac{2}{3}$, which means $F \leq \frac{5}{6}$.

Chapter summary.

- The dense coding protocol transmits two bits of information in qubit messages assisted by shared entanglement.
- The teleportation protocol uses shared entanglement and classical communication to transmit quantum information from one lab to another.
- The entanglement swapping protocol allows two systems that have never interacted to become entangled by teleporting one share of an entangled state.
- The quantum random access code shows that qubits can outperform classical bits in communication tasks.
- Quantum information cannot be cloned and the best possible approximate cloning has a fidelity per copy of $F = \frac{5}{6}$.

7. QUANTUM CHANNELS AND OPERATIONS

The quantum evolution postulate G2 in Chapter 1.1 stipulated that quantum transformations in closed systems are represented by unitary operators. In this chapter, we discuss quantum transformations in open systems, thereby completing the move from rays, projective measurements and unitaries in closed systems to density operators, POVMs and quantum channels in open systems. We provide a formalism and toolbox for how to describe and use the most general quantum transformations and operations.

7.1. Completely positive trace-preserving maps

We are interested in the most general physically consistent map from density operators on one Hilbert space, $\mathcal{H}_A$, to density operators on another Hilbert space, $\mathcal{H}_{A'}$. Such maps are referred to as *quantum channels* and we denote them as $\Lambda(\rho)$, or sometimes as $\Lambda_{A \rightarrow A'}(\rho)$ to emphasise also the Hilbert spaces that we map to and from. In order to define quantum channels, we need to introduce their properties.

The first property required for a quantum channel is linearity,

$$\textbf{Linearity:} \quad \Lambda(\alpha X + \beta Y) = \alpha \Lambda(X) + \beta \Lambda(Y), \quad (7.1)$$

where α and β are complex numbers, and X and Y are operators. This property is implied from the linearity of quantum theory. To also get some intuition, let us think of X and Y as states and set $\alpha = p$ and $\beta = 1 - p$ for $0 \leq p \leq 1$. Now we can think of $\alpha X + \beta Y$ as the density operator obtained from selecting the state X with probability p and the state Y with probability $1 - p$. On the one hand, if we run the channel on the mixture, we get the left-hand-side of Eq (7.1). On the other hand, we could first announce whether we selected X or Y and then run the channel. This would give us the right-hand-side of (7.1). However, these two procedures ought to yield the same result, so we expect Eq (7.1) to hold.

Secondly, Λ must preserve the trace of the input, i.e.

$$\textbf{Trace preservation:} \quad \text{tr}(\Lambda(X)) = \text{tr}(X). \quad (7.2)$$

This property ensures that the trace of the density operator remains the same after applying the quantum channel. Without this property, we would transform states into objects that are no longer states.

Thirdly, Λ should map positive semidefinite operators to positive semidefinite operators. This property is called positivity,

$$\Lambda(X) \geq 0 \quad \text{if} \quad X \geq 0 \quad (7.3)$$

However, positivity is actually not enough. The reason is that when a positive map is applied only to one part of a system, it does not guarantee that the total state remains positive. We have already seen an example of this, namely the partial transposition map. The map $\Lambda(\rho) = \rho^T$ is positive (transposition does not change the eigenvalues), but when we applied it to half of the maximally entangled state, namely $(\Lambda \otimes \mathbb{1})[\phi^+]$, we found in Chapter 4.3 that the result is no longer positive. To account for this, we need a stronger property that is called *complete positivity*,

$$\textbf{Complete positivity:} \quad (\Lambda \otimes \mathbb{1})[X] \geq 0 \quad \text{if} \quad X \geq 0, \quad (7.4)$$

where the identity operator acts on a reference system of arbitrary dimension. Putting together our three conditions, we have arrived at the definition of a quantum channel.

1: Quantum channel

A quantum channel is a completely positive and trace-preserving (CPTP) linear map.

Let us give two examples of quantum channels. Firstly, we can now think of the partial trace map as a quantum channel, namely $(\Lambda \otimes \mathbb{1}_B)(X_{AB}) = X_B$. It is clearly linear and trace-preserving. To see that it is also completely positive, we show that for every $|\psi\rangle_B$ it holds that ${}_B\langle\psi|(\Lambda \otimes \mathbb{1})[X_{AB}]|\psi\rangle_B \geq 0$ when $X_{AB} \geq 0$. Since $(\Lambda \otimes \mathbb{1})[X_{AB}] = \sum_i (\langle i| \otimes \mathbb{1}) X_{AB} (|i\rangle \otimes \mathbb{1})$ it follows that ${}_B\langle\psi|(\Lambda \otimes \mathbb{1})[X_{AB}]|\psi\rangle_B = \sum_i \langle i, \psi| X_{AB} |i, \psi\rangle \geq 0$, where the last step follows from $X_{AB} \geq 0$. A second example is the depolarisation channel. This channel corresponds to flipping a coin with distribution $(p, 1 - p)$. In the first event, the channel does nothing and in the second event it deletes the input and emits a maximally mixed state. Thus,

$$\Lambda(X_A) = p X_A + (1 - p) \frac{\mathbb{1}}{\dim(\mathcal{H}_A)} \text{tr}(X_A). \quad (7.5)$$

Again, the map is clearly linear and trace-preserving. To check complete positivity, note that $(\Lambda \otimes \mathbb{1})[X_{AB}] = pX_{AB} + (1 - p)\frac{\mathbb{1}}{\dim(\mathcal{H})} \text{tr}_A(X_{AB})$ which is positive whenever $X_{AB} \geq 0$, because the partial trace is completely positive. The depolarisation channel is effectively introducing noise into the input state: if we apply it to the maximally entangled state we obtain the isotropic state that we encountered in Eq (4.36).

7.2. Choi representation

Consider that we are given some linear map Λ . How can we determine whether it is completely positive? It might seem hard because we need to prove positivity for a reference system of potentially any size and we need to check Eq (7.4) for infinitely many positive semidefinite input operators. Fortunately, there exists a simple necessary and sufficient condition. For this condition, we need to introduce an important tool called the *Choi operator*.

2: Choi operator

Let $\Lambda_{A \rightarrow A'}$ be a linear map. The Choi operator corresponding to the map is

$$\eta_{A'A} = (\Lambda \otimes \mathbb{1})[\phi^+], \quad (7.6)$$

where ϕ^+ is the maximally entangled state of local dimension $d_A = \dim(\mathcal{H}_A)$.

If Λ is a quantum channel the Choi operator becomes a state, because it is by definition both positive semidefinite and has unit-trace. Physically, we can think of it as sending half of a d_A -dimensional maximally entangled state through the channel⁴⁰. To determine whether a linear map is completely positive, we need only to examine its Choi operator.

R16: Completely positive maps

A linear map $\Lambda_{A \rightarrow A'}$ is completely positive if and only if its Choi operator is positive semidefinite, i.e.

$$\Lambda_{A \rightarrow A'} \text{ is CP} \iff \eta_{A'A} \geq 0. \quad (7.7)$$

One direction is trivial: if Λ is completely positive then by definition $(\Lambda \otimes \mathbb{1})[X] \geq 0$ for every positive semidefinite X . By choosing $X = \phi^+$ it therefore follows that $\eta_{A'A} \geq 0$.

To show the converse direction, we begin with a useful observation. Every pure state $|\psi\rangle$ can be obtained by local measurement on ϕ^+ followed by a post-selection. To see this, use the Schmidt decomposition R4 to write $|\psi\rangle = \sum_i \sqrt{\mu_i} |ii\rangle$ for $\mu_i \geq 0$ and define the POVM-element $E = \sum_i \sqrt{\mu_i} |i\rangle\langle i|$. We then have

$$(\mathbb{1} \otimes E)|\phi^+\rangle = \mathbb{1} \otimes \left(\sum_i \sqrt{\mu_i} |i\rangle\langle i| \right) \left(\frac{1}{\sqrt{d}} \sum_j |jj\rangle \right) = \frac{1}{\sqrt{d}} \sum_i \sqrt{\mu_i} |ii\rangle = \frac{1}{\sqrt{d}} |\psi\rangle. \quad (7.8)$$

Now, we take an arbitrary bipartite state ρ and write it in its spectral decomposition, $\rho = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$. To each pure state $|\lambda_i\rangle$ in this decomposition, we apply the above reformulation as a post-selection of an E -outcome local measurement on a maximally entangled state. Specifically, we write Eq (7.8) as $|\lambda_i\rangle = \sqrt{d} (\mathbb{1} \otimes E_i) |\phi^+\rangle$ for a suitable E_i chosen as above. Now we have

$$\begin{aligned} (\Lambda \otimes \mathbb{1})[\rho] &= \sum_i \lambda_i (\Lambda \otimes \mathbb{1})[|\lambda_i\rangle\langle\lambda_i|] = d \sum_i \lambda_i (\Lambda \otimes \mathbb{1})[(\mathbb{1} \otimes E_i) \phi^+ (\mathbb{1} \otimes E_i)] \\ &= d \sum_i \lambda_i (\mathbb{1} \otimes E_i) \underbrace{(\Lambda \otimes \mathbb{1})[\phi^+]}_{\equiv \eta} (\mathbb{1} \otimes E_i) = d \sum_i \lambda_i \underbrace{(\mathbb{1} \otimes E_i) \eta (\mathbb{1} \otimes E_i)}_{\geq 0} \geq 0. \end{aligned} \quad (7.9)$$

First, we inserted the spectral decomposition. Second, we used the reformulation (7.8) for each $|\lambda_i\rangle$. Third, we used that E_i and Λ act on different Hilbert spaces and therefore commute, allowing us to identify the Choi operator. Fourth, we note that

⁴⁰ Some authors define the Choi state as $\eta = (\mathbb{1} \otimes \Lambda)[\phi^+]$. Whether the channel acts on the left or the right is merely a choice of convention.

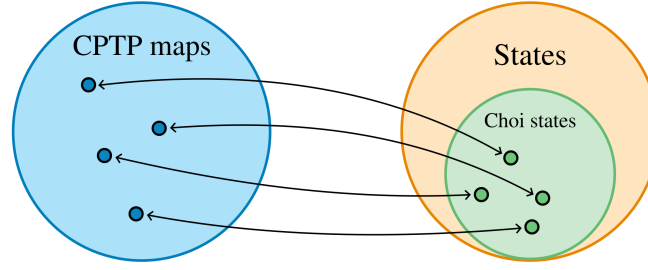


FIG. 7.14. State-channel duality. Every completely positive map can be uniquely associated with a Choi operator. If the map is also trace-preserving (CPTP) then the Choi operator can be interpreted as a bipartite state. The space of Choi states is a subset of all entangled states.

$O^\dagger X O \geq 0$ for any $X \geq 0$. Last, we have a convex combination of positive semidefinite operators. This proves that the map is completely positive. $\square$

Now we know that the Choi operator tells us whether a map is completely positive. Can we extract further useful information from the Choi operator? We certainly can. Due to linearity, the action of a channel is completely determined by its action on the set of basis elements $\{|i\rangle\langle j|\}_{i,j}$, i.e. we need only to know the set $\{\Lambda(|i\rangle\langle j|)\}_{i,j}$ in order to characterise Λ . With that in mind, let us write the Choi operator more explicitly as

$$\eta_{A'A} = (\Lambda \otimes \mathbb{1})[\phi^+] = \frac{1}{d} \sum_{i,j} \Lambda(|i\rangle\langle j|) \otimes |i\rangle\langle j|. \quad (7.10)$$

We see that all the information $\Lambda(|i\rangle\langle j|)$ is present in the Choi operator. We can think of it as the matrices $\Lambda(|i\rangle\langle j|)$ sitting as blocks in the larger matrix η^{41} . Therefore, η is a complete characterisation of Λ . In other words, there is an isomorphism between completely positive maps and Choi operators. This is known as *state-channel duality* and we illustrate it in Fig 7.14. Hence, if we want to recover the completely positive map description from the Choi operator, we must express the action $\Lambda(X)$ in terms of $\eta_{A'A}$. For this, we can use the linearity of the map: write the input operator as $X = \sum_{i,j} c_{ij} |i\rangle\langle j|$ for some coefficients c_{ij} and use that $\Lambda(X) = \sum_{i,j} c_{ij} \Lambda(|i\rangle\langle j|)$. Thus, we need only to know the action of the map on the elements $|i\rangle\langle j|$. This is given by

$$\Lambda(|i\rangle\langle j|) \stackrel{(7.10)}{=} d \langle i | \eta_{A'A} | j \rangle_A = d \operatorname{tr}_A \left((\mathbb{1}_{A'} \otimes |j\rangle\langle i|_A) \eta_{A'A} \right). \quad (7.11)$$

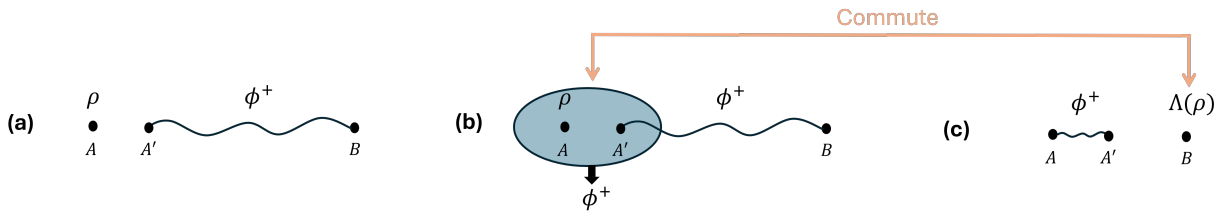
Hence, with the help of state-channel duality, we have expressed the action of a completely positive map only in terms of the Choi operator. The formula is given in Eq (7.12) and is known as the *Choi representation theorem*. While we have already shown this result by the above argument, we will also discuss a different proof below. This proof relies on quantum teleportation and provides an alternative way to think about state-channel duality.

R17: Choi representation theorem

The action of a completely positive map $\Lambda_{A \rightarrow A'}$ on any given state ρ of dimension d can be represented in terms of the map's Choi operator $\eta_{A'A}$. The action is

$$\Lambda_{A \rightarrow A'}(\rho) = d \operatorname{tr}_A \left((\mathbb{1}_{A'} \otimes \rho_A^T) \eta_{A'A} \right). \quad (7.12)$$

The main idea of the proof is illustrated in the three steps here below.



⁴¹ Strictly speaking, we would obtain blocks if we had had in the summand $|i\rangle\langle j| \otimes \Lambda(|i\rangle\langle j|)$. This is only a cosmetic difference. With our convention, the “blocks” are smeared out in the Choi matrix but the information is nevertheless there.

In step (a) we have placed our system, ρ , on the Hilbert space A and introduced the state $\phi_{BA'}^+$ on Hilbert space $A'B$. In step (b) we have teleported ρ_A into system B . We know from the teleportation protocol that if we project system AA' onto the maximally entangled state ϕ^+ , then system B is left in the state ρ (without the need for a unitary correction). In step (c) we have run the channel on the teleported state now residing in system B . The essential observation is that the channel is acting on B and the ϕ^+ -projection is acting on AA' : these are different Hilbert spaces and therefore the operations commute. Thus, without changing the action of the channel, we can swap the order and first run the channel on system B and then project AA' onto ϕ^+ . The probability of a successful projection is $1/d^2$. Renormalising for this and evaluating the above argument, we get

$$\begin{aligned}
\Lambda(\rho) &= d^2 \operatorname{tr}_{AA'} \left((\phi_{AA'}^+ \otimes \mathbb{1}_B) (\mathbb{1}_{AA'} \otimes \Lambda_B) [\rho_A \otimes \phi_{BA'}^+] (\phi_{AA'}^+ \otimes \mathbb{1}_B) \right) \\
&= d^2 \operatorname{tr}_{AA'} \left\{ (\phi_{AA'}^+ \otimes \mathbb{1}_B) \rho_A \otimes \underbrace{(\Lambda_B \otimes \mathbb{1}_{A'}) [\phi_{BA'}^+]}_{\equiv \eta_{BA'}} (\phi_{AA'}^+ \otimes \mathbb{1}_B) \right\} \\
&= d^2 \operatorname{tr}_{AA'} \left((\phi_{AA'}^+ \otimes \mathbb{1}_B) (\rho_A \otimes \eta_{BA'}) \right) \\
&\stackrel{(4.21)}{=} d^2 \operatorname{tr}_{AA'} \left((\phi_{AA'}^+ \otimes \mathbb{1}_B) (\mathbb{1}_{AB} \otimes \rho_{A'}^T) (\mathbb{1}_A \otimes \eta_{BA'}) \right) = d \operatorname{tr}_{A'} \left((\mathbb{1}_B \otimes \rho_{A'}^T) \eta_{BA'} \right). \tag{7.13}
\end{aligned}$$

First, we commuted the channel and teleportation operations. Second, we identified the Choi state. In the penultimate step we used the property (4.21) of the maximally entangled state which then allows us to trace out system A in the last step. Now, we only need to change the names of the subsystems ($A' \rightarrow A$ and $B \rightarrow A'$) to recover the expression in Eq (7.12). $\square$

CPTP maps correspond to Choi operators that can be interpreted as (possibly entangled) states over system A' and A . However, not all states over A' and A are valid Choi states; see Fig 7.14. The defining properties of a Choi state are

$$\eta_{A'A} \geq 0, \quad \text{and} \quad \operatorname{tr}_{A'}(\eta_{A'A}) = \frac{\mathbb{1}}{d}. \tag{7.14}$$

The first condition holds for all states but the second one is distinct for Choi states (note that it implies also $\operatorname{tr}(\eta_{A'A}) = 1$). We can understand the second condition in two ways. Firstly, consider that Alice and Bob share ϕ^+ . Bob's local state is therefore $\frac{\mathbb{1}}{d}$. Consider that Alice now either does nothing or applies the channel Λ to her system. If Alice's channel could change the local state of Bob, it would mean that he can gain information about whether or not Alice chose to apply the channel. This is a violation of the no-signalling principle. It therefore follows from the definition (7.6) that the reduction of the Choi state to system A must be the same as the reduction of ϕ^+ , namely the maximally mixed state. Secondly, we can derive this result directly from our formula in Eq (7.12). To do that, let us check the trace-preservation condition,

$$\operatorname{tr}(\Lambda(\rho)) \stackrel{(7.12)}{=} d \operatorname{tr} \left((\mathbb{1}_{A'} \otimes \rho_A^T) \eta_{A'A} \right) = d \operatorname{tr}_A \left(\rho_A^T \eta_A \right) \stackrel{(7.2)}{=} \operatorname{tr}(\rho) \quad \Rightarrow \quad \eta_A = \operatorname{tr}_{A'}(\eta_{A'A}) = \frac{\mathbb{1}}{d}. \tag{7.15}$$

Notice that for a completely positive map that is not a quantum channel (i.e. it is not trace-preserving) the Choi operator does not have to be proportional to the identity operator when it is reduced to system A .

7.3. Kraus representation

Choi representation is one among several ways of representing quantum channels. Another important representation is called *Kraus representation*. This representation is often a useful tool and it is given by the following theorem.

R18: Kraus representation of CPTP maps

Let $\Lambda_{A \rightarrow A'}$ be a linear map. It is CPTP if and only if it can be decomposed as

$$\Lambda_{A \rightarrow A'}(\rho) = \sum_{i=1}^D K_i \rho K_i^\dagger, \tag{7.16}$$

where the number D does not need to exceed $d_{A'} d_A$ and the *Kraus operators* $\{K_i\}_i$ satisfy

$$\sum_{i=1}^D K_i^\dagger K_i = \mathbb{1}_A. \tag{7.17}$$

To prove the result, we first assume that Λ has a Kraus decomposition and prove that it implies that Λ is CPTP. Linearity follows trivially from Eq (7.16). Trace-preservation holds because

$$\text{tr}(\Lambda(\rho_A)) \stackrel{(7.16)}{=} \text{tr}\left(\sum_i K_i \rho_A K_i^\dagger\right) = \text{tr}\left(\rho_A \sum_i K_i^\dagger K_i\right) \stackrel{(7.17)}{=} \text{tr}(\rho_A \mathbb{1}_A) = \text{tr}(\rho_A). \quad (7.18)$$

Complete positivity also follows straightforwardly from Eq (7.16) because $(\Lambda \otimes \mathbb{1})[X] = \sum_i (K_i \otimes \mathbb{1})X(K_i \otimes \mathbb{1})^\dagger$ and any operator of the form $OXO^\dagger$ is positive semidefinite when $X \geq 0$.

For the converse, we show that CPTP maps always have a Kraus decomposition. To that end, define the Choi state of the channel, $\eta_{A'A} = (\Lambda \otimes \mathbb{1})[\phi^+]$ and let $\eta = \sum_l \lambda_l |\lambda_l\rangle\langle\lambda_l|$ be its spectral decomposition. Recall now that we can recover $\Lambda(|i\rangle\langle j|)$ from the Choi state via Eq. (7.11). Inserting the spectral decomposition into that, we get

$$\Lambda(|i\rangle\langle j|) \stackrel{(7.11)}{=} d \sum_l \lambda_l \langle i|\eta_{A'A}|j\rangle_A = d \sum_l \lambda_l \langle i|\lambda_l\rangle\langle\lambda_l|j\rangle_A. \quad (7.19)$$

Next, consider a generic bipartite state expanded in the computational basis, $|\lambda\rangle_{A'A} = \sum_{i,j} c_{ij} |i\rangle|j\rangle$. In analogy with how we proved the Choi representation theorem, we can write the state as $|\lambda\rangle = \tilde{K} \otimes \mathbb{1} |\phi^+\rangle$ if we choose $\tilde{K} = \sqrt{d} \sum_{i,j} c_{ij} |i\rangle\langle j|$. Therefore, let us assign $\tilde{K}_l$ for each $|\lambda_l\rangle$. Then it follows from Eq (7.19) that

$$\begin{aligned} \Lambda(|i\rangle\langle j|) &= d \sum_l \lambda_l \langle i|\lambda_l\rangle\langle\lambda_l|j\rangle_A \\ &= d \sum_l \lambda_l \langle i|(\tilde{K}_l \otimes \mathbb{1}_A)|\phi^+\rangle\langle\phi^+|(\tilde{K}_l^\dagger \otimes \mathbb{1}_A)|j\rangle_A \\ &= d \sum_l \lambda_l \left(\frac{1}{\sqrt{d}} \tilde{K}_l |i\rangle\right) \left(\frac{1}{\sqrt{d}} \langle j|\tilde{K}_l^\dagger\right) = \sum_l K_l |i\rangle\langle j| K_l^\dagger, \end{aligned} \quad (7.20)$$

where we have defined $K_l = \sqrt{\lambda_l} \tilde{K}_l$. Since every linear operator, X , can be written as a linear combination of the basis elements $|i\rangle\langle j|_{i,j}$, i.e. $X = \sum_{i,j} r_{ij} |i\rangle\langle j|$, it follows from the linearity of the map that

$$\Lambda(X) = \Lambda\left(\sum_{i,j} r_{ij} |i\rangle\langle j|\right) \stackrel{(7.1)}{=} \sum_{i,j} r_{ij} \Lambda(|i\rangle\langle j|) \stackrel{(7.20)}{=} \sum_{i,j} r_{ij} \sum_l K_l |i\rangle\langle j| K_l^\dagger = \sum_l K_l X K_l^\dagger, \quad (7.21)$$

which is Eq (7.16). It is now immediate from our calculation in Eq (7.18) that trace-preservation corresponds to the condition in Eq (7.17). Finally, note that the range of the index l is the size of the Choi operator, which we know to be an operator of dimension $d_{A'} d_A$. Hence, we never require more than this number of Kraus operators in the Kraus decomposition. $\square$

Observe that above proof is constructive: if we are given a channel we therefore know how to construct its Kraus operators. In Exercises B.4.2, B.4.3 and B.4.4 we will compute Kraus representations for some selected quantum channels.

7.4. Schrödinger and Heisenberg picture

In quantum mechanics textbooks, we learn of the Schrödinger and Heisenberg pictures of time evolution. We can now understand them through the lens of quantum channels. In the Schrödinger picture, time evolution is represented by states changing in time and measurements remaining constant. We know that the most general evolution of a quantum state is represented by a quantum channel. Therefore, in the Schrödinger picture, Born's rule for some POVM $\{E_a\}_a$ and some state evolved from ρ becomes

$$p(a) = \text{tr}(E_a \Lambda(\rho)). \quad (7.22)$$

In contrast, the Heisenberg picture of quantum theory views states as constant and the measurements as evolving in time. From quantum mechanics studies, we know that the two pictures are operationally equivalent. In our formalism, we write the probabilities arising in the Heisenberg picture as an evolution of the initial POVM, namely

$$p(a) = \text{tr}(\Lambda^\dagger(E_a) \rho), \quad (7.23)$$

where we have introduced the object $\Lambda^\dagger$ ⁴². We need to define $\Lambda^\dagger$ appropriately, so that the probabilities predicted in the Schrödinger and Heisenberg pictures become the same. For that to happen, we must require $\Lambda^\dagger$ to satisfy the relation

$$\text{tr}(O \Lambda(X)) = \text{tr}(\Lambda^\dagger(O) X), \quad \forall O, X \quad (7.24)$$

⁴² Λ is not a matrix so do not think of $\Lambda^\dagger$ as the Hermitian transpose of Λ .

This is the defining condition for the map $\Lambda^\dagger$. Thus, for every Λ we can use this definition to derive $\Lambda^\dagger$ and we therefore call it the *dual map* (i.e. $\Lambda^\dagger$ is the dual map of Λ). The dual map always exists and it is unique. It allows us to “pass the quantum channel” from one operator (e.g. the state) to another operator (e.g. the POVM-element).

How do we represent the dual map? The answer in Choi representation is discussed in Exercise B.4.5. Here, we discuss the answer in Kraus representation. The Kraus decomposition of the dual channel is

$$\text{tr}(O \Lambda(X)) \stackrel{(7.16)}{=} \text{tr}\left(O \sum_i K_i X K_i^\dagger\right) = \text{tr}\left(\underbrace{\sum_i K_i^\dagger O K_i}_{=\Lambda^\dagger(O)} X\right) = \text{tr}(\Lambda^\dagger(O)X). \quad (7.25)$$

Thus, we need only to send the individual Kraus operators of Λ to their Hermitian conjugates in order to find the dual map, i.e.

$$\Lambda^\dagger(X) = \sum_i K_i^\dagger X K_i. \quad (7.26)$$

Observe that this is not identical to Eq (7.16). Sometimes the dual map is called the dual channel, but while it is completely positive it is actually not trace-preserving. To see this, recall that the property (7.17) was needed for trace-preservation in a completely positive map. However, taking the trace of the dual map, we instead find $\text{tr}(\Lambda^\dagger(X)) = \text{tr}(\sum_i K_i K_i^\dagger X)$ but $\sum_i K_i K_i^\dagger$ is not guaranteed to equal $\mathbb{1}$, which we would have needed for trace-preservation. Instead, the dual map has a different property: it preserves the identity operator. This follows directly from its Kraus decomposition,

$$\Lambda^\dagger(\mathbb{1}) \stackrel{(7.26)}{=} \sum_i K_i^\dagger \mathbb{1} K_i \stackrel{(7.17)}{=} \mathbb{1}. \quad (7.27)$$

Maps that preserve the identity are called *unital maps*⁴³. Thus, the dual of a CPTP map is a completely positive unital map, and the dual of a completely positive unital map is a CPTP map. This is no accident. In the Heisenberg picture, the dual map evolves the POVMs in time. It is therefore essential that the dual map transforms POVMs into POVMs. To ensure that this happens we must guarantee that the completeness relation is preserved. That is,

$$\sum_a \Lambda^\dagger(E_a) = \Lambda^\dagger\left(\sum_a E_a\right) \stackrel{(1.8)}{=} \Lambda^\dagger(\mathbb{1}) \stackrel{(7.27)}{=} \mathbb{1}, \quad (7.28)$$

which is the unital property.

7.5. Stinespring representation

In Chapter 1.1 we postulated that transformations in closed quantum theory are given by unitary operators. We must still reconcile this postulate with our use of quantum channels for describing transformations in open quantum theory. In analogy with how density operators admitted purifications and how POVMs admitted Naimark dilations, quantum channels can also be viewed as marginals of unitary transformations acting on a larger Hilbert space. This approach is called *Stinespring dilation*. We focus on a somewhat restricted variant of this representation, in which the input and output spaces are identical.

R19: Stinespring representation

Let Λ be a quantum channel with identical input and output spaces. Initialise ancilla system, E , in the state $|0\rangle_E$. For some unitary U acting on the joint system AE , the action of the channel takes the form

$$\Lambda(\rho) = \text{tr}_E\left(U(\rho_A \otimes |0\rangle\langle 0|_E)U^\dagger\right). \quad (7.29)$$

We first prove that the map in Eq (7.29) is CPTP. The essence of the Stinespring representation is that we can purify the evolution by adding the ancilla and then discarding it to obtain the action on only system A . Let U be a unitary acting jointly on A and E and let $\{|\lambda_i\rangle\}_i$ be some basis of $\mathcal{H}_E$ which we use for the trace operation. Then,

$$\Lambda(\rho_A) \stackrel{(7.29)}{=} \sum_i \langle \lambda_i | U(\rho_A \otimes |0\rangle\langle 0|_E)U^\dagger | \lambda_i \rangle_E = \sum_i \underbrace{\langle \lambda_i | U(\mathbb{1}_A \otimes |0\rangle\langle 0|_E)}_{=K_i} \rho_A \underbrace{(\mathbb{1}_A \otimes \langle 0|_E)U^\dagger}_{=K_i^\dagger} | \lambda_i \rangle = \sum_i K_i \rho_A K_i^\dagger, \quad (7.30)$$

⁴³ Do not confuse unital maps with unitary maps. All unitary maps are unital because $U\mathbb{1}U^\dagger = UU^\dagger = \mathbb{1}$, but not vice versa.

where we have defined operators $\{K_i\}_i$. These are valid Kraus operators because they satisfy the condition in Eq (7.17),

$$\sum_i K_i^\dagger K_i = \sum_i (\mathbb{1}_A \otimes \langle 0|_E) U^\dagger |\lambda_i\rangle \langle \lambda_i| U (\mathbb{1}_A \otimes |0\rangle_E) = (\mathbb{1} \otimes \langle 0|) \underbrace{U^\dagger \sum_i |\lambda_i\rangle \langle \lambda_i| U}_{=\mathbb{1}} (\mathbb{1} \otimes |0\rangle) = \mathbb{1}. \quad (7.31)$$

The Kraus decomposition is valid and therefore we know that the map from ρ to $\Lambda(\rho)$ is CPTP.

For the converse, we show how to find U for a given set of Kraus operators $\{K_i\}_i$. To this end, we define U by the relations

$$U|\varphi\rangle_A \otimes |0\rangle_E = \sum_i K_i |\varphi\rangle_A \otimes |i\rangle_E \quad (7.32)$$

and verify its unitarity by checking that the inner product is preserved,

$$\langle \varphi', 0 | U^\dagger U | \varphi, 0 \rangle \stackrel{(7.32)}{=} \sum_{i,j} \underbrace{\langle \varphi' | K_i^\dagger K_j | \varphi \rangle}_{=\delta_{i,j}} \underbrace{\langle i | j \rangle}_{=\mathbb{1}} = \langle \varphi' | \sum_i K_i^\dagger K_i | \varphi \rangle = \langle \varphi' | \varphi \rangle. \quad (7.33)$$

We now insert our unitary into Eq (7.29) while writing the state in spectral decomposition $\rho_A = \sum_j \lambda_j |\lambda_j\rangle \langle \lambda_j|_A$. This leads to

$$\begin{aligned} \text{tr}_E (U(\rho_A \otimes |0\rangle \langle 0|_E) U^\dagger) &= \sum_j \lambda_j \text{tr}_E (U(|\lambda_j\rangle \langle \lambda_j|_A \otimes |0\rangle \langle 0|_E) U^\dagger) = \sum_j \lambda_j \text{tr}_E (U(|\lambda_j\rangle_A \otimes |0\rangle_E) (\langle \lambda_j|_A \otimes \langle 0|_E) U^\dagger) \\ &\stackrel{(7.32)}{=} \sum_j \lambda_j \text{tr}_E \left(\left(\sum_p K_p |\lambda_j\rangle_A \otimes |p\rangle_E \right) \left(\sum_q \langle \lambda_j|_A K_q^\dagger \otimes \langle q|_E \right) \right) = \text{tr}_E \left(\sum_{p,q} K_p \rho_A K_q^\dagger \otimes |p\rangle \langle q|_E \right) \\ &= \sum_{p,q} K_p \rho_A K_q^\dagger \underbrace{\text{tr}_E (|p\rangle \langle q|)}_{=\delta_{p,q}} = \sum_p K_p \rho_A K_p^\dagger \stackrel{(7.16)}{=} \Lambda(\rho). \quad \square \end{aligned} \quad (7.34)$$

We have actually already encountered a quantum channel represented in via a Stinespring dilation. In Chapter 6.5 we introduced the universal quantum cloning machine and in constructing it we added not one but two ancilla systems. One ancilla qubit was meant for housing our clone but the other was simply discarded later. Having seen the Stinespring dilation, we now understand that it represented the ancilla register that allowed us to express our quantum channel as a unitary.

B9. A glance beyond

Positive maps and entanglement witnesses

We have seen in Chapter 4.3 that the transposition map is positive, but when applied to half of the maximally entangled state it leads to a non-positive output. This is an instance of a more general connection between maps that are positive but not completely positive and entanglement witnesses. Consider that ρ_{AB} is a separable state, as in Eq (4.31). Applying any positive map, Λ , to system A , we obtain $(\Lambda \otimes \mathbb{1})[\rho_{AB}] = \sum_i q_i \Lambda(\phi_i) \otimes \phi_i \geq 0$, where the inequality follows from $\Lambda(\phi_i) \geq 0$ by definition of a positive map. Let us define $O_\Lambda = (\Lambda^\dagger \otimes \mathbb{1})[\mathbb{1}]$. Then for any separable state

$$\text{tr}((\Lambda \otimes \mathbb{1})[\rho_{AB}]) \stackrel{(7.24)}{=} \text{tr}(O_\Lambda \rho_{AB}) \geq 0. \quad (7.35)$$

This is an entanglement witness because the inequality holds for all separable states but possibly not for entangled states. If Λ is positive but not completely positive, there exists by definition an entangled state ρ_{AB} so that $(\Lambda \otimes \mathbb{1})[\rho_{AB}] \not\geq 0$. In fact, separability can be fully characterised in terms of these maps: the state ρ_{AB} is separable if and only if it remains positive semidefinite after being passed through every Λ that is positive but not completely positive.

7.6. Quantum instruments

The language of completely positive maps can be used to describe all quantum operations. An example of this is quantum measurements. Quantum measurements, which we know as POVMs, can be viewed as quantum channels that map a quantum

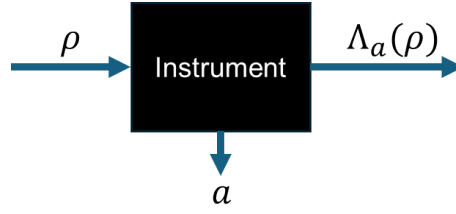


FIG. 7.15. A quantum instrument transforms a quantum state into a classical outcome, a , and a post-measurement quantum state proportional to $\Lambda_a(\rho)$. It is convenient to think of $\Lambda_a(\rho)$ as a sub-normalised state where the sub-normalisation represents the probability of observing a .

input into a classical output. Formally, we write it in the language of channels as⁴⁴

$$\Lambda(\rho) = \sum_a \text{tr}(\rho E_a) |a\rangle\langle a|, \quad (7.36)$$

where $\{E_a\}_a$ is the POVM. However, POVMs only describe the classical outcome of the measurement process. What if we are not only interested in the outcome of the measurement but also in the quantum state immediately afterwards? For projective measurements, this question of state updating was addressed by the quantum measurement postulate in Chapter 1.1, but for POVMs the state updating is richer. The mathematical object that describes the complete measurement process is called a *quantum instrument* and it is illustrated in Fig 7.15. Its action is described by the CPTP map

$$\rho \rightarrow \sum_a \Lambda_a(\rho) \otimes |a\rangle\langle a|. \quad (7.37)$$

The second register holds the classical outcome and when reading it the information about the corresponding post-measurement state is in the first register, namely $\Lambda_a(\rho)$. Since the post-measurement state should be conditioned on observing the outcome a , it is subnormalised by the probability of obtaining that outcome, i.e. $\Lambda_a(\rho) = p(a) \times [\text{post-measurement state}]$. Hence, Born's rule takes the form $p(a) = \text{tr}(\Lambda_a(\rho))$. For consistency we impose that $\Lambda \equiv \sum_a \Lambda_a$ is trace-preserving, so that the probabilities sum to one, namely

$$\text{tr} \left(\sum_a \Lambda_a(\rho) \otimes |a\rangle\langle a| \right) = \text{tr} \left(\sum_a \Lambda_a(\rho) \right) = \text{tr}(\Lambda(\rho)) = 1, \quad \forall \rho. \quad (7.38)$$

This leads us to the definition of a quantum instrument.

G19: Quantum instrument

A quantum instrument is a collection of completely positive trace non-increasing maps, $\{\Lambda_a\}_a$, with the property that $\sum_a \Lambda_a$ is a trace-preserving map.

We could represent each Λ_a via Kraus operators $\{K_{ab}\}_b$. Note that our classical outcome is now the pair (a, b) , so we have more Kraus operators than we have outcomes in the actual instrument. For given pair (a, b) , the subnormalised post-measurement state is $K_{ab}\rho K_{ab}^\dagger$ and the classical register now becomes $|a\rangle\langle a| \otimes |b\rangle\langle b|$. To recover our instrument, we are going to ignore (trace out) the register holding the outcome b , i.e.

$$\rho \rightarrow \text{tr}_C \left(\sum_{a,b} K_{ab}\rho K_{ab}^\dagger \otimes |a\rangle\langle a|_B \otimes |b\rangle\langle b|_C \right) = \sum_a \underbrace{\left(\sum_b K_{ab}\rho K_{ab}^\dagger \right)}_{=\Lambda_a(\rho)} \otimes |a\rangle\langle a| = \sum_a \Lambda_a(\rho) \otimes |a\rangle\langle a|. \quad (7.39)$$

This permits us to think of the instrument as performing a measurement and then forgetting about a part of the outcome.

Every instrument, $\{\Lambda_a\}_a$, is associated with a specific POVM given by $\{\Lambda_a^\dagger(\mathbb{I})\}_a$. Think of it as the POVM we are left with if we discard the post-measurement state in Fig 7.15. However, any given POVM can be implemented by many different instruments. The reason for this is that after our measurement we could apply some arbitrary quantum channel and call the result our post-measurement state. It would not change the already produced classical outcome but it would change the final quantum state. Hence, we cannot put POVMs and instruments in one-to-one correspondence. Nevertheless, for any given POVM there is one particularly important instrument known as the *Lüders instrument*.

⁴⁴ We can always write a classical state in the language of quantum theory by associating the classical integer outcome a to the state $|a\rangle$.

G20: Lüders instrument

The Lüders instrument, $\{\Omega_a\}_a$, that implements the POVM $\{E_a\}_a$ is defined as

$$\Omega_a(\rho) = \sqrt{E_a} \rho \sqrt{E_a}. \quad (7.40)$$

Notice the special case in which E_a is projective: then $\sqrt{E_a} = E_a$ and we recover from $\Omega_a(\rho)$ the (unnormalised) post-measurement state postulated in G3 in Chapter 1.1. In general, Ω_a is completely positive and $\Omega \equiv \sum_a \Omega_a$ is trace-preserving because

$$\text{tr}(\Omega(\rho)) = \text{tr}\left(\sum_a \sqrt{E_a} \rho \sqrt{E_a}\right) = \sum_a \text{tr}(\rho E_a) \stackrel{(1.8)}{=} \text{tr}(\rho). \quad (7.41)$$

It is natural to ask what makes the Lüders instrument stand out among all the other instruments that implement a given POVM. There is a simple answer: every instrument, $\{\Lambda_a\}_a$, that implements $\{E_a\}_a$ can be understood as first applying the Lüders instrument, $\{\Omega_a\}_a$, and then applying an outcome-dependent CPTP map Γ_a . To show that this is the case, let Π_a be the projector onto the subspace complementary to that on which E_a is supported. Then, define the maps

$$\Gamma_a(\rho) \equiv \Lambda_a\left(E_a^{-\frac{1}{2}} \rho E_a^{-\frac{1}{2}}\right) + \Theta(\Pi_a \rho \Pi_a), \quad (7.42)$$

where Θ is a CPTP map. Γ_a is completely positive by construction. Since the above definition is valid for every ρ , we can equivalently express it in terms of the dual map as

$$\Gamma_a^\dagger(\mathbb{1}) = \underbrace{E_a^{-\frac{1}{2}} \Lambda_a^\dagger(\mathbb{1}) E_a^{-\frac{1}{2}}}_{=E_a} + \underbrace{\Pi_a \Theta^\dagger(\mathbb{1}) \Pi_a}_{=\Pi_a \mathbb{1} \Pi_a = \Pi_a} = (\mathbb{1} - \Pi_a) + \Pi_a = \mathbb{1}. \quad (7.43)$$

In the first term, we have used that Λ_a implements E_a and then that $E_a^{-\frac{1}{2}} E_a E_a^{-\frac{1}{2}}$ is the identity on the support space of E_a , which we defined as $\mathbb{1} - \Pi_a$. In the second term, we have used that the dual map of a CPTP map always is unital, $\Theta^\dagger(\mathbb{1}) = \mathbb{1}$. Since Eq (7.43) shows that $\Gamma_a^\dagger$ is unital, it implies that Γ_a is trace-preserving. Thus, the construction in Eq (7.42) is CPTP. Now, we use Eq (7.42) to compose Γ_a and the Lüders instrument. This gives

$$(\Gamma_a \circ \Omega_a)[\rho] \stackrel{(7.40)}{\stackrel{(7.42)}{=}} \Lambda_a\left(E_a^{-\frac{1}{2}} E_a^{\frac{1}{2}} \rho E_a^{\frac{1}{2}} E_a^{-\frac{1}{2}}\right) + \underbrace{\Theta\left(\Pi_a E_a^{\frac{1}{2}} \rho E_a^{\frac{1}{2}} \Pi_a\right)}_{=0} = \Lambda_a((\mathbb{1} - \Pi_a) \rho (\mathbb{1} - \Pi_a)) = \Lambda_a(\rho). \quad (7.44)$$

Here, we have used that Π_a is orthogonal to E_a and therefore also to $E_a^{\frac{1}{2}}$. In the last step, we have used that for a completely positive map L it holds that $L(\rho) = L(P\rho P)$ for any ρ if P is the projector onto the support of $L^\dagger(\mathbb{1})$. Thus, we have recovered the generic instrument $\{\Lambda_a\}_a$ that realises $\{E_a\}_a$ by first applying a Lüders instrument and then a suitable quantum channel. Therefore, we can view the Lüders instrument as the elementary state update rule associated with any POVM.

7.7. Information-disturbance relations

The Lüders instrument shows us how to update a quantum state when implementing an arbitrary measurement. A crucial consequence is that whenever we want to extract information from a system, we have no option but to induce an irreversible disturbance in its state. This means that we have no hope to extract information and then deterministically return the state to what it was before we measured it.

R20: No information without disturbance

A Lüders update of a quantum state, as defined in Eq (7.40), leaves the state undisturbed if and only if $\Omega_a^\dagger(\mathbb{1}) \propto \mathbb{1}$.

Assume that $\Omega_a^\dagger(\mathbb{1}) \propto \mathbb{1}$. This means the POVM implemented by the instrument is $E_a = \Omega_a^\dagger(\mathbb{1}) = q_a \mathbb{1}$, where $\{q_a\}_a$ is a probability distribution. The corresponding Lüders update is $\Omega_a(\rho) = q_a \rho$. Thus, the instrument affects only the probability of the post-measurement state but not the post-measurement state itself. No disturbance has taken place.

Conversely, assume that no disturbance is induced by the Lüders update. From Eq (7.40) this means that we require $\Omega_a(\rho) = \sqrt{E_a} \rho \sqrt{E_a} \propto \rho$. Since this must hold for every ρ , the operator $\sqrt{E_a}$ must be proportional to the identity. $\square$

Note that this result applies also to instruments that are not of the Lüders type, because we have already seen that these are just compositions of a Lüders instrument and a CPTP map. To avoid disturbance, we first need the Lüders instrument to be as in R20, and then that the subsequent outcome-dependent CPTP map is reversible (for instance unitary). In Chapter 10 we will view the inability to extract information without leaving a disturbance as the fundamental feature which paves the way for quantum cryptography. In fact, the more information we extract the more disturbance we must induce. This leads to the notion of *information-disturbance* trade-offs and they manifest themselves in many different ways across quantum information theory. Here, in order to provide an intuition, we explore them from two complementary perspectives.

7.7.1. Gentle measurements

Say that we make trivial measurement $\{\mathbb{1}, 0\}$ on a state ρ . Clearly, we will observe the first outcome independently of ρ : we have neither extracted information nor disturbed the state. However, instead of being non-interacting, what if our measurement is interacting only “gently” with the system. By this, we mean that the outcome associated with the POVM element E is only almost guaranteed to be observed. Say that we implement E through its Lüders instrument. Can we limit the magnitude of disturbance induced in the post-measurement state just from knowing how likely the outcome is? This would imply an upper limitation on how much disturbance we have to accept when running our measurement. As we now show, the answer is positive.

Let ρ_E be the Lüders post-measurement state conditioned on the outcome associated with E . Assume that the probability of observing the outcome satisfies $\text{tr}(\rho E) \geq 1 - \epsilon$ for some $\epsilon \in [0, 1]$. The completely positive map associated with the Lüders instrument when considering only the measurement operator E is $\Omega_E(\rho) = \sqrt{E} \rho \sqrt{E}$. Hence, $\rho_E = \sqrt{E} \rho \sqrt{E} / \text{tr}(\rho E)$ where we can without loss of generality put $\text{tr}(\rho E) = 1 - \epsilon$. We now consider the fidelity $F(\rho, \rho_E)$ from Eq (2.17) to get

$$\sqrt{(1 - \epsilon)F(\rho, \rho_E)} = \text{tr} \left(\underbrace{(\sqrt{\rho} \sqrt{E} \rho \sqrt{E} \sqrt{\rho})^{1/2}}_{=(\sqrt{\rho} \sqrt{E} \sqrt{\rho})^2} \right) = \text{tr}(\sqrt{\rho} \sqrt{E} \sqrt{\rho}) = \text{tr}(\rho \sqrt{E}) \geq \text{tr}(\rho E) = 1 - \epsilon. \quad (7.45)$$

Re-arranging the the left- and right-hand-sides gives $F(\rho, \rho_E) \geq 1 - \epsilon$. Next, we use the Fuchs-van de Graaf inequalities from Eq (2.32) to bound the trace distance in terms of the fidelity. This implies

$$\|\rho - \rho_E\|_1 \leq 2\sqrt{1 - F(\rho, \rho_E)} \leq 2\sqrt{\epsilon}. \quad (7.46)$$

Hence, we conclude that if we are only ϵ -unsure of observing an outcome, the corresponding state update is no more disturbing than $2\sqrt{\epsilon}$ measured in trace-distance. In fact, the inequality (7.46) holds also if we re-define ρ_E as $\rho_E = \sqrt{E} \rho \sqrt{E}$, i.e. we do not re-normalise for the probability of the measurement outcome. This is shown in Exercise B.4.8.

7.7.2. State discrimination

Let us consider a more explicit information-disturbance trade-off inspired by a simple task. Consider that we randomly and uniformly draw a pure qubit state $|\psi\rangle$ with Bloch vector $\hat{n}$. Alice’s job is to tell us whether $\hat{n}$ sits in the northern (N) or the southern (S) Bloch hemisphere; see Fig 7.16a. This means that she must measure the qubit and return either one of the possible outcomes $a \in \{N, S\}$. The instrument that implements this measurement is denoted $\{\Lambda_N, \Lambda_S\}$. The post-measurement state is passed to Bob who measures its fidelity with the original state $|\psi\rangle$. Thus, the information extraction corresponds to learning the Bloch hemisphere and the disturbance corresponds to how much the fidelity is reduced as a consequence of the measurement.

The average success probability of hemisphere discrimination and the average fidelity of the post-measurement state are

$$p_{\text{win}} = \frac{1}{2} \int_N d\psi \text{tr}(\Lambda_N(\psi)) + \frac{1}{2} \int_S d\psi \text{tr}(\Lambda_S(\psi)), \quad \text{and} \quad F = \int d\psi \langle \psi | \sum_a \Lambda_a(\psi) | \psi \rangle, \quad (7.47)$$

where $d\psi$ represents a uniform density over the state space. In order to evaluate these quantities we must select the POVM. Since we are trying to learn the hemisphere, it appears reasonable to measure along the Z -axis. However, we do not want to measure too strongly because then it will imply a large reduction in the fidelity of the post-measurement state. The solution is to do an *unsharp* measurement, i.e. an instrument that implements σ_Z weakly, with a strength parameter that we call $\gamma \in [0, 1]$. Such an instrument can be written in Kraus representation as

$$K_N = \sqrt{\frac{1+\gamma}{2}} |0\rangle\langle 0| + \sqrt{\frac{1-\gamma}{2}} |1\rangle\langle 1|, \quad K_S = \sqrt{\frac{1-\gamma}{2}} |0\rangle\langle 0| + \sqrt{\frac{1+\gamma}{2}} |1\rangle\langle 1|. \quad (7.48)$$

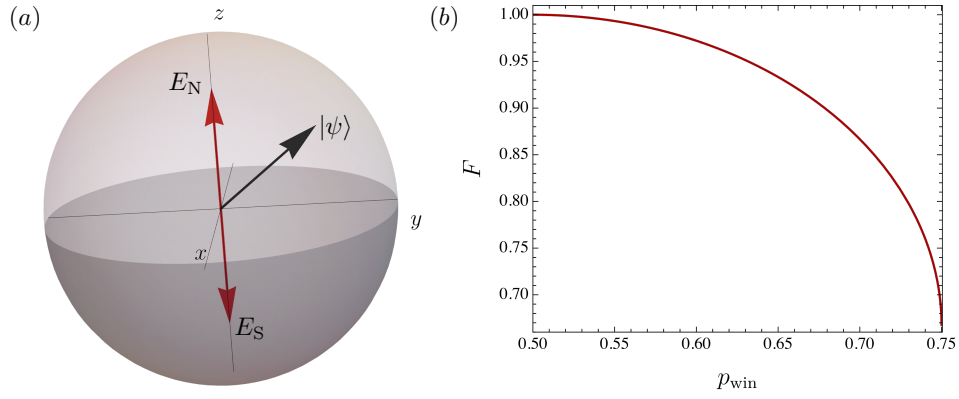


FIG. 7.16. a) Hemisphere discrimination task. We are given an arbitrary pure state $|\psi\rangle$ and asked to determine whether it belongs to the northern or southern Bloch hemisphere, while disturbing the state as little as possible. b) Information-disturbance trade-off for the task. The information is represented by the average discrimination probability p_{win} and the disturbance is represented in terms of the fidelity by $1 - F$.

The corresponding POVM becomes $E_N = K_N^\dagger K_N$ and $E_S = K_S^\dagger K_S$ which means

$$E_N = \frac{1+\gamma}{2}|0\rangle\langle 0| + \frac{1-\gamma}{2}|1\rangle\langle 1|, \quad \text{and} \quad E_S = \frac{1-\gamma}{2}|0\rangle\langle 0| + \frac{1+\gamma}{2}|1\rangle\langle 1| \quad (7.49)$$

If we set $\gamma = 1$ the POVM is just σ_Z , and as we decrease γ it becomes a mixture of σ_Z with white noise, to finally at $\gamma = 0$ become just a non-interacting measurement. In Exercise B.4.10 we compute the two parameters for the fidelity and discrimination probability as functions of γ . Their trade-off becomes

$$F = \frac{2}{3} + \frac{1}{3}\sqrt{16p_{\text{win}}(1-p_{\text{win}}) - 3}, \quad (7.50)$$

which is illustrated as the information-disturbance relation in Fig 7.16b. We see that the better we perform the discrimination, the smaller is the fidelity. In particular, if we set $p_{\text{win}} = 1/2$, which Alice can achieve by just guessing without interacting with the state, then $F = 1$ (no disturbance).

Chapter summary.

- Quantum channels are completely positive and trace-preserving linear maps. Complete positivity means that positive semidefinite operators are mapped into positive semidefinite operators even when the map acts only on a part of a system.
- There are several ways of representing a quantum channel. Three important examples are Choi, Kraus and Stinespring representations.
- Quantum instruments are channels that describe measurement processes. They generalise POVMs to include not only the classical outcome but also the quantum post-measurement state.
- Extracting information from a quantum system is associated with an unavoidable disturbance in its state. The more information that is extracted, the greater is the induced disturbance.

Further reading. For a more extensive discussion about quantum channels, see for example the book [Quantum Information Theory](#) by Wilde and the notes [Quantum Channels and Operations - Guided Tour](#) by Wolf.

8. ENTROPIC CONCEPTS

Entropy plays a central role in the analysis of information. The mathematical discipline of how to quantify, store and transmit information is called information theory, or sometimes Shannon theory after Claude Shannon who pioneered the field in the mid 20th century. In this chapter, we introduce entropic concepts and their applications, starting in classical information theory and then continuing to quantum theory.

8.1. Shannon entropy

Consider that we have a random variable, X , with a probability distribution $p_X(x) \equiv p_X(X = x)$. We sample a value, x^* , from this distribution. How surprised are we about the value of this sample? Clearly, if $p_X(x^*) = 1$ then we are not surprised at all because we already knew that we were going to find $X = x^*$. In contrast, if $p_X(x^*)$ is small we would have had little reason to bet on $X = x^*$. One way to quantify this “surprisal” is by defining $i(x) = -\log_2(p_X(x))$. This satisfies some basic properties: at $p_X(x) = 1$ we get $i(x) = 0$, then by decreasing $p_X(x)$ we have a monotonic growth in $i(x)$, and in the limit $p_X(x) \rightarrow 0$ we have $i(x)$ diverging. By using the logarithm⁴⁵ in the definition we have made sure that the surprisal of two independent variables, X and Y , is simply the sum of their surprisals, $i(x, y) = -\log(p_X(x)p_Y(y)) = -\log(p_X(x)) - \log(p_Y(y)) = i(x) + i(y)$.

Often, we do not only once sample a random variable but instead (perhaps infinitely) many times. For each sample in that sequence, we would obtain some value of the surprisal function. We can ask how surprised we are going to be on average, i.e. what is the expectation value of the surprisal? This quantity turns out to be important. It is called the *Shannon entropy*.

G21: Shannon entropy

The Shannon entropy of a random variable, X , with probability distribution $p_X(x)$ is

$$H(X) = - \sum_{x=1}^m p_X(x) \log(p_X(x)), \quad (8.1)$$

where m is the cardinality of X . To account for vanishing probabilities, we define $0 \log 0 = 0$.

The Shannon entropy is important for several reasons⁴⁶. Our focus will be its operational importance in a relevant information processing task discussed later in this Chapter. On our way there, we first need to establish some basic properties.

1. **Non-negativity.** The Shannon entropy satisfies $H(X) \geq 0$. This follows directly from the surprisal being non-negative and $H(X)$ being its expectation value. Moreover, $H(X) = 0$ if and only if $p_X(x) \in \{0, 1\}$.
2. **Label-invariance.** The Shannon entropy depends only on $p_X(x)$ and not on the labeling of the possible events of X .
3. **Largest value.** The Shannon entropy satisfies $H(X) \leq \log(m)$ with equality if and only if $p_X(x) = \frac{1}{m}$. This follows from Jensen’s inequality which says that for a concave function, $f(z)$, it holds that $\sum_i q_i f(z_i) \leq f(\sum_i q_i z_i)$ for any $q_i \geq 0$ such that $\sum_i q_i = 1$. The inequality is uniquely saturated for $z_1 = z_2 = \dots = z_m$. Since $\log(x)$ is concave, we have

$$H(X) = - \sum_{x=1}^m p_X(x) \log(p_X(x)) = \sum_{x=1}^m p_X(x) \log\left(\frac{1}{p_X(x)}\right) \stackrel{\text{Jensen ineq}}{\leq} \log\left(\sum_{x=1}^m \frac{p_X(x)}{p_X(x)}\right) = \log(m). \quad (8.2)$$

4. **Additivity.** The Shannon entropy of two independent random variables, X and Y , is the sum of the individual Shannon entropies. This follows from considering the distribution $p_{X,Y}(x, y) = p_X(x)p_Y(y)$,

$$\begin{aligned} H(X, Y) &\stackrel{(8.1)}{=} - \sum_{x,y} p_X(x)p_Y(y) \log(p_X(x)p_Y(y)) \\ &= - \sum_x p_X(x) \underbrace{\sum_y p_Y(y) \log(p_X(x))}_{=1} - \sum_y \underbrace{\sum_x p_X(x) p_Y(y) \log(p_Y(y))}_{=1} \stackrel{(8.1)}{=} H(X) + H(Y). \end{aligned} \quad (8.3)$$

⁴⁵ All logarithms are in base 2.

⁴⁶ It is also possible to derive the Shannon entropy as the unique function that satisfies a number of reasonable axioms for an information measure. We do not pursue that route here.

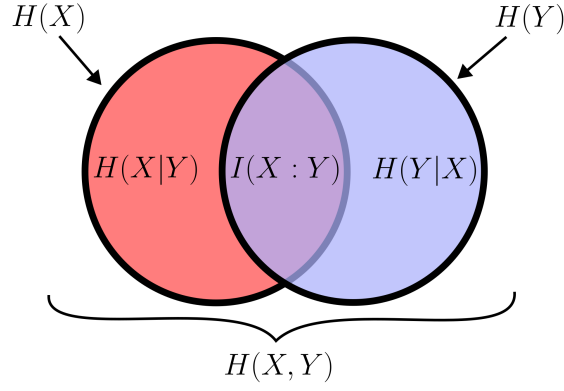


FIG. 8.17. Venn diagram for entropic quantities. The red and the blue circles represent $H(X)$ and $H(Y)$, respectively. Their intersection is the mutual information $I(X : Y)$. The conditional entropy is what is left of the circles once the mutual information is removed.

In the last property we assumed that X and Y are independent, but what if they are not? In general, if we sample y , it could give us some knowledge about X . This means we would have to update the Shannon entropy of X with the knowledge that $Y = y$,

$$H(X|Y = y) = - \sum_{x=1}^m p_{X|Y}(x|y) \log(p_{X|Y}(x|y)). \quad (8.4)$$

The knowledge provided by the event $Y = y$ can lead to $H(X|Y = y)$ being smaller than $H(X)$; see e.g. Exercise B.5.1 for an example from everyday life. However, since Y itself also is a random variable, we may consider the expectation value of this updated Shannon entropy. The resulting quantity is called the *conditional entropy*,

$$H(X|Y) = \sum_{y=1}^m p_Y(y) H(X|Y = y). \quad (8.5)$$

Because providing knowledge of Y should not increase our uncertainty about X , it appears natural that

$$H(X) \geq H(X|Y), \quad (8.6)$$

with equality if and only if X and Y are independent. In Exercise B.5.1 we prove this intuition to be correct.

We can now use the conditional entropy to extend the fourth property of the Shannon entropy to correlated random variables. The result is known as the *chain rule* of conditional entropy,

$$H(X, Y) = H(Y) + H(X|Y). \quad (8.7)$$

The chain rule follows directly from Eq (8.5) because

$$\begin{aligned} H(X|Y) &= \sum_y p_Y(y) H(X|Y = y) \stackrel{(8.4)}{=} - \sum_{x,y} p_Y(y) p_X(x|y) \log p_X(x|y) = \sum_{x,y} p_{X,Y}(x, y) \log \left(\frac{p_Y(y)}{p_{X,Y}(x, y)} \right) \\ &= \sum_y \left(\underbrace{\sum_x p_{X,Y}(x, y)}_{=p_Y(y)} \right) \log p_Y(y) - \sum_{x,y} p_{X,Y}(x, y) \log p_{X,Y}(x, y) = -H(Y) + H(X, Y). \end{aligned} \quad (8.8)$$

Note that if X and Y are independent then the chain rule (8.7) reduces to the additivity property in Eq (8.3).

An important distinction is that the conditional entropy gives the uncertainty about X after learning Y , but not the amount of information gained about X thanks to the knowledge of Y . To pinpoint the information gain, one needs to consider the difference between the uncertainty about X before learning Y and the uncertainty about X after learning Y . This quantity is called the *mutual information* and it is given by

$$I(X : Y) = H(X) - H(X|Y) \stackrel{(8.7)}{=} H(X) + H(Y) - H(X, Y). \quad (8.9)$$

Due to Eq (8.6), the mutual information is non-negative. From the right-hand-side, we also see that $I(X : Y) = I(Y : X)$, which is expected because what X gains us about Y should be the same as what Y gains us about X . The relations between the various entropic quantities are qualitatively illustrated in Fig 8.17.

8.2. Quantum entropy

While classical entropies are properties of probability distributions, quantum entropies are properties of density operators, ρ . The quantum version of the Shannon entropy is called the *von Neumann entropy*. We write it as $H(A)_\rho$, where A is the Hilbert space on which ρ is defined⁴⁷.

G22: von Neumann entropy

The von Neumann entropy of a quantum state, ρ , on a Hilbert space labelled by A is

$$H(A)_\rho = -\text{tr}(\rho_A \log \rho_A). \quad (8.10)$$

To use this definition, recall that functions of Hermitian operators (such as $\log \rho_A$) are defined as $f(O) = \sum_i f(x_i) |x_i\rangle\langle x_i|$, where $\{x_i\}_i$ and $\{|x_i\rangle\}_i$ are, respectively, the eigenvalues and eigenvectors of the operator O .

How do we motivate the definition of the von Neumann entropy? One answer is that, like Shannon entropy, the von Neumann entropy has a relevant operational meaning, which we will encounter later in this chapter. Another answer is that it can be related to the Shannon entropy in a simple way. For the latter, let us write ρ_A in its spectral decomposition, namely $\rho_A = \sum_x \lambda_x |\lambda_x\rangle\langle \lambda_x|$, and evaluate the von Neumann entropy. We obtain

$$H(A)_\rho \stackrel{(8.10)}{=} -\text{tr} \left(\sum_x \lambda_x |\lambda_x\rangle\langle \lambda_x| \log \left(\sum_y \lambda_y |\lambda_y\rangle\langle \lambda_y| \right) \right) = -\sum_x \lambda_x \sum_y \log(\lambda_y) \underbrace{|\langle \lambda_x | \lambda_y \rangle|^2}_{=\delta_{x,y}} = -\sum_x \lambda_x \log(\lambda_x) = H(X), \quad (8.11)$$

where X has been defined as a random variable with distribution $p_X(X=x) = \lambda_x$. Thus, the von Neumann entropy of ρ_A is the Shannon entropy of the probability distribution given by the eigenvalues of ρ_A . Using this, we can see that the properties 1–4 discussed for Shannon entropy have direct counterparts for the von Neumann entropy.

1. **Non-negativity.** The von Neumann entropy satisfies $H(\rho_A) \geq 0$ with equality if and only if ρ_A is pure. Pure states are analogous to deterministic distributions because their eigenvalues are either zero or one.
2. **Unitary invariance.** The von Neumann entropy is invariant under unitary transformations, U . That is, $H(\rho) = H(U\rho U^\dagger)$. This follows from Eq (8.11) because unitaries cannot change the spectrum of ρ_A .
3. **Largest value.** The von Neumann entropy satisfies $H(\rho_A) \leq \log(\dim(A))$ with equality if and only if ρ_A is the maximally mixed state. Hence, the Hilbert space dimension is the quantum analogy to the cardinality of the random variable. The maximally mixed state is the only state whose eigenvalues form a uniform distribution.
4. **Additivity.** Consider states ρ_A and σ_B . The von Neumann entropy of the joint is the sum of the individual entropies,

$$H(A, B)_{\rho \otimes \sigma} = H(A)_\rho + H(B)_\sigma. \quad (8.12)$$

This follows from Eq (8.11) because the eigenvalues of $\rho \otimes \sigma$ are $p(x, y) = \lambda_x \mu_y$ where $\{\lambda_x\}_x$ are the eigenvalues of ρ and $\{\mu_y\}_y$ are the eigenvalues of σ . This is a product distribution, which corresponds to independent random variables.

Inspired by the chain rule for classical entropy, we also define a conditional version of the von Neumann entropy

$$H(A|B)_{\rho_{AB}} = H(A, B)_{\rho_{AB}} - H(B)_{\rho_B}. \quad (8.13)$$

However, there are important differences between classical and quantum conditional entropy. We interpreted the classical conditional entropy as the average uncertainty about one variable given knowledge of another; see Eq (8.5). However, this interpretation does not hold in the quantum case, because there are no conditional probabilities for quantum states. Furthermore, the conditional Shannon entropy is a non-negative quantity, but the conditional von Neumann entropy is not. For example, consider the maximally entangled two-qubit state ϕ_{AB}^+ . Since it is pure we have $H(A, B)_{\phi^+} = 0$ and since its reduction is maximally mixed we have $H(B) = 1$. Inserting this in Eq (8.13) we obtain $H(A|B)_{\phi^+} = -1$. This negativity is a manifestation of how entanglement permits the subsystem to be more mixed than the joint state. In Exercise B.5.2 it is shown that entanglement is necessary for negative conditional entropy. In other words, the violation of the condition $H(A|B)_\rho \geq 0$ implies entanglement. This does not explain what it actually means to have a negative conditional entropy, but later when we arrive to Box B10 we will provide a brief sketch of an answer.

⁴⁷ We label quantum systems by A, B, C and classical random variables by X, Y, Z . When clear from context, we write $H(A)_\rho = H(A) = H(\rho)$.

8.2.1. Quantum relative entropy

An elegant way to represent many different entropic quantities is by viewing them as special cases of a single entropic quantity called the *quantum relative entropy*.

G23: Quantum relative entropy

The quantum relative entropy of a density operator ρ with respect to a positive semidefinite operator σ is defined as

$$D(\rho||\sigma) = \text{tr}(\rho \log \rho) - \text{tr}(\rho \log \sigma), \quad (8.14)$$

where the support of ρ needs to be contained in the support of σ ; otherwise, we define $D(\rho||\sigma) = \infty$.

Notice that the quantity is not symmetric, i.e. even if σ is a state we do have in general $D(\rho||\sigma) \neq D(\sigma||\rho)$. In Exercise B.5.3 we show that the quantum relative entropy between two density operators is non-negative and in Exercise B.5.4 we prove some more of its basic properties.

Let us select σ in three different ways: first as $\sigma = \mathbb{1}$, then as $\sigma = \mathbb{1}_A \otimes \rho_B$ and finally as $\sigma = \rho_A \otimes \rho_B$. Evaluating the quantum relative entropy in each case, we find that we can recover the von Neumann entropy, the quantum conditional entropy and the quantum mutual information, respectively. Specifically,

$$H(A)_\rho = -D(\rho_A||\mathbb{1}_A), \quad H(A|B)_\rho = -D(\rho_{AB}||\mathbb{1} \otimes \rho_B), \quad I(A : B)_\rho = D(\rho_{AB}||\rho_A \otimes \rho_B). \quad (8.15)$$

The last expression can be viewed as the definition of quantum mutual information. The reason we define it in this way becomes apparent if one evaluates the right-most quantum relative entropy. It gives

$$\begin{aligned} D(\rho_{AB}||\rho_A \otimes \rho_B) &\stackrel{(8.14)}{=} \text{tr}(\rho_{AB} \log \rho_{AB}) - \text{tr}(\rho_{AB} \log(\rho_A \otimes \rho_B)) \stackrel{(8.17)}{=} \text{tr}(\rho_{AB} \log \rho_{AB}) - \text{tr}(\rho_A \log \rho_A) - \text{tr}(\rho_B \log \rho_B) \\ &\stackrel{(8.10)}{=} -H(A, B)_\rho + H(A)_{\rho_A} + H(B)_{\rho_B} \stackrel{(8.13)}{=} H(A)_{\rho_A} - H(A|B)_\rho, \end{aligned} \quad (8.16)$$

which is analogous to the classical mutual information defined in Eq (8.9). In the second step above we have used that

$$\log(X \otimes Y) = \log(X) \otimes \mathbb{1} + \mathbb{1} \otimes \log(Y), \quad (8.17)$$

for positive semidefinite operators X and Y , as is shown in Exercise B.5.4.

From Eq (8.15), it may appear as if the quantum relative entropy is just a clever way of writing other entropic quantities in a compact way, but there is more to it than that. In particular, $D(\rho||\sigma)$ can be interpreted as a measure of distinguishability. To understand this, let us consider a variation of the state discrimination problem that we encountered in Chapter 3.3. Just like then, we have two states, ρ and σ , and we want to determine which state we have been given using a POVM, $\{E_0, E_1\}$. This time, however, we are going to assume that the probability of incorrectly identifying ρ is fixed to some value, γ , and then ask how small we can make the probability of incorrectly identifying σ . The minimal error probability for σ is

$$t_\gamma(\rho, \sigma) \equiv \min_{\{E_0, E_1\}} \left\{ \text{tr}(\sigma E_0) \text{ such that } \text{tr}(\rho E_1) = \gamma \right\}. \quad (8.18)$$

While this quantity can be computed by various methods, that is not our interest here. Our focus is what happens in the limit in which we are given infinitely many copies of ρ and σ . To that end, recall first that a small error probability, t_γ , corresponds to a large distinguishability between ρ and σ . To capture this distinguishability, we may focus on the quantity $-\log(t_\gamma)$. This is reasonable because the logarithm is monotonic and because it maps a zero probability for error to infinite distinguishability, while a unit probability for error maps to zero distinguishability. Moreover, if we have n copies of each state, $\rho^{\otimes n}$ and $\sigma^{\otimes n}$, we expect our distinguishability to scale with n . It is then convenient to consider the distinguishability we get per copy, namely $-\frac{1}{n} \log(t_\gamma)$. The *quantum Stein's lemma* tells us that in the limit of many copies, this distinguishability rate is given by the quantum relative entropy, namely

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log(t_\gamma(\rho^{\otimes n}, \sigma^{\otimes n})) = D(\rho||\sigma). \quad (8.19)$$

We do not prove this result⁴⁸ but focus instead on understanding its consequences.

⁴⁸ The proof is lengthy and it requires additional tools whose introduction would deviate from our main focus. The interested reader can find a complete proof for instance in the book *Quantum information theory* by Renes.

8.2.2. Monotonicity and strong sub-additivity

We are now going to use the quantum Stein's lemma to prove an essential property of the quantum relative entropy, namely its monotonicity under CPTP maps.

R21: Monotonicity of quantum relative entropy

For any quantum states ρ and σ and any CPTP map Λ it holds that

$$D(\rho\|\sigma) \geq D(\Lambda(\rho)\|\Lambda(\sigma)). \quad (8.20)$$

This is usually a complicated theorem to show but thanks to accepting the quantum Stein's lemma it becomes simple. The result follows from

$$D(\Lambda(\rho)\|\Lambda(\sigma)) \stackrel{(8.19)}{=} \lim_{n \rightarrow \infty} -\frac{1}{n} \log(t_\gamma(\Lambda(\rho)^{\otimes n}, \Lambda(\sigma)^{\otimes n})) \leq \lim_{n \rightarrow \infty} -\frac{1}{n} \log(t_\gamma(\rho^{\otimes n}, \sigma^{\otimes n})) \stackrel{(8.19)}{=} D(\rho\|\sigma). \quad (8.21)$$

The inequality is justified by the fact that applying a CPTP map before discrimination cannot reduce the minimal error probability. That is, $t_\gamma(\tau, \omega) \leq t_\gamma(\Gamma(\tau), \Gamma(\omega))$ for any states τ and ω and any CPTP map Γ . In our case, $\tau = \rho^{\otimes n}$, $\omega = \sigma^{\otimes n}$ and $\Gamma = \Lambda^{\otimes n}$. To see why this is true, observe that Eq (8.18) already contains an optimisation over the POVM, and that applying a CPTP map before applying a POVM effectively corresponds to applying another POVM. $\square$

An immediate consequence of this theorem is that the quantum relative entropy, when evaluated between states, is a non-negative quantity. This follows from Eq (8.20) because we can choose Λ to be the trace operation. That gives us

$$D(\rho\|\sigma) \geq D(\text{tr}(\rho)\|\text{tr}(\sigma)) = D(1\|1) = 0. \quad (8.22)$$

Another application of the result R21 is presented in Exercise B.5.5. There, we use it to prove a quantum uncertainty relation formulated with entropies instead of with the variances encountered before in the discussion of the Robertson uncertainty relation in Chapter 3.1.

The monotonicity of quantum relative entropy also yields a curious property of quantum entropy called *strong sub-additivity*. Consider a quantum system, ρ , with three subsystems A , B and C . Strong sub-additivity is the claim that

$$H(A|B, C) \leq H(A|B). \quad (8.23)$$

For classical systems, this is immediate because conditioning can only decrease the entropy, but for quantum systems it is less straightforward. To derive it, let Λ be the partial-trace channel over system C . Then it follows from Eq (8.20) that $D(\rho_{ABC}\|\rho_A \otimes \rho_{BC}) \geq D(\Lambda(\rho_{ABC})\|\Lambda(\rho_A \otimes \rho_{BC})) = D(\rho_{AB}\|\rho_A \otimes \rho_B)$. Hence,

$$\begin{aligned} 0 &\leq D(\rho_{ABC}\|\rho_A \otimes \rho_{BC}) - D(\rho_{AB}\|\rho_A \otimes \rho_B) \stackrel{(8.15)}{=} I(A : B, C) - I(A : B) \\ &\stackrel{(8.16)}{=} [H(A, B) - H(B)] - [H(A, B, C) - H(B, C)] \stackrel{(8.13)}{=} H(A|B) - H(A|B, C), \end{aligned} \quad (8.24)$$

which is equivalent to Eq (8.23).

An insightful way to express strong sub-additivity is by using the penultimate line above, namely $H(A, B) - H(B) - H(A, B, C) + H(B, C) \geq 0$, and then purify system ABC with an ancilla system, D . Since any bipartition of a system must have identical spectra in the reduced states (see Exercise B.1.2) it follows that $H(A, B) = H(C, D)$ and $H(A, B, C) = H(D)$. With these substitutions, and in the end re-naming D as A for simplicity, we obtain

$$\begin{aligned} \underbrace{H(A, B)}_{=H(C, D)} - H(B) - \underbrace{H(A, B, C)}_{=H(D)} + H(B, C) &\geq 0 \quad \Leftrightarrow \quad H(C, D) + H(B, C) \geq H(B) + H(D) \\ &\stackrel{\text{Rename}}{\Rightarrow} H(A) + H(B) \leq H(A, C) + H(B, C). \end{aligned} \quad (8.25)$$

Recall now that entanglement between systems A and C can imply a negative conditional entropy, which via the chain rule we write as $H(A) > H(A, C)$. Similarly, entanglement between systems B and C can lead to $H(B) > H(B, C)$. Nevertheless, Eq (8.25) tells us that these two inequalities cannot hold simultaneously. In conclusion, strong sub-additivity is a manifestation of entanglement monogamy, which we discussed in Chapter 4.6. It gives us a quantitative limitation on how much system C can be entangled with both systems A and B simultaneously.

8.3. Data compression

The Shannon entropy and the von Neumann entropy have operational meaning in an information processing task called data compression. More often than not, data sets contain redundancy. For example, even though one letter in this sentence has been deleted, the reader is very likely to fully understand it. In data compression, our goal is to represent a data set with as few symbols as possible while still being able to recover the original data with high probability⁴⁹.

Formally, write the data as a sequence $\bar{x} = x_1 \dots x_n$ in which each element, x_i , is sampled from a random variable X in independent rounds. We therefore write $\bar{x}$ as sampled from the random variable X^n . The data set has $|X|^n$ sequences and their probability distribution is $p_{X^n}(x_1, \dots, x_n) = \prod_{i=1}^n p_X(x_i)$. We aim to compress the data set: from the initial $n \log(|X|)$ bits⁵⁰ into as few bits as possible. Such a scheme can be characterised through the tuple (n, ϵ, L) . Here, n is the size of our sequence, ϵ is the error probability that we are willing to accept when recovering the original data and L is the average size of the compressed data. For instance, we could trivially achieve $(n, 0, |X|^n)$ by doing nothing, but we want to do better. To that end, it is more convenient to instead of L work with the bit-rate of the compressed data. To turn the number of symbols into the number of bits we apply a logarithm and to make it a rate we divide by the number of bits, thereby obtaining

$$R = \frac{1}{n} \log L. \quad (8.26)$$

Our goal is to determine the lowest possible bit-rate for an asymptotically large data set ($n \rightarrow \infty$) for which we can make the error vanish asymptotically ($\epsilon \rightarrow 0^+$) in the size of the data set. The solution to this foundational problem in the field is called *Shannon's source coding theorem* which we present here below.

R22: Shannon's source coding theorem

Let X be a random variable sampled n times in independent rounds. For asymptotically large sequences and asymptotically vanishing error, data compression is achievable if and only if the bit-rate satisfies $R > H(X)$.

We show that $R = H(X) + \epsilon$ is achievable for any $\epsilon > 0$, and provide only intuition for there being no better compression scheme. For the achievability, let us partition the space of sequences into two sets by posing the condition

$$2^{-n(H(X)+\epsilon)} \leq p(\bar{x}) \leq 2^{-n(H(X)-\epsilon)}. \quad (8.27)$$

If the probability of the sequence $\bar{x}$ satisfies this condition, then we say that the sequence belongs to the set T_ϵ . Thus, any given sequence either belongs to T_ϵ or it does not. We are going to name T_ϵ the *typical set* because in the limit of large n almost all the probability over the data set is concentrated in T_ϵ . This feature is built into the definition (8.27). To see that, we equivalently write the condition (8.27) as

$$H(X) - \epsilon \leq -\frac{1}{n} \log p(\bar{x}) \leq H(X) + \epsilon. \quad (8.28)$$

Consider now the random variable $Z = -\frac{1}{n} \log(X^n)$. Since our samples are independent and identically distributed, the expectation value⁵¹ of Z becomes

$$E[Z] = -\frac{1}{n} E[\log(X^n)] = -\frac{1}{n} E\left[\log\left(\prod_{i=1}^n p_X(x_i)\right)\right] = -\frac{1}{n} \sum_{i=1}^n E[\log(p_X(x_i))] = -\sum_x p_X(x) \log(p_X(x)) = H(X). \quad (8.29)$$

Thus, our condition in Eq (8.27) is actually saying that $-\frac{1}{n} \log p(\bar{x})$ is “ ϵ -close” to its expectation value. By the law of large numbers, we can select any $\delta > 0$ and find a sufficiently large n so that

$$\text{Prob}(|Z - H(X)| < \epsilon) \geq 1 - \delta. \quad (8.30)$$

This means that by making sufficiently many samples the probability of Z to be arbitrarily near its expectation value can be made arbitrarily large. To make things simple, let us pick $\delta = \epsilon$. Hence, a randomly sampled $\bar{x}$ of sufficient length belongs to

⁴⁹ Data compression is frequently used on standard computers. The common example is ZIP files. It is also common to use other types of compression, that do not perfectly reconstruct the file. For instance, many softwares shrink the size of an image at the expense of its resolution.

⁵⁰ If we have a data set of m elements we need $\lceil \log(m) \rceil$ bits to represent it. Since we are mainly interested in scaling, we ignore the upward rounding.

⁵¹ The expectation value of a random variable X is defined as $E[X] = \sum_x p_X(x)x$.

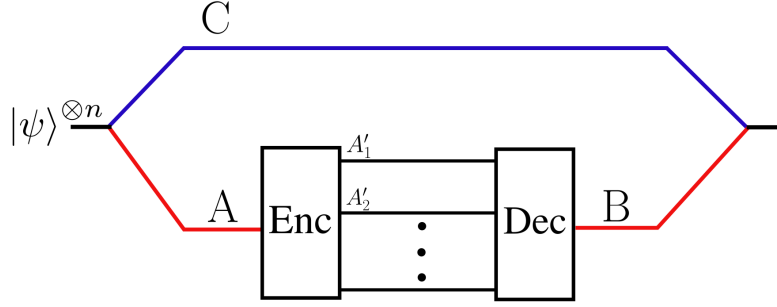


FIG. 8.18. Quantum data compression. The state $\text{tr}_C(\psi^{\otimes n}) = \rho^{\otimes n}$ is sent through an encoding channel. It compresses the system into a smaller one which is then de-compressed to its original size by a decoding channel. The final state must closely approximate the input state.

the typical set with probability at least $1 - \epsilon$. Since almost all the probability is associated with the typical set, it is interesting to know how large share of the data set is occupied by T_ϵ . We can bound it as follows,

$$1 = \sum_{\bar{x}} p(\bar{x}) \geq \sum_{\bar{x} \in T_\epsilon} p(\bar{x}) \stackrel{(8.27)}{\geq} \sum_{\bar{x} \in T_\epsilon} 2^{-n(H(X)+\epsilon)} = 2^{-n(H(X)+\epsilon)} |T_\epsilon| \Rightarrow |T_\epsilon| \leq 2^{n(H(X)+\epsilon)}. \quad (8.31)$$

The ratio between the typical set and the total data set is no more than $2^{n(H(X)+\epsilon)} / 2^{n \log |X|} = 2^{-n(\log |X| - H(X) - \epsilon)}$. Recalling that we want $\epsilon \rightarrow 0^+$, we see that if the Shannon entropy is not maximal, i.e. $H(X) < \log |X|$, the typical set is exponentially smaller than the data set.

The properties of the typical set suggest a simple compression scheme: we associate to every sequence in the typical set a symbol $(1, 2, 3, \dots, |T_\epsilon|)$. From Eq (8.31), we know that we need no more than $L = 2^{n(H(X)+\epsilon)}$ symbols, which via Eq (8.26) corresponds to the bit-rate $R = H(X) + \epsilon$. We can effectively ignore all the sequences that are not in the typical set because their probability vanishes in the limit of large n , thus fulfilling $\epsilon \rightarrow 0^+$. This shows that any bit-rate above $H(X)$ is asymptotically achievable at vanishing error.

There is an intuition for why we cannot reliably compress more than $R = H(X)$. Any further reduction of the bit-rate would mean that we would not assign distinct symbols to some (with n scaling) subset of the typical set. In the relevant limits of n and ϵ , all sequences in the typical set have essentially the same probability distribution, namely the uniform one $p(\bar{x}) \approx 2^{-nH(X)}$ for $\bar{x} \in T_\epsilon$. Therefore, there is no subset of the typical set whose elements are significantly more unlikely than the others, which we could have leveraged to further improve the bit-rate while keeping the error vanishing. $\square$

Naturally, we intend to extend the data compression scheme into the quantum domain. As it turns out, the classical reasoning can largely be recycled and we just need to substitute the classical concepts for the appropriate quantum ones.

8.3.1. Quantum compression

For quantum systems, we replace the n -fold sampling from the probability distribution $p_X(x)$ with n copies of a density operator, namely $\rho^{\otimes n} = \rho_{A_1} \otimes \dots \otimes \rho_{A_n}$, where each d -dimensional sub-system is labelled by $A_1, \dots, A_n$ and where A labels the total system. To compress this quantum data, we apply to A a CPTP map, called Enc, whose output lives on a smaller (compressed) Hilbert space called A' . System A' is then given to the decoder that de-compresses the data by applying another CPTP map, called Dec, whose output is named B . Actually, we can select a purification $|\psi\rangle$ of ρ and consider the encoding and decoding channels as acting on the relevant subsystem. The scheme is illustrated in Fig 8.18. Our goal is that the de-compressed state is essentially identical to the initial state. We quantify their closeness by the trace distance,

$$\left\| \psi_{AC}^{\otimes n} - \left(\text{Dec}^{A' \rightarrow B} \circ \text{Enc}^{A \rightarrow A'} \otimes \mathbb{1}_C \right) [\psi_{AC}^{\otimes n}] \right\|_1 < \epsilon, \quad (8.32)$$

where $\epsilon > 0$ represents the acceptable error. Compressing the quantum data means to simultaneously achieve two objectives. Firstly, we want the de-compressed state to be arbitrarily close to the original state. This means that the error asymptotically vanishes, namely $\epsilon \rightarrow 0^+$ when $n \rightarrow \infty$. Secondly, we want to find the smallest possible compressed Hilbert space, i.e. we want to minimise $d_{A'}$. We can think of $d_{A'}$ as is the quantum counterpart of L from the classical case. In further analogy with the classical case, we do not work directly with $d_{A'}$ but prefer instead to use the qubit-rate: following Eq (8.26) we define as $R = \frac{1}{n} \log d_{A'}$. The main question is therefore to find the smallest qubit-rate R compatible with an asymptotically vanishing error. The solution is provided by *Schumacher's source coding theorem* and it tells us that optimal compression rate is limited by the von Neumann entropy.

R23: Schumacher's source coding theorem

Let ρ be a density operator sampled n times in independent rounds. For asymptotically large sequences and asymptotically vanishing error, quantum data compression is achievable if and only if the qubit-rate satisfies $R > H(\rho)$.

The lesson behind classical data compression was to keep the typical sequences and discard the rest. We use the same idea for the quantum case. To this end, we promote the typical set to the *typical subspace*. We define it in analogy with Eq (8.27). Specifically, we compute the spectrum of $\rho^{\otimes n}$ and if an eigenvalue, λ_i , satisfies

$$2^{-n(H(\rho)+\epsilon)} \leq \lambda_i \leq 2^{-n(H(\rho)-\epsilon)}, \quad (8.33)$$

then we say that the corresponding eigenvector, $|\lambda_i\rangle$, belongs to the typical subspace, which we again denote T_ϵ . Once we have collected all the eigenvectors of $\rho^{\otimes n}$ belonging to the typical subspace, we define a measurement $\{E, \mathbb{1} - E\}$, where

$$E = \sum_{i \in \text{ind}(T_\epsilon)} |\lambda_i\rangle\langle\lambda_i|, \quad (8.34)$$

is the projector onto the typical subspace. Here we have let $\text{ind}(T_\epsilon)$ be the set of indices i for which $|\lambda_i\rangle \in T_\epsilon$. In Exercise B.5.6 it is shown that typical subspace has properties that are strongly analogous to the typical set. Firstly, for sufficiently large n it contains nearly all the support of $\rho^{\otimes n}$ since $\text{tr}(E\rho^{\otimes n}) \geq 1 - \epsilon$ for any $\epsilon > 0$. Secondly, the dimension of the typical subspace is no larger than $2^{n(H(\rho)+\epsilon)}$, making it in n exponentially smaller than the full state space.

The compression can be viewed as a quantum instrument. We first perform the typical measurement defined by Eq (8.34) on system A . In case the outcome corresponds to E (call it “0”) the post-measurement state is confined to the typical subspace. We then perform a unitary, U , that maps the basis vectors of the typical subspace to the computational basis of the compressed Hilbert space. This is defined as $U = \sum_{i \in \text{ind}(T_\epsilon)} |i\rangle\langle\lambda_i|$. Thus, we end up with a state proportional to $UE\rho^{\otimes n}EU^\dagger$. However, if the outcome instead corresponds to $\mathbb{1} - E$ (call it “1”) our best move is to discard the system and instead prepare a state $|\perp\rangle$ that is orthogonal to the typical subspace. This compression procedure is represented by the channel

$$\text{Enc}(\rho^{\otimes n}) = |0\rangle\langle 0| \otimes UE\rho^{\otimes n}EU^\dagger + |1\rangle\langle 1| \otimes \text{tr}((\mathbb{1} - E)\rho^{\otimes n}) |\perp\rangle\langle\perp|, \quad (8.35)$$

where the factor $\text{tr}((\mathbb{1} - E)\rho^{\otimes n})$ accounts for the probability of the outcome “1”. In the subsequent de-compression step, we must reverse this channel up to the allowed error. For this, we first read the outcome register. If it is “0” then we apply $U^\dagger$ to reverse the unitary. If it is “1” then we do nothing. Thus, de-compression returns

$$\text{Dec}(\text{Enc}(\rho^{\otimes n})) = |0\rangle\langle 0| \otimes E\rho^{\otimes n}E + |1\rangle\langle 1| \otimes \text{tr}((\mathbb{1} - E)\rho^{\otimes n}) |\perp\rangle\langle\perp|, \quad (8.36)$$

after which we can ignore the classical register. Next, we need to verify that this state satisfies the quantum compression condition in Eq (8.32). Evaluating the norm gives

$$\begin{aligned} \|\psi_{AC}^{\otimes n} - (\text{Dec} \circ \text{Enc} \otimes \mathbb{1}_C)[\psi_{AC}^{\otimes n}]\|_1 &\stackrel{\text{Triangle ineq}}{\leq} \|\psi_{AC}^{\otimes n} - E_A\psi_{AC}^{\otimes n}E_A\|_1 + \|\text{tr}_A((\mathbb{1} - E_A)\psi_{AC}^{\otimes n})|\perp\rangle\langle\perp|\|_1 \\ &\stackrel{\text{Prop 5 Ch. 2.2}}{=} \underbrace{\|\rho^{\otimes n} - E\rho^{\otimes n}E\|_1}_{\leq 2\sqrt{\epsilon} \text{ via (7.46)}} + \underbrace{\text{tr}((\mathbb{1} - E)\rho^{\otimes n})}_{\leq \epsilon \text{ due to } T_\epsilon} \leq 2\sqrt{\epsilon} + \epsilon. \end{aligned} \quad (8.37)$$

In the last step, we bound the second term using that for the typical subspace it holds that $\text{tr}(E\rho^{\otimes n}) \geq 1 - \epsilon$ for sufficiently large n . This allows us to bound also the first term by using the information-disturbance relation for gentle measurements (see Exercise B.4.8 and Chapter 7.7). The expression on the right-hand-side vanishes in the limit of small ϵ . The dimension of the compressed Hilbert space is therefore $\dim(A') = 2^{n(H(\rho)+\epsilon)}$ (ignoring the single state $|\perp\rangle$) which corresponds to a qubit-rate of $R = H(\rho) + \epsilon$. This shows that any qubit-rate above the von Neumann entropy is achievable for reliable compression. We do not prove that no better scheme exists but it is true for reasons analogous to the classical case⁵². $\square$

The optimality of both the classical and the quantum source coding theorem can be understood as follows: any code that compresses at a (qu)bit-rate lower than $H(X)$ and $H(\rho)$ respectively cannot achieve a vanishing error ($\epsilon \rightarrow 0$) in the limit of

⁵² The proof requires additional tools that go beyond the scope of this book. The interested reader can find a derivation for instance in [Quantum information theory](#) by Wilde.

many rounds ($n \rightarrow \infty$). Together, the two results establish the operational importance of Shannon and von Neumann entropy in information processing. In the quantum case, one can also view it as a way to give the concept of a qubit a clear information theoretical meaning. To a physicist, qubits are two-level quantum systems. To a Shannon theorist, qubits can now be viewed as units for measuring the quantum information content of a quantum source.

Let us conclude with a little perspective. The practical implementation of the quantum source coding theorem necessitates an advanced level of quantum technology. The theorem requires infinitely many copies of the state. One does not need this to meaningfully use it in practice due to good statistical confidence being reachable at large but finitely many copies, but at least a few million is to be expected. All these copies need to be coherently controlled: the typical measurement tells us to collectively measure them in a potentially highly complicated basis and the subsequent unitary tells us to transform them into the computational basis. These multi-million-qubit operations need not only to be realisable but they also need to be implemented at high quality. This implementation is well beyond the state-of-the-art of quantum technology at the time of writing.

B10. A glance beyond

The meaning of negative conditional quantum entropy

We have seen that entanglement may cause a negative conditional von Neumann entropy. What does that actually mean? One way to understand it is through a variation of the data compression task which is called state merging. In state merging, Alice and Bob share asymptotically many copies of a two-qubit state ρ_{AB} and Alice wants to send over her part of the state to Bob with vanishing error. To facilitate this, the parties can freely communicate classically, but sending qubits is expensive so they aim to send as few as possible. Naturally, Alice could just send all her n qubits to Bob but a better protocol is possible in which Alice only sends $nH(A|B)$ qubits. When the conditional entropy is negative, sending $nH(A|B)$ qubits means that no qubits are sent at all (e.g. as in teleportation) and that at the end of the protocol Alice and Bob generate an extra $-nH(A|B)$ copies of the maximally entangled state. In other words, they can merge their states without sending qubits and produce some unconsumed pairs of ϕ^+ in the process. These could be used later as communication resources in other quantum information protocols.

Beyond information theory, conditional quantum entropy also has a meaning in thermodynamics. Any process that erases information must come with a work cost associated with a heating of the environment. A well-known example of this is the Maxwell's demon thought experiment. To compute this work cost, one must know the information held by the observer. The cost is therefore largely determined by the conditional entropy between the system and observer. If the observer's information is classical, the conditional entropy is non-negative, so we must supply work to erase information. However, if the observer's information is quantum, the conditional entropy can be negative. Remarkably, it would allow us to gain work from the erasure process. In other words, we can cool down the environment while erasing information from the system.

8.4. Channel capacity

It is natural that a real-life channel is never perfect, meaning that it has some noise. The noise prevents Bob from reliably receiving the message that Alice sends. What can Alice and Bob do to improve the reliability of their communication? The answer is that they should use the channel many times. To illustrate this, imagine that Alice's message is a bit and that the channel flips its value with a probability $0 \leq p < \frac{1}{2}$. Thus, Bob will get a faulty message with probability p . To mitigate the error, Alice represents her message in three copies, either 000 or 111. She sends each symbol through the channel, which flips each of them with probability p . Once received by Bob, he outputs the most frequent symbol, i.e. the symbol which occurs at least twice. This means that he recovers Alice's message unless at least two bit-flip errors occur. Three bit-flip errors happen with probability p^3 whereas two bit-flip errors can happen in three different ways, each with probability $p^2(1-p)$. Hence, the total probability of error is $p' = 3p^2(1-p) + p^3 = 3p^2 - 2p^3$. This makes the communication more reliable since $p - p' = p(1-p)(\frac{1}{2} - p) \geq 0$. In conclusion, by using the channel three times instead of one time, we have reduced the error probability of the communication.

Inspired by the above example, let us formalise the problem of how to communicate reliably over a noisy classical channel. The general process is illustrated in Fig 8.19. Alice has a message, m , selected from the set $\{1, \dots, M\}$. We say that it is represented by $\log M$ logical bits. Then, Alice encodes m into a *codeword*, x . The codeword is represented as a sequence $\bar{x} = x_1 \dots x_n \in \{0, 1\}^n$ which we call *physical bits* because these are the bits that are sent through the channel in independent and identical uses. The channel is called Λ and the i 'th use of it is represented by a conditional probability distribution $p_{Y_i|X_i}(y_i|x_i)$. The n uses of the channel result in the output sequence $\bar{y} = y_1 \dots y_n$ which is called the *distorted codeword*. Bob must decode the distorted codeword to produce a guess, $\hat{m}$, for Alice's message. To measure the performance of this code, one relevant metric

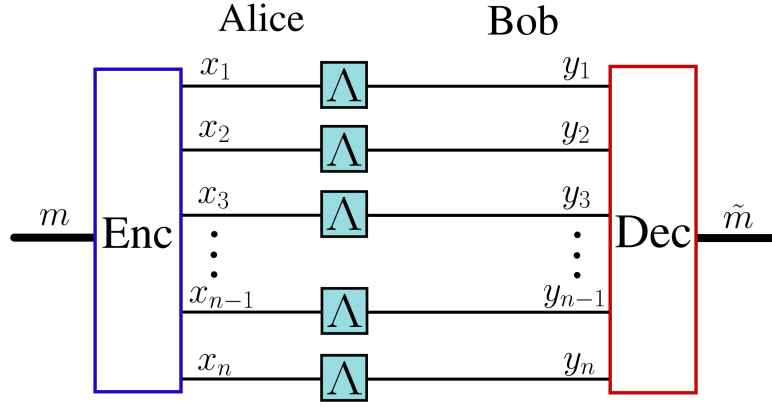


FIG. 8.19. Communication over classical channel. Alice's message, m , is encoded into a codeword, $x_1 \dots x_n$, and each letter in the codeword is sent over the channel Λ . The distorted codeword $y_1 \dots y_n$ is decoded by Bob to form his guess, $\tilde{m}$, for the message m .

is the probability of Bob failing to recover Alice's message. Another one is the transmission rate of the code, R , namely the number of channel uses we require to transmit each logical bit. The error probability and the transmission rate are defined as

$$p_{\text{error}} = \frac{1}{M} \sum_{m=1}^M p(m \neq \tilde{m}|m) \quad \text{and} \quad R = \frac{\log M}{n}. \quad (8.38)$$

We have for simplicity defined the error probability on average, but one can alternatively define it as the worst-case over all the messages⁵³. Just like in data compression, we are going to operate our code in the limit of many uses of the channel ($n \rightarrow \infty$). Our goal is to achieve the highest possible rate that is consistent with an asymptotically vanishing error probability. This supremum rate for reliable communication is called the *channel capacity* and we denote it $C(\Lambda)$.

Before seeing how to compute the channel capacity, let us gather some intuition by considering the above discussed bit-flip channel. Say that we send each letter in the codeword $\bar{x} = x_1 \dots x_n$ through the channel. This n -fold use of the channel could in principle map $\bar{x}$ into any one of the 2^n possible bit-strings. However, from the discussion about typicality, we know that a large share of those error patterns are unlikely to occur. Specifically, we expect that roughly np of the bits to get flipped and typicality told us that there is roughly $2^{nH(p)}$ such bit-strings, where $H(p) = -p \log p - (1-p) \log(1-p)$ is the Shannon entropy. Hence, to every $\bar{x}$ we can associate a set of "likely error patterns". In order for code to work, we need these sets to not overlap: in that way, if we see a distorted codeword, we would know from which $\bar{x}$ it likely originated. This means that we cannot hope to have more than $2^n / 2^{nH(p)} = 2^{n(1-H(p))}$ non-overlapping sets. This puts an upper limit on the capacity of the channel, at $C \leq 1 - H(p)$. Perhaps surprisingly, this turns out to actually be the exact value of the capacity of our bit-flip channel.

The above intuition can be extended to general classical channels. This is a foundational result known as *Shannon's noisy channel coding theorem* and it shows us how to compute the channel capacity.

R24: Shannon's noisy channel coding theorem

The capacity of a classical channel Λ is given by the largest possible mutual information between its input random variable, X , and its output random variable, Y ,

$$C(\Lambda) = \max_{\{p_X(x)\}} I(X : Y). \quad (8.39)$$

We only informally discuss the achievability of any rate below that in Eq (8.39)⁵⁴. The key idea is to assign the codewords in a stochastic way, instead of a perhaps more intuitive deterministic assignment. In other words, before any communication takes place, Alice and Bob randomly agree on a codebook and draw codewords with distribution $\prod_{i=1}^n p_X(x_i)$. This makes the codeword itself a random variable, X . We are going to combine this with typicality to construct our code. In Eq (8.27) we defined the typical set of X as all sequences whose probability is ϵ -close to $2^{-nH(X)}$. Consider now the joint variable (X, Y) .

⁵³ It is relevant to define a worst-case error probability, namely $p_{\text{error}} = \max_m p(m \neq \tilde{m})$. We employ the average error probability because it simplifies the analysis, but the two can be related in such a way that the results obtained for the average error apply also to the worst-case error with small modification.

⁵⁴ For a formal proof, see for example the book [Quantum Information Theory](#) by Wilde

We say that a sequence $\bar{x}\bar{y} = x_1 \dots x_n y_1 \dots y_n$ is *jointly typical* if (i) $\bar{x}$ is typical, (ii) $\bar{y}$ is typical, and (iii) $2^{-n(H(X,Y)+\epsilon)} \leq p(\bar{x}, \bar{y}) \leq 2^{-n(H(X,Y)-\epsilon)}$. Just like before, the law of large numbers tells us that for large enough n we can make the probability concentrated in the jointly typical set arbitrarily large. We are therefore going to assign distinct messages only to the typical codewords. When Bob receives $\bar{y}$, he asks if there is a unique codeword, $\bar{x}$, that is jointly typical with $\bar{y}$. If the answer is positive, Bob learns $\bar{x}$ and uses the agreed codebook to determine m . If the answer is negative, then Bob declares an error. Thus, an error can occur in three ways: either $\bar{y}$ is not typical, or $\bar{y}$ is typical but there is no jointly typical $\bar{x}$, or there is more than one $\bar{x}$ that is jointly typical with $\bar{y}$. Since we know that the probability of sequences outside either the typical set or the jointly typical set vanishes in the limit $n \rightarrow \infty$, we can focus only the latter type of error.

To deal with this, we need a bound on the number of jointly typical sequences. This is obtained in close analogy with Eq (8.31) and it gives that there is at most $2^{n(H(X,Y)+\epsilon)}$ jointly typical sequences. Next, we ask for the probability that a randomly drawn $\bar{x}$ and a randomly drawn $\bar{y}$ are jointly typical. For that, we obtain

$$\sum_{\bar{x}, \bar{y}} \text{Prob}(\bar{x} \text{ and } \bar{y} \text{ jointly typical}) \stackrel{\text{j. typ}}{\leq} \underbrace{2^{n(H(X,Y)+\epsilon)}}_{\geq \# \text{ j. typical}} \times \underbrace{2^{-n(H(X)-\epsilon)}}_{\geq \text{Prob}(\bar{x} \text{ typical seq})} \times \underbrace{2^{-n(H(Y)-\epsilon)}}_{\geq \text{Prob}(\bar{y} \text{ typical seq})} \stackrel{(8.9)}{=} 2^{-n(I(X:Y)-3\epsilon)}, \quad (8.40)$$

where we used that the probability of $\bar{x}$ being typical is no larger than $2^{-n(H(X)-\epsilon)}$ and analogously for $\bar{y}$. From Eq (8.38) it follows that we have 2^{nR} different messages and therefore the same number of codewords. The probability that at least two codewords will be jointly typical with $\bar{y}$ is therefore no larger than $2^{-n(I(X:Y)-R-3\epsilon)}$. Since we want this probability to vanish in the limit of large n , we must select a rate that essentially does not exceed the mutual information. Moreover, note that we are free to select the distribution of the codewords, namely $p_X(x)$, without changing any of the above arguments. Therefore, we ought to select the distribution that maximises the mutual information. This is what appears in Eq (8.39). $\square$

In conclusion, the capacity is the largest mutual information possible between the input and output of the channel. This provides us with an operational interpretation of the mutual information. What is more, the capacity is given by a so-called single-letter formula, i.e. Eq (8.39) involves only one use of the channel, and not the infinitely many uses of it appearing in the coding scheme. This facilitates the computation of the capacity. Moreover, it is worth appreciating that the above proof is not constructive, i.e. it does not tell us how to actually make a specific code that saturates the capacity. Making such codes is far from trivial and they only came along in information theory long after Shannon's original discovery.

8.4.1. Holevo's theorem

Let us now start to bring in quantum systems for the task of transmitting classical information. Consider that Alice samples a random variable, X , with probability distribution $p_X(x)$ and then encodes the classical information in the quantum state ρ_x . Alice's classical variable and her quantum encoding is represented by the ensemble $\mathcal{E} = \{p_X(x), \rho_x\}$. Bob is not given the classical label, x , but only the quantum state, ρ_x . To that, he applies some POVM $\{E_y\}_y$ with outcome y , yielding a probability distribution $p(y|x) = \text{tr}(\rho_x E_y)$. Bob's goal is to maximise the mutual information between Alice's random variable X and the random variable Y representing his outcome. This quantity is called the *accessible information* and we define it as

$$I_{\text{acc}}(\mathcal{E}) = \max_E I(X : Y). \quad (8.41)$$

In general, this is not an easy quantity to compute but *Holevo's theorem* places a useful bound on it.

R25: Holevo's theorem

The accessible information of an ensemble $\mathcal{E} = \{p_X(x), \rho_x\}$ is bounded by

$$I_{\text{acc}}(\mathcal{E}) \leq H(\rho) - \sum_x p_X(x) H(\rho_x), \quad (8.42)$$

where $\rho = \sum_x p_X(x) \rho_x$. The right-hand-side is labelled $\chi(\mathcal{E})$ and is called the *Holevo information*.

To prove this, we first write the classical mutual information as a quantum mutual information. This we can do trivially by associating the probability distribution $p_{X,Y}(x,y) = p(x)p(y|x)$ with the diagonal quantum state $\sum_{x,y} p(x,y)|x\rangle\langle x|_B \otimes |y\rangle\langle y|_C$. To emphasise that we adopt a quantum picture (even though the object is classical), we have named the systems that hold the classical variables X and Y by B and C respectively. Alice's quantum state ρ_x is assigned to system A . Hence, the joint state

of Alice's quantum state and her random variable X can now be expressed as

$$\tau_{AB} = \sum_x p(x) \rho_x \otimes |x\rangle\langle x|_B. \quad (8.43)$$

Bob has access only to system A . He can apply a quantum instrument with classical outcome y and obtain the post-measurement state $\rho_{y|x}$. Representing it as a quantum channel applied to τ_{AB} , the transformed state becomes

$$\sigma_{ABC} \equiv \sum_{x,y} p(x,y) \rho_{y|x} \otimes |x\rangle\langle x|_B \otimes |y\rangle\langle y|_C. \quad (8.44)$$

Our goal is to bound $I(B : C)_\sigma$. To achieve that, we first observe from Eq (8.24) that strong sub-additivity can be expressed (with appropriate re-naming of the systems) as $I(B : C)_\sigma \leq I(B : A, C)_\sigma$. Secondly, recall from the definition of quantum mutual information in Eq (8.15) that $I(B : A, C)_\sigma = D(\sigma_{ABC} \| \sigma_B \otimes \sigma_{AC})$. Since σ_{ABC} was obtained from τ_{AB} through a CPTP map, it follows from the monotonicity of quantum relative entropy R21 that

$$I(B : A, C)_\sigma \stackrel{(8.15)}{=} D(\sigma_{ABC} \| \sigma_B \otimes \sigma_{AC}) \stackrel{(8.20)}{\leq} D(\tau_{AB} \| \tau_A \otimes \tau_B) \stackrel{(8.15)}{=} I(A : B)_\tau. \quad (8.45)$$

Finally, we can compute $I(A : B)_\tau$. To do that from Eq (8.16) we need $H(A)_\tau$, $H(B)_\tau$ and $H(A, B)_\tau$. Since $\tau_B = \sum_x p(x) |x\rangle\langle x|$ we have $H(B) = H(X)$. Since $\tau_A = \sum_x p(x) \rho_x \equiv \rho$, we have $H(A)_\tau = H(\rho)$. Lastly,

$$\begin{aligned} H(A, B)_\tau &\stackrel{(8.43)}{=} -\text{tr} \left(\sum_x p(x) \rho_x \otimes |x\rangle\langle x| \log \left(\sum_z p(z) \rho_z \otimes |z\rangle\langle z| \right) \right) = -\text{tr} \left(\sum_x p(x) \rho_x \otimes |x\rangle\langle x| \sum_z \log(p(z) \rho_z) \otimes |z\rangle\langle z| \right) \\ &= -\text{tr} \left(\sum_x p(x) \rho_x \log(p(x) \rho_x) \right) = \underbrace{-\sum_x p(x) \log(p(x))}_{=H(X)} - \sum_x p(x) \underbrace{\text{tr}(\rho_x \log(\rho_x))}_{=-H(\rho_x)} \\ &= H(X) + \sum_x p(x) H(\rho_x), \end{aligned} \quad (8.46)$$

where in the second equality we have noticed that since $\sum_z p(z) \rho_z \otimes |z\rangle\langle z|$ is block-diagonal the logarithm acts independently on each block. Eq (8.16) lets us put this together as $I(A : B)_\tau = H(A) + H(B) - H(A, B)$, which returns the final result in Eq (8.42). $\square$

An interesting feature of the Holevo information is that it implies $I_{\text{acc}}(\mathcal{E}) \leq H(\rho) \leq \log d$, where d is the Hilbert space dimension of ρ . This means that a d -dimensional quantum ensemble cannot provide more classical information than can an ensemble of classical d -dimensional messages. We can view this as complementary to our result in Chapter 6.1, where it was shown that there is no quantum advantage in state discrimination by upgrading classical messages to quantum states.

8.4.2. Classical capacity of quantum channels

Consider that Alice can prepare any ensemble $\mathcal{E} = \{p(x), \rho_x\}$ but that she now must pass the states through a given quantum channel, Λ , before giving them to Bob. Having received $\Lambda(\rho_x)$, Bob again seeks to implement the POVM that maximises the mutual information with Alice's classical input. Following Holevo's theorem, we bound the accessible information associated with Λ through a quantity called the *Holevo information of Λ* ,

$$\chi(\Lambda) = \max_{\{p(x), \rho_x\}} \chi(\{p(x), \Lambda(\rho_x)\}). \quad (8.47)$$

This quantity is interesting because it is closely related to the quantum version of the noisy channel coding theorem R24: by an appropriate use of this Holevo information we can compute the classical capacity of a quantum channel.

Let us briefly return to the problem of noisy channel coding, but now with quantum instead of classical channels. Like in the classical case, Alice and Bob want to communicate classical information reliably over a given channel Λ . In further analogy, they will use the channel many times in order to surpress the probability of transmission errors. Thus, the scheme in Fig 8.19 still applies, with the substitution that the encoding, decoding and transmission channels now are CPTP maps. For intuition, note that if we fix encoding and decoding maps then the total channel from Alice to Bob is classical and its capacity is therefore given by the noisy channel coding theorem. This suggests that the accessible information of $\Lambda^{\otimes n}$ (in the large n limit) is central for the classical capacity of Λ . A crucial difference between the quantum and classical case is that Alice can associate her message, m , to an arbitrary n -partite quantum state. This "quantum codeword", ρ , can be entangled. The distorted

quantum codeword arriving to Bob, namely $\Lambda^{\otimes n}[\rho]$, can therefore also be entangled. If one forbids Alice from using entangled codewords, i.e. her channel inputs are just product states, then the capacity of the quantum channel becomes simply the Holevo information, $C_{\text{prod}}(\Lambda) = \chi(\Lambda)$. In Exercise B.5.8 we show that the Holevo information of Λ is additive under product codewords and therefore motivates the existence of such a single-letter formula. However, if Alice uses entanglement in her coding, the formula for the quantum channel capacity becomes more complicated. The *Holevo-Schumacher-Westmoreland theorem* shows that the classical capacity of Λ is given by⁵⁵

$$C(\Lambda) = \lim_{n \rightarrow \infty} \frac{1}{n} \chi(\Lambda^{\otimes n}). \quad (8.48)$$

In contrast to the noisy channel coding theorem R24, this capacity formula is not a single-letter statement, i.e. it does not involve only a single use of the channel but infinitely many. That is major drawback because it makes it very hard to evaluate the capacity for a general Λ . A convenient case is if Λ happens to have properties that make its Holevo information additive, i.e. if $\chi(\Gamma \otimes \Lambda) = \chi(\Gamma) + \chi(\Lambda)$ for any CPTP maps Γ and Λ . For such channels the formula in Eq (8.48) simplifies to the single-letter expression $C(\Lambda) = \chi(\Lambda)$. One example of a channel with additive Holevo information is the depolarisation channel encountered in Eq (7.5). Another example is so-called entanglement-breaking channels, which we discuss in Exercise B.5.9. However, in general the Holevo information of a quantum channel is super-additive, i.e. one can find Λ such that

$$\chi(\Lambda \otimes \Lambda) > 2 \chi(\Lambda). \quad (8.49)$$

An example of such a channel is discussed in Exercise B.5.10. In general, super-additivity forbids us from circumventing the difficult computation associated with Eq (8.48).

Chapter summary.

- Entropies are measures of uncertainty. For classical and quantum systems the most important example is Shannon entropy and von Neumann entropy respectively.
- Conditional von Neumann entropy can be negative as a consequence of entanglement.
- The Shannon and von Neumann entropies determine, respectively, the limiting rate at which classical and quantum data can be compressed.
- The capacity of a channel is the highest asymptotic rate at which it can reliably transmit information. The capacity of classical channels is associated with mutual information whereas for quantum channels it is associated with the Holevo information. The latter capacity admits no general single-letter formulation.

Further reading. For a thorough introduction to classical information theory, see for example the book [Elements of information theory](#) by Cover and Thomas. For a more extensive introduction to the quantum Shannon theory, see the notes [Quantum Shannon Theory](#) by Preskill. More in-depth coverage of quantum Shannon theory is found for instance in the books [Quantum information theory](#) by Wilde and [Quantum information theory](#) by Renes.

⁵⁵ For a proof see for example Chapter 8 in the book [The Theory of Quantum Information](#) by Watrous.

Part III**Specialised topics**

9. DEVICE-INDEPENDENT CERTIFICATION

In Chapter 5 we learned that Bell nonlocality constitutes a fundamental signature of how quantum theory breaks with classical theories. In this chapter we discuss how Bell nonlocality can be used as a tool for certifying the functionality of quantum devices while making very weak assumptions about what is going on under their hoods.

9.1. The idea of device-independence

Imagine that we are given a device with a sticker on it that says “ σ_Z measurement”. How can we be sure that the device actually performs a σ_Z measurement? We must somehow certify that it works as promised. Certification is an important routine for quantum devices: the success of most quantum information protocols relies on accurately implementing a given sequence of quantum operations. For instance, quantum teleportation tells Alice and Bob to generate the maximally entangled state, to perform a Bell state measurement and then to implement Pauli unitaries. Only if our devices precisely follow all these instructions are we guaranteed to end up with a successfully teleported state. However, there is a variety of reasons that a device might not operate as we want. One basic reason is that perfect control is an idealisation and therefore any quantum device unavoidably features some imperfections. Another reason is that sometimes our theoretical model for how a device should behave does not accurately match its actual physics. A third reason is that in some scenarios, such as cryptographic tasks, one may legitimately worry that an adversary could maliciously tamper with the quantum devices in order to attack the protocol.

To certify a quantum device, one must make some assumption about how the process works. For instance, if our device is a measurement, we can probe it with a number of different states, examine the outcome statistics and then use it to reconstruct the POVM implemented by the device. In doing this, we have assumed that our states are accurately prepared and that we know the Hilbert space in which the device operates. If either of these fail (for instance a tiny imperfection in the state) our certification could be flawed. It is therefore interesting to find the strongest possible way to certify quantum devices, i.e. to invoke only minimal assumptions. Such schemes are often associated with the concept of *device-independence*.

The core idea of device-independence is to guarantee the validity of a quantum information protocol without making any assumptions about the internal workings of the involved quantum devices. The main assumption needed is the causal independence of the devices. Hence, device-independence adopts a “black-box” picture where we do not need any physical modelling of what our devices are doing, apart from them not secretly exchanging information. The goal is to deduce the success of a protocol solely by inspecting the input-output statistics that these devices produce⁵⁶. In a sense, this is the opposite of what physics is usually about: normally we make a model and deduce its predictions, but in the device-independent picture we start with the observations and deduce the model. It is advisory to take a moment to marvel at this at all being possible.

Bell nonlocality is the essential ingredient for device-independent certification. Let us exemplify what goes wrong when nonlocality is not involved. For that, we return to the device labelled “ σ_Z measurement”. Imagine that we probe this device with some state and that we observe the outcomes “0” and “1” with uniform probability. In the spirit of device-independence, we do not know what the state is, but we nevertheless want to deduce that no other measurement than σ_Z could have been implemented by the measurement device. Consider now the following two possible explanations of our observations. In the first explanation, the device actually measures σ_Z and the incoming state was $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Indeed, this implies a uniform distribution of “0” and “1”. In the second explanation, the state is something else and the POVM implemented by the device is actually $\{\frac{1}{2}, \frac{1}{2}\}$. This too implies a uniform distribution of “0” and “1”. Thus, we cannot exclude the possibility that our measurement device is something totally different from σ_Z ⁵⁷. To make our attempt at device-independent certification work, we must find a way to narrow the possible explanations for our observed statistics to such an extent that there is essentially only one possible explanation left. For that to be possible, we must place our measurement device in a Bell scenario and use it to violate a Bell inequality. As it turns out, large violations of appropriate Bell inequalities are only possible with very particular selections of states and measurements. This is precisely what we need to achieve device-independent certification. In what follows, we show how to use violations of the CHSH inequality to certify properties of both the involved measurements and the shared state.

9.2. Certification of measurements

Consider that Alice and Bob have set up a Bell experiment for a test of the CHSH inequality. However, their end-goal is not to falsify local hidden variable theories (as was the case in Chapter 5). Instead, they are going to assume from the outset that quantum theory governs the physics of their experiment. Thus, they start the experiment assuming that they share some unknown

⁵⁶ Device-independent approaches are not limited to quantum certification problems but apply to a variety of problems in quantum information processing.

⁵⁷ In practice, the quantum state could enter the measurement device and then get thrown into its internal waste bin. Then, inside the device, there is a mouse who is told to choose between two equally appealing types of cheese (“0” and “1”) and whose random preference in each round decides the outcome seen by the experimenter.

quantum state and that their respective devices perform some unknown POVMs. Their task is to violate the CHSH inequality and solely from that observation infer which state they share and which measurements their respective devices implement. For simplicity, let us suppose that Alice and Bob do their CHSH test (see Box R8) and find the largest possible value of the CHSH parameter, namely the Tsirelson bound $\mathcal{S} = 2\sqrt{2}$ (see Box R10). We now show how this observation allows both Alice and Bob to largely determine the POVMs implemented by their devices.

9.2.1. Anticommuting measurements

Recall that in a CHSH test, Alice and Bob each have two binary-outcome measurements described by observables (A_0, A_1) and (B_0, B_1) respectively. Since we assume no limitation on the Hilbert space dimension, we can without loss of generality consider them as projective measurements, because otherwise we can use Naimark dilation (see Box R3) to make them projective. To proceed, it is convenient to define the operator $2\sqrt{2}\mathbb{1} - \mathcal{Z}$, where

$$\mathcal{Z} = A_0 \otimes B_0 + A_0 \otimes B_1 + A_1 \otimes B_0 - A_1 \otimes B_1 \quad (9.1)$$

is the Hermitian operator that describes the effective action of the observables in a quantum model of the CHSH test. Since no greater value than $2\sqrt{2}$ is possible for the CHSH parameter, i.e. $\langle \psi | \mathcal{Z} | \psi \rangle \leq 2\sqrt{2}$ for all bipartite states $|\psi\rangle$, it follows that $2\sqrt{2}\mathbb{1} - \mathcal{Z}$ is positive semidefinite. There exists a useful decomposition for it as a sum of two squared operators. Specifically,

$$2\sqrt{2}\mathbb{1} - \mathcal{Z} = \frac{1}{\sqrt{2}} \left[\left(\frac{A_0 + A_1}{\sqrt{2}} - B_0 \right)^2 + \left(\frac{A_0 - A_1}{\sqrt{2}} - B_1 \right)^2 \right]. \quad (9.2)$$

This decomposition is verified in Exercise C.1.1. Above, we have omitted the symbol $\otimes$ but we remember that Alice and Bob's measurements commute ($A_x B_y = B_y A_x$) since they are on separate Hilbert spaces. The reason we want this decomposition is that each term is necessarily positive semidefinite due to the square. It gives us the following implication: if we observe Tsirelson's bound, namely $\langle \psi | 2\sqrt{2}\mathbb{1} - \mathcal{Z} | \psi \rangle = 0$, the expectation value of each of the two terms must be zero. It implies

$$\left(\frac{A_0 + A_1}{\sqrt{2}} - B_0 \right) |\psi\rangle = 0, \quad \text{and} \quad \left(\frac{A_0 - A_1}{\sqrt{2}} - B_1 \right) |\psi\rangle = 0. \quad (9.3)$$

Consider now the anticommutator of Bob, $\{B_0, B_1\}$, on the support of our state $|\psi\rangle$. It evaluates to

$$\begin{aligned} \{B_0, B_1\} |\psi\rangle &= (B_0 B_1 + B_1 B_0) |\psi\rangle \stackrel{(9.3)}{=} \left(B_0 \frac{A_0 - A_1}{\sqrt{2}} + B_1 \frac{A_0 + A_1}{\sqrt{2}} \right) |\psi\rangle = \left(\frac{A_0 - A_1}{\sqrt{2}} B_0 + \frac{A_0 + A_1}{\sqrt{2}} B_1 \right) |\psi\rangle \\ &\stackrel{(9.3)}{=} \left(\frac{A_0 - A_1}{\sqrt{2}} \frac{A_0 + A_1}{\sqrt{2}} + \frac{A_0 + A_1}{\sqrt{2}} \frac{A_0 - A_1}{\sqrt{2}} \right) |\psi\rangle = (A_0^2 - A_1^2) |\psi\rangle = 0, \end{aligned} \quad (9.4)$$

where in the last step we used that $A^2 = \mathbb{1}$ for any binary-outcome observable associated with a projective measurement; recall Eq (3.5). Thus, Bob's observables anticommute on the support of the state. In an analogous way, one can show that the same holds for Alice, namely $\{A_0, A_1\} |\psi\rangle = 0$.

Recall that the measurements we used in Chapter 5.3 to construct the maximal violation of the CHSH inequality were $\{\sigma_X, \sigma_Z\}$ for Alice and $\{\frac{\sigma_Z + \sigma_X}{\sqrt{2}}, \frac{\sigma_Z - \sigma_X}{\sqrt{2}}\}$ for Bob. Indeed, both pairs are anticommuting. At this point, we may wonder why we are certifying anticommutation instead of the precise observables used by Alice and Bob. The answer is two-fold. Firstly, anti-commutation carries additional information. In Exercise C.1.2 we show that if two observables anticommute then there exists a unitary that maps them to $\sigma_X \otimes \mathbb{1}$ and $\sigma_Z \otimes \mathbb{1}$. Thus, up to the appropriate change of coordinates, we have certified a pair of Pauli operators and a trivial action on some appended Hilbert space. Secondly, the device-independent approach has some inherent limitations. To see that, imagine that Bob were to apply a unitary, U , that rotates his observables from $(\frac{\sigma_Z + \sigma_X}{\sqrt{2}}, \frac{\sigma_Z - \sigma_X}{\sqrt{2}})$ to (σ_X, σ_Z) . They are still anticommuting and we could still saturate Tsirelson's bound simply by making the corresponding rotation also on Bob's share of the maximally entangled state, namely we replace $|\phi^+\rangle$ with $\mathbb{1} \otimes U |\phi^+\rangle$. Hence, from a device-independent standpoint, we cannot distinguish the two realisations. This means that we cannot tell exactly what each measurement is but we can tell their Hilbert space dimension and how they relate to each other.

9.2.2. Quantitative certification

Let us take this further and consider that Alice and Bob test the CHSH inequality but now they observe a potentially non-maximal value of the CHSH parameter $\mathcal{S}$. Our ability to device-independently extract information about the measurements is tied to the

strength of the nonlocality, i.e. the larger is the CHSH parameter the stronger becomes our certification. It is therefore natural that for $\mathcal{S} < 2\sqrt{2}$ we can no longer certify perfect anticommution. A natural question becomes if we can make a quantitative certification about the measurements as a function of $\mathcal{S}$. To that end, we need to select a relevant quantifier. Focusing on Bob's pair of observables, an appealing choice is to follow in the approach of Robertson's uncertainty relation (see Box R2) and consider the extent to which B_0 and B_1 fail to commute on the support of Bob's state. We quantify this by

$$\gamma \equiv \frac{1}{2} \text{tr}(\psi_B [[B_0, B_1]]). \quad (9.5)$$

To bound this quantity, consider the operator $\mathcal{Z}$ in Eq (9.1). Let us compute its square,

$$\begin{aligned} \mathcal{Z}^2 &= (A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1))^2 \\ &= \underbrace{A_0^2 \otimes (B_0 + B_1)^2}_{=1\mathbb{1}} + \underbrace{A_1^2 \otimes (B_0 - B_1)^2}_{=2\mathbb{1} - \{B_0, B_1\}} + \underbrace{A_0 A_1 \otimes (B_0 + B_1)(B_0 - B_1)}_{=-[B_0, B_1]} + \underbrace{A_1 A_0 \otimes (B_0 - B_1)(B_0 + B_1)}_{=[B_0, B_1]} \\ &= 4\mathbb{1} \otimes \mathbb{1} - [A_0, A_1] \otimes [B_0, B_1] \leq 4\mathbb{1} \otimes \mathbb{1} + |[A_0, A_1]| \otimes |[B_0, B_1]| \\ &\leq 4\mathbb{1} \otimes \mathbb{1} + 2\mathbb{1} \otimes |[B_0, B_1]|, \end{aligned} \quad (9.6)$$

where we have used the triangle inequality and finally that $|[A_0, A_1]| \leq 2\mathbb{1}$ for any pair of observables (A_0, A_1) . To see how this is useful for us, we bound the CHSH parameter as

$$\mathcal{S}^2 = \text{tr}(\psi \mathcal{Z})^2 \stackrel{\text{CS ineq}}{\leq} \text{tr}(\psi \mathcal{Z}^2), \quad (9.7)$$

where we have used the operator version of the Cauchy-Schwarz inequality, namely $|\text{tr}(X^\dagger Y)|^2 \leq |\text{tr}(X^\dagger X)| |\text{tr}(Y^\dagger Y)|$ (see Exercise B.4.8 for a proof), with the selection $X = \mathcal{Z}\sqrt{\psi}$ and $Y = \sqrt{\psi}$. Combining this with Eq (9.6) we arrive at

$$\mathcal{S}^2 \leq 4 + 4\gamma \quad \Rightarrow \quad \gamma \geq \frac{\mathcal{S}^2}{4} - 1, \quad (9.8)$$

which bounds how much Bob's observables anticommute on the support of the state in terms of the CHSH parameter. Note that it trivially holds from the definition in Eq (9.5) that $0 \leq \gamma \leq 1$. If $\mathcal{S} = 2\sqrt{2}$, we therefore get an equality, namely $\gamma = 1$. Then, as $\mathcal{S}$ decreases our lower bound on γ also decreases. When we reach the local hidden variable bound of $\mathcal{S} = 2$, the bound becomes $\gamma \geq 0$, which is trivial. This illustrates how Bell nonlocality is a quantitative resource for device-independent certification.

9.3. Certification of states

The natural next step is to ask what a CHSH parameter of $\mathcal{S} = 2\sqrt{2}$ reveals to us about the state. However, before going down that road, let us elaborate on the inherent limitations of device-independent certification.

To see these limitations, say that our device is a source producing the state $|\psi\rangle_{AB}$. Let apply unitary operators $U \otimes V$ to $|\psi\rangle$ and simultaneously change the measurements of Alice and Bob from $\{A_{a|x}\}_{a,x}$ and $\{B_{b|y}\}_{b,y}$ to $\{UA_{a|x}U^\dagger\}_{a,x}$ and $\{VB_{b|y}V^\dagger\}_{b,y}$. The probabilities remain invariant under these transformations because

$$p(a, b|x, y) = \langle \psi | (U^\dagger \otimes V^\dagger) (UA_{a|x}U^\dagger \otimes VB_{b|y}V^\dagger) (U \otimes V) | \psi \rangle = \langle \psi | A_{a|x} \otimes B_{b|y} | \psi \rangle. \quad (9.9)$$

Thus, we cannot device-independently distinguish $|\psi\rangle$ from any state that is reachable from it by local unitaries. What is more, we also cannot distinguish it from a state that has additional degrees of freedom. This is because the uncharacterised measurements might act trivially on the corresponding Hilbert space. For example, consider that our state actually reads $|\psi\rangle_{AB} \otimes |\kappa\rangle_{A'B'}$, where $|\kappa\rangle$ is the extra degree of freedom shared between Alice and Bob. Consider also that the measurements on systems AA' and BB' are $\{A_{a|x} \otimes \mathbb{1}_{A'}\}_{a,x}$ and $\{B_{b|y} \otimes \mathbb{1}_{B'}\}_{b,y}$. This extension of the Hilbert space leaves the probabilities unchanged because

$$p(a, b|x, y) = \langle \psi | \langle \kappa | (A_{a|x} \otimes \mathbb{1}_{A'}) \otimes (B_{b|y} \otimes \mathbb{1}_{B'}) | \psi \rangle_{AB} | \kappa \rangle_{A'B'} = \langle \psi | A_{a|x} \otimes B_{b|y} | \psi \rangle. \quad (9.10)$$

Hence, we cannot hope to certify $|\psi\rangle$ beyond possible local unitaries and extra degrees of freedom. This motivates a notion of equivalence between states that takes local unitaries and additional degrees of freedom into account. This notion is called an *isometry*.

G24: Isometry

An isometry, Φ , from a Hilbert space $\mathcal{H}_1$ to a Hilbert space $\mathcal{H}_2$ is a linear transformation that preserves the inner product. It can be expressed as a unitary operation, U , acting on an enlarged Hilbert space

$$\Phi[X] = U (X \otimes |0\rangle\langle 0|) U^\dagger. \quad (9.11)$$

We now set out to device-independently certify that every state that can achieve $\mathcal{S} = 2\sqrt{2}$ is equivalent to the maximally entangled two-qubit state up to appropriate isometries on Alice and Bob's local spaces. Formally, we want to show that there exists isometries Φ_A and Φ_B such that

$$(\Phi_A \otimes \Phi_B)[\psi_{AB}] = |\phi^+\rangle\langle\phi^+| \otimes |\kappa\rangle\langle\kappa|, \quad (9.12)$$

for some auxiliary state $|\kappa\rangle$. At the moment we do not know how to choose these isometries. To address that, let us first find inspiration by collecting some useful observations.

9.3.1. SWAP unitary

Imagine momentarily that Alice and Bob each have an auxiliary qubit called $|0\rangle_{A'}$ and $|0\rangle_{B'}$ respectively. If Alice and Bob share the maximally entangled state, $|\phi^+\rangle_{AB}$, they can locally swap their systems A and B into the systems A' and B' respectively and be left with the state $|\phi^+\rangle_{A'B'}$. The swap operation corresponds to the unitary

$$\text{SWAP}_{AA'} |j\rangle_A |k\rangle_{A'} = |k\rangle_A |j\rangle_{A'} \quad \Leftrightarrow \quad \text{SWAP}_{AA'} = |00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11|. \quad (9.13)$$

One can decompose the SWAP unitary in terms of another two-qubit unitary called a quantum controlled-NOT (CNOT). The CNOT unitary is defined as

$$\text{CNOT}_{AA'} |j\rangle_A |k\rangle_{A'} = |j\rangle_A |j \oplus k\rangle_{A'} \quad \Leftrightarrow \quad \text{CNOT} = |0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes \sigma_X. \quad (9.14)$$

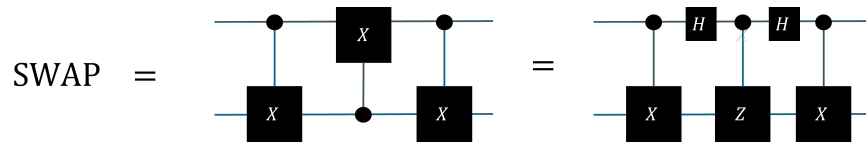
Thus, it leaves the first register untouched while it flips the second register only if the first register is turned on ($j = 1$). Therefore, the first qubit is called the control qubit and the second is called the target qubit. To relate the CNOT unitary to the SWAP unitary, we can decompose the latter as three instances of the former. Specifically,

$$\text{SWAP}_{AA'} = \text{CNOT}_{AA'} \times \text{CNOT}_{A'A} \times \text{CNOT}_{AA'}. \quad (9.15)$$

This can be verified from Eq (9.14), but note that we have interchanged the target and control qubit in the second CNOT. We would prefer to always have the first register as our control qubit. Therefore, we now go a step further and decompose $\text{CNOT}_{A'A}$ in terms of other unitaries. To this end, we can define a controlled phase-flip unitary,

$$\text{CZ}_{AA'} |j, k\rangle = (-1)^{jk} |j\rangle \otimes |k\rangle \quad \Leftrightarrow \quad \text{CZ} = |0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes \sigma_Z. \quad (9.16)$$

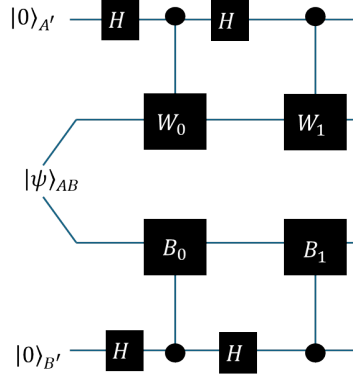
Hence, in analogy with the CNOT, this unitary reads the control qubit and if it is turned on it applies σ_Z to the target qubit. One can now verify that the second CNOT can be decomposed in terms of the CZ unitary as $\text{CNOT}_{A'A} = (H \otimes \mathbb{1}) \otimes \text{CZ}_{AA'} \otimes (H \otimes \mathbb{1})$, where we defined $H = \frac{\sigma_Z + \sigma_X}{\sqrt{2}}$. This decomposition and that in Eq (9.15) are verified in Exercise C.1.4. Putting this together, our SWAP unitary can be illustrated as in the diagram here below



9.3.2. SWAP isometry

We use the SWAP unitary to inspire our construction of the local isometries needed in Eq (9.12) to certify the state. Given that Tsirelson's bound is saturated, we would like Alice and Bob to each have measured (up to a unitary) the pair of observables

σ_Z and σ_X . To be specific, say that we would like Bob to have measured $B_0 = \sigma_Z$ and $B_1 = \sigma_X$ and Alice to have measured $A_0 = \frac{\sigma_Z + \sigma_X}{\sqrt{2}}$ and $A_1 = \frac{\sigma_Z - \sigma_X}{\sqrt{2}}$. However, in the device-independent picture we cannot assume that this is the case because our devices are uncharacterised. Instead, we note only that under ideal conditions we would have $W_0 \equiv \frac{A_0 + A_1}{\sqrt{2}} = \sigma_Z$ and $W_1 \equiv \frac{A_0 - A_1}{\sqrt{2}} = \sigma_X$. Inspired by this, we construct the local isometry illustrated here below⁵⁸.



On each of Alice and Bob's side, notice the similarity with our diagram for the SWAP unitary. On Alice's side, W_0 and W_1 play the role of σ_Z and σ_X , but notice that the first CNOT appearing in the SWAP diagram is missing in our local isometry. This is because we can discard it: as we are initialising the auxiliary states in $|0\rangle$ the control qubit is inactivated, implying that the first CNOT would have acted trivially. Let us now evaluate the above local isometry, $\Phi_{A'A} \otimes \Phi_{B'B}$, on the initially unknown state $|\psi\rangle_{AB}$ and the auxiliaries $|00\rangle_{A'B'}$. A calculation, whose entirety is discussed in Exercise C.1.5, reveals that

$$\begin{aligned}
 (\Phi_{A'A} \otimes \Phi_{B'B})[|\psi\rangle_{AB} \otimes |00\rangle_{A'B'}] &= \frac{1}{4} \left(|00\rangle_{A'B'} \otimes \underbrace{(\mathbb{1} + W_0)(\mathbb{1} + B_0)}_{=G_{00}} |\psi\rangle_{AB} + |01\rangle \otimes \underbrace{(\mathbb{1} + W_0)B_1(\mathbb{1} - B_0)}_{=G_{01}} |\psi\rangle \right. \\
 &\quad \left. + |10\rangle \otimes \underbrace{W_1(\mathbb{1} - W_0)(\mathbb{1} + B_0)}_{=G_{10}} |\psi\rangle + |11\rangle \otimes \underbrace{W_1(\mathbb{1} - W_0)B_1(\mathbb{1} - B_0)}_{=G_{11}} |\psi\rangle \right). \tag{9.17}
 \end{aligned}$$

Now we need to use our knowledge that the state reaches Tsirelson's bound. Before we have shown that it implies the relations in Eq (9.3). Using these, it follows that $G_{01}|\psi\rangle = G_{10}|\psi\rangle = 0$. The same relations also simplify the expression for $G_{11}|\psi\rangle$ to

$$G_{11}|\psi\rangle = \underbrace{W_1(\mathbb{1} - W_0)}_{\text{use anti-commutation}} \underbrace{B_1(\mathbb{1} - B_0)}_{\text{use anti-commutation}} |\psi\rangle = (\mathbb{1} + W_0) \underbrace{W_1(\mathbb{1} + B_0)}_{\text{use (9.3)}} \underbrace{B_1}_{=\mathbb{1}} |\psi\rangle = G_{00}|\psi\rangle. \tag{9.18}$$

Hence, the action of the local isometry has been simplified into

$$(\Phi_{A'A} \otimes \Phi_{B'B})[|\psi\rangle_{AB} \otimes |00\rangle_{A'B'}] = |00\rangle_{A'B'} \otimes G_{00}|\psi\rangle_{AB} + |11\rangle_{A'B'} \otimes G_{00}|\psi\rangle_{AB} = |\phi^+\rangle_{A'B'} \otimes \sqrt{2}G_{00}|\psi\rangle_{AB}. \tag{9.19}$$

Thus, we have extracted a two-qubit maximally entangled state as well as some auxiliary state on the extra degrees of freedom. This is a relation of the type sought after in Eq (9.12) and therefore it completes our device-independent certification.

Chapter summary.

- Device-independent quantum information protocols do not rely on assumptions about the internal workings of the devices. Their success rests on causal independence and input/output statistics that violates Bell inequalities.
- Observing the maximal violation of the CHSH inequality certifies a maximally entangled two-qubit state and local anticommuting measurements. The certification is valid up to local unitaries and extra degrees of freedom.

Further reading. A deeper introduction to device-independence is found in the book [Bell nonlocality](#) by Scarani.

⁵⁸ For simplicity, we overlook the possibility that W_0 and W_1 are not unitary.

10. QUANTUM CRYPTOGRAPHY

Cryptography is the study and practice of secure communication. It is used in scenarios where Alice communicates with Bob and seeks to protect the confidential content of her messages from a potential eavesdropper, referred to as Eve. To do this, Alice takes her message, which in cryptography is called the *plaintext*, and encodes it into an encrypted message called the *ciphertext* which is then sent to Bob over a public channel. A successful use of cryptography means that Bob can decrypt the ciphertext into the plaintext, whereas Eve cannot.

History has witnessed the use of many different cryptosystems and some have been very simple. An example is the Caesar cipher. In this cipher, each letter in the plaintext is shifted k steps down in the alphabet to create the ciphertext. For instance if, $k = 1$, Alice encodes the plaintext “hey” into the ciphertext “ifz”. To decrypt it, Bob shifts each letter in the ciphertext k steps up in the alphabet. To be able to do that Bob must know k , which is the crucial piece of information for encrypting and decrypting the cipher⁵⁹. Such information is in general called the *cipher key*. A famous variation of the Caesar cipher is the Vigenère cipher. In the Vigenère cipher, each letter in the plaintext is given a different shift in the encryption process, i.e. the key is a string $(k_1, \dots, k_n)$ telling us to shift the i ’th letter k_i steps down the alphabet⁶⁰. If the key is private, chosen uniformly at random and is used only once, the Vigenère cipher becomes information-theoretically secure in the sense that every single letter is randomly and independently scrambled. It is then better known as a *one-time pad*.

Formally, we can view the one-time pad as follows. Alice represents her plaintext as a bitstring $\mathbf{m} = (m_1, \dots, m_n) \in \{0, 1\}^n$. The key is a random sequence $\mathbf{k} = (k_1, \dots, k_n) \in \{0, 1\}^n$. Shifting the i ’th bit by k_i steps means that the ciphertext becomes $\mathbf{c} = (c_1, \dots, c_n)$ where $c_i = m_i \oplus k_i$ for $i = 1, \dots, n$. To decrypt $\mathbf{c}$ into the plaintext, Bob needs only to apply the key again, namely $m_i = c_i \oplus k_i$. However, executing the one-time pad presents a challenge because Alice and Bob must both have access to the key while making sure that Eve does not know it. If Alice generates the key and tries to send it to Bob, Eve could intercept it and write it down without Alice and Bob finding out about her intervention. Another option is for Alice and Bob to meet and agree on a key but this is unpractical⁶¹. This highlights a basic problem, namely the secure distribution of the keys needed for cryptographic protocols. This is where the use of quantum resources can make a difference. From Chapter 6.5 we know that quantum information cannot be cloned, i.e. if Eve intercepts a quantum state she cannot copy it without introducing errors. Therefore, it appears promising for Alice to encode the key into a quantum system and transmit that system to Bob, in hope that quantum theory keeps the information safe. This is a central task in the field of quantum cryptography and it is called *quantum key distribution* (QKD). Introducing the basic ideas behind QKD and its security is the focus of this chapter.

B11. A glance beyond

Public-key cryptosystems

The one-time pad is a so-called private-key cryptosystem. It is named in this way because the key needs to be known only by Alice and Bob. In contrast, there are also public-key cryptosystems, in which the key is public information. Specifically, the key consists in two parts: one that is public and one that is private but known only to Bob (no need to distribute it). Alice uses the public key for her encryption while Bob uses his private key for his decryption. Since Eve knows the public key, such a scheme cannot be information-theoretically secure. Instead, the security derives from the supposed computational hardness of certain mathematical problems. A common example is the Rivest-Shamir-Adleman (RSA) protocol. Its security is based on the believed difficulty of factoring large composite numbers into prime numbers. In other words, the security is based on the assumption that Eve has neither the knowledge or the technology needed to solve the factoring problem efficiently. For very large numbers ($\gtrsim 10^{100}$) the best known algorithm for factoring is called the General Number Field Sieve and it scales sub-exponentially but super-polynomially, roughly as $\exp((\log N)^{1/3}(\log \log N)^{2/3})$, where N is the number to be factored. Running this is very time-consuming. However, it is known that if Eve would have access to a fully functioning quantum computer, she could implement a quantum algorithm which can solve the factoring problem in polynomially many steps. This is dramatically faster. It paves the way for quantum attacks on this cryptosystem and it motivates the development of quantum-secure cryptography.

⁵⁹ This cipher is not particularly secure. If Eve does not know k , she can just try out every $k = 0, \dots, 25$ for all the 26 letters in the English alphabet and see when the result forms a plausible word.

⁶⁰ The original Vigenère cipher was developed in 1553 and broken only in 1863.

⁶¹ After the Cuban missile crisis, the governments of USA and the USSR setup a hotline that used the one-time pad for encryption. To distribute the key between Washington DC and Moscow, each country prepared keys that were delivered via their embassy.

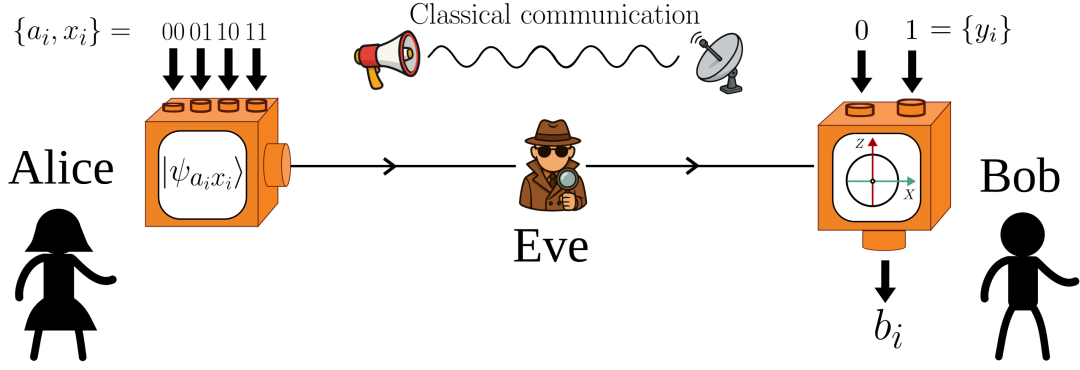


FIG. 10.20. The BB84 QKD protocol. Alice randomly selects two bits (a_i, x_i) and prepares the qubit states $|\psi_{a_i x_i}\rangle$. She sends the state over a quantum channel to Bob, not knowing whether Eve tampered with it on the way. Bob randomly selects a bit y_i and performs a corresponding measurement in one of two complementary bases. To generate their key and check its security, Alice and Bob use a public authenticated classical communication channel to process the data obtained from the quantum preparation and measurement.

10.1. Noiseless BB84 protocol

The promise of QKD is to securely distribute private keys between Alice and Bob. This can be achieved in many different ways but the main idea is captured by the original QKD protocol proposed by Bennett and Brassard in 1984 (BB84)⁶². It is illustrated in Fig 10.20 and below we describe it step by step. To facilitate its presentation, we are going to start out by assuming idealised conditions, namely that Alice and Bob are connected by a noiseless quantum channel, i.e whatever state Alice emits arrives undisturbed to Bob. In the next section we remove these assumptions.

Noiseless BB84 protocol

The BB84 protocol takes place over n rounds. We index the rounds by $i = 1, \dots, n$.

1. **State preparation.** In each round, Alice privately and randomly selects two bits, $(a_i, x_i) \in \{0, 1\}^2$. She then prepares a qubit state which we denote $|\psi_{a_i x_i}\rangle$. We may think of x_i as a basis index: if $x_i = 0$ she prepares eigenstates of σ_Z and if $x_i = 1$ she prepares eigenstates of σ_X . The bit a_i determines which of the two eigenstates to prepare in each case. Hence, her four possible states are

$$|\psi_{00}\rangle = |0\rangle, \quad |\psi_{10}\rangle = |1\rangle, \quad |\psi_{01}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |\psi_{11}\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \quad (10.1)$$

She sends $|\psi_{a_i x_i}\rangle$ over the channel with Bob as its intended destination.

2. **Measurement.** In every round, Bob randomly selects a bit $y_i \in \{0, 1\}$. He selects the observable σ_Z if $y_i = 0$ and σ_X if $y_i = 1$. This measurement is applied to the incoming state and the outcome is denoted $b_i \in \{0, 1\}$.
3. **Sifting.** When all rounds are finished, Alice and Bob publicly announce their respective basis choices, i.e the bit-strings $x = x_1 \dots x_n$ and $y = y_1 \dots y_n$. They keep the data from all rounds in which the bases match, whereas all rounds where the bases do not match are discarded. The set of kept rounds is labelled $R = \{i | x_i = y_i\}$.
- 4* **Testing.** The set R is partitioned into two sets which we label S and K . S is meant for testing the security and K is meant for generating the key. To test for security, Alice and Bob both announce the values of (a_i, b_i) for all $i \in S$. Since these two bits should be identical, we quantify the amount of error by the relative frequency of a mismatch between a_i and b_i ,

$$\delta = \frac{\#\text{events with } a_i \neq b_i}{|S|}. \quad (10.2)$$

If Alice and Bob find that $\delta > 0$, they decide to abort the protocol. However, this does not happen because we have assumed that the channel is noiseless and we have post-selected on rounds where Alice and Bob code in the same basis.

⁶² The protocol was conceived in 1984 but its security was proven only around the turn of the millennium.

In other words, Bob always measures in the eigenbasis of Alice's state and therefore we are guaranteed that $a_i = b_i$ for every $i \in S$, implying $\delta = 0$. If Alice and Bob indeed find $\delta = 0$, they keep the bit-strings associated with the set K , namely $\mathbf{a} = a_{K_1} a_{K_2} \dots$ and $\mathbf{b} = b_{K_1} b_{K_2} \dots$, and conclude that they are identical, $\mathbf{a} = \mathbf{b}$.

6 Privacy amplification.⁶³ Alice and Bob have now ensured that they share identical bit-strings. However, they still need to guarantee that these bit-strings are private, i.e. they are not known to Eve. Let us suppose that Eve has somehow obtained partial knowledge about the secret. For instance, maybe she has found out about some but not all of the bits in the strings $\mathbf{a} = \mathbf{b}$. Alice and Bob's goal is therefore to remove any partial knowledge Eve might have. This is achieved by an entirely classical procedure called privacy amplification. It uses classical randomness and public communication to process $\mathbf{a}$ and $\mathbf{b}$ into a new pair of identical bit-strings about which Eve will have no knowledge. The price for this is that the new strings are shorter, i.e. some of the shared bits of Alice and Bob are consumed in the process⁶⁴.

The central feature of the BB84 protocol is that Alice and Bob's use of superposition states allows them to detect the potential presence of Eve. That is the main overall contribution of QKD to cryptography: even if Eve would have unlimited quantum technological capabilities, any attack that gives her information about the key can be discovered by Alice and Bob. The reason that this is possible stems from the concept of information-disturbance trade-offs discussed in Chapter 7.7. Specifically, any quantum instrument Eve might use to extract information from Alice's states must induce an irreversible disturbance in the post-measurement states (recall the result R20). Therefore, the state that in the end arrives to Bob will not be the same state that Alice emitted. Consequently, if Bob measures in the same basis as Alice's preparation it will pass **Sifting**, but it will nevertheless have a non-zero probability of causing an error ($a_i \neq b_i$) in **Testing**. Since the protocol is repeated in many rounds, it will mean that $\delta > 0$ whereas we had expected $\delta = 0$. In conclusion, Eve has been detected.

We illustrate this through an explicit example of an intercept-resend attack. Eve intercepts the state $|\psi_{a_i x_i}\rangle$ on its way from Alice to Bob. She chooses randomly to measure either σ_Z or σ_X . The post-measurement state is then relayed to Bob, with the intention of being passed off as Alice's original preparation. This can play out in the following two ways.

1. Eve gets lucky and happens to measure in the eigenbasis of Alice's state $|\psi_{a_i x_i}\rangle$. Therefore, she learns a_i deterministically and leaves the state undisturbed. She then relays it to Bob as if nothing had happened.
2. Eve gets unlucky and happens to measure in the basis complementary to Alice's state $|\psi_{a_i x_i}\rangle$. Now, her outcome will be uniformly random. She could still get lucky and get the right result a_i , but in half the cases she will not. Moreover, regardless of the outcome, the post-measurement state passed to Bob will be an eigenstate of Eve's measurement basis, which is the opposite of that used by Alice. If this round survives the **Sifting** step, there is only a 50% chance that Bob's outcome matches Alice's eigenstate choice, i.e. that $a_i = b_i$.

Thus, we see that if Eve happens to choose the wrong basis, there is a 50% chance of the sifted round producing $a_i \neq b_i$ and thereby contributing to the error parameter in Eq (10.2). Since Eve ends up choosing the wrong basis in half the cases, Alice and Bob observe a total error probability of $\delta = 25\%$. In **Testing** this makes them conclude that the protocol is not secure and therefore they abort it. We note that this is not the best possible attack for Eve, but it illustrates how Eve's actions end up leaving a fingerprint that Alice and Bob can detect.

10.2. Noisy BB84 protocol

The assumption that in Eve's absence Alice and Bob do not observe errors ($a_i \neq b_i$) does not hold up in practice. Alice and Bob are likely going to be separated by hundreds or thousands of kilometers. The qubits that are used for quantum communication are usually carried by photons, because they can easily move over large distances. However, single photons can easily be lost. If Alice emits the photons in free space, other photons in the environment will blur the signal. If she emits them in an optical fiber, a large share of them will be lost every few dozen kilometers⁶⁵. Hence, even if Eve is absent, Alice and Bob are going to observe some errors, meaning that $\delta > 0$ in Eq (10.2). This presents us with a challenge: are the observed errors caused by harmless technological imperfections or are they caused by Eve attacking the communication? In general, we cannot know. However, in cryptography we are naturally paranoid and therefore we must take a worst-case approach in which any error observed is assumed to be due to Eve's activity. The central question becomes to what extent we can operate QKD under these conditions.

⁶³ The reader might notice that the previous step was called 4* instead of 4 and that we skipped step 5 entirely. The reason for this will become clear soon.

⁶⁴ While we do not discuss privacy amplification, let us provide a minimal intuition for why it is at all possible. Let Alice and Bob individually compute $a_{\text{end}} = \bigoplus_j a_{K_j}$ and $b_{\text{end}} = \bigoplus_j b_{K_j}$. If Eve knew some but not all of the bits in $\mathbf{a}$ and $\mathbf{b}$, she would have no idea of a_{end} and b_{end} because her known bits have been added to bits unknown to her. Thus, Alice and Bob now share a private bit. However, this is not a smart privacy amplification strategy, because it is very costly: Alice and Bob consumed all but one of their bits in the process.

⁶⁵ In a commercial optical fiber operating at telecom wavelength, roughly 50% of the photons are lost for every 15 kilometers.

Were we to follow the noiseless BB84 protocol, we would now always observe $\delta > 0$ and therefore abort the protocol in **Testing**. Clearly, this means that our protocol does not generate any key. To overcome this, we must hope to find a new threshold, $\delta_{\text{crit}} > 0$, below which we can still guarantee that protocol can operate securely and proceed to generate a key. For that, we need to upgrade the **Testing** step.

- 4 **Testing.** The set R is partitioned into two sets which we label S and K . S is meant for testing the security and K is meant for generating the key. To test for security, Alice and Bob both announce the values of (a_i, b_i) for all $i \in S$. Since these two bits should be identical, we quantify the amount of error by the relative frequency of a mismatch between a_i and b_i ,

$$\delta = \frac{\#\text{events with } a_i \neq b_i}{|S|}. \quad (10.3)$$

If Alice and Bob find that $\delta \geq \delta_{\text{crit}}$, they abort the protocol. If they find $\delta < \delta_{\text{crit}}$, they keep the bit-strings associated with the set K , namely $\mathbf{a} = a_{K_1} a_{K_2} \dots$ and $\mathbf{b} = b_{K_1} b_{K_2} \dots$.

Unlike in the noiseless case, if we now pass the **Testing** step the two bit-strings $\mathbf{a}$ and $\mathbf{b}$ are not identical. This is because a value of δ in the range $0 < \delta \leq \delta_{\text{crit}}$ means that some errors are present. This prevents us from applying **Privacy amplification** directly, because for that we need $\mathbf{a}$ and $\mathbf{b}$ to be identical. In view of this, we must first perform an intermediate step to reconcile the two bit-strings.

- 5 **Information reconciliation.** This is a classical procedure conducted over the public channel. Its purpose is to update Bob's bit-string so that it becomes identical to Alice's bit-string while leaking as little information as possible to Eve in the process⁶⁶.

Once this is completed, Alice and Bob can proceed to **Privacy amplification** and remove Eve's information about their shared bit-strings. In information reconciliation, the more similar $\mathbf{a}$ is to $\mathbf{b}$, the less information needs to leak to Eve in order to update the two bit-strings so they become identical. Therefore, it is important that Alice and Bob perform the quantum part of the BB84 protocol with as little noise as possible. If Eve's information about $\mathbf{a}$ or $\mathbf{b}$ is too large then **Information reconciliation** and **Privacy amplification** will fail in producing a non-zero sized secure key for Alice and Bob. A simple condition for Alice and Bob being able to extract a key is that their mutual information exceeds the mutual information with Eve, namely

$$I(\mathbf{a} : \mathbf{b}) > I(\mathbf{a} : \mathbf{e}). \quad (10.4)$$

where $\mathbf{e}$ is the bit-string of Eve⁶⁷. The central question now is how much imperfection can be tolerated in the BB84 protocol before Alice and Bob no longer can trust it to be secure. In other words, we want to find the value of δ_{crit} .

10.3. Eavesdropping attacks

The critical error rate δ_{crit} depends on the type of attacks that we consider for Eve. In general, attacks in QKD are classified into three different categories (see Box B12). Here, we focus on the simplest class known as *individual attacks*. Individual attacks are named in this way because Eve acts on each round of the protocol individually, i.e. she cannot store copies over the rounds to then measure them collectively. Hence, she must first intercept Alice's state and then apply a quantum channel to it. The output of this channel consists of two systems: one that is relayed to Bob (A) and one that Eve keeps for herself (E). Eve waits until Alice and Bob have proceeded through the classical parts of the BB84 protocol, including in particular their announcements of the bases in **Sifting**, before she measures her system with the aim of learning the key. Let us express Eve's action as first appending a qubit ancilla, $|0\rangle_E$, and then applying a unitary U jointly to $|\psi_{ax}\rangle_A \otimes |0\rangle_E$. Consider for instance a unitary such that

$$\begin{aligned} U(|0\rangle_A \otimes |0\rangle_E) &= |0\rangle_A \otimes \underbrace{(\sqrt{1-\delta}|0\rangle_E + \sqrt{\delta}|1\rangle_E)}_{=|e_0\rangle} = |0\rangle_A \otimes |e_0\rangle_E \\ U(|1\rangle_A \otimes |0\rangle_E) &= |1\rangle_A \otimes \underbrace{(\sqrt{1-\delta}|0\rangle_E - \sqrt{\delta}|1\rangle_E)}_{=|e_1\rangle} = |1\rangle_A \otimes |e_1\rangle_E. \end{aligned} \quad (10.5)$$

⁶⁶ For a discussion of information reconciliation, the interested reader may consider for instance the book [Introduction to Quantum Cryptography](#) by Vidick and Wehner.

⁶⁷ In slight abuse of notation, we have written $I(\mathbf{a} : \mathbf{b})$ where $\mathbf{a}$ and $\mathbf{b}$ now are the random variables representing Alice and Bob's outcome strings.

The role of the parameter δ is to tune how strongly Eve interacts Alice's incoming state with her ancilla. At $\delta = 0$, there is no interaction, so the channel's output is just a product state $|\psi_{ax}\rangle_A \otimes |0\rangle_E$. This means that Bob receives Alice's state unperturbed but also that Eve does not gain information since she holds $|0\rangle_E$ regardless of (a, x) . Let us now consider what happens in our example when Eve begins to increase δ .

For convenience, let us first define $\cos \frac{\theta}{2} = \sqrt{1 - \delta}$ and $\sin \frac{\theta}{2} = \sqrt{\delta}$ so that we can write $\langle e_0 | e_1 \rangle = 1 - 2\delta = \cos \theta$. Let us start by considering the eigenstates of σ_Z , namely $|\psi_{a0}\rangle$ for $a \in \{0, 1\}$. If we pass them through Eve's channel, the state of Bob is obtained by tracing out system E in Eq (10.5) and that leaves us with the state $|a\rangle_A$. This state is precisely that prepared by Alice. Hence, we see that Eve has induced no disturbance. Next, consider the state held by Eve. For that, we keep only system E in Eq (10.5), which is the state $|e_a\rangle_E$. After **Sifting**, Eve is informed that Alice prepared eigenstates of σ_Z and she can therefore try to read out the value of a from her state. However, note that $|e_0\rangle_E$ and $|e_1\rangle_E$ are not orthogonal, thereby preventing Eve to deterministically learn a . Instead, she can apply the quantum state discrimination protocol discussed in Chapter 3.3 to the pair of states $|e_0\rangle_E$ and $|e_1\rangle_E$. From Eq (3.15), we know that the largest possible probability of Eve to learn a is

$$p_{\text{win}} = \frac{1}{2} \left(1 + \sqrt{1 - |\langle e_0 | e_1 \rangle|^2} \right) = \frac{1}{2} \left(1 + \sqrt{1 - \cos^2 \theta} \right) = \frac{1}{2} (1 + \sin \theta). \quad (10.6)$$

We see that her probability of learning a beyond randomly guessing is determined by $\frac{\sin \theta}{2} = \sqrt{\delta(1 - \delta)}$, i.e. as we increase δ above $\delta = 0$ Eve can better learn the secret. So far, Eve's attack is successful because she manages to extract information while not disturbing the state relayed to Bob, but that is why the BB84 protocol also has a second basis. Consider the state relayed Bob when Alice prepares eigenstates of σ_X , namely $|\psi_{a1}\rangle$ for $a \in \{0, 1\}$. This state is

$$\begin{aligned} \text{tr}_E \left(U (|\psi_{a1}\rangle \otimes |0\rangle \langle 0|) U^\dagger \right) &\stackrel{(10.5)}{=} \frac{1}{2} \text{tr}_E \left(|0\rangle \langle 0| \otimes |e_0\rangle \langle e_0| + |1\rangle \langle 1| \otimes |e_1\rangle \langle e_1| + (-1)^a (|0\rangle \langle 1| \otimes |e_0\rangle \langle e_1| + |1\rangle \langle 0| \otimes |e_1\rangle \langle e_0|) \right) \\ &= \frac{1}{2} \left(|0\rangle \langle 0| + |1\rangle \langle 1| + (-1)^a (\langle e_1 | e_0 \rangle |0\rangle \langle 1| + \langle e_0 | e_1 \rangle |1\rangle \langle 0|) \right) = \frac{1}{2} (\mathbb{1} + (-1)^a \cos \theta \sigma_X). \end{aligned} \quad (10.7)$$

From the right-hand-side, we see that it is a noisy version of the eigenstate of σ_X , because the Bloch vector keeps its orientation but its length shrinks as $\cos \theta = 1 - 2\delta$. This means that Eve has disturbed Bob's state and therefore Alice and Bob are going to see errors in **Testing** with a probability of $\frac{1}{2} (1 - \cos \theta) = \delta$. Hence, Eve's actions have caused an information-disturbance relation. For small δ , the mutual information between Alice and Bob will still exceed that between Alice and Eve, allowing the honest parties to extract a non-zero amount of key via the condition (10.4). At a critical value of δ , labelled δ_{crit} , Eve's information becomes too large to permit a secure key and the protocol can no longer be used.

The question remains how Eve best designs her individual attack. It turns out that the strongest individual attack is closely related to the concept of a quantum cloning machine discussed in Chapter 6.5. The cloning attack we consider consists in selecting Eve's channel so that it implements a transformation that takes $|\psi_{ax}\rangle$ to a two-qubit state whose reductions are identical and equal to $\rho_{ax} = \frac{1}{2} (\mathbb{1} + \eta \hat{n}_{ax} \cdot \vec{\sigma})$, where $\hat{n}_{ax}$ is the Bloch vector of $|\psi_{ax}\rangle$ and $\eta \in [0, 1]$ is the shrinking factor made necessary by the quantum no-cloning theorem. The fidelity between each clone and the original state is $F(\psi, \rho) = \frac{1+\eta}{2}$. One might be tempted to use the universal quantum cloning machine from Chapter 6.5, which achieved $F = \frac{5}{6}$ and $\eta = \frac{2}{3}$. However, we can do better by exploiting that the four BB84 states in Eq (10.1) are confined to the equator of the Bloch sphere. Therefore, we do not need the ability to clone arbitrary qubit states but only the qubit states in the XZ -plane. In Exercise B.3.9 the best cloning machine of this restricted type is discussed and it is shown that it achieves a fidelity per copy of $F = \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}} \right)$, which corresponds to $\eta = \frac{1}{\sqrt{2}}$. By construction, Bob and Eve hold identical states in every round. Consequently, this attack represents the critical point at which the two mutual informations in Eq (10.4) are equal, i.e. it is the threshold at which we no longer can extract a key. By operating this cloning machine, the error introduced by Eve in **Testing** corresponds to

$$\delta_{\text{crit}} = p(a \neq b | \text{same basis}) = \frac{1 - \eta}{2} = \frac{1}{2} \left(1 - \frac{1}{\sqrt{2}} \right) \approx 14.6\%, \quad (10.8)$$

In conclusion, if the error exceeds this value, we cannot guarantee the security of the key, but if the error is smaller we can be sure that Eve did not perform the cloning attack. Importantly, since the cloning attack turns out to be the strongest individual attack on the BB84 protocol, we can also conclude that $\delta < 14.6\%$ is sufficient to extract a non-zero amount of key against any individual attack Eve might have performed⁶⁸.

⁶⁸ More general attacks that are not individual over the rounds reduce the error threshold from 14.6% to around 11%.

B12. A glance beyond

Attacking QKD protocols

When Alice emits her preparation and Eve intercepts it, she can apply to it an arbitrary quantum channel. We can view this as Eve appending an ancilla $|0\rangle_E$ and then applying a joint unitary, U_{AE} , yielding the state $U(|\psi_{ax}\rangle \otimes |0\rangle_E)$. Eve can then keep her system and relay Alice's system to Bob. Depending on what capabilities Eve has in this process, one traditionally speaks of three different categories of attacks in QKD.

Individual attacks: Eve attacks every round of the protocol in the same way. This means that she appends $|0\rangle$, performs U and then keeps only her subsystem, which we can write as $\rho_{ax} = \text{tr}_A(U(|\psi_{ax}\rangle \otimes |0\rangle_E)U^\dagger)$. She then measures her system with some POVM $\{E_e\}$. This means that Eve's distribution over the rounds factorises as $p(e_1, \dots, e_N) = p(e_1) \dots p(e_N)$ where $p(e_i) = \text{tr}(\rho_{ax} E_{e_i})$. Our example of the intercept-resend attack and the cloning attack are both instances of individual attacks.

Collective attacks: In every round $i = 1, \dots, N$, Eve appends $|0\rangle$, applies U_{AB} and keeps $\rho_{ax}^{(i)}$. However, in the end she takes the total state $\rho_{ax}^{(1)} \otimes \dots \otimes \rho_{ax}^{(N)}$ and measures it collectively with a POVM $\{E_e\}$, yielding the distribution $p(e_1, \dots, e_N) = \text{tr}((\rho_{a_1 x_1} \otimes \dots \otimes \rho_{a_N x_N}) E_e)$. Implementing such a POVM is in general more powerful than in the case of individual attacks.

Coherent attacks: These are the most powerful quantum attacks. Eve intercepts Alice's states for all the N rounds and appends an ancilla state to the total state. Then, she applies a unitary U on the whole state and keeps her part, namely $\text{tr}_{A_1 \dots A_N}(U((\rho_{a_1 x_1} \otimes \dots \otimes \rho_{a_N x_N}) \otimes |0\rangle\langle 0|_E)U^\dagger)$, and relays the rest to Bob. On this state she performs a collective measurement to extract information about the key.

10.4. Entanglement-based protocols

The BB84 protocol is based on transmitting quantum systems over the distance that separates Alice and Bob's labs; see Fig 10.20. However, an alternative is to achieve QKD by instead distributing entanglement between Alice and Bob. In this way, one may substitute quantum communication with shared entanglement as the main quantum resource for QKD protocols.

To exemplify how this is done, consider that Alice and Bob share the maximally entangled two-qubit state ϕ^+ . We can then interpret the four BB84 states (10.1) as being remotely prepared by Alice for Bob. To do that, Alice needs to measure either σ_Z or σ_X . The choice of observable corresponds to be bit x_i in the BB84 protocol. Similarly, the random outcome of the measurement corresponds to the bit a_i chosen by Alice in the BB84 protocol. Once Alice has selected her measurement and recorded the outcome, Bob's system is in one of the four BB84 states. Therefore, Bob can proceed just as in BB84, namely to measure either σ_Z or σ_X , knowing that if his basis choice matches that of Alice then their outcomes will be identical.

However, in practice we cannot assume that the shared state actually is ϕ^+ . The source of entanglement is favourably placed somewhere between the labs of Alice and Bob. This means not only that the qubits will be exposed to noise and loss during the distribution but also that the source could be controlled by Eve, i.e. she could choose the state that is emitted. The latter is the reason that we need two measurement bases in the protocol. If we had only one basis, say σ_Z , Alice and Bob's outcomes could be identical and locally random but also known to Eve. To achieve that, Eve only has to toss an unbiased coin and distribute the state $|00\rangle$ or $|11\rangle$ respectively. Since Alice and Bob are unaware of the coin, their local outcomes appear random whereas to Eve they are deterministic. However, when two bases are employed in the protocol we know from our discussion about entanglement witnesses in Chapter 4.5 that no separable state can achieve perfect correlations in both bases, namely $\langle \sigma_Z \otimes \sigma_Z \rangle = \langle \sigma_X \otimes \sigma_X \rangle = 1$. In fact, the only two-qubit state that does the job is ϕ^+ because one can verify that $|\phi^+\rangle$ is the unique eigenstate of $\sigma_Z \otimes \sigma_Z + \sigma_X \otimes \sigma_X$ associated with its maximal eigenvalue 2. Thus, if we observe these correlations, we can be sure that the state is maximally entangled.

In the BB84 protocol, the security was fundamentally guaranteed by the fact that Eve had to induce a disturbance in Alice's states in order to extract information. To achieve a mutual information with the key large enough to exceed that of Alice and Bob (see Eq (10.4)), she had to introduce a disturbance above a critical threshold. In the entanglement-based version of the BB84 protocol, the security is guaranteed by the monogamy of entanglement. Specifically, the source can be thought of as a three-partite state ρ_{ABE} where Eve keeps system E . Ideally, Eve would like Alice and Bob to observe $\langle \sigma_Z \otimes \sigma_Z \rangle = \langle \sigma_X \otimes \sigma_X \rangle = 1$ while she remains maximally entangled with for instance Alice. After **Sifting**, where the bases are announced, this would allow her to learn the secret bit without Alice and Bob detecting her presence. However, we know from Chapter 4.6 that entanglement monogamy forbids a three-qubit state where Alice is maximally entangled with both Bob and Eve. Instead, as system E becomes more strongly entangled with system A , the reduced state ρ_{AB} becomes more weakly entangled. That causes the expectation values $\langle \sigma_Z \otimes \sigma_Z \rangle$ and $\langle \sigma_X \otimes \sigma_X \rangle$ to decrease away from unity, which Alice and Bob detect in **Testing** through a higher error

rate between Alice and Bob's outcomes in the sifted bit-strings. By the same token as in standard BB84, a too large error rate in **Testing** prevents Alice and Bob from excluding the possibility of an attack and therefore they abort the protocol, whereas a not-too-large error rate still allows them to extract a non-zero amount of secure key.

Let us conclude by highlighting some noteworthy differences between the BB84 QKD protocol and its entanglement-based version. Firstly, in the former we must trust the source, namely Alice's state preparations, whereas in the latter the source is assumed to be under Eve's control. In contrast, the entanglement-based version requires Alice to perform trusted measurements, which are not present in the original BB84 protocol. Secondly, by avoiding the need for entanglement distribution, the BB84 protocol is technologically more simple and therefore easier to deploy, but its entanglement-based version is amenable to quantum network technology and has a pathway to device-independence (see Box B13). Thirdly, security proofs for QKD are usually based on re-formulating protocols based on sending quantum systems to their entanglement-based counterparts.

B13. A glance beyond

Device-independent QKD

The success of the BB84 protocol, whether implemented by sending qubits or by entanglement distribution, needs the assumption that Alice and Bob's quantum devices work according to the protocol's instructions. What happens if these devices feature imperfections, or worse if the internal working of the devices has been influenced by Eve? Such concerns can be overcome through the concept of device-independence encountered in Chapter 9. By leveraging quantum nonlocality, it is possible to perform QKD device-independently. In other words, by achieving an appropriate Bell inequality violation, the security of the key can be guaranteed without having to characterise the quantum operations implemented by the devices. This represents a gold standard for the security of QKD, mainly requiring Alice and Bob to have trusted random number generators (to be used as inputs in a Bell test) and to make sure that unwanted information (such as the key or the input to the Bell test) does not leak from their labs.

To understand the principle that propels device-independent QKD, consider that Alice and Bob observe a maximal quantum violation of the CHSH inequality, namely $S = 2\sqrt{2}$. From our discussion about quantum certification in Chapter 9, we know that this implies that the state is two-qubit maximally entangled up to local isometries and that the measurements are anticommuting on the support of the state. Hence, in addition to certifying the source, Alice and Bob have certified the functionality of their devices to be essentially those assumed in the entanglement-based version of the BB84 protocol. Therefore, by measuring the shared ϕ^+ state, Alice and Bob can generate identical outputs whose security is guaranteed from the fact that Eve cannot entangle herself with their pure state. While this illustrates the idea of device-independent QKD, any real-life implementation will find $S < 2\sqrt{2}$ due to noisy environment, lossy transmission and the potential presence of Eve. It is more challenging to determine the amount of secure key that can be generated as a function of S but useful bounds are known. For instance, a non-zero key can be extracted whenever $S \gtrsim 2.42$, which is associated with an error rate in the sifted bit-strings corresponding to $\delta_{\text{crit}} \approx 7.1\%$.

Chapter summary.

- Distributing cryptographic keys via classical technology is vulnerable to eavesdropping. By encoding the key into quantum resources, it can be protected against eavesdropping because extracting the information is associated with causing a disturbance in the quantum state. That disturbance can be detected by the users.
- The most well-known quantum key distribution protocol is called the BB84 protocol and it is based on preparing, transmitting and measuring qubits in the eigenstates of one two different Pauli observables.
- Quantum key distribution can be implemented without sending quantum states but instead using sharing entanglement between the users.

Further reading. For an overviewing discussion, see for example [Quantum Advantage in Cryptography](#) by Renner & Wolf. A more complete introduction to quantum cryptography is found in the textbooks [Introduction to Quantum Cryptography](#) by Vidick and Wehner and [Quantum Key Distribution](#) by Wolf.

11. MARKOVIAN MASTER EQUATIONS

Predicting the time evolution of a system is a core physics problem. In elementary quantum mechanics, one learns that the dynamics of a closed system is described by the Schrödinger equation and that it leads to a unitary time evolution. However, we have seen that open quantum systems motivate the more general notion of evolution represented by quantum channels. In this chapter, we begin to discuss time evolution in open quantum systems by deriving an equation that in relevant limits generalises the Schrödinger equation to account also for interactions between the system and its environment.

11.1. The Lindblad master equation

It is an idealisation that quantum systems can be isolated entirely from their environment. Many times, interaction between a system and its environment is not only unavoidable but also interesting and even useful. It motivates an open system view in which the dynamics is no longer described by unitary operators but instead by quantum channels. In general, determining the time evolution of an open quantum system is very challenging. However, under limits that reasonably approximate a fair share of actual physical systems, a generalisation of the Schrödinger equation can be derived.

Consider that we have a system, S , that interacts with an environment, E . We want to avoid using the Schrödinger equation on the total system, partly because we are only interested in S and partly because E can be complicated. Because we want to describe only the dynamics of system S , we should expect that its interaction with E causes the state of S to be mixed. Therefore, let us write the Schrödinger equation on a form that is compatible with density operators. To that end, we do a spectral decomposition of the state $\rho = \sum_k \lambda_k |\lambda_k\rangle\langle\lambda_k|$ and use the linearity of quantum theory to apply to each of the pure states in the decomposition the time-independent Schrödinger equation, $\frac{d}{dt}|\psi\rangle = -iH|\psi\rangle$ ⁶⁹. This gives us

$$\frac{d\rho}{dt} = \sum_k \lambda_k \frac{d}{dt}(|\lambda_k\rangle\langle\lambda_k|) = \sum_k \lambda_k \left(\underbrace{\frac{d}{dt}(|\lambda_k\rangle)}_{=-iH|\lambda_k\rangle} \langle\lambda_k| + |\lambda_k\rangle \underbrace{\frac{d}{dt}(\langle\lambda_k|)}_{=i\langle\lambda_k|H} \right) = -iH\rho + i\rho H = i[\rho, H]. \quad (11.1)$$

The result is called the *von Neumann equation*. In the same way that evolution under the Schrödinger equation transforms pure states to pure states, evolution under the von Neumann equation preserves the purity of the initial density operator. To see this, we check that the derivative of the purity function (see Eq (1.27)) vanishes at all times, namely

$$\frac{d}{dt} \text{tr}(\rho^2) = \text{tr}\left(\frac{d\rho^2}{dt}\right) = 2 \text{tr}\left(\rho \frac{d\rho}{dt}\right) \stackrel{(11.1)}{=} 2i \text{tr}(\rho[\rho, H]) = 2i \left(\text{tr}(\rho^2 H) - \underbrace{\text{tr}(\rho H \rho)}_{=\text{tr}(\rho^2 H)} \right) = 0. \quad (11.2)$$

We now set out to generalise the von Neumann equation so that it can account for system-environment interactions. In general, the limiting situation in which this can be achieved is when the dynamics has a property called *Markovianity*.

G25: Markovian dynamics

Markovian dynamics is time evolution without memory: the change in the state at time t depends only on the state at time t , and not on the state at any other time. Formally, this is written as

$$\frac{d\rho(t)}{dt} = \mathcal{L}(\rho), \quad (11.3)$$

where $\mathcal{L}$ is a linear map called the *Liouvillian*¹⁵ that transforms operators to operators.

A simple example of a Markovian process is a random walk: we flip a coin and depending on the result we take a step either to the left or to the right. Our position after the n 'th step is not going to depend on how we arrived at our current position but only on where we stand at the moment. While memory is often present in physical processes, it can sometimes be safely neglected, i.e. some systems quickly forget their history. Typically, we consider Markovian dynamics when excitations of the environment

⁶⁹ We have set $\hbar = 1$ for simplicity.

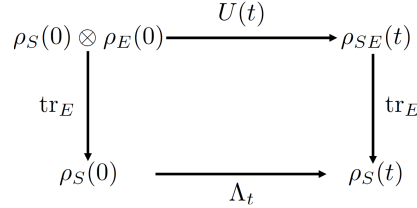
⁷⁰ It is called the Liouvillian because of the similarity with the Liouville equation in Hamiltonian mechanics.

caused by interaction with the system decay quickly (e.g. when the E is much larger than S) and the system-environment correlations develop only slowly over time⁷¹.

The memoryless state update equation in Eq (11.3) admits a solution of the form

$$\rho(t) = \Lambda_t[\rho(0)], \quad (11.4)$$

where Λ_t is a CPTP map. Because of Markovianity, $\rho(t)$ depends only on the specific CPTP map at time t (as opposed to earlier times). Thus, in order to evolve from $t = 0$ to $t = t_1 + t_2$, we can first evolve t_1 units and then another t_2 units, i.e. $\Lambda_{t_1+t_2} = \Lambda_{t_2} \circ \Lambda_{t_1}$. As illustrated in the Figure below, this type of evolution is equivalent to adopting the point of view of the Stinespring representation R19, namely to evolve jointly the state and the initially independent environment and then tracing out the latter.



Our task is to determine the most general Liouvillian with the above structure and with the property that Λ_t becomes CPTP. The solution is called the *Lindblad master equation*⁷².

R26: Lindblad master equation

Every Markovian quantum evolution is described by a master equation of the form

$$\frac{d\rho}{dt} = i[\rho, H] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right), \quad (11.5)$$

where L_k are operators called *jump operators* that represent the k 'th dissipation process and γ_k is a set of real numbers called *damping rates*. The damping rates have units of $[\gamma_k] = [\text{time}^{-1}]$.

Notice that by setting all the damping rates to $\gamma_k = 0$ we recover the von Neumann equation in Eq (11.1). Therefore, the second term in Eq (11.5) accounts for the exchange between the system and the environment. The form of this term might look rather arbitrary, especially since no constraints are placed on the jump operators. To lend it some initial credibility, let us do a sanity check and verify that the solution has a time-independent trace, i.e. that the derivative of the trace vanishes at all times,

$$\begin{aligned} \frac{d}{dt} \text{tr}(\rho) &= \text{tr} \left(\frac{d\rho}{dt} \right) \stackrel{(11.5)}{=} \underbrace{\text{tr}(i[\rho, H])}_{=0} + \text{tr} \left(\sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right) \right) \\ &= \sum_k \gamma_k \text{tr} \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right) = \sum_k \gamma_k \underbrace{\text{tr} \left(L_k \rho L_k^\dagger - \frac{1}{2} L_k^\dagger L_k \rho - \frac{1}{2} \rho L_k^\dagger L_k \right)}_{=0} = 0, \end{aligned} \quad (11.6)$$

where in the last step we have used the cyclicity of the trace. This check is of course not enough to prove the validity of the Lindblad equation. That is the focus of the next section.

11.2. Lindblad equation from Kraus representation

The Lindblad equation is relevant to several branches of quantum mechanics and hence there are several ways to derive it. Here, we take the quantum information route by derivinig it from the Kraus representation of CPTP maps discussed in Chapter 7.3.

⁷¹ For a detailed discussion on physical approximations needed to build Markovian master equations, see e.g. the notes [Quantum Thermodynamics](#) by Potts.

⁷² Master equation is a historical term. It means that it is the equation from which all properties can be obtained. The Lindblad equation is also referred to as the Gorini–Kossakowski–Sudarshan–Lindblad (GKLS) equation.

Let us start with introducing an orthonormal basis, $\{F_i\}_{i=1}^{d^2}$, of $d \times d$ operators. By definition, they satisfy $\text{tr}(F_i^\dagger F_j) = \delta_{i,j}$. We select the final element as $F_{d^2} = \frac{1}{\sqrt{d}}$ because this conveniently implies that all other elements, $F_1, \dots, F_{d^2-1}$ are traceless. Using Kraus representation of CPTP maps, we have $\Lambda_t(\rho) = \sum_l K_l \rho K_l^\dagger$, where the Kraus operators are implicitly time-dependent, $K_l = K_l(t)$. We proceed to expand each of them in our selected basis,

$$K_l = \sum_{i=1}^{d^2} \text{tr}(F_i^\dagger K_l) F_i. \quad (11.7)$$

The action of the channel is then

$$\begin{aligned} \Lambda_t(\rho) &= \sum_l K_l \rho K_l^\dagger \stackrel{(11.7)}{=} \sum_l \left(\sum_{i=1}^{d^2} \text{tr}(F_i^\dagger K_l) F_i \right) \rho \left(\sum_{j=1}^{d^2} \text{tr}(F_j^\dagger K_l)^* F_j^\dagger \right) \\ &= \sum_{i,j=1}^{d^2} F_i \rho F_j^\dagger \underbrace{\sum_l \text{tr}(F_i^\dagger K_l) \text{tr}(F_j^\dagger K_l)^*}_{=c_{i,j}(t)} = \sum_{i,j} c_{i,j}(t) F_i \rho F_j^\dagger, \end{aligned} \quad (11.8)$$

where we have defined time-dependent coefficients $c_{i,j}(t)$. Notice that the matrix of these coefficients, $[c]_{i,j}$, at any given time t , is Hermitian because $c_{i,j} = c_{j,i}^*$.

Next, we invoke the assumption of Markovian dynamics. Because Markovianity implies $\Lambda_{t+\delta t} = \Lambda_{\delta t} \circ \Lambda_t$, we express the change in the state as

$$\frac{d\rho}{dt} = \lim_{\delta t \rightarrow 0} \frac{\Lambda_{\delta t}(\rho) - \rho}{\delta t}, \quad (11.9)$$

where $\rho = \rho(t) = \Lambda_t(\rho(0))$. Inserting our expansion from Eq (11.8) we obtain

$$\frac{d\rho}{dt} = \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \left(\sum_{i,j=1}^{d^2-1} c_{i,j}(\delta t) F_i \rho F_j^\dagger + \sum_{i=1}^{d^2-1} c_{i,d^2}(\delta t) F_i \rho F_{d^2}^\dagger + \sum_{j=1}^{d^2-1} c_{d^2,j}(\delta t) F_{d^2} \rho F_j^\dagger + c_{d^2,d^2}(\delta t) F_{d^2} \rho F_{d^2}^\dagger - \rho \right). \quad (11.10)$$

Now we use $F_{d^2} = \frac{1}{\sqrt{d}}$ to simplify this expression. Also, we can get rid of the time dependence by defining new coefficients

$$r_{i,j} = \lim_{\delta t \rightarrow 0} \frac{c_{i,j}(\delta t)}{\delta t}, \quad r_{d^2,d^2} = \lim_{\delta t \rightarrow 0} \frac{c_{d^2,d^2}(\delta t) - d}{\delta t},$$

for $(i,j) \neq (d^2, d^2)$. The Hermiticity of the matrix of these coefficients, $[r]_{i,j}$, is inherited from the Hermiticity of the matrix $[c]_{i,j}$. Inserting the new coefficients in Eq (11.10) and using $F_{d^2} = \frac{1}{\sqrt{d}}$ gives us

$$\frac{d\rho}{dt} = \sum_{i,j=1}^{d^2-1} r_{i,j} F_i \rho F_j^\dagger + \underbrace{\frac{1}{\sqrt{d}} \sum_{i=1}^{d^2-1} r_{i,d^2} F_i \rho}_{=F} + \frac{1}{\sqrt{d}} \sum_{j=1}^{d^2-1} r_{d^2,j} \rho F_j^\dagger + \frac{r_{d^2,d^2}}{d} \rho = \sum_{i,j=1}^{d^2-1} r_{i,j} F_i \rho F_j^\dagger + F \rho + \rho F^\dagger + \frac{r_{d^2,d^2}}{d} \rho. \quad (11.11)$$

Next, focus on the newly defined matrix F . To it we apply a so-called Toeplitz decomposition. It allows us to split any matrix into a Hermitian and a skew-Hermitian part,

$$F = \underbrace{\frac{F + F^\dagger}{2}}_{\text{Hermitian}} + \underbrace{(-i) \frac{F^\dagger - F}{2i}}_{\text{skew-Hermitian}} = S' - iH, \quad (11.12)$$

where we have defined the Hermitian matrices $S' = \frac{F + F^\dagger}{2}$ and $H = \frac{F^\dagger - F}{2i}$. Inserted in Eq (11.11) this gives

$$\frac{d\rho}{dt} = \sum_{i,j=1}^{d^2-1} r_{i,j} F_i \rho F_j^\dagger + \{S', \rho\} + i[\rho, H] + \frac{r_{d^2,d^2}}{d} \rho. \quad (11.13)$$

To simplify this, we define $S = S' + \frac{1}{2d} r_{d^2, d^2} \mathbb{1}$ which yields

$$\frac{d\rho}{dt} = i[\rho, H] + \sum_{i,j=1}^{d^2-1} r_{i,j} F_i \rho F_j^\dagger + \{S, \rho\}. \quad (11.14)$$

So far, our map is completely positive thanks to the Kraus decomposition, but we also need to make sure that it is trace-preserving. That amounts to the condition $\frac{d}{dt} \text{tr}(\rho) = 0$. Using Eq (11.14) the condition becomes

$$0 = \sum_{i,j=1}^{d^2-1} r_{i,j} \text{tr}(\rho F_j^\dagger F_i) + 2 \text{tr}(\rho S) = \text{tr}\left(\rho \left(2S + \sum_{i,j} r_{i,j} F_j^\dagger F_i\right)\right), \quad \forall \rho. \quad (11.15)$$

Since this holds for every ρ , it is equivalent to the constraint

$$S = -\frac{1}{2} \sum_{i,j=1}^{d^2-1} r_{i,j} F_j^\dagger F_i. \quad (11.16)$$

Inserting this in Eq (11.14) we arrive at

$$\frac{d\rho}{dt} = i[\rho, H] + \sum_{i,j=1}^{d^2-1} r_{i,j} \left(F_i \rho F_j^\dagger - \frac{1}{2} \{F_j^\dagger F_i, \rho\} \right). \quad (11.17)$$

This starts to look like the Lindblad master equation (11.5) but it is still not quite the same. To transform it to the right form, recall that the matrix $R = [r_{i,j}]_{i,j}$ is Hermitian, which means it can be diagonalised as $URU^\dagger = D$, where D is the diagonal matrix holding the eigenvalues γ_i and U is the unitary matrix comprised of the corresponding eigenvectors. Thus,

$$\sum_{k,l} U_{m,k} r_{k,l} (U^\dagger)_{l,n} = \gamma_n \delta_{n,m}. \quad (11.18)$$

We can now define the jump operator as

$$L_k = \sum_{j'} (U^\dagger)_{j',k} F_{j'}. \quad (11.19)$$

Multiplying both sides by $U_{k,j}$ and summing over k gives

$$\sum_k U_{k,j} L_k = \sum_{j'} \underbrace{\left(\sum_k U_{k,j} (U^\dagger)_{j',k} \right)}_{=\langle j' | U^\dagger U | j \rangle = \delta_{j,j'}} F_{j'} = F_j. \quad (11.20)$$

Inserting this back into Eq (11.17) we obtain

$$\begin{aligned} \frac{d\rho}{dt} &= i[\rho, H] + \sum_{i,j=1}^{d^2-1} r_{i,j} \left(\left(\sum_k U_{k,i} L_k \right) \rho \left(\sum_{k'} U_{k',j}^* L_{k'}^\dagger \right) - \frac{1}{2} \left\{ \left(\sum_k U_{k,j}^* L_k^\dagger \right) \left(\sum_{k'} U_{k',i} L_{k'} \right), \rho \right\} \right) \\ &\stackrel{(11.18)}{=} i[\rho, H] + \sum_{k=1}^{d^2-1} \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \end{aligned} \quad (11.21)$$

which is the Lindblad equation (11.5). □

Notice that we have not equated H with the free Hamiltonian of our system but allowed it simply to be the Hermitian operator relevant for our proof. This means that H can contain additional terms to the free Hamiltonian, which may arise from interactions between the system and the environment. Moreover, the jump operators are not unique. In fact, one can show that the Lindblad equation remains invariant under particular transformations of the operators $\{L_k\}_k$ and H , as exemplified in Exercise C.3.1.

At this point, it is natural to ask how one in practice determines the values of the damping rates and the jump operators. The answer is that these must be determined from a physical description of the specific system under consideration. However, it is many times possible to make reasonable heuristic selections, without explicitly considering a microscopic physical model. In the next section we discuss an example of this based on spontaneous emission.

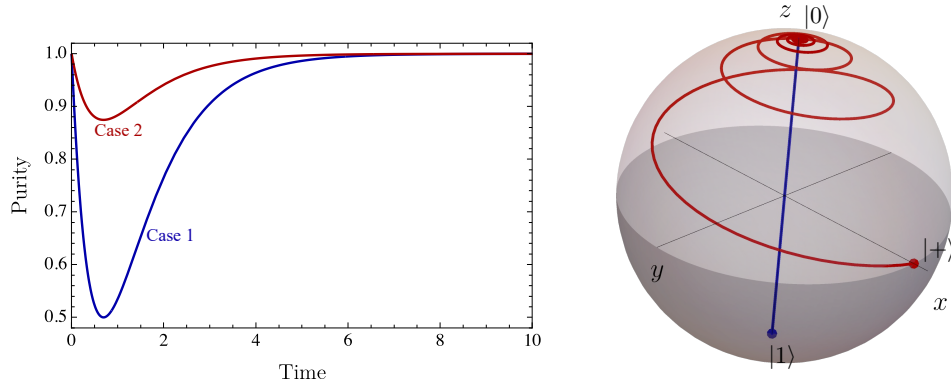


FIG. 11.21. (Left) Purity of state during evolution due to spontaneous emission. Case 1 corresponds to starting in the excited state and case 2 corresponds to starting in a uniform superposition of the ground and excited states. The plots are shown for $\Gamma = 1$. (Right) Trajectory of the time evolved state inside the Bloch sphere in cases 1 and 2.

11.3. Spontaneous emission

Spontaneous emission is caused by the coupling between the atom (system) and the vacuum state of the quantised electromagnetic field (environment). We use the Lindblad equation to construct a toy model of this process without having to analyse the full physics of the problem, i.e. we will not deal with any quantum optics here. Consider that we have a two-level atom where the ground state energy is zero and the excited state energy is E . The free Hamiltonian becomes $H_{\text{free}} = E|1\rangle\langle 1|$. With some rate Γ , the atom interacts with the vacuum state and loses a photon to the environment, thereby de-exciting to the ground state. To this dissipation process we associate the jump operator $\sigma_- = |0\rangle\langle 1|$: it takes us from the excited state to the ground state. The Lindblad master equation becomes

$$\frac{d\rho}{dt} = -\frac{iE}{2}[\rho, \sigma_Z] + \Gamma \left(\sigma_- \rho \sigma_-^\dagger - \frac{1}{2} \{ \sigma_-^\dagger \sigma_-, \rho \} \right) = -\frac{iE}{2}[\rho, \sigma_Z] + \Gamma \left(\rho_{11} |0\rangle\langle 0| - \frac{1}{2} \{ |1\rangle\langle 1|, \rho \} \right), \quad (11.22)$$

where we have used that $[\rho, H_{\text{free}}] = [\rho, E \frac{1-\sigma_Z}{2}] = -\frac{E}{2}[\rho, \sigma_Z]$. We have four entries in the matrix ρ but only two are free: one is fixed by normalisation, $\rho_{11} = 1 - \rho_{00}$, and the other is fixed by Hermiticity, $\rho_{10} = \rho_{01}^*$. Therefore, we break up the master equation (11.22) into two components; one for ρ_{00} and one for ρ_{01} . These become

$$\begin{aligned} \frac{d\rho_{00}}{dt} &= \Gamma(1 - \rho_{00}) & \Rightarrow & \rho_{00}(t) = 1 - (1 - \rho_{00}(0))e^{-\Gamma t}, \\ \frac{d\rho_{01}}{dt} &= \left(iE - \frac{\Gamma}{2} \right) \rho_{01} & \Rightarrow & \rho_{01}(t) = \rho_{01}(0) e^{(iE - \frac{\Gamma}{2})t}. \end{aligned} \quad (11.23)$$

Both these equations are first order ordinary differential equations to which the solutions are standard and given on the right-hand-side. We see that the ground state population, $\rho_{00}(t)$, is increasing exponentially in time independently of the energy gap. The coherence term, $\rho_{01}(t)$, has both a time-dependent phase and amplitude. A larger energy means that the phase rotates faster, whereas a higher damping rate means that the coherence magnitude dampens faster, approaching zero in the limit $t \rightarrow \infty$. That limit is necessary because when $t \rightarrow \infty$ we indeed recover the ground state $\rho = |0\rangle\langle 0|$. To better understand the evolution, let us look at two different initial conditions for our system.

- **Start the system in $\rho(0) \simeq |1\rangle$.** This means $\rho_{00}(0) = 0$ and $\rho_{01}(t) = 0$. From Eq (11.23), we see that the evolution is

$$\rho_{00}(t) = 1 - e^{-\Gamma t}, \quad \rho_{01}(t) = 0. \quad (11.24)$$

The energy gap is irrelevant and the coherence is always zero. Thus, at time $t = 0$ we start at the south-pole of the Bloch sphere and then evolve toward the north-pole ($\rho(\infty) = |0\rangle\langle 0|$) along a straight line; see Fig 11.21 (right, blue). To get an idea of how quickly we evolve along that straight path, consider the purity: it is computed to be $\text{tr}(\rho(t)^2) = 1 + 2e^{-2\Gamma t}(1 - e^{-\Gamma t})$ and illustrated in blue on the left panel. We see that it starts pure, then rapidly decays to maximally mixed and then begins ascending to a pure state. If we increase the damping rate, Γ , the decay and subsequent ascent of the purity is faster.

- **Start the system in $\rho(0) \simeq \frac{|0\rangle + |1\rangle}{\sqrt{2}}$.** This means $\rho_{00}(0) = \frac{1}{2}$ and $\rho_{01}(t) = \frac{1}{2}$. From Eq (11.23) we see that the evolution is

$$\rho_{00}(t) = 1 - \frac{e^{-\Gamma t}}{2}, \quad \rho_{01}(t) = \frac{1}{2} e^{-\frac{\Gamma}{2}t} e^{iEt}. \quad (11.25)$$

The coherence term is now non-zero but exponentially damped due to the factor $e^{-\frac{\Gamma}{2}t}$ and the phase is rotating with a frequency corresponding to E due to the factor e^{iEt} . This means that on the Bloch sphere, our evolution starts at the equator and spirals upwards towards the north-pole; see Fig 11.21 (right, red). If we increase the energy gap the frequency of the spiraling is larger and if we increase the damping rate the ascent to the north-pole is faster. We can compute the purity to be $\text{tr}(\rho(t)^2) = 1 - \frac{1}{2}(e^{-\Gamma t} - e^{-2\Gamma t})$, which indeed is independent of E , because E is only associated with the phase. The behaviour here is similar to the previous case but with the difference that the purity reaches its lowest value at $t = \frac{\log(2)}{\Gamma}$. This also reflects how Γ determines the speed of the process.

11.4. Steady state properties

In Eq (11.3) we encountered the Liouvillian, $\mathcal{L}$, which is a non-Hermitian map from operators to operators. This can be intuitively seen from the fact that the Lindblad equation is not purity-preserving, unlike the Schrödinger equation which is a representation of unitary dynamics (induced by a Hermitian Hamiltonian). Alternatively, it can be seen that $\mathcal{L}$, in general, has complex eigenvalues, $\{\lambda_i\}_i$, and a distinct set of right and left eigenmatrices⁷³ which we call φ_i and σ_i , respectively, determined by the following eigenvalue equations

$$\mathcal{L}(\varphi_i) = \lambda_i \varphi_i, \quad \text{and} \quad \mathcal{L}^\dagger(\sigma_i) = \lambda_i^* \sigma_i. \quad (11.26)$$

Unlike the Hamiltonian which has orthogonal eigenstates, the Liouvillian has bi-orthogonal eigenmatrices. With appropriate normalisation, this means that $\text{tr}(\sigma_i \varphi_j) = \delta_{ij}$. Liouvillian eigenvalues can be shown to have two key properties.

1. If the Liouvillian is time-independent and the dimension of the Hilbert space is finite, then there is at least one zero-eigenvalue.
2. All eigenvalues have a non-positive real part, i.e., $\text{Re}(\lambda_i) \leq 0$.

Assuming that $\lambda_0 = 0$, the zero-eigenvalue corresponds to the eigenvalue equation $\mathcal{L}(\varphi_0) = 0$, i.e., the corresponding eigenmatrix is non-evolving under the action of the Liouvillian. Such a state is known as a *steady state*. Let us restrict to the case where $\mathcal{L}$ has a unique zero-eigenvalue and therefore a unique steady state. If the Liouvillian is time-independent and diagonalisable⁷⁴ (i.e., it has d^2 distinct eigenmatrices) it is possible to solve the Lindblad equation and obtain the following expression for the time-dependent state,

$$\rho(t) = e^{\mathcal{L}t} \rho(0) = \sum_i c_i e^{\lambda_i t} \varphi_i = \varphi_{\text{steady}} + \sum_{i=1}^{d^2-1} c_i e^{\lambda_i t} \varphi_i. \quad (11.27)$$

φ_{steady} is the appropriately normalised φ_0 and $c_i = \text{tr}(\sigma_i \rho_0)$ are initial-state-dependent coefficients. The non-positivity of the real parts of the eigenvalues mean that at long times the state must evolve to the steady state. The time-scale for converging to the steady-state is determined by the eigenvalue with the largest non-zero real part. In practice it is convenient to represent the Liouvillian as a matrix. To be able to do that, a standard trick is to wrap the $d \times d$ density operator into a $d^2 \times 1$ vector and then wrap it back to a $d \times d$ density operator at the end of the calculation. We will encounter this in Exercise C.3.2.

Chapter summary.

- Markovian dynamics is time evolution without memory effects. For some systems, it is a good approximation of the actual physics.
- The Lindblad master equation represents the most general Markovian quantum dynamics. It can be derived from the theory of quantum channels and it can be used to heuristically model many open systems.
- Finite-dimensional Lindblad dynamics always has at least one steady state.

Further reading. A more extensive introduction to the Lindblad master equation is found in the article [A short introduction to the Lindblad Master Equation](#) by Manzano. For a deeper discussion, see for example [Open Quantum Systems: An Introduction](#) by Rivas & Huelga.

⁷³ Since $\mathcal{L}$ transforms operators to operators, it has eigenmatrices instead of eigenvectors. Moreover, if the Hilbert space is d -dimensional, $\mathcal{L}$ has d^2 eigenvalues.

⁷⁴ Since $\mathcal{L}$ is non-Hermitian, it may not always be diagonalisable. Such situations can be handled case-by-case.

12. QUANTUM ESTIMATION THEORY

Estimation theory is the study of how to make efficient estimates of unknown parameters from a given set of data. Imagine that we are interested in some parameter, θ , that we cannot directly measure. However, through some given rule, θ influences a random variable, X , that we can observe. By sampling from X we try to estimate θ . For example, θ could be the probability of observing Heads when flipping a coin and X could be the random variable associated with each coin flip. By flipping the coin many times, we could make increasingly precise estimates of θ . In what follows, we introduce the necessary background in estimation theory and then discuss how quantum resources can improve our ability to make more precise estimations.

12.1. Estimators

Consider that we are given a set of data $x = \{x_1, \dots, x_N\}$ which consists of independent samples, x_i , from identically distributed random variables, X_i . The parameter θ influences $X = (X_1, X_2, \dots)$ according to a given model and therefore the probability distribution of X can be written $p_X(x|\theta)$. We must convert our data into a guess for the value of θ ⁷⁵. This inference rule is called an *estimator* and we write it as $\hat{\theta}(x)$. For convenience, we sometimes omit the dependence on x and just write $\hat{\theta}(x) = \hat{\theta}$.

How should we choose our estimator? To answer this, we need a notion of what makes for a good estimator. Naturally, a good estimator should output estimates that are near the true parameter value. It is common to quantify this via the mean-square error,

$$(\Delta\hat{\theta})^2 \equiv E[(\hat{\theta} - \theta)^2] = \int dx p(x|\theta) (\hat{\theta} - \theta)^2. \quad (12.1)$$

A good estimator has a small mean-square error, but this quantity is θ -dependent so the estimator might be good around some values of θ and bad for other values of θ . Typically, we assume that we have a rough idea of what θ should be.

A convenient property to have in an estimator is that it is *unbiased*, namely that it on average returns the true value of θ ,

$$E[\hat{\theta}] = \int dx p(x|\theta) \hat{\theta}(x) = \theta. \quad (12.2)$$

Notice that for an unbiased estimator, the mean-squared error is the same as the estimator's variance because

$$E[(\hat{\theta} - \theta)^2] = E[\hat{\theta}^2] + \theta^2 - 2\theta E[\hat{\theta}] = \underbrace{E[\hat{\theta}^2] - \theta^2}_{=\text{Var}[\hat{\theta}]} = \text{Var}[\hat{\theta}]. \quad (12.3)$$

However, unbiased estimators are not always available for a given estimation problem, and even if they are it is not always immediate how to find them. Let us illustrate the latter with a simple example.

12.1.1. Example: Normal distribution

Let $X_1, \dots, X_N$ be independent and normally distributed random variables with expectation value μ and variance σ^2 . Consider now that we want to estimate the variance. An intuitive guess for a good estimator is the average mean-square deviation between each random variable and its sample average,

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2, \quad (12.4)$$

where $\bar{X} = \frac{1}{N} \sum_i X_i$ is the sample average. To check whether the estimator is unbiased, we compute the expectation value

$$E[\hat{\theta}] \stackrel{(12.4)}{=} \frac{1}{N} \sum_{i=1}^N E[(X_i - \bar{X})^2] = \frac{1}{N} \sum_{i=1}^N E\left[\left(\frac{1}{N} \sum_k (X_i - X_k)\right)^2\right] = \frac{1}{N^3} \sum_{i=1}^N E\left[\sum_{j,k} (X_i - X_k)(X_i - X_j)\right]. \quad (12.5)$$

⁷⁵ We use θ interchangeably for the parameter to be estimated and the true value of that parameter.

We can now expand the right-most expectation value. Note that since the random variables are independent and identically distributed we have $E[X_i X_j] = E[X_i]E[X_j]$ and $E[X_i] = E[X_1] \forall i$. Using these relations we get

$$\begin{aligned} E \left[\sum_{j,k} (X_i - X_k)(X_i - X_j) \right] &= \sum_{j,k} E[X_i^2 + X_j X_k - X_i X_k - X_i X_j] = N^2 E[X_i^2] + \left(\sum_j E[X_j^2] + \sum_{j \neq k} E[X_j X_k] \right) \\ &\quad - \left(\sum_j E[X_i^2] + N \sum_{k \neq i} E[X_i X_k] \right) - \left(\sum_k E[X_i^2] + N \sum_{j \neq i} E[X_i X_j] \right) \\ &= N^2 E[X_1^2] + (N E[X_1^2] + N(N-1)E[X_1]^2) - (N E[X_1^2] + N(N-1)E[X_1]^2) \\ &\quad - (N E[X_1^2] + N(N-1)E[X_1]^2) = N(N-1)(E[X_1^2] - E[X_1]^2) = N(N-1)\sigma^2, \end{aligned} \quad (12.6)$$

where we have written $\sigma^2 = \text{Var}[X_1]$. Inserted into Eq (12.5) we obtain

$$E[\hat{\theta}] = \frac{N-1}{N} \sigma^2. \quad (12.7)$$

For the estimator to be unbiased we would have needed $E[\hat{\theta}] = \sigma^2$. Hence, our estimator is biased by a factor $\frac{N-1}{N}$. However, this factor is only significant for small data sets, as it tends to one with large N . In contrast, if we wanted to make an unbiased estimator for every N , we would have had to make a not-so-intuitive modification to Eq (12.4), namely replacing it with

$$\hat{\theta} = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}). \quad (12.8)$$

Repeating the above calculation with this estimator, one finds that $E[\hat{\theta}] = \sigma^2$. This estimator is called the *unbiased sample variance* and the correction factor of $\frac{N}{N-1}$ is called Bessel's correction.

12.1.2. Maximum likelihood estimator

A popular systematic way of selecting estimators is the so-called *maximum likelihood estimator*. It tells us to first draw x and then guess on the θ that assigns the highest probability to our sample.

G26: Maximum likelihood estimator

Let the random variable X depend on the parameter θ and denote its conditional probability distribution $p_X(x|\theta)$. The maximum likelihood estimator is

$$\hat{\theta}(x) = \underset{\theta}{\operatorname{argmax}} p_X(x|\theta). \quad (12.9)$$

Actually, it is often convenient to work not directly with the likelihood function, namely $p_X(x|\theta)$, but instead with its logarithm. The log-likelihood function is therefore $\ln p_X(x|\theta)$ ⁷⁶. Note that since the logarithm is strictly increasing its argmax will remain the same as in Eq (12.9). The reason that one may prefer the log-likelihood function is that it is an additive quantity thanks to $\ln(xy) = \ln(x) + \ln(y)$. Let us explore the maximum log-likelihood estimator in a couple of concrete cases.

Example: Bernoulli distribution.— Consider a binary random variable, X , with distribution $p_X(0) = 1 - \theta$ and $p_X(1) = \theta$. Given data $x = \{x_1, \dots, x_N\}$, the likelihood function is

$$p_X(x|\theta) = \theta^{\#ones} (1 - \theta)^{\#zeros} = \theta^{\sum_i x_i} (1 - \theta)^{N - \sum_i x_i} = \theta^{N\bar{x}} (1 - \theta)^{N(1-\bar{x})}, \quad (12.10)$$

where we have defined the sample average $\bar{x} = \frac{1}{N} \sum_i x_i$. The log-likelihood function then becomes

$$\ln p_X(x|\theta) = N\bar{x} \ln(\theta) + N(1 - \bar{x}) \ln(1 - \theta). \quad (12.11)$$

⁷⁶ Here we use the natural logarithm.

We want to select the θ that maximises the log-likelihood. For that we find the roots of the derivative

$$0 = \frac{\partial \ln p_X(x|\theta)}{\partial \theta} = N \left(\frac{\bar{x}}{\theta} - \frac{1 - \bar{x}}{1 - \theta} \right) \Rightarrow \hat{\theta}(x) = \bar{x}. \quad (12.12)$$

Thus, our maximum likelihood estimator is just the relative frequency of observing the event “1”⁷⁷. We can also confirm that this estimator is unbiased because

$$E[\hat{\theta}] = \frac{1}{N} \sum_{i=1}^N E[X_i] = \frac{1}{N} \sum_{i=1}^N \theta = \theta. \quad (12.13)$$

Example: Normal distribution.— Let X_i be independent and normally distributed random variables with mean θ and variance σ^2 . The likelihood function is

$$p_X(x|\theta, \sigma^2) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \theta)^2}{2\sigma^2}\right). \quad (12.14)$$

The log-likelihood function becomes

$$\ln p_X(x|\theta, \sigma^2) = -\sum_i \frac{(x_i - \theta)^2}{2\sigma^2} - \frac{N}{2} \ln(2\pi\sigma^2). \quad (12.15)$$

Say that our goal is to estimate θ via the maximum likelihood estimator. Then the most likely value of θ is that which maximises the log-likelihood function. Hence we solve

$$0 = \frac{\partial \ln p_X(x|\theta, \sigma^2)}{\partial \theta} = \frac{N(\bar{x} - \theta)}{\sigma^2} \Rightarrow \hat{\theta}(x) = \bar{x}. \quad (12.16)$$

where we have again set $\bar{x} = \frac{1}{N} \sum_i x_i$. Thus, our maximum likelihood estimator is just the sample average.

Consider that we instead want to estimate the variance, σ^2 . For that, we again follow the maximum likelihood approach to get

$$0 = \frac{\partial \ln p_X(x|\theta, \sigma^2)}{\partial \sigma^2} = \frac{N}{\sigma^4} \left(\sigma^2 - \frac{1}{N} \sum_i (x_i - \theta)^2 \right) \Rightarrow \hat{\sigma}^2 = \frac{1}{N} \sum_i (x_i - \theta)^2. \quad (12.17)$$

If we were to substitute θ for our maximum likelihood estimate, namely $\hat{\theta} = \bar{x}$, we recover the average sample variance in Eq (12.4), but without Bessel’s correction. Consequently, the maximum likelihood estimator is not always unbiased for finite N .

12.2. Cramér-Rao bound and Fisher information

How good can we ever hope for an estimator to be? The *Cramér-Rao bound* provides a lower bound on the mean-square error for any unbiased estimator. Thus, it sets a limit for how well we could make our estimations.

R27: Cramér-Rao bound

Let $\hat{\theta}(x)$ be an unbiased estimator for θ . Its variance is bounded as

$$\text{Var}[\hat{\theta}] \geq \frac{1}{\mathcal{I}(p(x|\theta))}, \quad (12.18)$$

where the quantity $\mathcal{I}(p(x|\theta))$ is called the *Fisher information* and defined as

$$\mathcal{I}(p(x|\theta)) = \int dx \frac{1}{p(x|\theta)} \left(\frac{\partial p(x|\theta)}{\partial \theta} \right)^2 = \int dx p(x|\theta) \left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2 = E \left[\left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2 \right]. \quad (12.19)$$

⁷⁷ This is probably the estimator that most people would use without giving further thought to the matter: flip the coin N times and take the ratio of Heads to N .

To prove the Cramér-Rao bound, we re-write the definition of an unbiased estimator in Eq (12.2) in a form convenient for us,

$$0 = \int dx p(x|\theta) (\hat{\theta}(x) - \theta). \quad (12.20)$$

Differentiate both sides with respect to θ to obtain

$$0 = \frac{\partial}{\partial \theta} \int dx p(x|\theta) (\hat{\theta}(x) - \theta) = \int dx \left(\frac{\partial p(x|\theta)}{\partial \theta} (\hat{\theta}(x) - \theta) - p(x|\theta) \right) \quad (12.21)$$

Using that $\int dx p(x|\theta) = 1$ we re-write the above as

$$1 = \int dx p(x|\theta) \left[\frac{\partial \ln p(x|\theta)}{\partial \theta} (\hat{\theta}(x) - \theta) \right], \quad (12.22)$$

where we have expressed the derivative from Eq (12.21) in a somewhat unintuitive way, but by expanding it we see that it is indeed valid. The reason we choose to write (12.21) on the form in Eq (12.22) is that it is suitable for applying the Cauchy-Schwarz inequality. Recall that on square-integrable functions the Cauchy-Schwarz inequality is $(\int dx g(x)h(x))^2 \leq \int dx g(x)^2 \times \int dx h(x)^2$. We use it in the following way: set $g(x) = \sqrt{p(x|\theta)} \frac{\partial \ln p(x|\theta)}{\partial \theta}$ and $h(x) = \sqrt{p(x|\theta)} (\hat{\theta}(x) - \theta)$. This leads to

$$1 \leq \underbrace{\int dx p(x|\theta) \left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2}_{=\mathcal{I}(p(x|\theta)) \text{ via Eq (12.19)}} \times \underbrace{\int dx p(x|\theta) (\hat{\theta}(x) - \theta)^2}_{=\text{Var}[\hat{\theta}]}, \quad (12.23)$$

which can be re-arranged into the Cramér-Rao bound in Eq (12.18). $\square$

It may seem as if the definition of Fisher information is just a convenient name given to the factor appearing in Eq (12.23), but there is more to it than that. The Fisher information, as appearing in Eq (12.23), can be interpreted as the expectation value of $\left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2$; see the right-most step in Eq (12.19). Here, we see the slope of the log-likelihood function appearing, namely $\frac{\partial \ln p(x|\theta)}{\partial \theta}$. If the slope is large then it should intuitively be easier to estimate θ : it narrows the values of θ associated with a high probability for the observed data sample. This is regardless of whether the slope is positive or negative, and indeed this is eliminated by the square in our expression. In conclusion, the Fisher information is giving us the average magnitude of the slope of the log-likelihood function, which is commonly interpreted as the amount of information about θ that can be extracted from the probability distribution⁷⁸.

This informational interpretation of $\mathcal{I}$ is further served by two properties: the Fisher information is non-negative and additive. Non-negativity follows immediately from the right-most expression in Eq (12.19) but additivity is less immediate. To prove additivity, consider that we take N independent samples, associated with the distribution $p(x|\theta) = \prod_{i=1}^N p(x_i|\theta)$. Then,

$$\begin{aligned} \mathcal{I}(p(x|\theta)) &\stackrel{(12.19)}{=} \int dx p(x|\theta) \left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2 = \int dx \left(\prod_i p(x_i|\theta) \right) \left(\frac{\partial \ln \prod_j p(x_j|\theta)}{\partial \theta} \right)^2 \\ &= \int dx \left(\prod_i p(x_i|\theta) \right) \left(\sum_j \frac{\partial \ln p(x_j|\theta)}{\partial \theta} \right) \left(\sum_k \frac{\partial \ln p(x_k|\theta)}{\partial \theta} \right) \\ &= \sum_{j \neq k} \int dx \left(\prod_{i \neq j, i \neq k} p(x_i|\theta) \right) \frac{\partial p(x_j|\theta)}{\partial \theta} \frac{\partial p(x_k|\theta)}{\partial \theta} + \sum_{j=k} \int dx \left(\prod_{i \neq j} p(x_i|\theta) \right) \frac{1}{p(x_j|\theta)} \left(\frac{\partial p(x_j|\theta)}{\partial \theta} \right)^2. \end{aligned} \quad (12.24)$$

In the last line we have applied the chain rule of the derivative and for simplicity grouped terms according to $j \neq k$ and $j = k$. Also, note that $dx = dx_1 \dots dx_N$. In the first term on the last line, the integral over all x_i except $i = j$ and $i = k$ is simply $\int dx_i p(x_i|\theta) = 1$. Using this, the first term vanishes because

$$\int dx_j \frac{\partial p(x_j|\theta)}{\partial \theta} \int dx_k \frac{\partial p(x_k|\theta)}{\partial \theta} = \left(\frac{\partial}{\partial \theta} \underbrace{\int dx_j p(x_j|\theta)}_{=1} \right) \left(\frac{\partial}{\partial \theta} \underbrace{\int dx_k p(x_k|\theta)}_{=1} \right) = 0. \quad (12.25)$$

⁷⁸ In addition, one can also show that the Fisher information determines the lowest-order contribution to a suitable distance measure between $p(x|\theta)$ and $p(x|\theta + \delta\theta)$, for some small perturbation $\delta\theta$.

We are left only with the second term on the last line of Eq (12.24). For the same reason as above, when $i \neq j$ the integral over x_i is unity. It leaves us with

$$\mathcal{I}(p(x|\theta)) = \sum_j \int dx_j \frac{1}{p(x_j|\theta)} \left(\frac{\partial p(x_j|\theta)}{\partial \theta} \right)^2 \stackrel{(12.19)}{=} \sum_j \mathcal{I}(p(x_j|\theta)), \quad (12.26)$$

which is the sum of the Fisher information (12.19) over all the N random variables. Hence, the Fisher information is additive. For simplicity, consider that all the independent rounds are also identically distributed, namely $p(x_j|\theta) = p(x_1|\theta)$. Then, the Cramér-Rao bound implies that the variance of an unbiased estimator is limited by the scaling

$$\text{Var}[\hat{\theta}] \geq \frac{1}{N \times \mathcal{I}(p(x_1|\theta))}. \quad (12.27)$$

This scaling which is proportional to the inverse of N is known as the *standard quantum limit* or sometimes the *shot-noise limit*.

12.3. Quantum metrology

We are ready to treat parameter estimation in a quantum framework. To that end, consider that we prepare a multipartite state ρ and use a CPTP map Λ_θ to encode the parameter θ . Then, we perform a POVM $\{E_x\}_x$. The outcome is therefore sampled from the probability distribution

$$p(x|\theta) = \text{tr}(\Lambda_\theta[\rho]E_x). \quad (12.28)$$

Once we have our quantum probability distribution, we use our preferred estimator to infer θ just as before. Hence, the goal of quantum metrology is to optimise the quantum resources used in the sampling part when performing parameter estimation.

To emphasise the quantum nature of $p(x|\theta)$, we write the Fisher information as $\mathcal{I}(p(x|\theta)) = \mathcal{I}(\rho_\theta, E)$. Here, we label by $\rho_\theta = \Lambda_\theta[\rho]$ the parameter-encoded state. Since the Fisher information limits our precision, we want to find the POVM that maximises it, namely

$$\mathcal{I}^Q(\rho_\theta) \equiv \max_E \mathcal{I}(\rho_\theta, E) \stackrel{(12.19)}{\stackrel{(12.28)}}{=} \max_E \int dx \frac{\text{tr}(\dot{\rho}_\theta E_x)^2}{\text{tr}(\rho_\theta E_x)}. \quad (12.29)$$

where $\dot{\rho}_\theta = \frac{\partial \rho_\theta}{\partial \theta}$. This quantity, $\mathcal{I}^Q$, is called the *quantum Fisher information*. It represents the largest Fisher information achievable from a given quantum state and a given encoding rule for the parameter. This brings us to a quantum version of the Cramér-Rao bound.

R28: Quantum Cramér-Rao bound

Let θ be encoded in the state ρ via a CPTP map and let $\hat{\theta}(x)$ be an unbiased estimator for θ . The variance of the estimator is bounded as

$$\text{Var}[\hat{\theta}] \geq \frac{1}{\mathcal{I}^Q(\rho_\theta)}, \quad (12.30)$$

where the quantum Fisher information can be expressed as

$$\mathcal{I}^Q(\rho_\theta) = \sum_j \frac{\dot{\lambda}_j^2}{\lambda_j} + 2 \sum_{j,k} \frac{(\lambda_j - \lambda_k)^2}{\lambda_j + \lambda_k} |\langle \dot{\psi}_j | \dot{\psi}_k \rangle|^2. \quad (12.31)$$

The bound in Eq (12.30) can always be saturated.

The maximisation appearing in Eq (12.29) for the quantum Fisher information might look like a formidable task but we can prove that it admits the closed form in Eq (12.31). To that end, we first need to introduce a handy concept: the *symmetric logarithmic derivative*, L_θ , is defined as the self-adjoint operator that satisfies

$$L_\theta \rho_\theta + \rho_\theta L_\theta = 2\dot{\rho}_\theta. \quad (12.32)$$

The reason that we make this definition is because we now can write

$$\mathrm{tr}(\dot{\rho}_\theta E_x) \stackrel{(12.32)}{=} \frac{1}{2} \mathrm{tr}(L_\theta \rho_\theta E_x) + \frac{1}{2} \mathrm{tr}(\rho_\theta L_\theta E_x) = \mathrm{Re}(\mathrm{tr}(L_\theta \rho_\theta E_x)), \quad (12.33)$$

where in the last step we used that $\mathrm{tr}(\rho_\theta L_\theta E_x) = \mathrm{tr}((\rho_\theta L_\theta E_x)^\dagger)^* = \mathrm{tr}(E_x L_\theta \rho_\theta)^* = \mathrm{tr}(L_\theta \rho_\theta E_x)^*$. The Fisher information therefore admits the following bound,

$$\begin{aligned} \mathcal{I}(\rho_\theta, E) &\stackrel{(12.29)}{=} \int dx \frac{\mathrm{Re}(\mathrm{tr}(L_\theta \rho_\theta E_x))^2}{\mathrm{tr}(\rho_\theta E_x)} \leq \int dx \left| \frac{\mathrm{tr}(L_\theta \rho_\theta E_x)}{\sqrt{\mathrm{tr}(\rho_\theta E_x)}} \right|^2 = \int dx \left| \underbrace{\mathrm{tr}\left(\frac{\sqrt{\rho_\theta} \sqrt{E_x}}{\sqrt{\mathrm{tr}(\rho_\theta E_x)}}\right)}_{=A} \underbrace{\sqrt{E_x} L_\theta \sqrt{\rho_\theta}}_{=B} \right|^2 \\ &\stackrel{\text{CS-ineq}}{\leq} \int dx \underbrace{\mathrm{tr}(A^\dagger A)}_{=1} \times \underbrace{\mathrm{tr}(B^\dagger B)}_{=\mathrm{tr}(L_\theta \rho_\theta L_\theta E_x)} = \mathrm{tr}\left(L_\theta \rho_\theta L_\theta \underbrace{\int dx E_x}_{=1}\right) = \mathrm{tr}(L_\theta^2 \rho_\theta). \end{aligned} \quad (12.34)$$

In the last line, we have used the Cauchy-Schwarz inequality for operators and then the completeness relation for POVMs. The bound on the right-hand-side is independent of the POVM and therefore applies to the quantum Fisher information. It is saturated by choosing $\{E_x\}_x$ as the eigenbasis of L_θ since this saturates the Cauchy-Schwarz inequality.

However, the bound in Eq (12.34) still depends on the symmetric logarithmic derivative. We can substitute it with a closed expression. To this end, motivated by Eq (12.32), let us define $\Omega(A) = \frac{1}{2} A \rho_\theta + \frac{1}{2} \rho_\theta A$. In Exercise C.4.2 it is shown that the inverse of Ω is

$$\Omega^{-1}(A) = \sum_{j,k} \frac{2}{\lambda_j + \lambda_k} A_{jk} |\psi_j\rangle \langle \psi_k|, \quad (12.35)$$

where $\rho_\theta = \sum_l \lambda_l |\psi_l\rangle \langle \psi_l|$ is the spectral decomposition of ρ_θ and $A_{jk} = \langle \psi_j | A | \psi_k \rangle$. By the definition of the symmetric logarithmic derivative in Eq (12.32) we have $\Omega(L_\theta) = \dot{\rho}_\theta$. Using Eq (12.35), we invert this to $L_\theta = \Omega^{-1}(\dot{\rho}_\theta)$. Inserting this in Eq (12.34) we obtain $\mathcal{I}^Q(\rho_\theta) = \mathrm{tr}(\Omega^{-1}(\dot{\rho}_\theta)^2 \rho_\theta)$. In Exercise C.4.2 we show that simplifying this expression gives the final result in Eq (12.31). $\square$

The quantum Fisher information has properties that are analogous to the ordinary Fisher information. Firstly, it is by definition non-negative. Secondly, it is additive in the sense that

$$\mathcal{I}^Q(\rho_\theta \otimes \sigma_\theta) = \mathcal{I}^Q(\rho_\theta) + \mathcal{I}^Q(\sigma_\theta). \quad (12.36)$$

Also, the quantum Fisher information is convex,

$$\mathcal{I}^Q(\lambda \rho_\theta + (1 - \lambda) \sigma_\theta) \leq \lambda \mathcal{I}^Q(\rho_\theta) + (1 - \lambda) \mathcal{I}^Q(\sigma_\theta), \quad (12.37)$$

for any θ -independent parameter $\lambda \in [0, 1]$. These properties are proven in Exercise C.4.3.

When Λ_θ is unitary, convexity implies that the maximal value of the quantum Fisher information taken over all density matrices is achieved for a pure state, $\rho = |\psi\rangle \langle \psi|$. The unitary encoding maps this to another pure state, $\rho_\theta = |\psi_\theta\rangle \langle \psi_\theta|$. It is therefore interesting to consider the quantum Fisher information for pure states. As shown in Exercise C.4.3, pure states simplify the expression in Eq (12.31) to

$$\mathcal{I}^Q(\psi_\theta) = 4 \left(\langle \dot{\psi}_\theta | \dot{\psi}_\theta \rangle - |\langle \dot{\psi}_\theta | \psi_\theta \rangle|^2 \right). \quad (12.38)$$

Unitary channels can be written as $U = e^{-i\theta H}$, where H is a Hermitian operator that can be thought of as the system's Hamiltonian. Our estimation task now is to learn the angle θ . Using the Schrödinger equation, namely $|\dot{\psi}\rangle = -iH|\psi\rangle$, the quantum Fisher information becomes

$$\mathcal{I}^Q(\psi_\theta) \stackrel{(12.38)}{=} 4 \left(\langle \psi | H^2 | \psi \rangle - \langle \psi | H | \psi \rangle^2 \right) = 4 \mathrm{Var}[H]. \quad (12.39)$$

Inserting this in the quantum version of the Cramér-Rao bound in Eq (12.30), we obtain

$$\mathrm{Var}[\hat{\theta}] \mathrm{Var}[H] \geq \frac{1}{4}. \quad (12.40)$$

This is an uncertainty relation between the estimator and the generator of the transformation that encodes the parameter. It is reminiscent of Robertson's uncertainty relation from Chapter 3.1.

12.4. The Heisenberg limit

Entanglement is the key quantum resource that will allow us to enhance parameter estimation beyond the standard quantum limit encountered in Eq (12.27). In other words, entanglement lets us achieve a better scaling in the variance than $\sim \frac{1}{N}$. To see how entanglement helps, consider that we prepare the GHZ state that we first encountered in Chapter 4.6. For N -partite systems it is defined as $|GHZ\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + |1\rangle^{\otimes N})$. We independently send each qubit through a unitary channel of the form $U_\theta = e^{i\frac{\theta}{2}}e^{-i\frac{\theta}{2}\sigma_z} = |0\rangle\langle 0| + e^{i\theta}|1\rangle\langle 1|$. Therefore, the state after the qubits are sent through the channel becomes

$$|\psi_\theta\rangle = (U_\theta \otimes \dots \otimes U_\theta) |GHZ\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + e^{iN\theta}|1\rangle^{\otimes N}). \quad (12.41)$$

Consider now that we measure all the N qubits separately using the observable $\sigma_X^{\otimes N}$. This effectively corresponds to measuring σ_X on each qubit, obtaining the outcome $a_i \in \{+1, -1\}$ from the measurement on the i 'th qubit and finally outputting the parity $a = \prod_i a_i$. The probability distribution over positive and negative parity becomes

$$p_\pm = \frac{1}{2} (1 \pm \langle \psi_\theta | \sigma_X^{\otimes N} | \psi_\theta \rangle) = \frac{1}{2} \left(1 \pm \frac{e^{iN\theta} + e^{-iN\theta}}{2} \right) = \frac{1 \pm \cos(N\theta)}{2}. \quad (12.42)$$

We compute the Fisher information of this distribution from Eq (12.19). Using that $\left| \frac{dp_\pm}{d\theta} \right| = \left| \frac{dp_-}{d\theta} \right|$, we obtain

$$\mathcal{I}(p_\pm) = \frac{1}{p_+} \left(\frac{dp_+}{d\theta} \right)^2 + \frac{1}{p_-} \left(\frac{dp_-}{d\theta} \right)^2 = \left(\frac{dp_+}{d\theta} \right)^2 \left(\frac{1}{p_+} + \frac{1}{p_-} \right) = \frac{4}{\sin^2(N\theta)} \times \frac{1}{4} N^2 \sin^2(N\theta) = N^2. \quad (12.43)$$

Furthermore, we would like to check that no larger Fisher information is possible. To that end, we must compute the quantum Fisher information. From Eq (12.34) we have that $\mathcal{I}^Q = \text{tr}(L_\theta^2 \rho_\theta)$, where $\rho_\theta = |\psi_\theta\rangle\langle\psi_\theta|$. Since ρ_θ is pure, we can determine the symmetric logarithmic derivative directly from Eq (12.32) as

$$L_\theta \rho_\theta + \rho_\theta L_\theta = 2 \frac{d\rho_\theta}{d\theta} \stackrel{\text{pure state}}{=} 2 \frac{d\rho_\theta^2}{d\theta} = 2(\dot{\rho}_\theta \rho_\theta + \rho_\theta \dot{\rho}_\theta) \Rightarrow L_\theta = 2\dot{\rho}_\theta. \quad (12.44)$$

Using the encoded state (12.41), we calculate that $\dot{\rho}_\theta^2 = \frac{N^2}{4} ((|0\rangle\langle 0|)^{\otimes N} + (|1\rangle\langle 1|)^{\otimes N})$. Hence, the quantum Fisher information is

$$\mathcal{I}^Q \stackrel{(12.34)}{=} \text{tr}(L_\theta^2 \rho_\theta) \stackrel{(12.44)}{=} 4 \text{tr}(\dot{\rho}_\theta^2 \rho_\theta) = N^2, \quad (12.45)$$

which is identical to Eq (12.43). We conclude that our measurement of the parity, $\sigma_X^{\otimes N}$, is optimal for parameter estimation. Thanks to our use of entanglement, we have achieved a quadratic scaling in N for the Fisher information, corresponding to the variance scaling as $\frac{1}{N^2}$. This is called the *Heisenberg limit*. It outperforms the $\frac{1}{N}$ scaling for the variance that we encountered for classical systems through the standard quantum limit.

Chapter summary.

- Estimation theory is the study of how to infer an unknown parameter from observed data. The Cramér-Rao bound imposes a limit on the best achievable precision for the estimation in terms of the Fisher information.
- For classical systems, the Fisher information scales linearly in the number of independent and identically distributed data samples.
- The use of multipartite entanglement makes possible a quadratic scaling for the Fisher information in the number of independent and identically distributed samples. This constitutes a quantum advantage in parameter estimation.

Further reading. For an extensive introduction, see for example the review paper [Quantum metrology from a quantum information science perspective](#) by Tóth and Apellaniz and Chapter 24 in [Modern Quantum Theory](#), by Bertlmann & Friis.

Part IV

Exercises

A. BASICS OF QUANTUM THEORY - EXERCISES

A.1. Elementary concepts

A.1.1 Trace ●

Recall that $\text{tr}(A) = \sum_i \langle i|A|i \rangle$. Prove the following properties:

- **Linearity**

(a) $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$,

(b) $\text{tr}(cA) = c \text{tr}(A)$,

- **Cyclicity**

$\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$,

- **Transpose**

$\text{tr}(A^T) = \text{tr}(A)$,

- **Composition**

$\text{tr}(P_A \otimes Q_B) = \text{tr}(P_A) \text{tr}(Q_B)$.

Solution (p. 122)

A.1.2 Hermitian matrices ●●●

(a) Consider an Hermitian matrix $A = A^\dagger$. Show that its eigenvalues are real.

(b) Let the spectral decomposition be $A = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$. Show that eigenvalues λ_j and λ_k , such that $\lambda_j \neq \lambda_k$, correspond to orthogonal eigenvectors $|\lambda_j\rangle$ and $|\lambda_k\rangle$.

(c) Compute the spectral decomposition for each of the Pauli matrices σ_X , σ_Y and σ_Z .

Solution (p. 122)

A.1.3 Qubit unitaries ●

Let $\hat{n} = (n_X, n_Y, n_Z)$ be a unit Bloch vector and let $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$ be the vector of Pauli matrices. Show that

$$U = e^{i\varphi} (\cos \theta \mathbb{1} + i \sin \theta \hat{n} \cdot \vec{\sigma}), \quad (\text{A.1})$$

is a unitary operator and that it can be interpreted as a rotation on the Bloch sphere.

Solution (p. 124)

A.1.4 Orthogonal states on the Bloch sphere ●

Let $|\phi_\pm\rangle = \cos \frac{\theta_\pm}{2} |0\rangle + e^{i\varphi_\pm} \sin \frac{\theta_\pm}{2} |1\rangle$ be two pure qubit states with Bloch vectors $\hat{m}_\pm = (\sin \theta_\pm \cos \varphi_\pm, \sin \theta_\pm \sin \varphi_\pm, \cos \theta_\pm)$. Assume that the states are orthogonal, namely $\langle\phi_+|\phi_-\rangle = 0$. Show that their Bloch vectors satisfy $\hat{m}_- = -\hat{m}_+$.

Solution (p. 125)

A.1.5 Probabilities on the Bloch sphere ●

Let ρ be a qubit state with Bloch vector $\vec{n}$ and let $\{|\phi_+\rangle, |\phi_-\rangle\}$ be a basis whose states are associated with unit-length Bloch vectors $\pm\hat{n}$. Show that the probability for the two possible outcomes is

$$p(\pm) = \frac{1 \pm \vec{n} \cdot \hat{n}}{2}. \quad (\text{A.2})$$

Solution (p. 125)

A.1.6 Dimension of quantum state space ●●

A pure qubit state, $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$, is characterised by two real parameters, θ and φ . From the Bloch sphere representation, we know that a mixed qubit state has also a third parameter, namely the length of the Bloch vector. Now, consider a quantum state in a d -dimensional Hilbert space.

(a) How many real parameters are needed to describe a pure d -dimensional quantum state? How many real parameters are needed for a pure state of m qubits?

(b) How many real parameters are needed to describe a mixed d -dimensional quantum state?

Hint: Density matrices are positive-semidefinite, Hermitian and have unit trace. Use this to count the parameters.

Solution (p. 125)

A.2. Purification and distance measures

A.2.1 Purification ●

Find all purifications of the maximally mixed state ρ in dimension d ,

$$\rho = \frac{1}{d} \sum_{i=1}^d |\lambda_i\rangle\langle\lambda_i|, \quad (\text{A.3})$$

for an arbitrary basis $\{|\lambda_i\rangle\}_i$ of the Hilbert space $\mathcal{H}_A$.

Solution (p. 126)

A.2.2 Trace-distance ●●●

The trace-distance between two arbitrary quantum states ρ, σ over $\mathcal{H}$ is defined as

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1, \quad (\text{A.4})$$

or equivalently as

$$D(\rho, \sigma) = \max_{\Pi} \text{tr}(\Pi(\rho - \sigma)), \quad (\text{A.5})$$

for any arbitrary operator Π such that $0 \leq \Pi \leq \mathbb{1}$.

(a) Show that the trace-distance satisfies properties 3, 4 and 5 from Chapter 2.2, namely

- $D(\rho, \sigma) \leq D(\rho, \tau) + D(\tau, \sigma), \quad \forall \rho, \sigma, \tau;$
- $D(\rho, \sigma) = D(U\rho U^\dagger, U\sigma U^\dagger), \quad \forall \rho, \sigma \text{ and any unitary } U;$
- $D(\rho \otimes \tau, \sigma \otimes \tau) = D(\rho, \sigma), \quad \forall \rho, \sigma, \tau;$

(b) For any positive-semidefinite operator A and any projector Π , show that $\text{tr}(\Pi A) \leq \text{tr}(A)$.

(c) Let $\mathcal{E}$ be a linear map with the property $\text{tr}(\mathcal{E}(\rho)) = \text{tr}(\rho)$ for any state ρ . Use the result derived in (a) and the two definitions of trace-distance (A.4) and (A.5) to show that

$$D(\mathcal{E}(\rho), \mathcal{E}(\sigma)) \leq D(\rho, \sigma). \quad (\text{A.6})$$

Hint: Decompose $\rho - \sigma$ into a positive and negative part and use that their traces are equal in magnitude.

Solution (p. 126)

A.2.3 Properties of fidelity ●●●

Prove that fidelity satisfies properties 2, 4 and 5 from Chapter 2.2, namely

- (a) $F(\rho, \sigma) = F(\sigma, \rho), \quad \forall \rho, \sigma;$
- (b) $F(\rho, \sigma) = F(U\rho U^\dagger, U\sigma U^\dagger), \quad \forall \rho, \sigma \text{ and any unitary } U;$
- (c) $F(\rho \otimes \tau, \sigma \otimes \tau) = F(\rho, \sigma), \quad \forall \rho, \sigma, \tau.$

Solution (p. 128)

A.2.4 Fidelity is not a proper distance measure ●

Using the fidelity defined in Chapter 2.2, it is appealing to think of $d(\rho, \sigma) = 1 - F(\rho, \sigma)$ as a distance. However, this is not possible since $d(\rho, \sigma)$ does not satisfy the triangle inequality. Consider the states $|\psi\rangle = |0\rangle$, $|\phi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$ and $|\varphi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$ and verify that $d(\psi, \varphi) \not\leq d(\psi, \phi) + d(\phi, \varphi)$.

Solution (p. 129)

A.2.5 A useful inequality ●

Use polar decomposition to show that $|\text{tr}(AU)| \leq \text{tr}|A|$ for any operator A and unitary U .

Solution (p. 129)

A.2.6 Fuchs - van de Graaf inequality ●●●●

Trace-distance and fidelity between two quantum states ρ, σ over $\mathcal{H}$, with $\dim(\mathcal{H}) = d$, are related by the Fuchs - van de Graaf inequalities,

$$1 - \sqrt{F(\rho, \sigma)} \leq D(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)}. \quad (\text{A.7})$$

Through several steps, we will prove the first of these two inequalities.

- (a) Given two probability distributions p, q with values $\{p_i, q_i\}$ where $i \in \{1, \dots, n\}$, show that

$$\|p - q\|_1 \geq 2 - 2 \sum_{i=1}^n \sqrt{p_i q_i}, \quad (\text{A.8})$$

using the definition $\|p - q\|_1 = \sum_{i=1}^n |p_i - q_i|$.

Hint: use that $|a - b| \geq (\sqrt{a} - \sqrt{b})^2$, for $a, b \geq 0$.

- (b) Consider the case where ρ, σ are invertible. Show that for the operator

$$Y = \rho^{-\frac{1}{2}} \sqrt{\rho^{\frac{1}{2}} \sigma \rho^{\frac{1}{2}}} \rho^{-\frac{1}{2}}, \quad (\text{A.9})$$

it holds that $\sqrt{F(\rho, \sigma)} = \text{tr}(Y \rho)$ where $\sigma = Y \rho Y$.

- (c) Consider the spectral decomposition of Y ,

$$Y = \sum_{i=1}^d \lambda_i |\lambda_i\rangle \langle \lambda_i|, \quad (\text{A.10})$$

where $\{\lambda_i \geq 0\}_i$ and $\{|\lambda_i\rangle\}_i$ are the eigenvalues and eigenvectors, respectively. Define $p_i = \langle \lambda_i | \rho | \lambda_i \rangle$ and $q_i = \langle \lambda_i | \sigma | \lambda_i \rangle$, and show that

$$\sum_{i=1}^d \sqrt{p_i} \sqrt{q_i} = \sqrt{F(\rho, \sigma)}. \quad (\text{A.11})$$

- (d) Use the trace-preserving quantum operation

$$\mathcal{E}(X) = \sum_{i=1}^d \langle \lambda_i | X | \lambda_i \rangle |\lambda_i\rangle \langle \lambda_i|, \quad (\text{A.12})$$

and the property (A.6) to prove that

$$\|\rho - \sigma\|_1 \geq 2 - 2\sqrt{F(\rho, \sigma)}, \quad (\text{A.13})$$

which is the first inequality of (A.7).

Solution (p. 129)

A.2.7 Fidelity vs trace-distance ●●●●

Consider that we have three states, ψ , φ_1 and φ_2 . We will show that the fidelity and the trace-distance can disagree on whether ψ is closer to φ_1 or to φ_2 .

- (a) Show that for qubit states, the trace-distance is

$$D(\psi, \varphi) = \frac{1}{2} |\hat{n}_\psi - \hat{n}_\varphi|, \quad (\text{A.14})$$

where $\hat{n}_\psi$ and $\hat{n}_\varphi$ are the Bloch vectors of the qubit state ψ and φ , respectively.

- (b) Consider the pure qubit state $\psi = \frac{1+\sigma_X}{2}$ and show that the fidelity with a state φ defined by the Bloch vector $\hat{n}_\varphi = (n_X^{(\varphi)}, n_Y^{(\varphi)}, n_Z^{(\varphi)})$ is

$$F(\psi, \varphi) = \frac{1}{2} (1 + n_X^{(\varphi)}). \quad (\text{A.15})$$

- (c) Define $d(\rho, \sigma) = 1 - F(\rho, \sigma)$ and consider the two states $\varphi_1 = \frac{1}{2} (\mathbb{1} - \frac{1}{4} \sigma_X)$ and $\varphi_2 = \frac{1}{2} (\mathbb{1} + \sigma_Z)$. Show that $d(\psi, \varphi_1) > d(\psi, \varphi_2)$, while $D(\psi, \varphi_1) < D(\psi, \varphi_2)$, i.e. the fidelity sees ψ closer to φ_2 than φ_1 , while the trace-distance sees the opposite.

Solution (p. 130)

A.3. Quantum measurements

A.3.1 Qubit manipulation ●●●●

Consider a bipartite two-qubit system in the state $|\psi\rangle = |00\rangle$.

- (a) Compute the state $|\psi_1\rangle = (H \otimes \mathbb{1})|\psi\rangle$ obtained from applying the local unitary

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (\text{A.16})$$

- (b) Consider the two-qubit unitary (known as a CNOT gate)

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (\text{A.17})$$

What is the state $|\psi_2\rangle = \text{CNOT}|\psi_1\rangle$?

- (c) Apply to the second qubit the measurement σ_X . What is the two-qubit state, $|\psi_3\rangle$, after the measurement if the outcome was “+1”?
- (d) Trace out the second qubit from the state $|\psi_3\rangle$ to determine the state of the first qubit.

Solution (p. 131)

A.3.2 Basis measurements are not enough ●

Prove that there exists no basis measurement (i.e. no rank-1 projective measurement) which when applied to every state returns $p(0) = 1$ and $p(1) = 0$. Find a non-projective POVM that can achieve this.

Solution (p. 132)

A.3.3 Binary qubit measurements ●●

In Chapter 3.2, we have seen that every qubit POVM-element can be written in the form $E_a = \lambda_a (\mathbb{1} + \vec{n}_a \cdot \vec{\sigma})$, where $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$ is the vector of the Pauli matrices, $0 \leq \lambda_a \leq 1$ and some Bloch vector $|\vec{n}_a| \leq 1$.

- (a) If there are only two possible outcomes, what are the conditions on $\{\lambda_a\}_a$ and $\{\vec{n}_a\}_a$ so that we obtain a valid POVM? What is the relative orientation of the two Bloch vectors?
- (b) Show that any two-outcome qubit POVM can be written as a convex combination of projective measurements.

Solution (p. 132)

A.3.4 Isoceles POVM ●

Consider a three-outcome qubit POVM represented by an isosceles triangle in the XZ -plane of the Bloch sphere. Its Bloch vectors are given by

$$\hat{n}_1 = (1, 0, 0), \quad \hat{n}_2 = (\cos \theta, 0, \sin \theta), \quad \hat{n}_3 = (\cos \theta, 0, -\sin \theta), \quad (\text{A.18})$$

for some $\theta \in (0, \pi)$. The qubit POVM-elements can be written $E_a = \lambda_a (\mathbb{1} + \hat{n}_a \cdot \vec{\sigma})$. Determine the coefficients $\{\lambda_a\}_a$ such that $\{E_a\}_a$ becomes a POVM. For which values of θ is it possible to construct a POVM?

Solution (p. 133)

A.3.5 POVM vs projective measurement ●●●

On a given Hilbert space, we can think of a POVM, $\{E_a\}_a$, as being reducible to projective measurements if it can be written as a convex combination of them. However, some POVMs cannot be reduced to projective measurements. Consider the three-outcome qubit POVM $\{E_a\}_{a=1}^3$ characterised by the Bloch vectors

$$\vec{n}_1 = (1, 0, 0), \quad \vec{n}_2 = \left(-\frac{1}{2}, 0, \frac{\sqrt{3}}{2}\right), \quad \vec{n}_3 = \left(-\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}\right), \quad (\text{A.19})$$

where $E_a = \lambda_a(\mathbb{1} + \vec{n}_a \cdot \vec{\sigma})$ and $\lambda_1 = \lambda_2 = \lambda_3 = \frac{1}{3}$. We show that it cannot be reduced to projective measurements by proving that there exists a state discrimination problem where it outperforms any projective measurement. To this end, let $\psi_a = \frac{1}{2}(\mathbb{1} + \vec{n}_a \cdot \vec{\sigma})$ for $a = 1, 2, 3$ be three quantum states. Consider that we want to discriminate them using some measurement $\{M_a\}_a$. The average winning probability is

$$p_{\text{win}} = \frac{1}{3} \sum_{a=1}^3 \text{tr}(\psi_a M_a). \quad (\text{A.20})$$

- (a) Choose $\{M_a\}_{a=1}^3$ as $\{E_a\}_{a=1}^3$ and compute p_{win} .
- (b) Show that the operators $(\psi_1 - \psi_2)$, $(\psi_1 - \psi_3)$ and $(\psi_2 - \psi_3)$ have the same spectrum.
- (c) Choose $\{M_a\}_{a=1}^3$ as a projective measurement. Since qubit projective measurements, $\{\Pi_a\}_{a=1}^3$, only have two outcomes, one of the three possible outcomes must be associated with the zero-projector. For instance $M_1 = \Pi_1$, $M_2 = \Pi_2$, $M_3 = 0$. Then,

$$p_{\text{win}}^{\text{proj}} = \frac{1}{3} [\text{tr}(\psi_1 \Pi_1) + \text{tr}(\psi_2 \Pi_2)]. \quad (\text{A.21})$$

Compute the maximum winning probability for any $\{\Pi_1, \Pi_2\}$ and confirm that it is smaller than the result in (a). Would the result have changed if we had assigned the zero-projector to one of the other outcomes?

Solution (p. 133)

A.3.6 Symmetric informationally complete POVMs ●●●

Assume that there exists a set of d -dimensional rays $\{|\psi_a\rangle\}_{a=1}^{d^2}$ with the property that for any $i \neq j$

$$|\langle\psi_i|\psi_j\rangle|^2 = c. \quad (\text{A.22})$$

- (a) Define $\Pi_a = |\psi_a\rangle\langle\psi_a|$ and $E_a = \frac{1}{d}\Pi_a$ and assume that $\{E_a\}_a$ is a POVM. Show that $c = \frac{1}{d+1}$.
Hint: Square the completeness condition for measurements.
- (b) One can show that any $\{E_a\}_a$ with the above property is a POVM that spans the full space of density matrices. Verify that any density matrix can be decomposed in terms of the POVM elements as

$$\rho = \sum_{a=1}^{d^2} \left((d+1)p_a - \frac{1}{d} \right) \Pi_a, \quad (\text{A.23})$$

where $p_a = \text{tr}(E_a \rho)$.

Hint: Use that $\rho = \sum_j c_j \Pi_j$ for some coefficients $\{c_j\}_j$ since $\{\Pi_j\}_j$ span the Hermitian matrices.

Solution (p. 134)

A.3.7 Minimal error discrimination with biased inputs ●●●

Consider two quantum states $\{\rho_1, \rho_2\}$ that are chosen with probability $\{q_1, q_2\}$, respectively. Suppose that we want to discriminate between these two states using a two-outcomes measurement $\{E_1, E_2\}$, such that E_i is associated with guessing that the state is ρ_i . The maximal winning probability is given by the Helstrom bound,

$$p_{\text{win}} = \sum_{i=1}^2 q_i \text{tr}(E_i \rho_i) \leq \frac{1}{2} + \frac{1}{2} \|q_1 \rho_1 - q_2 \rho_2\|_1. \quad (\text{A.24})$$

Prove this result by following the steps below.

- (a) Define an operator $X = q_1 \rho_1 - q_2 \rho_2$ and show that p_{win} can be equivalently expressed in two ways as

$$p_{\text{win}} = \text{tr}(q_2 \rho_2 + E_1 X) = \text{tr}(q_1 \rho_1 - E_2 X). \quad (\text{A.25})$$

Therefore, we can express $p_{\text{win}} = \text{tr}(K)$, with the operator K having either of the two possible definitions

$$K = \begin{cases} q_2 \rho_2 + E_1 X, \\ q_1 \rho_1 - E_2 X. \end{cases} \quad (\text{A.26})$$

(b) Define the observable $E = E_1 - E_2$. Show that it is possible to express K in terms of E as

$$K = \frac{1}{2}(q_1 \rho_1 + q_2 \rho_2) + \frac{1}{2}EX, \quad (\text{A.27})$$

(c) Using (A.27), derive the Helstrom bound in (A.24). What do you get when choosing $q_1 = q_2 = \frac{1}{2}$?

Hint: What is the operator E that maximises $\text{tr}(EX)$? How is the trace-norm defined?

Solution (p. 135)

A.3.8 Unambiguous discrimination POVM ●

Consider unambiguous state discrimination of the two pure qubit states $|\psi\rangle$ and $|\phi\rangle$. In Chapter 3.3, we proposed to use the following three-outcome POVM

$$E_0 = \frac{\mathbb{1} - |\phi\rangle\langle\phi|}{1 + |\langle\psi|\phi\rangle|}, \quad E_1 = \frac{\mathbb{1} - |\psi\rangle\langle\psi|}{1 + |\langle\psi|\phi\rangle|}, \quad E_\emptyset = \mathbb{1} - E_0 - E_1. \quad (\text{A.28})$$

Prove that this is a valid POVM.

Hint: Use that $|\psi\rangle$ and $|\phi\rangle$ span a two-dimensional subspace and parameterise them appropriately when calculating the necessary eigenvalues.

Solution (p. 136)

A.3.9 Pretty good state discrimination ●●

Consider that we are randomly given one of n possible states $\rho_1, \dots, \rho_n$ and that we are asked to guess which state it was. Finding the best solution is in general hard but a pretty good strategy can easily be constructed. Let us define $S = \frac{1}{n} \sum_{a=1}^n \rho_a$ and $E_a = \frac{1}{n} S^{-1/2} \rho_a S^{-1/2}$.

(a) Show that $\{E_a\}_{a=1}^n$ is a POVM.

(b) Consider the case $\rho_1 = |0\rangle\langle 0|$, $\rho_2 = |1\rangle\langle 1|$ and $\rho_3 = |+\rangle\langle +|$, where $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$. Compute the $\{E_a\}_a$ and the success probability of the state discrimination $p_{\text{win}} = \frac{1}{3} \sum_a \text{tr}(\rho_a E_a)$.

Solution (p. 136)

A.3.10 Naimark dilation ●

Consider the three-outcome qubit POVM $\{E_a\}_{a=1}^3$ where $E_a = \frac{1}{3}(\mathbb{1} + \vec{n}_a \cdot \vec{\sigma})$ and

$$\vec{n}_1 = (1, 0, 0), \quad \vec{n}_2 = \frac{1}{2}(-1, 0, \sqrt{3}), \quad \vec{n}_3 = \frac{1}{2}(-1, 0, -\sqrt{3}), \quad (\text{A.29})$$

Construct a Naimark dilation of this POVM by adding an ancilla and finding a suitable global unitary, U , such that $E_a = \text{tr}_B(U^\dagger (\mathbb{1} \otimes |a\rangle\langle a|) U)$.

Solution (p. 137)

BASICS OF QUANTUM THEORY - SOLUTIONS

A.1. Elementary concepts

A.1.1 Solution (Text at p. 116)

- **Linearity**

(a) Consider two operators $A = \sum_{k,l} a_{kl} |k\rangle\langle l|$ and $B = \sum_{p,q} b_{pq} |p\rangle\langle q|$ expressed in the same basis $\{|k\rangle\}_{k=0}^{d-1}$ of a d -dimensional Hilbert space. Then,

$$\text{tr}(A + B) = \text{tr} \left(\sum_{k,l} a_{kl} |k\rangle\langle l| + \sum_{p,q} b_{pq} |p\rangle\langle q| \right) = \text{tr} \left(\sum_{k,l} (a_{kl} + b_{kl}) |k\rangle\langle l| \right), \quad (\text{A.30})$$

where we used the freedom to rename the indices $(p, q) \rightarrow (k, l)$. Therefore, we get

$$\text{tr}(A + B) = \sum_i \sum_{k,l} (a_{kl} + b_{kl}) \langle i|k\rangle\langle l|i\rangle = \sum_i (a_{ii} + b_{ii}) = \sum_i a_{ii} + \sum_i b_{ii} = \text{tr}(A) + \text{tr}(B). \quad (\text{A.31})$$

(b) Consider an operator $A = \sum_{k,l} a_{kl} |k\rangle\langle l|$ and a scalar c . Then,

$$\text{tr}(cA) = \text{tr} \left(c \sum_{k,l} a_{kl} |k\rangle\langle l| \right) = \sum_i \sum_{k,l} \langle i|c a_{kl} |k\rangle\langle l|i\rangle = c \sum_i a_{ii} = c \text{tr}(A). \quad (\text{A.32})$$

- **Cyclicity**

Consider two operators $A = \sum_{k,l} a_{kl} |k\rangle\langle l|$ and $B = \sum_{p,q} b_{pq} |p\rangle\langle q|$ expressed in the same basis $\{|k\rangle\}_{k=0}^{d-1}$ of a d -dimensional Hilbert space. Then,

$$\text{tr}(AB) = \text{tr} \left(\sum_{k,l} a_{kl} |k\rangle\langle l| \sum_{p,q} b_{pq} |p\rangle\langle q| \right) = \text{tr} \left(\sum_{k,p,q} a_{kp} b_{pq} |k\rangle\langle q| \right) = \sum_i \sum_{k,p,q} a_{kp} b_{pq} \langle i|k\rangle\langle q|i\rangle = \sum_{i,p} a_{ip} b_{pi}, \quad (\text{A.33})$$

where we can change the summation order and also write

$$\text{tr}(AB) = \sum_{i,p} a_{ip} b_{pi} = \sum_{p,i} b_{pi} a_{ip} = \text{tr}(BA). \quad (\text{A.34})$$

In general, it holds that

$$\text{tr}(A_1 A_2 \dots A_n) = \text{tr}(A_n A_1 A_2 \dots A_{n-1}) = \dots = \text{tr}(A_2 A_3 \dots A_{n-1} A_n A_1). \quad (\text{A.35})$$

- **Transpose**

Given an operator $A = \sum_{k,l} a_{kl} |k\rangle\langle l|$, its transpose is $A^T = \sum_{k,l} a_{lk} |k\rangle\langle l|$. Therefore, the trace is

$$\text{tr}(A^T) = \text{tr} \left(\sum_{k,l} a_{lk} |k\rangle\langle l| \right) = \sum_i \sum_{k,l} a_{lk} \langle i|k\rangle\langle l|i\rangle = \sum_i a_{ii} = \text{tr}(A). \quad (\text{A.36})$$

- **Composition**

Consider an operator $H = P_A \otimes Q_B$ living in the bipartite system $\mathcal{H}_A \otimes \mathcal{H}_B$. Define $\{|\alpha_k\rangle\}_{k=1}^{d_A}$ and $\{|\beta_l\rangle\}_{l=1}^{d_B}$ as some orthonormal bases of $\mathcal{H}_A$ and $\mathcal{H}_B$ respectively, then $\{|\alpha_k\rangle \otimes |\beta_l\rangle \equiv |\alpha_k \beta_l\rangle\}_{k,l}$ is an orthonormal basis for $\mathcal{H}_A \otimes \mathcal{H}_B$. Therefore, the trace on $\mathcal{H}_A \otimes \mathcal{H}_B$ can be written as

$$\text{tr}(H) = \sum_{k=1}^{d_A} \sum_{l=1}^{d_B} \langle \alpha_k \beta_l | P \otimes Q | \alpha_k \beta_l \rangle = \sum_{k=1}^{d_A} \langle \alpha_k | P | \alpha_k \rangle \sum_{l=1}^{d_B} \langle \beta_l | Q | \beta_l \rangle = \text{tr}(P) \text{tr}(Q). \quad (\text{A.37})$$

A.1.2 Solution (Text at p. 116)

- (a) Let λ be an eigenvalue of A and let $|\lambda\rangle$ be the corresponding eigenvector. Then

$$\langle \lambda | A | \lambda \rangle = \lambda \langle \lambda | \lambda \rangle = \lambda. \quad (\text{A.38})$$

Let us conjugate both sides of this equation. Since both sides are numbers, we can equally well apply the conjugate-transpose operation to both sides. This gives

$$\lambda^* = (\langle \lambda | A | \lambda \rangle)^\dagger = \langle \lambda | A^\dagger | \lambda \rangle = \langle \lambda | A | \lambda \rangle = \lambda, \quad (\text{A.39})$$

where we used that $A^\dagger = A$. Hence, λ is real.

- (b) Since $|\lambda_k\rangle$ is an eigenvector of A we have

$$\langle \lambda_j | \underbrace{A | \lambda_k \rangle}_{= \lambda_k | \lambda_k \rangle} = \lambda_k \langle \lambda_j | \lambda_k \rangle. \quad (\text{A.40})$$

Due to Hermiticity, we also have

$$\langle \lambda_j | A | \lambda_k \rangle = \langle \lambda_j | A^\dagger | \lambda_k \rangle = \langle \underbrace{A | \lambda_j \rangle}_{= \lambda_j | \lambda_j \rangle} | \lambda_k \rangle = \lambda_j \langle \lambda_j | \lambda_k \rangle. \quad (\text{A.41})$$

Therefore, from (A.41) and (A.40), we get

$$\lambda_k \langle \lambda_j | \lambda_k \rangle = \lambda_j \langle \lambda_j | \lambda_k \rangle \implies (\lambda_k - \lambda_j) \langle \lambda_j | \lambda_k \rangle = 0, \quad (\text{A.42})$$

which holds only if $\langle \lambda_j | \lambda_k \rangle = 0$, since $\lambda_j \neq \lambda_k$.

- (c) Using the definition of the Pauli matrices in (1.14), we notice that σ_Z is already in its diagonal form with eigenvalues $\lambda_+(\sigma_Z) = 1$ and $\lambda_-(\sigma_Z) = -1$. The corresponding eigenvectors are $|\lambda_+\rangle = [1, 0]^T$ and $|\lambda_-\rangle = [0, 1]^T$.

Using that the eigenvalues of a 2×2 matrix can be computed as

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longrightarrow \lambda_\pm = \frac{1}{2} \left[(a + d) \pm \sqrt{(a - d)^2 + 4bc} \right], \quad (\text{A.43})$$

we can compute the spectral decomposition of σ_X and σ_Y . Then,

$$\lambda_\pm(\sigma_X) = \pm \frac{1}{2} \sqrt{4} = \pm 1 \implies \begin{pmatrix} -\lambda_\pm(\sigma_X) & 1 \\ 1 & -\lambda_\pm(\sigma_X) \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies -\lambda_\pm(\sigma_X)x + y = 0, \quad (\text{A.44})$$

from which we get

$$y = \lambda_\pm(\sigma_X)x \implies |\lambda_\pm\rangle = x \begin{pmatrix} 1 \\ \lambda_\pm(\sigma_X) \end{pmatrix} = x \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad (\text{A.45})$$

where x can be derived from the normalisation condition, i.e.

$$1 = \langle \lambda_\pm | \lambda_\pm \rangle = |x|^2 \begin{pmatrix} 1 & \pm 1 \end{pmatrix} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix} = |x|^2 (1 + 1) = 2|x|^2 \implies |x| = \frac{1}{\sqrt{2}}. \quad (\text{A.46})$$

Therefore $|\lambda_\pm\rangle = \frac{1}{\sqrt{2}} [1, \pm 1]^T$. In the same way, we can compute the eigenvalues of σ_Y ,

$$\lambda_\pm(\sigma_Y) = \pm \frac{1}{2} \sqrt{4(-i)i} = \pm 1 \implies \begin{pmatrix} -\lambda_\pm(\sigma_Y) & -i \\ i & -\lambda_\pm(\sigma_Y) \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies -\lambda_\pm(\sigma_Y)x - iy = 0, \quad (\text{A.47})$$

from which we get

$$y = i\lambda_\pm(\sigma_Y)x \implies |\lambda_\pm\rangle = x \begin{pmatrix} 1 \\ i\lambda_\pm(\sigma_Y) \end{pmatrix} = x \begin{pmatrix} 1 \\ \pm i \end{pmatrix}, \quad (\text{A.48})$$

where x can be derived from the normalisation condition, i.e.

$$1 = \langle \lambda_\pm | \lambda_\pm \rangle = |x|^2 \begin{pmatrix} 1 & \mp i \end{pmatrix} \begin{pmatrix} 1 \\ \pm i \end{pmatrix} = |x|^2 (1 + 1) = 2|x|^2 \implies |x| = \frac{1}{\sqrt{2}}. \quad (\text{A.49})$$

Therefore $|\lambda_\pm\rangle = \frac{1}{\sqrt{2}} [1, \pm i]^T$.

A.1.3 Solution (Text at p. 116)

We first show that $U = e^{i\varphi} (\cos \theta \mathbb{1} + i \sin \theta \hat{n} \cdot \vec{\sigma})$ is a unitary matrix, i.e. that $U^\dagger U = U U^\dagger = \mathbb{1}$. Recall that the Pauli matrices $\sigma_X, \sigma_Y, \sigma_Z$ satisfy the following relations

$$[\sigma_i, \sigma_j] = i \epsilon_{ijk} \sigma_k, \quad \text{with } \epsilon_{ijk} \text{ being the Levi-Civita tensor,} \quad (\text{A.50})$$

$$\{\sigma_i, \sigma_j\} = 2 \mathbb{1} \delta_{ij}, \quad (\text{A.51})$$

$$\sigma_i^\dagger = \sigma_i. \quad (\text{A.52})$$

Therefore,

$$U^\dagger = e^{-i\varphi} (\cos \theta \mathbb{1} - i \sin \theta \hat{n} \cdot \vec{\sigma}^\dagger) = e^{-i\varphi} (\cos \theta \mathbb{1} - i \sin \theta \hat{n} \cdot \vec{\sigma}), \quad (\text{A.53})$$

from which

$$\begin{aligned} U^\dagger U &= [e^{-i\varphi} (\cos \theta \mathbb{1} - i \sin \theta \hat{n} \cdot \vec{\sigma})] [e^{i\varphi} (\cos \theta \mathbb{1} + i \sin \theta \hat{n} \cdot \vec{\sigma})] \\ &= \cos^2 \theta \mathbb{1} + \sin^2 \theta (\hat{n} \cdot \vec{\sigma})^2 + \underbrace{i \cos \theta \sin \theta \hat{n} \cdot \vec{\sigma} - i \cos \theta \sin \theta \hat{n} \cdot \vec{\sigma}}_{=0} \end{aligned} \quad (\text{A.54})$$

Using the anticommutation relation (A.51) we get

$$\begin{aligned} (\hat{n} \cdot \vec{\sigma})^2 &= \left(\sum_i n_i \sigma_i \right) \left(\sum_j n_j \sigma_j \right) = \sum_{i,j} n_i n_j \sigma_i \sigma_j = \sum_i n_i^2 \sigma_i^2 + \sum_{i \neq j} n_i n_j \sigma_i \sigma_j \\ &= \mathbb{1} + \sum_{i < j} n_i n_j \sigma_i \sigma_j + \sum_{j < i} n_i n_j \sigma_i \sigma_j = \mathbb{1} + \sum_{i < j} n_i n_j (\sigma_i \sigma_j + \sigma_j \sigma_i) = \mathbb{1} + \sum_{i < j} n_i n_j \underbrace{\{\sigma_i, \sigma_j\}}_{=0}, \end{aligned} \quad (\text{A.55})$$

where we relabeled the indices $(i, j) \rightarrow (j, i)$. Therefore,

$$U^\dagger U = \cos^2 \theta \mathbb{1} + \sin^2 \theta (\hat{n} \cdot \vec{\sigma})^2 = (\cos^2 \theta + \sin^2 \theta) \mathbb{1} = \mathbb{1}, \quad (\text{A.56})$$

which proves that U is unitary.

More formally, it is possible to show that the expression for U presented in Chapter 1.2 is general for all unitary operations in dimension $d = 2$. Indeed, consider the set of matrices in dimension $d = 2$ that are unitary and have unit determinant, i.e. $U U^\dagger = U^\dagger U = \mathbb{1}$ and $\det(U) = 1$. It can be shown that these matrices form the special unitary group $\text{SU}(2)$, with Lie algebra given by the Pauli matrices $\sigma_X, \sigma_Y, \sigma_Z$ that satisfy the relations (A.50), (A.51) and (A.52). Practically, this means that any unitary in dimension $d = 2$ can be expressed as

$$U = e^{i\theta \hat{n} \cdot \vec{\sigma}}, \quad (\text{A.57})$$

with $\hat{n} = (n_X, n_Y, n_Z)$ being a vector of real coefficients with $|\hat{n}|^2 = 1$, and $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$ being the vector of the Pauli matrices. We can expand the exponential as

$$U = e^{i\theta \hat{n} \cdot \vec{\sigma}} = \sum_{k=0}^{\infty} \frac{(i\theta \hat{n} \cdot \vec{\sigma})^k}{k!}. \quad (\text{A.58})$$

We can split the sum into even and odd exponents,

$$\sum_{k=0}^{\infty} \frac{(i\theta \hat{n} \cdot \vec{\sigma})^k}{k!} = \sum_{k=0}^{\infty} \frac{(i\theta \hat{n} \cdot \vec{\sigma})^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{(i\theta \hat{n} \cdot \vec{\sigma})^{2k+1}}{(2k+1)!}, \quad (\text{A.59})$$

and using (A.55) we get

$$\sum_{k=0}^{\infty} \frac{(i\theta \hat{n} \cdot \vec{\sigma})^{2k}}{(2k)!} = \sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k}}{(2k)!} = \cos \theta \mathbb{1}, \quad (\text{A.60})$$

$$\sum_{k=0}^{\infty} \frac{(i\theta \hat{n} \cdot \vec{\sigma})^{2k+1}}{(2k+1)!} = i \sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k+1}}{(2k+1)!} (\hat{n} \cdot \vec{\sigma}) = i (\hat{n} \cdot \vec{\sigma}) \sin \theta, \quad (\text{A.61})$$

where we used the Taylor expansions of cosine and sine. Therefore, the generic unitary $U \in \text{SU}(2)$ can be written as

$$U = e^{i\theta \hat{n} \cdot \vec{\sigma}} = \cos \theta \mathbb{1} + i(\hat{n} \cdot \vec{\sigma}) \sin \theta, \quad (\text{A.62})$$

which is the form of U presented in Chapter 1.2, up to a phase factor. This is the expression for a clockwise rotation of the angle 2θ around the $\hat{n}$ axis in the Bloch sphere. For example, fix $\hat{n} = (0, 1, 0)$. Then, we have

$$U = e^{i\theta \sigma_Y} = \cos \theta \mathbb{1} + i \sin \theta \sigma_Y = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}. \quad (\text{A.63})$$

Now consider the quantum state $|0\rangle$ along the z -axis. By applying the unitary U in (A.63) with $\theta = \frac{\pi}{4}$, we get

$$U|0\rangle = \begin{pmatrix} \cos \frac{\pi}{4} & \sin \frac{\pi}{4} \\ -\sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \cos \frac{\pi}{4} |0\rangle - \sin \frac{\pi}{4} |1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle, \quad (\text{A.64})$$

i.e. the unitary performs a clockwise rotation of $\frac{\pi}{2}$ around the y -axis in the Bloch sphere, bringing the state $|0\rangle$ to $|-\rangle$.

A.1.4 Solution (Text at p. 116)

Imposing the orthogonality of the states, we find

$$\begin{aligned} 0 = \langle \phi_+ | \phi_- \rangle &= \left(\cos \frac{\theta_+}{2} \langle 0| + e^{-i\varphi_+} \sin \frac{\theta_+}{2} \langle 1| \right) \left(\cos \frac{\theta_-}{2} |0\rangle + e^{i\varphi_-} \sin \frac{\theta_-}{2} |1\rangle \right) \\ &= \cos \frac{\theta_+}{2} \cos \frac{\theta_-}{2} \underbrace{\langle 0|0\rangle}_{=1} + e^{i(\varphi_- - \varphi_+)} \sin \frac{\theta_+}{2} \sin \frac{\theta_-}{2} \underbrace{\langle 1|1\rangle}_{=1} + \cos \frac{\theta_+}{2} e^{i\varphi_-} \sin \frac{\theta_-}{2} \underbrace{\langle 0|1\rangle}_{=0} + \cos \frac{\theta_-}{2} e^{-i\varphi_+} \sin \frac{\theta_+}{2} \underbrace{\langle 1|0\rangle}_{=0} \\ &= \cos \frac{\theta_+}{2} \cos \frac{\theta_-}{2} + e^{i(\varphi_- - \varphi_+)} \sin \frac{\theta_+}{2} \sin \frac{\theta_-}{2}, \end{aligned} \quad (\text{A.65})$$

where we used the orthonormality of the computational basis, $\langle i|j\rangle = \delta_{i,j}$. Since we are imposing that $\langle \phi_+ | \phi_- \rangle = 0$, the imaginary part in (A.65) must disappear, i.e. $e^{i(\varphi_- - \varphi_+)} = \pm 1$, which is possible only for $\varphi_- = \varphi_+$ or $\varphi_- = \varphi_+ + \pi$.

We get

$$\begin{aligned} \varphi_- = \varphi_+ &\rightarrow \langle \phi_+ | \phi_- \rangle = \cos \frac{\theta_+}{2} \cos \frac{\theta_-}{2} + \sin \frac{\theta_+}{2} \sin \frac{\theta_-}{2} = \cos \left(\frac{\theta_+ - \theta_-}{2} \right) = 0, \\ \varphi_- = \varphi_+ + \pi &\rightarrow \langle \phi_+ | \phi_- \rangle = \cos \frac{\theta_+}{2} \cos \frac{\theta_-}{2} - \sin \frac{\theta_+}{2} \sin \frac{\theta_-}{2} = \cos \left(\frac{\theta_+ + \theta_-}{2} \right) = 0, \end{aligned} \quad (\text{A.66})$$

where we have used the trigonometric identities, $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$. The first possibility in (A.66) implies that $\theta_- = \theta_+ \pm \pi$, which is not allowed since θ_- and θ_+ live in the range $[0, \pi]$. The second possibility implies that $\theta_- = \pi - \theta_+$, which is allowed in the range $[0, \pi]$. Inserting $\theta_- = \pi - \theta_+$ and $\varphi_- = \varphi_+ + \pi$ for the Bloch vector $\hat{m}_-$ we obtain

$$\begin{aligned} \hat{m}_- &= (\sin \theta_- \cos \varphi_-, \sin \theta_- \sin \varphi_-, \cos \theta_-) = (\sin(\pi - \theta_+) \cos(\varphi_+ + \pi), \sin(\pi - \theta_+) \sin(\varphi_+ + \pi), \cos(\pi - \theta_+)) \\ &= -(\sin \theta_+ \cos \varphi_+, \sin \theta_+ \sin \varphi_+, \cos \theta_+) = -\hat{m}_+, \end{aligned} \quad (\text{A.67})$$

where we have used the trigonometric relations $\cos(\pi \pm \alpha) = -\cos \alpha$ and $\sin(\pi \pm \alpha) = \mp \sin \alpha$.

A.1.5 Solution (Text at p. 116)

Using (1.39), we can represent both ρ and $|\phi_{\pm}\rangle$ in terms of their Bloch vectors. Therefore, from Born's rule,

$$p(\pm) = \text{tr}(\rho |\phi_{\pm}\rangle\langle\phi_{\pm}|) = \text{tr} \left(\frac{\mathbb{1} + \vec{n} \cdot \vec{\sigma}}{2} \frac{\mathbb{1} \pm \hat{m} \cdot \vec{\sigma}}{2} \right) = \frac{1}{4} \text{tr} (\mathbb{1} + \vec{n} \cdot \vec{\sigma} \pm \hat{m} \cdot \vec{\sigma} \pm (\vec{n} \cdot \vec{\sigma})(\hat{m} \cdot \vec{\sigma})) = \frac{1 \pm \vec{n} \cdot \hat{m}}{2}, \quad (\text{A.68})$$

where in the last step we have used that the Pauli operators are traceless and that $\text{tr}((\vec{n} \cdot \vec{\sigma})(\hat{m} \cdot \vec{\sigma})) = 2(\vec{n} \cdot \hat{m})$, which can be shown using the same approach followed in (A.55).

A.1.6 Solution (Text at p. 116)

- (a) Each pure d -dimensional quantum state is a complex vector $|\psi\rangle$ in a d -dimensional Hilbert space,

$$|\psi\rangle = \begin{pmatrix} z_1 \\ \vdots \\ z_d \end{pmatrix}, \quad \text{where } z_k = x_k + iy_k, \quad \forall k = 1, \dots, d. \quad (\text{A.69})$$

Therefore, $|\psi\rangle$ is characterised by $2d$ real parameters. By imposing the two constraints of normalisation and invariance under global phase factor, we are left with $2d - 2$ free parameters. For example, if $d = 2$ we recover $2(2) - 2 = 2$ free parameters, which corresponds to the pair of angles describing the surface of the Bloch sphere.

If we consider a pure state of m qubit, the density operator lives in a 2^m -dimensional Hilbert space. Therefore, the number of free real parameter is $2(2^m) - 2 = 2^{m+1} - 2$.

- (b) Mixed states are described by d -dimensional density operators. Complex matrices of size d have 2 real components for each element, i.e. $2d^2$ real parameters in total. From the properties characterising density operators (Hermitian, positive-semidefinite and unit trace) we get a set of constraints that fix some of the free parameters:

- Hermitian

$$\rho = \rho^\dagger \Leftrightarrow \rho_{ij} = \rho_{ji}^*, \quad \forall i, j. \quad (\text{A.70})$$

For the diagonal elements this means that ρ_{ii} are all real (d equations). For the off-diagonal elements, it is enough to consider only the lower-triangle part (since the upper-triangle part would give the same equations). The number of elements in this case is $\frac{d(d-1)}{2}$. However, since ρ_{ij} are complex numbers, we need to impose the condition (A.70) both on the real and imaginary parts, i.e.

$$\rho_{ij} = \rho_{ji}^* \implies 2 \frac{d(d-1)}{2} = d(d-1) \text{ equations}, \quad \forall i \neq j, \quad (\text{A.71})$$

$$\rho_{ii} = \rho_{ii}^* \implies d \text{ equations}, \quad \forall i. \quad (\text{A.72})$$

- Unit trace

$$\text{tr}(\rho) = 1 \Leftrightarrow \sum_i \rho_{ii} = 1 \longrightarrow 1 \text{ equation}. \quad (\text{A.73})$$

Therefore, a density operator in dimension d is characterised by

$$2d^2 - d - d(d-1) - 1 = d^2 - 1 \quad \text{free real parameters}. \quad (\text{A.74})$$

For example, mixed qubit states are described by $(2^2) - 1 = 3$ real free parameters, which is the dimension of the Bloch sphere.

A.2. Purification and distance measures

A.2.1 Solution (Text at p. 117)

Consider the state ρ in the Hilbert space $\mathcal{H}_A$ and introduce an ancillary system B . Let $\{|i\rangle\}_i$ be an orthonormal basis for $\mathcal{H}_B$. We can construct any other possible basis applying a unitary U , i.e. $\{U|i\rangle\}_i$. Since ρ is maximally mixed, its eigenvalues are $\lambda_i = \frac{1}{d}$ for $i = 1, \dots, d$. Therefore, following Chapter 2.1, all possible purifications $|\psi^{(U)}\rangle_{AB}$ of the state ρ are

$$|\psi^{(U)}\rangle_{AB} = \sum_{i=1}^d \sqrt{\lambda_i} |\lambda_i\rangle_A \otimes U|i\rangle_B = \frac{1}{\sqrt{d}} \sum_{i=1}^d |\lambda_i\rangle_A \otimes U|i\rangle_B. \quad (\text{A.75})$$

A.2.2 Solution (Text at p. 117)

- (a) • By adding and subtracting an arbitrary density operator τ , we get

$$\begin{aligned} D(\rho, \sigma) &= \max_{\Pi} \text{tr}(\Pi(\rho - \sigma)) = \max_{\Pi} \text{tr}(\Pi(\rho - \tau + \tau - \sigma)) = \max_{\Pi} (\text{tr}(\Pi(\rho - \tau)) + \text{tr}(\Pi(\tau - \sigma))), \\ &\leq \max_{\Pi} \text{tr}(\Pi(\rho - \tau)) + \max_{\Pi} \text{tr}(\Pi(\tau - \sigma)) = D(\rho, \tau) + D(\tau, \sigma), \end{aligned} \quad (\text{A.76})$$

where in the first inequality we split the maximisation into two separate maximisations and in the last step we identified the definition of the trace-distance.

- For any unitary U we can write

$$D(U\rho U^\dagger, U\sigma U^\dagger) = \frac{1}{2} \|U\rho U^\dagger - U\sigma U^\dagger\|_1 = \frac{1}{2} \|U(\rho - \sigma)U^\dagger\|_1 = \sum_i |\lambda_i(U(\rho - \sigma)U^\dagger)|, \quad (\text{A.77})$$

where we used the definition of the trace-norm, i.e. $\|A\|_1 = \text{tr} \sqrt{A^\dagger A} = \sum_i |\lambda_i(A)|$, with $\{\lambda_i(A)\}$ being the eigenvalues of A for any square matrix A . The spectrum of any matrix A is invariant under unitary transformations since

$$A = \sum_i \lambda_i(A) |\lambda_i\rangle \langle \lambda_i| \implies UAU^\dagger = \sum_i \lambda_i(A) U|\lambda_i\rangle \langle \lambda_i|U^\dagger, \quad (\text{A.78})$$

then $\sum_i |\lambda_i(A)| = \sum_i |\lambda_i(UAU^\dagger)|$, from which we get

$$\sum_i |\lambda_i(U(\rho - \sigma)U^\dagger)| = \sum_i |\lambda_i(\rho - \sigma)| \implies D(U\rho U^\dagger, U\sigma U^\dagger) = D(\rho, \sigma). \quad (\text{A.79})$$

- Using the definition of the trace-distance, we can write

$$D(\rho \otimes \tau, \sigma \otimes \tau) = \frac{1}{2} \|\rho \otimes \tau - \sigma \otimes \tau\|_1 = \frac{1}{2} \|(\rho - \sigma) \otimes \tau\|_1 = \frac{1}{2} \|(\rho - \sigma)\|_1 \underbrace{\|\tau\|_1}_{=1} = D(\rho, \sigma). \quad (\text{A.80})$$

In the third step we have used that for arbitrary operators A and B it holds that

$$\|A \otimes B\|_1 = \text{tr} \sqrt{(A \otimes B)^\dagger (A \otimes B)} = \text{tr} \sqrt{A^\dagger A \otimes B^\dagger B} = \text{tr} \sqrt{A^\dagger A} \text{tr} \sqrt{B^\dagger B} = \|A\|_1 \|B\|_1. \quad (\text{A.81})$$

- (b) Let $\Pi = \sum_i \lambda_i |i\rangle \langle i|$ be the spectral decomposition of Π . Projectors are defined by $\Pi^2 = \Pi$. Therefore, all the eigenvalues λ_i must be either 0 or 1. Consequently, for any positive-semidefinite operator A ,

$$\text{tr}(\Pi A) = \sum_i \lambda_i \text{tr}(|i\rangle \langle i| A) = \sum_i \lambda_i \langle i| A |i\rangle = \sum_{i \in \{\lambda_i=1\}} 1 \langle i| A |i\rangle + \underbrace{\sum_{i \in \{\lambda_i=0\}} 0 \langle i| A |i\rangle}_{=0} \leq \sum_i \langle i| A |i\rangle = \text{tr}(A). \quad (\text{A.82})$$

- (c) Consider two quantum states $\rho, \sigma \in \mathcal{H}$ and a trace-preserving quantum operation $\mathcal{E}$ such that $\text{tr}(\mathcal{E}(\rho)) = \text{tr}(\rho)$. The trace-distance between the two states, $D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1$, depends on the difference $\rho - \sigma$. Consider the spectral decomposition of this operator, $\rho - \sigma = \sum_i \lambda_i |i\rangle \langle i|$. We can split the sum into terms that are positive and terms that are negative, i.e.

$$\rho - \sigma = \underbrace{\sum_i \lambda_i^{(+)} |i\rangle \langle i|}_R + \underbrace{\sum_i \lambda_i^{(-)} |i\rangle \langle i|}_{-S}. \quad (\text{A.83})$$

Therefore, $\|\rho - \sigma\|_1 = \text{tr}(R + S)$, since for any operator $\|A\|_1 = \sum_i |\lambda_i|$. The trace-distance becomes

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1 = \frac{1}{2} (\text{tr}(R) + \text{tr}(S)) = \text{tr}(R). \quad (\text{A.84})$$

The last step uses that $\text{tr}(R) = \text{tr}(S)$ which follows from $\text{tr}(R) - \text{tr}(S) = \text{tr}(R - S) = \text{tr}(\rho - \sigma) = \text{tr}(\rho) - \text{tr}(\sigma) = 0$. Now, we can use that $\mathcal{E}$ is a trace-preserving quantum operation

$$D(\rho, \sigma) = \text{tr}(R) = \text{tr}(\mathcal{E}(R)) \geq \max_{\Pi} \text{tr}(\Pi \mathcal{E}(R)) \geq \max_{\Pi} [\text{tr}(\Pi \mathcal{E}(R)) - \text{tr}(\Pi \mathcal{E}(S))], \quad (\text{A.85})$$

where we used the trace-decreasing property of projectors proved in (a). Now we get

$$\begin{aligned} D(\rho, \sigma) &\geq \max_{\Pi} [\text{tr}(\Pi \mathcal{E}(R)) - \text{tr}(\Pi \mathcal{E}(S))] = \max_{\Pi} \text{tr}(\Pi (\mathcal{E}(R - S))) \\ &= \max_{\Pi} \text{tr}(\Pi (\mathcal{E}(\rho - \sigma))) = D(\mathcal{E}(\rho), \mathcal{E}(\sigma)), \end{aligned} \quad (\text{A.86})$$

which proves that trace-distance is decreasing under trace-preserving quantum operations.

A.2.3 Solution (Text at p. 117)

(a) We can prove symmetry of fidelity using two different methods.

- **Method 1.** To prove that fidelity is symmetric, we show that for two Hermitian operators A and B ,

$$\|BA\|_1^2 = \|AB\|_1^2, \quad (\text{A.87})$$

which is equivalent to showing that $(\text{tr} \sqrt{ABBA})^2 = (\text{tr} \sqrt{BAAB})^2$. This would follow if we prove that $ABBA$ and $BAAB$ have the same eigenvalues. To that end, assume that $ABBA$ has the spectral decomposition

$$ABBA|\psi_i\rangle = \lambda_i|\psi_i\rangle. \quad (\text{A.88})$$

Consequently,

$$(BA)ABBA|\psi_i\rangle = \lambda_i BA|\psi_i\rangle, \quad (\text{A.89})$$

where we multiplied both sides of (A.88) by BA . Since $(BA)ABBA|\psi_i\rangle = BAAB(BA)|\psi_i\rangle$, we get

$$BAAB(BA)|\psi_i\rangle = \lambda_i BA|\psi_i\rangle. \quad (\text{A.90})$$

This means that $BAAB$ can be spectrally decomposed in terms of eigenvectors $\{BA|\psi_i\rangle\}$ with eigenvalues $\{\lambda_i\}$. Since the eigenvalues are the same for both $ABBA$ and $BAAB$, then (A.87) holds, which means that the fidelity is symmetric.

- **Method 2.** From Uhlmann's theorem we know that

$$F(\rho, \sigma) = \max_{\{|\psi_\rho\rangle, |\phi_\sigma\rangle\}} |\langle\psi_\rho|\phi_\sigma\rangle|^2, \quad (\text{A.91})$$

where $|\psi_\rho\rangle$ and $|\phi_\sigma\rangle$ are purifications of ρ and σ respectively. Since the inner product is symmetric in its arguments, $|\langle\psi_\rho|\phi_\sigma\rangle|^2 = |\langle\phi_\sigma|\psi_\rho\rangle|^2$, then also $F(\rho, \sigma) = F(\sigma, \rho)$.

(b) Given any unitary U , the fidelity in terms of the trace-norm is

$$F(U\rho U^\dagger, U\sigma U^\dagger) = \left\| \sqrt{U\rho U^\dagger} \sqrt{U\sigma U^\dagger} \right\|_1^2. \quad (\text{A.92})$$

For any operator $A \geq 0$, it holds that $\sqrt{UAU^\dagger} = U\sqrt{A}U^\dagger$. Indeed, considering the spectral decomposition, we get

$$A = \sum_i \lambda_i(A) |\lambda_i\rangle\langle\lambda_i| \implies \sqrt{A} = \sum_i \sqrt{\lambda_i(A)} |\lambda_i\rangle\langle\lambda_i|, \quad (\text{A.93})$$

from which

$$\sqrt{UAU^\dagger} = \sum_i \sqrt{\lambda_i(UAU^\dagger)} U|\lambda_i\rangle\langle\lambda_i|U^\dagger, \quad U\sqrt{A}U^\dagger = \sum_i \sqrt{\lambda_i(A)} U|\lambda_i\rangle\langle\lambda_i|U^\dagger, \quad (\text{A.94})$$

that are equal once we recall that the spectrum of an operator is invariant under unitary transformations, as shown in (A.78). Therefore, the fidelity can be written as

$$F(U\rho U^\dagger, U\sigma U^\dagger) = \left\| U \underbrace{\sqrt{\rho} U^\dagger U}_{=1} \sqrt{\sigma} U^\dagger \right\|_1^2 = \left\| U \sqrt{\rho} \sqrt{\sigma} U^\dagger \right\|_1^2. \quad (\text{A.95})$$

From the definition of the trace-norm, $\|A\|_1 = \sum_i |\lambda_i(A)|$ and using again the invariance of the spectrum under unitary transformations, then we get

$$F(U\rho U^\dagger, U\sigma U^\dagger) = \left\| U \sqrt{\rho} \sqrt{\sigma} U^\dagger \right\|_1^2 = \left\| \sqrt{\rho} \sqrt{\sigma} \right\|_1^2 = F(\rho, \sigma). \quad (\text{A.96})$$

(c) Using the definition of the fidelity in terms of the trace-norm,

$$\begin{aligned} F(\rho \otimes \tau, \sigma \otimes \tau) &= \left\| \sqrt{\rho \otimes \tau} \sqrt{\sigma \otimes \tau} \right\|_1^2 = \left\| (\sqrt{\rho} \otimes \sqrt{\tau})(\sqrt{\sigma} \otimes \sqrt{\tau}) \right\|_1^2 = \left\| \sqrt{\rho} \sqrt{\sigma} \otimes \tau \right\|_1^2 \\ &= \left\| \sqrt{\rho} \sqrt{\sigma} \right\|_1^2 \underbrace{\left\| \tau \right\|_1}_{=1} = F(\rho, \sigma). \end{aligned} \quad (\text{A.97})$$

A.2.4 Solution (Text at p. 117)

Given the three states

$$|\psi\rangle = |0\rangle, \quad |\phi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle, \quad |\varphi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle, \quad (\text{A.98})$$

we have

$$\begin{aligned} d(\psi, \varphi) &= 1 - |\langle\psi|\varphi\rangle|^2 = 1 - \left|\frac{1}{2}\right|^2 = \frac{3}{4}, \\ d(\psi, \phi) &= 1 - |\langle\psi|\phi\rangle|^2 = 1 - \left|\frac{\sqrt{3}}{2}\right|^2 = \frac{1}{4}, \\ d(\phi, \varphi) &= 1 - |\langle\phi|\varphi\rangle|^2 = 1 - \left|\frac{\sqrt{3}}{2}\frac{1}{2} + \frac{1}{2}\frac{\sqrt{3}}{2}\right|^2 = \frac{1}{4}, \end{aligned} \quad (\text{A.99})$$

from which we find that

$$d(\psi, \varphi) \not\leq d(\psi, \phi) + d(\phi, \varphi). \quad (\text{A.100})$$

A.2.5 Solution (Text at p. 117)

Consider the polar decomposition of the operator $A = |A|V$, with $|A|$ being positive semidefinite and V being unitary⁷⁹. Therefore,

$$|\operatorname{tr}(AU)| = |\operatorname{tr}(|A|VU)| = |\operatorname{tr}\left(|A|^{\frac{1}{2}}|A|^{\frac{1}{2}}VU\right)| \leq \sqrt{\operatorname{tr}|A|}\sqrt{\operatorname{tr}(U^\dagger V^\dagger |A| VU)}, \quad (\text{A.101})$$

where in the last step we used Cauchy-Schwarz inequality for operators (see Preliminaries). From cyclicity of trace and unitarity we have $\operatorname{tr}(U^\dagger V^\dagger |A| VU) = \operatorname{tr}|A|$. Therefore,

$$|\operatorname{tr}(AU)| \leq \sqrt{\operatorname{tr}|A|}\sqrt{\operatorname{tr}(U^\dagger V^\dagger |A| VU)} \leq \sqrt{\operatorname{tr}|A|}\sqrt{\operatorname{tr}|A|} = \operatorname{tr}|A|. \quad (\text{A.102})$$

A.2.6 Solution (Text at p. 118)

(a) Starting from the definition, we obtain

$$\|p - q\|_1 = \sum_{i=1}^n |p_i - q_i| \geq \sum_{i=1}^n (\sqrt{p_i} - \sqrt{q_i})^2, \quad (\text{A.103})$$

where we used that $|a - b| \geq (\sqrt{a} - \sqrt{b})^2$, for $a, b \geq 0$. Expanding the square, we get

$$\sum_{i=1}^n (\sqrt{p_i} - \sqrt{q_i})^2 = \sum_{i=1}^n (p_i + q_i - 2\sqrt{p_i q_i}) = 2 - 2 \sum_{i=1}^n \sqrt{p_i q_i}, \quad (\text{A.104})$$

where we used that $\sum_{i=1}^n p_i = \sum_{i=1}^n q_i = 1$.

(b) From the definition of Y in (A.9), we get

$$\operatorname{tr}(Y\rho) = \operatorname{tr}\left(\rho^{-\frac{1}{2}}\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}}\rho^{-\frac{1}{2}}\rho\right) = \operatorname{tr}\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}} = \sqrt{F(\rho, \sigma)}, \quad (\text{A.105})$$

where we used the cyclicity of the trace. In the same way,

$$Y\rho Y = \rho^{-\frac{1}{2}}\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}}\underbrace{\rho^{-\frac{1}{2}}\rho\rho^{-\frac{1}{2}}}_{=\mathbb{I}}\sqrt{\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}}\rho^{-\frac{1}{2}} = \rho^{-\frac{1}{2}}\sqrt{(\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}})^2}\rho^{-\frac{1}{2}} = \underbrace{\rho^{-\frac{1}{2}}\rho^{\frac{1}{2}}}_{=\mathbb{I}}\sigma\underbrace{\rho^{\frac{1}{2}}\rho^{-\frac{1}{2}}}_{=\mathbb{I}} = \sigma. \quad (\text{A.106})$$

⁷⁹ You can think at polar decomposition as the operator analogue of the decomposition $z = |z|e^{i\varphi}$ for complex numbers.

(c) Using the definitions $p_i = \langle \lambda_i | \rho | \lambda_i \rangle$ and $q_i = \langle \lambda_i | \sigma | \lambda_i \rangle$, we get

$$\sum_{i=1}^d \sqrt{p_i} \sqrt{q_i} = \sum_{i=1}^d \sqrt{\langle \lambda_i | \rho | \lambda_i \rangle} \sqrt{\langle \lambda_i | \sigma | \lambda_i \rangle} = \sum_{i=1}^d \sqrt{\langle \lambda_i | \rho | \lambda_i \rangle} \sqrt{\langle \lambda_i | Y \rho Y | \lambda_i \rangle}. \quad (\text{A.107})$$

Using the spectral decomposition (A.10) of Y , we get

$$\sum_{i=1}^d \sqrt{p_i} \sqrt{q_i} = \sum_{i=1}^d \lambda_i \langle \lambda_i | \rho | \lambda_i \rangle = \text{tr}(Y \rho) = \sqrt{F(\rho, \sigma)}. \quad (\text{A.108})$$

(d) Considering the action of $\mathcal{E}$ on the states ρ and σ , we get

$$\mathcal{E}(\rho) - \mathcal{E}(\sigma) = \sum_{i=1}^d \langle \lambda_i | \rho | \lambda_i \rangle |\lambda_i\rangle \langle \lambda_i| - \sum_{i=1}^d \langle \lambda_i | \sigma | \lambda_i \rangle |\lambda_i\rangle \langle \lambda_i| = \sum_{i=1}^d (\langle \lambda_i | \rho | \lambda_i \rangle - \langle \lambda_i | \sigma | \lambda_i \rangle) |\lambda_i\rangle \langle \lambda_i|, \quad (\text{A.109})$$

from which

$$\|\mathcal{E}(\rho) - \mathcal{E}(\sigma)\|_1 = \left\| \sum_{i=1}^d (\langle \lambda_i | \rho | \lambda_i \rangle - \langle \lambda_i | \sigma | \lambda_i \rangle) |\lambda_i\rangle \langle \lambda_i| \right\|_1 = \left\| \sum_{i=1}^d (p_i - q_i) |\lambda_i\rangle \langle \lambda_i| \right\|_1 = \|p - q\|_1. \quad (\text{A.110})$$

Using that the trace-distance is decreasing under trace-preserving quantum operations (A.6) we get

$$\|\rho - \sigma\|_1 \geq \|\mathcal{E}(\rho) - \mathcal{E}(\sigma)\|_1 = \|p - q\|_1 \geq 2 - 2 \sum_{i=1}^d \sqrt{p_i q_i} = 2 - 2\sqrt{F(\rho, \sigma)}, \quad (\text{A.111})$$

where we used (A.8) and (A.11).

A.2.7 Solution (Text at p. 118)

(a) Since ψ and φ are two qubit states, we can write their density operator in terms of the Bloch vectors as

$$\psi = \frac{\mathbb{1} + \hat{n}_\psi \cdot \vec{\sigma}}{2}, \quad \text{and} \quad \varphi = \frac{\mathbb{1} + \hat{n}_\varphi \cdot \vec{\sigma}}{2}, \quad (\text{A.112})$$

where $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$ is the vector of the Pauli matrices. Therefore,

$$\psi - \varphi = \frac{1}{2} (\hat{n}_\psi - \hat{n}_\varphi) \cdot \vec{\sigma}. \quad (\text{A.113})$$

Let us define a new Bloch vector $\hat{m} \equiv (\hat{n}_\psi - \hat{n}_\varphi)$, from which

$$\psi - \varphi = \frac{1}{2} \hat{m} \cdot \vec{\sigma} = \frac{1}{2} (m_X \sigma_X + m_Y \sigma_Y + m_Z \sigma_Z) = \frac{1}{2} \begin{pmatrix} m_Z & m_X - im_Y \\ m_X + im_Y & -m_Z \end{pmatrix}. \quad (\text{A.114})$$

Using (A.43) to compute the eigenvalues of a 2×2 matrix, we get

$$\begin{aligned} \lambda_{\pm}(\psi - \varphi) &= \frac{1}{2} \left[\frac{1}{2} (m_Z - m_Z) \pm \sqrt{\frac{1}{4} (m_Z + m_Z)^2 + 4(m_X - im_Y)(m_X + im_Y)} \right] \\ &= \pm \frac{1}{2} \sqrt{m_Z^2 + m_X^2 + m_Y^2} = \pm \frac{1}{2} |\hat{m}|. \end{aligned} \quad (\text{A.115})$$

Then, using the definition of the trace-distance from (2.14) in Chapter 2.2, we find

$$D(\psi, \varphi) = \sum_{\lambda_i > 0} \lambda_i = \frac{1}{2} |\hat{m}| = \frac{1}{2} |\hat{n}_\psi - \hat{n}_\varphi|. \quad (\text{A.116})$$

(b) Recall from Chapter 2.2 that the fidelity between a pure state and a mixed state is

$$F(|\psi\rangle\langle\psi|, \varphi) = \langle \psi | \varphi | \psi \rangle = \text{tr}(\varphi |\psi\rangle\langle\psi|), \quad (\text{A.117})$$

from which

$$\begin{aligned}
 F(|\psi\rangle\langle\psi|, \varphi) &= \text{tr} \left(\left(\frac{\mathbb{1} + \sigma_X}{2} \right) \left(\frac{\mathbb{1} + \hat{n}_\varphi \cdot \vec{\sigma}}{2} \right) \right) \\
 &= \frac{1}{4} \text{tr} (\mathbb{1} + \sigma_X + \hat{n}_\varphi \cdot \vec{\sigma} + \sigma_X \hat{n}_\varphi \cdot \vec{\sigma}) = \frac{1}{4} (\underbrace{\text{tr}(\mathbb{1})}_{=0} + \underbrace{\text{tr}(\sigma_X)}_{=0} + \underbrace{\text{tr}(\hat{n}_\varphi \cdot \vec{\sigma})}_{=0} + \underbrace{\text{tr}(\hat{n}_\varphi \cdot \vec{\sigma} \sigma_X)}_{=2n_X^{(\varphi)})}) \quad (\text{A.118})
 \end{aligned}$$

where in the last equality we have used that the product of two Pauli matrices is traceless unless they are equal. Therefore only the X -component of $\hat{n}_\varphi$ survives. The fidelity becomes

$$F(|\psi\rangle\langle\psi|, \varphi) = \frac{1}{4} (2 + 2n_X^{(\varphi)}) = \frac{1}{2} (1 + n_X^{(\varphi)}), \quad (\text{A.119})$$

(c) The Bloch vectors of the states φ_1 and φ_2 are

$$\hat{n}_{\varphi_1} = \left(-\frac{1}{4}, 0, 0 \right), \quad \text{and} \quad \hat{n}_{\varphi_2} = (0, 0, 1). \quad (\text{A.120})$$

Using the expressions for the trace-distance derived in (A.116), we can compute

$$\begin{aligned}
 D(\psi, \varphi_1) &= \frac{1}{2} |(1, 0, 0) - (-1/4, 0, 0)| = \frac{1}{2} \left(1 + \frac{1}{4} \right) = 0.6250, \\
 D(\psi, \varphi_2) &= \frac{1}{2} |(1, 0, 0) - (0, 0, 1)| = \frac{1}{2} \sqrt{2} = \frac{1}{\sqrt{2}} = 0.7071,
 \end{aligned} \quad (\text{A.121})$$

from which we find that $D(\psi, \varphi_1) < D(\psi, \varphi_2)$. Similarly, using the expression (A.119), we can evaluate the “fidelity-distance”,

$$\begin{aligned}
 d(\psi, \varphi_1) &= 1 - F(\psi, \varphi_1) = 1 - \frac{1}{2} \left(1 + \frac{1}{4} \right) = 0.6250, \\
 d(\psi, \varphi_2) &= 1 - F(\psi, \varphi_2) = 1 - \frac{1}{2} (1 + 0) = \frac{1}{2} = 0.5,
 \end{aligned} \quad (\text{A.122})$$

from which we find that $d(\psi, \varphi_1) > d(\psi, \varphi_2)$.

A.3. Quantum measurements

A.3.1 Solution (Text at p. 119)

(a) The state $|\psi\rangle = |00\rangle$ can be represented as the matrix

$$|\psi\rangle = |00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}. \quad (\text{A.123})$$

The unitary can be written as

$$H \otimes \mathbb{1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}. \quad (\text{A.124})$$

Therefore, the state $|\psi_1\rangle = (H \otimes \mathbb{1})|\psi\rangle$ is

$$|\psi_1\rangle = (H \otimes \mathbb{1})|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle). \quad (\text{A.125})$$

(b) The application of the CNOT gate can be represented as

$$|\psi_2\rangle = \text{CNOT}|\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \quad (\text{A.126})$$

(c) The “+1” outcome of the observable σ_X corresponds to the ray $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. The projection associated with this outcome is

$$M = \mathbb{1} \otimes |+\rangle\langle+| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}. \quad (\text{A.127})$$

Applying this to $|\psi_2\rangle$ gives,

$$M|\psi_2\rangle = \frac{1}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{2\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}. \quad (\text{A.128})$$

To get the post-measurement state we must normalise this vector by computing

$$\sqrt{\langle\psi_2|M|\psi_2\rangle} = \sqrt{\frac{1}{4} \begin{pmatrix} 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}} = \sqrt{\frac{2}{4}} = \frac{1}{\sqrt{2}}. \quad (\text{A.129})$$

Therefore, the post-measurement state $|\psi_3\rangle$ is

$$|\psi_3\rangle = \frac{M|\psi_2\rangle}{\sqrt{\langle\psi_2|M|\psi_2\rangle}} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle). \quad (\text{A.130})$$

(d) To perform the partial trace on the second qubit it is more convenient to work with the density operator $\rho = |\psi_3\rangle\langle\psi_3|$:

$$\rho = |\psi_3\rangle\langle\psi_3| = \frac{1}{4} [|00\rangle + |01\rangle + |10\rangle + |11\rangle][\langle 00| + \langle 01| + \langle 10| + \langle 11|] = \frac{1}{4} \sum_{i,j,k,l=0}^1 |ij\rangle\langle kl|. \quad (\text{A.131})$$

Therefore, the partial trace is

$$\rho_A = \text{tr}_B(\rho) = \frac{1}{4} \sum_{i,j,k,l=0}^1 |i\rangle\langle k| \otimes \underbrace{\sum_{p=0}^1 \langle p|j\rangle \langle l|p\rangle}_{=\delta_{p,j}=\delta_{p,l}} = \frac{1}{4} \sum_{j=0}^1 \sum_{i,k=0}^1 |i\rangle\langle k| = \frac{1}{2} \sum_{i,k=0}^1 |i\rangle\langle k| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (\text{A.132})$$

A.3.2 Solution (Text at p. 119)

A basis measurement is a projection onto a set of orthogonal rays: $\{|\phi_a\rangle\}_a$ with $\langle\phi_a|\phi_{a'}\rangle = \delta_{a,a'}$. Born's rule is $p(a) = \langle\phi_a|\rho|\phi_a\rangle$ for any given state ρ . It is impossible to have $p(0) = 1$ for every ρ because $p(0) < 1$ whenever $\rho \neq |\phi_0\rangle\langle\phi_0|$. In order to satisfy $p(0) = 1$ we need every ρ to be an eigenstate of the zero-outcome POVM-element with eigenvalue “+1”. The only possible choice is the trivial POVM $\{E_0 = \mathbb{1}, E_1 = 0\}$.

A.3.3 Solution (Text at p. 119)

- (a) For a two-outcome qubit POVM, we have two POVM-elements. We write them as $E_0 = \lambda_0 (\mathbb{1} + \vec{n}_0 \cdot \vec{\sigma})$ and $E_1 = \lambda_1 (\mathbb{1} + \vec{n}_1 \cdot \vec{\sigma})$. The completeness condition, namely $E_0 + E_1 = \mathbb{1}$, becomes

$$\mathbb{1} = (\lambda_0 + \lambda_1) \mathbb{1} + (\lambda_0 \vec{n}_0 + \lambda_1 \vec{n}_1) \cdot \vec{\sigma}. \quad (\text{A.133})$$

For this to hold, we need to set $\lambda_0 + \lambda_1 = 1$ and $\lambda_0 \vec{n}_0 + \lambda_1 \vec{n}_1 = 0$. From the former, we get the parameter $\lambda_0 \equiv \lambda$ and $\lambda_1 = 1 - \lambda$, for $0 \leq \lambda \leq 1$. From the latter, we get that the orientation of $\vec{n}_1$ must be opposite to that of $\vec{n}_0$. We can write the vectors as $\vec{n}_0 = l_0 \hat{n}$ and $\vec{n}_1 = -l_1 \hat{n}$, where $\hat{n}_0 = \hat{n}$ is the unit vector for $\vec{n}_0$. Then, our second condition becomes $\lambda l_0 - (1 - \lambda) l_1 = 0$, which implies $\lambda = \frac{l_1}{l_0 + l_1}$. Thus, the POVM becomes

$$E_0 = \frac{l_1}{l_0 + l_1} (\mathbb{1} + l_0 \hat{n} \cdot \vec{\sigma}) \quad E_1 = \frac{l_0}{l_0 + l_1} (\mathbb{1} - l_1 \hat{n} \cdot \vec{\sigma}) \quad (\text{A.134})$$

Note that for a projective measurement we have $l_0 = l_1 = 1$ and hence $\lambda = 1/2$.

- (b) Let us without loss of generality assume that $l_1 \geq l_0$. Define the projective measurements $P_{\pm} = \frac{1}{2} (\mathbb{1} \pm \hat{n} \cdot \vec{\sigma})$ and $Q_{\pm} = \frac{1}{2} (\mathbb{1} \mp \hat{n} \cdot \vec{\sigma})$. Define also the trivial projective measurement $\{1, 0\}$. We run P with probability q_1 , Q with probability q_2 and the trivial measurement with probability q_3 . We seek choices of (q_1, q_2, q_3) such that we recover the POVM in (A.134). To this end, we need

$$E_0 = q_1 P_+ + q_2 Q_+ + q_3 \mathbb{1} \quad \text{and} \quad E_1 = q_1 P_- + q_2 Q_- + q_3 0. \quad (\text{A.135})$$

From completeness, if the first condition is satisfied then also the second condition is satisfied. Hence, we examine only the first condition. Using (A.134) this gives us three equations

$$\frac{q_1 - q_2}{2} = \frac{l_0 l_1}{l_0 + l_1}, \quad q_3 + \frac{q_1 + q_2}{2} = \frac{l_1}{l_0 + l_1}, \quad q_1 + q_2 + q_3 = 1 \quad (\text{A.136})$$

This is a linear and complete equation system that can be solved straightforwardly. The solution is

$$q_1 = \frac{l_0 + l_0 l_1}{l_0 + l_1}, \quad q_2 = \frac{l_0 - l_0 l_1}{l_0 + l_1}, \quad q_3 = \frac{l_1 - l_0}{l_0 + l_1}. \quad (\text{A.137})$$

This is a valid probability distribution since all the numbers are non-negative and sum to unit.

A.3.4 Solution (Text at p. 119)

From Eq (3.8) in Chapter 3.2, any 3-outcome qubit POVM must satisfy $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and $\lambda_1 \hat{n}_1 + \lambda_2 \hat{n}_2 + \lambda_3 \hat{n}_3 = 0$. The former gives $\lambda_1 = 1 - \lambda_2 - \lambda_3$. The latter gives us

$$(\lambda_1 + (\lambda_2 + \lambda_3) \cos \theta, 0, (\lambda_2 - \lambda_3) \sin \theta) = (0, 0, 0), \quad (\text{A.138})$$

which is a system of three equations. From the third component, we get $\lambda \equiv \lambda_2 = \lambda_3$ and thus we have $\lambda_1 = 1 - 2\lambda$. Therefore, from the first component, we get

$$\lambda_1 + (\lambda_2 + \lambda_3) \cos \theta = 0 \implies 1 - 2\lambda + 2\lambda \cos \theta = 0, \quad (\text{A.139})$$

whose solution is $\lambda = (2 - 2 \cos \theta)^{-1}$. This satisfies $0 \leq \lambda \leq 1$ for every θ . However, since we also require $0 \leq \lambda_1 \leq 1$ we must enforce that $0 \leq 1 - 2\lambda \leq 1$. This gives us an additional constraint on the allowed values of θ . Solving $1 - 2\lambda = 0$ gives $\theta = \frac{\pi}{2}$. Thus, a POVM is only possible in the range $\frac{\pi}{2} \leq \theta < \pi$.

A.3.5 Solution (Text at p. 119)

- (a) The winning probability becomes

$$p_{\text{win}} = \frac{1}{3} \sum_{a=1}^3 \text{tr}(\psi_a E_a) = \frac{2}{9} \sum_{a=1}^3 \text{tr}(\psi_a^2) = \frac{2}{9} \sum_{a=1}^3 \underbrace{\text{tr}(\psi_a)}_{=1} = \frac{2}{3}, \quad (\text{A.140})$$

where we have used that $E_a = \frac{2}{3} \psi_a$ by definition and that ψ_a is a pure state.

(b) Let us write $(\psi_1 - \psi_2)$, $(\psi_1 - \psi_3)$ and $(\psi_2 - \psi_3)$ in their matrix form:

$$\psi_1 = \frac{1}{2}(\mathbb{1} + \sigma_X), \quad \psi_2 = \frac{1}{2}\left(\mathbb{1} - \frac{1}{2}\sigma_X + \frac{\sqrt{3}}{2}\sigma_Z\right), \quad \psi_3 = \frac{1}{2}\left(\mathbb{1} - \frac{1}{2}\sigma_X - \frac{\sqrt{3}}{2}\sigma_Z\right), \quad (\text{A.141})$$

from which

$$\begin{aligned} \psi_1 - \psi_2 &= \frac{1}{2}\left(\frac{3}{2}\sigma_X - \frac{\sqrt{3}}{2}\sigma_Z\right) = \frac{1}{2}\begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{3}{2} \\ \frac{3}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}, \\ \psi_1 - \psi_3 &= \frac{1}{2}\left(\frac{3}{2}\sigma_X + \frac{\sqrt{3}}{2}\sigma_Z\right) = \frac{1}{2}\begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{3}{2} \\ \frac{3}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}, \\ \psi_2 - \psi_3 &= \frac{1}{2}\sqrt{3}\sigma_Z = \frac{1}{2}\begin{pmatrix} \sqrt{3} & 0 \\ 0 & -\sqrt{3} \end{pmatrix}. \end{aligned} \quad (\text{A.142})$$

We can notice that $(\psi_2 - \psi_3)$ is already expressed in its diagonal form, with $\lambda_1 = \frac{\sqrt{3}}{2}$ and $\lambda_2 = -\frac{\sqrt{3}}{2}$. We can calculate the spectrum of $(\psi_1 - \psi_2)$ and $(\psi_1 - \psi_3)$ using (A.43) and find the same eigenvalues.

(c) If we are allowed to use only projective measurements for the discrimination, then the winning probability is

$$p_{\text{win}}^{\text{proj}} = \frac{1}{3} [\text{tr}(\psi_1 \Pi_1) + \text{tr}(\psi_2 \Pi_2)] = \frac{1}{3} \underbrace{(\text{tr}(\psi_2) + \text{tr}((\psi_1 - \psi_2) \Pi_1))}_{=1}, \quad (\text{A.143})$$

where we have used that $\Pi_2 = \mathbb{1} - \Pi_1$. Maximising over Π_1 , we get

$$p_{\text{win}}^{\text{proj}} \leq \frac{1}{3} \left(1 + \max_{\Pi_1} \text{tr}((\psi_1 - \psi_2) \Pi_1) \right). \quad (\text{A.144})$$

From (b) we know that the eigenvalues of $(\psi_1 - \psi_2)$ are $\lambda_{1,2} = \pm \frac{\sqrt{3}}{2}$. Since $\Pi_1 \geq 0$, the optimal choice is to select Π_1 to be the eigenvector associated to the positive eigenvalue, from which

$$p_{\text{win}}^{\text{proj}} \leq \frac{1}{3} \left(1 + \max_{\Pi_1} \text{tr}((\psi_1 - \psi_2) \Pi_1) \right) = \frac{1}{3} (1 + \lambda_{\max}(\psi_1 - \psi_2)) = \frac{1}{3} \left(1 + \frac{\sqrt{3}}{2} \right) \simeq 0.6220. \quad (\text{A.145})$$

This is smaller than the result in (a). Moreover, since the result in (A.145) depends only on the maximum eigenvalue of $(\psi_1 - \psi_2)$, which is equal to the one of $(\psi_1 - \psi_3)$ and $(\psi_2 - \psi_3)$ from (b), we can choose any one of the three outcomes as corresponding to the zero-projector and obtain the same result.

A.3.6 Solution (Text at p. 120)

(a) Assuming that $\{E_a\}_a$ is a POVM means $\mathbb{1} = \sum_a E_a$. This implies $\mathbb{1} = (\sum_a E_a)^2$. If we trace both side we get

$$\begin{aligned} d &= \text{tr} \left(\left(\sum_a E_a \right)^2 \right) = \text{tr} \left(\sum_{a,a'} E_a E_{a'} \right) = \text{tr} \left(\sum_a E_a E_a + \sum_{a \neq a'} E_a E_{a'} \right) \\ &= \sum_a \text{tr} \left(\frac{1}{d} |\psi_a\rangle \langle \psi_a| \frac{1}{d} |\psi_a\rangle \langle \psi_a| \right) + \sum_{a \neq a'} \text{tr} \left(\frac{1}{d} |\psi_a\rangle \langle \psi_a| \frac{1}{d} |\psi_{a'}\rangle \langle \psi_{a'}| \right) = \frac{1}{d^2} (d^2 + (d^4 - d^2)c). \end{aligned} \quad (\text{A.146})$$

Solving for c gives the result $c = \frac{1}{d+1}$.

(b) Since the projectors $\{\Pi_a\}_a$ span the Hermitian space, we can write any Hermitian operator as a linear combination of them, namely

$$\rho = \sum_{j=1}^{d^2} c_j \Pi_j, \quad (\text{A.147})$$

for some coefficients c_j . Consider now the probability of measuring outcome a .

$$\begin{aligned} p_a &= \text{tr}(\rho E_a) = \frac{1}{d} \sum_j c_j \text{tr}(\Pi_j \Pi_a) = \frac{1}{d} \left(c_a \text{tr}(\Pi_a \Pi_a) + \underbrace{\sum_{j \neq a} c_j \text{tr}(\Pi_j \Pi_a)}_{= \frac{1}{d+1}} \right) = \frac{1}{d(d+1)} \left((d+1)c_a + \sum_{j \neq a} c_j \right) \\ &= \frac{1}{d(d+1)} \left((d+1)c_a - c_a + c_a + \sum_{j \neq a} c_j \right) = \frac{1}{d(d+1)} (dc_a + 1), \end{aligned} \quad (\text{A.148})$$

where we used that $\sum_j c_j = 1$, since the state in (A.147) is normalized. Solving this for c_a gives

$$c_a = (d+1)p_a - \frac{1}{d}, \quad (\text{A.149})$$

which is the sought coefficient.

A.3.7 Solution (Text at p. 120)

(a) The winning probability is

$$p_{\text{win}} = \sum_{i=1}^2 q_i \text{tr}(E_i \rho_i) = q_1 \text{tr}(E_1 \rho_1) + q_2 \text{tr}(E_2 \rho_2) = \text{tr}(q_1 E_1 \rho_1 + q_2 E_2 \rho_2). \quad (\text{A.150})$$

Since $E_1 + E_2 = \mathbb{1}$, the operator in the trace can be written as

$$q_1 E_1 \rho_1 + q_2 E_2 \rho_2 = \begin{cases} q_1 E_1 \rho_1 + q_2 (\mathbb{1} - E_1) \rho_2 = q_2 \rho_2 + E_1 (q_1 \rho_1 - q_2 \rho_2), \\ q_1 (\mathbb{1} - E_2) \rho_1 + q_2 E_2 \rho_2 = q_1 \rho_1 - E_2 (q_1 \rho_1 - q_2 \rho_2), \end{cases} \quad (\text{A.151})$$

which gives the two possible definitions of K in (A.26), once we define $X = q_1 \rho_1 - q_2 \rho_2$.

(b) Using the two definitions of K in (A.26), we can write

$$K = \frac{1}{2} (q_1 \rho_1 - E_2 X + q_2 \rho_2 + E_1 X) = \frac{1}{2} (q_1 \rho_1 + q_2 \rho_2) + \frac{1}{2} (E_1 - E_2) X = \frac{1}{2} (q_1 \rho_1 + q_2 \rho_2) + \frac{1}{2} E X, \quad (\text{A.152})$$

where we used that $E = E_1 - E_2$.

(c) Using (A.27), the winning probability becomes

$$\begin{aligned} p_{\text{win}} &= \text{tr}(K) = \frac{1}{2} \text{tr}(q_1 \rho_1 + q_2 \rho_2) + \frac{1}{2} \text{tr}(E X) = \frac{1}{2} (q_1 \underbrace{\text{tr}(\rho_1)}_{=1} + q_2 \underbrace{\text{tr}(\rho_2)}_{=1}) + \frac{1}{2} \text{tr}(E X) \\ &= \frac{1}{2} (q_1 + q_2) + \frac{1}{2} \text{tr}(E X) = \frac{1}{2} + \frac{1}{2} \text{tr}(E X), \end{aligned} \quad (\text{A.153})$$

where we used that $q_1 + q_2 = 1$. Therefore, the maximum winning probability is

$$\frac{1}{2} + \max_E \frac{1}{2} \text{tr}(E X). \quad (\text{A.154})$$

Any Hermitian operator X can be spectrally decomposed in

$$X = \sum_{i \in \lambda_i \geq 0} \lambda_i^{(+)} X_i^{(+)} + \sum_{i \in \lambda_i < 0} \lambda_i^{(-)} X_i^{(-)}, \quad (\text{A.155})$$

where $\{\lambda_i^{(+)}\}$ and $\{\lambda_i^{(-)}\}$ are the positive and negative eigenvalues, respectively, and $\{X_i^{(+)}\}$ and $\{X_i^{(-)}\}$ are the associated eigenvectors. Therefore, the optimal operator E which maximises $\text{tr}(E X)$ is exactly the one defined as

$$E_{\text{opt}} = \sum_{i \in \lambda_i \geq 0} X_i^{(+)} - \sum_{i \in \lambda_i < 0} X_i^{(-)}, \quad (\text{A.156})$$

i.e. the one for which

$$\text{tr}(E_{\text{opt}}X) = \sum_{i \in \lambda_i \geq 0} \lambda_i^{(+)} - \sum_{i \in \lambda_i < 0} \lambda_i^{(-)} = \|X\|_1, \quad (\text{A.157})$$

that is exactly the trace-norm of the operator, $\|X\|_1 = \text{tr} \sqrt{X^\dagger X} = \text{tr} |X| = \sum_i |\lambda_i(X)|$. Therefore, inserting (A.157) into (A.154), we get

$$p_{\text{win}} \leq \frac{1}{2} + \max_E \frac{1}{2} \text{tr}(EX) = \frac{1}{2} + \frac{1}{2} \|X\|_1 = \frac{1}{2} + \frac{1}{2} \|q_1 \rho_1 - q_2 \rho_2\|_1, \quad (\text{A.158})$$

where we used the definition of X . This is the Helstrom bound.

Interestingly, for $q_1 = q_2 = \frac{1}{2}$, the bound becomes

$$p_{\text{win}} \leq \frac{1}{2} + \underbrace{\frac{1}{2} \frac{1}{2} \|\rho_1 - \rho_2\|_1}_{=D(\rho_1, \rho_2)} = \frac{1}{2} + \frac{1}{2} D(\rho_1, \rho_2), \quad (\text{A.159})$$

where we used the definition of the trace-distance between two states. This is exactly the bound derived in Chapter 3.3 for the quantum state discrimination of two randomly chosen states.

A.3.8 Solution (Text at p. 121)

We must show that $E_0 + E_1 + E_\emptyset = \mathbb{1}$ and $E_0 \geq 0$, $E_1 \geq 0$ and $E_\emptyset \geq 0$. The first condition is satisfied by construction. Also $E_0 \geq 0$ and $E_1 \geq 0$ are clearly satisfied. Thus, we need only to show that also $E_\emptyset \geq 0$. We can explicitly write $E_\emptyset$,

$$E_\emptyset = \mathbb{1} - E_0 - E_1 = \mathbb{1} - \frac{\mathbb{1} - |\phi\rangle\langle\phi|}{1 + |\langle\psi|\phi\rangle|} - \frac{\mathbb{1} - |\psi\rangle\langle\psi|}{1 + |\langle\psi|\phi\rangle|} = \frac{1}{1 + |\langle\psi|\phi\rangle|} ((|\langle\psi|\phi\rangle| - 1)\mathbb{1} + |\psi\rangle\langle\psi| + |\phi\rangle\langle\phi|). \quad (\text{A.160})$$

Thus, showing $E_\emptyset \geq 0$ is equivalent to showing that

$$\lambda_{\min} [|\psi\rangle\langle\psi| + |\phi\rangle\langle\phi|] + |\langle\psi|\phi\rangle| - 1 \geq 0. \quad (\text{A.161})$$

Since we are free to apply a collective unitary, we can parameterise $|\psi\rangle = |0\rangle$ and $|\phi\rangle = \cos \theta |0\rangle + \sin \theta |1\rangle$ and write

$$|\psi\rangle\langle\psi| + |\phi\rangle\langle\phi| = \begin{pmatrix} 1 + \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{pmatrix}, \quad (\text{A.162})$$

from which, using (A.43), we get the eigenvalues

$$\lambda_{\pm} = \frac{1}{2} \left[\underbrace{(1 + \cos^2 \theta + \sin^2 \theta)}_{=1} \pm \sqrt{(1 + \cos^2 \theta - \sin^2 \theta)^2 + 4 \cos^2 \theta \sin^2 \theta} \right] = 1 \pm |\cos \theta| = 1 \pm |\langle\psi|\phi\rangle|. \quad (\text{A.163})$$

Thus, $\lambda_{\min} [|\psi\rangle\langle\psi| + |\phi\rangle\langle\phi|] + |\langle\psi|\phi\rangle| - 1 = 0$ which shows the result.

A.3.9 Solution (Text at p. 121)

(a) Note that S is a Hermitian operator. Therefore, we can write $E_a = \frac{1}{n} S^{-1/2} \rho_a^{1/2} \rho_a^{1/2} S^{-1/2} = \frac{1}{n} X X^\dagger$, where $X = S^{-1/2} \rho_a^{1/2}$. For any X it holds that $X X^\dagger \geq 0$. Completeness follows from

$$\sum_{a=1}^n E_a = \sum_{a=1}^n \frac{1}{n} S^{-1/2} \rho_a S^{-1/2} = S^{-1/2} \left(\underbrace{\frac{1}{n} \sum_{a=1}^n \rho_a}_{=S} \right) S^{-1/2} = S^{-1/2} S S^{-1/2} = \mathbb{1}. \quad (\text{A.164})$$

(b) Computing the operator S we get

$$S = \frac{1}{3} (\rho_1 + \rho_2 + \rho_3) = \begin{pmatrix} \frac{1}{2} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{2} \end{pmatrix}. \quad (\text{A.165})$$

We can use that for any 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \implies A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \quad (\text{A.166})$$

Then, we can compute the square root to get

$$S^{-1/2} = \frac{\sqrt{3}}{4} \begin{pmatrix} 2 + \sqrt{2} & \sqrt{2} - 2 \\ \sqrt{2} - 2 & 2 + \sqrt{2} \end{pmatrix}. \quad (\text{A.167})$$

We can now evaluate the success probability to

$$p_{\text{win}} = \frac{1}{9} \sum_a \text{tr}(\rho_a S^{-1/2} \rho_a S^{-1/2}) = \frac{1}{12} (5 + 2\sqrt{2}) \approx 0.65. \quad (\text{A.168})$$

A.3.10 Solution (Text at p. 121)

The POVM-elements in matrix form are

$$\begin{aligned} E_1 &= \frac{1}{3}(\mathbb{1} + \sigma_X) = \frac{1}{3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \\ E_2 &= \frac{1}{3} \left(\mathbb{1} - \frac{1}{2} \sigma_X + \frac{\sqrt{3}}{2} \sigma_Z \right) = \frac{1}{3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{6} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0.6220 & -0.1667 \\ -0.1667 & 0.0447 \end{pmatrix}, \\ E_3 &= \frac{1}{3} \left(\mathbb{1} - \frac{1}{2} \sigma_X - \frac{\sqrt{3}}{2} \sigma_Z \right) = \frac{1}{3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{6} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0.0447 & -0.1667 \\ -0.1667 & 0.6220 \end{pmatrix}. \end{aligned} \quad (\text{A.169})$$

To find a suitable unitary U for the Naimark dilation, we follow the method in Chapter 3.4. Define

$$U|i, 0\rangle_{AB} = \sum_a \sqrt{E_a} |i\rangle_A \otimes |a\rangle_B, \quad \text{for } i = 1, 2. \quad (\text{A.170})$$

Since the POVM is in dimension $d = 2$ and it has three outcomes, this is a 6×6 unitary. However, only two columns will be fixed by (A.170). Indeed,

$$U = \underbrace{\left(\sum_a \sqrt{E_a} |0, a\rangle \right) \langle 0, 0| + \left(\sum_a \sqrt{E_a} |1, a\rangle \right) \langle 1, 0|}_{\tilde{U}} + \text{orthogonal columns}. \quad (\text{A.171})$$

It is easier to use a computer for this calculation: first we have

$$\begin{aligned} |\psi_0\rangle &= \sum_a \sqrt{E_a} |0, a\rangle = \sqrt{\begin{pmatrix} 0.3333 & 0.3333 \\ 0.3333 & 0.3333 \end{pmatrix}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sqrt{\begin{pmatrix} 0.6220 & -0.1667 \\ -0.1667 & 0.0447 \end{pmatrix}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &\quad + \sqrt{\begin{pmatrix} 0.0447 & -0.1667 \\ -0.1667 & 0.6220 \end{pmatrix}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.4082 \\ 0.7618 \\ 0.0547 \\ 0.4082 \\ -0.2041 \\ -0.2041 \end{pmatrix}. \end{aligned} \quad (\text{A.172})$$

Similarly, we can get

$$|\psi_1\rangle = \sum_a \sqrt{E_a} |1, a\rangle = \begin{pmatrix} 0.4082 \\ -0.2041 \\ -0.2041 \\ 0.4082 \\ 0.0547 \\ 0.7618 \end{pmatrix}. \quad (\text{A.173})$$

Therefore, we get

$$\begin{aligned}
 \tilde{U} = |\psi_0\rangle\langle 0,0| + |\psi_1\rangle\langle 1,0| &= \begin{pmatrix} 0.4082 \\ 0.7618 \\ 0.0547 \\ 0.4082 \\ -0.2041 \\ -0.2041 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0.4082 \\ -0.2041 \\ -0.2041 \\ 0.4082 \\ 0.0547 \\ 0.7618 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0.4082 & 0 & 0.4082 & 0 & 0 & 0 \\ 0.7618 & 0 & -0.2041 & 0 & 0 & 0 \\ 0.0547 & 0 & -0.2041 & 0 & 0 & 0 \\ 0.4082 & 0 & 0.4082 & 0 & 0 & 0 \\ -0.2041 & 0 & 0.0547 & 0 & 0 & 0 \\ -0.2041 & 0 & 0.7618 & 0 & 0 & 0 \end{pmatrix}, \tag{A.174}
 \end{aligned}$$

where the remaining columns of U can be selected in any way consistent with the unitarity condition $UU^\dagger = \mathbb{1}$, i.e. the columns must be pairwise orthogonal.

B. CORE TOPICS - EXERCISES

B.1. Entanglement

B.1.1 Maximally entangled d -dimensional state ●●●

In Chapter 4.2 we showed that the two-qubit state $\frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$ can be transformed into any other two-qubit state by LOCC. By Schmidt decomposition, any pure bipartite d -dimensional state can be expressed as $|\psi\rangle_{AB} = \sum_{k=0}^{d-1} \sqrt{\lambda_k} |kk\rangle_{AB}$. Here, we will show that the state

$$|\phi^+\rangle_{AB} = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |kk\rangle_{AB} \quad (\text{B.1})$$

can be transformed via LOCC into any other bipartite d -dimensional state.

To this end, Alice generates the auxiliary state $|0\rangle_{A'}$ in her lab, such that the total state becomes

$$|0\rangle_{A'} \otimes |\phi^+\rangle_{AB} = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |0k\rangle_{A'A} \otimes |k\rangle_B. \quad (\text{B.2})$$

Define the generalised Pauli- X operator $X = \sum_{k=0}^{d-1} |k+1\rangle\langle k|$. Alice transforms $A'A$ as

$$|0p\rangle_{A'A} \rightarrow (X^p \otimes \mathbb{1}) \sum_{k=0}^{d-1} \sqrt{\lambda_k} |kk\rangle_{A'A} \quad \forall p = 0, \dots, d-1, \quad (\text{B.3})$$

where X^p is the p -power of X .

- Find the general expression for X^p and show that (B.3) is a legitimate unitary transformation.
- Find the total state after Alice applies the transformation (B.3) to the state (B.2).
- Alice measures A' in the computational basis and sends the outcome, l , to Bob who performs the unitary

$$U_l = \sum_{k=0}^{d-1} |k\rangle\langle l-k \pmod{d}|, \quad \forall l = 0, \dots, d-1. \quad (\text{B.4})$$

Show that the final state is $|\psi\rangle_{AB}$.

Solution (p. 155)

B.1.2 Schmidt decomposition ●●

- Show that for any bipartite pure state $|\psi\rangle_{AB}$ the reductions $\rho_A = \text{tr}_B(|\psi\rangle\langle\psi|)$ and $\rho_B = \text{tr}_A(|\psi\rangle\langle\psi|)$ have the same eigenvalues.
- Compute the Schmidt decomposition of the following two states

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad |\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle). \quad (\text{B.5})$$

Solution (p. 156)

B.1.3 Properties of the maximally entangled state ●●

Consider the maximally entangled state, $|\phi^+\rangle = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |kk\rangle$. Show that the following properties hold.

- For any linear operator O

$$(O \otimes \mathbb{1})|\phi^+\rangle = (\mathbb{1} \otimes O^T)|\phi^+\rangle. \quad (\text{B.6})$$

- For any unitary operator U

$$(U \otimes U^*)|\phi^+\rangle = |\phi^+\rangle. \quad (\text{B.7})$$

Solution (p. 158)

B.1.4 Partial transpose ●●

Consider a bipartite state ρ_{AB} .

- (a) Prove that the spectra of ρ_{AB} and ρ_{AB}^T are identical, i.e. that transposition does not change the eigenvalues.
- (b) Show that the spectra of $\rho_{AB}^{T_A}$ and $\rho_{AB}^{T_B}$ are identical.

Solution (p. 158)

B.1.5 PPT for two-qubit ●

Consider the following two-qubit state

$$\rho_{AB} = p|\psi\rangle\langle\psi|_{AB} + \frac{1-p}{4}\mathbb{1}, \quad \text{where } |\psi\rangle_{AB} = \sqrt{\frac{2}{5}}|01\rangle + \sqrt{\frac{3}{5}}|10\rangle \quad \text{and } p \in [0, 1]. \quad (\text{B.8})$$

Use the PPT criterion to find the critical value of p for which ρ_{AB} is entangled.

Solution (p. 159)

B.1.6 Critical visibility for maximally entangled state ●

Let $|\phi^+\rangle = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |kk\rangle$ be the maximally entangled state in dimension d and consider its mixture with white noise,

$$\rho = p |\phi^+\rangle\langle\phi^+| + \frac{1-p}{d^2}\mathbb{1}, \quad (\text{B.9})$$

where the parameter $p \in [0, 1]$ is called visibility. Using the PPT criterion, find the smallest p above which ρ is entangled.

Solution (p. 159)

B.1.7 Entanglement witness for qubit states ●

In the main text we derived the entanglement witness for qubit systems

$$\langle \sigma_X \otimes \sigma_X + \sigma_Z \otimes \sigma_Z \rangle \stackrel{\text{SEP}}{\leq} 1 \stackrel{\text{ENT}}{\leq} 2, \quad (\text{B.10})$$

where σ_X and σ_Z are Pauli matrices. When applied to the noisy state (B.9) in dimension $d = 2$, we found a violation for $p > \frac{1}{2}$. Can you think of another entanglement witness that is able to detect entanglement for $p > \frac{1}{3}$?

Hint: The witness in (B.10) only measures two Pauli matrices. Extend it with one more measurement.

Solution (p. 160)

B.1.8 Entanglement witness with anticommuting operators ●●

Let $\{X_i\}_i$ be any set of n Hermitian operators with eigenvalues ± 1 such that $\{X_i, X_j\} = 0$ for $i \neq j$.

- (a) Prove that

$$\sum_{i=1}^n \langle X_i \rangle^2 \leq 1. \quad (\text{B.11})$$

Hint: Define $B = \sum_{i=1}^n \lambda_i X_i$ for a set of real coefficients $\{\lambda_i\}_i$ such that $\sum_{i=1}^n \lambda_i^2 = 1$ and compute B^2 .

- (b) Use (a) to prove that for all separable states ρ_{AB}

$$\sum_{i=1}^n \text{tr}(X_i \otimes X_i \rho_{AB}) \leq 1. \quad (\text{B.12})$$

Solution (p. 161)

B.1.9 GHZ and W states ●

Consider the following three-qubit GHZ and W states,

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle), \quad \text{and} \quad |W\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle). \quad (\text{B.13})$$

Compute in both cases the reduced state of two qubits. Are the reduced states entangled?

Solution (p. 162)

B.1.10 Stochastic transformations ●

Consider two pure bipartite entangled states $|\psi\rangle_{AB}$ and $|\phi\rangle_{AB}$ on the same Hilbert space whose local dimension is d . Show that it always is possible, with a non-zero success probability, to transform $|\psi\rangle_{AB}$ into $|\phi\rangle_{AB}$ by performing an appropriate POVM $\{E, E_\perp\}$ on system A and post select on the outcome associated with E .

Hint: Use the Schmidt decomposition for $|\psi\rangle$ and $|\phi\rangle$ and draw inspiration to make a good ansatz for the POVM $\{E, E_\perp\}$.

Solution (p. 163)

B.2. Bell nonlocality

B.2.1 Convexity of quantum Bell correlations ●

Show that the set of quantum correlations in a bipartite Bell experiment is convex.

Hint: Use that there is no limit on the Hilbert space dimension.

Solution (p. 163)

B.2.2 Facets of the local polytope ●●●●

In Chapter 5.2 we stated that the CHSH inequality is a facet of the local polytope in the scenario in which Alice and Bob have two possible inputs, $(x, y) \in \{0, 1\}$, and two outputs each, $(a, b) \in \{-1, 1\}$. In this exercise, we will prove this result.

For each choice of settings (x, y) , define the correlation coefficient

$$E_{xy} = p(a = b|x, y) - p(a \neq b|x, y) = \langle a_x b_y \rangle, \quad (\text{B.14})$$

and the correlation vectors $u = (E_{00}, E_{01}, E_{10}, E_{11})$. In a deterministic local input/output strategy, we can write $E_{xy} = a_x b_y$, where a_x and b_y are Alice and Bob's outputs for their given inputs.

- What is the range of possible values for E_{xy} ? How many correlation vectors are there such that $|u|^2 = 4$?
- Show that the product of the correlation coefficients $E_{00}E_{01}E_{10}E_{11}$ for local deterministic strategies is always 1. Find the eight possible tuples $(E_{00}, E_{01}, E_{10}, E_{11})$ that lead to this result and show that they span a real four-dimensional space. Each such tuple represents an extremal point of the local correlation polytope.
- In (b) we have found that the smallest embedding of the local polytope is $\mathbb{R}^4$, i.e. we need four linearly independent vectors to define a facet. There are sixteen sets of these vectors $V_{\vec{s}} = \{s_1 v_1, s_2 v_2, s_3 v_3, s_4 v_4\}$, with $\vec{s} = (s_1, s_2, s_3, s_4) \in \{-1, +1\}^4$. Find the normal $n_{\vec{s}}$ to each hyperplane generated by $V_{\vec{s}}$.
Hint: The normal to the hyperplane of the facet should satisfy $n_{\vec{s}} \cdot (s_k v_k) = 4$ for $k = 1, \dots, 4$.
- Each of the sixteen sets $V_{\vec{s}}$ defines a facet through

$$n_{\vec{s}} \cdot u \leq 4. \quad (\text{B.15})$$

Show that all the facet inequalities are either the CHSH inequality (up to relabelings) or $E_{xy} \leq 1$.

Solution (p. 163)

B.2.3 Gisin's theorem ●●●

In a slightly different way than in Chapter 5.2, the CHSH inequality can be stated as

$$\mathcal{S} = \langle A_0 B_0 \rangle - \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle + \langle A_1 B_1 \rangle \leq 2, \quad (\text{B.16})$$

where $\{A_0, A_1\}$ denote two measurement settings on Alice's side, $\{B_0, B_1\}$ denote two measurement settings on Bob's side and $\langle A_i B_j \rangle = \langle \psi | \vec{a}^{(i)} \cdot \vec{\sigma} \otimes \vec{b}^{(j)} \cdot \vec{\sigma} | \psi \rangle$ is the expectation value of those measurements on the state $|\psi\rangle_{AB}$, where $\vec{a}^{(i)}$ and $\vec{b}^{(j)}$ are the Bloch vectors associated to the measurements. In Schmidt decomposition, any pure two-qubit state is written,

$$|\psi\rangle = c_1|00\rangle + c_2|11\rangle, \quad c_1, c_2 > 0, \quad c_1^2 + c_2^2 = 1. \quad (\text{B.17})$$

We will show through several steps that the inequality (B.16) is violated by all pure qubit entangled states.

- (a) Recall that $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$ is the vector of the Pauli matrices and that for two Bloch vectors $\vec{a} = (a_X, a_Y, a_Z)$ and $\vec{b} = (b_X, b_Y, b_Z)$. Compute the effect of $\vec{a} \cdot \vec{\sigma}$ and $\vec{b} \cdot \vec{\sigma}$ on the computational basis states $|0\rangle, |1\rangle$, i.e.

$$\vec{a} \cdot \vec{\sigma}|0\rangle, \quad \vec{a} \cdot \vec{\sigma}|1\rangle, \quad \vec{b} \cdot \vec{\sigma}|0\rangle, \quad \vec{b} \cdot \vec{\sigma}|1\rangle. \quad (\text{B.18})$$

- (b) Using the expression (B.17) and the results derived in (a), compute $\langle A_i B_j \rangle$.

- (c) Without loss of generality, we can choose the measurement directions in the XZ plane, i.e. $a_Y^{(i)} = b_Y^{(j)} = 0$ and $a_X^{(i)} = \sin(\alpha_i), a_Z^{(i)} = \cos(\alpha_i), b_X^{(j)} = \sin(\beta_j), b_Z^{(j)} = \cos(\beta_j)$, where $\{\alpha_i, \beta_j\}$ for $i \in \{0, 1\}$ and $j \in \{0, 1\}$ are the angles related to the x axis. Furthermore, fix $\alpha_0 = 0$ and $\alpha_1 = \frac{\pi}{2}$ and compute (B.16). What is the maximal value of $\mathcal{S}$ with respect to β_0 and β_1 ?

Solution (p. 164)

B.2.4 Biased CHSH inequality ●●●

Consider a Bell scenario where Alice and Bob have binary inputs $x \in \{0, 1\}$ and $y \in \{0, 1\}$ respectively and where they have binary outcomes $a, b \in \{0, 1\}$. Unlike the CHSH game, we assume that Alice and Bob draw their inputs from biased distributions, $p_A(x = 0) = p$ and $p_B(y = 0) = q$ respectively. The biased CHSH parameter becomes

$$\mathcal{S} = pq\langle A_0 B_0 \rangle + p(1-q)\langle A_0 B_1 \rangle + (1-p)q\langle A_1 B_0 \rangle - (1-p)(1-q)\langle A_1 B_1 \rangle. \quad (\text{B.19})$$

Without loss of generality we assume $p \geq \frac{1}{2}$ and $q \geq \frac{1}{2}$. Note that for $p = q = \frac{1}{2}$ this is just the ordinary CHSH parameter with an extra factor of $\frac{1}{4}$.

- (a) Prove that for local hidden variable models

$$\mathcal{S} \leq 1 - 2(1-p)(1-q) \quad (\text{B.20})$$

- (b) Derive an upper bound on $\mathcal{S}$ valid for any quantum model. When is this bound tight?

Hint: Follow the approach leading to Tsirelson's bound.

Solution (p. 165)

B.2.5 Horodecki criterion ●●●

Consider that we are given an arbitrary two-qubit state, ρ_{AB} , on the form in Eq (5.18). The Horodecki criterion states that the largest CHSH value achievable with this state is given by $\mathcal{S}(\rho) = 2\sqrt{\lambda_1 + \lambda_2}$, where λ_1 and λ_2 are the two largest eigenvalues of the 3×3 matrix $T^T T$, where $T_{ij} = \text{tr}(\rho_{AB} \sigma_i \otimes \sigma_j)$ for $i, j = X, Y, Z$. Our task is to prove this result.

- (a) Start from Eq (5.18) and let A_0, A_1, B_0 and B_1 be observables with Bloch vectors $\vec{a}_0, \vec{a}_1, \vec{b}_0$ and $\vec{b}_1$ respectively. Show that the joint expectation values can be written as $\langle A_x, B_y \rangle_\rho = \vec{a}_x \cdot (T \vec{b}_y)$.
- (b) Evaluate the largest possible value of the CHSH parameter over the Bloch vectors $\vec{a}_0$ and $\vec{a}_1$.
- (c) Derive the Horodecki criterion by optimising the expression from (b).

Hint: Use that $\vec{b}_0$ and $\vec{b}_1$ span a two-dimensional plane of the Bloch sphere.

Solution (p. 166)

B.2.6 I_{3322} Bell inequality ●●

The CHSH inequality is the simplest Bell inequality because Alice and Bob have only binary inputs and outputs. The next-simplest case, called I_{3322} Bell inequality, gives them three possible inputs each. It reads

$$-2p_B(0|1) - p_B(0|2) - p_A(0|1) + \sum_{x,y=1}^3 c_{xy} p(0, 0|x, y) \leq 0, \quad (\text{B.21})$$

where p_A and p_B are the marginal probabilities of Alice and Bob and the coefficients are

$$c = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 0 \end{pmatrix}. \quad (\text{B.22})$$

We will show that there are entangled states that cannot violate the CHSH inequality but can violate the I_{3322} inequality.

(a) Define $|\phi\rangle = \frac{2|00\rangle + |11\rangle}{\sqrt{5}}$ and consider the state

$$\rho = \frac{17}{20}|\phi\rangle\langle\phi| + \frac{3}{20}|01\rangle\langle 01|. \quad (\text{B.23})$$

Use the Horodecki criterion from Chapter 5.3 to show that it cannot violate the CHSH inequality.

(b) Consider the following three qubit measurements for each party,

$$A_{a|x} = \frac{1}{2}(\mathbb{1} + (-1)^a \vec{a}_x \cdot \vec{\sigma}), \quad B_{b|y} = \frac{1}{2}(\mathbb{1} + (-1)^b \vec{b}_y \cdot \vec{\sigma}), \quad (\text{B.24})$$

where $(a, b) \in \{0, 1\}$ and $\vec{a}_x, \vec{b}_y$ are Alice and Bob's measurement Bloch vectors. Choose these in the XZ -plane as

$$\vec{a}_x = (\sin \theta_x, \cos \theta_x), \quad \vec{b}_y = (\sin \phi_y, \cos \phi_y), \quad (\text{B.25})$$

with

$$\begin{aligned} \theta_1 &= \eta, & \theta_2 &= -\eta, & \theta_3 &= -\frac{\pi}{2}, \\ \phi_1 &= -\chi, & \phi_2 &= \chi, & \phi_3 &= \frac{\pi}{2}, \end{aligned} \quad (\text{B.26})$$

where $\eta = \arccos\left(\sqrt{\frac{7}{8}}\right)$ and $\chi = \arccos\left(\sqrt{\frac{2}{3}}\right)$. Use Born's rule to calculate the quantum correlations $p(a, b|x, y)$ and show that the I_{3322} Bell inequality is violated.

Hint: It is easier to use a computer to calculate the probabilities appearing in (B.21).

Solution (p. 167)

B.2.7 Hardy's nonlocality argument ●●●

Consider a bipartite Bell scenario where Alice and Bob each have binary inputs and binary outputs. Their joint probability distribution is $p(a, b|x, y)$. Consider the following conditions,

$$p(+1, +1|1, 1) = 0, \quad p(-1, -1|1, 2) = 0, \quad p(-1, -1|2, 1) = 0, \quad p(-1, -1|2, 2) > 0. \quad (\text{B.27})$$

(a) Show that the conditions (B.27) cannot be satisfied in a local hidden variable model.

Hint: List all the deterministic strategies and consider which of them can occur with a non-zero probability in a local hidden variable model that is compatible with the constraints.

(b) Select the state $|\psi\rangle = \frac{1}{\sqrt{3}}(|01\rangle + |10\rangle + |11\rangle)$ and the measurements σ_Z and σ_X for both Alice and Bob. Show that all conditions in (B.27) are satisfied in this quantum model.

(c) Consider now that we add to the conditions (B.27) also the condition

$$p(-1, -1|1, 1) = 0. \quad (\text{B.28})$$

Show that all the conditions cannot be satisfied in quantum theory.

Solution (p. 168)

B.2.8 GHZ paradox ●●

Consider a three-party Bell scenario in which Alice, Bob and Charlie each choose one of two possible measurements, labelled 0 and 1. Their respective inputs are denoted x, y and z . They are asked to return binary outputs a, b and c with values ± 1 respectively. These should have the following properties. If exactly one party selects 1 then they output $abc = -1$. If all three parties select 1, then they output $abc = +1$.

(a) Prove that no local hidden variable theory permits Alice, Bob and Charlie to win the game.

(b) Consider a quantum model in which all three parties choose between σ_Y and σ_X . Their shared state is the GHZ state $|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$. Show that the game is won.

Solution (p. 169)

B.2.9 Mermin's Bell inequality ●●●

The GHZ paradox from Exercise B.2.8 inspires a three-party Bell inequality called Mermin's inequality. It reads

$$\mathcal{S} = \langle A_1 B_1 C_1 \rangle - \langle A_0 B_0 C_1 \rangle - \langle A_0 B_1 C_0 \rangle - \langle A_1 B_0 C_0 \rangle \leq 2, \quad (\text{B.29})$$

where the bound holds for all local hidden variable models. The winning strategy in the GHZ paradox achieves the violation $\mathcal{S} = 4$.

(a) Apply the measurements from the GHZ paradox to the state

$$\rho = v|\text{GHZ}\rangle\langle\text{GHZ}| + \frac{1-v}{8}\mathbb{1}. \quad (\text{B.30})$$

What is the smallest value of $v \in [0, 1]$ for which Mermin's inequality is violated?

(b) Apply the given measurements to the state

$$\rho = \phi^+ \otimes |s\rangle\langle s|, \quad (\text{B.31})$$

where $|s\rangle\langle s| = \frac{1}{2} \left(\mathbb{1} - \frac{1}{\sqrt{2}} (\sigma_X + \sigma_Y) \right)$. Does this violate Mermin's inequality?

(c) Assume that the three-partite state is pure and only has entanglement between Alice and Bob, i.e. it is of the form $|\psi\rangle_{AB} \otimes |\phi\rangle_C$. Show that Mermin's inequality can be reduced to the CHSH inequality.

Solution (p. 169)

B.2.10 No-signaling in local hidden variable theories and quantum theory ●●

Consider a bipartite Bell scenario where Alice and Bob have input/output pairs (a, x) and (b, y) respectively.

(a) Show that every local hidden variable model respects the no-signaling principle.

(b) Show that every quantum model respects the no-signaling principle.

Solution (p. 170)

B.3. Quantum information primitives

B.3.1 d -dimensional dense coding ●

Consider a dense coding scenario in which Alice and Bob share a maximally entangled state $|\phi^+\rangle = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |ii\rangle$ of local dimension d . Alice draws uniformly at random a symbol x selected from the alphabet $\{1, \dots, d^2\}$. She then applies to her state the unitary $U_x = X^{x_1} Z^{x_2}$ before relaying the state to Bob. Here, we have defined

$$X = \sum_{k=0}^{d-1} |k+1\rangle\langle k|, \quad Z = \sum_{k=0}^{d-1} e^{\frac{2\pi i}{d}k} |k\rangle\langle k|, \quad (\text{B.32})$$

where $k+1$ is computed modulo d , and where we have written $x = x_1 x_2 \in \{0, \dots, d-1\}^2$. Show that there exists a measurement that allows Bob to deterministically read out x .

Solution (p. 170)

B.3.2 Dense coding with separable measurement ●

Consider that Alice and Bob share the entangled state, $|\psi\rangle_{AB}$, of local dimension d . Alice selects $x \in \{1, \dots, n\}$ and uses a unitary operation, U_x , to encode her input x into her share of $|\psi\rangle$. The total state is given to Bob who performs a separable measurement $E_b = \sum_{\lambda} M_{b,\lambda}^A \otimes M_{b,\lambda}^B$, where $\{M_{b,\lambda}^A\}$ and $\{M_{b,\lambda}^B\}$ are suitable single-system operators. Prove that the winning probability of minimal error state discrimination cannot exceed the classical limit, namely

$$p_{\text{win}} = \frac{1}{n} \sum_{x=1}^n p(b=x|x) \leq \frac{d}{n}. \quad (\text{B.33})$$

Solution (p. 171)

B.3.3 Teleportation with non-maximally entangled state ●●●

Consider the teleportation protocol introduced in Chapter 6.2, but instead of sharing the maximally entangled state Alice and Bob now have the shared state

$$|\varphi\rangle_{AB} = \lambda_1|00\rangle_{AB} + \lambda_2|11\rangle_{AB}, \quad \text{where } \lambda_1^2 + \lambda_2^2 = 1. \quad (\text{B.34})$$

- (a) Alice wants to teleport the state $|\psi\rangle_{A'} = \alpha|0\rangle_{A'} + \beta|1\rangle_{A'}$, where $\alpha = \cos \frac{\theta}{2}$ and $\beta = e^{i\varphi} \sin \frac{\theta}{2}$. Compute the state of combined three qubit system

$$|\psi\rangle_{A'} \otimes |\varphi\rangle_{AB}. \quad (\text{B.35})$$

Express the states on $A'A$ in terms of the basis of Bell states $\{|\phi^\pm\rangle, |\psi^\pm\rangle\}$ defined in G18.

- (b) As in the standard teleportation protocol, Bob performs the unitaries $U_k = \{\mathbb{1}, \sigma_Z, \sigma_X, \sigma_X \sigma_Z\}$ on his share of the entangled state $|\varphi\rangle_{AB}$ after Alice sends him the classical outcome k of her Bell state measurement. Compute the fidelity of Bob's states with the state Alice wants to teleport.
- (c) Show that the average fidelity of the protocol is

$$F = \int d\psi \langle \psi | \rho | \psi \rangle = \frac{2 + 2\lambda_1\lambda_2}{3}, \quad (\text{B.36})$$

where ρ is the state Bob gets after performing the unitaries.

Solution (p. 171)

B.3.4 Remote state preparation of real-valued qubit ●

Assume that Alice prepares a state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ that is unknown to Bob, but with the promise that it is confined to the real plane of the Bloch sphere ($\alpha, \beta \in \mathbb{R}$). Let Alice and Bob share a maximally entangled state and permit one bit of communication from Alice to Bob which can use her knowledge of $|\psi\rangle$. Find a way to use these resources to create $|\psi\rangle$ in Bob's lab.

Solution (p. 173)

B.3.5 Entanglement swapping with any entangled measurement ●

Consider four d -dimensional systems (qudits). Both qudits A and B , and C and D are in the maximally entangled state,

$$|\phi^+\rangle_{AB} = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |kk\rangle_{AB}, \quad \text{and} \quad |\phi^+\rangle_{CD} = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |kk\rangle_{CD}. \quad (\text{B.37})$$

As explained in Chapter 6.3, if we project qudits B and C onto $|\phi^+\rangle$ then the qudits A and D become maximally entangled. Consider now that we instead project the qudits B and C onto $E = \rho^T$, where ρ is some arbitrary pure state. Show that the post-measurement state of qudits A and D is ρ .

Hint: Use the property of the maximally entangled state in Eq (4.21).

Solution (p. 174)

B.3.6 Greenberger-Horne-Zeilinger measurement ●●

One way to generalise the Bell state measurement is to consider a basis of n -qubit GHZ states. The measurement is defined as a projection onto the basis

$$|\psi_a\rangle = (\sigma_Z^{a_1} \otimes \sigma_X^{a_2} \otimes \dots \otimes \sigma_X^{a_n}) |GHZ\rangle, \quad (\text{B.38})$$

where $a = a_1 \dots a_n \in \{0, 1\}^n$ labels the basis vectors and $|GHZ\rangle = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}$ is the GHZ state.

- (a) Show that $\{|\psi_a\rangle\}_a$ is a basis of n -qubit Hilbert space.
- (b) Imagine that we have n parties and that party number k , for $k = 1, \dots, n$, shares the state $\phi_{A_k B_k}^+$ with a single party called B . Show that if B measures all the n systems given to him in the transpose of the GHZ basis and then communicates the outcome to the n parties, they can perform local unitaries so that they now share the GHZ state.

Solution (p. 174)

B.3.7 3-bit random access code ●●●

Consider that Alice has three bits $x_1, x_2, x_3 \in \{0, 1\}$ which she encodes into a single qubit $\rho_{x_1 x_2 x_3}$ that is sent to Bob. Bob wants to learn one of the three bits, which he marks by $y \in \{1, 2, 3\}$. His guess for its value is denoted b . If all selections are uniformly random, the average success probability becomes

$$p_{\text{win}} = \frac{1}{24} \sum_{x_1, x_2, x_3} \sum_y p(b = x_y | x_1, x_2, x_3, y). \quad (\text{B.39})$$

Show the following.

- (a) The winning probability can be expressed as

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{48} \sum_{x_1, x_2, x_3} \sum_y (-1)^{x_y} \text{tr}(\rho_{x_1 x_2 x_3} E_y), \quad (\text{B.40})$$

where $E_y = E_{0|y} - E_{1|y}$ is Bob's observable.

- (b) Let Alice prepare her eight states so that their Bloch vectors, $\hat{n}_{x_1 x_2 x_3}$, form the vertices of a cube in on the Bloch sphere, $\hat{n}_{x_1 x_2 x_3} = \frac{1}{\sqrt{3}} [(-1)^{x_1}, (-1)^{x_2}, (-1)^{x_3}]$. Compute p_{win} as a function of Bob's measurement Bloch vectors.

Hint: Define vectors $\vec{r}_y = \sum_{x_1, x_2, x_3} (-1)^{x_y} \hat{n}_{x_1 x_2 x_3}$ and use them to express p_{win} .

- (c) How should we select Bob's three measurements in order to find the largest possible success probability?

Solution (p. 174)

B.3.8 Simulating measurement statistics on a qubit ●●

Teleportation shows that a qubit can be substituted for two bits of communication and a maximally entangled state. Consider now that we are interested only in simulating the outcome statistics obtained from measuring the qubit: Alice sends a qubit, ρ , to Bob who performs a measurement $\{E_b\}_b$ on that qubit, yielding the distribution $p(b|\rho, M) = \text{tr}(\rho E_b)$.

- (a) Show that if Bob's measurement is projective the outcome statistics can be simulated by sharing a maximally entangled state and Alice sending just one bit to Bob.

Hint: Draw inspiration from the solution to Exercise B.3.4.

- (b) Show that the protocol from (a) can be modified so that it also can simulate the qubit's statistics if Bob's measurement is any POVM. What is the conceptual difference between the two simulation protocols?

Solution (p. 175)

B.3.9 Phase-covariant cloning machine ●●●

Consider a qubit cloning machine that is not universal. Its task is only to clone states in the XY -plane of the Bloch sphere, i.e. any state of the form $|\psi_\theta\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{e^{i\theta}}{\sqrt{2}}|1\rangle$. Let the machine implement the transformation

$$|00\rangle \rightarrow |00\rangle, \quad |10\rangle \rightarrow \cos \varphi |10\rangle + \sin \varphi |01\rangle, \quad (\text{B.41})$$

for $\varphi \in [0, \frac{\pi}{2}]$.

- (a) Show that the transformation (B.41) is unitary.
- (b) Compute the action of the transformation on the state $|\psi_\theta\rangle \otimes |0\rangle$.
- (c) Select φ so that both clones have the same fidelity with the input state. Show that their fidelity is $F(\psi_\theta, \rho_\theta) = \frac{2+\sqrt{2}}{4}$.

Solution (p. 176)

B.3.10 Cloning fidelity from no-signaling principle ●●●●

We will prove that a universal quantum cloning machine (QCM) achieving a cloning fidelity larger than $F = \frac{5}{6}$ would violate the no-signaling principle. Denote the qubit input and the two-qubit output of the QCM as $\rho_{\text{in}}(\hat{m})$ and $\rho_{\text{out}}(\hat{m})$, respectively. Express them in their Bloch representation as

$$\rho_{\text{in}}(\hat{m}) = \frac{1}{2}(\mathbb{1} + \hat{m} \cdot \vec{\sigma}), \quad \text{and} \quad \rho_{\text{out}}(\hat{m}) = \frac{1}{4} \left(\mathbb{1} \otimes \mathbb{1} + \eta (\hat{m} \cdot \vec{\sigma} \otimes \mathbb{1} + \mathbb{1} \otimes \hat{m} \cdot \vec{\sigma}) + \sum_{i,j=X,Y,Z} T_{ij} \sigma_i \otimes \sigma_j \right), \quad (\text{B.42})$$

where $\hat{m}$ is the unit Bloch vector of the input state and η is the shrinking factor of $\hat{m}$ which characterises the cloned states residing in the reductions of $\rho_{\text{out}}(\hat{m})$.

- (a) Compute $\text{tr}_B(\rho_{\text{out}}(\hat{m}))$ and show that its fidelity with the input state is

$$F = \frac{1 + \eta}{2}. \quad (\text{B.43})$$

- (b) Since the cloning machine is universal, its action on a rotated state $U\rho_{\text{in}}(\hat{m})U^\dagger$ is the same as its action on $\rho_{\text{in}}(\hat{m})$ followed by the rotation U , i.e., defining R_U as the corresponding rotation matrix for the Bloch vector $\hat{m}$ we have

$$\rho_{\text{out}}(R_U\hat{m}) = U \otimes U \rho_{\text{out}}(\hat{m}) U^\dagger \otimes U^\dagger. \quad (\text{B.44})$$

- i. Show that $(R_V\hat{m}) \cdot \vec{\sigma} = \hat{m} \cdot V\vec{\sigma}V^\dagger$ for any unitary V .
 ii. Consider $U = e^{-i\alpha\sigma_Z}$, define $f(i, j) = U\sigma_i U^\dagger \otimes U\sigma_j U^\dagger$ and prove that

$$\begin{cases} f(X, X) + f(Y, Y) = \sigma_X \otimes \sigma_X + \sigma_Y \otimes \sigma_Y, \\ f(X, Y) - f(Y, X) = \sigma_X \otimes \sigma_Y - \sigma_Y \otimes \sigma_X, \\ f(Z, Z) = \sigma_Z \otimes \sigma_Z, \end{cases} \quad (\text{B.45})$$

which explicitly shows that U acts non-trivially only on the 2-dimensional subspace XY . Therefore, (B.44) implies that $T_{XX} = T_{YY}$, $T_{XY} = -T_{YX}$ and $T_{XZ} = T_{ZX} = T_{YZ} = T_{ZY} = 0$. Using the constraints and assuming $\hat{m} = (0, 0, 1)$, write the expression for $\rho_{\text{out}}(\hat{m})$.

- (c) Let Alice and Bob share the bipartite state ρ_{AB} and let $\rho_i^{(B)}$ for $i \in \{0, 1+, -\}$ denote Bob's reduced state when Alice measures in either the Z or X basis and obtains the outcome 0/1 and $+/-$ respectively. The no-signaling condition requires that

$$\rho_0^{(B)} + \rho_1^{(B)} = \rho_+^{(B)} + \rho_-^{(B)}, \quad (\text{B.46})$$

Show that applying (B.46) on $\rho_{\text{out}}(\hat{m})$ implies that $T_{XX} = T_{ZZ}$ and $T_{XY} = 0$.

- (d) Find the maximum value of η for which $\rho_{\text{out}}(\hat{m})$ is positive semidefinite and use it in (B.43) to show that the maximum cloning fidelity is $F = \frac{5}{6}$.

Solution (p. 176)

B.4. Quantum channels and operations

B.4.1 Random Pauli channel ●

Consider a qubit channel that randomly applies one of the unitaries $\{\mathbb{1}, \sigma_X, \sigma_Y, \sigma_Z\}$, i.e.

$$\Lambda(\rho) = \frac{1}{4}[\mathbb{1}\rho\mathbb{1} + \sigma_X\rho\sigma_X + \sigma_Y\rho\sigma_Y + \sigma_Z\rho\sigma_Z]. \quad (\text{B.47})$$

Show that the action of this channel on any qubit state ρ is the maximally mixed state, i.e.

$$\Lambda(\rho) = \frac{\mathbb{1}}{2}, \quad \forall \rho. \quad (\text{B.48})$$

Hint: Use the Bloch representation for the qubit state ρ .

Solution (p. 179)

B.4.2 Kraus representation ●●●●

For each of the following quantum channels, verify that the Kraus operators satisfy $\Lambda(\rho) = \sum_i K_i \rho K_i^\dagger$ and $\sum_i K_i^\dagger K_i = \mathbb{1}$.

- (a) A measurement is a special type of channel that maps a quantum system into a classical system. Let $\{E_a\}_a$ be the implemented POVM and let $\{|a\rangle\}_a$ be a basis, then the channel is

$$\Lambda(\rho) = \sum_a \text{tr}(\rho E_a) |a\rangle\langle a|, \quad (\text{B.49})$$

with Kraus operators are $K_{a,i} = |a\rangle\langle i| \sqrt{E_a}$.

- (b) The partial trace is a quantum channel in which we discard a part of a quantum system. This channel is defined through $\Lambda : \rho_{AB} \rightarrow \rho_B$ and the Kraus operators are $K_i = \langle i|_A$, where $\{|i\rangle_A\}_i$ is a basis of $\mathcal{H}_A$.
- (c) The tensor product channel is defined as $\Lambda : \rho_A \rightarrow \rho_A \otimes \sigma_B$. Taken the spectral decomposition of the appended state, $\sigma = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$, the Kraus operators are defined as $K_i = \mathbb{1}_A \otimes \sqrt{\lambda_i} |\lambda_i\rangle$.
- (d) We measure a state ρ in some basis $\{|\phi_a\rangle\}_a$ and then prepare a pure quantum state based on the outcome. This corresponds to the channel $\Lambda : \rho \rightarrow \sum_a \langle\phi_a|\rho|\phi_a\rangle \sigma_a$. The Kraus operators are $K_a = |\psi_a\rangle\langle\phi_a|$, where $\sigma_a = |\psi_a\rangle\langle\psi_a|$.

Solution (p. 179)

B.4.3 Dephasing channel ●●

The qubit dephasing channel dampens the coherence terms in the computational basis. It is defined as

$$\Lambda(\rho) = v\rho + (1-v)\sigma_Z\rho\sigma_Z. \quad (\text{B.50})$$

- (a) Verify that the Kraus operators are

$$K_1 = \sqrt{v}\mathbb{1}, \quad K_2 = \sqrt{1-v}\sigma_Z. \quad (\text{B.51})$$

- (b) Compute the effect of the dephasing channel on the Bloch vector.

Solution (p. 180)

B.4.4 Choi, Kraus and dual ●●●

Consider the qubit depolarisation channel,

$$\Lambda(\rho) = v\rho + (1-v)\text{tr}(\rho)\frac{\mathbb{1}}{2}. \quad (\text{B.52})$$

- (a) Compute the Choi representation of the channel.
- (b) Compute the Kraus representation of the channel.
- (c) Compute the dual channel $\Lambda^\dagger$.

Solution (p. 181)

B.4.5 Choi representation of dual channel ●

Let $\Lambda_{A \rightarrow A'}$ be a channel and let $\eta_{A'A}$ be its Choi operator. Also, let $\Lambda_{A' \rightarrow A}^\dagger$ be the channel dual to Λ and let γ be its Choi operator. Show that

$$\eta_{A'A} = \frac{d_{A'}}{d_A} P^\dagger \gamma_{AA'}^T P, \quad (\text{B.53})$$

where P is the swap operator defined as $P|i\rangle \otimes |j\rangle = |j\rangle \otimes |i\rangle$.

Hint: Use (7.12) to express the action of a channel $\Lambda_{A \rightarrow A'}$ in terms of the Choi state $\eta_{A'A}$ and use that, given two operators $O_A \in \mathcal{H}_A$ and $S_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$, $\text{tr}(O_A \text{tr}_B(S_{AB})) = \text{tr}((O_A \otimes \mathbb{1}_B)S_{AB})$.

Solution (p. 182)

B.4.6 Measure-and-prepare channels ●●

A channel Λ is called entanglement-breaking if when applied to half of any bipartite state it outputs a separable state, i.e.

$$(\Lambda \otimes \mathbb{1})[\rho_{AB}] \text{ is separable, } \forall \rho_{AB}. \quad (\text{B.54})$$

Consider a POVM $\{E_i\}_i$ and a set of states $\{\sigma_i\}_i$. A measure-and-prepare channel is defined as

$$\Lambda(\rho) = \sum_i \text{tr}(\rho E_i) \sigma_i, \quad \forall \rho, \quad (\text{B.55})$$

i.e. it performs a measurement on the state ρ and if outcome i is observed it prepares a new state σ_i .

- (a) Show that measure-and-prepare channels are entanglement breaking.
- (b) Compute the Choi state η of the measure-and-prepare channel. Is it separable or entangled?

Solution (p. 183)

B.4.7 Choi state of entanglement-breaking channel ●●

Prove that

- (a) if a channel Λ is entanglement-breaking (as defined in Exercise B.4.6), its Choi state is separable.
- (b) if the Choi state of a channel Λ is separable then the channel is entanglement-breaking.

Hint: Use (7.12) to express the action of a channel $\Lambda_{A \rightarrow A'}$ in terms of the Choi state $\eta_{A'A}$.

Solution (p. 183)

B.4.8 Gentle measurement ●●

Let ρ be a state and E be a POVM-element. Also, let $\rho_E = \sqrt{E}\rho\sqrt{E}/\text{tr}(\rho E)$ be the Lüders post-measurement state conditioned on the outcome associated with E . In Chapter 7.7 we have proven that, if $\text{tr}(\rho E) \geq 1 - \epsilon$, then

$$\|\rho - \rho_E\|_1 \leq 2\sqrt{\epsilon}. \quad (\text{B.56})$$

Now we will prove that the same inequality holds also if we redefine $\rho_E = \sqrt{E}\rho\sqrt{E}$.

- (a) Prove the operator form of the Cauchy-Schwarz inequality, namely that for any operators A and B

$$|\text{tr}(A^\dagger B)|^2 \leq \text{tr}(A^\dagger A) \text{tr}(B^\dagger B). \quad (\text{B.57})$$

Hint: Define a polynomial $f(\lambda) = \text{tr}((A - \lambda B)^\dagger (A - \lambda B))$ and find its minimum.

- (b) Use the operator Cauchy-Schwarz inequality to prove (B.56).

Hint: Use the triangle inequality $\|A + B\|_1 \leq \|A\|_1 + \|B\|_1$.

Solution (p. 183)

B.4.9 Information-disturbance trade-off for qubits ●●

Consider that we randomly draw a state $\rho_0 = |0\rangle\langle 0|$ or $\rho_1 = |+\rangle\langle +|$, where $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$. On the one hand, we are interested in the probability of discriminating between the two states by measuring them. This is quantified by the discrimination probability $p_{\text{win}} \equiv \frac{1}{2}p(0|\rho_0) + \frac{1}{2}p(1|\rho_1)$. On the other hand, we are interested in the fidelity between the original state and the state after the discrimination measurement has been performed.

- (a) Denote by $\{\Lambda_0, \Lambda_1\}$ the instrument implementing the measurement that discriminates ρ_0 and ρ_1 and let $\{\eta_0^{A'A}, \eta_1^{A'A}\}$ denote the associated Choi operators. Show that

$$p_{\text{win}} = \frac{1}{2} + \text{tr}(|0\rangle\langle 0| + |+\rangle\langle +|)\eta_0^A, \quad (\text{B.58})$$

where $\eta_0^A = \text{tr}_{A'}(\eta_0^{A'A})$.

Hint: For two operators $O_A \in \mathcal{H}_A$ and $S_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ we have $\text{tr}(O_A \text{tr}_B(S_{AB})) = \text{tr}((O_A \otimes \mathbb{I}_B)S_{AB})$.

- (b) The map $\sum_a \Lambda_a$ is CPTP. We therefore write $\sum_a \Lambda_a^{A \rightarrow A'}(\psi) = 2 \text{tr}_A((\mathbb{I}_{A'} \otimes \psi_A^T)\eta^{A'A})$, where $\eta^{A'A} = \eta_0^{A'A} + \eta_1^{A'A}$ is the Choi state associated to the map $\sum_a \Lambda_a$. Show that the average fidelity of the post-measurement state with the initial state is given by

$$F = \frac{1}{2} \sum_{a=0,1} [\langle 0|\Lambda_a(\rho_0)|0\rangle + \langle +|\Lambda_a(\rho_1)|+\rangle] = \text{tr}(|00\rangle\langle 00| + |++\rangle\langle ++|)\eta^{A'A}. \quad (\text{B.59})$$

Solution (p. 184)

B.4.10 Information-disturbance trade-off in hemisphere discrimination ●●●●

In Chapter 7.7, we have considered the following information-disturbance problem. We apply a binary-outcome instrument represented by completely positive maps $\{\Lambda_N, \Lambda_S\}$ on a randomly drawn qubit state $|\psi\rangle$ with Bloch vector $\hat{n}$. The goal is to maximise the average probability, p_{win} , of determining whether it is located in the northern (N) or the

southern (S) Bloch hemisphere. For the post-measurement state, we evaluate its average fidelity, F , with $|\psi\rangle$. These quantities are given by

$$p_{\text{win}} = \frac{1}{2} \int_N d\psi_N \text{tr}(\Lambda_N(\psi)) + \frac{1}{2} \int_S d\psi_S \text{tr}(\Lambda_S(\psi)), \quad \text{and} \quad F = \int d\psi \langle \psi | \sum_{a=N,S} \Lambda_a(\psi) | \psi \rangle, \quad (\text{B.60})$$

where $d\psi_N$ ($d\psi_S$) and $d\psi$ are uniform densities over the northern (southern) hemisphere and the whole Bloch sphere, respectively, and they are normalised such that $\int_N d\psi_N = \int_S d\psi_S = \int d\psi = 1$.

- (a) Let $\{\eta_N^{A'A}, \eta_S^{A'A}\}$ be the Choi operators representing $\{\Lambda_N, \Lambda_S\}$ and let $\eta^{A'A} = \eta_N^{A'A} + \eta_S^{A'A}$ be the Choi state associated to the CPTP map $\Lambda_N + \Lambda_S$. Show that

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{2} \text{tr}(\eta_N^A \sigma_Z), \quad (\text{B.61})$$

where $\eta_N^A = \text{tr}_{A'}(\eta_N^{A'A})$.

Hint: The integral in the S hemisphere can be expressed as $\int_S d\psi_S \psi = \int_N d\psi_N (\sigma_X \psi \sigma_X)^$.*

- (b) Assume that $|\psi\rangle$ is invariant under a unitary U , i.e. $U|\psi\rangle = |\psi\rangle$ and show that $U \otimes U^*(\psi \otimes \psi^T)U^\dagger \otimes U^T = \psi \otimes \psi^T$.
(c) A well-known result, called Schur-Weyl duality, states that each state invariant under $U \otimes U^*$ can be written as

$$U \otimes U^* \rho_{AB} U^\dagger \otimes U^T = \rho_{AB} \quad \forall U \quad \implies \quad \rho_{AB} = \alpha \phi^+ + \beta \mathbb{1}, \quad (\text{B.62})$$

where ϕ^+ is the maximally entangled state. Rewrite the average fidelity in (B.60) as

$$F = 2 \int d\psi \text{tr}(\psi_{A'} \otimes \psi_A^T \eta^{A'A}), \quad (\text{B.63})$$

and use (B.62) to show that

$$F = 2 \int d\psi \text{tr}(\psi_{A'} \otimes \psi_A^T \eta^{A'A}) = \frac{1}{3} + \frac{2}{3} \text{tr}(\phi^+ \eta^{A'A}). \quad (\text{B.64})$$

Hint: Determine α and β using that $\int d\psi \text{tr}(\psi_{A'} \otimes \psi_A^T) = 1$ and computing $\int d\psi \text{tr}(\phi^+ \psi_{A'} \otimes \psi_A^T)$.

- (d) Recall that the normalised Choi operators associated with instruments $\{\Lambda_a\}_a$ are $\eta_a = (K_a \otimes \mathbb{1})\phi^+(K_a^\dagger \otimes \mathbb{1})$, where $\{K_a\}_a$ are the Kraus operators. In the main text we used for this task the Kraus operators

$$K_N = \sqrt{\frac{1+\gamma}{2}} |0\rangle\langle 0| + \sqrt{\frac{1-\gamma}{2}} |1\rangle\langle 1|, \quad K_S = \sqrt{\frac{1-\gamma}{2}} |0\rangle\langle 0| + \sqrt{\frac{1+\gamma}{2}} |1\rangle\langle 1|. \quad (\text{B.65})$$

Use these to compute η_a and then the values of p_{win} and F . Show that they become

$$p_{\text{win}} = \frac{1}{2} + \frac{\gamma}{4}, \quad F = \frac{2}{3} + \frac{\sqrt{1-\gamma^2}}{3}. \quad (\text{B.66})$$

Solution (p. 185)

B.5. Entropic concepts

B.5.1 Properties of Shannon entropy ●●●●

- (a) As this book is authored in Sweden, consider that we sample a random adult from Sweden's population and let the random variable X represent whether or not this person primarily sympathises with the far-right party named the Sweden Democrats (SD). According to data from May 2025 provided by Sweden's Central Bureau of Statistics, $p_X(X = \text{SD}) = 17.5\%$. Let Y be another random variable representing whether or not the person has at least three years of higher education (EDU). The updated probability of voting for SD provided by the Central Bureau of Statistics is $p_{X|Y}(X = \text{SD}|Y = \text{EDU}) = 6.0\%$. Compute the Shannon entropy of voting SD and the updated Shannon entropy after knowing $Y = \text{EDU}$. Which one is larger? Why?

- (b) In the main text we discussed the chain rule (8.7) for the Shannon entropy of two random variables X and Y . Use it to show that the chain rule can be extended to n random variables,

$$H(X_1, \dots, X_n) = H(X_1) + H(X_2|X_1) + H(X_3|X_2, X_1) + \dots + H(X_n|X_{n-1}, \dots, X_1). \quad (\text{B.67})$$

- (c) In the main text we claimed that conditioning cannot increase the Shannon entropy, namely $H(X) \geq H(X|Y)$. Prove this inequality and that equality holds if and only if X and Y are independent.

Hint: Use the log sum inequality, $\sum_i a_i \log \frac{a_i}{b_i} \geq (\sum_i a_i) \log \frac{\sum_{i'} a_{i'}}{\sum_{i''} b_{i''}}$, where equality holds if and only if $a_i = c b_i$ for some constant c .

- (d) Use (b) together with the chain rule to prove that the Shannon entropy is sub-additive, namely

$$H(X_1, \dots, X_n) \leq \sum_{i=1}^n H(X_i), \quad (\text{B.68})$$

with equality if and only if all the random variables are independent.

Solution (p. 188)

B.5.2 Quantum conditional entropy and mutual information ●●●●●●

- (a) Consider a separable state $\rho_{AB} = \sum_x q_x \phi_x \otimes \varphi_x$. Show that the conditional von Neumann entropy is non-negative,

$$H(A|B)_\rho \geq 0. \quad (\text{B.69})$$

Hint: Classically extend the state as $\rho_{ABX} = \sum_x q_x \phi_x \otimes \varphi_x \otimes |x\rangle\langle x|$.

- (b) Use the above reasoning to show that the quantum mutual information of a separable state satisfies

$$I(A : B) \leq \min \{H(A), H(B)\}. \quad (\text{B.70})$$

- (c) Compute the quantum mutual information for any pure bipartite entangled state. When is the inequality in (b) violated?
- (d) Compute the conditional von Neumann entropy of the two-qubit state $p \phi^+ + (1-p) \frac{1}{4}$. Numerically determine the values of p for which it is negative.
- (e) Show that for three quantum systems A, B and C , the following equalities hold

$$I(A : B, C) = I(A : B) + I(A, B : C) - I(C : B), \quad (\text{B.71})$$

$$I(A, B : C) = I(A : C) + I(B : C|A). \quad (\text{B.72})$$

- (f) Show that for product probability distributions $p(x_1, x_2) = p(x_1)p(x_2)$ and $p(y_1, y_2) = p(y_1)p(y_2)$,

$$I(X_1, X_2 : Y_1, Y_2) = I(X_1 : Y_1) + I(X_2 : Y_2). \quad (\text{B.73})$$

Solution (p. 188)

B.5.3 Klein's inequality ●

In the main text we have seen that the quantum relative entropy satisfies $D(\rho||\sigma) \geq 0$ when ρ and σ are states, with equality if and only if $\rho = \sigma$. This was shown by relying on quantum Stein's lemma. Prove it again, but now working directly from the definition of quantum relative entropy. To this end, use Jensen's inequality for a concave function $f(x)$,

$$f\left(\frac{1}{A} \sum_k a_k x_k\right) \geq \frac{1}{A} \sum_k a_k f(x_k), \quad (\text{B.74})$$

where $a_k \geq 0$ and $A = \sum_k a_k$.

Solution (p. 190)

B.5.4 Quantum relative entropy ●●●●●●

- (a) Let U be a unitary, ρ be a state and σ be a positive semidefinite operator. Show that the quantum relative entropy is invariant under the unitary,

$$D(\rho||\sigma) = D(U\rho U^\dagger||U\sigma U^\dagger) \quad (\text{B.75})$$

(b) Let X and Y be two positive semidefinite operators and show that

$$\log(X \otimes Y) = \log(X) \otimes \mathbb{1} + \mathbb{1} \otimes \log(Y). \quad (\text{B.76})$$

(c) Now let ρ and ρ' be states and let σ and σ' be positive semidefinite operators. Use the property derive in (b) to show that

$$D(\rho \otimes \rho' \| \sigma \otimes \sigma') = D(\rho \| \sigma) + D(\rho' \| \sigma'). \quad (\text{B.77})$$

(d) Show that a unital channel, Λ , cannot decrease the von Neumann entropy,

$$H(\Lambda(\rho)) \geq H(\rho). \quad (\text{B.78})$$

Can you show a simple violation of this inequality using a quantum channel that is not unital?

(e) Let ρ and σ be positive semidefinite and let $\rho = \sum_x q_x \rho_x$ and $\sigma = \sum_x q_x \sigma_x$ where $\{q_x\}_x$ is a probability distribution and ρ_x and σ_x are states. Show that

$$\sum_x q_x D(\rho_x \| \sigma_x) \geq D(\rho \| \sigma). \quad (\text{B.79})$$

Hint: Append a classical register to ρ and σ and use monotonicity of relative entropy.

Solution (p. 190)

B.5.5 Maassen-Uffink uncertainty relation ●●●

Let X and Y be random variables representing the outcome distribution obtained from measuring a quantum state, ρ , in the basis $A_x = |a_x\rangle\langle a_x|$ and $B_y = |b_y\rangle\langle b_y|$ respectively. Hence $p_X(x) = \langle a_x | \rho | a_x \rangle$ and $p_Y(y) = \langle b_y | \rho | b_y \rangle$. We will show that for any ρ , the sum of the entropies of the two distributions satisfies

$$H(X) + H(Y) \geq -\log(c), \quad (\text{B.80})$$

where $c = \max_{x,y} |\langle a_x | b_y \rangle|^2$.

(a) Define the channel

$$\Lambda^{A \rightarrow X}(\rho) = \sum_x \langle a_x | \rho | a_x \rangle |a_x\rangle\langle a_x| \quad (\text{B.81})$$

and show that

$$D(\rho \| \Lambda(\rho)) + H(A) = H(X). \quad (\text{B.82})$$

(b) Define the channel

$$\Gamma^{B \rightarrow Y}(\rho) = \sum_y \langle b_y | \rho | b_y \rangle |b_y\rangle\langle b_y|. \quad (\text{B.83})$$

Use (B.82) together with the monotonicity of quantum relative entropy to show that

$$H(X) - H(A) \geq \text{tr}(\Gamma(\rho) \log(\Gamma(\rho))) - \text{tr}(\Gamma(\rho) \log(\Gamma(\Lambda(\rho)))). \quad (\text{B.84})$$

(c) Evaluate each of the two terms on the right-hand-side of (B.84) and prove the result in Eq (B.80).

Solution (p. 191)

B.5.6 Typical subspace ●●●

Let E be the projector onto the typical subspace of $\rho^{\otimes n}$ under allowed error ϵ , where d is the dimension of ρ .

(a) Show that $[\rho^{\otimes n}, E] = 0$.

(b) Show that for large n a projection of $\rho^{\otimes n}$ onto E nearly always succeeds, i.e. that $\text{tr}(E \rho^{\otimes n}) > 1 - \epsilon$.

(c) Show that $\text{tr}(E) \leq 2^{n(H(\rho) + \epsilon)}$. Is the typical subspace exponentially smaller than the state space?

Solution (p. 192)

B.5.7 Additivity of classical capacity ●●

Given two independent classical channels Λ_1 and Λ_2 , the classical capacity of the channels satisfies

$$C(\Lambda_1 \otimes \Lambda_2) = C(\Lambda_1) + C(\Lambda_2). \quad (\text{B.85})$$

We will prove this result in two steps.

- (a) Show that $C(\Lambda_1 \otimes \Lambda_2) \geq C(\Lambda_1) + C(\Lambda_2)$.

Hint: Use Shannon's noisy channel coding theorem, Eq (B.73) and independent inputs for the two channels.

- (b) Show that $C(\Lambda_1 \otimes \Lambda_2) \leq C(\Lambda_1) + C(\Lambda_2)$.

Hint: Use Shannon's noisy channel coding theorem, the subadditivity of the Shannon entropy and the independence of the channels.

Solution (p. 193)

B.5.8 Holevo information of product codewords ●

Consider the Holevo information defined in Chapter 8.4,

$$\chi(\mathcal{E}) = H(\rho) - \sum_x p(x) H(\rho_x), \quad (\text{B.86})$$

where $\mathcal{E} = \{p(x), \rho_x\}_x$ denotes the ensemble of states and $\rho = \sum_x p(x) \rho_x$. Assume that $\rho_x = \rho(x_1) \otimes \rho(x_2) \otimes \cdots \otimes \rho(x_n)$ and $p(x) = p(x_1, x_2, \dots, x_n)$. Show that in this case the Holevo information is subadditive.

Solution (p. 193)

B.5.9 Holevo information of entanglement-breaking channel ●●●

Here we show that if Λ_1 and Λ_2 are entanglement-breaking channels (defined in Exercise B.4.6), the Holevo information is subadditive, i.e.

$$\chi(\Lambda_1 \otimes \Lambda_2) \leq \chi(\Lambda_1) + \chi(\Lambda_2). \quad (\text{B.87})$$

- (a) Consider the state $\rho_{XA_1A_2} = \sum_x p(x) |x\rangle\langle x| \otimes \rho(x)_{A_1A_2}$, and apply Λ_1 and Λ_2 to obtain $\rho'_{XB_1B_2}$. Show that

$$I(X : B_1, B_2)_{\rho'} \leq \chi(\Lambda_1) + I(X, B_1 : B_2)_{\rho'}. \quad (\text{B.88})$$

Hint: Use the property (B.71).

- (b) Use the definition of entanglement-breaking channel to show that the state $\rho'_{XB_1B_2}$ can be written as

$$\rho'_{XB_1B_2} = \sum_{x,y} p(x) p(y|x) |x\rangle\langle x| \otimes \sigma(x, y)_{B_1} \otimes \Lambda_2(\tau(x, y))_{B_2}, \quad (\text{B.89})$$

for some states $\sigma(x, y)$ and $\tau(x, y)$.

- (c) We can view $\rho'_{XB_1B_2}$ as the marginal of the state

$$\omega'_{XYB_1B_2} = \sum_{x,y} p(x, y) |xy\rangle\langle xy| \otimes \sigma(x, y)_{B_1} \otimes \Lambda_2(\tau(x, y))_{B_2}, \quad (\text{B.90})$$

over system Y . Use this to show that $I(X, B_1 : B_2)_{\rho'} \leq \chi(\Lambda_2)$ and prove (B.87).

Hint: Use strong subadditivity of the mutual information, i.e. $I(A : C) \leq I(A, B : C)$, and Eq (B.72).

Solution (p. 194)

B.5.10 Superadditivity of Holevo information of channel ●●

Consider that Alice uniformly at random sends to Bob one of the states

$$|\psi_0\rangle = |0\rangle, \quad |\psi_1\rangle = \frac{|0\rangle + \sqrt{3}|1\rangle}{2}, \quad |\psi_2\rangle = \frac{|0\rangle - \sqrt{3}|1\rangle}{2}. \quad (\text{B.91})$$

Bob tries to determine the classical label, $m \in \{0, 1, 2\}$, by performing a qubit POVM $\{E_0, E_1, E_2\}$. We will use this task to show that the Holevo information is superadditive.

- (a) It is possible to show that the accessible information of the ensemble $\mathcal{E} = \{\frac{1}{3}, |\psi_x\rangle\}_x$ is $I_{\text{acc}}(\mathcal{E}) = \log_2(3) - 1$. Show that this is saturated by choosing $E_m = \frac{2}{3}|\phi(t + r_m)\rangle\langle\phi(t + r_m)|$ where $t = \frac{\pi}{3}$ and $r_m = \frac{m\pi}{3}$ and $|\phi(t)\rangle = \cos(t/2)|0\rangle + \sin(t/2)|1\rangle$.
- (b) Now consider that Alice sends the same qubit twice, i.e. she prepares the states $|\beta_x\rangle = |\psi_x\rangle \otimes |\psi_x\rangle$ for $x = 0, 1, 2$, and that Bob performs the POVM

$$E_y = \frac{1}{3}\rho^{-\frac{1}{2}}|\beta_y\rangle\langle\beta_y|\rho^{-\frac{1}{2}}, \quad (\text{B.92})$$

for $y = 0, 1, 2$ where $\rho = \frac{1}{3} \sum_{x=0}^2 |\beta_x\rangle\langle\beta_x|$. Compute the mutual information $I(X : Y)$ and show that it is larger than double the accessible information $I_{\text{acc}}(\mathcal{E})$ in (a).

Hint: Let $\rho = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$ be the spectral decomposition of ρ . Then, define the pseudo-inverse as $\rho^{-1} = \sum_i \lambda_i^{-1} |\lambda_i\rangle\langle\lambda_i|$. Use a computer to evaluate $\rho^{-\frac{1}{2}}$ and the probabilities $p(y|x)$.

Solution (p. 194)

CORE TOPICS - SOLUTIONS

B.1. Entanglement

B.1.1 Solution (Text at p. 139)

(a) The p -power becomes

$$\begin{aligned} X^p &= \underbrace{XX \dots X}_{p \text{ times}} = \sum_{k_1, \dots, k_p=0}^{d-1} |k_1 + 1\rangle \underbrace{\langle k_1 | k_2 + 1 \rangle \langle k_2 | \dots \langle k_{p-1} | k_p + 1 \rangle}_{=\delta_{k_1, k_2+1}} \underbrace{\langle k_p |}_{=\delta_{k_{p-1}, k_p+1}} \\ &= \sum_{k_1, \dots, k_p=0}^{d-1} \delta_{k_1, k_2+1} \delta_{k_2, k_3+1} \dots \delta_{k_{p-1}, k_p+1} |k_1 + 1\rangle \langle k_p|. \end{aligned} \quad (\text{B.93})$$

From the product of the Dirac deltas, we get $k_1 = k_2 + 1 = k_3 + 2 = \dots = k_p + 1 + p - 1 = k_p + p$. Hence,

$$X^p = \sum_{k=0}^{d-1} |k + p \pmod{d}\rangle \langle k|. \quad (\text{B.94})$$

Now we can compute the effect of Alice's operations on her two qudits $A'A$, i.e. for each p

$$\begin{aligned} |0p\rangle_{A'A} &\rightarrow (X^p \otimes \mathbb{1}) \sum_{k=0}^{d-1} \sqrt{\lambda_k} |kk\rangle_{A'A} = \sum_{k=0}^{d-1} \sqrt{\lambda_k} \left(\sum_{l=0}^{d-1} |l + p \pmod{d}\rangle \langle l|_k \right) \otimes |k\rangle_A \\ &= \sum_{k=0}^{d-1} \sqrt{\lambda_k} |k + p \pmod{d}\rangle_{A'} \otimes |k\rangle_A. \end{aligned} \quad (\text{B.95})$$

To prove that Alice's transformation is unitary, we show that it preserves the inner product. To that end, we compute

$$\begin{aligned} \langle 0k|0p\rangle_{A'A} &\rightarrow \left(\sum_{l=0}^{d-1} \sqrt{\lambda_l} \langle l + k \pmod{d} |_{A'} \otimes \langle l |_A \right) \left(\sum_{q=0}^{d-1} \sqrt{\lambda_q} |q + p \pmod{d}\rangle_{A'} \otimes |q\rangle_A \right) \\ &= \sum_{l,q=0}^{d-1} \sqrt{\lambda_l} \sqrt{\lambda_q} \underbrace{\langle l + k \pmod{d} | q + p \pmod{d} \rangle_{A'}}_{=\delta_{l+k \pmod{d}, q+p \pmod{d}}} \underbrace{\langle l | q \rangle_A}_{=\delta_{l,q}} = \delta_{k,p}. \end{aligned} \quad (\text{B.96})$$

This is zero for any $p \neq k$.

(b) Let us call $U_{A'A}$ the unitary operation (B.3) performed by Alice. Then, the state (B.2) after the transformation is

$$\begin{aligned} (U_{A'A} \otimes \mathbb{1}_B) |0\rangle_{A'} \otimes |\phi^+\rangle_{AB} &= \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} U_{A'A} |0k\rangle_{A'A} \otimes |k\rangle_B \\ &= \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} \left(\sum_{p=0}^{d-1} \sqrt{\lambda_p} |p + k \pmod{d}\rangle_{A'} \otimes |p\rangle_A \right) \otimes |k\rangle_B \\ &= \frac{1}{\sqrt{d}} \sum_{k,p=0}^{d-1} \sqrt{\lambda_p} |p + k \pmod{d}\rangle_{A'} \otimes |pk\rangle_{AB}. \end{aligned} \quad (\text{B.97})$$

(c) Since the sum $p + k \pmod{d}$ assumes values only in $\{0, \dots, d-1\}$, we can define $l = p + k \pmod{d}$ and rewrite (B.97) as

$$(U_{A'A} \otimes \mathbb{1}_B) |0\rangle_{A'} \otimes |\phi^+\rangle_{AB} = \frac{1}{\sqrt{d}} \sum_{l=0}^{d-1} |l\rangle_{A'} \otimes \sum_{p=0}^{d-1} \sqrt{\lambda_p} |p\rangle_A \otimes |l - p \pmod{d}\rangle_B. \quad (\text{B.98})$$

Therefore, if Alice measures $\{|l\rangle\}_{l=0}^{d-1}$ on A' and finds the outcome l , the normalised post-measurement state on AB becomes

$$|\varphi_l\rangle = \sum_{p=0}^{d-1} \sqrt{\lambda_p} |p\rangle_A \otimes |l - p \pmod{d}\rangle_B \quad (\text{B.99})$$

Applying the unitary U_l to this state returns the final result

$$(\mathbb{1}_A \otimes U_l)|\varphi_l\rangle = \sum_{p=0}^{d-1} \sqrt{\lambda_p} |p\rangle_A \otimes \left(\sum_{k=0}^{d-1} |k\rangle \underbrace{\langle l-k \pmod{d} | l-p \pmod{d} \rangle_B}_{=\delta_{k,p}} \right) = \sum_{k=0}^{d-1} \sqrt{\lambda_k} |kk\rangle_{AB}. \quad (\text{B.100})$$

B.1.2 Solution (Text at p. 139)

- (a) Any pure state $|\psi\rangle_{AB}$ on a bipartite Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$, where $\dim(\mathcal{H}_A) = d_A$ and $\dim(\mathcal{H}_B) = d_B$, can be expressed in its Schmidt decomposition as

$$|\psi\rangle_{AB} = \sum_{i=1}^{\min(d_A, d_B)} \sqrt{\lambda_i} |\alpha_i\rangle \otimes |\beta_i\rangle, \quad (\text{B.101})$$

where $\{|\alpha_j\rangle\}_{j=1}^{d_A}$ and $\{|\beta_k\rangle\}_{k=1}^{d_B}$ are orthonormal bases of $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively, and $\lambda_i \geq 0$ for all i , with $\sum_{i=1}^{\min(d_A, d_B)} \lambda_i = 1$. Then, the density operator is

$$\rho_{AB} = |\psi\rangle\langle\psi| = \sum_{i,j=1}^{\min(d_A, d_B)} \sqrt{\lambda_i} \sqrt{\lambda_j} |\alpha_i\rangle\langle\alpha_j| \otimes |\beta_i\rangle\langle\beta_j|. \quad (\text{B.102})$$

The reductions can be expressed as

$$\rho_A = \text{tr}_B(|\psi\rangle\langle\psi|) = \sum_{i,j=1}^{\min(d_A, d_B)} \sqrt{\lambda_i} \sqrt{\lambda_j} |\alpha_i\rangle\langle\alpha_j| \underbrace{\text{tr}(|\beta_i\rangle\langle\beta_j|)}_{=\delta_{i,j}} = \sum_{i=1}^{\min(d_A, d_B)} \lambda_i |\alpha_i\rangle\langle\alpha_i|, \quad (\text{B.103})$$

$$\rho_B = \text{tr}_A(|\psi\rangle\langle\psi|) = \sum_{i,j=1}^{\min(d_A, d_B)} \sqrt{\lambda_i} \sqrt{\lambda_j} \underbrace{\text{tr}(|\alpha_i\rangle\langle\alpha_j|)}_{=\delta_{i,j}} |\beta_i\rangle\langle\beta_j| = \sum_{i=1}^{\min(d_A, d_B)} \lambda_i |\beta_i\rangle\langle\beta_i|. \quad (\text{B.104})$$

We see that they are diagonal with the same eigenvalues $\{\lambda_i\}_i$.

- (b) Observe that the first state is already in Schmidt decomposition:

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle. \quad (\text{B.105})$$

It corresponds to Eq (B.101) with Schmidt basis $\{|0\rangle, |1\rangle\}$ for both $\mathcal{H}_A$ and $\mathcal{H}_B$, and Schmidt coefficients $\lambda_1 = \lambda_2 = \frac{1}{2}$.

The state

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle) \quad (\text{B.106})$$

is not in its Schmidt decomposition. Indeed, from (B.101) we require the states $\{|\alpha_i\rangle\}$ (and $\{|\beta_i\rangle\}$) in the sum to be orthogonal, which does not happen in (B.106).

Let us recall from Chapter 4.1 the recipe for how to construct the Schmidt decomposition:

- i. Define $\rho = |\psi\rangle\langle\psi|$ and compute $\rho_A = \text{tr}_B(\rho)$.
- ii. Compute the eigenvalues of ρ_A . These eigenvalues are the Schmidt coefficients, $\{\lambda_i\}_i$.
- iii. Compute the eigenvectors of ρ_A . These form the Schmidt basis, $\{|\alpha_j\rangle\}$, on $\mathcal{H}_A$.
- iv. Compute the eigenvectors of $\rho_B = \text{tr}_A(\rho)$. These form the Schmidt basis, $\{|\beta_j\rangle\}$, on $\mathcal{H}_B$.

Therefore, in our case

- i. The density operator is

$$\begin{aligned} \rho &= |\psi\rangle\langle\psi| = \frac{1}{3} (|00\rangle + |01\rangle + |10\rangle) (\langle 00| + \langle 01| + \langle 10|) \\ &= \frac{1}{3} [|00\rangle\langle 00| + |00\rangle\langle 01| + |00\rangle\langle 10| + |01\rangle\langle 00| + |01\rangle\langle 01| + |01\rangle\langle 10| + |10\rangle\langle 00| + |10\rangle\langle 01| + |10\rangle\langle 10|]. \end{aligned} \quad (\text{B.107})$$

The reduction $\rho_A = \text{tr}_B(\rho)$ is

$$\begin{aligned}
 \rho_A &= \text{tr}_B(\rho) = \sum_{i=0}^1 (\mathbb{1}_A \otimes \langle i|_B) |\psi\rangle \langle \psi| (\mathbb{1}_A \otimes |i\rangle_B) \\
 &= \frac{1}{3} (\mathbb{1}_A \otimes \langle 0|_B) [(|00\rangle + |01\rangle + |10\rangle) (\langle 00| + \langle 01| + \langle 10|)] (\mathbb{1}_A \otimes |0\rangle_B) \\
 &\quad + \frac{1}{3} (\mathbb{1}_A \otimes \langle 1|_B) [(|00\rangle + |01\rangle + |10\rangle) (\langle 00| + \langle 01| + \langle 10|)] (\mathbb{1}_A \otimes |1\rangle_B) \\
 &= \frac{1}{3} (|0\rangle + |1\rangle) (\langle 0| + \langle 1|) + \frac{1}{3} |0\rangle \langle 0| \\
 &= \frac{1}{3} [2|0\rangle \langle 0| + |0\rangle \langle 1| + |1\rangle \langle 0| + |1\rangle \langle 1|] = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.
 \end{aligned} \tag{B.108}$$

ii. Using that for any 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longrightarrow \lambda_{1,2} = \frac{1}{2} \left[(a+d) \pm \sqrt{(a-d)^2 + 4bc} \right], \tag{B.109}$$

we can compute the eigenvalues from the matrix form of ρ_A (B.108)

$$\lambda_1 = \frac{1}{6}(3 - \sqrt{5}), \quad \lambda_2 = \frac{1}{6}(3 + \sqrt{5}). \tag{B.110}$$

These are the coefficients of their Schmidt decomposition.

iii. The eigenvectors of ρ_A are

$$\lambda_1 = \frac{1}{6}(3 - \sqrt{5}) : \begin{pmatrix} \frac{2}{3} - \lambda_1 & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} - \lambda_1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \left(\frac{2}{3} - \lambda_1 \right) x + \frac{1}{3} y = 0, \tag{B.111}$$

from which

$$x = -\frac{\frac{1}{3}y}{\frac{1}{6}(1 + \sqrt{5})} = \frac{2}{1 + \sqrt{5}} y \implies |\alpha_1\rangle = y \begin{pmatrix} \frac{2}{1+\sqrt{5}} \\ -1 \end{pmatrix} \implies |\alpha_1\rangle = \sqrt{\frac{5+\sqrt{5}}{10}} \begin{pmatrix} \frac{2}{1+\sqrt{5}} \\ -1 \end{pmatrix}, \tag{B.112}$$

where the value of y is derived from the normalisation condition $|\langle \alpha_1 | \alpha_1 \rangle| = 1$.

Similarly, we can find

$$\lambda_2 = \frac{1}{6}(3 + \sqrt{5}) : \begin{pmatrix} \frac{2}{3} - \lambda_2 & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} - \lambda_2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \left(\frac{2}{3} - \lambda_2 \right) x + \frac{1}{3} y = 0, \tag{B.113}$$

from which

$$x = -\frac{\frac{1}{3}y}{\frac{1}{6}(1 - \sqrt{5})} = \frac{2}{1 - \sqrt{5}} y \implies |\alpha_2\rangle = y \begin{pmatrix} \frac{2}{1-\sqrt{5}} \\ -1 \end{pmatrix} \implies |\alpha_2\rangle = \sqrt{\frac{5-\sqrt{5}}{10}} \begin{pmatrix} \frac{2}{1-\sqrt{5}} \\ -1 \end{pmatrix}. \tag{B.114}$$

iv. To find the Schmidt basis on $\mathcal{H}_B$, we need to compute the eigenvectors of the reduction $\rho_B = \text{tr}_A(\rho)$:

$$\begin{aligned}
 \rho_B &= \text{tr}_A(\rho) = \sum_{i=0}^1 (\langle i|_A \otimes \mathbb{1}) |\psi\rangle \langle \psi| (|i\rangle_A \otimes \mathbb{1}_B) \\
 &= \frac{1}{3} (\langle 0|_A \otimes \mathbb{1}_B) [(|00\rangle + |01\rangle + |10\rangle) (\langle 00| + \langle 01| + \langle 10|)] (|0\rangle_A \otimes \mathbb{1}_B) \\
 &\quad + \frac{1}{3} (\langle 1|_A \otimes \mathbb{1}_B) [(|00\rangle + |01\rangle + |10\rangle) (\langle 00| + \langle 01| + \langle 10|)] (|1\rangle_A \otimes \mathbb{1}_B) \\
 &= \frac{1}{3} (|0\rangle + |1\rangle) (\langle 0| + \langle 1|) + \frac{1}{3} |0\rangle \langle 0| \\
 &= \frac{1}{3} [2|0\rangle \langle 0| + |0\rangle \langle 1| + |1\rangle \langle 0| + |1\rangle \langle 1|] = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},
 \end{aligned} \tag{B.115}$$

which is the same matrix found in (B.108). Therefore, $|\alpha_1\rangle = |\beta_1\rangle$ and $|\alpha_2\rangle = |\beta_2\rangle$.

The Schmidt decomposition of $|\psi\rangle$ is:

$$|\psi\rangle = \sqrt{\frac{1}{6}(3 - \sqrt{5})} |\alpha_1\rangle \otimes |\alpha_1\rangle + \sqrt{\frac{1}{6}(3 + \sqrt{5})} |\alpha_2\rangle \otimes |\alpha_2\rangle. \quad (\text{B.116})$$

B.1.3 Solution (Text at p. 139)

(a) A linear operator O can be written as $O = \sum_{p,q=0}^{d-1} O_{pq} |p\rangle\langle q|$, while the identity operator is $\mathbb{1} = \sum_{j=0}^{d-1} |j\rangle\langle j|$. Using these two expressions, we can write

$$(O \otimes \mathbb{1}) = \sum_{p,q,j=0}^{d-1} O_{pq} |p\rangle\langle q| \otimes |j\rangle\langle j|, \quad (\text{B.117})$$

$$(\mathbb{1} \otimes O^T) = \sum_{j,p,q=0}^{d-1} |j\rangle\langle j| \otimes O_{qp} |p\rangle\langle q|, \quad (\text{B.118})$$

where we have used that $(O^T)_{pq} = O_{qp}$. Therefore,

$$\begin{aligned} (O \otimes \mathbb{1})|\phi^+\rangle &= \frac{1}{\sqrt{d}} \left(\sum_{p,q,j=0}^{d-1} O_{pq} |p\rangle\langle q| \otimes |j\rangle\langle j| \right) \sum_{k=0}^{d-1} |k\rangle \otimes |k\rangle \\ &= \frac{1}{\sqrt{d}} \sum_{p,q,j,k=0}^{d-1} O_{pq} |p\rangle \underbrace{\langle q|k\rangle}_{=\delta_{q,k}} \otimes |j\rangle \underbrace{\langle j|k\rangle}_{=\delta_{j,k}} \\ &= \frac{1}{\sqrt{d}} \sum_{p,k=0}^{d-1} O_{pk} |p\rangle \otimes |k\rangle. \end{aligned} \quad (\text{B.119})$$

In the same way,

$$(\mathbb{1} \otimes O^T)|\phi^+\rangle = \frac{1}{\sqrt{d}} \sum_{k,p=0}^{d-1} O_{kp} |k\rangle \otimes |p\rangle, \quad (\text{B.120})$$

where we can relabel $(k, p) \rightarrow (p, k)$ to get exactly the expression in (B.119), which proves the result.

(b) By rewriting the action of $U \otimes U^*$ as

$$(U \otimes U^*)|\phi^+\rangle = (U \otimes \mathbb{1})(\mathbb{1} \otimes U^*)|\phi^+\rangle, \quad (\text{B.121})$$

we can use Eq (B.6) on $(\mathbb{1} \otimes U^*)|\phi^+\rangle$. Recalling that $(U^*)^T = U^\dagger$, we get

$$(U \otimes U^*)|\phi^+\rangle = (U \otimes \mathbb{1})(\mathbb{1} \otimes U^*)|\phi^+\rangle = (U \otimes \mathbb{1})(U^\dagger \otimes \mathbb{1})|\phi^+\rangle = (UU^\dagger \otimes \mathbb{1})|\phi^+\rangle = |\phi^+\rangle, \quad (\text{B.122})$$

where we used the unitarity condition $UU^\dagger = \mathbb{1}$.

B.1.4 Solution (Text at p. 140)

(a) The spectra of ρ_{AB} and ρ_{AB}^T can be derived solving the determinant secular equations,

$$\det(\rho_{AB} - \lambda \mathbb{1}) = 0, \quad \det(\rho_{AB}^T - \lambda \mathbb{1}) = 0. \quad (\text{B.123})$$

Since $\mathbb{1}^T = \mathbb{1}$ and $\det(P) = \det(P^T)$ for any matrix P , we can write

$$\det(\rho_{AB}^T - \lambda \mathbb{1}) = \det((\rho_{AB} - \lambda \mathbb{1})^T) = \det(\rho_{AB} - \lambda \mathbb{1}). \quad (\text{B.124})$$

Therefore, since the determinant secular equations for ρ_{AB} and ρ_{AB}^T are the same, their spectra coincide.

- (b) Define the global transpose map as $T = T_A T_B = T_B T_A$, such that $\rho_{AB}^T = (\rho_{AB}^{T_A})^{T_B} = (\rho_{AB}^{T_B})^{T_A}$. Then, the spectrum of $\rho_{AB}^{T_A}$ can be derived solving the determinant secular equation, i.e.

$$\det(\rho_{AB}^{T_A} - \lambda \mathbb{1}) = 0. \quad (\text{B.125})$$

Using the result in (a), the spectra of $\rho_{AB}^{T_A}$ and $(\rho_{AB}^{T_A})^T$ are the same. Therefore, we get

$$0 = \det(\rho_{AB}^{T_A} - \lambda \mathbb{1}) = \det((\rho_{AB}^{T_A})^T - \lambda \mathbb{1}) = \det(\rho_{AB}^{T_B} - \lambda \mathbb{1}), \quad (\text{B.126})$$

where we used that $(\rho_{AB}^{T_A})^T = \rho_{AB}^{T_B}$.

B.1.5 Solution (Text at p. 140)

Let us compute the density operator associated to the state $|\psi\rangle_{AB}$,

$$\begin{aligned} |\psi\rangle\langle\psi|_{AB} &= \left(\sqrt{\frac{2}{5}}|01\rangle + \sqrt{\frac{3}{5}}|10\rangle \right) \left(\sqrt{\frac{2}{5}}\langle 01| + \sqrt{\frac{3}{5}}\langle 10| \right) \\ &= \frac{2}{5}|01\rangle\langle 01| + \frac{3}{5}|10\rangle\langle 10| + \frac{\sqrt{6}}{5}|01\rangle\langle 10| + \frac{\sqrt{6}}{5}|10\rangle\langle 01| = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{2}{5} & \frac{\sqrt{6}}{5} & 0 \\ 0 & \frac{\sqrt{6}}{5} & \frac{3}{5} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \end{aligned} \quad (\text{B.127})$$

from which

$$\rho_{AB} = p|\psi\rangle\langle\psi|_{AB} + \frac{1-p}{4}\mathbb{1} = \begin{pmatrix} \frac{1-p}{4} & 0 & 0 & 0 \\ 0 & \frac{2}{5}p + \frac{1-p}{4} & \frac{\sqrt{6}}{5}p & 0 \\ 0 & \frac{\sqrt{6}}{5}p & \frac{3}{5}p + \frac{1-p}{4} & 0 \\ 0 & 0 & 0 & \frac{1-p}{4} \end{pmatrix}. \quad (\text{B.128})$$

Applying the partial transpose gives

$$\rho_{AB}^{T_B} = \begin{pmatrix} \frac{1-p}{4} & 0 & 0 & \frac{\sqrt{6}}{5}p \\ 0 & \frac{2}{5}p + \frac{1-p}{4} & 0 & 0 \\ 0 & 0 & \frac{3}{5}p + \frac{1-p}{4} & 0 \\ \frac{\sqrt{6}}{5}p & 0 & 0 & \frac{1-p}{4} \end{pmatrix}. \quad (\text{B.129})$$

The eigenvalues of this matrix can be computed to

$$\lambda(\rho_{AB}^{T_B}) = \left\{ \frac{3p+5}{20}, \frac{7p+5}{20}, \frac{5+4\sqrt{6}p-5}{20}, \frac{5-4\sqrt{6}p-5p}{20} \right\} \quad (\text{B.130})$$

Only the last eigenvalue can be negative. The condition for that is

$$\frac{5-4\sqrt{6}p-5p}{20} < 0 \implies p > \frac{5}{5+4\sqrt{6}}. \quad (\text{B.131})$$

In conclusion, the state ρ_{AB} is entangled for all $p > \frac{5}{5+4\sqrt{6}}$.

B.1.6 Solution (Text at p. 140)

To compute the critical visibility for which the noisy state in (B.9) is detected to be entangled by the PPT criterion, we need to find the value p^* below which $\rho_{AB}^{T_B} \not\geq 0$, i.e. the critical value at which at least one of the eigenvalues of $\rho_{AB}^{T_B}$ is no longer positive. Using (B.9), we can compute

$$\rho^{T_B} = \frac{p}{d} \sum_{k,l=0}^{d-1} (|kk\rangle\langle ll|)^{T_B} + \frac{1-p}{d^2} \sum_{k,l=0}^{d-1} (|kl\rangle\langle kl|)^{T_B} = \frac{p}{d} \sum_{k,l=0}^{d-1} |kl\rangle\langle lk| + \frac{1-p}{d^2} \sum_{k,l=0}^{d-1} |kl\rangle\langle kl|, \quad (\text{B.132})$$

where we used $\mathbb{1} = \sum_{k,l=0}^{d-1} |kl\rangle\langle kl|$. To calculate the eigenvalues of this matrix, we can split the cases in which $k = l$ from the ones in which $k \neq l$, i.e.

$$\rho^{T_B} = \left(\frac{p}{d} + \frac{1-p}{d^2} \right) \sum_{k=0}^{d-1} |kk\rangle\langle kk| + \sum_{k \neq l} \left(\frac{p}{d} |kl\rangle\langle lk| + \frac{1-p}{d^2} |kl\rangle\langle kl| \right). \quad (\text{B.133})$$

Let us define $\{|kk\rangle\}_{k=0}^{d-1} \rightarrow \{|k'\rangle\}_{k'=0}^{d-1}$: these vectors are orthonormal and span a d -dimensional space. In the same way, for each pair (k, l) with $k \neq l$ we can define a two-dimensional basis $\{|kl\rangle, |lk\rangle\}$ that we denote as $\{|k''\rangle\}_{k''=0}^1$; we will have a total of $\frac{d(d-1)}{2}$ bases⁸⁰. Therefore, since these spaces do not overlap, we can write ρ^{T_B} as a direct sum,

$$\rho^{T_B} = \underbrace{\left(\frac{p}{d} + \frac{1-p}{d^2} \right) \sum_{k'=0}^{d-1} |k'\rangle\langle k'|}_{M_1} \oplus_{n=1}^{\frac{d(d-1)}{2}} \underbrace{\left(\frac{p}{d} \sum_{k'' \neq q''} |k''\rangle\langle q''| + \frac{1-p}{d^2} \sum_{k''=0}^1 |k''\rangle\langle k''| \right)}_{M_2}. \quad (\text{B.134})$$

To compute the eigenvalues of ρ^{T_B} , we need to calculate the eigenvalues of both M_1 and M_2 . M_1 is already expressed in its diagonal form with eigenvalues $\lambda_{k'}(M_1) = \frac{p}{d} + \frac{1-p}{d^2}$, that are always positive for all possible p . Each of the $\frac{d(d-1)}{2}$ matrices of the type M_2 are 2×2 , therefore we can compute their eigenvalues using (B.109) that in our case gives

$$M_2 = \begin{pmatrix} \frac{1-p}{d^2} & \frac{p}{d} \\ \frac{p}{d} & \frac{1-p}{d^2} \end{pmatrix} \implies \lambda_{1,2}(M_2) = \frac{1-p}{d^2} \pm \frac{p}{d}. \quad (\text{B.135})$$

The first eigenvalue, $\lambda_1(M_2)$ is always non-negative, while the second one, $\lambda_2(M_2) = \frac{1-p}{d^2} - \frac{p}{d}$, is negative for $p > \frac{1}{d+1}$. Therefore, the critical value of p for which the state is no longer detected as entangled by the PPT criterion is $p^* = \frac{1}{d+1}$. For example, for $d = 2$ we get $p^* = \frac{1}{3}$, which is the result found in Chapter 4.3.

B.1.7 Solution (Text at p. 140)

The witness operator in (B.10) uses only two of the three Pauli matrices in dimension $d = 2$. Therefore, it is reasonable to think that a better witness operator can be constructed using all Pauli matrices. For example, let us consider

$$W = \sigma_X \otimes \sigma_X - \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z. \quad (\text{B.136})$$

Using the same approach presented in Chapter 4.5, we can derive a bound on the expectation value over separable states, $\rho = \sum_i q_i \psi_i \otimes \phi_i$, i.e.

$$\begin{aligned} \text{tr}(\rho W) &= \sum_i q_i \text{tr}(\psi_i \otimes \phi_i (\sigma_X \otimes \sigma_X - \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z)) \\ &= \sum_i q_i (\langle \psi_i | \sigma_X | \psi_i \rangle \langle \phi_i | \sigma_X | \phi_i \rangle - \langle \psi_i | \sigma_Y | \psi_i \rangle \langle \phi_i | \sigma_Y | \phi_i \rangle + \langle \psi_i | \sigma_Z | \psi_i \rangle \langle \phi_i | \sigma_Z | \phi_i \rangle). \end{aligned} \quad (\text{B.137})$$

We can express this as an inner product between two vectors and bound it using Cauchy-Schwarz inequality,

$$\begin{aligned} \text{tr}(\rho W) &= \sum_i q_i \left(\langle \psi_i | \sigma_X | \psi_i \rangle, -\langle \psi_i | \sigma_Y | \psi_i \rangle, \langle \psi_i | \sigma_Z | \psi_i \rangle \right) \cdot \begin{pmatrix} \langle \phi_i | \sigma_X | \phi_i \rangle \\ \langle \phi_i | \sigma_Y | \phi_i \rangle \\ \langle \phi_i | \sigma_Z | \phi_i \rangle \end{pmatrix} \\ &\stackrel{\text{CS-ineq}}{\leq} \sum_i q_i \sqrt{\langle \psi_i | \sigma_X | \psi_i \rangle^2 + \langle \psi_i | \sigma_Y | \psi_i \rangle^2 + \langle \psi_i | \sigma_Z | \psi_i \rangle^2} \sqrt{\langle \phi_i | \sigma_X | \phi_i \rangle^2 + \langle \phi_i | \sigma_Y | \phi_i \rangle^2 + \langle \phi_i | \sigma_Z | \phi_i \rangle^2}. \end{aligned} \quad (\text{B.138})$$

Any qubit state can be represented in terms of its Bloch vector as $\psi_i = \frac{1}{2}(\mathbb{1} + \vec{n}_i \cdot \vec{\sigma})$, where each component of $\vec{n}_i$ is $n_{(X,Z,Y),i} = \langle \psi_i | \sigma_{(X,Z,Y)} | \psi_i \rangle$, we get that

$$\langle \psi_i | \sigma_X | \psi_i \rangle^2 + \langle \psi_i | \sigma_Y | \psi_i \rangle^2 + \langle \psi_i | \sigma_Z | \psi_i \rangle^2 = n_{X,i}^2 + n_{Y,i}^2 + n_{Z,i}^2 = 1, \quad (\text{B.139})$$

⁸⁰ We have $d(d-1)$ contributions from $(k \neq l)$. By grouping together the contributions $\{|kl\rangle, |lk\rangle\}$, we end up with only $\frac{d(d-1)}{2}$ different spaces. For example, for $d = 3$, we have three possible groups: $\{|01\rangle, |10\rangle\}$, $\{|02\rangle, |20\rangle\}$ and $\{|12\rangle, |21\rangle\}$.

since the Bloch vectors are normalised. Therefore, inserting this result into (B.138), we get

$$\text{tr}(\rho W) \stackrel{\text{SEP}}{\leq} 1. \quad (\text{B.140})$$

However, if $\rho = \phi^+$ we get $\langle \sigma_X \otimes \sigma_X - \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z \rangle_{\phi^+} = 3$. From this it also becomes clear why we introduce a minus sign for the $\sigma_Y \otimes \sigma_Y$ operator in (B.136). Without the minus sign, the expectation value $\langle \sigma_X \otimes \sigma_X + \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z \rangle_{\phi^+} = 1$, that is exactly the same bound we get for separable states. Therefore, with such an operator it would not be possible to detect this state.

Now we can compute the critical visibility for the detection of entanglement using the witness (B.136).

$$\begin{aligned} \langle \sigma_X \otimes \sigma_X - \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z \rangle_{\rho} &= \text{tr} \left((\sigma_X \otimes \sigma_X - \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z) \left(p\phi^+ + (1-p)\frac{\mathbb{1}}{4} \right) \right) \\ &= p \text{tr}((\sigma_X \otimes \sigma_X - \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z)\phi^+) + \frac{1-p}{4} \text{tr}((\sigma_X \otimes \sigma_X - \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z)\mathbb{1}) = 3p, \end{aligned} \quad (\text{B.141})$$

where we used that $\langle \sigma_X \otimes \sigma_X - \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z \rangle_{\phi^+} = 3$ and that $\text{tr}(\sigma_X) = \text{tr}(\sigma_Y) = \text{tr}(\sigma_Z) = 0$. Therefore, inserting this into (B.140), we get

$$\langle \sigma_X \otimes \sigma_X - \sigma_Y \otimes \sigma_Y + \sigma_Z \otimes \sigma_Z \rangle_{\rho} = 3p \leq 1 \quad \implies \quad p^* = \frac{1}{3}, \quad (\text{B.142})$$

i.e. the state is detected to be entangled for any $p > \frac{1}{3}$, which is exactly what we get from the PPT criterion.

B.1.8 Solution (Text at p. 140)

(a) We define $B = \sum_{i=1}^n \lambda_i X_i$ for a set of real coefficients $\{\lambda_i\}_i$ such that $\sum_{i=1}^n \lambda_i^2 = 1$. Therefore,

$$\begin{aligned} B^2 &= \left(\sum_{i=1}^n \lambda_i X_i \right)^2 = \sum_{i,j=1}^n \lambda_i \lambda_j X_i X_j = \sum_{i=1}^n \lambda_i^2 X_i^2 + \sum_i \sum_{j \neq i} \lambda_i \lambda_j X_i X_j \\ &= \sum_{i=1}^n \lambda_i^2 X_i^2 + \sum_i \sum_{j < i} \lambda_i \lambda_j X_i X_j + \sum_i \sum_{j > i} \lambda_i \lambda_j X_i X_j \\ &\stackrel{i \leftrightarrow j}{=} \sum_{i=1}^n \lambda_i^2 \underbrace{X_i^2}_{=1} + \sum_i \sum_{j < i} \lambda_i \lambda_j \underbrace{\{X_i, X_j\}}_{=0} = \sum_{i=1}^n \lambda_i^2 \mathbb{1} = \mathbb{1}, \end{aligned} \quad (\text{B.143})$$

where in the third line we have relabeled the indices $i \leftrightarrow j$ to recover the commutator $\{X_i, X_j\}$ and we have used that $X_i^2 = \mathbb{1}$ since the eigenvalues are ± 1 . From (B.143) we get that $|\langle B \rangle|^2 \leq 1$. Therefore, if we define the vectors $\vec{\lambda} = (\lambda_1, \dots, \lambda_n)$ and $\vec{X} = (\langle X_1 \rangle, \dots, \langle X_n \rangle)$, we can write

$$|\langle B \rangle|^2 = \left| \sum_{i=1}^n \lambda_i \langle X_i \rangle \right|^2 = |\vec{\lambda} \cdot \vec{X}|^2 \leq 1. \quad (\text{B.144})$$

Since this is true for any $\vec{\lambda}$, then

$$|\vec{X}|^2 = \sum_{i=1}^n \langle X_i \rangle^2 \leq 1. \quad (\text{B.145})$$

(b) The expectation value over separable states $\rho_{AB} = \sum_i q_i \phi_i \otimes \varphi_i$ is

$$\begin{aligned} \sum_{i=1}^n \text{tr}(X_i \otimes X_i \rho_{AB}) &= \sum_j q_j \sum_{i=1}^n \langle X_i \rangle_{\phi_j} \langle X_i \rangle_{\varphi_j} \leq \sum_{i=1}^n \langle X_i \rangle_{\phi} \langle X_i \rangle_{\varphi} \\ &\stackrel{\text{CS-ineq}}{\leq} \sqrt{\sum_{i=1}^n \langle X_i \rangle_{\phi}^2} \sqrt{\sum_{i=1}^n \langle X_i \rangle_{\varphi}^2} \leq \max_{\phi} \sum_{i=1}^n \langle X_i \rangle_{\phi}^2 \stackrel{(\text{B.11})}{\leq} 1 \end{aligned} \quad (\text{B.146})$$

where we have used convexity to restrict from separable states to product states $\phi \otimes \varphi$, then the Cauchy-Schwarz inequality and finally the result of part (a).

B.1.9 Solution (Text at p. 140)

The three-qubit density operators are

$$\rho_{ABC}^{(GHZ)} = \frac{1}{2}(|000\rangle + |111\rangle)(\langle 000| + \langle 111|) = \frac{1}{2}(|000\rangle\langle 000| + |000\rangle\langle 111| + |111\rangle\langle 000| + |111\rangle\langle 111|), \quad (\text{B.147})$$

$$\begin{aligned} \rho_{ABC}^{(W)} &= \frac{1}{3}(|001\rangle + |010\rangle + |100\rangle)(\langle 001| + \langle 010| + \langle 100|) = \frac{1}{3}(|001\rangle\langle 001| + |001\rangle\langle 010| + |001\rangle\langle 100| \\ &\quad + |010\rangle\langle 001| + |010\rangle\langle 010| + |010\rangle\langle 100| + |100\rangle\langle 001| + |100\rangle\langle 010| + |100\rangle\langle 100|). \end{aligned} \quad (\text{B.148})$$

Since both states are permutation-invariant, we can without loss of generality choose to discard system C to compute the two-qubit reduced state. For GHZ we get

$$\begin{aligned} \rho_{AB}^{(GHZ)} &= \text{tr}_C(\rho_{ABC}^{(GHZ)}) = \sum_{i=0}^1 (\mathbb{1}_A \otimes \mathbb{1}_B \otimes \langle i|_C) \rho_{ABC}^{(GHZ)} (\mathbb{1}_A \otimes \mathbb{1}_B \otimes |i\rangle_C) \\ &= \frac{1}{2} \sum_{i=0}^1 (|00\rangle\langle 00|_{AB} \underbrace{\langle i|0\rangle\langle 0|i\rangle_C}_{=0} + |00\rangle\langle 11|_{AB} \underbrace{\langle i|0\rangle\langle 1|i\rangle_C}_{=0} + |11\rangle\langle 00|_{AB} \underbrace{\langle i|1\rangle\langle 0|i\rangle_C}_{=0} + |11\rangle\langle 11|_{AB} \underbrace{\langle i|1\rangle\langle 1|i\rangle_C}_{=0}), \end{aligned} \quad (\text{B.149})$$

where we used the orthonormality of the computational basis $\langle i|j\rangle = \delta_{i,j}$. Therefore, we get

$$\rho_{AB}^{(GHZ)} = \frac{1}{2}(|00\rangle\langle 00|_{AB} + |11\rangle\langle 11|_{AB}) = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (\text{B.150})$$

which is a separable state (since it is a convex combination of two product states). For the W state we get

$$\begin{aligned} \rho_{AB}^{(W)} &= \text{tr}_C(\rho_{ABC}^{(W)}) = \sum_{i=0}^1 (\mathbb{1}_A \otimes \mathbb{1}_B \otimes \langle i|_C) \rho_{ABC}^{(W)} (\mathbb{1}_A \otimes \mathbb{1}_B \otimes |i\rangle_C) \\ &= \frac{1}{3} \sum_{i=0}^1 (|00\rangle\langle 00|_{AB} \underbrace{\langle i|1\rangle\langle 1|i\rangle_C}_{=0} + |00\rangle\langle 01|_{AB} \underbrace{\langle i|1\rangle\langle 0|i\rangle_C}_{=0} + |00\rangle\langle 10|_{AB} \underbrace{\langle i|1\rangle\langle 0|i\rangle_C}_{=0} \\ &\quad + |01\rangle\langle 00|_{AB} \underbrace{\langle i|0\rangle\langle 1|i\rangle_C}_{=0} + |01\rangle\langle 01|_{AB} \underbrace{\langle i|0\rangle\langle 0|i\rangle_C}_{=0} + |01\rangle\langle 10|_{AB} \underbrace{\langle i|0\rangle\langle 0|i\rangle_C}_{=0} \\ &\quad + |10\rangle\langle 00|_{AB} \underbrace{\langle i|0\rangle\langle 1|i\rangle_C}_{=0} + |10\rangle\langle 01|_{AB} \underbrace{\langle i|0\rangle\langle 0|i\rangle_C}_{=0} + |10\rangle\langle 10|_{AB} \underbrace{\langle i|0\rangle\langle 0|i\rangle_C}_{=0}) \\ &= \frac{1}{3}(|00\rangle\langle 00|_{AB} + |01\rangle\langle 01|_{AB} + |01\rangle\langle 10|_{AB} + |01\rangle\langle 01|_{AB} + |10\rangle\langle 10|_{AB}) = \frac{1}{3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (\text{B.151})$$

We can check if this state is entangled using the PPT criterion. To do that, we need to compute the partial transpose of the density operator,

$$(\rho_{AB}^{(W)})^{T_B} = \frac{1}{3} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{B.152})$$

One finds that its eigenvalues are

$$\lambda((\rho_{AB}^{(W)})^{T_B}) = \left\{ \frac{1}{6}(1 + \sqrt{5}), \frac{1}{3}, \frac{1}{3}, \frac{1}{6}(1 - \sqrt{5}) \right\}. \quad (\text{B.153})$$

Since the last eigenvalue is negative we conclude that the reduced W state is entangled.

B.1.10 Solution (Text at p. 141)

The states $|\psi\rangle_{AB}$ and $|\phi\rangle_{AB}$ can be expressed in their Schmidt decomposition as

$$|\psi\rangle_{AB} = \sum_i \sqrt{\lambda_i} |ii\rangle_{AB}, \quad \text{and} \quad |\phi\rangle_{AB} = \sum_i \sqrt{\gamma_i} |ii\rangle_{AB}, \quad \text{where} \quad \sum_i \lambda_i = \sum_i \gamma_i = 1, \quad (\text{B.154})$$

up to some unitary transformations. We can overlook these unitaries since they are local operations. Now define the operator on $\mathcal{H}_A$

$$\tilde{E} \equiv \sum_j \sqrt{\frac{\gamma_j}{\lambda_j}} |j\rangle\langle j|_A. \quad (\text{B.155})$$

In order to build a proper POVM $\{E, E_\perp\}$ from it, we need to ensure that $E, E_\perp \leq \mathbb{1}$. Therefore, call $\{\tilde{\lambda}_i(E)\}_i$ the eigenvalues of E and define

$$E \equiv \frac{\tilde{E}}{\tilde{\lambda}_{\max}(\tilde{E})} = \frac{1}{\max_i \sqrt{\frac{\gamma_i}{\lambda_i}}} \sum_j \sqrt{\frac{\gamma_j}{\lambda_j}} |j\rangle\langle j|_A = \sqrt{\frac{\lambda_{\max}}{\gamma_{\max}}} \sum_j \sqrt{\frac{\gamma_j}{\lambda_j}} |j\rangle\langle j|_A, \quad \text{and} \quad E_\perp = \mathbb{1} - E. \quad (\text{B.156})$$

Now we can perform this POVM on $|\psi\rangle_{AB}$ and post-select on the outcome associated with E . The updated state is

$$|\tilde{\psi}\rangle_{AB} = E \otimes \mathbb{1} |\psi\rangle_{AB} = \sum_i \sqrt{\lambda_i} E |i\rangle_A \otimes |i\rangle_B = \sum_i \sqrt{\lambda_i} \sqrt{\frac{\lambda_{\max}}{\gamma_{\max}}} \sum_j \sqrt{\frac{\gamma_j}{\lambda_j}} |j\rangle\langle j|_A \otimes |i\rangle_B = \sqrt{\frac{\lambda_{\max}}{\gamma_{\max}}} \sum_i \underbrace{\sqrt{\gamma_i} |ii\rangle_{AB}}_{=\delta_{i,j}}, \quad (\text{B.157})$$

but we must still normalise it. For that, we compute

$$\langle \tilde{\psi} | \tilde{\psi} \rangle = \frac{\lambda_{\max}}{\gamma_{\max}} \sum_{i,j} \sqrt{\gamma_i} \sqrt{\gamma_j} \underbrace{\langle ii | jj \rangle_{AB}}_{=\delta_{i,j}} = \frac{\lambda_{\max}}{\gamma_{\max}} \sum_i \gamma_i = \frac{\lambda_{\max}}{\gamma_{\max}}, \quad (\text{B.158})$$

where we used that $\sum_i \gamma_i = 1$. Thus, the updated state becomes

$$|\psi'\rangle_{AB} = \frac{|\tilde{\psi}\rangle_{AB}}{\sqrt{\langle \tilde{\psi} | \tilde{\psi} \rangle}} = \sum_i \sqrt{\gamma_i} |ii\rangle_{AB}, \quad (\text{B.159})$$

which is exactly the state $|\phi\rangle_{AB}$ defined in (B.154). The probability that the transformation succeeds is

$$p(E|\psi) = \text{tr}(E \otimes \mathbb{1} |\psi\rangle\langle\psi|_{AB}) = \langle \psi | E \otimes \mathbb{1} | \psi \rangle = \langle \psi | \tilde{\psi} \rangle = \sqrt{\frac{\lambda_{\max}}{\gamma_{\max}}} \sum_{i,j} \sqrt{\lambda_i} \sqrt{\gamma_j} \underbrace{\langle ii | jj \rangle_{AB}}_{=\delta_{i,j}} = \sqrt{\frac{\lambda_{\max}}{\gamma_{\max}}} \sum_i \sqrt{\lambda_i \gamma_i}. \quad (\text{B.160})$$

B.2. Bell nonlocality

B.2.1 Solution (Text at p. 141)

Let p_1 and p_2 be quantum probability distributions obtained from Bell experiments on the state $|\psi_1\rangle_{AB}$ and $|\psi_2\rangle_{AB}$ respectively. To show convexity, we must prove that every mixture of these two distributions, namely $p_3 = \alpha p_1 + (1 - \alpha) p_2$, for any $\alpha \in [0, 1]$, also admits a quantum model in the Bell scenario. To show that, let Alice and Bob add a shared local variable $\lambda \in \{1, 2\}$ to the quantum state emitted by the source such that the total state becomes

$$\alpha |\psi_1\rangle\langle\psi_1|_{AB} \otimes |11\rangle\langle 11|_{A'B'} + (1 - \alpha) |\psi_2\rangle\langle\psi_2|_{AB} \otimes |22\rangle\langle 22|_{A'B'}. \quad (\text{B.161})$$

Alice and Bob first observe the value of λ by respectively measuring the registers A' and B' in the computational basis. It leaves them with the shared state $|\psi_\lambda\rangle$. Then they can apply the corresponding measurements that generate p_λ . Since the probability of observing λ is distributed as $\{\alpha, 1 - \alpha\}$, they replicate the distribution p_3 .

B.2.2 Solution (Text at p. 141)

- (a) Since E_{xy} is defined as a difference between two probabilities $0 \leq p \leq 1$, then $-1 \leq E_{xy} \leq 1$. Therefore, in order to have $|u|^2 = 4$, we need all E_{xy} to be either +1 or -1. Since there are two possible values for each of the four E_{xy} in the vector u , there are $2^4 = 16$ possible vectors such that $|u|^2 = 4$.
- (b) Since for $(a_x, b_y) \in \{-1, +1\}$ the correlation coefficients for local deterministic strategies are $E_{xy} = a_x b_y$, then

$$E_{00}E_{01}E_{10}E_{11} = (a_0b_0)(a_0b_1)(a_1b_0)(a_1b_1) = a_0^2 b_0^2 a_1^2 b_1^2 = 1. \quad (\text{B.162})$$

The eight vectors which represent the extremal points of the correlation polytope for local deterministic strategies are given by all possible combinations of E_{xy} which satisfy (B.162), i.e.

$$\begin{aligned} v_1 &= (+1, +1, +1, +1), & v_5 &= -v_1 = (-1, -1, -1, -1), \\ v_2 &= (+1, +1, -1, -1), & v_6 &= -v_2 = (-1, -1, +1, +1), \\ v_3 &= (+1, -1, +1, -1), & v_7 &= -v_3 = (-1, +1, -1, +1), \\ v_4 &= (+1, -1, -1, +1), & v_8 &= -v_4 = (-1, +1, +1, -1). \end{aligned} \quad (\text{B.163})$$

Note that the vectors $\{v_1, v_2, v_3, v_4\}$ are mutually orthogonal. Since they are linearly independent, $\mathbb{R}^4$ is the smallest embedding of the local polytope.

- (c) In order to find the vectors $n_{\vec{s}}$ normal to each hyperplane generated by $V_{\vec{s}}$, we need to solve the four linearly independent equations

$$n_{\vec{s}} \cdot (s_k v_k) = 4, \quad \text{for } k = 1, \dots, 4. \quad (\text{B.164})$$

Since $v_i \cdot v_j = 4\delta_{i,j}$, we can check that $n_{\vec{s}} = \sum_{k=1}^4 s_k v_k$ is a solution. Indeed,

$$n_{\vec{s}} \cdot (s_k v_k) = \sum_{l=1}^4 s_l v_l \cdot (s_k v_k) = \sum_{l=1}^4 s_l s_k \underbrace{v_l \cdot v_k}_{=4\delta_{l,k}} = 4 \underbrace{s_k^2}_{=1} = 4. \quad (\text{B.165})$$

- (d) We can find all facets by considering all possible values of $\vec{s} = (s_1, s_2, s_3, s_4) \in \{-1, +1\}^4$. There are 16 possible values, but, since $n_{-\vec{s}} = -n_{\vec{s}}$, we can just look at eight of them. For each $n_{\vec{s}}$ we find a related inequality considering $n_{\vec{s}} \cdot u \leq 4$. Therefore, using the values of $\{v_k\}_k$ found in (B.163), we get

$$\begin{aligned} \vec{s} = (+1, +1, +1, +1) &\longrightarrow n_{\vec{s}} = v_1 + v_2 + v_3 + v_4 = (4, 0, 0, 0) \longrightarrow 4E_{00} \leq 4, \\ \vec{s} = (+1, +1, +1, -1) &\longrightarrow n_{\vec{s}} = v_1 + v_2 + v_3 - v_4 = (2, 2, 2, -2) \longrightarrow 2E_{00} + 2E_{01} + 2E_{10} - 2E_{11} \leq 4, \\ \vec{s} = (+1, +1, -1, +1) &\longrightarrow n_{\vec{s}} = v_1 + v_2 - v_3 + v_4 = (2, 2, -2, 2) \longrightarrow 2E_{00} + 2E_{01} - 2E_{10} + 2E_{11} \leq 4, \\ \vec{s} = (+1, -1, +1, +1) &\longrightarrow n_{\vec{s}} = v_1 + v_2 - v_3 + v_4 = (2, -2, 2, 2) \longrightarrow 2E_{00} - 2E_{01} + 2E_{10} + 2E_{11} \leq 4, \\ \vec{s} = (+1, +1, -1, -1) &\longrightarrow n_{\vec{s}} = v_1 + v_2 - v_3 - v_4 = (0, 4, 0, 0) \longrightarrow 4E_{01} \leq 4, \\ \vec{s} = (+1, -1, +1, -1) &\longrightarrow n_{\vec{s}} = v_1 - v_2 + v_3 - v_4 = (0, 0, 4, 0) \longrightarrow 4E_{10} \leq 4, \\ \vec{s} = (+1, -1, -1, +1) &\longrightarrow n_{\vec{s}} = v_1 - v_2 - v_3 + v_4 = (0, 0, 0, 4) \longrightarrow 4E_{11} \leq 4, \\ \vec{s} = (+1, -1, +1, +1) &\longrightarrow n_{\vec{s}} = v_1 - v_2 + v_3 + v_4 = (-2, 2, 2, 2) \longrightarrow -2E_{00} + 2E_{01} + 2E_{10} + 2E_{11} \leq 4. \end{aligned} \quad (\text{B.166})$$

Up to relabeling of inputs and outputs, four facets describe the constraint $E_{00} \leq 1$, which is what we have already found in part (a). The other facets describe the constraint

$$E_{00} + E_{01} + E_{10} - E_{11} \leq 2, \quad (\text{B.167})$$

which is exactly the CHSH inequality.

B.2.3 Solution (Text at p. 141)

- (a) A direct calculation gives

$$\vec{a} \cdot \vec{\sigma} = \begin{pmatrix} a_Z & a_X - ia_Y \\ a_X + ia_Y & -a_Z \end{pmatrix}, \quad \vec{b} \cdot \vec{\sigma} = \begin{pmatrix} b_Z & b_X - ib_Y \\ b_X + ib_Y & -b_Z \end{pmatrix}. \quad (\text{B.168})$$

Therefore, since the computational basis states are $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, we get

$$\begin{aligned} \vec{a} \cdot \vec{\sigma}|0\rangle &= a_Z|0\rangle + (a_X + ia_Y)|1\rangle, & \vec{a} \cdot \vec{\sigma}|1\rangle &= (a_X - ia_Y)|0\rangle - a_Z|1\rangle, \\ \vec{b} \cdot \vec{\sigma}|0\rangle &= b_Z|0\rangle + (b_X + ib_Y)|1\rangle, & \vec{b} \cdot \vec{\sigma}|1\rangle &= (b_X - ib_Y)|0\rangle - b_Z|1\rangle. \end{aligned} \quad (\text{B.169})$$

(b) Using the results in (B.169), we can now compute

$$\begin{aligned}
\langle A_i B_j \rangle &= \langle \psi | \vec{a}^{(i)} \cdot \vec{\sigma} \otimes \vec{b}^{(j)} \cdot \vec{\sigma} | \psi \rangle = (c_1 \langle 00 | + c_2 \langle 11 |) (\vec{a}^{(i)} \cdot \vec{\sigma} \otimes \vec{b}^{(j)} \cdot \vec{\sigma}) (c_1 | 00 \rangle + c_2 | 11 \rangle) \\
&= (c_1 \langle 00 | + c_2 \langle 11 |) \left[c_1 (a_Z^{(i)} | 0 \rangle + (a_X^{(i)} + i a_Y^{(i)}) | 1 \rangle) \otimes (b_Z^{(j)} | 0 \rangle + (b_X^{(j)} + i b_Y^{(j)}) | 1 \rangle) \right. \\
&\quad \left. + c_2 ((a_X^{(i)} - i a_Y^{(i)}) | 0 \rangle - a_Z^{(i)} | 1 \rangle) \otimes ((b_X^{(j)} - i b_Y^{(j)}) | 0 \rangle - b_Z^{(j)} | 1 \rangle) \right] \\
&= c_1^2 a_Z^{(i)} b_Z^{(j)} + c_2 c_1 (a_X^{(i)} + i a_Y^{(i)}) (b_X^{(j)} + i b_Y^{(j)}) + c_1 c_2 (a_X^{(i)} - i a_Y^{(i)}) (b_X^{(j)} - i b_Y^{(j)}) + c_2^2 a_Z^{(i)} b_Z^{(j)} \\
&= \underbrace{(c_1^2 + c_2^2)}_{=1} a_Z^{(i)} b_Z^{(j)} + 2 c_1 c_2 (a_X^{(i)} b_X^{(j)} - a_Y^{(i)} b_Y^{(j)}) = a_Z^{(i)} b_Z^{(j)} + 2 c_1 c_2 (a_X^{(i)} b_X^{(j)} - a_Y^{(i)} b_Y^{(j)}),
\end{aligned} \tag{B.170}$$

where we have used the orthogonality of the computational basis and the property of Schmidt coefficient $c_1^2 + c_2^2 = 1$.

(c) We can fix $a_Y^{(i)} = b_Y^{(j)} = 0$ and $a_X^{(i)} = \sin(\alpha_i)$, $a_Z^{(i)} = \cos(\alpha_i)$, $b_X^{(j)} = \sin(\beta_j)$, $b_Z^{(j)} = \cos(\beta_j)$, where $\{\alpha_i, \beta_j\}$ for $i \in \{0, 1\}$ and $j \in \{0, 1\}$ are the angles related to the x axis. Therefore,

$$\langle A_i B_j \rangle = a_Z^{(i)} b_Z^{(j)} + 2 c_1 c_2 \underbrace{(a_X^{(i)} b_X^{(j)} - a_Y^{(i)} b_Y^{(j)})}_{=0} = \cos(\alpha_i) \cos(\beta_j) + 2 c_1 c_2 \sin(\alpha_i) \sin(\beta_j). \tag{B.171}$$

Inserting this expression into (B.16), we get

$$\begin{aligned}
\mathcal{S} &= \langle A_0 B_0 \rangle - \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle + \langle A_1 B_1 \rangle \\
&= \cos(\alpha_0) (\cos(\beta_0) - \cos(\beta_1)) + 2 c_1 c_2 \sin(\alpha_0) (\sin(\beta_0) - \sin(\beta_1)) \\
&\quad + \cos(\alpha_1) (\cos(\beta_0) + \cos(\beta_1)) + 2 c_1 c_2 \sin(\alpha_1) (\sin(\beta_0) + \sin(\beta_1)).
\end{aligned} \tag{B.172}$$

Fixing $\alpha_0 = 0$ and $\alpha_1 = \frac{\pi}{2}$, we get

$$\mathcal{S} = \cos(\beta_0) - \cos(\beta_1) + 2 c_1 c_2 (\sin(\beta_0) + \sin(\beta_1)). \tag{B.173}$$

Using that $\sin(\beta_0) = (1 - \cos(\beta_0)^2)^{\frac{1}{2}}$ and $\sin(\beta_1) = (1 - \cos(\beta_1)^2)^{\frac{1}{2}}$ (we choose both of them positive, since we want to maximise the expression), we can compute the maximum with respect to $\cos(\beta_0)$ and $\cos(\beta_1)$, i.e.

$$0 = \frac{\partial \mathcal{S}}{\partial (\cos(\beta_0))} = 1 - 2 c_1 c_2 \frac{\cos(\beta_0)}{\sqrt{1 - \cos(\beta_0)^2}} \implies \cos(\beta_0) = \frac{1}{\sqrt{1 + 4(c_1 c_2)^2}}, \tag{B.174}$$

$$0 = \frac{\partial \mathcal{S}}{\partial (\cos(\beta_1))} = -1 - 2 c_1 c_2 \frac{\cos(\beta_1)}{\sqrt{1 - \cos(\beta_1)^2}} \implies \cos(\beta_1) = -\frac{1}{\sqrt{1 + 4(c_1 c_2)^2}}. \tag{B.175}$$

Therefore, the maximum of (B.173) becomes

$$\mathcal{S} \leq \frac{2}{\sqrt{1 + 4(c_1 c_2)^2}} + 4 c_1 c_2 \sqrt{1 - \frac{1}{1 + 4(c_1 c_2)^2}} = 2 \sqrt{1 + 4(c_1 c_2)^2}, \tag{B.176}$$

which is always greater than 2 if the Schmidt rank exceeds 1 (i.e. the state is not product).

B.2.4 Solution (Text at p. 142)

(a) Following the same steps as in the proof of the CHSH inequality (see below Eq (5.6)) we arrive at

$$\mathcal{S} = p A_0^\lambda (q B_0^\lambda + (1 - q) B_1^\lambda) + (1 - p) A_1^\lambda (q B_0^\lambda - (1 - q) B_1^\lambda). \tag{B.177}$$

Using the triangle inequality and recalling that $|A_0^\lambda| = |A_1^\lambda| = +1$ gives

$$\mathcal{S} \leq p |q B_0^\lambda + (1 - q) B_1^\lambda| + (1 - p) |q B_0^\lambda - (1 - q) B_1^\lambda|. \tag{B.178}$$

Since we have $p \geq 1 - p$ and $q \geq 1 - q$, the optimal choice is $B_0^\lambda = B_1^\lambda = +1$. This gives the local hidden variable bound $\mathcal{S} \leq 1 - 2(1 - p)(1 - q)$.

- (b) To determine an upper bound on the maximal quantum violation of the inequality, we proceed in analogy with the proof of Tsirelson's bound in Chapter 5.3. In the same notations as there, we have

$$\mathcal{S} = p \langle \alpha_0 | (q|\beta_0\rangle + (1-q)|\beta_1\rangle) + (1-p) \langle \alpha_1 | (q|\beta_0\rangle - (1-q)|\beta_1\rangle). \quad (\text{B.179})$$

Applying Cauchy-Schwarz inequality to each term gives

$$\langle \alpha_x | (q|\beta_0\rangle \pm (1-q)|\beta_1\rangle) \leq \sqrt{q^2 + (1-q)^2 \pm q(1-q)\gamma}, \quad (\text{B.180})$$

where we have $\gamma = \langle \beta_0 | \beta_1 \rangle + \langle \beta_1 | \beta_0 \rangle = \langle \psi | \mathbb{1} \otimes \{B_0, B_1\} | \psi \rangle$. We treat this as a free parameter. Thus, we have

$$\mathcal{S} \leq p\sqrt{q^2 + (1-q)^2 + q(1-q)\gamma} + (1-p)\sqrt{q^2 + (1-q)^2 - q(1-q)\gamma}. \quad (\text{B.181})$$

Solving the equation $\frac{d\mathcal{S}}{d\gamma} = 0$ gives the solution

$$\gamma = \frac{(2p-1)(1-2q+2q^2)}{q(1-q)(1-2p+2p^2)}. \quad (\text{B.182})$$

At this value of γ the bound becomes

$$\mathcal{S} \leq \sqrt{2}\sqrt{(1-2p+2p^2)(1-2q+2q^2)}. \quad (\text{B.183})$$

Note however that (B.182) does not always respect the relevant range of γ , which is confined to $-2 \leq \gamma \leq 2$. Therefore, we expect this to be a tight bound only in the region of (p, q) for which the range is respected.

B.2.5 Solution (Text at p. 142)

- (a) Use the general form $\rho_{AB} = \frac{1}{4} (\mathbb{1} \otimes \mathbb{1} + \vec{r} \cdot \vec{\sigma} \otimes \mathbb{1} + \mathbb{1} \otimes \vec{s} \cdot \vec{\sigma} + \sum_{i,j=X,Y,Z} T_{ij} \sigma_i \otimes \sigma_j)$ to evaluate the expectation values. They become

$$\begin{aligned} \langle A_x, B_y \rangle_\rho &= \frac{1}{4} \left[\underbrace{\text{tr}(A_x)}_{=0} \underbrace{\text{tr}(B_y)}_{=0} + \underbrace{\text{tr}((\vec{r} \cdot \vec{\sigma}) A_x)}_{=0} \underbrace{\text{tr}(B_y)}_{=0} + \text{tr}(A_x) \text{tr}((\vec{s} \cdot \vec{\sigma}) B_y) + \sum_{i,j=X,Y,Z} T_{ij} \text{tr}(\sigma_i A_x) \text{tr}(\sigma_j B_y) \right] \\ &= \frac{1}{4} \sum_{i,j=X,Y,Z} T_{ij} \sum_{k,k'} a_{x,k} b_{y,k'} \underbrace{\text{tr}(\sigma_i \sigma_k)}_{=2\delta_{i,k}} \underbrace{\text{tr}(\sigma_j \sigma_{k'})}_{=2\delta_{j,k'}} = \sum_{i,j} T_{i,j} a_{x,i} b_{y,j} = \vec{a}_x \cdot (T \vec{b}_y), \end{aligned} \quad (\text{B.184})$$

where we have denoted the observables for Alice as $A_x = \sum_k a_{x,k} \sigma_k$ and analogously for Bob.

- (b) To find Alice's best pair of Bloch vectors, we express the CHSH parameter (5.6) on the convenient form made possible from the previous step, namely

$$\mathcal{S} = \vec{a}_0 \cdot T (\vec{b}_0 + \vec{b}_1) + \vec{a}_1 \cdot T (\vec{b}_0 - \vec{b}_1). \quad (\text{B.185})$$

Clearly, the best choice for Alice is to align her respective vectors with the vectors $T(\vec{b}_0 + \vec{b}_1)$ and $T(\vec{b}_0 - \vec{b}_1)$ respectively. This gives

$$\max_{\vec{a}_0, \vec{a}_1} \mathcal{S}(\rho) = \max_{\vec{a}_0, \vec{a}_1} [\vec{a}_0 \cdot T(\vec{b}_0 + \vec{b}_1) + \vec{a}_1 \cdot T(\vec{b}_0 - \vec{b}_1)] = |T(\vec{b}_0 + \vec{b}_1)| + |T(\vec{b}_0 - \vec{b}_1)|. \quad (\text{B.186})$$

- (c) Bob's two Bloch vectors span a plane. We can therefore write them on the form

$$\vec{b}_0 = \cos \theta \hat{c}_1 + \sin \theta \hat{c}_2 \quad \vec{b}_1 = \cos \theta \hat{c}_1 - \sin \theta \hat{c}_2, \quad (\text{B.187})$$

for some angle θ and a pair of orthogonal unit vectors $\{\hat{c}_1, \hat{c}_2\}$. This simplifies the maximisation of the right-hand-side of (B.186) into

$$\max_{\theta, \hat{c}_1, \hat{c}_2} 2 [\cos \theta |T \cdot \hat{c}_1| + \sin \theta |T \cdot \hat{c}_2|]. \quad (\text{B.188})$$

By computing the derivative with respect to $\cos \theta$ and $\sin \theta$ and setting it to zero, we find that the optimal angle corresponds to

$$\begin{aligned} 0 = \frac{\partial \mathcal{S}}{\partial \cos \theta} &= 2 \left[|T \cdot \hat{c}_1| - \frac{\cos \theta}{\sqrt{1 - \cos^2 \theta}} |T \cdot \hat{c}_2| \right] \implies \cos \theta = \frac{|T \cdot \hat{c}_1|}{\sqrt{|T \cdot \hat{c}_1|^2 + |T \cdot \hat{c}_2|^2}}, \\ 0 = \frac{\partial \mathcal{S}}{\partial \sin \theta} &= 2 \left[-\frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}} |T \cdot \hat{c}_1| + |T \cdot \hat{c}_2| \right] \implies \sin \theta = \frac{|T \cdot \hat{c}_2|}{\sqrt{|T \cdot \hat{c}_1|^2 + |T \cdot \hat{c}_2|^2}}. \end{aligned} \quad (\text{B.189})$$

Hence, inserting these expressions into (B.188), we are left with

$$\max_{\hat{c}_1, \hat{c}_2} \mathcal{S}(\rho) = \max_{\hat{c}_1, \hat{c}_2} 2\sqrt{|T \cdot \hat{c}_1|^2 + |T \cdot \hat{c}_2|^2} = \max_{\hat{c}_1, \hat{c}_2} 2\sqrt{\hat{c}_1 \cdot T^T T \cdot \hat{c}_1 + \hat{c}_2 \cdot T^T T \cdot \hat{c}_2} \leq 2\sqrt{\lambda_1 + \lambda_2}, \quad (\text{B.190})$$

where to solve the remaining maximisation we have chosen $\hat{c}_1$ and $\hat{c}_2$ as the two eigenvectors of $T^T T$ corresponding to its two largest eigenvalues, denoting them by λ_1 and λ_2 . Therefore, we have obtained the Horodecki criterion.

B.2.6 Solution (Text at p. 142)

(a) For a general two-qubit state,

$$\rho = \frac{1}{4} \left(\mathbb{1} \otimes \mathbb{1} + \vec{r} \cdot \hat{\sigma} \otimes \mathbb{1} + \mathbb{1} \otimes \vec{s} \cdot \hat{\sigma} + \sum_{i,j=x,y,z} T_{ij} \sigma_i \otimes \sigma_j \right), \quad (\text{B.191})$$

where $T_{ij} = \text{tr}(\rho \sigma_i \otimes \sigma_j)$, the Horodecki criterion states that the maximal value of the CHSH parameter, is given by

$$\max \mathcal{S}(\rho) = 2\sqrt{\lambda_1 + \lambda_2}, \quad (\text{B.192})$$

where λ_1 and λ_2 are the two largest eigenvalues of $T^T T$. Let us explicitly write the state ρ ,

$$\rho = \frac{17}{20} \frac{1}{5} [4|00\rangle\langle 00| + |11\rangle\langle 11| + 2|00\rangle\langle 11| + 2|11\rangle\langle 00|] + \frac{3}{20} |01\rangle\langle 01|. \quad (\text{B.193})$$

Recalling the action of the Pauli matrices $\sigma_X, \sigma_Y, \sigma_Z$ on the computational basis states,

$$\begin{aligned} \sigma_X|0\rangle &= |1\rangle, & \sigma_X|1\rangle &= |0\rangle, \\ \sigma_Y|0\rangle &= i|1\rangle, & \sigma_Y|1\rangle &= -i|0\rangle, \\ \sigma_Z|0\rangle &= |0\rangle, & \sigma_Z|1\rangle &= -|1\rangle, \end{aligned} \quad (\text{B.194})$$

we can calculate each element of the correlation tensor as

$$\begin{aligned} T_{xx} &= \text{tr}(\sigma_X \otimes \sigma_X \rho) = \frac{17}{20} \frac{1}{5} \text{tr} \left[\underbrace{4|11\rangle\langle 00|}_{=0} + \underbrace{|00\rangle\langle 11|}_{=0} + 2|11\rangle\langle 11| + 2|00\rangle\langle 00| \right] + \frac{3}{20} \text{tr}(\underbrace{|10\rangle\langle 01|}_{=0}) \\ &= \frac{17}{20} \frac{4}{5} = 0.6800. \end{aligned} \quad (\text{B.195})$$

Using the same approach, we can calculate all the other elements T_{ij} to get

$$T = \begin{pmatrix} 0.6800 & 0 & 0 \\ 0 & -0.6800 & 0 \\ 0 & 0 & 0.7000 \end{pmatrix} \implies T^T T = \begin{pmatrix} 0.4624 & 0 & 0 \\ 0 & 0.4624 & 0 \\ 0 & 0 & 0.4900 \end{pmatrix}. \quad (\text{B.196})$$

$T^T T$ is already in its diagonal form, therefore $\sqrt{\lambda_1 + \lambda_2} = \sqrt{0.4900 + 0.4624} = 0.9759 < 1$, i.e. the state cannot violate the CHSH inequality.

(b) According to the Born rule,

$$\begin{aligned} p(a, b|x, y) &= \text{tr}(A_{a|x} \otimes B_{b|y} \rho) = \frac{1}{4} \text{tr}[(\mathbb{1} + (-1)^a \vec{a}_x \cdot \vec{\sigma}) \otimes (\mathbb{1} + (-1)^b \vec{b}_y \cdot \vec{\sigma}) \rho], \\ p_A(a|x) &= \text{tr}(A_{a|x} \otimes \mathbb{1} \rho) = \frac{1}{2} \text{tr}[(\mathbb{1} + (-1)^a \vec{a}_x \cdot \vec{\sigma}) \otimes \mathbb{1} \rho], \\ p_B(b|y) &= \text{tr}(\mathbb{1} \otimes B_{b|y} \rho) = \frac{1}{2} \text{tr}[\mathbb{1} \otimes (\mathbb{1} + (-1)^b \vec{b}_y \cdot \vec{\sigma}) \rho]. \end{aligned} \quad (\text{B.197})$$

We can compute all the probabilities appearing in the I_{3322} inequality. As an example, let us explicitly write $p_B(0|1)$,

$$p_B(0|1) = \frac{1}{2} \text{tr}[\mathbb{1} \otimes (\mathbb{1} + (-1)^0 \vec{b}_1 \cdot \vec{\sigma}) \rho] = \frac{1}{2} \text{tr}[\mathbb{1} \otimes (\mathbb{1} + \sin \phi_1 \sigma_x + \cos \phi_1 \sigma_z) \rho] = 0.6470. \quad (\text{B.198})$$

Further similar calculations give

$$\begin{aligned} p_B(0|2) &= 0.6470, \\ p_A(0|1) &= 0.8087, \\ p(0, 0|x, y) &= \begin{pmatrix} 0.5768 & 0.6462 & 0.4644 \\ 0.6462 & 0.5768 & 0.3442 \\ 0.4216 & 0.2253 & 0.0800 \end{pmatrix}, \end{aligned} \quad (\text{B.199})$$

from which we can evaluate (B.21) to get $0.0129 > 0$, i.e. the state (B.23) violates the I_{3322} inequality.

B.2.7 Solution (Text at p. 143)

- (a) In Bell scenarios with binary inputs and outputs there are 16 deterministic input/output strategies. Let us label them as $A_1^\pm A_2^\pm B_1^\pm B_2^\pm$, meaning that when Alice makes her first measurement she outputs + or -, when she makes her second measurement she outputs + or -, etc. Thus, the hidden variable is the string of four $\pm$ symbols, $\lambda \in \{(+, +, +, +), (+, +, +, -), \dots, (-, -, -, -)\}$. In order for a probability of a specific event to be exactly zero in a local hidden variable model, none of the deterministic strategies that are relevant for that event can have a non-zero probability in the distribution of the local hidden variable, $q(\lambda)$. Specifically,

- i. $p(+1, +1|1, 1) = 0$, is relevant to $A_1^+ A_2^+ B_1^+ B_2^+$. Hence, these four strategies have $q(\lambda) = 0$.
- ii. $p(-1, -1|1, 2) = 0$, is relevant to $A_1^- A_2^+ B_1^+ B_2^-$. Hence, these four strategies have $q(\lambda) = 0$.
- iii. $p(-1, -1|2, 1) = 0$, is relevant to $A_1^+ A_2^- B_1^- B_2^+$. Hence, these four strategies have $q(\lambda) = 0$.

However, to satisfy $p(-1, -1|2, 2) > 0$, we need at least one of the strategies $A_1^\pm A_2^- B_1^\pm B_2^-$ to have a non-zero value of $q(\lambda)$. Writing these strategies out, they are $(+, -, +, -)$, $(-, -, +, -)$, $(-, -, -, -)$ and $(+, -, -, -)$. However, the first one is forbidden by the first constraint, the second and third are forbidden by the second constraint, and the fourth is forbidden by the third constraint. Hence, we have arrived at a contradiction.

- (b) By direct verification, we have

$$p(+1, +1|1, 1) = |\langle 00|\psi \rangle|^2 = 0, \quad \text{and} \quad p(-1, -1|2, 2) = |\langle -, -|\psi \rangle|^2 = \frac{1}{12}. \quad (\text{B.200})$$

By symmetry we have $p(-1, -1|1, 2) = p(-1, -1|2, 1)$ and we find that $p(-1, -1|2, 1) = |\langle 1, -|\psi \rangle|^2 = 0$. Hence, all conditions are satisfied.

- (c) Let us write A_1, A_2, B_1, B_2 for the observables of Alice and Bob given their different measurements, e.g. $A_{\pm|1} = \frac{1}{2}(\mathbb{1} \pm A_1)$. Let $|\psi\rangle$ be the state shared between Alice and Bob. The four conditions in (B.27) imply that

$$(\mathbb{1} + A_1)(\mathbb{1} + B_1)|\psi\rangle = 0, \quad (\text{B.201})$$

$$(\mathbb{1} - A_1)(\mathbb{1} - B_2)|\psi\rangle = 0, \quad (\text{B.202})$$

$$(\mathbb{1} - A_2)(\mathbb{1} - B_1)|\psi\rangle = 0, \quad (\text{B.203})$$

$$(\mathbb{1} - A_2)(\mathbb{1} - B_2)|\psi\rangle \neq 0. \quad (\text{B.204})$$

The additional constraint $p(-1, -1|1, 1) = 0$ becomes

$$(\mathbb{1} - A_1)(\mathbb{1} - B_1)|\psi\rangle = 0. \quad (\text{B.205})$$

We have omitted the tensor product for simplicity of presentation but keep in mind that $A_x B_y = B_y A_x$ and that for all observables it holds that $A_x^2 = B_y^2 = \mathbb{1}$. Now, add (B.201) to (B.205). It gives

$$A_1 B_1 |\psi\rangle = -|\psi\rangle. \quad (\text{B.206})$$

Multiply both sides by A_1 and we get $B_1 |\psi\rangle = -A_1 |\psi\rangle$. Use this in Eqs (B.202) and (B.203) to obtain respectively

$$(\mathbb{1} - B_2) B_1 |\psi\rangle = -(\mathbb{1} - B_2) |\psi\rangle, \quad (\text{B.207})$$

$$(\mathbb{1} - A_2) A_1 |\psi\rangle = -(\mathbb{1} - A_2) |\psi\rangle. \quad (\text{B.208})$$

For example,

$$(\mathbb{1} - A_1)(\mathbb{1} - B_2)|\psi\rangle = |\psi\rangle - \underbrace{A_1|\psi\rangle}_{=B_1|\psi\rangle} - \underbrace{B_2|\psi\rangle}_{=B_2B_1|\psi\rangle} + A_1B_2|\psi\rangle = (\mathbb{1} - B_2)|\psi\rangle + (\mathbb{1} - B_2)B_1|\psi\rangle. \quad (\text{B.209})$$

Consider now (B.204) and use the obtained relations to get

$$\begin{aligned} (\mathbb{1} - A_2)(\mathbb{1} - B_2)|\psi\rangle &= (\mathbb{1} - B_2)(\mathbb{1} - A_2)|\psi\rangle \stackrel{(\text{B.206})}{=} -(\mathbb{1} - B_2)(\mathbb{1} - A_2)A_1B_1|\psi\rangle \\ &= -(\mathbb{1} - B_2)B_1(\mathbb{1} - A_2)A_1|\psi\rangle \stackrel{(\text{B.208})}{=} (\mathbb{1} - B_2)B_1(\mathbb{1} - A_2)|\psi\rangle \end{aligned} \quad (\text{B.210})$$

$$= (\mathbb{1} - A_2)(\mathbb{1} - B_2)B_1|\psi\rangle \stackrel{(\text{B.207})}{=} -(\mathbb{1} - A_2)(\mathbb{1} - B_2)|\psi\rangle. \quad (\text{B.211})$$

Due to the minus sign in the end, this is clearly a contradiction.

B.2.8 Solution (Text at p. 143)

- (a) Let a_x , b_y and c_z denote the output assigned by Alice, Bob and Charlie to their respective inputs. The winning conditions are

$$a_0b_0c_1 = -1, \quad a_0b_1c_0 = -1, \quad a_1b_0c_0 = -1, \quad a_1b_1c_1 = +1. \quad (\text{B.212})$$

These four conditions are not consistent because multiplying all right-hand-sides gives -1 whereas multiplying all left-hand-sides gives

$$(a_0b_0c_1)(a_0b_1c_0)(a_1b_0c_0)(a_1b_1c_1) = a_0^2b_0^2c_0^2a_1^2b_1^2c_1^2 = +1. \quad (\text{B.213})$$

- (b) In the quantum model, we win the game because the GHZ-state is an eigenstate of all four observables and with the correct eigenvalue. Specifically, we can verify the four relations

$$\begin{aligned} \sigma_Y \otimes \sigma_Y \otimes \sigma_X |\text{GHZ}\rangle &= -|\text{GHZ}\rangle \\ \sigma_Y \otimes \sigma_X \otimes \sigma_Y |\text{GHZ}\rangle &= -|\text{GHZ}\rangle \\ \sigma_X \otimes \sigma_Y \otimes \sigma_Y |\text{GHZ}\rangle &= -|\text{GHZ}\rangle \\ \sigma_X \otimes \sigma_X \otimes \sigma_X |\text{GHZ}\rangle &= |\text{GHZ}\rangle. \end{aligned} \quad (\text{B.214})$$

This can easily be seen because both σ_Y and σ_X are bit-flip operations, but σ_Y also leads to a global sign-flip when applied an even number of times on $|\text{GHZ}\rangle$.

B.2.9 Solution (Text at p. 144)

- (a) We already know from the GHZ paradox (Exercise B.2.8) that $\mathcal{S} = 4$ when selecting the GHZ state. If we select the maximally mixed state $\frac{1}{8}\mathbb{1}$ we instead obtain $\langle A_x B_y C_z \rangle = \text{tr}(A_x \otimes B_y \otimes C_z \frac{1}{8}\mathbb{1}) = 0$ because all the observables have zero trace. Therefore, the Mermin parameter for the mixed state $v|\text{GHZ}\rangle\langle\text{GHZ}| + \frac{1-v}{8}\mathbb{1}$ becomes simply $4v + 0 \times (1-v) = 4v$. The violation is obtained for $v > 1/2$.
- (b) If we consider the state $\phi^+ \otimes |s\rangle\langle s|$ then our expectation values become $\langle A_x B_y C_z \rangle = \text{tr}(A_x \otimes B_y \phi^+) \text{tr}(C_z |s\rangle\langle s|)$. Since $C_0 = \sigma_Y$ and $C_1 = \sigma_X$, we have $\text{tr}(C_z |s\rangle\langle s|) = -\frac{1}{\sqrt{2}}$. Hence, Mermin's inequality simplifies to

$$\mathcal{S} = \langle A_0 B_0 \rangle_{\phi^+} + \langle A_0 B_1 \rangle_{\phi^+} + \langle A_1 B_0 \rangle_{\phi^+} - \langle A_1 B_1 \rangle_{\phi^+} \leq 2\sqrt{2}. \quad (\text{B.215})$$

This is just the Tsirelson bound of the CHSH parameter. The inequality therefore cannot be violated in quantum theory.

- (c) From the previous step, we may suspect that there is a systematic way to boil down Mermin's inequality to the CHSH inequality when there is no entanglement with Charlie. This is indeed the case. For a state on the form $|\psi\rangle_{AB} \otimes |\phi\rangle_C$ we obtain the expectation values $\langle A_x B_y C_z \rangle = \langle A_x B_y \rangle_{\psi} \langle C_z \rangle_{\phi}$. Mermin's inequality becomes

$$\mathcal{S} = (\langle A_1 B_1 \rangle_{\psi} - \langle A_0 B_0 \rangle_{\psi}) \langle C_1 \rangle_{\phi} + (-\langle A_0 B_1 \rangle_{\psi} - \langle A_1 B_0 \rangle_{\psi}) \langle C_0 \rangle_{\phi} \leq 2. \quad (\text{B.216})$$

Let Charlie measure $C_z = -1$ which gives $\langle C_0 \rangle_{\phi} = \langle C_1 \rangle_{\phi} = -1$. This reduces the above to the CHSH inequality.

B.2.10 Solution (Text at p. 144)

(a) A local hidden variable model is by definition

$$p(a, b|x, y) = \int d\lambda q(\lambda) p(a|x, \lambda) p(b|y, \lambda). \quad (\text{B.217})$$

To check for no-signaling, we verify that the marginal distributions of each party are independent of the other party's input. This follows directly from

$$\begin{aligned} \sum_b p(a, b|x, y) &= \int d\lambda q(\lambda) p(a|x, \lambda) \underbrace{\sum_b p(b|y, \lambda)}_{=1} = \int d\lambda q(\lambda) p(a|x, \lambda) = p(a|x), \\ \sum_a p(a, b|x, y) &= \int d\lambda q(\lambda) \underbrace{\sum_a p(a|x, \lambda)}_{=1} p(b|y, \lambda) = \int d\lambda q(\lambda) p(b|y, \lambda) = p(b|y). \end{aligned} \quad (\text{B.218})$$

(b) Quantum models take the form

$$p(a, b|x, y) = \text{tr}(\rho_{AB} A_{a|x} \otimes B_{b|y}), \quad (\text{B.219})$$

where ρ_{AB} is the shared state and $\{A_{a|x}\}$ and $\{B_{b|y}\}$ are the collections of POVMs used by Alice and Bob. No-signaling holds because

$$\begin{aligned} \sum_b p(a, b|x, y) &= \text{tr}(\rho_{AB} A_{a|x} \otimes \underbrace{\sum_b B_{b|y}}_{=1}) = \text{tr}(\rho_{AB} A_{a|x} \otimes \mathbb{1}) = \text{tr}(\rho_A A_{a|x}) = p(a|x), \\ \sum_a p(a, b|x, y) &= \text{tr}(\rho_{AB} \underbrace{\sum_a A_{a|x}}_{=1} \otimes B_{b|y}) = \text{tr}(\rho_{AB} \mathbb{1} \otimes B_{b|y}) = \text{tr}(\rho_B B_{b|y}) = p(b|y). \end{aligned} \quad (\text{B.220})$$

B.3. Quantum information primitives

B.3.1 Solution (Text at p. 144)

The state of the system arriving to Bob is given by

$$|\psi_x\rangle = U_x \otimes \mathbb{1} |\phi^+\rangle = X^{x_1} Z^{x_2} \otimes \mathbb{1} |\phi^+\rangle. \quad (\text{B.221})$$

To show that Bob can deterministically learn $x = (x_1, x_2)$, we must show that the set $\{|\psi_x\rangle\}_{x=1}^{d^2}$ forms an orthonormal basis. This would allow Bob to measure in that basis and distinguish all the elements. Clearly, the elements $|\psi_x\rangle$ are already normalised so we need only to check that they are orthogonal. To that end,

$$\begin{aligned} \langle \psi_{x'} | \psi_x \rangle &= \langle \phi^+ | (X^{x'_1} Z^{x'_2} \otimes \mathbb{1})^\dagger (X^{x_1} Z^{x_2} \otimes \mathbb{1}) | \phi^+ \rangle = \langle \phi^+ | (Z^{-x'_2} X^{-x'_1} \otimes \mathbb{1}) (X^{x_1} Z^{x_2} \otimes \mathbb{1}) | \phi^+ \rangle \\ &= \langle \phi^+ | Z^{-x'_2} X^{x_1-x'_1} Z^{x_2} \otimes \mathbb{1} | \phi^+ \rangle = \text{tr}(Z^{-x'_2} X^{x_1-x'_1} Z^{x_2} \otimes \mathbb{1} |\phi^+\rangle) = \text{tr}\left(Z^{-x'_2} X^{x_1-x'_1} Z^{x_2} \frac{\mathbb{1}}{d}\right) \\ &= \frac{1}{d} \text{tr}\left(X^{x_1-x'_1} Z^{x_2-x'_2}\right), \end{aligned} \quad (\text{B.222})$$

where we have used that $X^\dagger = X^{-1}$, $Z^\dagger = Z^{-1}$ and

$$\text{tr}(A \otimes \mathbb{1} |\phi^+\rangle) = \frac{1}{d} \sum_{k,l=0}^{d-1} \text{tr}(A \otimes \mathbb{1} |k\rangle\langle l|) = \frac{1}{d} \sum_{k,l=0}^{d-1} \text{tr}(A |k\rangle\langle l|) \underbrace{\text{tr}(|k\rangle\langle l|)}_{=\delta_{k,l}} = \frac{1}{d} \text{tr}\left(A \sum_{k=0}^{d-1} |k\rangle\langle k|\right) = \text{tr}\left(A \frac{\mathbb{1}}{d}\right), \quad (\text{B.223})$$

for any operator A . Consider now the matrix $X^{x_1-x'_1} Z^{x_2-x'_2}$,

$$X^{x_1-x'_1} Z^{x_2-x'_2} = \underbrace{\sum_{k=0}^{d-1} |k+x_1-x'_1\rangle\langle k|}_{X^{x_1-x'_1}} \underbrace{\sum_{l=0}^{d-1} e^{\frac{2\pi i}{d} l(x_2-x'_2)} |l\rangle\langle l|}_{Z^{x_2-x'_2}} = \sum_{k=0}^{d-1} e^{\frac{2\pi i}{d} k(x_2-x'_2)} |k+x_1-x'_1\rangle\langle k|. \quad (\text{B.224})$$

Taking the trace, we obtain

$$\langle \psi_{x'} | \psi_x \rangle = \frac{1}{d} \text{tr} \left(X^{x_1 - x'_1} Z^{x_2 - x'_2} \right) = \frac{1}{d} \sum_{k=0}^{d-1} e^{\frac{2\pi i}{d} k(x_2 - x'_2)} \underbrace{\text{tr}(|k + x_1 - x'_1\rangle\langle k|)}_{=\delta_{x_1, x'_1}} = \frac{\delta_{x_1, x'_1}}{d} \sum_{k=0}^{d-1} e^{\frac{2\pi i}{d} k(x_2 - x'_2)} \underbrace{= d\delta_{x_2, x'_2}}_{(B.225)},$$

where in the last step of the calculation, we have used that if $x_2 \neq x'_2$ then $\sum_{k=0}^{d-1} e^{\frac{2\pi i}{d} k(x_2 - x'_2)}$ is the sum of all roots of unity, which is equals zero⁸¹. Hence $\{|\psi_x\rangle\}_{x=1}^{d^2}$ forms a basis. Note that all the basis elements are maximally entangled.

B.3.2 Solution (Text at p. 144)

Denote the total state given to Bob by

$$\tau_x^{AB} = U_x \otimes \mathbb{1} |\psi\rangle\langle\psi| U_x^\dagger \otimes \mathbb{1} \quad (B.226)$$

The winning probability in state discrimination is

$$p_{\text{win}} = \frac{1}{n} \sum_{x=1}^n \text{tr}(\tau_x^{AB} E_x) = \frac{1}{n} \sum_{x=1}^n \sum_{\lambda} \text{tr}(\tau_x^{AB} M_{x,\lambda}^A \otimes M_{x,\lambda}^B) \quad (B.227)$$

Use that $\tau_x^{AB} \leq \mathbb{1}^A \otimes \tau_x^B$ and that $\tau_x^B = \text{tr}_A(\tau_x^{AB}) = \psi^B$ is independent of x . This gives

$$p_{\text{win}} \leq \frac{1}{n} \sum_{x=1}^n \sum_{\lambda} \text{tr}(\mathbb{1} \otimes \tau_x^B M_{x,\lambda}^A \otimes M_{x,\lambda}^B) = \frac{1}{n} \sum_{x=1}^n \sum_{\lambda} \text{tr}((\mathbb{1} \otimes \psi^B)(M_{x,\lambda}^A \otimes M_{x,\lambda}^B)) = \frac{1}{n} \text{tr}(\mathbb{1} \otimes \psi^B) = \frac{d}{n}, \quad (B.228)$$

where we used the completeness relation for POVMs.

B.3.3 Solution (Text at p. 145)

(a) The state of the combined three qubit system is

$$\begin{aligned} |\psi\rangle_{A'} \otimes |\varphi\rangle_{AB} &= (\alpha|0\rangle_{A'} + \beta|1\rangle_{A'}) \otimes (\lambda_1|00\rangle_{AB} + \lambda_2|11\rangle_{AB}) \\ &= \alpha(\lambda_1|000\rangle_{A'AB} + \lambda_2|011\rangle_{A'AB}) + \beta(\lambda_1|100\rangle_{A'AB} + \lambda_2|111\rangle_{A'AB}) \\ &= \alpha(\lambda_1|00\rangle_{A'A} \otimes |0\rangle_B + \lambda_2|01\rangle_{A'A} \otimes |1\rangle_B) + \beta(\lambda_1|10\rangle_{A'A} \otimes |0\rangle_B + \lambda_2|11\rangle_{A'A} \otimes |1\rangle_B) \\ &= \alpha \left(\frac{\lambda_1}{\sqrt{2}} (|\phi^+\rangle_{A'A} + |\phi^-\rangle_{A'A}) \otimes |0\rangle_B + \frac{\lambda_2}{\sqrt{2}} (|\psi^+\rangle_{A'A} + |\psi^-\rangle_{A'A}) \otimes |1\rangle_B \right) \\ &\quad + \beta \left(\frac{\lambda_1}{\sqrt{2}} (|\psi^+\rangle_{A'A} - |\psi^-\rangle_{A'A}) \otimes |0\rangle_B + \frac{\lambda_2}{\sqrt{2}} (|\phi^+\rangle_{A'A} - |\phi^-\rangle_{A'A}) \otimes |1\rangle_B \right), \end{aligned} \quad (B.229)$$

where we have used the following basis transformation

$$|00\rangle = \frac{1}{\sqrt{2}} (|\phi^+\rangle + |\phi^-\rangle), \quad |01\rangle = \frac{1}{\sqrt{2}} (|\psi^+\rangle + |\psi^-\rangle), \quad (B.230)$$

$$|10\rangle = \frac{1}{\sqrt{2}} (|\psi^+\rangle - |\psi^-\rangle), \quad |11\rangle = \frac{1}{\sqrt{2}} (|\phi^+\rangle - |\phi^-\rangle). \quad (B.231)$$

We can properly separate the contributions on the $A'A$ and the B subsystems, rewriting (B.229) as

$$\begin{aligned} |\psi\rangle_{A'} \otimes |\varphi\rangle_{AB} &= \frac{|\phi^+\rangle_{A'A}}{2} (\sqrt{2}\alpha\lambda_1|0\rangle_B + \sqrt{2}\beta\lambda_2|1\rangle_B) + \frac{|\phi^-\rangle_{A'A}}{2} (\sqrt{2}\alpha\lambda_1|0\rangle_B - \sqrt{2}\beta\lambda_2|1\rangle_B) \\ &= \frac{|\psi^+\rangle_{A'A}}{2} (\sqrt{2}\alpha\lambda_2|1\rangle_B + \sqrt{2}\beta\lambda_1|0\rangle_B) + \frac{|\psi^-\rangle_{A'A}}{2} (\sqrt{2}\alpha\lambda_2|1\rangle_B - \sqrt{2}\beta\lambda_1|0\rangle_B). \end{aligned} \quad (B.232)$$

⁸¹ You can show this by noticing that $\sum_{k=0}^{d-1} e^{\frac{2\pi i}{d} k(x_2 - x'_2)}$ is the sum of a geometric series.

- (b) From (B.232), we can extract the four possible states for Bob's system conditioned on each of the four possible outcomes of Alice's Bell state measurement,

$$\begin{aligned} |\psi_1\rangle_B &= \sqrt{2}\alpha\lambda_1|0\rangle_B + \sqrt{2}\beta\lambda_2|1\rangle_B, & |\psi_2\rangle_B &= \sqrt{2}\alpha\lambda_1|0\rangle_B - \sqrt{2}\beta\lambda_2|1\rangle_B, \\ |\psi_3\rangle_B &= \sqrt{2}\alpha\lambda_2|1\rangle_B + \sqrt{2}\beta\lambda_1|0\rangle_B, & |\psi_4\rangle_B &= \sqrt{2}\alpha\lambda_2|1\rangle_B - \sqrt{2}\beta\lambda_1|0\rangle_B. \end{aligned} \quad (\text{B.233})$$

The effect of the unitaries $U_k = \{\mathbb{1}, \sigma_Z, \sigma_X, \sigma_X \sigma_Z\}$ on these states is

$$\begin{aligned} U_1|\psi_1\rangle_B &= \mathbb{1}(\sqrt{2}\alpha\lambda_1|0\rangle_B + \sqrt{2}\beta\lambda_2|1\rangle_B) = \sqrt{2}\alpha\lambda_1|0\rangle_B + \sqrt{2}\beta\lambda_2|1\rangle_B, \\ U_2|\psi_2\rangle_B &= \sigma_Z(\sqrt{2}\alpha\lambda_1|0\rangle_B - \sqrt{2}\beta\lambda_2|1\rangle_B) = \sqrt{2}\alpha\lambda_1|0\rangle_B + \sqrt{2}\beta\lambda_2|1\rangle_B, \\ U_3|\psi_3\rangle_B &= \sigma_X(\sqrt{2}\alpha\lambda_2|1\rangle_B + \sqrt{2}\beta\lambda_1|0\rangle_B) = \sqrt{2}\alpha\lambda_2|0\rangle_B + \sqrt{2}\beta\lambda_1|1\rangle_B, \\ U_4|\psi_4\rangle_B &= \sigma_X \sigma_Z(\sqrt{2}\alpha\lambda_2|1\rangle_B - \sqrt{2}\beta\lambda_1|0\rangle_B) = -\sqrt{2}\alpha\lambda_2|0\rangle_B - \sqrt{2}\beta\lambda_1|1\rangle_B, \end{aligned} \quad (\text{B.234})$$

from which the fidelity with the state $|\psi\rangle_{A'}$ Alice wants to teleport is

$$\begin{aligned} |\langle\psi|U_1|\psi_1\rangle|^2 &= |(\alpha^*\langle 0| + \beta^*\langle 1|)(\sqrt{2}\alpha\lambda_1|0\rangle + \sqrt{2}\beta\lambda_2|1\rangle)|^2 = |\sqrt{2}\alpha^2\lambda_1 + \sqrt{2}\beta^2\lambda_2|^2, \\ |\langle\psi|U_2|\psi_2\rangle|^2 &= |(\alpha^*\langle 0| + \beta^*\langle 1|)(\sqrt{2}\alpha\lambda_1|0\rangle - \sqrt{2}\beta\lambda_2|1\rangle)|^2 = |\sqrt{2}\alpha^2\lambda_1 - \sqrt{2}\beta^2\lambda_2|^2, \\ |\langle\psi|U_3|\psi_3\rangle|^2 &= |(\alpha^*\langle 0| + \beta^*\langle 1|)(\sqrt{2}\alpha\lambda_2|0\rangle + \sqrt{2}\beta\lambda_1|1\rangle)|^2 = |\sqrt{2}\alpha^2\lambda_2 + \sqrt{2}\beta^2\lambda_1|^2, \\ |\langle\psi|U_4|\psi_4\rangle|^2 &= |(\alpha^*\langle 0| + \beta^*\langle 1|)(-\sqrt{2}\alpha\lambda_2|0\rangle - \sqrt{2}\beta\lambda_1|1\rangle)|^2 = |\sqrt{2}\alpha^2\lambda_2 - \sqrt{2}\beta^2\lambda_1|^2. \end{aligned} \quad (\text{B.235})$$

- (c) To compute the average fidelity, we need to compute

$$F = \int d\psi \langle\psi|\rho|\psi\rangle = \frac{1}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \langle\psi|\rho|\psi\rangle, \quad (\text{B.236})$$

where we have used the Bloch sphere representation to parametrise the space of all qubit states (as in Chapter 6.2). The state ρ obtained by Bob at the end of the protocol can be expressed as

$$\rho = \sum_{k=1}^4 p_k U_k |\psi_k\rangle \langle\psi_k| U_k^\dagger, \quad (\text{B.237})$$

where $p_k = \frac{1}{4}$ is the probability of Alice measuring one of the four Bell states in (B.232). Therefore, the average fidelity becomes

$$\begin{aligned} F &= \frac{1}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \langle\psi|\rho|\psi\rangle = \frac{1}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \frac{1}{4} \sum_{k=1}^4 |\langle\psi|U_k|\psi_k\rangle|^2 \\ &= \frac{1}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \frac{1}{2} \left[|\sqrt{2}\alpha^2\lambda_1 + \sqrt{2}\beta^2\lambda_2|^2 + |\sqrt{2}\alpha^2\lambda_2 + \sqrt{2}\beta^2\lambda_1|^2 \right], \end{aligned} \quad (\text{B.238})$$

where we have used the fidelities computed in (B.235). For simplicity, let us split the integral in two parts:

$$\begin{aligned} F_1 &= \frac{1}{8\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta |\sqrt{2}\alpha^2\lambda_1 + \sqrt{2}\beta^2\lambda_2|^2, \\ F_2 &= \frac{1}{8\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta |\sqrt{2}\alpha^2\lambda_2 + \sqrt{2}\beta^2\lambda_1|^2, \end{aligned} \quad (\text{B.239})$$

where for simplicity we are denoting $|\alpha|^2 = \alpha^2$ and $|\beta|^2 = \beta^2$.

Let us start computing F_1 . Recalling that, from the Bloch sphere parametrisation, $\alpha = \cos \frac{\theta}{2}$ and $\beta = e^{i\varphi} \sin \frac{\theta}{2}$,

$$\begin{aligned} F_1 &= \frac{1}{8\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \left(\sqrt{2} \cos^2 \frac{\theta}{2} \lambda_1 + \sqrt{2} \sin^2 \frac{\theta}{2} \lambda_2 \right)^2 \\ &= \frac{1}{8\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \left(\sqrt{2} \frac{1 + \cos\theta}{2} \lambda_1 + \sqrt{2} \frac{1 - \cos\theta}{2} \lambda_2 \right)^2, \end{aligned} \quad (\text{B.240})$$

where we have used the trigonometric identities $\cos^2 \frac{\theta}{2} = \frac{1+\cos \theta}{2}$ and $\sin^2 \frac{\theta}{2} = \frac{1-\cos \theta}{2}$. Therefore,

$$\begin{aligned} F_1 &= \frac{1}{8\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \left(\sqrt{2} \frac{1+\cos \theta}{2} \lambda_1 + \sqrt{2} \frac{1-\cos \theta}{2} \lambda_2 \right)^2 \\ &= \frac{1}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \left(\lambda_1^2 \left(\frac{1+\cos \theta}{2} \right)^2 + \lambda_2^2 \left(\frac{1-\cos \theta}{2} \right)^2 + 2\lambda_1 \lambda_2 \left(\frac{1+\cos \theta}{2} \right) \left(\frac{1-\cos \theta}{2} \right) \right) \\ &= \frac{1}{2} \left[\lambda_1^2 \int_0^\pi d\theta \sin \theta \left(\frac{1+\cos \theta}{2} \right)^2 + \lambda_2^2 \int_0^\pi d\theta \sin \theta \left(\frac{1-\cos \theta}{2} \right)^2 + 2\lambda_1 \lambda_2 \int_0^\pi d\theta \sin \theta \left(\frac{1+\cos \theta}{2} \right) \left(\frac{1-\cos \theta}{2} \right) \right]. \end{aligned}$$

We can use that

$$\int_0^\pi d\theta \sin \theta (1+\cos \theta)^2 = \int_0^\pi d\theta \sin \theta (1-\cos \theta)^2 = \frac{8}{3}, \quad \int_0^\pi d\theta \sin \theta (1+\cos \theta)(1-\cos \theta) = \frac{4}{3}, \quad (\text{B.241})$$

and we get

$$F_1 = \frac{1}{8} \left[\lambda_1^2 \frac{8}{3} + \lambda_2^2 \frac{8}{3} + 2\lambda_1 \lambda_2 \frac{4}{3} \right] = \frac{1}{8} \left[\frac{8}{3} + \lambda_1 \lambda_2 \frac{8}{3} \right] = \frac{1}{3} + \frac{\lambda_1 \lambda_2}{3}, \quad (\text{B.242})$$

where we have used that $\lambda_1^2 + \lambda_2^2 = 1$, since they are the Schmidt coefficients of the entangled state $|\varphi\rangle_{AB}$.

Similarly, we can compute F_2 ,

$$\begin{aligned} F_2 &= \frac{1}{8\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \left(\sqrt{2} \cos^2 \frac{\theta}{2} \lambda_2 + \sqrt{2} \sin^2 \frac{\theta}{2} \lambda_1 \right)^2 \\ &= \frac{1}{8\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \left(\sqrt{2} \frac{1+\cos \theta}{2} \lambda_2 + \sqrt{2} \frac{1-\cos \theta}{2} \lambda_1 \right)^2, \\ &= \frac{1}{8\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \left(\sqrt{2} \frac{1+\cos \theta}{2} \lambda_2 + \sqrt{2} \frac{1-\cos \theta}{2} \lambda_1 \right)^2 \\ &= \frac{1}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \left(\lambda_2^2 \left(\frac{1+\cos \theta}{2} \right)^2 + \lambda_1^2 \left(\frac{1-\cos \theta}{2} \right)^2 + 2\lambda_1 \lambda_2 \left(\frac{1+\cos \theta}{2} \right) \left(\frac{1-\cos \theta}{2} \right) \right) \\ &= \frac{1}{2} \left[\lambda_2^2 \int_0^\pi d\theta \sin \theta \left(\frac{1+\cos \theta}{2} \right)^2 + \lambda_1^2 \int_0^\pi d\theta \sin \theta \left(\frac{1-\cos \theta}{2} \right)^2 + 2\lambda_1 \lambda_2 \int_0^\pi d\theta \sin \theta \left(\frac{1+\cos \theta}{2} \right) \left(\frac{1-\cos \theta}{2} \right) \right] \\ &= \frac{1}{8} \left[\lambda_2^2 \frac{8}{3} + \lambda_1^2 \frac{8}{3} + 2\lambda_1 \lambda_2 \frac{4}{3} \right] = \frac{1}{8} \left[\frac{8}{3} + \lambda_1 \lambda_2 \frac{8}{3} \right] = \frac{1}{3} + \frac{\lambda_1 \lambda_2}{3}. \end{aligned} \quad (\text{B.243})$$

Therefore, the total average fidelity (B.236) becomes

$$F = F_1 + F_2 = \frac{2 + 2\lambda_1 \lambda_2}{3}. \quad (\text{B.244})$$

Note that, if $|\varphi\rangle_{AB}$ is maximally entangled, $\lambda_1 = \lambda_2 = \frac{1}{\sqrt{2}}$. Then, the average fidelity becomes

$$F = \frac{2 + 2\lambda_1 \lambda_2}{3} = \frac{2 + 2\frac{1}{2}}{3} = 1. \quad (\text{B.245})$$

B.3.4 Solution (Text at p. 145)

Let $\hat{n}$ be the Bloch vector of $|\psi\rangle$ and define $\vec{n}' = (n_X, -n_Y, n_Z)$ as the Bloch vector of a measurement $E = \vec{n}' \cdot \vec{\sigma}$, whose outcomes $\{E_0, E_1\}$ correspond to Bloch vectors $\{\vec{n}', -\vec{n}'\}$. Let Alice and Bob share $|\phi^+\rangle$. Alice measures E on her share of ϕ^+ , obtaining either E_0 or E_1 . Due to the action at distance relation (see Eq (4.21)),

$$E \otimes \mathbb{1}|\phi^+\rangle = \mathbb{1} \otimes E^T|\phi^+\rangle, \quad (\text{B.246})$$

where $E^T = \vec{n}' \cdot (\sigma_X^T, \sigma_Y^T, \sigma_Z^T) = \vec{n}' \cdot (\sigma_X, -\sigma_Y, \sigma_Z) = \hat{n} \cdot \vec{\sigma}$. Therefore, this prepares for Bob either the state $|\psi\rangle$ or the state corresponding to $-\hat{n}$, which is $|\psi_\perp\rangle = \beta|0\rangle - \alpha|1\rangle$. Note that there are no complex conjugates since we have assumed that $\hat{n}$ is restricted to the XZ -plane (real plane) of the Bloch sphere. Alice can send one bit to communicate to Bob the binary outcome of her measurement. If the outcome is 0, Bob does nothing since he already has $|\psi\rangle$ in his lab. If the outcome

is 1, he applies an operation that maps the Bloch vector $-\hat{n}$ to $\hat{n}$ to recover the state $|\psi\rangle$. The unitary that does the job is $U = \sigma_Z \sigma_X$ because

$$U|\psi_\perp\rangle = \sigma_Z \sigma_X (\beta|0\rangle - \alpha|1\rangle) = \sigma_Z (\beta|1\rangle - \alpha|0\rangle) = -\beta|1\rangle - \alpha|0\rangle = -|\psi\rangle, \quad (\text{B.247})$$

where the negative sign is a global phase and can therefore be ignored.

B.3.5 Solution (Text at p. 145)

Recall the entanglement swapping protocol from Chapter 6.3. The unnormalised post-measurement state on AD after post-selecting on the measurement operator $E = \rho^T$ on the BC subsystems is

$$\begin{aligned} \tilde{\sigma}_{AD} &= \text{tr}_{BC} ((\mathbb{1}_A \otimes E_{BC} \otimes \mathbb{1}_D)(\phi_{AB}^+ \otimes \phi_{CD}^+)(\mathbb{1}_A \otimes E_{BC} \otimes \mathbb{1}_D)) \\ &= \text{tr}_{BC} ((\mathbb{1}_B \otimes E_{AC}^{TA} \otimes \mathbb{1}_D)(\phi_{AB}^+ \otimes \phi_{CD}^+)(\mathbb{1}_A \otimes E_{BD}^{TD} \otimes \mathbb{1}_C)) \\ &= \text{tr}_{BC} ((\mathbb{1}_B \otimes E_{AD}^{TATD} \otimes \mathbb{1}_C)(\phi_{AB}^+ \otimes \phi_{CD}^+)(\mathbb{1}_B \otimes E_{AD}^{TD} \otimes \mathbb{1}_C)) \\ &= E_{AD}^{TATD} \text{tr}_B(\phi_{AB}^+) \otimes \text{tr}_C(\phi_{CD}^+) E_{AD}^{TD}, \end{aligned} \quad (\text{B.248})$$

where in the first two equalities we used the action at distance property in Eq (4.21). From the partial traces of the maximally entangled states we get $\frac{1}{d}\mathbb{1}_A$ and $\frac{1}{d}\mathbb{1}_D$, from which, recalling that $T_A T_D = T$ and that ρ is pure, the unnormalised post-measurement state becomes

$$\tilde{\sigma}_{AD} = \frac{1}{d^2} E_{AD}^T E_{AD}^T = \frac{1}{d^2} \rho_{AD}. \quad (\text{B.249})$$

B.3.6 Solution (Text at p. 145)

(a) We show that $|\psi_a\rangle = (\sigma_Z^{a_1} \otimes \sigma_X^{a_2} \otimes \dots \otimes \sigma_X^{a_n})|GHZ\rangle$ forms a basis. To this end, we compute the inner products,

$$\begin{aligned} \langle\psi_a|\psi_b\rangle &= \langle GHZ | \sigma_Z^{a_1+b_1} \otimes \sigma_X^{a_2+b_2} \otimes \dots \otimes \sigma_X^{a_n+b_n} | GHZ \rangle \\ &= \frac{1}{2} (\langle 0, \dots, 0 | + \langle 1, \dots, 1 |) (|0, a_2 \oplus b_2, \dots, a_n \oplus b_n\rangle + (-1)^{a_1+b_1} |1, 1 \oplus a_2 \oplus b_2, \dots, 1 \oplus a_n \oplus b_n\rangle) \\ &= \frac{1}{2} (\delta_{a_2, b_2} \dots \delta_{a_n, b_n} + (-1)^{a_1+b_1} \delta_{a_2, b_2} \dots \delta_{a_n, b_n}) = \delta_{a, b}, \end{aligned} \quad (\text{B.250})$$

where $a \oplus b$ denotes the sum modulo-2.

(b) The unnormalised post-measurement state conditioned on outcome a is

$$\text{tr}_{B_1 \dots B_n} ((\phi_{A_1 B_1}^+ \otimes \dots \otimes \phi_{A_n B_n}^+) (\mathbb{1}_{A_1 \dots A_n} \otimes |\psi_a\rangle \langle \psi_a|_{B_1 \dots B_n}^T)). \quad (\text{B.251})$$

Using action at distance relation from Eq (4.21) we obtain

$$\text{tr}_{B_1 \dots B_n} ((\phi_{A_1 B_1}^+ \otimes \dots \otimes \phi_{A_n B_n}^+) (\mathbb{1}_{B_1 \dots B_n} \otimes |\psi_a\rangle \langle \psi_a|_{A_1 \dots A_n})) = \frac{1}{2^n} |\psi_a\rangle \langle \psi_a|, \quad (\text{B.252})$$

where we have used that $\text{tr}_B(\phi_{AB}^+(\mathbb{1}_A \otimes R_B)) = \frac{1}{2} R$ for any operator R . Note that the probability distribution of the outcome a is uniform. Therefore, when the n parties are given the outcome a , they can implement the local unitary $\sigma_Z^{a_1}$ for party 1, and $\sigma_X^{a_k}$ for party $k = 2, \dots, n$, to rotate the state back to $|GHZ\rangle$.

B.3.7 Solution (Text at p. 146)

(a) The winning probability is

$$p_{\text{win}} = \frac{1}{24} \sum_{x_1, x_2, x_3} \sum_y \text{tr}(\rho_{x_1 x_2 x_3} M_{x_y | y}). \quad (\text{B.253})$$

Use the completeness relation $E_{0|y} + E_{1|y} = \mathbb{1}$ to write

$$E_{0|y} = \frac{\mathbb{1} + E_y}{2}, \quad E_{1|y} = \frac{\mathbb{1} - E_y}{2}. \quad (\text{B.254})$$

Substituting this in the expression, we get

$$\begin{aligned}
 p_{\text{win}} &= \frac{1}{24} \sum_{\substack{x_1, x_2, x_3, y \\ x_y=0}} \text{tr}(\rho_{x_1 x_2 x_3} E_{0|y}) + \frac{1}{24} \sum_{\substack{x_1, x_2, x_3, y \\ x_y=1}} \text{tr}(\rho_{x_1 x_2 x_3} E_{1|y}) \\
 &= \frac{1}{24} \sum_{\substack{x_1, x_2, x_3, y \\ x_y=0}} \text{tr}\left(\rho_{x_1 x_2 x_3} \frac{1 + E_y}{2}\right) + \frac{1}{24} \sum_{\substack{x_1, x_2, x_3, y \\ x_y=1}} \text{tr}\left(\rho_{x_1 x_2 x_3} \frac{1 - E_y}{2}\right) \\
 &= \frac{12}{48} + \frac{1}{48} \sum_{\substack{x_1, x_2, x_3, y \\ x_y=0}} \text{tr}(\rho_{x_1 x_2 x_3} E_y) + \frac{12}{48} - \frac{1}{48} \sum_{\substack{x_1, x_2, x_3, y \\ x_y=1}} \text{tr}(\rho_{x_1 x_2 x_3} E_y) \\
 &= \frac{1}{2} + \frac{1}{48} \sum_{x_1, x_2, x_3} \sum_y (-1)^{x_y} \text{tr}(\rho_{x_1 x_2 x_3} E_y). \tag{B.255}
 \end{aligned}$$

Note that for a given y , out of the eight tuples (x_1, x_2, x_3) four have $x_y = 0$ and four have $x_y = 1$.

(b) Alice's states are given by

$$\rho_{x_1 x_2 x_3} = \frac{\mathbb{1} + \hat{n}_{x_1 x_2 x_3} \cdot \vec{\sigma}}{2}. \tag{B.256}$$

For $E_y = \vec{m}_y \cdot \vec{\sigma}$, we have

$$\text{tr}(\rho_{x_1 x_2 x_3} E_y) = \hat{n}_{x_1 x_2 x_3} \cdot \vec{m}_y. \tag{B.257}$$

Therefore,

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{48} \sum_{x_1, x_2, x_3} \sum_y (-1)^{x_y} \hat{n}_{x_1 x_2 x_3} \cdot \vec{m}_y. \tag{B.258}$$

Define vectors $\vec{r}_y = \sum_{x_1, x_2, x_3} (-1)^{x_y} \hat{n}_{x_1 x_2 x_3}$. These are

$$\vec{r}_y = \frac{1}{\sqrt{3}} \sum_{x_1, x_2, x_3} (-1)^{x_y} [(-1)^{x_1}, (-1)^{x_2}, (-1)^{x_3}] = \begin{cases} r_{i,y} = \frac{8}{\sqrt{3}} & \text{if } i = y \\ r_{i,y} = 0 & \text{otherwise} \end{cases}. \tag{B.259}$$

Hence,

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{48} \sum_y \vec{r}_y \cdot \hat{m}_y = \frac{1}{2} + \frac{1}{6\sqrt{3}} \sum_y m_{y,y}, \tag{B.260}$$

where $m_{y,y}$ denotes the y 'th component of the vector $\vec{m}_y$. We have obtained the winning probability as function of Bob's Bloch vectors.

(c) If we want to maximise the success probability, we want to make the y 'th component of Bob's y 'th Bloch vector as large as possible. This means selecting $m_{y,y} = 1$, which is equivalent to

$$\hat{m}_1 = (1, 0, 0), \quad \hat{m}_2 = (0, 1, 0), \quad \hat{m}_3 = (0, 0, 1), \tag{B.261}$$

i.e., Bob performs the measurements of σ_X , σ_Y and σ_Z . This returns

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{6\sqrt{3}} 3 \times 1 = \frac{1}{2} \left(1 + \frac{1}{\sqrt{3}}\right). \tag{B.262}$$

B.3.8 Solution (Text at p. 146)

(a) The goal is to obtain the probabilities $\text{tr}(\rho E_b)$ for a projective qubit measurement $\{E_0, E_1\}$ by replacing the qubit ρ with a shared maximally entangled two-qubit state ϕ^+ and one bit of communication. To do this, Alice uses a protocol similar to the one in Exercise B.3.4. She defines the POVM $\{E_0 = \rho^T, E_1 = \mathbb{1} - \rho^T\}$ and measures her share of ϕ^+ . Due to the action at distance property (Eq (4.21)), this prepares for Bob the state ρ or the state $\mathbb{1} - \rho$, depending on whether the outcome of Alice's measurement is 0 or 1. Since the outcome is binary, she can send it over to Bob. On the one hand, if Bob receives 0, he already has ρ in his lab and can therefore simply apply the measurement $\{E_0, E_1\}$ to obtain the desired statistics $\text{tr}(\rho E_b)$. On the other hand, if Bob receives 1, he flips the outcomes in his measurement, i.e. $\{E_0, E_1\}$ is replaced with $\{E_1, E_0\}$. The probability of outcome b becomes $\text{tr}((\mathbb{1} - \rho) E_{\bar{b}}) = 1 - \text{tr}(\rho E_{\bar{b}}) = \text{tr}(\rho E_b)$, where $\bar{b}$ denotes bit-flip. We again obtain the correct probabilities.

- (b) We write Bob's POVM as $E_b = \lambda_b (\mathbb{1} + \vec{n}_b \cdot \vec{\sigma})$. As before, Alice starts by measuring her share of ϕ^+ with the POVM $\{E_0 = \rho^T, E_1 = \mathbb{1} - \rho^T\}$, preparing for Bob the state ρ (if outcome 0) and the state $\mathbb{1} - \rho$ (if outcome 1). In further analogy, Alice sends the outcome to Bob who simply measures $\{E_b\}_b$ if the outcome reads 0. However, if he receives outcome 1, then he replaces his measurement with a new one defined as $\tilde{E}_b = \lambda_b (\mathbb{1} - \vec{n}_b \cdot \vec{\sigma})$. Note that if $\{E_b\}$ is a POVM then so is $\{\tilde{E}_b\}_b$ because completeness and positive semidefiniteness are both preserved. Its outcome statistics becomes

$$\text{tr}((\mathbb{1} - \rho) \tilde{E}_b) = \text{tr}(\tilde{E}_b) - \text{tr}(\rho \tilde{E}_b) = 2\lambda_b - \lambda_b (1 - \text{tr}(\rho \vec{n}_b \cdot \vec{\sigma})) = \lambda_b (1 + \text{tr}(\rho \vec{n}_b \cdot \vec{\sigma})) = \text{tr}(\rho E_b). \quad (\text{B.263})$$

Thus, the simulation is successful.

The main difference between the first and second simulation protocol is that in the first one, Bob does not have to wait for Alice's message. He simply always performs $\{E_b\}$ and once the message arrives, he either keeps his outcome or flips it. In the second protocol, he can no longer do this, because the measurements $\{E_b\}$ and $\{\tilde{E}_b\}$ do not commute in general. Thus, Bob must wait for Alice's message and then adapt his measurement.

B.3.9 Solution (Text at p. 146)

- (a) The transformation is unitary because the orthogonal states $|00\rangle$ and $|10\rangle$ are transformed into orthogonal states $|00\rangle$ and $\cos \varphi |10\rangle + \sin \varphi |01\rangle$.
- (b) Applying the transformation to the state $|\psi_\theta\rangle \otimes |0\rangle$ gives

$$|\psi_\theta\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{e^{i\theta}}{\sqrt{2}}|10\rangle \rightarrow \frac{1}{\sqrt{2}}|00\rangle + \frac{e^{i\theta}}{\sqrt{2}}(\cos \varphi |10\rangle + \sin \varphi |01\rangle) = |\phi\rangle. \quad (\text{B.264})$$

- (c) First, let us compute the density operator ϕ ,

$$\begin{aligned} \phi &= |\phi\rangle\langle\phi| = \frac{1}{2} [|00\rangle + e^{i\theta}(\cos \varphi |10\rangle + \sin \varphi |01\rangle)] [\langle 00| + e^{-i\theta}(\cos \varphi \langle 10| + \sin \varphi \langle 01|)] \\ &= \frac{1}{2} [|00\rangle\langle 00| + e^{-i\theta}(\cos \varphi |00\rangle\langle 10| + \sin \varphi |00\rangle\langle 01|) + e^{i\theta}(\cos \varphi |10\rangle\langle 00| + \sin \varphi |01\rangle\langle 00|) \\ &\quad + \cos^2 \varphi |10\rangle\langle 10| + \cos \varphi \sin \varphi |10\rangle\langle 01| + \sin \varphi \cos \varphi |01\rangle\langle 10| + \sin^2 \varphi |01\rangle\langle 01|]. \end{aligned} \quad (\text{B.265})$$

Then, the state of ρ_θ^A is obtained from tracing out the second qubit, i.e.

$$\begin{aligned} \rho_\theta^A &= \text{tr}_B(\phi) = \sum_k (\mathbb{1}_A \otimes \langle k|_B) \phi (\mathbb{1}_A \otimes |k\rangle_B) \\ &= \frac{1}{2} [|0\rangle\langle 0| + e^{-i\theta} \cos \varphi |0\rangle\langle 1| + e^{i\theta} \cos \varphi |1\rangle\langle 0| + \cos^2 \varphi |1\rangle\langle 1| + \sin^2 \varphi |0\rangle\langle 0|]. \end{aligned} \quad (\text{B.266})$$

Computing the fidelity gives

$$\begin{aligned} F_A &= \langle \psi_\theta | \text{tr}_B(\phi) | \psi_\theta \rangle \\ &= \frac{1}{4} [\langle 0| + e^{-i\theta} \langle 1|] [|0\rangle\langle 0| + e^{-i\theta} \cos \varphi |0\rangle\langle 1| + e^{i\theta} \cos \varphi |1\rangle\langle 0| + \cos^2 \varphi |1\rangle\langle 1| + \sin^2 \varphi |0\rangle\langle 0|] [|0\rangle + e^{i\theta} |1\rangle] \\ &= \frac{1}{4} [\langle 0| + e^{-i\theta} \cos \varphi \langle 1| + \sin^2 \varphi \langle 0| + \cos \varphi \langle 0| + e^{-i\theta} \cos^2 \varphi \langle 1|] [|0\rangle + e^{i\theta} |1\rangle] \\ &= \frac{1}{4} [1 + \sin^2 \varphi + \cos \varphi + \cos \varphi + \cos^2 \varphi] = \frac{1}{2} (1 + \cos \varphi), \end{aligned} \quad (\text{B.267})$$

where we used that $\sin^2 \varphi + \cos^2 \varphi = 1$. In a similar way, the state of ρ_θ^B is obtained from tracing out the first qubit, i.e. $\rho_\theta^B = \text{tr}_A(\phi)$. In this case, computing the fidelity gives

$$F_B = \langle \psi_\theta | \text{tr}_A(\phi) | \psi_\theta \rangle = \frac{1}{2} (1 + \sin \varphi). \quad (\text{B.268})$$

To make the fidelities equal, we choose $\varphi = \frac{\pi}{4}$, which gives $F(\psi_\theta, \rho_\theta) = \frac{2+\sqrt{2}}{4}$.

B.3.10 Solution (Text at p. 146)

(a) Let us start computing density operator of the cloned state using (B.42)

$$\text{tr}_B(\rho_{\text{out}}(\hat{m})) = \frac{1}{4} \left(\underbrace{\mathbb{1} \text{tr}(\mathbb{1})}_{=2} + \eta \hat{m} \cdot \vec{\sigma} \underbrace{\text{tr}(\mathbb{1})}_{=2} + \eta \underbrace{\mathbb{1} \text{tr}(\hat{m} \cdot \vec{\sigma})}_{=0} + \sum_{i,j=X,Y,Z} T_{ij} \sigma_i \text{tr}(\sigma_j) \right) = \frac{1}{2} (\mathbb{1} + \eta \hat{m} \cdot \vec{\sigma}), \quad (\text{B.269})$$

where we have used that $\text{tr}(\sigma_i) = 0$ for $i = X, Y, Z$. Therefore, the fidelity with the input state $\rho_{\text{in}}(\hat{m})$ is

$$\begin{aligned} F &= \text{tr}(\rho_{\text{in}}(\hat{m}) \text{tr}_B(\rho_{\text{out}}(\hat{m}))) = \frac{1}{4} \text{tr}((\mathbb{1} + \hat{m} \cdot \vec{\sigma})(\mathbb{1} + \eta \hat{m} \cdot \vec{\sigma})) \\ &= \frac{1}{4} \left(\underbrace{\text{tr}(\mathbb{1})}_{=0} + \underbrace{\text{tr}(\hat{m} \cdot \vec{\sigma})}_{=0} + \eta \underbrace{\text{tr}(\hat{m} \cdot \vec{\sigma})}_{=0} + \eta \underbrace{\text{tr}((\hat{m} \cdot \vec{\sigma})^2)}_{=2} \right) = \frac{1+\eta}{2}, \end{aligned} \quad (\text{B.270})$$

where we have used (A.55) to show that $\text{tr}((\hat{m} \cdot \vec{\sigma})^2) = 2$.

(b) i. Since $V\sigma_k V^\dagger$ is an Hermitian matrix, it can be expressed in terms of the Pauli matrices as

$$V\sigma_k V^\dagger = \sum_j \gamma_{kj} \sigma_j, \quad \forall k, \quad (\text{B.271})$$

Therefore, expanding the scalar product, we get

$$\hat{m} \cdot V\vec{\sigma}V^\dagger = \sum_k m_k V\sigma_k V^\dagger = \sum_k m_k \sum_j \gamma_{kj} \sigma_j = \sum_j \left(\sum_k \gamma_{kj} m_k \right) \sigma_j = \sum_j (R_V \hat{m})_j \sigma_j = (R_V \hat{m}) \cdot \vec{\sigma}, \quad (\text{B.272})$$

where we have defined $(R_V)_{jk} = \gamma_{kj}$ the rotation operator associated to V and acting on the Bloch vector $\hat{m}$.

ii. Use that

$$U = e^{-i\alpha\sigma_Z} = \begin{pmatrix} e^{-i\alpha} & 0 \\ 0 & e^{i\alpha} \end{pmatrix} \quad (\text{B.273})$$

and compute the following matrices

$$U\sigma_X U^\dagger = \begin{pmatrix} e^{-i\alpha} & 0 \\ 0 & e^{i\alpha} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{-i\alpha} \end{pmatrix} = \begin{pmatrix} 0 & e^{-2i\alpha} \\ e^{2i\alpha} & 0 \end{pmatrix}, \quad (\text{B.274})$$

$$U\sigma_Y U^\dagger = \begin{pmatrix} e^{-i\alpha} & 0 \\ 0 & e^{i\alpha} \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{-i\alpha} \end{pmatrix} = \begin{pmatrix} 0 & -ie^{-2i\alpha} \\ ie^{2i\alpha} & 0 \end{pmatrix}, \quad (\text{B.275})$$

$$U\sigma_Z U^\dagger = \begin{pmatrix} e^{-i\alpha} & 0 \\ 0 & e^{i\alpha} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{-i\alpha} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (\text{B.276})$$

Therefore, by explicit computation, we get

$$\begin{aligned} f(X, X) + f(Y, Y) &= U\sigma_X U^\dagger \otimes U\sigma_X U^\dagger + U\sigma_Y U^\dagger \otimes U\sigma_Y U^\dagger \\ &= \begin{pmatrix} 0 & e^{-2i\alpha} \\ e^{2i\alpha} & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & e^{-2i\alpha} \\ e^{2i\alpha} & 0 \end{pmatrix} + \begin{pmatrix} 0 & -ie^{-2i\alpha} \\ ie^{2i\alpha} & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & -ie^{-2i\alpha} \\ ie^{2i\alpha} & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 & e^{-4i\alpha} \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ e^{4i\alpha} & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & -e^{-4i\alpha} \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -e^{4i\alpha} & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \end{aligned} \quad (\text{B.277})$$

which is exactly

$$\begin{aligned} \sigma_X \otimes \sigma_X + \sigma_Y \otimes \sigma_Y &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (\text{B.278})$$

In a similar way, we get

$$\begin{aligned}
 f(X, Y) - f(Y, X) &= U\sigma_X U^\dagger \otimes U\sigma_Y U^\dagger - U\sigma_Y U^\dagger \otimes U\sigma_X U^\dagger \\
 &= \begin{pmatrix} 0 & e^{-2i\alpha} \\ e^{2i\alpha} & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & -ie^{-2i\alpha} \\ ie^{2i\alpha} & 0 \end{pmatrix} - \begin{pmatrix} 0 & -ie^{-2i\alpha} \\ ie^{2i\alpha} & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & e^{-2i\alpha} \\ e^{2i\alpha} & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 & 0 & -ie^{-4i\alpha} \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ ie^{4i\alpha} & 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & -ie^{-4i\alpha} \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ ie^{4i\alpha} & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2i & 0 \\ 0 & -2i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{B.279}
 \end{aligned}$$

which is exactly

$$\begin{aligned}
 \sigma_X \otimes \sigma_Y - \sigma_Y \otimes \sigma_X &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2i & 0 \\ 0 & -2i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{B.280}
 \end{aligned}$$

The condition $f(Z, Z) = \sigma_Z \otimes \sigma_Z$ follows directly from (B.276). Therefore, the universality condition

$$\rho_{\text{out}}(R_U \hat{m}) = U \otimes U \rho_{\text{out}}(\hat{m}) U^\dagger \otimes U^\dagger \tag{B.281}$$

becomes

$$\begin{aligned}
 &\frac{1}{4} \left(\mathbb{1} \otimes \mathbb{1} + \eta((R_U \hat{m}) \cdot \vec{\sigma} \otimes \mathbb{1} + \mathbb{1} \otimes (R_U \hat{m}) \cdot \vec{\sigma}) + \sum_{i,j=X,Y,Z} T_{ij} \sigma_i \otimes \sigma_j \right) \\
 &= \frac{1}{4} \left(\mathbb{1} \otimes \mathbb{1} + \eta(\hat{m} \cdot U \vec{\sigma} U^\dagger \otimes \mathbb{1} + \mathbb{1} \otimes \hat{m} \cdot U \vec{\sigma} U^\dagger) + \sum_{i,j=X,Y,Z} T_{ij} U \sigma_i U^\dagger \otimes U \sigma_j U^\dagger \right). \tag{B.282}
 \end{aligned}$$

Using i. and ii. on (B.282), we get that $T_{XX} = T_{YY}$ and $T_{XY} = -T_{YX}$. Moreover, since from (B.276), we know that the unitary U acts non-trivially only on the XY subspace, all the off-diagonal terms mixing X (or Y) with Z must be zero, i.e. $T_{XZ} = T_{ZX} = T_{YZ} = T_{ZY} = 0$.

Then, assuming $\hat{m} = (0, 0, 1)$ and imposing all the constraints we get from the invariance under $U = e^{-i\alpha\sigma_Z}$, the density operator reads as

$$\begin{aligned}
 \rho_{\text{out}}(\hat{m}) &= \frac{1}{4} (\mathbb{1} \otimes \mathbb{1} + \eta(\sigma_Z \otimes \mathbb{1} + \mathbb{1} \otimes \sigma_Z) + T_{XX}(\sigma_X \otimes \sigma_X + \sigma_Y \otimes \sigma_Y) + T_{XY}(\sigma_X \otimes \sigma_Y - \sigma_Y \otimes \sigma_X) + T_{ZZ}\sigma_Z \otimes \sigma_Z) \\
 &= \frac{1}{4} \begin{pmatrix} 1 + 2\eta + T_{ZZ} & 0 & 0 & 0 \\ 0 & 1 - T_{ZZ} & 2T_{XX} + 2iT_{XY} & 0 \\ 0 & 2T_{XX} - 2iT_{XY} & 1 - T_{ZZ} & 0 \\ 0 & 0 & 0 & 1 - 2\eta + T_{ZZ} \end{pmatrix}. \tag{B.283}
 \end{aligned}$$

- (c) So far we have assumed $\hat{m} = (0, 0, 1)$, i.e. our density operator was expressed in the Z -basis. In order to apply the no-signaling condition, first we need to rotate the state in the X -basis. Therefore, define the unitary

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \text{such that} \quad \begin{cases} U\sigma_Z U^\dagger = \sigma_X, \\ U\sigma_X U^\dagger = \sigma_Z, \\ U\sigma_Y U^\dagger = -\sigma_Y. \end{cases} \tag{B.284}$$

Then, applying U on (B.283), we get

$$\begin{aligned}
 &U \otimes U \rho_{\text{out}}(\hat{m}) U^\dagger \otimes U^\dagger \\
 &= \frac{1}{4} (\mathbb{1} \otimes \mathbb{1} + \eta(\sigma_X \otimes \mathbb{1} + \mathbb{1} \otimes \sigma_X) + T_{XX}(\sigma_Z \otimes \sigma_Z + \sigma_Y \otimes \sigma_Y) + T_{XY}(-\sigma_Z \otimes \sigma_Y + \sigma_Y \otimes \sigma_Z) + T_{ZZ}\sigma_X \otimes \sigma_X). \tag{B.285}
 \end{aligned}$$

The no-signaling condition reads as

$$\rho_{\text{out}}(\hat{m}_Z) + \rho_{\text{out}}(-\hat{m}_Z) = \rho_{\text{out}}(\hat{m}_X) + \rho_{\text{out}}(-\hat{m}_X), \quad (\text{B.286})$$

where we have defined $\hat{m}_Z = (0, 0, 1)$ and $\hat{m}_X = (1, 0, 0)$. Using (B.283) and (B.285) in (B.286), we get

$$\frac{1}{2} \begin{pmatrix} T_{ZZ} + 1 & 0 & 0 & 0 \\ 0 & 1 - T_{ZZ} & 2T_{XX} + 2iT_{XY} & 0 \\ 0 & 2T_{XX} - 2iT_{XY} & 1 - T_{ZZ} & 0 \\ 0 & 0 & 0 & T_{ZZ} + 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} T_{XX} + 1 & iT_{XY} & -iT_{XY} & T_{ZZ} - T_{XX} \\ -iT_{XY} & 1 - T_{XX} & T_{XX} + T_{ZZ} & iT_{XY} \\ iT_{XY} & T_{XX} + T_{ZZ} & 1 - T_{XX} & -iT_{XY} \\ T_{ZZ} - T_{XX} & -iT_{XY} & iT_{XY} & T_{XX} + 1 \end{pmatrix}, \quad (\text{B.287})$$

which implies that $T_{ZZ} = T_{XX}$ and $T_{XY} = 0$.

- (d) To impose that $\rho_{\text{out}}(\hat{m})$ is positive semidefinite, we need to compute its eigenvalues. Given the constraints above, we can rewrite (B.283) imposing $T_{XX} = T_{ZZ} = t$ and $T_{XY} = 0$. Then, using any software, we get

$$\lambda_{1,2} = \frac{1}{4}(1 \pm 2\eta + t), \quad \lambda_{3,4} = \frac{1}{4}(1 - t \pm 2t). \quad (\text{B.288})$$

They must all be non-negative. We can check that this is satisfied for λ_3 and λ_4 when

$$\lambda_4 = \frac{1}{4}(1 - 3t) \geq 0 \implies t \leq \frac{1}{3}. \quad (\text{B.289})$$

Therefore,

$$\lambda_2^{\max} = \frac{1}{4}\left(1 - 2\eta + \frac{1}{3}\right) = \frac{1}{4}\left(\frac{4}{3} + 2\eta\right) \geq 0 \implies \eta \leq \frac{2}{3}. \quad (\text{B.290})$$

Then, using (B.270), we get that the maximum cloning fidelity is

$$F = \frac{1 + \eta}{2} \leq \frac{1 + \frac{2}{3}}{2} = \frac{5}{6}. \quad (\text{B.291})$$

B.4. Quantum channels and operations

B.4.1 Solution (Text at p. 147)

Any qubit state ρ in its Bloch representation is written $\rho = \frac{\mathbb{1} + \vec{n} \cdot \vec{\sigma}}{2}$. Since $\{\sigma_i, \sigma_j\} = 0$ for $i \neq j$ and $\sigma_i^2 = \mathbb{1}$ then

$$\sigma_i \sigma_j = -\sigma_j \sigma_i \implies \sigma_i \sigma_j \sigma_i = -\sigma_j, \quad \forall i \neq j. \quad (\text{B.292})$$

Using these relations we can compute the action of the channel (B.47) on the density operator ρ to be

$$\Lambda(\rho) = \frac{1}{4} \left[\frac{\mathbb{1} + n_X \sigma_X + n_Y \sigma_Y + n_Z \sigma_Z}{2} + \frac{\mathbb{1} + n_X \sigma_X - n_Y \sigma_Y - n_Z \sigma_Z}{2} + \frac{\mathbb{1} - n_X \sigma_X + n_Y \sigma_Y - n_Z \sigma_Z}{2} + \frac{\mathbb{1} - n_X \sigma_X - n_Y \sigma_Y + n_Z \sigma_Z}{2} \right] = \frac{1}{4} 4 \frac{\mathbb{1}}{2} = \frac{\mathbb{1}}{2}. \quad (\text{B.293})$$

B.4.2 Solution (Text at p. 147)

- (a) Using the definition of the Kraus representation of a channel,

$$\begin{aligned} \sum_{a,i} K_{a,i} \rho K_{a,i}^\dagger &= \sum_{a,i} (|a\rangle\langle i| \sqrt{E_a}) \rho (\sqrt{E_a} \langle i| \langle a|) = \sum_a \left(\sum_i \langle i| \sqrt{E_a} \rho \sqrt{E_a} |i\rangle \right) |a\rangle\langle a| \\ &= \sum_a \text{tr}(\sqrt{E_a} \rho \sqrt{E_a}) |a\rangle\langle a| = \sum_a \text{tr}(\rho E_a) |a\rangle\langle a|, \end{aligned} \quad (\text{B.294})$$

where we used the definition of the trace and its cyclicity. Moreover,

$$\sum_{a,i} K_{a,i}^\dagger K_{a,i} = \sum_{a,i} (\sqrt{E_a} \langle i| \langle a|) (|a\rangle\langle i| \sqrt{E_a}) = \sum_a \sqrt{E_a} \underbrace{\sum_i |i\rangle\langle i|}_{=\mathbb{1}} \sqrt{E_a} = \sum_a E_a = \mathbb{1}. \quad (\text{B.295})$$

(b) Using the definition of the Kraus representation of a channel

$$\sum_i (K_i \otimes \mathbb{1}) \rho_{AB} (K_i^\dagger \otimes \mathbb{1}) = \sum_i (\langle i|_A \otimes \mathbb{1}_B) \rho_{AB} (|i\rangle_A \otimes \mathbb{1}_B) = \text{tr}_A(\rho_{AB}). \quad (\text{B.296})$$

Moreover, $\sum_i K_i^\dagger K_i = \sum_i |i\rangle\langle i|_A = \mathbb{1}_A$ holds from the completeness of the basis $\{|i\rangle_A\}_i$.

(c) Using the definition of the Kraus representation of a channel

$$\sum_i K_i \rho_A K_i^\dagger = \sum_i (\mathbb{1}_A \otimes \sqrt{\lambda_i} |\lambda_i\rangle_B) \rho_A (\mathbb{1}_A \otimes \sqrt{\lambda_i} \langle \lambda_i|_B) = \rho_A \otimes \sum_i \lambda_i |\lambda_i\rangle\langle \lambda_i|_B = \rho_A \otimes \sigma_B, \quad (\text{B.297})$$

where we used the spectral decomposition of $\sigma_B = \sum_i \lambda_i |\lambda_i\rangle\langle \lambda_i|_B$. Moreover,

$$\sum_i K_i^\dagger K_i = \sum_i (\mathbb{1}_A \otimes \sqrt{\lambda_i} \langle \lambda_i|_B) (\mathbb{1}_A \otimes \sqrt{\lambda_i} |\lambda_i\rangle_B) = \mathbb{1}_A \otimes \sum_i \lambda_i \underbrace{\langle \lambda_i|_B |\lambda_i\rangle_B}_{=1} = \mathbb{1}_A, \quad (\text{B.298})$$

where we used that $\sum_i \lambda_i = 1$, since σ_B is a state and therefore it is normalised.

(d) Using the definition of the Kraus representation of a channel

$$\sum_a K_a \rho K_a^\dagger = \sum_a |\psi_a\rangle\langle \phi_a| \rho |\phi_a\rangle\langle \psi_a| = \sum_a \langle \phi_a| \rho |\phi_a\rangle \underbrace{|\psi_a\rangle\langle \psi_a|}_{=\sigma_a} = \sum_a \langle \phi_a| \rho |\phi_a\rangle \sigma_a. \quad (\text{B.299})$$

Moreover, $\sum_a K_a^\dagger K_a = \sum_a |\phi_a\rangle\langle \psi_a| \langle \psi_a| \langle \phi_a| = \sum_a |\phi_a\rangle\langle \phi_a| = \mathbb{1}$ since $\{|\phi_a\rangle\}$ is a basis.

B.4.3 Solution (Text at p. 148)

(a) The action of the Kraus operators is

$$\sum_i K_i \rho K_i^\dagger = K_1 \rho K_1^\dagger + K_2 \rho K_2^\dagger = \sqrt{v} \rho \sqrt{v} + \sqrt{1-v} \sigma_Z \rho \sqrt{1-v} \sigma_Z^\dagger = v \rho + (1-v) \sigma_Z \rho \sigma_Z, \quad (\text{B.300})$$

where we used that $\sigma_Z^\dagger = \sigma_Z$. We can also check that

$$\sum_i K_i^\dagger K_i = K_1^\dagger K_1 + K_2^\dagger K_2 = v \mathbb{1} + (1-v) \underbrace{\sigma_Z^2}_{=\mathbb{1}} = \mathbb{1}. \quad (\text{B.301})$$

(b) Consider the Bloch representation of a qubit density operator,

$$\rho = \frac{\mathbb{1} + \vec{n} \cdot \vec{\sigma}}{2}, \quad (\text{B.302})$$

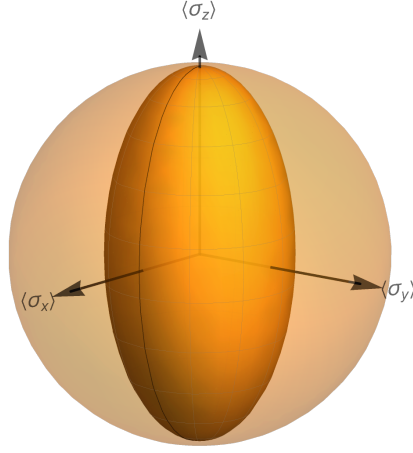
where $\vec{\sigma} = \{\sigma_X, \sigma_Y, \sigma_Z\}$ are the Pauli matrices. Applying the dephasing channel on this state we get

$$\begin{aligned} \Lambda(\rho) &= v \left(\frac{\mathbb{1} + \vec{n} \cdot \vec{\sigma}}{2} \right) + (1-v) \sigma_Z \left(\frac{\mathbb{1} + \vec{n} \cdot \vec{\sigma}}{2} \right) \sigma_Z \\ &= v \left(\frac{\mathbb{1}}{2} + \frac{n_X \sigma_X + n_Y \sigma_Y + n_Z \sigma_Z}{2} \right) + (1-v) \sigma_Z \left(\frac{\mathbb{1}}{2} + \frac{n_X \sigma_X + n_Y \sigma_Y + n_Z \sigma_Z}{2} \right) \sigma_Z \\ &= v \left(\frac{\mathbb{1}}{2} + \frac{n_X \sigma_X + n_Y \sigma_Y + n_Z \sigma_Z}{2} \right) + (1-v) \left(\frac{\mathbb{1}}{2} + \frac{-n_X \sigma_X - n_Y \sigma_Y + n_Z \sigma_Z}{2} \right), \end{aligned} \quad (\text{B.303})$$

where we used that $\sigma_Z \sigma_X \sigma_Z = -\sigma_X$, $\sigma_Z \sigma_Y \sigma_Z = -\sigma_Y$ and $\sigma_Z \sigma_Z \sigma_Z = \sigma_Z$. Therefore, we get

$$\Lambda(\rho) = \frac{\mathbb{1}}{2} + \frac{(2v-1)n_X \sigma_X + (2v-1)n_Y \sigma_Y + n_Z \sigma_Z}{2} = \frac{\mathbb{1} + \vec{n}' \cdot \vec{\sigma}}{2}, \quad (\text{B.304})$$

where we defined the modified Bloch vector $\vec{n}' = \{(2v-1)n_X, (2v-1)n_Y, n_Z\}$. We see that the dephasing channel shrinks the components n_X and n_Y of the Bloch vector by a factor $(2v-1)$, as illustrated in the figure.



B.4.4 Solution (Text at p. 148)

(a) Compute the action of the depolarisation channel on the following basis of 2×2 matrices,

$$\begin{aligned} \Lambda(|0\rangle\langle 0|) &= v|0\rangle\langle 0| + (1-v) \underbrace{\text{tr}(|0\rangle\langle 0|)}_{=1} \frac{\mathbb{1}}{2}, & \Lambda(|0\rangle\langle 1|) &= v|0\rangle\langle 1| + (1-v) \underbrace{\text{tr}(|0\rangle\langle 1|)}_{=0} \frac{\mathbb{1}}{2}, \\ \Lambda(|1\rangle\langle 0|) &= v|1\rangle\langle 0| + (1-v) \underbrace{\text{tr}(|1\rangle\langle 0|)}_{=0} \frac{\mathbb{1}}{2}, & \Lambda(|1\rangle\langle 1|) &= v|1\rangle\langle 1| + (1-v) \underbrace{\text{tr}(|1\rangle\langle 1|)}_{=1} \frac{\mathbb{1}}{2}. \end{aligned} \quad (\text{B.305})$$

To obtain the Choi representation of Λ , we send half of the maximally entangled state through Λ and evaluate the post-transformation state. Write the maximally entangled state as

$$\phi^+ = \frac{1}{2} [|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|], \quad (\text{B.306})$$

Using (B.305), the Choi representation becomes

$$\begin{aligned} \eta = (\Lambda \otimes \mathbb{1})[\phi^+] &= \frac{1}{2} \left[\left(v|0\rangle\langle 0| + \frac{1-v}{2} \mathbb{1} \right) \otimes |0\rangle\langle 0| + v|0\rangle\langle 1| \otimes |0\rangle\langle 1| \right. \\ &\quad \left. + v|1\rangle\langle 0| \otimes |1\rangle\langle 0| + \left(v|1\rangle\langle 1| + \frac{1-v}{2} \mathbb{1} \right) \otimes |1\rangle\langle 1| \right] = \frac{1}{2} \begin{pmatrix} \frac{1+v}{2} & 0 & 0 & v \\ 0 & \frac{1-v}{2} & 0 & 0 \\ 0 & 0 & \frac{1-v}{2} & 0 \\ v & 0 & 0 & \frac{1+v}{2} \end{pmatrix}. \end{aligned} \quad (\text{B.307})$$

(b) In order to find the Kraus operators K_i , we can compute the spectral decomposition of the Choi state associated to the channel, i.e.

$$\eta = (\Lambda \otimes \mathbb{1})[\phi^+] = \sum_{i=1}^d (K_i \otimes \mathbb{1}) \phi^+ (K_i^\dagger \otimes \mathbb{1}) = \sum_{i=1}^d \lambda_i |\lambda_i\rangle \langle \lambda_i|, \quad (\text{B.308})$$

from which we find that $(K_i \otimes \mathbb{1})|\phi^+\rangle = \sqrt{\lambda_i} |\lambda_i\rangle$. Diagonalising the Choi state found in (B.307), we find

$$\lambda_i(\eta) = \left\{ \frac{3v+1}{4}, \frac{1-v}{4}, \frac{1-v}{4}, \frac{1-v}{4} \right\}, \quad |\lambda_i\rangle = \{ |\phi^+\rangle, |\psi^+\rangle, |\phi^-\rangle, |\psi^-\rangle \}, \quad (\text{B.309})$$

where $\{ |\psi^+\rangle, |\psi^-\rangle, |\phi^-\rangle, |\phi^+\rangle \}$ are the four Bell states. It is possible to rewrite these states in terms of $|\phi^+\rangle$, using the Pauli matrices,

$$\begin{aligned} (\sigma_X \otimes \mathbb{1})|\phi^+\rangle &= |\psi^+\rangle, \\ (\sigma_Z \otimes \mathbb{1})|\phi^+\rangle &= |\phi^-\rangle, \\ (i\sigma_Y \otimes \mathbb{1})|\phi^+\rangle &= |\psi^-\rangle. \end{aligned} \quad (\text{B.310})$$

Therefore, we can write

$$\begin{aligned}
\sqrt{\lambda_1}|\lambda_1\rangle &= \sqrt{\frac{3v+1}{4}}|\phi^+\rangle = \sqrt{\frac{3v+1}{4}}(\mathbb{1} \otimes \mathbb{1})|\phi^+\rangle = (K_1 \otimes \mathbb{1})|\phi^+\rangle, \\
\sqrt{\lambda_2}|\lambda_2\rangle &= \sqrt{\frac{1-v}{4}}|\psi^+\rangle = \sqrt{\frac{1-v}{4}}(\sigma_X \otimes \mathbb{1})|\phi^+\rangle = (K_2 \otimes \mathbb{1})|\phi^+\rangle, \\
\sqrt{\lambda_3}|\lambda_3\rangle &= \sqrt{\frac{1-v}{4}}|\phi^-\rangle = \sqrt{\frac{1-v}{4}}(\sigma_Z \otimes \mathbb{1})|\phi^+\rangle = (K_3 \otimes \mathbb{1})|\phi^+\rangle, \\
\sqrt{\lambda_4}|\lambda_4\rangle &= \sqrt{\frac{1-v}{4}}|\psi^-\rangle = \sqrt{\frac{1-v}{4}}(i\sigma_Y \otimes \mathbb{1})|\phi^+\rangle = (K_4 \otimes \mathbb{1})|\phi^+\rangle,
\end{aligned} \tag{B.311}$$

from which the Kraus operators are

$$\begin{aligned}
K_1 &= \sqrt{\frac{3v+1}{4}}\mathbb{1}, & K_2 &= \sqrt{\frac{1-v}{4}}\sigma_X, \\
K_3 &= \sqrt{\frac{1-v}{4}}\sigma_Z, & K_4 &= \sqrt{\frac{1-v}{4}}i\sigma_Y.
\end{aligned} \tag{B.312}$$

We can that these are valid Kraus operators because

$$\sum_{i=1}^4 K_i K_i^\dagger = \frac{3v+1}{4}\mathbb{1} + \frac{1-v}{4} \underbrace{\sigma_X^2}_{=\mathbb{1}} + \frac{1-v}{v} \underbrace{\sigma_Z^2}_{=\mathbb{1}} + \frac{1-v}{4} \underbrace{i(-i)\sigma_Y^2}_{=\mathbb{1}} = \mathbb{1}. \tag{B.313}$$

- (c) The dual channel can be obtained directly from the Kraus decomposition, $\Lambda^\dagger(O) = \sum_i K_i^\dagger O K_i$, with the operators defined in (B.312), which satisfy $K_1^\dagger = K_1$, $K_2^\dagger = K_2$, $K_3^\dagger = K_3$, and $K_4^\dagger = -K_4$. Inserted above, we see that the dual of the depolarisation channel is the depolarisation channel itself. Another way of finding this result is by using directly the definition of channel duality

$$\text{tr}(O\Lambda(\rho)) = \text{tr}\left(O\left(v\rho + \frac{1-v}{d}\text{tr}(\rho)\mathbb{1}\right)\right) = \text{tr}\left(\left(vO + \frac{1-v}{d}\text{tr}(O)\mathbb{1}\right)\rho\right), \tag{B.314}$$

where we used that $\text{tr}(O\text{tr}(\rho)) = \text{tr}(\text{tr}(O)\rho)$.

B.4.5 Solution (Text at p. 148)

The action of a channel Λ and its dual $\Lambda^\dagger$ can be written as

$$\Lambda_{A \rightarrow A'}(X) = d_A \text{tr}_A((\mathbb{1}_{A'} \otimes X_A^T)\eta_{A'A}), \quad \text{and} \quad \Lambda_{A' \rightarrow A}^\dagger(Y) = d_{A'} \text{tr}_{A'}((\mathbb{1}_A \otimes Y_{A'}^T)\gamma_{AA'}), \tag{B.315}$$

for any operator $X \in \mathcal{H}_A$ and $Y \in \mathcal{H}_{A'}$. The definition of dual channel $\text{tr}(\Lambda(X)Y) = \text{tr}(\Lambda^\dagger(Y)X)$ becomes

$$d_A \text{tr}_{A'}(\text{tr}_A((\mathbb{1}_{A'} \otimes X_A^T)\eta_{A'A})Y_{A'}) = d_{A'} \text{tr}_A(\text{tr}_{A'}((\mathbb{1}_A \otimes Y_{A'}^T)\gamma_{AA'})X_A). \tag{B.316}$$

Using that, given two operators $O_A \in \mathcal{H}_A$ and $S_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$, $\text{tr}(O_A \text{tr}_B(S_{AB})) = \text{tr}((O_A \otimes \mathbb{1}_B)S_{AB})$, we can rewrite (B.316) as

$$d_A \text{tr}((Y_{A'} \otimes X_A^T)\eta_{A'A}) = d_{A'} \text{tr}((X_A \otimes Y_{A'}^T)\gamma_{AA'}) = d_{A'} \text{tr}((X_A^T \otimes Y_{A'})\gamma_{AA'}^T), \tag{B.317}$$

where we have used that $\text{tr}(O) = \text{tr}(O^T)$ for any operator O (see Exercise A.1.1) and that $(X_A \otimes Y_{A'}^T)^{T_A T_{A'}} = (X_A^T \otimes Y_{A'})$. Define the swap operator P such that $P|i\rangle \otimes |j\rangle = |j\rangle \otimes |i\rangle$, we can write

$$d_A \text{tr}((Y_{A'} \otimes X_A^T)\eta_{A'A}) = d_{A'} \text{tr}(P(Y_{A'} \otimes X_A^T)P^\dagger \gamma_{AA'}^T) = d_{A'} \text{tr}((Y_{A'} \otimes X_A^T)P^\dagger \gamma_{AA'}^T P). \tag{B.318}$$

From the last equality we get

$$\eta_{A'A} = \frac{d_{A'}}{d_A} P^\dagger \gamma_{AA'}^T P. \tag{B.319}$$

B.4.6 Solution (Text at p. 148)

- (a) Using the expression (B.55) for the channel $\Lambda_{A \rightarrow A'}(\rho)$, we can write

$$(\Lambda_{A \rightarrow A'} \otimes \mathbb{1}_B)[\rho_{AB}] = \sum_i \sigma_{A'}^i \otimes \text{tr}_A((E_A^i \otimes \mathbb{1}_B)\rho_{AB}) = \sum_i q_i \sigma_{A'}^i \otimes \varphi_B^i, \quad (\text{B.320})$$

where we have defined $\tilde{\varphi}_B^i = \text{tr}_A((E_A^i \otimes \mathbb{1}_B)\rho_{AB})$, $\varphi_B^i = \frac{\tilde{\varphi}_B^i}{\text{tr}(\tilde{\varphi}_B^i)}$ and $q_i = \text{tr}(\tilde{\varphi}_B^i)$. Indeed, using that $\{E_A^i\}_i$ is a POVM, we get

$$q_i \geq 0, \quad \forall i \quad \text{and} \quad \sum_i q_i = \sum_i \text{tr}_B \text{tr}_A((E_A^i \otimes \mathbb{1}_B)\rho_{AB}) = \text{tr}\left(\left(\sum_i E_A^i \otimes \mathbb{1}_B\right)\rho_{AB}\right) = 1, \quad (\text{B.321})$$

which means that $\{q_i\}_i$ is a probability distribution. Therefore, the output state (B.320) is separable for any possible ρ_{AB} , i.e. Λ is entanglement breaking.

- (b) Computing the Choi state of the measure-and-prepare channel gives

$$\eta = \frac{1}{d} \sum_{k,l} \Lambda[|k\rangle\langle l|] \otimes |k\rangle\langle l| = \frac{1}{d} \sum_{k,l} \text{tr}(|k\rangle\langle l| E_i) \sigma_i \otimes |k\rangle\langle l| = \frac{1}{d} \sum_i \sigma_i \otimes \left(\sum_{k,l} \langle l| E_i |k\rangle |k\rangle\langle l| \right), \quad (\text{B.322})$$

where we have used that $\text{tr}(|k\rangle\langle l| E_i) = \langle l| E_i |k\rangle$ is a scalar. Notice that $\sum_{k,l} \langle l| E_i |k\rangle |k\rangle\langle l| = E_i^T$, from which

$$\eta = \frac{1}{d} \sum_i \sigma_i \otimes E_i^T, \quad (\text{B.323})$$

which is a separable operator.

B.4.7 Solution (Text at p. 149)

- (a) The Choi state associated to Λ is defined as $\eta = (\Lambda \otimes \mathbb{1})[\phi^+]$. From the definition (B.54) of entanglement-breaking channel, the state η is immediately separable.
- (b) If the Choi state $\eta_{A'A}$ is separable, it can be written as

$$\eta_{A'A} = \sum_i p_i \psi^i \otimes \phi^i. \quad (\text{B.324})$$

Therefore, using that the action of a channel $\Lambda_{A \rightarrow A'}$ from $\mathcal{H}_A$ to $\mathcal{H}_{A'}$ can be written as

$$\Lambda_{A \rightarrow A'}(\rho) = d_A \text{tr}_A((\mathbb{1}_{A'} \otimes \rho_A^T) \eta_{A'A}), \quad (\text{B.325})$$

we get

$$\begin{aligned} (\Lambda_{A \rightarrow A'} \otimes \mathbb{1}_B)[\rho_{AB}] &= d_A \text{tr}_A((\mathbb{1}_{A'} \otimes \rho_{AB}^T) (\eta_{A'A} \otimes \mathbb{1}_B)) \\ &= d_A \sum_i p_i \text{tr}_A((\mathbb{1}_{A'} \otimes \rho_{AB}^T) (\psi_{A'}^i \otimes \phi_A^i \otimes \mathbb{1}_B)) \\ &= d_A \sum_i p_i \psi_{A'}^i \otimes \text{tr}_A(\rho_{AB}^T (\phi_A^i \otimes \mathbb{1}_B)), \end{aligned} \quad (\text{B.326})$$

which is a separable state over $A'B$. We conclude that a channel is entanglement-breaking if and only if it has a separable Choi state.

B.4.8 Solution (Text at p. 149)

- (a) Define the following polynomial in λ

$$f(\lambda) = \text{tr}((A - \lambda B)^\dagger (A - \lambda B)), \quad (\text{B.327})$$

By expanding it, we find

$$\begin{aligned} f(\lambda) &= \text{tr}((A - \lambda B)^\dagger (A - \lambda B)) \\ &= |\lambda|^2 \text{tr}(B^\dagger B) - \lambda \text{tr}(A^\dagger B) - \lambda^* \text{tr}(B^\dagger A) + \text{tr}(A^\dagger A) \\ &= |\lambda|^2 \text{tr}(B^\dagger B) - \lambda \text{tr}(A^\dagger B) - (\lambda \text{tr}(A^\dagger B))^* + \text{tr}(A^\dagger A), \end{aligned} \quad (\text{B.328})$$

where we have used that $B^\dagger A = (A^\dagger B)^\dagger$ and $\text{tr}(B^\dagger A) = \text{tr}((A^\dagger B)^T)^* = \text{tr}(A^\dagger B)^*$ (see Exercise A.1.1). Since, by definition in (B.327), $f(\lambda)$ is a non-negative quadratic polynomial in λ , then $f(\lambda) \geq 0$ for any λ . Now we can find the minimum of $f(\lambda)$ computing the derivatives with respect to λ and λ^* ,

$$\begin{aligned} \frac{\partial}{\partial \lambda} f(\lambda) &= \lambda^* \text{tr}(B^\dagger B) - \text{tr}(A^\dagger B) = 0 & \implies & \lambda_{\min}^* = \frac{\text{tr}(A^\dagger B)}{\text{tr}(B^\dagger B)}, \\ \frac{\partial}{\partial \lambda^*} f(\lambda) &= \lambda \text{tr}(B^\dagger B) - \text{tr}(A^\dagger B)^* = 0 & \implies & \lambda_{\min} = \frac{\text{tr}(A^\dagger B)^*}{\text{tr}(B^\dagger B)}. \end{aligned} \quad (\text{B.329})$$

from which

$$f(\lambda_{\min}) = \frac{|\text{tr}(A^\dagger B)|^2}{\text{tr}(B^\dagger B)^2} \text{tr}(B^\dagger B) - \frac{\text{tr}(A^\dagger B)^*}{\text{tr}(B^\dagger B)} \text{tr}(A^\dagger B) - \frac{\text{tr}(A^\dagger B)}{\text{tr}(B^\dagger B)} \text{tr}(A^\dagger B)^* + \text{tr}(A^\dagger A) \geq 0. \quad (\text{B.330})$$

By simplifying this expression, we get

$$|\text{tr}(A^\dagger B)|^2 \leq \text{tr}(A^\dagger A) \text{tr}(B^\dagger B). \quad (\text{B.331})$$

(b) From the triangle inequality, $\|A + B\|_1 \leq \|A\|_1 + \|B\|_1$, we get

$$\begin{aligned} \|\rho - \rho_E\|_1 &= \|\rho - \sqrt{E}\rho\sqrt{E}\|_1 = \|(\mathbb{1} - \sqrt{E} + \sqrt{E})\rho - \sqrt{E}\rho\sqrt{E}\|_1 \\ &\leq \|(\mathbb{1} - \sqrt{E})\rho\|_1 + \|\sqrt{E}\rho(\mathbb{1} - \sqrt{E})\|_1 \\ &= \text{tr} \left| (\mathbb{1} - \sqrt{E})\sqrt{\rho}\sqrt{\rho} \right| + \text{tr} \left| \sqrt{E}\sqrt{\rho}\sqrt{\rho}(\mathbb{1} - \sqrt{E}) \right| \\ &= \text{tr} \left((\mathbb{1} - \sqrt{E})\sqrt{\rho}\sqrt{\rho}U^\dagger \right) + \text{tr} \left(\sqrt{E}\sqrt{\rho}\sqrt{\rho}(\mathbb{1} - \sqrt{E})U^\dagger \right), \end{aligned} \quad (\text{B.332})$$

where in the third line we have used that $\|X\|_1 = \text{tr} \sqrt{X^\dagger X} = \text{tr} |X|$ and that $\rho = \sqrt{\rho}\sqrt{\rho}$ since $\rho \geq 0$. The last equality follows from the polar decomposition $X = |X|U$ for some unitary U . Defining $A = (\mathbb{1} - \sqrt{E})\sqrt{\rho}$ and $B = \sqrt{\rho}U^\dagger$ for the first term, and $A = \sqrt{E}\sqrt{\rho}$ and $B = \sqrt{\rho}(\mathbb{1} - \sqrt{E})U^\dagger$ for the second term, we can use the Cauchy-Schwarz inequality (B.331) to get

$$\begin{aligned} \|\rho - \rho_E\|_1 &\leq \sqrt{\text{tr}((\mathbb{1} - \sqrt{E})\sqrt{\rho}\sqrt{\rho}(\mathbb{1} - \sqrt{E})) \text{tr}(U\rho U^\dagger)} + \sqrt{\text{tr}(\sqrt{E}\rho\sqrt{E}) \text{tr}(U\sqrt{\rho}(\mathbb{1} - \sqrt{E})^2\sqrt{\rho}U^\dagger)} \\ &= \sqrt{\text{tr}((\mathbb{1} - \sqrt{E})\sqrt{\rho}\sqrt{\rho}(\mathbb{1} - \sqrt{E})) \text{tr}(\rho)} + \sqrt{\text{tr}(\sqrt{E}\rho\sqrt{E}) \text{tr}(\sqrt{\rho}(\mathbb{1} - \sqrt{E})^2\sqrt{\rho})} \\ &= \sqrt{\text{tr}((\mathbb{1} - \sqrt{E})^2\rho)} + \sqrt{\text{tr}(E\rho) \text{tr}((\mathbb{1} - \sqrt{E})^2\rho)} \\ &\leq \sqrt{\text{tr}((\mathbb{1} - E)\rho)} + \sqrt{\text{tr}((\mathbb{1} - E)\rho)} = 2\sqrt{\text{tr}((\mathbb{1} - E)\rho)} \leq 2\sqrt{1 - 1 + \epsilon} = 2\sqrt{\epsilon}, \end{aligned} \quad (\text{B.333})$$

where we have used the cyclicity of the trace and that $U^\dagger U = \mathbb{1}$ in the first two lines. The third equality follows again from the cyclicity of the trace and from $\text{tr}(\rho) = 1$. In the last line, we have used that $\text{tr}(E\rho) \leq 1$ and $(1 - \sqrt{x})^2 \leq 1 - x$ for $x \in [0, 1]$. In the final inequality, we have used that $\text{tr}(E\rho) \geq 1 - \epsilon$.

B.4.9 Solution (Text at p. 149)

(a) The success probability of discriminating between two randomly drawn states $\{\rho_0, \rho_1\}$ is

$$p_{\text{win}} = \frac{1}{2} (\text{tr}(E_0\rho_0) + \text{tr}(E_1\rho_1)) = \frac{1}{2} + \frac{1}{2} \text{tr}((\rho_0 - \rho_1)E_0), \quad (\text{B.334})$$

where we used that POVMs satisfy $E_0 + E_1 = \mathbb{1}$. Each element of the POVM can be associated with a quantum instrument $\{\Lambda_0, \Lambda_1\}$ such that $E_0 = \Lambda_0^\dagger(\mathbb{1})$ and $E_1 = \Lambda_1^\dagger(\mathbb{1})$. Then, in terms of the instruments the probability is

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{2} \text{tr}((\rho_0 - \rho_1)\Lambda_0^\dagger(\mathbb{1})) = \frac{1}{2} + \frac{1}{2} \text{tr}(\Lambda_0(\rho_0 - \rho_1)), \quad (\text{B.335})$$

where we used the definition of dual channel. Recalling that $\Lambda_a^{A \rightarrow A'}(\psi) = 2 \text{tr}_A(\mathbb{1}_{A'} \otimes \psi_A^T \eta_a^{A'A})$, where $\{\eta_a^{A'A}\}_a$ are the Choi operators associated to the instruments, we get

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{2} \text{tr}_{A'} 2 \text{tr}_A((\mathbb{1}_{A'} \otimes (\rho_0 - \rho_1)_A) \eta_0^{A'A}), \quad (\text{B.336})$$

where we used that $(\rho_0 - \rho_1)^T = \rho_0 - \rho_1$ for our specific states. Given $O_A \in \mathcal{H}_A$ and $S_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ we have

$$\text{tr}(O_A \text{tr}_B(S_{AB})) = \text{tr}((O_A \otimes \mathbb{1}_B)S_{AB}), \quad (\text{B.337})$$

which gives

$$p_{\text{win}} = \frac{1}{2} + \text{tr}_A((\rho_0 - \rho_1) \text{tr}_{A'}(\eta_0^{A'A})), \quad (\text{B.338})$$

where we can define $\eta_0^A = \text{tr}_{A'}(\eta_0^{A'A})$, to get the result in (B.58).

(b) The average fidelity of the post-measurement state is

$$F = \frac{1}{2} \sum_{a=0,1} [\langle 0 | \Lambda_a(\rho_0) | 0 \rangle + \langle + | \Lambda_a(\rho_1) | + \rangle]. \quad (\text{B.339})$$

Using that $\sum_a \Lambda_a(\psi) = 2 \text{tr}_A((\mathbb{1}_{A'} \otimes \psi_A^T) \eta^{A'A})$ and recalling that $\rho_0^T = \rho_0$ and $\rho_1^T = \rho_1$, we get

$$\begin{aligned} F &= \frac{1}{2} \left[\langle 0 | 2 \text{tr}_A((\mathbb{1}_{A'} \otimes \rho_0) \eta^{A'A}) | 0 \rangle + \langle + | 2 \text{tr}_A((\mathbb{1}_{A'} \otimes \rho_1) \eta^{A'A}) | + \rangle \right] \\ &= \text{tr}_{A'}(|0\rangle\langle 0| \text{tr}_A((\mathbb{1}_{A'} \otimes \rho_0) \eta^{A'A})) + \text{tr}_{A'}(|+\rangle\langle +| \text{tr}_A((\mathbb{1}_{A'} \otimes \rho_1) \eta^{A'A})) \\ &= \text{tr}(|0\rangle\langle 0| \otimes \mathbb{1}_A)(\mathbb{1}_{A'} \otimes \rho_0) \eta^{A'A} + \text{tr}(|+\rangle\langle +| \otimes \mathbb{1}_A)(\mathbb{1}_{A'} \otimes \rho_1) \eta^{A'A}, \end{aligned} \quad (\text{B.340})$$

where we used the property (B.337). Recalling that $\rho_0 = |0\rangle\langle 0|$ and $\rho_1 = |+\rangle\langle +|$, we get

$$F = \text{tr}(|00\rangle\langle 00| + |++\rangle\langle ++|) \eta^{A'A}. \quad (\text{B.341})$$

B.4.10 Solution (Text at p. 149)

(a) Let us use that $\Lambda_a^{A \rightarrow A'}(\psi) = 2 \text{tr}_A((\mathbb{1}_{A'} \otimes \psi_A^T) \eta_a^{A'A})$. Therefore,

$$\begin{aligned} \int_{\text{N}} d\psi_{\text{N}} \text{tr}(\Lambda_{\text{N}}(\psi)) &= 2 \int_{\text{N}} d\psi_{\text{N}} \text{tr}(\text{tr}_A((\mathbb{1}_{A'} \otimes \psi_A^T) \eta_{\text{N}}^{A'A})) \\ &= 2 \int_{\text{N}} d\psi_{\text{N}} \text{tr}(\psi_A^T \text{tr}_{A'}(\eta_{\text{N}}^{A'A})) = 2 \int_{\text{N}} d\psi_{\text{N}} \text{tr}(\psi_A^T \eta_{\text{N}}^A), \end{aligned} \quad (\text{B.342})$$

where in the last step we have used the property (B.337) and we have defined $\eta_{\text{N}}^A = \text{tr}_{A'}(\eta_{\text{N}}^{A'A})$. Similarly,

$$\int_{\text{S}} d\psi_{\text{S}} \text{tr}(\Lambda_{\text{S}}(\psi)) = 2 \int_{\text{S}} d\psi_{\text{S}} \text{tr}(\psi_A^T \text{tr}_{A'}(\eta - \eta_{\text{N}}^{A'A})) = 2 \int_{\text{S}} d\psi_{\text{S}} \text{tr}\left(\psi_A^T \left(\frac{1}{2} \mathbb{1} - \eta_{\text{N}}^A\right)\right), \quad (\text{B.343})$$

where we have used that $\eta^{A'A} = \eta_{\text{N}}^{A'A} + \eta_{\text{S}}^{A'A}$ and $\text{tr}_{A'}(\eta) = \frac{1}{2} \mathbb{1}$. The average success probability becomes

$$p_{\text{win}} = \int_{\text{N}} d\psi_{\text{N}} \text{tr}(\psi_A^T \eta_{\text{N}}^A) + \int_{\text{S}} d\psi_{\text{S}} \text{tr}\left(\psi_A^T \left(\frac{1}{2} \mathbb{1} - \eta_{\text{N}}^A\right)\right) = \underbrace{\frac{1}{2} \int_{\text{S}} d\psi_{\text{S}}}_{=1} + \int_{\text{N}} d\psi_{\text{N}} \text{tr}((\psi_A^T - (\sigma_X \psi_A^T \sigma_X)^*) \eta_{\text{N}}^A), \quad (\text{B.344})$$

where we have used that $\int_{\text{S}} d\psi_{\text{S}} \psi = \int_{\text{N}} d\psi_{\text{N}} (\sigma_X \psi \sigma_X)^*$ and that the density over the northern (southern) hemisphere is $d\psi_{\text{N}} = d\psi_{\text{S}} = \sin \theta d\theta d\varphi / 2\pi$. Now we can use the Bloch representation of a generic qubit state, $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{-i\varphi} \sin \frac{\theta}{2} |1\rangle$ to compute the integral over the N hemisphere,

$$\begin{aligned} \int_{\text{N}} d\psi_{\text{N}} \psi^T &= \frac{1}{2\pi} \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \sin \theta \psi^T \\ &= \frac{1}{2\pi} \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \sin \theta \left[\cos^2 \frac{\theta}{2} |0\rangle\langle 0| + \sin^2 \frac{\theta}{2} |1\rangle\langle 1| + e^{-i\varphi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} |0\rangle\langle 1| + e^{i\varphi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} |1\rangle\langle 0| \right] \\ &= \frac{1}{2\pi} [I_1 |0\rangle\langle 0| + I_2 |1\rangle\langle 1| + I_3 |0\rangle\langle 1| + I_4 |1\rangle\langle 0|], \end{aligned} \quad (\text{B.345})$$

where for simplicity we have divided the integral into four terms. We can realise that $I_3 + I_4 = 0$, since

$$I_3 + I_4 \propto \int_0^{2\pi} d\varphi (e^{-\varphi} + e^{i\varphi}) = \int_0^{2\pi} d\varphi \cos \varphi = 0. \quad (\text{B.346})$$

For I_1 and I_2 we can use the trigonometric identities $\cos^2 \frac{\theta}{2} = \left(\frac{1+\cos \theta}{2}\right)$ and $\sin^2 \frac{\theta}{2} = \left(\frac{1-\cos \theta}{2}\right)$. Therefore, we get

$$\begin{aligned} I_1 &= \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \sin \theta \cos^2 \frac{\theta}{2} = \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \sin \theta \left(\frac{1+\cos \theta}{2}\right) \\ &= 2\pi \left[\frac{1}{2} \int_0^{\pi/2} d\theta \sin \theta + \frac{1}{2} \int_0^{\pi/2} d\theta \sin \theta \cos \theta \right] = \pi \left[[-\cos \theta]_0^{\pi/2} + \left[\frac{\sin^2 \theta}{2} \right]_0^{\pi/2} \right] = \frac{3}{2}\pi. \end{aligned} \quad (\text{B.347})$$

Similarly,

$$\begin{aligned} I_2 &= \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \sin \theta \sin^2 \frac{\theta}{2} = \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \sin \theta \left(\frac{1-\cos \theta}{2}\right) \\ &= 2\pi \left[\frac{1}{2} \int_0^{\pi/2} d\theta \sin \theta - \frac{1}{2} \int_0^{\pi/2} d\theta \sin \theta \cos \theta \right] = \pi \left[[-\cos \theta]_0^{\pi/2} - \left[\frac{\sin^2 \theta}{2} \right]_0^{\pi/2} \right] = \frac{\pi}{2}. \end{aligned} \quad (\text{B.348})$$

Inserting I_1 and I_2 into (B.345) we get

$$\int_{\mathbf{N}} d\psi_{\mathbf{N}} \psi^T = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1| = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{4}\mathbb{1}, \quad \text{and} \quad \int_{\mathbf{N}} d\psi_{\mathbf{N}} (\sigma_X \psi^T \sigma_X)^* = \frac{1}{2}|1\rangle\langle 1| + \frac{1}{4}\mathbb{1}. \quad (\text{B.349})$$

Therefore, the average success probability (B.344) becomes

$$p_{\text{win}} = \frac{1}{2} + \text{tr} \left(\left(\frac{1}{2}|0\rangle\langle 0| + \frac{1}{4}\mathbb{1} - \frac{1}{2}|1\rangle\langle 1| - \frac{1}{4}\mathbb{1} \right) \eta_{\mathbf{N}}^A \right) = \frac{1}{2} + \frac{1}{2} \text{tr}(\sigma_Z \eta_{\mathbf{N}}^A). \quad (\text{B.350})$$

(b) If $|\psi\rangle$ is invariant under the action of U , i.e $U\psi U^\dagger = \psi$, then $\psi^T = (U\psi U^\dagger)^T = U^* \psi^T U^T$. Therefore,

$$U \otimes U^* (\psi \otimes \psi^T) U^\dagger \otimes U^T = U\psi U^\dagger \otimes U^* \psi^T U^T = \psi \otimes \psi^T. \quad (\text{B.351})$$

(c) Using that $\sum_a \Lambda_a^{A \rightarrow A'}(\psi) = 2 \text{tr}_A((\mathbb{1}_{A'} \otimes \psi_A^T) \eta^{A'A})$, we get

$$F = \int d\psi \langle \psi | \sum_a \Lambda_a(\psi) | \psi \rangle = 2 \int d\psi \text{tr}_{A'}(\text{tr}_A((\mathbb{1}_{A'} \otimes \psi_A^T) \eta^{A'A}) \psi_{A'}) = 2 \int d\psi \text{tr}(\psi_{A'} \otimes \psi_A^T \eta^{A'A}), \quad (\text{B.352})$$

where we have used again the property (B.337). Since we are integrating over the whole state space, we have that $\int d\psi = \int d(U\psi U^\dagger)$ for any unitary U . Therefore, we can assume that ψ is invariant under the action of U and use (b) with the Schur-Weyl duality to rewrite the integral as

$$F = 2 \int d\psi \text{tr}(\psi_{A'} \otimes \psi_A^T \eta^{A'A}) = 2 \int d\psi \text{tr}((\alpha \phi^+ + \beta \mathbb{1}) \eta^{A'A}). \quad (\text{B.353})$$

To derive the values of α and β we can impose that

$$1 = \int d\psi \text{tr}(\psi_{A'} \otimes \psi_A^T) = \int d\psi \text{tr}(\alpha \phi^+ + \beta \mathbb{1}) = \int d\psi [\alpha \text{tr}(\phi^+) + \beta \text{tr}(\mathbb{1})] = \alpha + 4\beta. \quad (\text{B.354})$$

Moreover, for a generic pure state density matrix $\psi = \sum_{i,j} c_{ij} |i\rangle\langle j|$, we can compute

$$\text{tr}(\phi^+ \psi \otimes \psi^T) = \frac{1}{d} \sum_{k,l} \sum_{i,j} \sum_{p,q} c_{ij} c_{qp} \underbrace{\text{tr}(|kk\rangle\langle ll|ip\rangle\langle jq|)}_{=\delta_{l,i}\delta_{l,p}\delta_{k,j}\delta_{k,q}} = \frac{1}{d} \sum_{k,l} c_{lk} c_{kl} = \frac{1}{d} \sum_{k,l} c_{lk} c_{lk}^* = \frac{1}{d} \sum_{k,l} |c_{lk}|^2 = \frac{1}{d}, \quad (\text{B.355})$$

$= \text{tr}(\psi^2) = 1$

where we have used the definition of the purity of a state. Therefore, in our case

$$\frac{1}{2} = \int d\psi \text{tr}(\phi^+ \psi_{A'} \otimes \psi_A^T) = \int d\psi [\alpha \text{tr}(\phi^+ \phi^+) + \beta \text{tr}(\phi^+)] = \alpha + \beta. \quad (\text{B.356})$$

We get a system of linear equations in α and β ,

$$\begin{cases} \alpha + 4\beta = 1, \\ \alpha + \beta = \frac{1}{2} \end{cases} \implies \alpha = \frac{1}{3}, \quad \beta = \frac{1}{6}. \quad (\text{B.357})$$

Therefore, the average fidelity (B.352) becomes

$$F = 2 \int d\psi \operatorname{tr}(\psi_{A'} \otimes \psi_A^T \eta^{A'A}) = \operatorname{tr}\left(\frac{2}{3}\phi^+ \eta^{A'A} + \frac{2}{6}\mathbb{1} \eta^{A'A}\right) = \frac{1}{3} + \frac{2}{3} \operatorname{tr}(\phi^+ \eta^{A'A}). \quad (\text{B.358})$$

(d) Using the Kraus operators in (B.65), we can compute the Choi states $\eta_N^{A'A}$ and $\eta_S^{A'A}$. Define $\beta_{\pm} = \sqrt{\frac{1 \pm \gamma}{2}}$. Then

$$\begin{aligned} \eta_N^{A'A} &= (K_N \otimes \mathbb{1})\phi^+(K_N^\dagger \otimes \mathbb{1}) \\ &= \frac{1}{2} [(\beta_+|0\rangle\langle 0| + \beta_-|1\rangle\langle 1|) \otimes \mathbb{1}] (|00\rangle\langle 00| + |11\rangle\langle 11| + |00\rangle\langle 11| + |11\rangle\langle 00|) [(\beta_+|0\rangle\langle 0| + \beta_-|1\rangle\langle 1|) \otimes \mathbb{1}] \\ &= \frac{1}{2} [\beta_+|00\rangle\langle 00| + \beta_+|00\rangle\langle 11| + \beta_-|11\rangle\langle 11| + \beta_-|11\rangle\langle 00|] [(\beta_+|0\rangle\langle 0| + \beta_-|1\rangle\langle 1|) \otimes \mathbb{1}] \\ &= \frac{1}{2} [\beta_+^2|00\rangle\langle 00| + \beta_+\beta_-|00\rangle\langle 11| + \beta_-\beta_+|11\rangle\langle 00| + \beta_-^2|11\rangle\langle 11|]. \end{aligned} \quad (\text{B.359})$$

Similarly,

$$\begin{aligned} \eta_S^{A'A} &= (K_S \otimes \mathbb{1})\phi^+(K_S^\dagger \otimes \mathbb{1}) \\ &= \frac{1}{2} [\beta_-^2|00\rangle\langle 00| + \beta_+\beta_-|00\rangle\langle 11| + \beta_-\beta_+|11\rangle\langle 00| + \beta_+^2|11\rangle\langle 11|]. \end{aligned} \quad (\text{B.360})$$

Using (B.359), we can compute

$$\eta_N^A = \operatorname{tr}_{A'}(\eta_N^{A'A}) = \frac{1+\gamma}{4}|0\rangle\langle 0| + \frac{1-\gamma}{4}|1\rangle\langle 1|, \quad (\text{B.361})$$

that inserted into (B.350) gives

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{2} \operatorname{tr}\left((|0\rangle\langle 0| - |1\rangle\langle 1|) \left(\frac{1+\gamma}{4}|0\rangle\langle 0| + \frac{1-\gamma}{4}|1\rangle\langle 1|\right)\right) = \frac{1}{2} + \frac{\gamma}{4}. \quad (\text{B.362})$$

To compute the average fidelity from (B.358), we first need to compute

$$\begin{aligned} \eta^{A'A} &= \eta_N^{A'A} + \eta_S^{A'A} \\ &= \frac{1}{2} [\beta_+^2|00\rangle\langle 00| + \beta_+\beta_-|00\rangle\langle 11| + \beta_-\beta_+|11\rangle\langle 00| + \beta_-^2|11\rangle\langle 11|] + \\ &\quad + \frac{1}{2} [\beta_-^2|00\rangle\langle 00| + \beta_+\beta_-|00\rangle\langle 11| + \beta_-\beta_+|11\rangle\langle 00| + \beta_+^2|11\rangle\langle 11|] \\ &= \frac{1}{2} [(\beta_+^2 + \beta_-^2)|00\rangle\langle 00| + 2\beta_+\beta_-|00\rangle\langle 11| + 2\beta_-\beta_+|11\rangle\langle 00| + (\beta_-^2 + \beta_+^2)|11\rangle\langle 11|]. \end{aligned} \quad (\text{B.363})$$

Then we get

$$\begin{aligned} \phi^+ \eta^{A'A} &= \frac{1}{2} (|00\rangle\langle 00| + |11\rangle\langle 11| + |00\rangle\langle 11| + |11\rangle\langle 00|) \cdot \\ &\quad \cdot \frac{1}{2} [(\beta_+^2 + \beta_-^2)|00\rangle\langle 00| + 2\beta_+\beta_-|00\rangle\langle 11| + 2\beta_-\beta_+|11\rangle\langle 00| + (\beta_-^2 + \beta_+^2)|11\rangle\langle 11|] \\ &= \frac{1}{4} (1 + \sqrt{1 - \gamma^2}) (|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|). \end{aligned} \quad (\text{B.364})$$

Therefore, the average fidelity becomes

$$\begin{aligned} F &= \frac{1}{3} + \frac{2}{3} \operatorname{tr}(\phi^+ \eta^{A'A}) \\ &= \frac{1}{3} + \frac{2}{3} \operatorname{tr}\left(\frac{1}{4}(1 + \sqrt{1 - \gamma^2})(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)\right) \\ &= \frac{1}{3} + \frac{2}{3} \left(\frac{1}{2}(1 + \sqrt{1 - \gamma^2})\right) = \frac{2}{3} + \frac{\sqrt{1 - \gamma^2}}{3}. \end{aligned} \quad (\text{B.365})$$

B.5. Entropic concepts

B.5.1 Solution (Text at p. 150)

- (a) The possible events are $X = \text{SD}$ and $X = \text{SD}$, corresponding to the sampled adult primarily sympathising with SD or not. Since $p_X(X = \text{SD}) = 17.5\%$ and $p_X(X = \text{SD}) = 100\% - 17.5\% = 82.5\%$, the Shannon entropy is

$$H(X) = - \sum_{x \in \{\text{SD}, \text{SD}\}} p_X(x) \log(p_X(x)) = -0.175 \log(0.175) - 0.825 \log(0.825) \approx 0.67 \text{ bits.} \quad (\text{B.366})$$

After knowing $Y = \text{EDU}$, we have $p_{X|Y}(X = \text{SD}|Y = \text{EDU}) = 6.0\%$ and $p_{X|Y}(X = \text{SD}|Y = \text{EDU}) = 100\% - 6.0\% = 94\%$. Therefore, the updated Shannon entropy is

$$H(X|Y = \text{EDU}) = - \sum_{x \in \{\text{SD}, \text{SD}\}} p_{X|Y}(x|y) \log(p_{X|Y}(x|y)) = -0.06 \log(0.06) - 0.94 \log(0.94) \approx 0.33 \text{ bits.} \quad (\text{B.367})$$

We see that $H(X|Y = \text{EDU}) < H(X)$ because the level of education correlates with political sympathies, i.e. knowledge of Y reduces our uncertainty about X .

- (b) The chain rule is $H(X, Y) = H(Y) + H(X|Y)$. Use it repeatedly to show the result, namely

$$H(X_1, \dots, X_n) = H(X_1) + H(X_2, \dots, X_n|X_1) = H(X_1) + H(X_2|X_1) + H(X_3, \dots, X_n|X_2, X_1) \quad (\text{B.368})$$

$$= H(X_1) + H(X_2|X_1) + H(X_3|X_2, X_1) + H(X_4, \dots, X_n|X_3, X_2, X_1) \quad (\text{B.369})$$

$$= \dots = H(X_1) + H(X_2|X_1) + H(X_3|X_2, X_1) + \dots + H(X_n|X_{n-1}, \dots, X_1) \quad (\text{B.370})$$

- (c) From (8.4) and (8.5), the conditional entropy is

$$\begin{aligned} H(X|Y) &= - \sum_{x,y} p_Y(y) p_{X|Y}(x|y) \log(p_{X|Y}(x|y)) = - \sum_{x,y} p_Y(y) p_{X|Y}(x|y) \log \left(\frac{p_Y(y) p_{X|Y}(x|y)}{p_Y(y)} \right) \\ &\leq - \sum_x \left[\left(\sum_y p_Y(y) p_{X|Y}(x|y) \right) \log \frac{\sum_{y'} p_Y(y') p_{X|Y}(x|y')}{\sum_{y''} p_Y(y'')} \right] \\ &= - \sum_x \left[\sum_y p_{X,Y}(x, y) \log \left(\sum_{y'} p_{X,Y}(x, y') \right) \right] = - \sum_x p_X(x) \log(p_X(x)) = H(X), \end{aligned} \quad (\text{B.371})$$

where in the first step we have added $1 = \frac{p_Y(y)}{p_Y(y)}$ in the logarithm. Then, calling $a_y = p_Y(y) p_{X|Y}(x|y)$ and $b_y = p_Y(y)$, we have used the log sum inequality. In the third line, we have used that $p_{X,Y}(x, y) = p_Y(y) p_{X|Y}(x|y)$. Finally, we have used the definition of the Shannon entropy (8.1). The equality holds if and only if $a_y = c b_y$ for all y and some constant c . In our notation, this means

$$\frac{p_Y(y) p_{X|Y}(x|y)}{p_Y(y)} = c, \quad \forall y \implies p_{X|Y}(x|y) = c, \quad \forall y. \quad (\text{B.372})$$

Therefore the probability conditioned on Y is constant, i.e. X and Y are independent.

- (d) Using the result in (c) we have

$$H(X_2|X_1) \leq H(X_2), \quad H(X_3|X_2, X_1) \leq H(X_3), \quad \dots, \quad H(X_n|X_{n-1}, \dots, X_1) \leq H(X_n). \quad (\text{B.373})$$

To have equality in the first case above, we need that X_2 is independent of X_1 . Equality in the second case means X_3 is independent of X_2 and X_1 . Continuing this means that all the random variables must be independent to have equality in all the above relations. Inserted into the result of (b) we get

$$H(X_1, \dots, X_n) = H(X_1) + H(X_2|X_1) + H(X_3|X_2, X_1) + \dots + H(X_n|X_{n-1}, \dots, X_1) \leq \sum_i H(X_i). \quad (\text{B.374})$$

B.5.2 Solution (Text at p. 151)

- (a) Append a classical register to the state, obtaining $\rho_{ABX} = \sum_x q_x \phi_x \otimes \varphi_x \otimes |x\rangle\langle x|$. Then, use that $H(A|B) \geq H(A|B, X)$. Since X is a classical register, we can write the conditional entropy as

$$H(A|B, X) = \sum_x q_x H(A|B, X=x) = \sum_x q_x (H(A, B|X=x) - H(B|X=x)). \quad (\text{B.375})$$

Since the joint entropy $H(A, B|X=x)$ is conditioned on a specific value of x , it is evaluated over the product state $\phi_x \otimes \varphi_x$. We know from Eq (8.12) that the von Neumann entropy of a product state is additive. Hence,

$$\begin{aligned} H(A|B) &\geq H(A|B, X) = \sum_x q_x (H(A, B|X=x) - H(B|X=x)) \\ &= \sum_x q_x (H(A|X=x) + H(B|X=x) - H(B|X=x)) = \sum_x q_x H(A|X=x) \geq 0, \end{aligned} \quad (\text{B.376})$$

where the last step is because the entropy when conditioned on a classical random variable is non-negative.

- (b) For the quantum mutual information we have

$$I(A : B) = H(A) - H(A|B) \leq H(A) - H(A|B, X) = H(A) - \sum_x q_x H(A|X=x) = H(A) - H(A|X) \leq H(A). \quad (\text{B.377})$$

where we first used $H(A|B) \geq H(A|B, X)$ and then assumed a separable state and used Eq (B.376). Since the mutual information satisfies $I(A : B) = I(B : A)$, we can repeat the same argument to also get $I(A : B) \leq H(B)$.

- (c) A pure bipartite state can in Schmidt decomposition be written as

$$|\psi\rangle = \sum_k \sqrt{\lambda_k} |kk\rangle, \quad (\text{B.378})$$

for $\lambda_k \geq 0$ and $\sum_k \lambda_k = 1$. The mutual information is

$$I(A : B) = H(A) + H(B) - H(A, B) = H(A) + H(B) = 2H(A), \quad (\text{B.379})$$

where we first used that the global entropy of the pure state is zero and then that $H(A) = H(B)$ since the reductions of a bipartite state have identical spectra. The eigenvalues of ψ_A are given by $\{\lambda_k\}_k$ and therefore

$$I(A : B) = 2H(\{\lambda_k\}) = -2 \sum_k \lambda_k \log(\lambda_k). \quad (\text{B.380})$$

To violate the inequality in (b), we require $H(A) \geq 0$ which is the case if and only if the state is entangled.

- (d) The conditional von Neumann entropy is $H(A|B) = H(A, B) - H(A)$. The reduced state of $p \phi^+ + (1-p) \frac{\mathbb{1}}{4}$ is maximally mixed and hence $H(A) = 1$. A direct calculation shows that the eigenvalues of $p \phi^+ + (1-p) \frac{\mathbb{1}}{4}$ are $\{\frac{1-p}{4}, \frac{1-p}{4}, \frac{1-p}{4}, \frac{1+3p}{4}\}$. Computing the Shannon entropy of this distribution gives

$$H(A, B) = -\frac{3(1-p)}{4} \log\left(\frac{1-p}{4}\right) - \frac{1+3p}{4} \log\left(\frac{1+3p}{4}\right). \quad (\text{B.381})$$

Hence the conditional entropy is simply $H(A|B) = H(A, B) - 1$. Numerically solving $H(A|B) = 0$ gives $p \approx 0.748$.

- (e) Using the definition of quantum mutual information, we obtain

$$\begin{aligned} I(A : B) + I(A, B : C) - I(C : B) &= [H(A) + H(B) - H(A, B)] + [H(A, B) + H(C) - H(A, B, C)] \\ &\quad - [H(C) + H(B) - H(C, B)] = H(A) + H(B, C) - H(A, B, C), \end{aligned} \quad (\text{B.382})$$

where we have used that $H(C, B) = H(B, C)$. The right-hand-side is again the definition of a mutual information,

$$I(A : B, C) = H(A) + H(B, C) - H(A, B, C), \quad (\text{B.383})$$

Similarly,

$$\begin{aligned} I(A : C) + I(B : C|A) &= [H(A) + H(C) - H(A, C)] + [H(B, A) + H(C, A) - H(B, C, A) - H(A)] \\ &= H(C) + H(B, A) - H(B, C, A), \end{aligned} \quad (\text{B.384})$$

where we have used that $H(B|A) = H(B, A) - H(A)$ and $H(C|A) = H(C, A) - H(A)$. This is exactly

$$I(A, B : C) = H(A, B) + H(C) - H(A, B, C). \quad (\text{B.385})$$

(f) Using the definition of the mutual information,

$$I(X_1, X_2 : Y_1, Y_2) = H(X_1, X_2) + H(Y_1, Y_2) - H(X_1, X_2, Y_1, Y_2). \quad (\text{B.386})$$

Shannon entropy of independent variables is additive. Hence,

$$H(X_1, X_2) = H(X_1) + H(X_2), \quad H(Y_1, Y_2) = H(Y_1) + H(Y_2), \quad (\text{B.387})$$

$$H(X_1, X_2, Y_1, Y_2) = H(X_1, Y_1) + H(X_2, Y_2). \quad (\text{B.388})$$

Using this, we get

$$\begin{aligned} I(X_1 : Y_1) + I(X_2 : Y_2) &= [H(X_1) + H(Y_1) - H(X_1, Y_1)] + [H(X_2) + H(Y_2) - H(X_2, Y_2)] \\ &= \underbrace{H(X_1) + H(X_2)}_{=H(X_1, X_2)} + \underbrace{H(Y_1) + H(Y_2)}_{=H(Y_1, Y_2)} - \underbrace{(H(X_1, Y_1) + H(X_2, Y_2))}_{=H(X_1, X_2, Y_1, Y_2)} \\ &= I(X_1, X_2 : Y_1, Y_2). \end{aligned} \quad (\text{B.389})$$

B.5.3 Solution (Text at p. 151)

Write the states in spectral decomposition, $\rho = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$ and $\sigma = \sum_j \mu_j |\mu_j\rangle\langle\mu_j|$. Insert this in the definition of quantum relative entropy to get

$$D(\rho||\sigma) = \underbrace{\text{tr}(\rho \log \rho)}_{\text{use Eq (8.11)}} - \text{tr}(\rho \log \sigma) = \sum_i \lambda_i \log(\lambda_i) - \sum_i \lambda_i \left(\sum_j \log(\mu_j) |\langle\lambda_i|\mu_j\rangle|^2 \right) \quad (\text{B.390})$$

$$\begin{aligned} &\stackrel{\text{Jensen ineq}}{\geq} \sum_i \lambda_i \log(\lambda_i) - \sum_i \lambda_i \log \left(\underbrace{\sum_j \mu_j |\langle\lambda_i|\mu_j\rangle|^2}_{=\nu_i} \right) = - \sum_i \lambda_i \log \left(\frac{\nu_i}{\lambda_i} \right) \stackrel{\text{Jensen ineq}}{\geq} - \log \left(\sum_i \frac{\nu_i}{\lambda_i} \lambda_i \right) = - \log(1) = 0. \end{aligned} \quad (\text{B.391})$$

Here, we have used that $\nu_i = \sum_j \mu_j |\langle\lambda_i|\mu_j\rangle|^2$ is a probability distribution; $\nu_i \geq 0$ and $\sum_i \nu_i = 1$. To saturate the inequality we need to achieve equality in both instances of Jensen's inequality. This corresponds precisely to $\rho = \sigma$.

B.5.4 Solution (Text at p. 151)

(a) From the definition of quantum relative entropy we have

$$D(\rho||\sigma) = \text{tr}(U \rho U^\dagger \log(U \rho U^\dagger)) - \text{tr}(U \rho U^\dagger \log(U \sigma U^\dagger)). \quad (\text{B.392})$$

For any positive operator, $X \geq 0$ with spectral decomposition $X = \sum_k \lambda_k |\lambda_k\rangle\langle\lambda_k|$, we have

$$\log(U X U^\dagger) = \log(U \sum_k \lambda_k |\lambda_k\rangle\langle\lambda_k| U^\dagger) = \log(\sum_k \lambda_k |\tilde{\lambda}_k\rangle\langle\tilde{\lambda}_k|) = \sum_k \log(\lambda_k) |\tilde{\lambda}_k\rangle\langle\tilde{\lambda}_k| = U (\log X) U^\dagger, \quad (\text{B.393})$$

where $|\tilde{\lambda}_k\rangle = U |\lambda_k\rangle$ is the rotated basis. Using this, we obtain

$$D(\rho||\sigma) = \text{tr}(U \rho U^\dagger U \log(\rho) U^\dagger) - \text{tr}(U \rho U^\dagger U \log(\sigma) U^\dagger) = \text{tr}(\rho \log(\rho)) - \text{tr}(\rho \log(\sigma)) = D(\rho||\sigma). \quad (\text{B.394})$$

(b) Consider the spectral decomposition $X = \sum_i x_i |x_i\rangle\langle x_i|$ and $Y = \sum_j y_j |y_j\rangle\langle y_j|$. Then, using the definition of the logarithm of an operator,

$$\begin{aligned} \log(X \otimes Y) &= \sum_{i,j} \underbrace{\log(x_i y_j)}_{=\log(x_i) + \log(y_j)} |x_i\rangle\langle x_i| \otimes |y_j\rangle\langle y_j| \\ &= \sum_i \log(x_i) |x_i\rangle\langle x_i| \otimes \underbrace{\sum_j |y_j\rangle\langle y_j|}_{=\mathbb{1}} + \sum_i \underbrace{|x_i\rangle\langle x_i|}_{=\mathbb{1}} \otimes \sum_j \log(y_j) |y_j\rangle\langle y_j| = \log(X) \otimes \mathbb{1} + \log(Y) \otimes \mathbb{1}, \end{aligned} \quad (\text{B.395})$$

where we have used that $\log(X) = \sum_i \log(x_i) |x_i\rangle\langle x_i|$ and $\log(Y) = \sum_j \log(y_j) |y_j\rangle\langle y_j|$.

(c) From the definition of quantum relative entropy we have

$$D(\rho \otimes \rho' \| \sigma \otimes \sigma') = \text{tr}(\rho \otimes \rho' \log(\rho \otimes \rho')) - \text{tr}(\rho \otimes \rho' \log(\sigma \otimes \sigma')). \quad (\text{B.396})$$

Applying the simplification of the logarithm (B.76) gives

$$\begin{aligned} D(\rho \otimes \rho' \| \sigma \otimes \sigma') &= \text{tr}(\rho \otimes \rho' \log(\rho) \otimes \mathbb{1}) + \text{tr}(\rho \otimes \rho' \mathbb{1} \otimes \log(\rho')) \\ &\quad - \text{tr}(\rho \otimes \rho' \log(\sigma) \otimes \mathbb{1}) - \text{tr}(\rho \otimes \rho' \mathbb{1} \otimes \log(\sigma')) \\ &= \text{tr}(\rho \log(\rho)) + \text{tr}(\rho' \log(\rho')) - \text{tr}(\rho \log(\sigma)) - \text{tr}(\rho' \log(\sigma')) \\ &= D(\rho \| \sigma) + D(\rho' \| \sigma'). \end{aligned} \quad (\text{B.397})$$

(d) Express the entropies in terms of quantum relative entropy: $H(\rho) = -D(\rho \| \mathbb{1})$ and $H(\Lambda(\rho)) = -D(\Lambda(\rho) \| \mathbb{1})$. Because the channel is unital, it holds that $\mathbb{1} = \Lambda(\mathbb{1})$. Hence,

$$H(\Lambda(\rho)) - H(\rho) = -D(\Lambda(\rho) \| \Lambda(\mathbb{1})) + D(\rho \| \mathbb{1}) \geq 0, \quad (\text{B.398})$$

where the last step follows from monotonicity of quantum relative entropy. To violate this inequality with a non-unital channel, we can for example select $\Lambda(\rho) = |0\rangle\langle 0|$.

(e) We append a classical register to each of ρ and σ . This gives

$$\rho_{AX} = \sum_x q_x \rho_x \otimes |x\rangle\langle x|, \quad \sigma_{AX} = \sum_x q_x \sigma_x \otimes |x\rangle\langle x|. \quad (\text{B.399})$$

Note that $\text{tr}_X(\rho_{AX}) = \rho$ and $\text{tr}_X(\sigma_{AX}) = \sigma$. Using the block-diagonal structure of the states in (B.399),

$$\begin{aligned} D(\rho_{AX} \| \sigma_{AX}) &= \text{tr}(\rho_{AX} (\log \rho_{AX} - \log \sigma_{AX})) \\ &= \text{tr} \left(\sum_y q_y \rho_y \sum_x (\log(q_x \rho_x) - \log(q_x \sigma_x)) \right) \text{tr}(|y\rangle\langle y|) \\ &= \sum_x q_x \text{tr}(\rho_x (\log \rho_x - \log \sigma_x)) = \sum_x q_x D(\rho_x \| \sigma_x). \end{aligned} \quad (\text{B.400})$$

By the monotonicity of quantum relative entropy we have that

$$\sum_x q_x D(\rho_x \| \sigma_x) = D(\rho_{AX} \| \sigma_{AX}) \geq D(\rho \| \sigma). \quad (\text{B.401})$$

B.5.5 Solution (Text at p. 152)

(a) Use the definition of quantum relative entropy to get

$$D(\rho \| \Lambda(\rho)) + H(A) = [\text{tr}(\rho \log(\rho)) - \text{tr}(\rho \log(\Lambda(\rho)))] - \text{tr}(\rho \log(\rho)) = -\text{tr}(\rho \log(\Lambda(\rho))). \quad (\text{B.402})$$

Next, use the definition of Λ to get

$$-\text{tr}(\rho \log(\Lambda(\rho))) = -\text{tr} \left(\rho \sum_x \log(\langle a_x | \rho | a_x \rangle) |a_x\rangle\langle a_x| \right) = -\sum_x \underbrace{\langle a_x | \rho | a_x \rangle}_{=p_X(x)} \log(\langle a_x | \rho | a_x \rangle) = H(X). \quad (\text{B.403})$$

(b) Re-arrange the result of (a) and apply the monotonicity of quantum relative entropy to obtain

$$\begin{aligned} H(X) - H(A) &\stackrel{(\text{B.82})}{=} D(\rho \| \Lambda(\rho)) \stackrel{(\text{8.20})}{\geq} D(\Gamma(\rho) \| \Gamma(\Lambda(\rho))) \\ &= \text{tr}(\Gamma(\rho) \log(\Gamma(\rho))) - \text{tr}(\Gamma(\rho) \log(\Gamma(\Lambda(\rho)))). \end{aligned} \quad (\text{B.404})$$

(c) Consider the final two terms in (b) individually. For the first term we have

$$\begin{aligned} -H(\Gamma(\rho)) &= \text{tr}(\Gamma(\rho) \log(\Gamma(\rho))) = \text{tr} \left(\sum_y \langle b_y | \rho | b_y \rangle |b_y\rangle\langle b_y| \sum_{y'} \log(\langle b_{y'} | \rho | b_{y'} \rangle) |b_{y'}\rangle\langle b_{y'}| \right) \\ &= \sum_{y, y'} \underbrace{\langle b_y | \rho | b_y \rangle \log(\langle b_{y'} | \rho | b_{y'} \rangle)}_{=\delta_{y, y'}} \underbrace{\text{tr}(|b_y\rangle\langle b_y| |b_{y'}\rangle\langle b_{y'}|)}_{=p_Y(y)} = -H(Y). \end{aligned}$$

For the second term we have

$$\begin{aligned}
\text{tr}(\Gamma(\rho) \log(\Gamma(\Lambda(\rho)))) &= \text{tr} \left(\sum_{y'} \langle b_{y'} | \rho | b_{y'} \rangle |b_{y'}\rangle \langle b_{y'}| \log \left(\sum_x \langle a_x | \rho | a_x \rangle \Gamma(|a_x\rangle \langle a_x|) \right) \right) \\
&= \sum_{y'} \langle b_{y'} | \rho | b_{y'} \rangle \text{tr} \left(|b_{y'}\rangle \langle b_{y'}| \log \left(\sum_y \sum_x \langle a_x | \rho | a_x \rangle |\langle a_x | b_y \rangle|^2 |b_y\rangle \langle b_y| \right) \right) \\
&= \sum_{y'} \langle b_{y'} | \rho | b_{y'} \rangle \sum_y \log \left(\sum_x \langle a_x | \rho | a_x \rangle |\langle a_x | b_y \rangle|^2 \right) \underbrace{\text{tr}(|b_{y'}\rangle \langle b_{y'}| |b_y\rangle \langle b_y|)}_{=\delta_{y',y}} \\
&= \sum_y \langle b_y | \rho | b_y \rangle \log \left(\sum_x \langle a_x | \rho | a_x \rangle |\langle a_x | b_y \rangle|^2 \right). \tag{B.405}
\end{aligned}$$

Inserting this back into Eq (B.404) we get

$$\begin{aligned}
H(X) - H(A) &= -H(Y) - \sum_y \langle b_y | \rho | b_y \rangle \log \left(\sum_x \langle a_x | \rho | a_x \rangle |\langle a_x | b_y \rangle|^2 \right) \\
&\geq -H(Y) - \underbrace{\sum_y \langle b_y | \rho | b_y \rangle}_{=1} \log \left(\max_{x',y'} |\langle a_{x'} | b_{y'} \rangle|^2 \underbrace{\sum_x \langle a_x | \rho | a_x \rangle}_{=1} \right) \\
&= -H(Y) - \log \left(\max_{x',y'} |\langle a_{x'} | b_{y'} \rangle|^2 \right) = -\log(c) - H(Y) \geq -\log(c). \tag{B.406}
\end{aligned}$$

Re-arranging this inequality gives Maassen-Uffink uncertainty relation.

B.5.6 Solution (Text at p. 152)

- (a) Given the spectral decomposition $\rho^{\otimes n} = \sum_i \lambda_i |\lambda_i\rangle \langle \lambda_i|$, the projector onto the typical subspace T_ϵ is defined as $E = \sum_{i \in \text{ind}(T_\epsilon)} |\lambda_i\rangle \langle \lambda_i|$ for all the indices i corresponding to eigenvalues λ_i that satisfy (8.33). Therefore, since $\rho^{\otimes n}$ and E are diagonal in the same basis, they commute, i.e. $[\rho^{\otimes n}, E] = 0$.
- (b) Using the spectral decomposition of $\rho^{\otimes n}$ and the definition of E , we can write

$$\text{tr}(E \rho^{\otimes n}) = \text{tr} \left(\sum_{i \in \text{ind}(T_\epsilon)} |\lambda_i\rangle \langle \lambda_i| \sum_j \lambda_j |\lambda_j\rangle \langle \lambda_j| \right) = \sum_{i \in \text{ind}(T_\epsilon)} \sum_j \lambda_j \underbrace{|\langle \lambda_i | \lambda_j \rangle|^2}_{=\delta_{i,j}} = \sum_{i \in \text{ind}(T_\epsilon)} \lambda_i, \tag{B.407}$$

where each λ_i satisfies (8.33). Following the same procedure used for the typical set in Chapter 8.3, we can show that the condition (8.33) implies

$$\text{Prob}(|Z - H(\rho)| < \epsilon) \geq 1 - \epsilon, \tag{B.408}$$

for the random variable $Z = -\frac{1}{n} \log(\lambda_i)$. This guarantees that, for a sufficiently large n , $|\lambda_i\rangle$ belongs to the typical subspace with a probability at least $1 - \epsilon$. Using this into (B.407), we get $\text{tr}(E \rho^{\otimes n}) \geq 1 - \epsilon$.

- (c) From the definition of E , we get

$$\text{tr}(E) = \text{tr} \left(\sum_{i \in \text{ind}(T_\epsilon)} |\lambda_i\rangle \langle \lambda_i| \right) = \sum_{i \in \text{ind}(T_\epsilon)} \underbrace{\langle \lambda_i | \lambda_i \rangle}_{=1} = |T_\epsilon|, \tag{B.409}$$

where $|T_\epsilon|$ denotes the size of the typical subspace. From the spectral decomposition of $\rho^{\otimes n}$ and Eq (8.33) we get

$$1 = \sum_i \lambda_i \geq \sum_{i \in \text{ind}(T_\epsilon)} \lambda_i \geq |T_\epsilon| 2^{-n(H(\rho)+\epsilon)} \implies |T_\epsilon| \leq 2^{n(H(\rho)+\epsilon)}. \tag{B.410}$$

Inserting this inequality into (B.409), we get

$$\text{tr}(E) = |T_\epsilon| \leq 2^{n(H(\rho)+\epsilon)}. \tag{B.411}$$

The state space is of size $2^{n \log d}$, which is exponentially larger whenever $H(\rho) < \log(d) - \epsilon$.

B.5.7 Solution (Text at p. 153)

(a) By Shannon's noisy channel coding theorem the capacity of a classical channel is

$$C(\Lambda_1 \otimes \Lambda_2) = \max_{\{p(x_1, x_2)\}} I(X_1, X_2 : Y_1, Y_2). \quad (\text{B.412})$$

By restricting to product probability distributions $p(x_1, x_2) = p(x_1)p(x_2)$, we can use the property in (B.73) to get

$$\begin{aligned} C(\Lambda_1 \otimes \Lambda_2) &\geq \max_{\{p(x_1), p(x_2)\}} I(X_1, X_2 : Y_1, Y_2) = \max_{\{p(x_1), p(x_2)\}} [I(X_1 : Y_1) + I(X_2 : Y_2)] \\ &= \underbrace{\max_{\{p(x_1)\}} I(X_1 : Y_1)}_{=C(\Lambda_1)} + \underbrace{\max_{\{p(x_2)\}} I(X_2 : Y_2)}_{=C(\Lambda_2)} = C(\Lambda_1) + C(\Lambda_2), \end{aligned} \quad (\text{B.413})$$

where in the last equality we used Shannon's noisy channel coding theorem.

(b) Using Shannon's noisy channel coding theorem and elementary properties of entropy, we can write the channel capacity as

$$\begin{aligned} C(\Lambda_1 \otimes \Lambda_2) &= \max_{\{p(x_1, x_2)\}} I(X_1, X_2 : Y_1, Y_2) = \max_{\{p(x_1, x_2)\}} I(Y_1, Y_2 : X_1, X_2) \\ &= \max_{\{p(x_1, x_2)\}} [H(Y_1, Y_2) - H(Y_1, Y_2 | X_1, X_2)] \leq \max_{\{p(x_1, x_2)\}} [H(Y_1) + H(Y_2) - H(Y_1, Y_2 | X_1, X_2)]. \end{aligned} \quad (\text{B.414})$$

From the definition of the conditional Shannon entropy we obtain

$$\begin{aligned} C(\Lambda_1 \otimes \Lambda_2) &\leq \max_{\{p(x_1, x_2)\}} [H(Y_1) + H(Y_2) - \sum_{x_1, x_2} p(x_1, x_2) H(Y_1, Y_2 | X_1 = x_1, X_2 = x_2)] \\ &= \max_{\{p(x_1, x_2)\}} [H(Y_1) + H(Y_2) - \sum_{x_1, x_2} p(x_1, x_2) (H(Y_1 | X_1 = x_1) + H(Y_2 | X_2 = x_2))], \end{aligned} \quad (\text{B.415})$$

where we have used that, since the two channels are independent, $p(y_1, y_2 | x_1, x_2) = p(y_1 | x_1)p(y_2 | x_2)$ the conditional Shannon entropy becomes additive, i.e.

$$\begin{aligned} H(Y_1, Y_2 | X_1 = x_1, X_2 = x_2) &= - \sum_{y_1, y_2} p(y_1, y_2 | x_1, x_2) \log(p(y_1, y_2 | x_1, x_2)) \\ &= - \sum_{y_1, y_2} p(y_1 | x_1) p(y_2 | x_2) \log(p(y_1 | x_1) p(y_2 | x_2)) \\ &= - \sum_{y_1, y_2} p(y_1 | x_1) p(y_2 | x_2) (\log(p(y_1 | x_1)) + \log(p(y_2 | x_2))) \\ &= H(Y_1 | X_1 = x_1) + H(Y_2 | X_2 = x_2). \end{aligned} \quad (\text{B.416})$$

Therefore, using that $p(x_1, x_2) = p(x_1)p(x_2)$ from the channels independence, we get

$$\begin{aligned} C(\Lambda_1 \otimes \Lambda_2) &\leq \max_{\{p(x_1, x_2)\}} [H(Y_1) + H(Y_2) - \sum_{x_1} p(x_1) H(Y_1 | X_1 = x_1) - \sum_{x_2} p(x_2) H(Y_2 | X_2 = x_2)] \\ &= \max_{\{p(x_1, x_2)\}} [H(Y_1) + H(Y_2) - H(Y_1 | X_1) - H(Y_2 | X_2)] \\ &= \max_{\{p(x_1, x_2)\}} [I(X_1 : Y_1) + I(X_2 : Y_2)] = C(\Lambda_1) + C(\Lambda_2). \end{aligned} \quad (\text{B.417})$$

B.5.8 Solution (Text at p. 153)

The reduction of the state ρ to subsystem number i is $\sum_{x_i} p_i(x_i) \rho(x_i)$, where $p_i(x_i)$ is the i 'th marginal of $p(x)$. From the subadditivity of von Neumann entropy we therefore have

$$H(\rho) \leq \sum_{i=1}^n H\left(\sum_{x_i} p_i(x_i) \rho(x_i)\right). \quad (\text{B.418})$$

For product states the von Neumann entropy is additive. Hence

$$H(\rho(x_1) \otimes \cdots \otimes \rho(x_n)) = \sum_{i=1}^n H(\rho(x_i)), \quad (\text{B.419})$$

The second term in (B.86) then becomes

$$\sum_x p(x) H(\rho_x) = \sum_{x_1, x_2, \dots, x_n} p(x_1, x_2, \dots, x_n) \sum_{i=1}^n H(\rho(x_i)) = \sum_{i=1}^n \sum_{x_i} p_i(x_i) H(\rho(x_i)), \quad (\text{B.420})$$

Putting together (B.418) and (B.420), we get

$$\chi(\mathcal{E}) \leq \sum_{i=1}^n \left(H \left(\sum_{x_i} p_i(x_i) \rho(x_i) \right) - \sum_{x_i} p_i(x_i) H(\rho(x_i)) \right) = \sum_{i=1}^n \chi(\mathcal{E}_i), \quad (\text{B.421})$$

where $\mathcal{E}_i = \{p_i(x_i), \rho(x_i)\}_{x_i}$.

B.5.9 Solution (Text at p. 153)

(a) We have $\rho'_{XB_1B_2} = \sum_x p(x) |x\rangle\langle x| \otimes (\Lambda_1 \otimes \Lambda_2)(\rho(x)_{A_1A_2})$. By tracing out subsystem B_2 , we get

$$\rho'_{XB_1} = \sum_x p(x) |x\rangle\langle x| \otimes \Lambda_1(\rho(x)_{A_1}). \quad (\text{B.422})$$

Using the property (B.71) of the mutual information, we can write

$$I(X : B_1, B_2)_{\rho'} = I(X : B_1)_{\rho'} + I(X, B_1 : B_2)_{\rho'} - I(B_2 : B_1)_{\rho'} \leq I(X : B_1)_{\rho'} + I(X, B_1 : B_2)_{\rho'}. \quad (\text{B.423})$$

By definition of the Holevo information, we get

$$I(X : B_1)_{\rho'} \leq \chi(\Lambda_1), \quad (\text{B.424})$$

from which

$$I(X : B_1, B_2)_{\rho'} \leq \chi(\Lambda_1) + I(X, B_1 : B_2)_{\rho'}. \quad (\text{B.425})$$

(b) Since Λ_1 is an entanglement-breaking channel, the following is a separable state,

$$(\Lambda_1 \otimes \mathbb{I})(\rho(x)_{A_1A_2}) = \sum_y p(y|x) \sigma(x, y)_{B_1} \otimes \tau(x, y)_{A_2}. \quad (\text{B.426})$$

Therefore, we can write

$$\rho'_{XB_1B_2} = \sum_{x,y} p(x) p(y|x) |x\rangle\langle x| \otimes \sigma(x, y)_{B_1} \otimes \Lambda_2(\tau(x, y))_{B_2}. \quad (\text{B.427})$$

(c) Since $\rho'_{XB_1B_2}$ can be seen as the marginal of $\omega'_{XYB_1B_2}$, we have $I(X, B_1 : B_2)_{\rho'} = I(X, B_1 : B_2)_{\omega'}$. Now we can use strong subadditivity of the mutual information to write

$$I(X, B_1 : B_2)_{\omega'} \stackrel{\text{SSA}}{\leq} I(X, Y, B_1 : B_2)_{\omega'} \stackrel{(\text{B.72})}{\leq} I(X, Y : B_2)_{\omega'} + \underbrace{I(B_1 : B_2 | X, Y)_{\omega'}}_{=0} \leq I(X, Y : B_2)_{\omega'}, \quad (\text{B.428})$$

where $I(B_1 : B_2 | X, Y)_{\omega'} = 0$ since ω' is a separable state by definition. Now we can consider the reduction

$$\omega'_{XYB_2} = \text{tr}_{B_1} (\omega'_{XYB_1B_2}) = \sum_{x,y} p(x, y) |xy\rangle\langle xy| \otimes \Lambda_2(\tau(x, y)), \quad (\text{B.429})$$

and by definition of the Holevo information we get

$$I(X, B_1 : B_2)_{\omega'} \leq \chi(\Lambda_2). \quad (\text{B.430})$$

Therefore, using the definition of the Holevo information, we get

$$\chi(\Lambda_1 \otimes \Lambda_2) = \max_{\{\rho'_{XB_1B_2}\}} I(X : B_1, B_2)_{\rho'} \stackrel{(\text{B.423})}{\leq} I(X : B_1)_{\rho'} + I(X, B_1 : B_2)_{\rho'} \stackrel{\substack{(\text{B.424}) \\ (\text{B.430})}}{\leq} \chi(\Lambda_1) + \chi(\Lambda_2). \quad (\text{B.431})$$

B.5.10 Solution (Text at p. 153)

(a) The joint state of the classical input register X and the classical output register Y after the measurements is

$$\rho_{XY} = \sum_{x,y=0}^2 p(x,y)|x\rangle\langle x| \otimes |y\rangle\langle y|, \quad \text{where} \quad p(x,y) = p(x)p(y|x) = \frac{1}{3} \text{tr}(\psi_x E_y). \quad (\text{B.432})$$

The marginals are

$$\rho_X = \sum_{x=0}^2 p(x)|x\rangle\langle x| = \frac{1}{3} \sum_{x=0}^2 |x\rangle\langle x|, \quad \rho_Y = \sum_{y=0}^2 p(x)p(y|x)|y\rangle\langle y| = \sum_{y=0}^2 \left(\frac{1}{3} \sum_{x=0}^2 \text{tr}(\psi_x E_y) \right) |y\rangle\langle y|. \quad (\text{B.433})$$

Using Born's rule, we can compute

$$\begin{aligned} p(0|0) &= \text{tr}(|\psi_0\rangle\langle\psi_0|E_0) = \frac{2}{3} \langle 0| \left(\cos\left(\frac{\pi}{6}\right)|0\rangle + \sin\left(\frac{\pi}{6}\right)|1\rangle \right) \left(\cos\left(\frac{\pi}{6}\right)\langle 0| + \sin\left(\frac{\pi}{6}\right)\langle 1| \right) |0\rangle \\ &= \frac{2}{3} \langle 0| \left(\cos^2\left(\frac{\pi}{6}\right)|0\rangle\langle 0| + \sin^2\left(\frac{\pi}{6}\right)|1\rangle\langle 1| + \cos\left(\frac{\pi}{6}\right)\sin\left(\frac{\pi}{6}\right)|0\rangle\langle 1| + \sin\left(\frac{\pi}{6}\right)\cos\left(\frac{\pi}{6}\right)|1\rangle\langle 0| \right) |0\rangle = \frac{1}{2}. \end{aligned} \quad (\text{B.434})$$

In the same way, we can compute all the other probabilities. We can group them into a matrix C whose elements are $\{C_{y,x} = p(y|x)\}_{y,x=0}^2$,

$$C = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}. \quad (\text{B.435})$$

Now we can compute the eigenvalues of ρ_Y in (B.433), from which

$$\rho_Y = \sum_{y=0}^2 \left(\frac{1}{3} \sum_{x=0}^2 \text{tr}(\psi_x E_y) \right) |y\rangle\langle y| = \frac{1}{3} \sum_{y=0}^2 |y\rangle\langle y| = \frac{\mathbb{1}}{3}. \quad (\text{B.436})$$

Hence $H(Y)_{\rho_Y} = \log(3)$. To compute the mutual information we also need

$$H(Y|X)_{\rho_{XY}} = - \sum_{x=0}^2 p(x) \sum_{y=0}^2 p(y|x) \log(p(y|x)) = - \frac{1}{3} \sum_{x=0}^2 \left(\sum_{y=0}^2 p(y|x) \log(p(y|x)) \right) = - \frac{1}{3} \sum_{x=0}^2 \log\left(\frac{1}{2}\right) = 1. \quad (\text{B.437})$$

Therefore, we get

$$I(X : Y) = I(Y : X) = H(Y)_{\rho_Y} - H(Y|X)_{\rho_{XY}} = \log(3) - 1. \quad (\text{B.438})$$

(b) Now the joint state XY after the measurements will be

$$\rho_{XY} = \sum_{x,y=0}^2 p(x,y)|x\rangle\langle x| \otimes |y\rangle\langle y|, \quad \text{where} \quad p(x,y) = p(x)p(y|x) = \frac{1}{3} \text{tr}(\beta_x E_y). \quad (\text{B.439})$$

Using a computer to diagonalise $\rho = \frac{1}{3} \sum_{x=0}^2 |\beta_x\rangle\langle\beta_x|$, we get that

$$\rho = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i| \implies \{\lambda_i\}_i = \left\{0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right\}, \quad \{|\lambda_i\rangle\}_i = \{|\psi^-\rangle, |\phi^-\rangle, |\psi^+\rangle, |\phi^+\rangle\}, \quad (\text{B.440})$$

where $\{|\psi^-\rangle, |\phi^-\rangle, |\psi^+\rangle, |\phi^+\rangle\}$ is the Bell basis. Therefore, we can write

$$\rho^{-\frac{1}{2}} = 2|\phi^-\rangle\langle\phi^-| + 2|\psi^+\rangle\langle\psi^+| + \sqrt{2}|\phi^+\rangle\langle\phi^+|. \quad (\text{B.441})$$

The probabilities can be computed using Born's rule as $p(y|x) = \text{tr}(\beta_x E_y)$. Using a computer, we can calculate all of them and group them into a matrix $\{C_{y,x} = p(y|x)\}_{y,x=0}^2$,

$$C = \begin{pmatrix} 0.9714 & 0.0143 & 0.0143 \\ 0.0143 & 0.9714 & 0.0143 \\ 0.0143 & 0.0143 & 0.9714 \end{pmatrix}. \quad (\text{B.442})$$

The marginal ρ_Y after performing the measurement is

$$\rho_Y = \sum_{x,y=0}^2 p(x)p(y|x)|y\rangle\langle y| = \frac{1}{3} \sum_{y=0}^2 \left(\sum_{x=0}^2 p(y|x) \right) |y\rangle\langle y| = \frac{1}{3} \sum_{y=0}^2 |y\rangle\langle y|. \quad (\text{B.443})$$

Hence $H(Y)_{\rho_Y} = \log(3)$. To compute the mutual information we also need

$$H(Y|X)_{\rho_{XY}} = -\frac{1}{3} \sum_{x=0}^2 (0.9714 \log(0.9714) + 0.0143 \log(0.0143) + 0.0143 \log(0.0143)) \approx 0.2159. \quad (\text{B.444})$$

Therefore, we get

$$I(X : Y) = I(Y : X) = H(Y)_{\rho_Y} - H(Y|X)_{\rho_{XY}} \approx \log(3) - 0.2159 \approx 1.3690, \quad (\text{B.445})$$

which is larger than double accessible information $I_{\text{acc}}(\mathcal{E})$ in (a).

C. SPECIALISED TOPICS - EXERCISES

C.1. Device-independence

C.1.1 Sum of squares decomposition ●

In Chapter 9.2 we used the following decomposition,

$$2\sqrt{2}\mathbb{1} - \mathcal{Z} = \frac{1}{\sqrt{2}} \left[\left(\frac{A_0 + A_1}{\sqrt{2}} - B_0 \right)^2 + \left(\frac{A_0 - A_1}{\sqrt{2}} - B_1 \right)^2 \right], \quad (\text{C.1})$$

where $\mathcal{Z} = A_0 \otimes B_0 + A_0 \otimes B_1 + A_1 \otimes B_0 - A_1 \otimes B_1$ is the CHSH operator, to certify the observables certified by the maximal quantum violation of the CHSH inequality. Show that the decomposition (C.1) holds for arbitrary observables representing projective measurements.

Hint: Note that in Eq (C.1) we have omitted the symbol $\otimes$ for simplicity and that Alice's and Bob's measurements commute, since they are on separate Hilbert spaces.

Solution (p. 204)

C.1.2 Anticommutation and Pauli matrices ●●●●

Let A and B be two Hermitian operators such that $A^2 = B^2 = \mathbb{1}$ and $\{A, B\} = 0$. We will prove that there exists a unitary transformation, U , such that

$$U^\dagger AU = \sigma_X \otimes \mathbb{1}, \quad \text{and} \quad U^\dagger BU = \sigma_Z \otimes \mathbb{1}. \quad (\text{C.2})$$

(a) Define the projectors $\Pi_\pm = \frac{1}{2}(\mathbb{1} \pm B)$ and show that

$$\Pi_\pm A \Pi_\pm = 0 \quad \text{and} \quad A = \Pi_+ A \Pi_- + \Pi_- A \Pi_+. \quad (\text{C.3})$$

Use that $A^2 = \mathbb{1}$ to show that $\Pi_\pm A \Pi_\mp A \Pi_\pm = \Pi_\pm$.

Hint: Multiply both sides of $\{A, B\} = 0$ by $\Pi_\pm$.

(b) Express the projectors as $\Pi_\pm = \sum_j |u_j^\pm\rangle\langle u_j^\pm|$ and let $S = \sum_j (|u_j^+\rangle\langle u_j^-| + |u_j^-\rangle\langle u_j^+|)$. Show that $S = S^\dagger$, $S^2 = \mathbb{1}$ and $S\Pi_\pm = \Pi_\mp S$.

(c) Define the operator

$$U = \Pi_+ A \Pi_- S + \Pi_-, \quad (\text{C.4})$$

and show that it is unitary, i.e. $U^\dagger U = U U^\dagger = \mathbb{1}$.

(d) Show that $U^\dagger BU = B$ and $U^\dagger AU = S$ and that this corresponds to

$$U^\dagger BU = \sigma_Z \otimes \mathbb{1} \quad \text{and} \quad U^\dagger AU = \sigma_X \otimes \mathbb{1}, \quad (\text{C.5})$$

once we relabel the basis elements as $|0, j\rangle := |u_j^+\rangle$ and $|1, j\rangle := |u_j^-\rangle$.

Solution (p. 204)

C.1.3 Mayers-Yao certification ●●

Consider a Bell scenario in which Alice has two inputs and Bob has three inputs, each with binary outcomes. Consider that the expectation values of their observables are found to be

$$\langle A_0 B_0 \rangle = \langle A_1 B_1 \rangle = 1, \quad \langle A_0 B_2 \rangle = \langle A_1 B_2 \rangle = \frac{1}{\sqrt{2}}, \quad \langle A_0 B_1 \rangle = \langle A_1 B_0 \rangle = 0, \quad \langle A_x \rangle = \langle B_y \rangle = 0, \quad (\text{C.6})$$

where $\langle A_x B_y \rangle = \langle \psi | A_x B_y | \psi \rangle$, $\langle A_x \rangle = \langle \psi | A_x | \psi \rangle$ and $\langle B_y \rangle = \langle \psi | B_y | \psi \rangle$ for some shared state pure state $|\psi\rangle$. We assume that the measurements are projective and we omit the tensor product for simplicity of the presentation.

(a) Show that the CHSH inequality is violated by the expectation values (C.6) if we suitably select which two of Bob's three measurements to use for the CHSH test. Does the violation reach Tsirelson's bound?

(b) Show that the relations (C.6) imply that

$$\{A_0, A_1\}|\psi\rangle = 0 \quad \text{and} \quad \{B_0, B_1\}|\psi\rangle = 0. \quad (\text{C.7})$$

Solution (p. 206)

C.1.4 Circuit identity ●●

The two-qubit CNOT gate and the SWAP gate are defined by the actions $\text{CNOT}_{AA'}(|a\rangle_A \otimes |b\rangle_{A'}) = |a\rangle_A \otimes |a \oplus b\rangle_{A'}$ and $\text{SWAP}_{AA'}|j\rangle_A |k\rangle_{A'} = |k\rangle_A |j\rangle_{A'}$, respectively. Prove the decompositions used in Chapter 9.2, i.e.

$$(a) \text{CNOT}_{A'A} = (H \otimes \mathbb{1}) \times \text{CZ}_{AA'} \times (H \otimes \mathbb{1}).$$

Hint: Notice that the control qubit and the target qubit are interchanged.

$$(b) \text{SWAP}_{AA'} = \text{CNOT}_{AA'} \times \text{CNOT}_{A'A} \times \text{CNOT}_{AA'}.$$

Solution (p. 207)

C.1.5 Local isometry for certification of maximally entangled state ●

In Chapter 9.3 we certified the maximally entangled two-qubit state from the maximal violation of the CHSH inequality. There we used and illustrated a local isometry $\Phi_{A'A} \otimes \Phi_{B'B}$ which we applied to the initially unknown state $|\psi\rangle_{AB}$ and the auxiliary state $|00\rangle_{A'B'}$. Show that this isometry leads to the formula stated in the main text, namely

$$\begin{aligned} (\Phi_{A'A} \otimes \Phi_{B'B})[|\psi\rangle_{AB} \otimes |00\rangle_{A'B'}] = \frac{1}{4} & \left(|00\rangle \otimes (\mathbb{1} + W_0) \otimes (\mathbb{1} + B_0) |\psi\rangle + |01\rangle \otimes (\mathbb{1} + W_0) \otimes B_1 (\mathbb{1} - B_0) |\psi\rangle \right. \\ & \left. + |10\rangle \otimes W_1 (\mathbb{1} - W_0) \otimes (\mathbb{1} + B_0) |\psi\rangle + |11\rangle \otimes W_1 (\mathbb{1} - W_0) \otimes B_1 (\mathbb{1} - B_0) |\psi\rangle \right), \end{aligned} \quad (\text{C.8})$$

where W_0, W_1 are some observables measured by Alice and B_0, B_1 are observables measured by Bob.

Solution (p. 207)

C.2. Quantum cryptography

C.2.1 Candidate QKD protocol ●

Consider the BB84 protocol described in Chapter 10.1 and assume that now Alice sends two qubit states $|\psi_{a_i x_i}\rangle = |x_i\rangle \otimes |a_i(x_i)\rangle$, where

$$\begin{aligned} |a_i(x_i)\rangle &= |a_i\rangle, & \text{if } x_i = 0, \\ |a_i(x_i)\rangle &= H|a_i\rangle, & \text{if } x_i = 1, \end{aligned} \quad (\text{C.9})$$

where $H = \frac{1}{\sqrt{2}}(\sigma_X + \sigma_Z)$ is the Hadamard unitary matrix. Is this protocol secure?

Solution (p. 209)

C.2.2 Mutual information in cloning attack ●●●

Eve attacks the BB84 protocol by intercepting Alice's state $|\psi\rangle$ and passing it through a quantum cloning machine. Suppose that the cloning machine is such that any one of the BB84 states is transformed into two identical copies, $\rho = \frac{1}{2}(\mathbb{1} + \eta \hat{n} \cdot \vec{\sigma})$, where $\hat{n}$ is the Bloch vector of $|\psi\rangle$, $\eta \in [0, 1]$ is some shrinking factor and $\vec{\sigma}$ is the vector of the Pauli matrices.

(a) Assume that Alice and Bob both select the Z -basis. Calculate the mutual information between Alice choice of eigenstate and Bob's outcome and show that it is

$$I(\mathbf{a} : \mathbf{b}) = 1 - h(F), \quad (\text{C.10})$$

where $h(X) = -p \log(p) - (1-p) \log(1-p)$ is the binary entropy and F is the fidelity between Alice's state ψ and the cloned state ρ .

(b) Repeat this calculation but now for the X -basis.

(c) Find the maximum value achievable of the mutual information.

Hint: Use the maximum cloning fidelity discussed in Chapter 10.3.

Solution (p. 209)

C.2.3 Six-state protocol ●●●

The six-state protocol is an extension of the BB84 protocol. In this case, instead of using only the four eigenstates of σ_Z and σ_X , Alice and Bob use also the two eigenstates of σ_Y . The steps of the protocol are the following:

- Alice privately and randomly selects inputs $x_i \in \{0, 1, 2\}$ and $a_i \in \{0, 1\}$ for $i = 1, \dots, N$ and prepares a qubit state $|\psi_{a_i x_i}\rangle$. She associates $x_i = 0$ with the Z -basis, $x_i = 1$ with the X -basis and $x_i = 2$ with the Y -basis, i.e. she prepares the and sends to Bob one of the states

$$\begin{aligned} |\psi_{00}\rangle &= |0\rangle, & |\psi_{10}\rangle &= |1\rangle, \\ |\psi_{01}\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), & |\psi_{11}\rangle &= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \\ |\psi_{02}\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle), & |\psi_{12}\rangle &= \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle). \end{aligned} \quad (\text{C.11})$$

- Bob privately and randomly selects $y_i \in \{0, 1, 2\}$ for $i = 1, \dots, N$ and measures the incoming state in the Z , X or Y basis if $y_i = 0, 1$ and 2 , respectively.
- Alice and Bob publicly announce their bases and, as in BB84, the set of rounds where the bases match are kept. The set of kept round indices is $R = \{i | x_i = y_i\}$.
- Alice and Bob quantify their errors through the parameter

$$\delta = \frac{\#\text{events with } a_i \neq b_i}{|S|}, \quad (\text{C.12})$$

where S is the subset of R used for testing security.

Consider the case in which the eavesdropper Eve performs a cloning attack on this protocol.

- Calculate δ under intercept-resend attack.
- From Chapter 10.2, we know that the secure error threshold for a cloning on the BB84 protocol is $\delta \approx 14.6\%$. What is the corresponding threshold for the six-state protocol? Which cloning attack should Eve use?
- For the cloning attack used in (b), calculate the mutual information $I(\mathbf{a} : \mathbf{e})$ as a function of δ . What is the mutual information at the the maximum error threshold?

Hint: Use that the error δ corresponds to the probability of Alice and Eve finding a different outcome, i.e. $p(a \neq e|e) = \delta$.

Solution (p. 210)

C.3. Markovian master equations

C.3.1 Invariant transformations and jump operators ●

Show that the Lindblad equation,

$$\frac{d\rho}{dt} = i[\rho, H] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (\text{C.13})$$

is invariant under the following transformations of the jump operators $\{L_k\}_k$ and of the Hamiltonian,

$$\begin{aligned} L_k &\longrightarrow L_k + a_k \mathbb{1}, & a_k &\in \mathbb{C}, \\ H &\longrightarrow H + \frac{1}{2i} \sum_k \gamma_k (a_k L_k^\dagger - a_k^* L_k). \end{aligned} \quad (\text{C.14})$$

Solution (p. 211)

C.3.2 Dissipation with finite temperature bath ●●●

Consider a two-level atom coupled to a thermal bath with finite temperature T . Fix the ground state energy to zero and the excited state energy to E . Set the free Hamiltonian to $H_{\text{free}} = E|1\rangle\langle 1|$. With some rate γ^- the atom interacts with the environment and loses a photon, de-exciting to the ground state; with some other rate γ^+ , the atom gains a photon and excites to the first level. To these two processes we associate the jump operators $\sigma_- = |0\rangle\langle 1|$ and $\sigma_+ = |1\rangle\langle 0|$, respectively. The master equation becomes

$$\frac{d\rho}{dt} = \frac{iE}{2}[\sigma_Z, \rho] + \gamma^+ \mathcal{D}[\sigma_+]\rho + \gamma^- \mathcal{D}[\sigma_-]\rho, \quad (\text{C.15})$$

where the dissipator is $\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$, and we used that $[H_{\text{free}}, \rho] = E\left[\frac{1-\sigma_Z}{2}, \rho\right] = -\frac{E}{2}[\sigma_Z, \rho]$.

(a) Determine the action of the Liouvillian, $\mathcal{L}\rho$. Then, vectorize the density matrix,

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \longrightarrow \rho_v = \begin{pmatrix} \rho_{00} \\ \rho_{01} \\ \rho_{10} \\ \rho_{11} \end{pmatrix}, \quad (\text{C.16})$$

and compute the Liouvillian superoperator matrix, i.e. the matrix $\mathcal{L}_{\text{sup}}$ for which $\mathcal{L}_{\text{sup}}\rho_v$ gives the vectorized $\mathcal{L}\rho$.

(b) Consider that the atom is initially in the ground state, i.e. $\rho(0) = |0\rangle\langle 0|$. Using the Liouvillian superoperator matrix, derive $\rho(t)$, recalling that $\rho(t) = e^{\mathcal{L}t}\rho(0)$.

Hint: Use a computer to calculate the exponential of the Liouvillian.

(c) Take the long-time limit ($t \rightarrow +\infty$) of $\rho(t)$ and compare it with the Gibbs state,

$$\rho_G = \frac{e^{-\beta H_{\text{free}}}}{\text{tr}(e^{-\beta H_{\text{free}}})}, \quad (\text{C.17})$$

where $\beta = \frac{1}{k_B T}$ and k_B is the Boltzmann constant. Show that in the long-time limit $\rho(t)$ becomes the Gibbs state if $\frac{\gamma^+}{\gamma^-} = e^{-\beta E}$. This is called the *detailed balance condition*.

Solution (p. 211)

C.3.3 Energy currents in open quantum systems ●●●

The average energy of a quantum system is $\langle E \rangle = \text{tr}(\rho H)$.

(a) Use the Lindblad equation to show that

$$\frac{d\langle E \rangle}{dt} = \sum_k \gamma_k \text{tr}(\mathcal{D}[L_k]\rho H), \quad (\text{C.18})$$

where $\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$.

(b) It is often useful to define the average currents between the system and a particular bath as $\langle J(t) \rangle = \sum_k \gamma_k \text{tr}(\mathcal{D}[L_k]\rho(t)H)$, where k runs over all dissipator contributions with the bath. Consider the state

$$\rho(t) = \begin{pmatrix} \frac{\gamma^- + \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} & 0 \\ 0 & \frac{\gamma^+ - \gamma^- e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} \end{pmatrix} \quad (\text{C.19})$$

derived in part (b) of Exercise C.3.2 and compute the average current using the Hamiltonian $H = E|1\rangle\langle 1|$.

(c) Compute the average current for the state in (C.19) when $t \rightarrow +\infty$. How do you interpret this result?

Solution (p. 212)

C.3.4 Damped coupled resonators ●●●

Consider two coupled resonators. Their modes are described by two bosonic field operators a and b , such that $[a, a^\dagger] = [b, b^\dagger] = 1$. Let ω_a and ω_b be their resonance frequencies. The Hamiltonian can be written as

$$H = \omega_a a^\dagger a + \omega_b b^\dagger b + g(ab^\dagger + a^\dagger b) \quad (\text{C.20})$$

in the rotating wave approximation, i.e. when $g \ll \omega_a, \omega_b$, with g being the coupling constant. The Lindblad equation describing the evolution of the system is

$$\frac{d\rho}{dt} = i[\rho, H] + \gamma_a \mathcal{D}[a]\rho + \gamma_b \mathcal{D}[b]\rho, \quad (\text{C.21})$$

where $\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$. Note that now ρ is infinite dimensional.

- (a) Write the equation of motion for $\langle a(t) \rangle$ and $\langle b(t) \rangle$.

Hint: Use $\langle a \rangle = \text{tr}(\rho a)$, the Lindblad equation (C.21) and the commutation relations between bosonic field operators.

- (b) Define a vector $\begin{pmatrix} \langle a(t) \rangle \\ \langle b(t) \rangle \end{pmatrix}$ and express the equations of motion as a matrix differential equation of the form

$$\begin{pmatrix} \dot{\langle a(t) \rangle} \\ \dot{\langle b(t) \rangle} \end{pmatrix} = M \begin{pmatrix} \langle a(t) \rangle \\ \langle b(t) \rangle \end{pmatrix}. \quad (\text{C.22})$$

Compute the eigenvalues and the eigenvectors of M at resonance $\omega = \omega_a = \omega_b$.

- (c) Define $\langle A \rangle = \langle a \rangle e^{i\omega t}$ and $\langle B \rangle = \langle b \rangle e^{i\omega t}$ and show that the equations of motion of part (a) can be expressed as

$$\frac{d\langle A \rangle}{dt} = -ig\langle B \rangle - \frac{\gamma_a}{2}\langle A \rangle, \quad \frac{d\langle B \rangle}{dt} = -ig\langle A \rangle - \frac{\gamma_b}{2}\langle B \rangle. \quad (\text{C.23})$$

- (d) Rewrite (C.23) as a single second-order differential equation of the form

$$\frac{d^2\langle A \rangle}{dt^2} + 2\zeta\omega_0 \frac{d\langle A \rangle}{dt} + \omega_0^2\langle A \rangle = 0. \quad (\text{C.24})$$

This is the differential equation for a damped harmonic oscillator. Find the expressions of ζ and ω_0 . The critical damping corresponds to $\zeta = 1$. What does this imply for the eigenvalues and the eigenvectors of M found in (b)?

Solution (p. 213)

C.4. Quantum estimation theory

C.4.1 Fisher information for a binary random variable ●●

Consider the binary random variable X with distribution $p_X(0) = 1 - \theta$ and $p_X(1) = \theta$.

- (a) Compute the Fisher information. For which value of θ do we get the minimum? What is the value of the Fisher information for $\theta \in \{0, 1\}$?
- (b) Compute the Cramér-Rao bound for unbiased estimators of θ over N independent and identical samples.

Solution (p. 215)

C.4.2 Inverse relations ●●

In our derivation of the quantum version of the Cramér-Rao bound, we defined in Chapter 12.3 the map $\Omega(A) = \frac{1}{2}A\rho_\theta + \frac{1}{2}\rho_\theta A$ and claimed that its inverse is

$$\Omega^{-1}(A) = \sum_{j,k} \frac{2}{\lambda_j + \lambda_k} A_{jk} |\psi_j\rangle\langle\psi_k|, \quad (\text{C.25})$$

where $\rho_\theta = \sum_l \lambda_l |\psi_l\rangle\langle\psi_l|$ is the spectral decomposition of ρ_θ and $A_{jk} = \langle\psi_j|A|\psi_k\rangle$.

- (a) Verify that this is the inverse of $\Omega(A)$, i.e. that $\Omega^{-1}(\Omega(A)) = A$.
- (b) Use that $L_\theta = \Omega^{-1}(\dot{\rho}_\theta)$ and $\mathcal{I}^Q(\rho_\theta) = \text{tr}(L_\theta^2 \rho_\theta)$ to derive the expression (12.31) for the quantum Fisher information.
Hint: Show that $\mathcal{I}^Q(\rho_\theta) = \sum_{j,k} \lambda_j |(\dot{L}_\theta)_{jk}|^2$ and compute $\dot{\rho}_\theta$ in terms of the eigenvalues $\{\lambda_j\}_j$.

Solution (p. 216)

C.4.3 Properties of quantum Fisher information ●●●●

Prove the following properties of the quantum Fisher information:

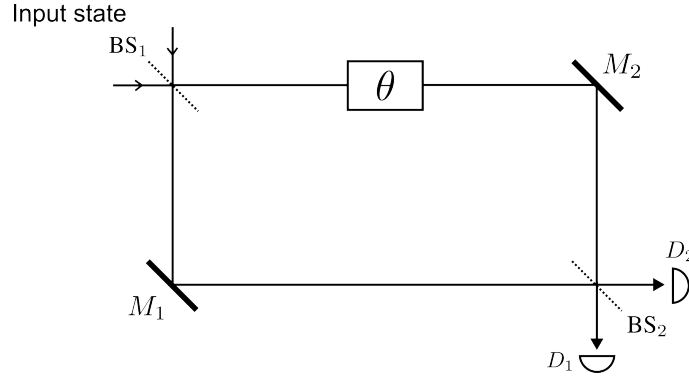
- (a) $\mathcal{I}^Q(\rho_\theta) = \text{tr}(L_\theta \dot{\rho}_\theta)$.
- (b) Additivity: $\mathcal{I}^Q(\rho_1 \otimes \rho_2) = \mathcal{I}^Q(\rho_1) + \mathcal{I}^Q(\rho_2)$.
- (c) Convexity: $\mathcal{I}^Q(\lambda \rho_1 + (1 - \lambda) \rho_2) \leq \lambda \mathcal{I}^Q(\rho_1) + (1 - \lambda) \mathcal{I}^Q(\rho_2)$, for all $\lambda \in [0, 1]$.
- (d) For pure states $\psi_\theta = |\psi_\theta\rangle\langle\psi_\theta|$, it simplifies to $\mathcal{I}^Q(\psi_\theta) = 4(\langle\dot{\psi}_\theta|\dot{\psi}_\theta\rangle - |\langle\psi_\theta|\dot{\psi}_\theta\rangle|^2)$.

Hint: Use that $\dot{\psi}_\theta = |\dot{\psi}_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\dot{\psi}_\theta|$ and project this expression onto $|\psi_\theta\rangle$ and $|\dot{\psi}_\theta\rangle$. Use also that the derivative, when applied to a ray, is anti-Hermitian, i.e. $(\frac{d}{d\theta}\langle\psi|)|\psi\rangle = -\langle\psi|(\frac{d}{d\theta}|\psi\rangle)$.

Solution (p. 217)

C.4.4 Interferometry with photons ●●●●

Consider a Mach-Zender interferometer, which consists of two beam splitters, BS_1 and BS_2 , and a pair of mirrors, M_1 and M_2 , as represented in the Figure below.



In the upper arm is placed a phase shifter, that may be understood as a unitary operator that introduces a relative phase shift θ between the two paths. Two detectors, D_1 and D_2 , are at the end of each path. Let a and b be two possible modes for a photon, and denote as $|1\rangle_a|0\rangle_b$ the associated one-photon state in the Fock space (i.e. it denotes the number of photons in mode a and mode b , respectively). For different input states, it is possible to calculate the probabilities of D_1 and D_2 clicking. Using these probabilities, compute the Fisher information $\mathcal{I}(p(x|\theta))$ in the following cases.

- (a) Input state 1:

$$|1\rangle_a|0\rangle_b, \quad p(1|\theta) = \frac{1}{2}(1 - \cos\theta), \quad p(2|\theta) = \frac{1}{2}(1 + \cos\theta), \quad (\text{C.26})$$

where $p(1|\theta)$ and $p(2|\theta)$ are the probabilities of D_1 and D_2 clicking, respectively;

- (b) Input state 2:

$$\frac{1}{\sqrt{2}}(|2\rangle_a|0\rangle_b + |0\rangle_a|2\rangle_b), \quad p(+|\theta) = \frac{1}{2}(1 + \cos(2\theta)), \quad p(-|\theta) = \frac{1}{2}(1 - \cos(2\theta)), \quad (\text{C.27})$$

where $p(+|\theta)$ is the probability that one detector clicks for both photons, and $p(-|\theta)$ is the probability that each detector clicks for one of the photons;

- (c) Input state 3:

$$\frac{1}{\sqrt{2}}(|N\rangle_a|0\rangle_b + |0\rangle_a|N\rangle_b), \quad p(+|\theta) = \frac{1}{2}(1 + \cos(N\theta)), \quad p(-|\theta) = \frac{1}{2}(1 - \cos(N\theta)), \quad (\text{C.28})$$

where $p(+|\theta)$ is the probability that all N photons are detected by the same detector, and $p(-|\theta)$ is the probability of this not happening. What is the scaling of the Fisher information for these states?

Solution (p. 218)

C.4.5 Quantum Fisher information ●

Consider the unitary $U = e^{-i\theta H}$ where H is a Hermitian operator and θ is a real parameter that we want to estimate. Determine the largest possible value of the quantum Fisher information over all pure states and show that this value is achieved by any state of the form

$$\frac{1}{\sqrt{2}}|\lambda_{\min}\rangle + \frac{e^{i\phi}}{\sqrt{2}}|\lambda_{\max}\rangle, \quad (\text{C.29})$$

where $|\lambda_{\max}\rangle$ and $|\lambda_{\min}\rangle$ are the eigenstates of H corresponding to the largest and smallest eigenvalues.

Solution (p. 219)

C.4.6 Multi-copy quantum Fisher information ●●

Consider that we have N identical non-interacting copies of a system and that to each copy we apply the unitary $U = e^{-i\theta H}$, where H is a finite-dimensional Hermitian operator and θ is a real parameter that we want to estimate. The total unitary is therefore $U^{(N)} = e^{-i\theta H^{(N)}}$, where $H^{(N)} = H_1 + \dots + H_N$ and $H_1 = H \otimes \mathbb{1} \otimes \dots \otimes \mathbb{1}$, $H_2 = \mathbb{1} \otimes H \otimes \mathbb{1} \otimes \dots \otimes \mathbb{1}$ etc.

- (a) Consider that we send into $U^{(N)}$ the product state $|\psi^{(N)}\rangle = |\psi\rangle \otimes \dots \otimes |\psi\rangle$. Compute the quantum Fisher information.
- (b) Show that the optimal entangled input state to $U^{(N)}$ is

$$\frac{1}{\sqrt{2}}|\lambda_{\min}, \dots, \lambda_{\min}\rangle + \frac{e^{i\phi}}{\sqrt{2}}|\lambda_{\max}, \dots, \lambda_{\max}\rangle \quad (\text{C.30})$$

for any ϕ , where $|\lambda_{\max}\rangle$ and $|\lambda_{\min}\rangle$ are the eigenstates of H associated with the largest and smallest eigenvalue.

Hint: Use the result of Exercise C.4.5.

Solution (p. 219)

SPECIALISED TOPICS - SOLUTIONS

C.1. Device-independence

C.1.1 Solution (Text at p. 197)

Expanding the right hand side of (C.1) and recalling that Alice and Bob's measurements commute, we obtain

$$\begin{aligned} \frac{1}{\sqrt{2}} \left[\left(\frac{A_0 + A_1}{\sqrt{2}} - B_0 \right)^2 + \left(\frac{A_0 - A_1}{\sqrt{2}} - B_1 \right)^2 \right] &= \frac{1}{\sqrt{2}} \left[\left(\frac{A_0 + A_1}{\sqrt{2}} \right)^2 + \underbrace{B_0^2}_{=1} - 2 \left(\frac{A_0 + A_1}{\sqrt{2}} \right) \otimes B_0 \right] \\ &+ \frac{1}{\sqrt{2}} \left[\left(\frac{A_0 - A_1}{\sqrt{2}} \right)^2 + \underbrace{B_1^2}_{=1} - 2 \left(\frac{A_0 - A_1}{\sqrt{2}} \right) \otimes B_1 \right], \end{aligned} \quad (\text{C.31})$$

where we have used that A_0, A_1, B_0, B_1 are observables representing projective binary-outcome measurements, from which $A_0^2 = A_1^2 = B_0^2 = B_1^2 = \mathbb{1}$. Note that we have re-inserted the tensor product notation when convenient. Continuing with the simplifications,

$$\begin{aligned} &\frac{1}{\sqrt{2}} \left[\left(\frac{A_0 + A_1}{\sqrt{2}} \right)^2 + \mathbb{1} - 2 \left(\frac{A_0 + A_1}{\sqrt{2}} \right) \otimes B_0 \right] + \frac{1}{\sqrt{2}} \left[\left(\frac{A_0 - A_1}{\sqrt{2}} \right)^2 + \mathbb{1} - 2 \left(\frac{A_0 - A_1}{\sqrt{2}} \right) \otimes B_1 \right] \\ &= \frac{1}{\sqrt{2}} \left[\frac{1}{2} (A_0^2 + A_1^2 + A_0 A_1 + A_1 A_0) + \mathbb{1} - 2 \left(\frac{A_0 + A_1}{\sqrt{2}} \right) \otimes B_0 \right] \\ &\quad + \frac{1}{\sqrt{2}} \left[\frac{1}{2} (A_0^2 + A_1^2 - A_0 A_1 - A_1 A_0) + \mathbb{1} - 2 \left(\frac{A_0 - A_1}{\sqrt{2}} \right) \otimes B_1 \right] \\ &= \frac{1}{\sqrt{2}} \left[\frac{1}{2} (2\mathbb{1}) + \mathbb{1} - 2 \left(\frac{A_0 + A_1}{\sqrt{2}} \right) \otimes B_0 \right] + \frac{1}{\sqrt{2}} \left[\frac{1}{2} (2\mathbb{1}) + \mathbb{1} - 2 \left(\frac{A_0 - A_1}{\sqrt{2}} \right) \otimes B_1 \right] \\ &= \frac{1}{\sqrt{2}} \left[4\mathbb{1} - 2 \left(\frac{A_0 + A_1}{\sqrt{2}} \right) \otimes B_0 - 2 \left(\frac{A_0 - A_1}{\sqrt{2}} \right) \otimes B_1 \right]. \end{aligned} \quad (\text{C.32})$$

Next, notice that, using the definition of $\mathcal{Z}$, we get

$$-2 \left(\frac{A_0 + A_1}{\sqrt{2}} \right) \otimes B_0 - 2 \left(\frac{A_0 - A_1}{\sqrt{2}} \right) \otimes B_1 = -\sqrt{2} (A_0 \otimes B_0 + A_1 \otimes B_0 + A_0 \otimes B_1 - A_1 \otimes B_1) = -\sqrt{2} \mathcal{Z}. \quad (\text{C.33})$$

Therefore, we get

$$\frac{1}{\sqrt{2}} \left[4\mathbb{1} - 2 \left(\frac{A_0 + A_1}{\sqrt{2}} \right) \otimes B_0 - 2 \left(\frac{A_0 - A_1}{\sqrt{2}} \right) \otimes B_1 \right] = 2\sqrt{2}\mathbb{1} - \mathcal{Z}. \quad (\text{C.34})$$

C.1.2 Solution (Text at p. 197)

(a) First notice that, from the definition of the projectors $\Pi_{\pm}$, the following relations hold

$$\Pi_+ + \Pi_- = \mathbb{1}, \quad \Pi_+ - \Pi_- = B, \quad \Pi_+ \Pi_- = \Pi_- \Pi_+ = 0. \quad (\text{C.35})$$

Now consider the anticommutation relation and multiply both sides by Π_+ ,

$$\Pi_+ \{A, B\} \Pi_+ = 0 \quad \implies \quad \Pi_+ A B \Pi_+ + \Pi_+ B A \Pi_+ = 0. \quad (\text{C.36})$$

Using (C.35), we can rewrite it as

$$\begin{aligned} 0 &= \Pi_+ A B \Pi_+ + \Pi_+ B A \Pi_+ = \Pi_+ A (\Pi_+ - \Pi_-) \Pi_+ + \Pi_+ (\Pi_+ - \Pi_-) A \Pi_+ \\ &= \Pi_+ A \Pi_+^2 - \underbrace{\Pi_+ A \Pi_- \Pi_+}_{=0} + \underbrace{\Pi_+^2 A \Pi_+ - \Pi_+ \Pi_- A \Pi_+}_{=0} \stackrel{(\text{C.35})}{=} 2\Pi_+ A \Pi_+, \end{aligned} \quad (\text{C.37})$$

from which $\Pi_+ A \Pi_+ = 0$. Similarly, we can multiply both sides of the anticommutator by Π_- to get

$$\begin{aligned} 0 &= \Pi_- A B \Pi_- + \Pi_- B A \Pi_- = \Pi_- A (\Pi_+ - \Pi_-) \Pi_- + \Pi_- (\Pi_+ - \Pi_-) A \Pi_- \\ &= \Pi_- A \underbrace{\Pi_+ \Pi_-}_{=0} - \Pi_- A \Pi_-^2 + \Pi_- \Pi_+ \underbrace{A \Pi_-}_{=0} - \Pi_-^2 A \Pi_- = -2 \Pi_- A \Pi_-, \end{aligned} \quad (\text{C.38})$$

from which $\Pi_- A \Pi_- = 0$. Moreover, using that $B^2 = \mathbb{1}$, we can write

$$\begin{aligned} AB + BA = 0 &\implies A = -BAB \stackrel{(\text{C.35})}{=} -(\Pi_+ - \Pi_-)A(\Pi_+ - \Pi_-) \\ &= -\underbrace{\Pi_+ A \Pi_+}_{=0} + \Pi_+ A \Pi_- + \Pi_- A \Pi_+ - \underbrace{\Pi_- A \Pi_-}_{=0} \\ &= \Pi_+ A \Pi_- + \Pi_- A \Pi_+. \end{aligned} \quad (\text{C.39})$$

Therefore,

$$\begin{aligned} \mathbb{1} &= A^2 = (\Pi_+ A \Pi_- + \Pi_- A \Pi_+)(\Pi_+ A \Pi_- + \Pi_- A \Pi_+) \\ &= \Pi_+ A \underbrace{\Pi_- \Pi_+}_{=0} A \Pi_- + \Pi_+ A \Pi_-^2 A \Pi_+ + \Pi_- A \Pi_+^2 A \Pi_- + \Pi_- A \underbrace{\Pi_+ \Pi_-}_{=0} A \Pi_+ \\ &= \Pi_+ A \Pi_- A \Pi_+ + \Pi_- A \Pi_+ A \Pi_-. \end{aligned} \quad (\text{C.40})$$

Since from (C.35), we know that $\Pi_+ + \Pi_- = \mathbb{1}$, we get

$$\Pi_{\pm} = \Pi_{\pm} A \Pi_{\mp} A \Pi_{\pm}. \quad (\text{C.41})$$

(b) By definition, we have $S^{\dagger} = \sum_j (|u_j^-\rangle\langle u_j^+| + |u_j^+\rangle\langle u_j^-|) = S$. Moreover,

$$\begin{aligned} S^2 &= \sum_j (|u_j^+\rangle\langle u_j^-| + |u_j^-\rangle\langle u_j^+|) \sum_k (|u_k^+\rangle\langle u_k^-| + |u_k^-\rangle\langle u_k^+|) \\ &= \sum_{j,k} \underbrace{(|u_j^+\rangle\langle u_j^-|u_k^+\rangle\langle u_k^-| + |u_j^-\rangle\langle u_j^+|u_k^-\rangle\langle u_k^+|)}_{=0} + \underbrace{|u_j^-\rangle\langle u_j^+|u_k^-\rangle\langle u_k^+|}_{=\delta_{j,k}} + \underbrace{|u_j^+\rangle\langle u_j^-|u_k^+\rangle\langle u_k^-|}_{=\delta_{j,k}} + \underbrace{|u_j^-\rangle\langle u_j^+|u_k^-\rangle\langle u_k^+|}_{=0} \\ &= \sum_j (|u_j^+\rangle\langle u_j^+| + |u_j^-\rangle\langle u_j^-|) \stackrel{(\text{C.35})}{=} \mathbb{1}, \end{aligned} \quad (\text{C.42})$$

where we have used that $\langle u_j^{\pm}|u_k^{\mp}\rangle = 0$, since $\Pi_{\pm}\Pi_{\mp} = 0$. Using again the spectral decomposition of $\Pi_{\pm}$, we get

$$\begin{aligned} S \Pi_+ S &= \sum_j (|u_j^+\rangle\langle u_j^-| + |u_j^-\rangle\langle u_j^+|) \sum_k |u_k^+\rangle\langle u_k^+| \sum_{j'} (|u_{j'}^+\rangle\langle u_{j'}^-| + |u_{j'}^-\rangle\langle u_{j'}^+|) \\ &= \sum_{j,k,j'} \underbrace{(|u_j^+\rangle\langle u_j^-|u_k^+\rangle\langle u_k^+|u_{j'}^-\rangle\langle u_{j'}^+| + |u_j^-\rangle\langle u_j^+|u_k^-\rangle\langle u_k^+|u_{j'}^+\rangle\langle u_{j'}^-|)}_{=0} \\ &\quad + \underbrace{|u_j^-\rangle\langle u_j^+|u_k^-\rangle\langle u_k^+|u_{j'}^-\rangle\langle u_{j'}^+|}_{=\delta_{j,k}} + \underbrace{|u_j^+\rangle\langle u_j^-|u_k^+\rangle\langle u_k^+|u_{j'}^+\rangle\langle u_{j'}^-|}_{=\delta_{k,j'}} + \underbrace{|u_j^-\rangle\langle u_j^+|u_k^-\rangle\langle u_k^+|u_{j'}^+\rangle\langle u_{j'}^-|}_{=0} = \sum_j |u_j^-\rangle\langle u_j^-| = \Pi_-, \end{aligned} \quad (\text{C.43})$$

from which $S \Pi_+ = \Pi_- S$, since $S^2 = \mathbb{1}$.

(c) Since Π_+ , Π_- , A and S are all Hermitian operators, $U^{\dagger} = (S \Pi_- A \Pi_+ + \Pi_-)$. Therefore,

$$\begin{aligned} U^{\dagger} U &= (S \Pi_- A \Pi_+ + \Pi_-)(\Pi_+ A \Pi_- S + \Pi_-) \\ &= S \Pi_- A \Pi_+^2 A \Pi_- S + S \Pi_- A \underbrace{\Pi_+ \Pi_-}_{=0} + \underbrace{\Pi_- \Pi_+}_{=0} A \Pi_- S + \Pi_-^2 \\ &\stackrel{(\text{C.41})}{=} S \Pi_- S + \Pi_- = \Pi_+ + \Pi_- \stackrel{(\text{C.35})}{=} \mathbb{1}, \end{aligned} \quad (\text{C.44})$$

where we have used that $S \Pi_- S = \Pi_+$.

(d) We can start computing

$$\begin{aligned}
 U^\dagger BU &= (S\Pi_- A\Pi_+ + \Pi_-)B(\Pi_+ A\Pi_- S + \Pi_-) \\
 &= S\Pi_- A\Pi_+ B\Pi_+ A\Pi_- S + S\Pi_- A\Pi_+ B\Pi_- + \Pi_- B\Pi_+ A\Pi_- S + \Pi_- B\Pi_- \\
 &\stackrel{(C.35)}{=} S\Pi_- A\Pi_+ (\Pi_+ - \Pi_-)\Pi_+ A\Pi_- S + S\Pi_- A\Pi_+ (\Pi_+ - \Pi_-)\Pi_- + \Pi_- (\Pi_+ - \Pi_-)\Pi_+ A\Pi_- S + \Pi_- (\Pi_+ - \Pi_-)\Pi_- \\
 &= \underbrace{S\Pi_- A\Pi_+ A\Pi_- S}_{=\Pi_+} - \Pi_-^2 \Pi_- = \Pi_+ - \Pi_- \stackrel{(C.35)}{=} B,
 \end{aligned} \tag{C.45}$$

where we have used that $B = \Pi_+ - \Pi_-$, $\Pi_\pm \Pi_\mp = 0$ and (C.41). In the same way, we can compute

$$\begin{aligned}
 U^\dagger AU &= (S\Pi_- A\Pi_+ + \Pi_-)A(\Pi_+ A\Pi_- S + \Pi_-) \\
 &= S\Pi_- A \underbrace{\Pi_+ A\Pi_+}_{=0} A\Pi_- S + S \underbrace{\Pi_- A\Pi_+ A\Pi_-}_{=\Pi_-} + \underbrace{\Pi_- A\Pi_+ A\Pi_-}_{=\Pi_-} S + \underbrace{\Pi_- A\Pi_-}_{=0} \\
 &= \underbrace{S\Pi_- + \Pi_- S}_{=\Pi_+ S} \stackrel{(C.35)}{=} (\underbrace{\Pi_+ + \Pi_-}_{=\mathbb{1}}) S = S,
 \end{aligned} \tag{C.46}$$

where we have used that $\Pi_\pm A\Pi_\pm = 0$ and (C.41). Now, using the spectral decomposition of $\Pi_\pm$, we can write

$$\begin{aligned}
 U^\dagger BU &= B \stackrel{(C.41)}{=} \Pi_+ - \Pi_- = \sum_j (|u_j^+\rangle\langle u_j^+| - |u_j^-\rangle\langle u_j^-|) \\
 &= \sum_j (|0, j\rangle\langle 0, j| - |1, j\rangle\langle 1, j|) = (|0\rangle\langle 0| - |1\rangle\langle 1|) \otimes \sum_j |j\rangle\langle j| = \sigma_Z \otimes \mathbb{1},
 \end{aligned} \tag{C.47}$$

where we have introduced the relabeling of the bases, $|0, j\rangle := |u_j^+\rangle$ and $|1, j\rangle := |u_j^-\rangle$. In the same way,

$$\begin{aligned}
 U^\dagger AU &= S = \sum_j (|u_j^+\rangle\langle u_j^-| + |u_j^-\rangle\langle u_j^+|) = \sum_j (|0, j\rangle\langle 1, j| + |1, j\rangle\langle 0, j|) \\
 &= (|0\rangle\langle 1| + |1\rangle\langle 0|) \otimes \sum_j |j\rangle\langle j| = \sigma_X \otimes \mathbb{1}.
 \end{aligned} \tag{C.48}$$

C.1.3 Solution (Text at p. 197)

- (a) In the CHSH test, both Alice and Bob have two measurements whereas in the given case Bob has three measurements. Simply ignore the setting B_1 , and consider B_2 as Bob's first setting in CHSH and B_0 as his second setting in a CHSH test. Then the CHSH inequality becomes

$$\langle A_0 B_2 \rangle + \langle A_0 B_0 \rangle + \langle A_1 B_2 \rangle - \langle A_1 B_0 \rangle = \frac{1}{\sqrt{2}} + 1 + \frac{1}{\sqrt{2}} - 0 = 1 + \sqrt{2}. \tag{C.49}$$

This violates the CHSH inequality but does not reach Tsirelson's bound.

- (b) First use that $\langle A_0 B_0 \rangle = \langle \psi | A_0 B_0 | \psi \rangle = 1$ and that $\langle A_1 B_1 \rangle = \langle \psi | A_1 B_1 | \psi \rangle = 1$. These imply the two relations⁸²

$$A_0 |\psi\rangle = B_0 |\psi\rangle, \quad \text{and} \quad A_1 |\psi\rangle = B_1 |\psi\rangle. \tag{C.50}$$

Now combine this with the relation $\langle A_0 B_1 \rangle = 0$. This gives

$$\langle \psi | A_0 A_1 | \psi \rangle = 0, \tag{C.51}$$

which means that $A_0 |\psi\rangle$ is orthogonal to $A_1 |\psi\rangle$. Next consider the relations $\langle A_0 B_2 \rangle = \langle A_1 B_2 \rangle = \frac{1}{\sqrt{2}}$. This tells us that $B_2 |\psi\rangle$ when projected onto $A_0 |\psi\rangle$ or $A_1 |\psi\rangle$ has length $1/\sqrt{2}$. Since the norm of $B_2 |\psi\rangle$ is one, it must follow that $B_2 |\psi\rangle$ is supported on the space spanned by $A_0 |\psi\rangle$ and $A_1 |\psi\rangle$, implying

$$B_2 |\psi\rangle = \frac{A_0 + A_1}{\sqrt{2}} |\psi\rangle. \tag{C.52}$$

⁸² We are omitting the tensor product for simplicity of the presentation, i.e. we write $\langle \psi | A_0 B_0 | \psi \rangle$ to denote $\langle \psi | A_0 \otimes B_0 | \psi \rangle$ and $A_0 |\psi\rangle$ to denote $(A_0 \otimes \mathbb{1}) |\psi\rangle$.

Multiply both sides by B_2 to obtain

$$B_2^2|\psi\rangle = B_2 \frac{A_0 + A_1}{\sqrt{2}}|\psi\rangle = \frac{A_0 + A_1}{\sqrt{2}}B_2|\psi\rangle \stackrel{\text{(C.52)}}{=} \left(\frac{A_0 + A_1}{\sqrt{2}}\right)^2 |\psi\rangle = \frac{1}{2}(A_0^2 + A_1^2 + \{A_0, A_1\})|\psi\rangle. \quad (\text{C.53})$$

Since $A_x^2 = \mathbb{1}$ and $B_y^2 = \mathbb{1}$, this is equivalent to

$$\{A_0, A_1\}|\psi\rangle = 0. \quad (\text{C.54})$$

Via (C.50) it then follows that also $\{B_0, B_1\}|\psi\rangle = 0$.

C.1.4 Solution (Text at p. 198)

(a) Recall the definition of the CZ-gate, the Hadamard gate and the CNOT gate,

$$\text{CZ}_{AA'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \text{CNOT}_{AA'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (\text{C.55})$$

In order to prove that

$$\text{CNOT}_{A'A} = (H \otimes \mathbb{1}) \times \text{CZ}_{AA'} \times (H \otimes \mathbb{1}), \quad (\text{C.56})$$

we can start computing

$$H \otimes \mathbb{1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}. \quad (\text{C.57})$$

Therefore, we get

$$(H \otimes \mathbb{1}) \times \text{CZ}_{AA'} \times (H \otimes \mathbb{1}) = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \quad (\text{C.58})$$

This is the CNOT gate in (C.56), where its control and target qubit registers interchanged, i.e. its action is $\text{CNOT}_{A'A}(|a\rangle \otimes |b\rangle) = |a \oplus b\rangle \otimes |b\rangle$.

(b) Using the expressions for the CNOT gate (C.55) and for the CNOT gate with control and target qubit registers interchanged (C.58), we can directly compute

$$\text{CNOT}_{AA'} \times \text{CNOT}_{A'A} \times \text{CNOT}_{AA'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (\text{C.59})$$

which is the SWAP gate with action $\text{SWAP}_{AA'} |j\rangle_A |k\rangle_{A'} = |k\rangle_A |j\rangle_{A'}$.

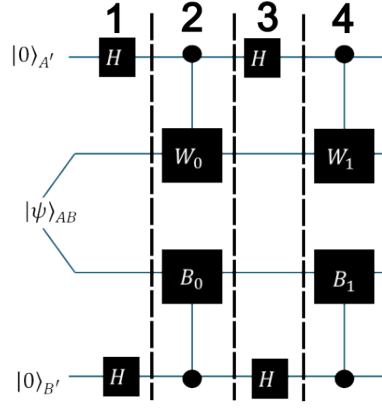
C.1.5 Solution (Text at p. 198)

To simplify the calculations, let us split the action of the circuit into four steps, as shown in Figure below.

Recall that the action of the Hadamard gate on the computational basis states is

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (\text{C.60})$$

We will write the control gates as CG where G is the gate implemented on the target register if the control register is turned on. We can compute each step in the circuit as follows.



(a) Step 1

$$\begin{aligned}
 |\varphi_1\rangle_{ABA'B'} &= (H_{A'} \otimes \mathbb{1}_A \otimes \mathbb{1}_B \otimes H_{B'}) |\psi\rangle_{AB} \otimes |00\rangle_{A'B'} = |\psi\rangle_{AB} \otimes \left(\frac{|0\rangle_{A'} + |1\rangle_{A'}}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle_{B'} + |1\rangle_{B'}}{\sqrt{2}} \right) \\
 &= \frac{1}{2} [|00\rangle_{A'B'} \otimes |\psi\rangle_{AB} + |01\rangle_{A'B'} \otimes |\psi\rangle_{AB} + |10\rangle_{A'B'} \otimes |\psi\rangle_{AB} + |11\rangle_{A'B'} \otimes |\psi\rangle_{AB}],
 \end{aligned} \tag{C.61}$$

where we used (C.60) on subsystems A' and B' .

(b) Step 2

$$\begin{aligned}
 |\varphi_2\rangle_{ABA'B'} &= ((CW_0)_{A'A} \otimes (CB_0)_{B'B}) |\varphi_1\rangle_{ABA'B'} \\
 &= \frac{1}{2} [|00\rangle_{A'B'} \otimes |\psi\rangle_{AB} + |01\rangle_{A'B'} \otimes (\mathbb{1} \otimes B_0) |\psi\rangle_{AB} \\
 &\quad + |10\rangle_{A'B'} \otimes (W_0 \otimes \mathbb{1}) |\psi\rangle_{AB} + |11\rangle_{A'B'} \otimes (W_0 \otimes B_0) |\psi\rangle_{AB}].
 \end{aligned} \tag{C.62}$$

(c) Step 3

Using again (C.60), we get

$$\begin{aligned}
 |\varphi_3\rangle_{ABA'B'} &= (H_{A'} \otimes \mathbb{1}_A \otimes \mathbb{1}_B \otimes H_{B'}) |\varphi_2\rangle_{ABA'B'} \\
 &= \frac{1}{2} \left[\left(\frac{|0\rangle_{A'} + |1\rangle_{A'}}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle_{B'} + |1\rangle_{B'}}{\sqrt{2}} \right) \otimes |\psi\rangle_{AB} + \left(\frac{|0\rangle_{A'} + |1\rangle_{A'}}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle_{B'} - |1\rangle_{B'}}{\sqrt{2}} \right) \otimes (\mathbb{1} \otimes B_0) |\psi\rangle_{AB} \right. \\
 &\quad \left. + \left(\frac{|0\rangle_{A'} - |1\rangle_{A'}}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle_{B'} + |1\rangle_{B'}}{\sqrt{2}} \right) \otimes (W_0 \otimes \mathbb{1}) |\psi\rangle_{AB} + \left(\frac{|0\rangle_{A'} - |1\rangle_{A'}}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle_{B'} - |1\rangle_{B'}}{\sqrt{2}} \right) \otimes (W_0 \otimes B_0) |\psi\rangle_{AB} \right],
 \end{aligned} \tag{C.63}$$

that becomes

$$\begin{aligned}
 |\varphi_3\rangle_{ABA'B'} &= (H_{A'} \otimes \mathbb{1}_A \otimes \mathbb{1}_B \otimes H_{B'}) |\varphi_2\rangle_{ABA'B'} \\
 &= \frac{1}{4} [|00\rangle_{A'B'} \otimes |\psi\rangle_{AB} + |01\rangle_{A'B'} \otimes |\psi\rangle_{AB} + |10\rangle_{A'B'} \otimes |\psi\rangle_{AB} + |11\rangle_{A'B'} \otimes |\psi\rangle_{AB} \\
 &\quad + |00\rangle_{A'B'} \otimes (\mathbb{1} \otimes B_0) |\psi\rangle_{AB} - |01\rangle_{A'B'} \otimes (\mathbb{1} \otimes B_0) |\psi\rangle_{AB} + |10\rangle_{A'B'} \otimes (\mathbb{1} \otimes B_0) |\psi\rangle_{AB} \\
 &\quad - |11\rangle_{A'B'} \otimes (\mathbb{1} \otimes B_0) |\psi\rangle_{AB} + |00\rangle_{A'B'} \otimes (W_0 \otimes \mathbb{1}) |\psi\rangle_{AB} + |01\rangle_{A'B'} \otimes (W_0 \otimes \mathbb{1}) |\psi\rangle_{AB} \\
 &\quad - |10\rangle_{A'B'} \otimes (W_0 \otimes \mathbb{1}) |\psi\rangle_{AB} - |11\rangle_{A'B'} \otimes (W_0 \otimes \mathbb{1}) |\psi\rangle_{AB} + |00\rangle_{A'B'} \otimes (W_0 \otimes B_0) |\psi\rangle_{AB} \\
 &\quad - |01\rangle_{A'B'} \otimes (W_0 \otimes B_0) |\psi\rangle_{AB} - |10\rangle_{A'B'} \otimes (W_0 \otimes B_0) |\psi\rangle_{AB} + |11\rangle_{A'B'} \otimes (W_0 \otimes B_0) |\psi\rangle_{AB}].
 \end{aligned} \tag{C.64}$$

(d) Step 4

$$\begin{aligned}
|\varphi_4\rangle_{ABA'B'} &= ((CW_1)_{A'A} \otimes (CB_1)_{B'B})|\varphi_3\rangle_{ABA'B'} \\
&= \frac{1}{4} [|00\rangle_{A'B'} \otimes |\psi\rangle_{AB} + |01\rangle_{A'B'} \otimes (\mathbb{1} \otimes B_1)|\psi\rangle_{AB} + |10\rangle_{A'B'} \otimes (W_1 \otimes \mathbb{1})|\psi\rangle_{AB} \\
&\quad + |11\rangle_{A'B'} \otimes (W_1 \otimes B_1)|\psi\rangle_{AB} + |00\rangle_{A'B'} \otimes (\mathbb{1} \otimes B_0)|\psi\rangle_{AB} - |01\rangle_{A'B'} \otimes (\mathbb{1} \otimes B_1B_0)|\psi\rangle_{AB} \\
&\quad + |10\rangle_{A'B'} \otimes (W_1 \otimes B_0)|\psi\rangle_{AB} - |11\rangle_{A'B'} \otimes (W_1 \otimes B_1B_0)|\psi\rangle_{AB} + |00\rangle_{A'B'} \otimes (W_0 \otimes \mathbb{1})|\psi\rangle_{AB} \\
&\quad + |01\rangle_{A'B'} \otimes (W_0 \otimes B_1)|\psi\rangle_{AB} - |10\rangle_{A'B'} \otimes (W_1W_0 \otimes \mathbb{1})|\psi\rangle_{AB} - |11\rangle_{A'B'} \otimes (W_1W_0 \otimes B_1)|\psi\rangle_{AB} \\
&\quad + |00\rangle_{A'B'} \otimes (W_0 \otimes B_0)|\psi\rangle_{AB} - |01\rangle_{A'B'} \otimes (W_0 \otimes B_1B_0)|\psi\rangle_{AB} - |10\rangle_{A'B'} \otimes (W_1W_0 \otimes B_0)|\psi\rangle_{AB} \\
&\quad + |11\rangle_{A'B'} \otimes (W_1W_0 \otimes B_1B_0)|\psi\rangle_{AB}].
\end{aligned} \tag{C.65}$$

Factorising the $A'B'$ subsystems we get

$$\begin{aligned}
|\varphi_4\rangle_{ABA'B'} &= \frac{1}{4} [|00\rangle_{A'B'} \otimes (\mathbb{1} \otimes \mathbb{1} + \mathbb{1} \otimes B_0 + W_0 \otimes \mathbb{1} + W_0 \otimes B_0)|\psi\rangle_{AB} \\
&\quad + |01\rangle_{A'B'} \otimes (\mathbb{1} \otimes B_1 - \mathbb{1} \otimes B_1B_0 + W_0 \otimes B_1 - W_0 \otimes B_1B_0)|\psi\rangle_{AB} \\
&\quad + |10\rangle_{A'B'} \otimes (W_1 \otimes \mathbb{1} + W_1 \otimes B_0 - W_1W_0 \otimes \mathbb{1} - W_1W_0 \otimes B_0)|\psi\rangle_{AB} \\
&\quad + |11\rangle_{A'B'} \otimes (W_1 \otimes B_1 - W_1 \otimes B_1B_0 - W_1W_0 \otimes B_1 + W_1W_0 \otimes B_1B_0)|\psi\rangle_{AB}].
\end{aligned} \tag{C.66}$$

In each line we can make the simplification

$$\begin{aligned}
(\mathbb{1} \otimes \mathbb{1} + \mathbb{1} \otimes B_0 + W_0 \otimes \mathbb{1} + W_0 \otimes B_0) &= (\mathbb{1} + W_0) \otimes (\mathbb{1} + B_0), \\
(\mathbb{1} \otimes B_1 - \mathbb{1} \otimes B_1B_0 + W_0 \otimes B_1 - W_0 \otimes B_1B_0) &= (\mathbb{1} + W_0) \otimes B_1(\mathbb{1} - B_0), \\
(W_1 \otimes \mathbb{1} + W_1 \otimes B_0 - W_1W_0 \otimes \mathbb{1} - W_1W_0 \otimes B_0) &= W_1(\mathbb{1} - W_0) \otimes (\mathbb{1} + B_0), \\
(W_1 \otimes B_1 - W_1 \otimes B_1B_0 - W_1W_0 \otimes B_1 + W_1W_0 \otimes B_1B_0) &= W_1(\mathbb{1} - W_0) \otimes B_1(\mathbb{1} - B_0),
\end{aligned} \tag{C.67}$$

from which the result (C.8) follows.

C.2. Quantum key distribution

C.2.1 Solution (Text at p. 198)

Alice is preparing two qubit states $|\psi_{a_i x_i}\rangle = |x_i\rangle \otimes |a_i(x_i)\rangle$, where the first qubit encodes the basis in which the second qubit is prepared. Eve can intercept both the qubits, measure the first one with σ_Z to deterministically learn the basis in which the second qubit state is diagonal. Then, she can measure the second qubit in that basis and deterministically learn a_i . Both qubits are left undisturbed by Eve and are then relayed to Bob. Hence, the attack cannot be detected and the protocol is not secure.

C.2.2 Solution (Text at p. 198)

(a) Using Born's rule and the Bloch representation of the cloned state and the measurements in the Z -basis, we can compute

$$\begin{aligned}
p(b|a) &= \text{tr}(\rho_{a0} E_b) = \text{tr} \left(\frac{\mathbb{1} + \eta(-1)^a \sigma_Z}{2} \frac{\mathbb{1} + (-1)^b \sigma_Z}{2} \right) \\
&= \frac{1}{4} (\text{tr}(\mathbb{1}) + \eta(-1)^a \underbrace{\text{tr}(\sigma_Z)}_{=0} + (-1)^b \underbrace{\text{tr}(\sigma_Z)}_{=0} + \eta(-1)^{a+b} \text{tr}(\sigma_Z^2)) = \frac{1 + \eta(-1)^{a+b}}{2},
\end{aligned} \tag{C.68}$$

where we have used that the σ_Z is traceless and $\sigma_Z^2 = \mathbb{1}$. Since Alice randomly chooses her basis, $p(a) = \frac{1}{2}$. Therefore,

$$p(a, b) = p(a)p(b|a) = \frac{1 + \eta(-1)^{a+b}}{4}. \tag{C.69}$$

From (8.4) and (8.5), we can compute

$$\begin{aligned} H(\mathbf{b}|\mathbf{a}) &= - \sum_{a,b=0,1} \underbrace{p(a)p(b|a)}_{=p(a,b)} \log(p(b|a)) \\ &= - \sum_{a,b=0,1} \frac{1+\eta(-1)^{a+b}}{4} \log\left(\frac{1+\eta(-1)^{a+b}}{2}\right) = -\frac{1+\eta}{2} \log\left(\frac{1+\eta}{2}\right) - \frac{1-\eta}{2} \log\left(\frac{1-\eta}{2}\right) = h(F), \end{aligned} \quad (\text{C.70})$$

where we have used that $F = \langle \psi | \rho | \psi \rangle = \frac{1+\eta}{2}$ for any input state $|\psi\rangle$. Therefore, the mutual information is

$$I(\mathbf{a} : \mathbf{b}) = \underbrace{H(\mathbf{a})}_{=1} - H(\mathbf{b}|\mathbf{a}) = 1 - h(F), \quad (\text{C.71})$$

where we have used that $H(\mathbf{a}) = -\sum_a p(a) \log(p(a)) = -\sum_a \frac{1}{2} \log\left(\frac{1}{2}\right) = 1$.

(b) If Alice and Bob both select the X -basis, the conditional probability becomes

$$\begin{aligned} p(b|a) &= \text{tr}(\rho_{a1} E_b) = \text{tr}\left(\frac{\mathbb{1} + \eta(-1)^a \sigma_X}{2} \frac{\mathbb{1} + (-1)^b \sigma_X}{2}\right) \\ &= \frac{1}{4} (\text{tr}(\mathbb{1}) + \underbrace{\eta(-1)^a \text{tr}(\sigma_X)}_{=0} + \underbrace{(-1)^b \text{tr}(\sigma_X)}_{=0} + \eta(-1)^{a+b} \text{tr}(\sigma_X^2)) = \frac{1+\eta(-1)^{a+b}}{2}, \end{aligned} \quad (\text{C.72})$$

which is the same result obtained in (C.68). Therefore, also the mutual information is the same as in part (a).

(c) From Chapter 10.3 and Exercise B.3.9, we know that the best cloning machine for the BB84 states achieves a fidelity of $F = \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}}\right) \approx 0.854$. Then, using (C.10), we get $I(\mathbf{a} : \mathbf{b}) = 1 - h(0.854) \approx 0.40$ bits.

C.2.3 Solution (Text at p. 199)

(a) If Eve chooses the wrong basis, she has a probability of $\frac{1}{2}$ of generating an outcome such that $a_i \neq b_i$. Since Alice can prepare the states in three different bases, Eve has a probability $\frac{2}{3}$ of picking the wrong basis. Therefore,

$$\delta = \frac{1}{2} \cdot \frac{2}{3} \approx 33.3\%. \quad (\text{C.73})$$

(b) The main difference with BB84 is that Alice can now prepare states in three different bases. Therefore, Eve should not anymore use a phase-covariant cloning machine for her attack: she needs to clone states in all three dimensions of the Bloch sphere. Therefore, she should base her attack on the universal cloning machine. From Chapter 6.5, we know that for any qubit state $|\psi\rangle$ the universal cloning machine outputs a two-qubit state whose reductions have a cloning fidelity of $F = \frac{5}{6}$. The error introduced by this attack is

$$\delta = 1 - F = 1 - \frac{5}{6} \approx 16.7\%. \quad (\text{C.74})$$

(c) The mutual information $I(\mathbf{a} : \mathbf{e})$ is defined as

$$I(\mathbf{a} : \mathbf{e}) = H(\mathbf{a}) - H(\mathbf{a}|\mathbf{e}). \quad (\text{C.75})$$

Since Alice's outcomes in the protocol are uniformly random, we have that $p(0) = p(1) = \frac{1}{2}$. Therefore $H(\mathbf{a}) = -\sum_{a=0}^1 p(a) \log(p(a)) = -2 \cdot \frac{1}{2} \log\left(\frac{1}{2}\right) = 1$.

The conditional entropy $H(\mathbf{a}|\mathbf{e})$ can be expressed in terms of the error δ introduced by Eve. Indeed, since $p(a \neq e|e) = \delta$ and $p(a = e|e) = 1 - \delta$ for each value of e , we can write

$$\begin{aligned} H(\mathbf{a}|\mathbf{e}) &= - \sum_{a,e} p(e) p(a|e) \log(p(a|e)) \\ &= - \sum_e p(e) [p(a \neq e|e) \log(p(a \neq e|e)) + p(a = e|e) \log(p(a = e|e))] \\ &= -\delta \log(\delta) - (1 - \delta) \log(1 - \delta). \end{aligned} \quad (\text{C.76})$$

Therefore, the mutual information is

$$I(\mathbf{a} : \mathbf{e}) = \underbrace{H(\mathbf{a})}_{=1} - H(\mathbf{a}|\mathbf{e}) = 1 + \delta \log(\delta) + (1 - \delta) \log(1 - \delta), \quad (\text{C.77})$$

which for $\delta \approx 16.7\%$ corresponds to $I(\mathbf{a} : \mathbf{e}) \approx 0.349$.

C.3. Markovian master equations

C.3.1 Solution (Text at p. 199)

Applying the transformation for the jump operators, we get

$$\begin{aligned}
 \frac{d\rho}{dt} &= i[\rho, H] + \sum_k \gamma_k \left((L_k + a_k \mathbb{1}) \rho (L_k^\dagger + a_k^* \mathbb{1}) - \frac{1}{2} \{ (L_k^\dagger + a_k^* \mathbb{1}) (L_k + a_k \mathbb{1}), \rho \} \right) \\
 &= i[\rho, H] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger + a_k^* L_k \rho + a_k \rho L_k^\dagger + a_k \rho a_k^* - \frac{1}{2} \{ L_k^\dagger L_k + L_k^\dagger a_k + a_k^* L_k + |a_k|^2 \mathbb{1}, \rho \} \right) \\
 &= i[\rho, H] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} + a_k^* L_k \rho + a_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger a_k + a_k^* L_k, \rho \} + |a_k|^2 \rho - \frac{1}{2} \{ |a_k|^2 \mathbb{1}, \rho \} \right) \quad (\text{C.78}) \\
 &= i[\rho, H] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} + \frac{1}{2} a_k^* L_k \rho + \frac{1}{2} a_k \rho L_k^\dagger - \frac{1}{2} a_k L_k^\dagger \rho - \frac{1}{2} a_k^* \rho L_k \right) \\
 &= i[\rho, H] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} + \frac{1}{2} a_k [\rho, L_k^\dagger] - \frac{1}{2} a_k^* [\rho, L_k] \right).
 \end{aligned}$$

We can define a new Hamiltonian

$$H' = H + \frac{1}{2i} \sum_k \gamma_k (a_k L_k^\dagger - a_k^* L_k), \quad (\text{C.79})$$

therefore we get

$$\begin{aligned}
 \frac{d\rho}{dt} &= i[\rho, H] + \sum_k \gamma_k \left(\frac{1}{2} a_k [\rho, L_k^\dagger] - \frac{1}{2} a_k^* [\rho, L_k] \right) + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right) \\
 &= i \left[\rho, H + \frac{1}{2i} \sum_k \gamma_k (a_k L_k^\dagger - a_k^* L_k) \right] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right) \quad (\text{C.80}) \\
 &= i[\rho, H'] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right),
 \end{aligned}$$

which is the Lindblad equation with a redefined Hamiltonian.

C.3.2 Solution (Text at p. 200)

(a) We determine the action of the Liouvillian $\mathcal{L}$ from the master equation, namely

$$\mathcal{L}\rho = \frac{iE}{2} [\sigma_Z, \rho] + \gamma^+ \mathcal{D}[\sigma_+] \rho + \gamma^- \mathcal{D}[\sigma_-] \rho. \quad (\text{C.81})$$

As an example, let us expand the first term on the right hand side,

$$\frac{iE}{2} [\sigma_Z, \rho] = \frac{iE}{2} (\sigma_Z \rho - \rho \sigma_Z). \quad (\text{C.82})$$

Use that

$$\sigma_Z \rho = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} = \begin{pmatrix} \rho_{00} & \rho_{01} \\ -\rho_{10} & -\rho_{11} \end{pmatrix}, \quad \rho \sigma_Z = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \rho_{00} & -\rho_{01} \\ \rho_{10} & -\rho_{11} \end{pmatrix}, \quad (\text{C.83})$$

to write

$$\frac{iE}{2} [\sigma_Z, \rho] = \frac{iE}{2} \left(\begin{pmatrix} \rho_{00} & \rho_{01} \\ -\rho_{10} & -\rho_{11} \end{pmatrix} - \begin{pmatrix} \rho_{00} & -\rho_{01} \\ \rho_{10} & -\rho_{11} \end{pmatrix} \right) = \begin{pmatrix} 0 & iE\rho_{01} \\ -iE\rho_{10} & 0 \end{pmatrix}. \quad (\text{C.84})$$

Similarly, the other two terms on the right hand side of (C.81) become

$$\mathcal{D}[\sigma_+] \rho = \begin{pmatrix} -\rho_{00} & -\frac{\rho_{01}}{2} \\ -\frac{\rho_{10}}{2} & \rho_{00} \end{pmatrix}, \quad \mathcal{D}[\sigma_-] \rho = \begin{pmatrix} \rho_{11} & -\frac{\rho_{01}}{2} \\ -\frac{\rho_{10}}{2} & -\rho_{11} \end{pmatrix}. \quad (\text{C.85})$$

Re-inserted in (C.81), the action of the Liouvillian on the state ρ becomes

$$\mathcal{L}\rho = \begin{pmatrix} \gamma^- \rho_{11} - \gamma^+ \rho_{00} & iE\rho_{01} - \frac{\gamma^- \rho_{01}}{2} - \frac{\gamma^+ \rho_{01}}{2} \\ -iE\rho_{10} - \frac{\gamma^- \rho_{10}}{2} - \frac{\gamma^+ \rho_{10}}{2} & \gamma^+ \rho_{00} - \gamma^- \rho_{11} \end{pmatrix}. \quad (\text{C.86})$$

The vectorisation of $\mathcal{L}\rho$ gives

$$(\mathcal{L}\rho)_v = \begin{pmatrix} \gamma^- \rho_{11} - \gamma^+ \rho_{00} \\ iE\rho_{01} - \frac{\gamma^- \rho_{01}}{2} - \frac{\gamma^+ \rho_{01}}{2} \\ -iE\rho_{10} - \frac{\gamma^- \rho_{10}}{2} - \frac{\gamma^+ \rho_{10}}{2} \\ \gamma^+ \rho_{00} - \gamma^- \rho_{11} \end{pmatrix}, \quad (\text{C.87})$$

from which the Liouvillian superoperator matrix becomes

$$\mathcal{L}_{\text{sup}} = \begin{pmatrix} -\gamma^+ & 0 & 0 & \gamma^- \\ 0 & iE - \frac{\gamma^-}{2} - \frac{\gamma^+}{2} & 0 & 0 \\ 0 & 0 & -iE - \frac{\gamma^-}{2} - \frac{\gamma^+}{2} & 0 \\ \gamma^+ & 0 & 0 & -\gamma^- \end{pmatrix}. \quad (\text{C.88})$$

(b) We can use an appropriate software to calculate the matrix exponential,

$$e^{\mathcal{L}t} = \begin{pmatrix} \frac{\gamma^-}{\gamma^- + \gamma^+} + \frac{\gamma^+ e^{-(\gamma^- + \gamma^+)t}}{\gamma^- + \gamma^+} & 0 & 0 & \frac{\gamma^-}{\gamma^- + \gamma^+} - \frac{\gamma^- e^{-(\gamma^- + \gamma^+)t}}{\gamma^- + \gamma^+} \\ 0 & e^{\frac{1}{2}it(i\gamma^- + i\gamma^+ + 2E)} & 0 & 0 \\ 0 & 0 & e^{-\frac{1}{2}it(-i\gamma^- - i\gamma^+ + 2E)} & 0 \\ \frac{\gamma^+}{\gamma^- + \gamma^+} - \frac{\gamma^+ e^{-(\gamma^- + \gamma^+)t}}{\gamma^- + \gamma^+} & 0 & 0 & \frac{\gamma^+}{\gamma^- + \gamma^+} + \frac{\gamma^- e^{-(\gamma^- + \gamma^+)t}}{\gamma^- + \gamma^+} \end{pmatrix}. \quad (\text{C.89})$$

Therefore, starting from the atom in the ground state, i.e. $\rho(0) = [1, 0, 0, 0]^T$, the time-dependent state is

$$\rho(t) = e^{\mathcal{L}t} \rho(0) = \begin{pmatrix} \frac{\gamma^- + \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} \\ 0 \\ 0 \\ \frac{\gamma^+ - \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} \end{pmatrix} \longrightarrow \begin{pmatrix} \frac{\gamma^- + \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} & 0 \\ 0 & \frac{\gamma^+ - \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} \end{pmatrix}, \quad (\text{C.90})$$

after we bring it back to the matrix form.

(c) Taking the long-time limit, we get

$$\lim_{t \rightarrow +\infty} \rho(t) = \begin{pmatrix} \frac{\gamma^-}{\gamma^- + \gamma^+} & 0 \\ 0 & \frac{\gamma^+}{\gamma^- + \gamma^+} \end{pmatrix}. \quad (\text{C.91})$$

Recalling that $H_{\text{free}} = E|1\rangle\langle 1|$, we can write the Gibbs state as

$$\rho_G = \frac{e^{-\beta H_{\text{free}}}}{\text{tr}(e^{-\beta H_{\text{free}}})} = \frac{1}{1 + e^{-\beta E}} \begin{pmatrix} 1 & 0 \\ 0 & e^{-\beta E} \end{pmatrix}. \quad (\text{C.92})$$

Therefore, the long-time limit state in (C.91) is exactly of the form of a Gibbs state if

$$\frac{\gamma^-}{\gamma^- + \gamma^+} = \frac{e^{-\beta E}}{1 + e^{-\beta E}} \implies 1 + e^{-\beta E} = \frac{\gamma^- + \gamma^+}{\gamma^-} e^{-\beta E} \implies \frac{\gamma^+}{\gamma^-} = e^{-\beta E}. \quad (\text{C.93})$$

Indeed,

$$\frac{\gamma^- + \gamma^+}{\gamma^-} = 1 + \frac{\gamma^+}{\gamma^-} \stackrel{(\text{C.93})}{=} 1 + e^{-\beta E} \implies \frac{\gamma^-}{\gamma^- + \gamma^+} = \frac{1}{1 + e^{-\beta E}}. \quad (\text{C.94})$$

- (a) Notice that $\frac{d\langle E \rangle}{dt} = \frac{d}{dt} \text{tr}(\rho H) = \text{tr}\left(\frac{d\rho}{dt} H\right)$ when the Hamiltonian is time-independent. Then, using the Lindblad equation in (11.5),

$$\frac{d\langle E \rangle}{dt} = \text{tr}\left(\frac{d\rho}{dt} H\right) = i \text{tr}([\rho, H]H) + \sum_k \gamma_k \text{tr}(\mathcal{D}[L_k]\rho H), \quad (\text{C.95})$$

where we have defined the dissipator $\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$. By explicit computation and using the cyclicity of the trace, we get

$$\text{tr}([\rho, H]H) = \text{tr}(\rho H H - H \rho H) = \text{tr}(\rho H H) - \text{tr}(\rho H H) = 0. \quad (\text{C.96})$$

- (b) From (C.15), the expression for the average current becomes

$$\langle J(t) \rangle = \gamma^+ \text{tr}(\mathcal{D}[\sigma^+]\rho(t)H) + \gamma^- \text{tr}(\mathcal{D}[\sigma^-]\rho(t)H). \quad (\text{C.97})$$

Using the results in (C.85) and the state in (C.19), we get

$$\begin{aligned} \mathcal{D}[\sigma^+]\rho(t) &= -\frac{\gamma^- + \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} |0\rangle\langle 0| + \frac{\gamma^- + \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} |1\rangle\langle 1| \\ \mathcal{D}[\sigma^-]\rho(t) &= \frac{\gamma^+ - \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} |0\rangle\langle 0| - \frac{\gamma^+ - \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} |1\rangle\langle 1| \end{aligned} \quad (\text{C.98})$$

from which

$$\mathcal{D}[\sigma^+]\rho(t)H = \left(-\frac{\gamma^- + \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} |0\rangle\langle 0| + \frac{\gamma^- + \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} |1\rangle\langle 1| \right) E|1\rangle\langle 1| = \frac{E(\gamma^- + \gamma^+ e^{-t(\gamma^- + \gamma^+)})}{\gamma^- + \gamma^+} |1\rangle\langle 1|, \quad (\text{C.99})$$

$$\mathcal{D}[\sigma^-]\rho(t)H = \left(\frac{\gamma^+ - \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} |0\rangle\langle 0| - \frac{\gamma^+ - \gamma^+ e^{-t(\gamma^- + \gamma^+)}}{\gamma^- + \gamma^+} |1\rangle\langle 1| \right) E|1\rangle\langle 1| = -\frac{E(\gamma^+ - \gamma^+ e^{-t(\gamma^- + \gamma^+)})}{\gamma^- + \gamma^+} |1\rangle\langle 1|. \quad (\text{C.100})$$

Therefore, we get

$$\langle J(t) \rangle = \gamma^+ \frac{E(\gamma^- + \gamma^+ e^{-t(\gamma^- + \gamma^+)})}{\gamma^- + \gamma^+} - \gamma^- \frac{E(\gamma^+ - \gamma^+ e^{-t(\gamma^- + \gamma^+)})}{\gamma^- + \gamma^+} = E \gamma^+ e^{-t(\gamma^- + \gamma^+)} \quad (\text{C.101})$$

- (c) For $t \rightarrow +\infty$, (C.101) goes to zero. This means that the system is at equilibrium, which is what we expect for the steady state.

C.3.4 Solution (Text at p. 201)

- (a) Since $\langle a \rangle = \text{tr}(\rho a)$, we can express $\langle \dot{a} \rangle = \frac{d}{dt} \text{tr}(\rho a) = \text{tr}\left(\frac{d\rho}{dt} a\right)$. Therefore, using (C.21), we get

$$\langle \dot{a} \rangle = \text{tr}\left(\frac{d\rho}{dt} a\right) = i \text{tr}([\rho, H]a) + \gamma_a \text{tr}(\mathcal{D}[a]\rho a) + \gamma_b \text{tr}(\mathcal{D}[b]\rho a). \quad (\text{C.102})$$

We can compute each term separately. Using the cyclicity of the trace, we can rewrite

$$\text{tr}([\rho, H]a) = \text{tr}(\rho H a - H \rho a) = \text{tr}(\rho H a - \rho a H) = \text{tr}(\rho[H, a]). \quad (\text{C.103})$$

Using the Hamiltonian in (C.20), we can write

$$[H, a] = \omega_a [a^\dagger a, a] + \omega_b [b^\dagger b, a] + g([ab^\dagger, a] + [a^\dagger b, a]). \quad (\text{C.104})$$

From the bosonic commutation relations $[a, a^\dagger] = 1$ and $[a, b] = [a^\dagger, b^\dagger] = [a, b^\dagger] = [a^\dagger, b] = 0$, we get

$$\begin{aligned} [a^\dagger a, a] &= \underbrace{a^\dagger a}_{=aa^\dagger-1} a - aa^\dagger a = -a, & [a^\dagger b, a] &= a^\dagger ba - aa^\dagger b = \underbrace{a^\dagger a}_{=aa^\dagger-1} b - aa^\dagger b = -b, \\ [b^\dagger b, a] &= b^\dagger ba - ab^\dagger b = b^\dagger ba - b^\dagger ba = 0, & [ab^\dagger, a] &= ab^\dagger a - aab^\dagger = ab^\dagger a - ab^\dagger a = 0. \end{aligned} \quad (\text{C.105})$$

Therefore,

$$\text{tr}([\rho, H]a) = \text{tr}(\rho[H, a]) = -\omega_a \text{tr}(\rho a) - g \text{tr}(\rho b) = -\omega_a \langle a \rangle - g \langle b \rangle. \quad (\text{C.106})$$

In the same way, using the cyclicity of the trace, we can compute

$$\begin{aligned} \text{tr}(\mathcal{D}[a]\rho a) &= \text{tr}(a\rho a^\dagger a) - \frac{1}{2} \text{tr}(\{a^\dagger a, \rho\}a) = \text{tr}(a\rho a^\dagger a) - \frac{1}{2} \text{tr}(a^\dagger a\rho a) - \frac{1}{2} \text{tr}(\rho a^\dagger a a) \\ &= \frac{1}{2} \text{tr}(\rho a^\dagger a a) - \frac{1}{2} \text{tr}(\rho a a^\dagger a) = \frac{1}{2} \text{tr}(\rho[a^\dagger a, a]) \stackrel{(\text{C.105})}{=} -\frac{1}{2} \text{tr}(\rho a) = -\frac{1}{2} \langle a \rangle, \end{aligned} \quad (\text{C.107})$$

$$\begin{aligned} \text{tr}(\mathcal{D}[b]\rho a) &= \text{tr}(b\rho b^\dagger a) - \frac{1}{2} \text{tr}(\{b^\dagger b, \rho\}a) = \text{tr}(b\rho b^\dagger a) - \frac{1}{2} \text{tr}(b^\dagger b\rho a) - \frac{1}{2} \text{tr}(\rho b^\dagger b a) \\ &= \text{tr}(b\rho b^\dagger a) - \frac{1}{2} \text{tr}(b\rho a b^\dagger) - \frac{1}{2} \text{tr}(b a \rho b^\dagger) = \text{tr}(b\rho b^\dagger a) - \frac{1}{2} \text{tr}(b\rho b^\dagger a) - \frac{1}{2} \text{tr}(a b \rho b^\dagger) = 0. \end{aligned} \quad (\text{C.108})$$

Inserting (C.106), (C.107) and (C.108) into (C.102), we get

$$\langle \dot{a} \rangle = -i\omega_a \langle a \rangle - ig \langle b \rangle - \frac{\gamma_a}{2} \langle a \rangle. \quad (\text{C.109})$$

With the same procedure, we get

$$\langle \dot{b} \rangle = -i\omega_b \langle b \rangle - ig \langle a \rangle - \frac{\gamma_b}{2} \langle b \rangle. \quad (\text{C.110})$$

(b) Define the matrix

$$M = -i \begin{pmatrix} \omega_a - i\frac{\gamma_a}{2} & g \\ g & \omega_b - i\frac{\gamma_b}{2} \end{pmatrix}. \quad (\text{C.111})$$

Then, we can rewrite (C.109) and (C.110) as

$$\begin{pmatrix} \langle \dot{a}(t) \rangle \\ \langle \dot{b}(t) \rangle \end{pmatrix} = -i \begin{pmatrix} \omega_a - i\frac{\gamma_a}{2} & g \\ g & \omega_b - i\frac{\gamma_b}{2} \end{pmatrix} \begin{pmatrix} \langle a(t) \rangle \\ \langle b(t) \rangle \end{pmatrix}. \quad (\text{C.112})$$

For a 2×2 matrix, the eigenvalues can be computed as

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longrightarrow \lambda_{1,2} = \frac{1}{2} \left[(a + d) \pm \sqrt{(a - d)^2 + 4bc} \right]. \quad (\text{C.113})$$

Therefore, at resonance $\omega_a = \omega_b = \omega$,

$$\lambda_{1,2}(M) = \frac{1}{2} \left[-2i\omega - \frac{\gamma_a + \gamma_b}{2} \pm \sqrt{\left(\frac{-\gamma_a + \gamma_b}{2} \right)^2 - 4g^2} \right]. \quad (\text{C.114})$$

We can compute the eigenvectors,

$$\begin{pmatrix} -i\omega - \frac{\gamma_a}{2} - \lambda_1(M) & -ig \\ -ig & -i\omega - \frac{\gamma_b}{2} - \lambda_1(M) \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \left(-i\omega - \frac{\gamma_a}{2} - \lambda_1(M) \right) x - igy = 0, \quad (\text{C.115})$$

from which, replacing the value of $\lambda_1(M)$, we get

$$x = \frac{4igy}{-\gamma_a + \gamma_b - \sqrt{(\gamma_a - \gamma_b)^2 - 16g^2}} \implies |\lambda_1\rangle = y \begin{pmatrix} \frac{4ig}{-\gamma_a + \gamma_b - \sqrt{(\gamma_a - \gamma_b)^2 - 16g^2}} \\ 1 \end{pmatrix}, \quad (\text{C.116})$$

where y can be derived from the normalisation condition $\langle \lambda_1 | \lambda_1 \rangle = 1$. In the same way, we can derive

$$\begin{pmatrix} -i\omega - \frac{\gamma_a}{2} - \lambda_2(M) & -ig \\ -ig & -i\omega - \frac{\gamma_b}{2} - \lambda_2(M) \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \left(-i\omega - \frac{\gamma_a}{2} - \lambda_2(M) \right) x - igy = 0, \quad (\text{C.117})$$

from which, replacing the value of $\lambda_2(M)$, we get

$$x = \frac{4igy}{-\gamma_a + \gamma_b + \sqrt{(\gamma_a - \gamma_b)^2 - 16g^2}} \implies |\lambda_2\rangle = y \left(\frac{\frac{4ig}{-\gamma_a + \gamma_b + \sqrt{(\gamma_a - \gamma_b)^2 - 16g^2}}}{1} \right), \quad (\text{C.118})$$

where y can be derived from the normalisation condition $\langle \lambda_2 | \lambda_2 \rangle = 1$.

(c) We can compute

$$\frac{d\langle A \rangle}{dt} = \langle \dot{a} \rangle e^{i\omega t} + i\omega \langle a \rangle e^{i\omega t}, \quad \frac{d\langle B \rangle}{dt} = \langle \dot{b} \rangle e^{i\omega t} + i\omega \langle b \rangle e^{i\omega t}, \quad (\text{C.119})$$

from which it follows that

$$\begin{aligned} \frac{d\langle A \rangle}{dt} = -ig\langle B \rangle - \frac{\gamma_a}{2}\langle A \rangle &\implies \langle \dot{a} \rangle e^{i\omega t} + i\omega \langle a \rangle e^{i\omega t} = -ig\langle \dot{b} \rangle e^{i\omega t} - \frac{\gamma_a}{2}\langle a \rangle e^{i\omega t} \\ \frac{d\langle B \rangle}{dt} = -ig\langle A \rangle - \frac{\gamma_b}{2}\langle B \rangle &\implies \langle \dot{b} \rangle e^{i\omega t} + i\omega \langle b \rangle e^{i\omega t} = -ig\langle \dot{a} \rangle e^{i\omega t} - \frac{\gamma_a}{2}\langle b \rangle e^{i\omega t}, \end{aligned} \quad (\text{C.120})$$

which are exactly the equations of motion (C.109) and (C.110) derived in (a) once we cancel the exponential $e^{i\omega t}$.

(d) Compute the second order derivative of $\langle A \rangle$ using (C.23),

$$\frac{d^2\langle A \rangle}{dt^2} = -ig \frac{d\langle B \rangle}{dt} - \frac{\gamma_a}{2} \frac{d\langle A \rangle}{dt} = -ig \left(-ig\langle A \rangle - \frac{\gamma_b}{2}\langle B \rangle \right) - \frac{\gamma_a}{2} \frac{d\langle A \rangle}{dt}, \quad (\text{C.121})$$

where in the second equality we have replaced $\frac{d\langle B \rangle}{dt}$ using (C.23). Using the expression of $\langle B \rangle$ from the first equation of (C.23), we get

$$\begin{aligned} \frac{d^2\langle A \rangle}{dt^2} &= -ig \left[-ig\langle A \rangle - \frac{\gamma_b}{2} \frac{1}{-ig} \left(\frac{d\langle A \rangle}{dt} + \frac{\gamma_a}{2}\langle A \rangle \right) \right] - \frac{\gamma_a}{2} \frac{d\langle A \rangle}{dt} \\ &= -\left(\frac{\gamma_a}{2} + \frac{\gamma_b}{2} \right) \frac{d\langle A \rangle}{dt} - \left(g^2 + \frac{\gamma_a\gamma_b}{4} \right) \langle A \rangle, \end{aligned} \quad (\text{C.122})$$

from which the equation becomes

$$\frac{d^2\langle A \rangle}{dt^2} + \left(\frac{\gamma_a}{2} + \frac{\gamma_b}{2} \right) \frac{d\langle A \rangle}{dt} + \left(g^2 + \frac{\gamma_a\gamma_b}{4} \right) \langle A \rangle = 0. \quad (\text{C.123})$$

Comparing it with the equation of motion of the damped harmonic oscillator (C.24), we get

$$\omega_0 = \left(g^2 + \frac{\gamma_a\gamma_b}{4} \right)^{\frac{1}{2}}, \quad \zeta = \frac{1}{2\omega_0} \left(\frac{\gamma_a + \gamma_b}{2} \right) = \frac{1}{2\sqrt{g^2 + \frac{\gamma_a\gamma_b}{4}}} \left(\frac{\gamma_a + \gamma_b}{2} \right). \quad (\text{C.124})$$

In the critical damping case, we have

$$\zeta = \frac{1}{2\sqrt{g^2 + \frac{\gamma_a\gamma_b}{4}}} \left(\frac{\gamma_a + \gamma_b}{2} \right) = 1 \implies \frac{(\gamma_a + \gamma_b)^2}{16} = \frac{\gamma_a\gamma_b}{4} + g^2 \implies (\gamma_a - \gamma_b)^2 = 16g^2. \quad (\text{C.125})$$

This condition implies that the eigenvalues (C.114) and the corresponding eigenvectors (C.116) and (C.118) become the same. Therefore, in this case the matrix M is not diagonalisable. Such behaviour is unique to open quantum systems.

C.4. Quantum estimation theory

C.4.1 Solution (Text at p. 201)

(a) Using the definition of the Fisher information (12.19), we get

$$\mathcal{I} = \frac{1}{p(0)} \left(\frac{dp(0)}{d\theta} \right)^2 + \frac{1}{p(1)} \left(\frac{dp(1)}{d\theta} \right)^2 = \frac{1}{1-\theta} \left(\frac{\theta}{d\theta} (1-\theta) \right)^2 + \frac{1}{\theta} \left(\frac{d}{d\theta} \theta \right)^2 = \frac{1}{\theta(1-\theta)}. \quad (\text{C.126})$$

This is unbounded for $\theta \in \{0, 1\}$. The minimal value is found at $\theta = \frac{1}{2}$. This is expected because if the parameter is more biased it should be easier to estimate it accurately within a few trials.

(b) Using (12.27), we get $\text{Var}[\hat{\theta}] \geq \frac{\theta(1-\theta)}{N}$.

C.4.2 Solution (Text at p. 201)

(a) We verify the inverse through a direct calculation,

$$\begin{aligned}\Omega^{-1}(\Omega(A)) &= \sum_{j,k} \frac{2}{\lambda_j + \lambda_k} \Omega(A)_{jk} |\psi_j\rangle\langle\psi_k| = \sum_{j,k} \frac{2}{\lambda_j + \lambda_k} \langle\psi_j| \left(\frac{1}{2} A \rho_\theta + \frac{1}{2} \rho_\theta A \right) |\psi_k\rangle |\psi_j\rangle\langle\psi_k| \\ &= \sum_{j,k} \frac{2}{\lambda_j + \lambda_k} \left(\frac{\lambda_k}{2} \langle\psi_j|A|\psi_k\rangle + \frac{\lambda_j}{2} \langle\psi_j|A|\psi_k\rangle \right) |\psi_j\rangle\langle\psi_k| = \sum_{j,k} A_{jk} |\psi_j\rangle\langle\psi_k| = A,\end{aligned}\quad (\text{C.127})$$

where we have used that $\rho_\theta |\psi_l\rangle = \lambda_l |\psi_l\rangle$ and $\langle\psi_l|\rho_\theta = \lambda_l \langle\psi_l|$.

(b) Use the spectral decomposition of $\rho_\theta = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j|$ to rewrite the quantum Fisher information as

$$\mathcal{I}^Q(\rho_\theta) = \text{tr}(L_\theta^2 \rho_\theta) = \sum_j \lambda_j \langle\psi_j|L_\theta^2|\psi_j\rangle = \sum_j \lambda_j \langle\psi_j|L_\theta \underbrace{\sum_k |\psi_k\rangle\langle\psi_k|}_{=\mathbb{I}} L_\theta|\psi_j\rangle = \sum_{j,k} \lambda_j |\langle\psi_j|L_\theta|\psi_k\rangle|^2 = \sum_{j,k} \lambda_j |(L_\theta)_{jk}|^2, \quad (\text{C.128})$$

where in the third step we have inserted the spectral decomposition of the identity. Using that $L_\theta = \Omega^{-1}(\dot{\rho}_\theta)$, we can compute

$$L_\theta = \Omega^{-1}(\dot{\rho}_\theta) = \sum_{j,k} \frac{2}{\lambda_j + \lambda_k} (\dot{\rho}_\theta)_{jk} |\psi_j\rangle\langle\psi_k| \implies |(L_\theta)_{jk}|^2 = \frac{4}{(\lambda_j + \lambda_k)^2} |(\dot{\rho}_\theta)_{jk}|^2. \quad (\text{C.129})$$

Using the spectral decomposition of ρ_θ , we can compute

$$\dot{\rho}_\theta = \frac{\partial}{\partial \theta} \sum_j \lambda_j |\psi_j\rangle\langle\psi_j| = \sum_j (\dot{\lambda}_j |\psi_j\rangle\langle\psi_j| + \lambda_j |\dot{\psi}_j\rangle\langle\psi_j| + \lambda_j |\psi_j\rangle\langle\dot{\psi}_j|), \quad (\text{C.130})$$

from which

$$\begin{aligned}(\dot{\rho}_\theta)_{jk} &= \sum_l \langle\psi_j|(\dot{\lambda}_l |\psi_l\rangle\langle\psi_l| + \lambda_l |\dot{\psi}_l\rangle\langle\psi_l| + \lambda_l |\psi_l\rangle\langle\dot{\psi}_l|)|\psi_k\rangle \\ &= \sum_l \dot{\lambda}_l \underbrace{\langle\psi_j|\psi_l\rangle}_{=\delta_{j,l}} \underbrace{\langle\psi_l|\psi_k\rangle}_{=\delta_{l,k}} + \sum_l \lambda_l \langle\psi_j|\dot{\psi}_l\rangle \underbrace{\langle\psi_l|\psi_k\rangle}_{=\delta_{l,k}} + \sum_l \lambda_l \underbrace{\langle\psi_j|\psi_l\rangle}_{=\delta_{j,l}} \langle\dot{\psi}_l|\psi_k\rangle \\ &= \dot{\lambda}_j \delta_{j,k} + \lambda_k \langle\psi_j|\dot{\psi}_k\rangle + \lambda_j \langle\dot{\psi}_j|\psi_k\rangle.\end{aligned}\quad (\text{C.131})$$

Since $\langle\psi_j|\psi_k\rangle = \delta_{j,k}$,

$$\frac{\partial}{\partial \theta} \langle\psi_j|\psi_k\rangle = \langle\dot{\psi}_j|\psi_k\rangle + \langle\psi_j|\dot{\psi}_k\rangle = 0 \implies \langle\dot{\psi}_j|\psi_k\rangle = -\langle\psi_j|\dot{\psi}_k\rangle. \quad (\text{C.132})$$

Therefore, (C.131) becomes

$$(\dot{\rho}_\theta)_{jk} = \dot{\lambda}_j \delta_{j,k} + (\lambda_j - \lambda_k) \langle\dot{\psi}_j|\psi_k\rangle. \quad (\text{C.133})$$

Inserting this into (C.129) and (C.128), we get

$$\begin{aligned}\mathcal{I}^Q(\rho_\theta) &= \sum_{j,k} \lambda_j |(L_\theta)_{jk}|^2 = \sum_{j,k} \lambda_j \frac{4}{(\lambda_j + \lambda_k)^2} |(\dot{\rho}_\theta)_{jk}|^2 \\ &= \sum_{j,k} \lambda_j \frac{4}{(\lambda_j + \lambda_k)^2} (\dot{\lambda}_j^2 \delta_{j,k} + (\lambda_j - \lambda_k)^2 |\langle\dot{\psi}_j|\psi_k\rangle|^2) \\ &= \sum_j 4\lambda_j \frac{\dot{\lambda}_j^2}{4\lambda_j^2} + \sum_{j \neq k} \frac{4\lambda_j}{(\lambda_j + \lambda_k)^2} (\lambda_j - \lambda_k)^2 |\langle\dot{\psi}_j|\psi_k\rangle|^2 \\ &= \sum_j \frac{\dot{\lambda}_j^2}{\lambda_j} + \frac{1}{2} \left(\sum_{j \neq k} \frac{4\lambda_j}{(\lambda_j + \lambda_k)^2} (\lambda_j - \lambda_k)^2 |\langle\dot{\psi}_j|\psi_k\rangle|^2 + \sum_{k \neq j} \frac{4\lambda_k}{(\lambda_k + \lambda_j)^2} (\lambda_k - \lambda_j)^2 |\langle\dot{\psi}_k|\psi_j\rangle|^2 \right) \\ &= \sum_j \frac{\dot{\lambda}_j^2}{\lambda_j} + 2 \sum_{j,k} \frac{(\lambda_j - \lambda_k)^2}{\lambda_j + \lambda_k} |\langle\dot{\psi}_j|\psi_k\rangle|^2,\end{aligned}\quad (\text{C.134})$$

where in the fourth line we have symmetrised the sum exchanging $j \leftrightarrow k$.

C.4.3 Solution (Text at p. 202)

(a) Using the definition of the symmetric logarithmic derivative (12.32), we can write

$$\text{tr}(L_\theta \dot{\rho}_\theta) = \frac{1}{2} \text{tr}(L_\theta(L_\theta \rho_\theta + \rho_\theta L_\theta)) = \frac{1}{2} (\text{tr}(L_\theta^2 \rho_\theta) + \text{tr}(L_\theta \rho_\theta L_\theta)) = \text{tr}(L_\theta^2 \rho_\theta) \stackrel{(12.34)}{=} \mathcal{I}^Q(\rho_\theta). \quad (\text{C.135})$$

(b) Define $\sigma = \rho_1 \otimes \rho_2$. Using the result found in (a), $\mathcal{I}^Q(\sigma) = \text{tr}(L\dot{\sigma})$. Define a symmetric logarithmic derivative operator L_1 and L_2 for ρ_1 and ρ_2 , respectively. Then, we can compute

$$\begin{aligned} \dot{\sigma} &= \dot{\rho}_1 \otimes \rho_2 + \rho_1 \otimes \dot{\rho}_2 \stackrel{(12.32)}{=} \frac{1}{2}(L_1 \rho_1 + \rho_1 L_1) \otimes \rho_2 + \rho_1 \otimes \frac{1}{2}(L_2 \rho_2 + \rho_2 L_2) \\ &= \frac{1}{2}(L_1 \otimes \mathbb{1} + \mathbb{1} \otimes L_2) \rho_1 \otimes \rho_2 + \frac{1}{2} \rho_1 \otimes \rho_2 (L_1 \otimes \mathbb{1} + \mathbb{1} \otimes L_2) = \frac{1}{2}(L\sigma + \sigma L), \end{aligned} \quad (\text{C.136})$$

from which we find that the symmetric logarithmic derivative operator $L = L_1 \otimes \mathbb{1} + \mathbb{1} \otimes L_2$. The quantum Fisher information becomes

$$\begin{aligned} \mathcal{I}^Q(\sigma) &= \text{tr}(L\dot{\sigma}) = \text{tr}((L_1 \otimes \mathbb{1} + \mathbb{1} \otimes L_2)(\dot{\rho}_1 \otimes \rho_2 + \rho_1 \otimes \dot{\rho}_2)) \\ &= \text{tr}(L_1 \dot{\rho}_1 \otimes \rho_2) + \text{tr}(L_1 \rho_1 \otimes \dot{\rho}_2) + \text{tr}(\dot{\rho}_1 \otimes L_2 \rho_2) + \text{tr}(\rho_1 \otimes L_2 \dot{\rho}_2) \\ &= \text{tr}(L_1 \dot{\rho}_1) \underbrace{\text{tr}(\rho_2)}_{=1} + \text{tr}(L_1 \rho_1) \underbrace{\text{tr}(\dot{\rho}_2)}_{=0} + \underbrace{\text{tr}(\dot{\rho}_1)}_{=0} \text{tr}(L_2 \rho_2) + \underbrace{\text{tr}(\rho_1)}_{=1} \text{tr}(L_2 \dot{\rho}_2) \\ &= \mathcal{I}^Q(\rho_1) + \mathcal{I}^Q(\rho_2), \end{aligned} \quad (\text{C.137})$$

where we have used that $\text{tr}(\dot{\rho}) = \frac{\partial}{\partial \theta} \text{tr}(\rho) = \frac{\partial}{\partial \theta} 1 = 0$ for any state ρ .

(c) Using the definition of the quantum Fisher information in (12.34),

$$\mathcal{I}^Q(\lambda \rho_1 + (1 - \lambda) \rho_2) = \text{tr}(L^2(\lambda \rho_1 + (1 - \lambda) \rho_2)) = \lambda \text{tr}(L^2 \rho_1) + (1 - \lambda) \text{tr}(L^2 \rho_2). \quad (\text{C.138})$$

Since L is not necessarily the operator which saturated (12.34) for ρ_1 and ρ_2 , we have in general

$$\lambda \text{tr}(L^2 \rho_1) + (1 - \lambda) \text{tr}(L^2 \rho_2) \leq \lambda \text{tr}(L_1^2 \rho_1) + (1 - \lambda) \text{tr}(L_2^2 \rho_2) = \lambda \mathcal{I}^Q(\rho_1) + (1 - \lambda) \mathcal{I}^Q(\rho_2). \quad (\text{C.139})$$

(d) Using the result found in (a), we can write

$$\mathcal{I}^Q(\psi_\theta) = \text{tr}(L_\theta \dot{\psi}_\theta) = \text{tr}(L_\theta(|\dot{\psi}_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\dot{\psi}_\theta|)) = \langle\psi_\theta|L_\theta|\dot{\psi}_\theta\rangle + \langle\dot{\psi}_\theta|L_\theta|\psi_\theta\rangle. \quad (\text{C.140})$$

From (12.32), we can compute

$$\dot{\psi}_\theta = |\dot{\psi}_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\dot{\psi}_\theta| = \frac{1}{2}(L_\theta|\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\psi_\theta|L_\theta). \quad (\text{C.141})$$

By projecting (C.141) onto $|\psi_\theta\rangle$, we get

$$\langle\psi_\theta|\dot{\psi}_\theta|\psi_\theta\rangle = \langle\psi_\theta|\dot{\psi}_\theta\rangle \underbrace{\langle\psi_\theta|\psi_\theta\rangle}_{=1} + \underbrace{\langle\psi_\theta|\psi_\theta\rangle}_{=1} \langle\dot{\psi}_\theta|\psi_\theta\rangle = \frac{1}{2}(\langle\psi_\theta|L_\theta|\psi_\theta\rangle \underbrace{\langle\psi_\theta|\psi_\theta\rangle}_{=1} + \underbrace{\langle\psi_\theta|\psi_\theta\rangle}_{=1} \langle\psi_\theta|L_\theta|\psi_\theta\rangle) = \langle\psi_\theta|L_\theta|\psi_\theta\rangle. \quad (\text{C.142})$$

Applying $\langle\dot{\psi}_\theta|$ and $|\psi_\theta\rangle$ on (C.141), we get

$$\begin{aligned} \langle\dot{\psi}_\theta|\dot{\psi}_\theta|\psi_\theta\rangle &= \langle\dot{\psi}_\theta|\dot{\psi}_\theta\rangle \underbrace{\langle\psi_\theta|\psi_\theta\rangle}_{=1} + \langle\dot{\psi}_\theta|\psi_\theta\rangle^2 = \frac{1}{2}(\langle\dot{\psi}_\theta|L_\theta|\psi_\theta\rangle \underbrace{\langle\psi_\theta|\psi_\theta\rangle}_{=1} + \underbrace{\langle\dot{\psi}_\theta|\psi_\theta\rangle}_{=1} \langle\psi_\theta|L_\theta|\psi_\theta\rangle) \\ &= \frac{1}{2}(\langle\dot{\psi}_\theta|L_\theta|\psi_\theta\rangle + \langle\dot{\psi}_\theta|\psi_\theta\rangle\langle\psi_\theta|\dot{\psi}_\theta\rangle + \langle\dot{\psi}_\theta|\psi_\theta\rangle^2), \end{aligned} \quad (\text{C.143})$$

where we have used (C.142) to rewrite $\langle\psi_\theta|L_\theta|\psi_\theta\rangle$. In the same way, we can apply $\langle\psi_\theta|$ and $|\dot{\psi}_\theta\rangle$ on (C.141) to get

$$\begin{aligned} \langle\psi_\theta|\dot{\psi}_\theta|\dot{\psi}_\theta\rangle &= \langle\psi_\theta|\dot{\psi}_\theta\rangle^2 + \underbrace{\langle\psi_\theta|\psi_\theta\rangle}_{=1} \langle\dot{\psi}_\theta|\dot{\psi}_\theta\rangle = \frac{1}{2}(\langle\psi_\theta|L_\theta|\psi_\theta\rangle \underbrace{\langle\psi_\theta|\dot{\psi}_\theta\rangle}_{=1} + \underbrace{\langle\psi_\theta|\dot{\psi}_\theta\rangle}_{=1} \langle\psi_\theta|L_\theta|\dot{\psi}_\theta\rangle) \\ &= \frac{1}{2}(\langle\psi_\theta|\dot{\psi}_\theta\rangle^2 + \langle\dot{\psi}_\theta|\psi_\theta\rangle\langle\psi_\theta|\dot{\psi}_\theta\rangle + \langle\psi_\theta|L_\theta|\dot{\psi}_\theta\rangle). \end{aligned} \quad (\text{C.144})$$

Using (C.143) and (C.144), we arrive at

$$\begin{aligned}
\langle \dot{\psi}_\theta | L_\theta | \psi_\theta \rangle + \langle \psi_\theta | L_\theta | \dot{\psi}_\theta \rangle &= [2(\langle \dot{\psi}_\theta | \dot{\psi}_\theta \rangle + \langle \dot{\psi}_\theta | \psi_\theta \rangle^2) - (\langle \dot{\psi}_\theta | \psi_\theta \rangle \langle \psi_\theta | \dot{\psi}_\theta \rangle + \langle \dot{\psi}_\theta | \psi_\theta \rangle^2)] \\
&\quad + [2(\langle \psi_\theta | \dot{\psi}_\theta \rangle^2 + \langle \dot{\psi}_\theta | \dot{\psi}_\theta \rangle) - (\langle \psi_\theta | \dot{\psi}_\theta \rangle^2 + \langle \dot{\psi}_\theta | \psi_\theta \rangle \langle \psi_\theta | \dot{\psi}_\theta \rangle)] \\
&= 4\langle \dot{\psi}_\theta | \dot{\psi}_\theta \rangle + \langle \dot{\psi}_\theta | \psi_\theta \rangle^2 + \langle \psi_\theta | \dot{\psi}_\theta \rangle^2 - 2\langle \dot{\psi}_\theta | \psi_\theta \rangle \langle \psi_\theta | \dot{\psi}_\theta \rangle \\
&= 4\langle \dot{\psi}_\theta | \dot{\psi}_\theta \rangle + (\langle \dot{\psi}_\theta | \psi_\theta \rangle - \langle \psi_\theta | \dot{\psi}_\theta \rangle)^2 \\
&= 4\langle \dot{\psi}_\theta | \dot{\psi}_\theta \rangle + (2i \operatorname{Im}(\langle \dot{\psi}_\theta | \psi_\theta \rangle))^2 \\
&= 4(\langle \dot{\psi}_\theta | \dot{\psi}_\theta \rangle - |\langle \dot{\psi}_\theta | \psi_\theta \rangle|^2),
\end{aligned} \tag{C.145}$$

where we have used that $\langle \dot{\psi}_\theta | \psi_\theta \rangle$ is in general a complex number but since the derivative is anti-Hermitian when it is applied to a ray, then $\langle \dot{\psi}_\theta | \psi_\theta \rangle^* = \langle \psi_\theta | \dot{\psi}_\theta \rangle = -\langle \dot{\psi}_\theta | \psi_\theta \rangle$, i.e. it is purely imaginary. Therefore, $\operatorname{Im}(\langle \dot{\psi}_\theta | \psi_\theta \rangle)^2 = |\langle \dot{\psi}_\theta | \psi_\theta \rangle|^2$. Inserting this in (C.140), gives

$$\mathcal{I}^Q(\psi_\theta) = 4(\langle \dot{\psi}_\theta | \dot{\psi}_\theta \rangle - |\langle \dot{\psi}_\theta | \psi_\theta \rangle|^2). \tag{C.146}$$

C.4.4 Solution (Text at p. 202)

Recall that the Fisher information is defined as

$$\mathcal{I}(p(x|\theta)) = \int dx \frac{1}{p(x|\theta)} \left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2, \tag{C.147}$$

where in our case the integral becomes a sum, since we always deal with only two possible cases.

(a) For input state 1, the derivatives are

$$\frac{\partial p(1|\theta)}{\partial \theta} = \frac{1}{2} \sin \theta \quad \text{and} \quad \frac{\partial p(2|\theta)}{\partial \theta} = -\frac{1}{2} \sin \theta, \tag{C.148}$$

from which we get

$$\mathcal{I}(\theta) = \frac{1}{\frac{1}{2}(1 - \cos \theta)} \left(\frac{1}{2} \sin \theta \right)^2 + \frac{1}{\frac{1}{2}(1 + \cos \theta)} \left(-\frac{1}{2} \sin \theta \right)^2 = \frac{1}{2} \sin^2 \theta \left(\frac{2}{1 - \cos^2 \theta} \right) = 1, \tag{C.149}$$

where in the last equality we used that $\sin^2 \theta = 1 - \cos^2 \theta$.

(b) For input state 2, the derivatives are

$$\frac{\partial p(+|\theta)}{\partial \theta} = -\sin(2\theta) \quad \text{and} \quad \frac{\partial p(-|\theta)}{\partial \theta} = \sin(2\theta), \tag{C.150}$$

from which we get

$$\mathcal{I}(\theta) = \frac{1}{\frac{1}{2}(1 + \cos(2\theta))} (-\sin(2\theta))^2 + \frac{1}{\frac{1}{2}(1 - \cos(2\theta))} (\sin(2\theta))^2 = 4 \frac{\sin^2(2\theta)}{1 - \cos^2(2\theta)} = 4, \tag{C.151}$$

where we used again that $\sin^2 \theta = 1 - \cos^2 \theta$.

(c) For input state 3, the derivatives are

$$\frac{\partial p(+|\theta)}{\partial \theta} = -\frac{N}{2} \sin(N\theta) \quad \text{and} \quad \frac{\partial p(-|\theta)}{\partial \theta} = \frac{N}{2} \sin(N\theta), \tag{C.152}$$

from which we get

$$\begin{aligned}
\mathcal{I}(\theta) &= \frac{1}{\frac{1}{2}(1 + \cos(N\theta))} \left(-\frac{N}{2} \sin(N\theta) \right)^2 + \frac{1}{\frac{1}{2}(1 - \cos(N\theta))} \left(\frac{N}{2} \sin(N\theta) \right)^2 \\
&= \frac{1}{2} \frac{N^2 \sin^2(N\theta)}{\frac{1}{2}(1 + \cos(N\theta))} + \frac{1}{2} \frac{N^2 \sin^2(N\theta)}{\frac{1}{2}(1 - \cos(N\theta))} = \frac{N^2}{2} \frac{2 \sin^2(N\theta)}{1 - \cos^2(N\theta)} = N^2.
\end{aligned} \tag{C.153}$$

Notice that the Fisher information $\mathcal{I} = N^2$ matches the Heisenberg limit discussed in Chapter 12.4. This is another example of how entanglement can enhance parameter estimation.

C.4.5 **Solution** (Text at p. 203)

From Eq (12.38) the quantum Fisher information for a pure state ψ is given by

$$\mathcal{I}^Q(\psi) = 4 \left(\langle \psi | H^2 | \psi \rangle - \langle \psi | H | \psi \rangle^2 \right). \quad (\text{C.154})$$

Since H is Hermitian, we let $\{|\lambda_k\rangle\}_k$ denote its eigenbasis. Next, we write ψ in this basis as $|\psi\rangle = \sum_k c_k |\lambda_k\rangle$. Therefore,

$$\mathcal{I}^Q(\psi) = 4 \sum_k \underbrace{|c_k|^2}_{=p(k)} \lambda_k^2 - 4 \left(\sum_k |c_k|^2 \lambda_k \right)^2 = 4 \sum_k p(k) \lambda_k^2 - 4 \underbrace{\left(\sum_k p(k) \lambda_k \right)^2}_{=\bar{\lambda}^2} = 4 \sum_k p(k) (\lambda_k - \bar{\lambda})^2, \quad (\text{C.155})$$

where we have written $p(k) = |c_k|^2$ to indicate that this can be viewed as a probability distribution. To maximise this expression, we should evidently put as much of the weight of the probability distribution as possible in front of the largest deviation from the mean $\bar{\lambda}$. This means that the two relevant values of k are those that correspond to $\lambda_{\min}$ and $\lambda_{\max}$. Hence we can write the probability of the former and latter as p and $1-p$ respectively. Therefore, we have $\bar{\lambda} = p\lambda_{\min} + (1-p)\lambda_{\max}$. Using this, the quantum Fisher information becomes

$$\begin{aligned} \mathcal{I}^Q(\psi) &= 4p(\lambda_{\min} - \bar{\lambda})^2 + 4(1-p)(\lambda_{\max} - \bar{\lambda})^2 \\ &= 4p(1-p)^2(\lambda_{\max} - \lambda_{\min})^2 + 4(1-p)p^2(\lambda_{\max} - \lambda_{\min})^2 = 4p(1-p)(\lambda_{\max} - \lambda_{\min})^2. \end{aligned} \quad (\text{C.156})$$

To maximise this expression, we must maximise the factor $p(1-p)$, which means choosing $p = \frac{1}{2}$. Then the quantum Fisher information becomes

$$\mathcal{I}^Q = (\lambda_{\max} - \lambda_{\min})^2 \quad (\text{C.157})$$

and the states which achieve this becomes $|\psi\rangle = \frac{1}{\sqrt{2}}|\lambda_{\max}\rangle + \frac{e^{i\phi}}{\sqrt{2}}|\lambda_{\min}\rangle$ for any ϕ . The freedom to choose the phase ϕ stems from the absolute value when defining $p(k) = |c_k|^2$.

C.4.6 **Solution** (Text at p. 203)

(a) To compute the quantum Fisher information for a pure state we follow Eq (12.38). We need to compute

$$\langle \psi^{(N)} | (H^{(N)})^2 | \psi^{(N)} \rangle = \langle \psi, \dots, \psi | (H_1 + \dots + H_N)^2 | \psi, \dots, \psi \rangle = \sum_{k=1}^N \langle \psi | H_k^2 | \psi \rangle + \sum_{k \neq l} \langle \psi | H_k | \psi \rangle \langle \psi | H_l | \psi \rangle, \quad (\text{C.158})$$

and

$$\begin{aligned} \langle \psi^{(N)} | H^{(N)} | \psi^{(N)} \rangle^2 &= \langle \psi, \dots, \psi | H_1 + \dots + H_N | \psi, \dots, \psi \rangle^2 = \left(\sum_{k=1}^N \langle \psi | H_k | \psi \rangle \right)^2 \\ &= \sum_{k=1}^N \langle \psi | H_k | \psi \rangle^2 + \sum_{k \neq l} \langle \psi | H_k | \psi \rangle \langle \psi | H_l | \psi \rangle. \end{aligned} \quad (\text{C.159})$$

Thus, in the N -partite quantum Fisher information, $\mathcal{I}_N^Q$, the cross-terms cancel and we get

$$\mathcal{I}_N^Q = \sum_{k=1}^N \langle \psi | H_k^2 | \psi \rangle - \sum_{k=1}^N \langle \psi | H_k | \psi \rangle^2 = N\mathcal{I}^Q, \quad (\text{C.160})$$

where $\mathcal{I}^Q$ is the quantum Fisher information of the single copy system.

(b) Since the operators $H_1, \dots, H_N$ all commute with each other, they all have the same eigenbasis. Hence, the smallest and largest eigenvalues of $H^{(N)}$ are $N\lambda_{\min}$ and $N\lambda_{\max}$, with associated eigenstates $|\lambda_{\min}, \dots, \lambda_{\min}\rangle$ and $|\lambda_{\max}, \dots, \lambda_{\max}\rangle$. The result then follows directly from the result in Exercise C.4.5.

INDEX

- I_{3322} Bell inequality, 142
- accessible information, 86
- action at distance property, 36
- associativity, 18
- basis measurement, 24
- BB84 protocol, 96
- Bell
 - basis, 57
 - inequality, 46
 - state, 57
 - state measurement, 57
- Bell's theorem, 49
- Bennett-Brassard 1984, 96
- Bessel's correction, 109
- biased CHSH inequality, 142
- bilinearity, 18
- bipartite state, 33
- Bloch
 - representation, 15
 - sphere, 12, 15
 - vector, 12, 15
- Born's rule, 10, 26, 31
- bra, 9
- capacity of quantum channel, 88
- ceasar cipher, 95
- chain rule, 77, 151
- channel capacity, 85
- Choi operator, 66
- Choi representation, 67
- Choi state, 66
- CHSH inequality, 47
- church of larger Hilbert space, 19
- ciphertext, 95
- classical data compression, 81
- Clauser-Horne-Shimony-Holt inequality, 47
- cloning attack, 98
- coherent attack, 100
- collective attack, 100
- compatible quantity, 25
- complete positivity, 65
- completely positive trace-preserving map, 65
- completeness, 10, 24
- computational basis, 11
- concurrence, 41
- conditional entropy, 77
- conditional quantum entropy, 78
- conditional von Neumann entropy, 78
- convex
 - combination, 13
- CPTP map, 65
- Cramér-Rao bound, 110
- damping rate, 103
- dense coding, 57, 144
- density
 - matrix, 13
 - operator, 13
- dephasing channel, 148
- Descartes' rule of signs, 17
- detailed balance, 200
- device-independent
 - certification, 90
 - QKD, 101
- distance measure, 19
- dual channel, 69
- E91 protocol, 100
- eavesdropping attack, 98
- Einstein-Podolsky-Rosen paradox, 45
- Ekert 91 protocol, 100
- entanglement
 - based QKD, 100
 - concurrence, 41
 - cost, 38
 - distillation, 38
 - formation, 38
 - measure, 40
 - monogamy, 44
 - negativity, 40
 - swapping, 60, 145
 - witness, 41, 71
- entanglement-breaking channel, 148
- entropy, 76
- EPR
 - paradox, 45
 - steering, 52
- error correction, 85
- expectation value, 12
- facet, 47, 141
- fidelity, 21
- fidelity witness, 43
- Fisher information, 110
- Fuchs-van de Graaf inequalities, 23
- gentle measurement, 74, 149
- GHZ
 - measurement, 145
 - paradox, 143
- GHZ state, 43
- Gisin's theorem, 141
- GKLS, 103
- Gleason's theorem, 31
- Greenberger-Horne-Zeilinger state, 43
- Hölder's inequality, 21
- Hahn-Banach theorem, 41
- Hardy's paradox, 143
- Heisenberg limit, 114
- Heisenberg picture, 69
- hemisphere discrimination, 74, 150

- Hermitian conjugate, 9
- Hilbert space, 9
- Holevo
 - bound, 86
 - information, 86
 - information of channel, 87
 - theorem, 86
- Holevo-Schumacher-Westmoreland theorem, 88
- Horodecki criterion, 50, 142
- individual attack, 100
- information reconciliation, 98
- information-disturbance, 73, 98
- inner product, 9
- isometry, 92
- isotropic state, 39, 140
- Jensen's inequality, 151
- joint typicality, 86
- jump operator, 103
- ket, 9
- Klein's inequality, 151
- Kraus operator, 69
- Kraus representation, 69
- Kronecker
 - delta, 19
 - product, 18
- Lüders
 - instrument, 72
 - measurement, 72
- Lagrange multiplier, 15
- latent variable, 108
- LHV model, 46
- Lindblad master equation, 103
- Liouville operator, 102, 107
- Liouvillian, 102, 107
- local hidden state model, 52
- local hidden variable, 46
- local isometry, 92
- local operations and classical communication, 35
- local polytope, 46
- LOCC, 35
- logarithmic negativity, 41
- logical bit, 85
- Maassen-Uffink uncertainty relation, 152
- Mach-Zehnder interferometer, 26
- majorisation, 36
- Markovian dynamics, 102
- Markovianity, 102
- master equation, 103
- maximally
 - entangled state, 33, 35, 139
 - mixed state, 15
- maximum likelihood estimator, 109
- Mayers-Yao self-test, 197
- mean-square error, 108
- measure-and-prepare channel, 148
- measurement certification, 91
- Mermin's Bell inequality, 144
- minimal error state discrimination, 28
- mixed state, 14
- monotonicity of relative entropy, 80
- mutual information, 77
- Naimark dilation, 30
- Naimark's theorem, 30
- negativity, 40
- Nielsen's majorisation theorem, 36
- no-cloning theorem, 63
- no-signaling principle, 53
- non-commutativity, 18, 25
- normal distribution, 108
- observable, 24
- observable variable, 108
- one-norm, 20
- one-time pad, 95
- open quantum system, 102
- orthogonal projector, 24
- partial
 - trace, 18
 - transpose, 39
- Pauli
 - matrix, 12
 - operator, 12
- phase-covariant cloning, 146
- physical bit, 85
- plaintext, 95
- polarisation measurement, 25
- Popescu-Rohrlich box, 53
- positive
 - operator-valued measure, 26
 - partial-transpose criterion, 39
 - semidefinite, 13
- POVM, 26
- PPT criterion, 39
- PR box, 53
- pre-order, 37
- pretty good measurement, 121
- privacy amplification, 97
- private key cryptosystem, 95
- product state, 33
- projection operator, 10
- projective measurement, 24
- public key cryptosystem, 95
- pure state, 14
- purification, 19
- purity function, 14
- QKD, 95
- quantum
 - certification, 90
 - composition, 18
 - Cramér-Rao bound, 112

- data compression, 82
- entropy, 78
- fidelity, 21
- Fisher information, 112, 113
- instrument, 72
- key distribution, 95
- measurement, 10, 24, 26
- mutual information, 79
- nonlocality, 49
- random access code, 62, 146
- relative entropy, 79
- repeater, 61
- state, 9
- state merging, 84
- state space, 17
- steering, 52
- Stein's lemma, 79
- qubit, 11
- random access code, 62
- rank-one projective measurement, 24
- ray, 9
- regularisation, 88
- remote state preparation, 145
- Robertson uncertainty relation, 25
- robust self-testing, 92
- sample average, 108
- Schmidt
 - coefficients, 35
 - decomposition, 34
 - rank, 35
 - vector, 35
- Schrödinger picture, 69
- Schumacher compression, 82
- Schumacher's source coding theorem, 82
- self-testing, 92
- separability, 39
- separable state, 39
- Shannon
 - entropy, 76
 - theory, 76
- Shannon's
 - source coding theorem, 81
- Shannon's noisy channel coding theorem, 85
- shot-noise limit, 112
- SIC-POVM, 27
- sifting, 96
- singular-value decomposition, 34
- six-state protocol, 199
- SLD, 112
- spectral decomposition, 13
- standard quantum limit, 112
- state
 - normalisation, 9
 - update, 10
- state-channel duality, 67
- steady state, 107
- Stern-Gerlach experiment, 24
- Stinespring
 - dilation, 70
 - representation, 70
- strong sub-additivity, 80
- sum-of-squared decomposition, 91
- superdense coding, 57
- SVD, 34
- SWAP isometry, 94, 198
- symmetric
 - informationally complete POVM, 27
 - logarithmic derivative, 112
- teleportation
 - fidelity, 58
 - protocol, 59
- tensor product, 18
- trace
 - distance, 20
 - norm, 20
- trine measurement, 27
- Tsirelson's bound, 49
- typical
 - set, 81
 - subspace, 83
- typical subspace, 152
- Uhlmann's theorem, 22
- unambiguous state discrimination, 29
- unbiased
 - estimator, 108
 - sample average, 109
- uncertainty relation, 25
- unital map, 70
- unitary operator, 10
- universal quantum cloning machine, 63
- UQCM, 63
- variance, 25
- vectorisation, 107
- Vigenère cipher, 95
- von Neumann
 - entropy, 78
 - equation, 102
 - measurement, 24
- W state, 43
- Werner's model, 51